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Quantum Physics

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Chapter 1

Background Toolkits

1.1 Prelude

Quantum physics and general relativity are the two foundations of modern sciences, which overcome the fundamental failure of Newtonian physics in explaining very basic observations in two limiting domains. One is in the limit of small particles and another is in the limit of large velocities. These two revolutionary theories change the understanding on what realistic world ought to be and how physic objects behave. It is worth noting that quantum physics deals with not only small particles (such as electrons, photons, atoms and molecules), but also the related properties and functionalities of macroscopic matter systems.

This textbook deals with nonrelativity quantum physics. As any emerging fields of science, the birth and development of quantum physics follows the basic principles of natural philosophy and the scientific aesthetics, involving *reduction*, *induction* and *deduction* methods. This was the case when Newton presented his three laws of motion in the “Principia Mathematica Philosophiae Naturalis” (Mathematical Principles of Natural Philosophy) in 1686. The birth of quantum physics started with solving the puzzles observed in experiments that beyond the understanding of classical physics. These include the blackbody radiations, photoelectric current phenomenon, the spectroscopy of hydrogen atoms, Compton scattering experiment, and the Stern–Gerlach experiment, which you may have more or less learnt before.

Further establishment of quantum physics engaged the *reductionism*. This is to pin down the fundamental deficiency in classical physics, identify the true nature about matter and wave, and construct *inductive* postulations for further *deductive* developments and experimental verifications. As

the reductionism is concerned, the **wave–particle–duality** is most fundamental. This true nature about matter and wave had been identified, upon solving the puzzles observed in the aforementioned experiments beyond the understanding of classical physics. The subsequent inductions and related terminologies are now daily used, such as the “photon” and “electronic microscopy”, reflecting the particle nature of light wave and the wave nature of electron, respectively. The Newton’s second law, with the trajectory description of the motion of particles, is not longer valid in microscopic world. Instead, a wave function description is called for. This is the Schrödinger equation proposed in 1926, together with the Born’s probability postulation on the wave function amplitude square, for the underlying wave–particle–duality.

Logic and further mathematic *deductions* are indispensable in the development of quantum physics. It is important to be active learners: If you were a college student in early years of 1900, witnessed those basic results that unambiguously head to a revolutionary field of science, would you had gained enough inspirations and passions to make your contributions? This is the strongly recommended attitude.

I would also emphasize the learning strategy, as reflected in this textbook, toward optimizing memory, conceptual understanding, enhancing motivated and self-directed learning. This is the reason that we present this first chapter an comprehensive overview on the related knowledge you had learnt before, so that the later chapters about quantum physics can go with minimal disturbance. To that end, this background chapter presents the basic tools and skills, with some simple but conceptual remarks that are mainly for your later references and comparisons.

1.2 Mathematical Toolkits

1.2.1 Basic skills

A. Ordinary differential equation

For example,

$$\frac{\partial}{\partial x} f(x) = cf(x) \implies f(x) = e^{cx} f(x=0)$$

Consider now

$$\frac{\partial^2}{\partial x^2} f(x) = -k^2 f(x) \quad \text{where } k \text{ is a real parameter}$$

(i) e^{ikx} and e^{-ikx} are two solutions to the above ODE, so that

$$f(x) = a_1 e^{ikx} + a_2 e^{-ikx}$$

with the initial values $f(0) = a_1 + a_2$ and $f'(0) = i(a_1 - a_2)k$, where $f'(x) \equiv \partial f(x)/\partial x$.

(ii) In terms of $\cos(kx)$ and $\sin(kx)$ that are also solutions,

$$f(x) = c_1 \cos(kx) + c_2 \sin(kx)$$

with $f(0) = c_1$ and $f'(0) = kc_2$.

B. Matrix/vector algebra

- Matrix/vector sum

$$\text{If } \mathbf{A} = \begin{bmatrix} A_{11} & \cdots & A_{1N} \\ \cdots & \cdots & \cdots \\ A_{N1} & \cdots & A_{NN} \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} B_{11} & \cdots & B_{1N} \\ \cdots & \cdots & \cdots \\ B_{N1} & \cdots & B_{NN} \end{bmatrix}, \text{ then}$$

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} A_{11} + B_{11} & \cdots & A_{1N} + B_{1N} \\ \cdots & \cdots & \cdots \\ A_{N1} + B_{N1} & \cdots & A_{NN} + B_{NN} \end{bmatrix}$$

Abbreviate $(\mathbf{A} + \mathbf{B})_{jk} = A_{jk} + B_{jk}$

- Matrix and c-number multiplication:

$$\lambda \mathbf{A} = \begin{bmatrix} \lambda A_{11} & \cdots & \lambda A_{1N} \\ \cdots & \cdots & \cdots \\ \lambda A_{N1} & \cdots & \lambda A_{NN} \end{bmatrix}$$

The c-numbers refer to convention commutable numbers or variables that can be complex, satisfying $yx = xy$.

- Matrixes multiplication,

$$\mathbf{A}\mathbf{v} = \begin{bmatrix} A_{11} & \cdots & A_{1N} \\ \cdots & \cdots & \cdots \\ A_{N1} & \cdots & A_{NN} \end{bmatrix} \begin{bmatrix} v_1 \\ \cdots \\ v_N \end{bmatrix} = \begin{bmatrix} A_{11}v_1 + \cdots + A_{1N}v_N \\ \cdots \\ A_{N1}v_1 + \cdots + A_{NN}v_N \end{bmatrix}$$

Denote $\mathbf{C} = \mathbf{AB}$, then

$$C_{jk} = (\mathbf{AB})_{jk} = \sum_i A_{ji}B_{ik}$$

- Unit matrix versus Kronecker delta notation

$$\mathbf{I}_{jk} = \delta_{jk} \equiv \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

- Matrix inverse $\mathbf{A}^{-1}$, satisfying

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I} \quad \text{and} \quad (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

- Matrix/vector transpose (T) and adjoint († , “dagger”)

$$\mathbf{A}^T = \begin{bmatrix} A_{11} & \cdots & A_{N1} \\ \cdots & \cdots & \cdots \\ A_{1N} & \cdots & A_{NN} \end{bmatrix} \quad \text{and} \quad \mathbf{A}^\dagger \equiv (\mathbf{A}^*)^T = \begin{bmatrix} A_{11}^* & \cdots & A_{N1}^* \\ \cdots & \cdots & \cdots \\ A_{1N}^* & \cdots & A_{NN}^* \end{bmatrix}$$

Evidently,

$$(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T \quad \text{and} \quad (\mathbf{AB})^\dagger = \mathbf{B}^\dagger\mathbf{A}^\dagger$$

- **Matrix trace:**

$$\boxed{\text{Tr}\mathbf{A} \equiv A_{11} + A_{22} + \cdots + A_{NN}}$$

- Matrix determinant, such as

$$\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} \quad \text{and} \quad \begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix}$$

In general,

$$|\mathbf{A}| \equiv \det \mathbf{A} = \sum_{k=1}^{N!} (-1)^{p_k} \mathcal{P}_k(A_{11}A_{22} \cdots A_{NN})$$

where $\mathcal{P}_k$ denotes permutation of the column indices, $k = 1, 2, \cdots, N$ and the sum runs all $N!$ permutations; p_k is the number of transpositions requires to restore the given $k_1, k_2, \cdots, k_N$ to the nature order $1, 2, \cdots, N$.

Properties: (1) A swap of any two columns (or rows) results in a negative sign; (2) $|\mathbf{A}^\dagger| = |\mathbf{A}|^*$; (3) $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$.

• **Unitary matrix versus unitary transformation:**

(i) Unitary matrix $\mathbf{U}$ satisfies $\mathbf{U}^\dagger \mathbf{U} = \mathbf{I} = \mathbf{U} \mathbf{U}^\dagger$; i.e.,

$$\sum_i U_{ij}^* U_{ik} = \delta_{jk} \quad \text{or} \quad \langle U_j | U_k \rangle = \delta_{jk}$$

When $\mathbf{U}$ is real, it becomes an orthogonal matrix such as

$$\mathbf{U} = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

that describes a rotation. Different reference coordinates representations are related each other by unitary transformations, which are irrelevant to physics and geometry. This feature is named the *representation invariance*.

(ii) Unitary transformations on matrix: $\mathbf{A}' = \mathbf{U}^\dagger \mathbf{A} \mathbf{U}$. Matrix trace and determinant are unitary transformation invariance; i.e.,

$$\text{Tr}(\mathbf{U}^\dagger \mathbf{A} \mathbf{U}) = \text{Tr} \mathbf{A} \quad \text{and} \quad |\mathbf{U}^\dagger \mathbf{A} \mathbf{U}| = |\mathbf{A}|$$

C. Basis set versus representation

First of all, a representation refers to the matrix/vector being specified with their elements. This is related to the choice of a set of reference vectors that are orthogonal, normalized and completeness. For example,

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = v_1 \mathbf{i} + v_2 \mathbf{j}, \quad \text{where} \quad \begin{aligned} v_1 &= \mathbf{i}^T \mathbf{v} \equiv \mathbf{i} \cdot \mathbf{v} \\ v_2 &= \mathbf{j}^T \mathbf{v} \equiv \mathbf{j} \cdot \mathbf{v} \end{aligned}$$

Moreover, for a 2×2 matrix $\mathbf{A} = \{A_{ij}\}$, we have $A_{12} = \mathbf{i}^T \mathbf{A} \mathbf{j}$ and so on. The involving basis set $(\mathbf{i}, \mathbf{j})$ satisfies

$$\begin{aligned} \text{(a) } \mathbf{i} \cdot \mathbf{j} &= 0 \quad (\text{orthogonal}), & \text{(b) } \mathbf{i} \cdot \mathbf{i} &= \mathbf{j} \cdot \mathbf{j} = 1 \quad (\text{normalized}) \\ \text{(c) } \mathbf{i} \mathbf{i}^T + \mathbf{j} \mathbf{j}^T &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (\text{completeness}) \end{aligned}$$

The completeness here is concerned with two-dimension space.

1.2.2 Hilbert-space algebra and Dirac notations

The *Hilbert-space* is a linear space in mathematics and extensively used in quantum mechanics. To facilitate the later study of quantum physics, it is worth to review the related algebra in this subsection. The Hilbert-space comprises three basic ingredients: (i) **Vectors** (call also *wave vectors* or *wave functions*) that are complex in general; (ii) **Operators** (refer also to *matrixes*), with *matrix-vector multiplication* being the rule of operation, to bring one vector to another. (iii) **Inner product** of any two vectors be

a commutable *c-number*; (iv) The self-inner product is the vector's length square that must be positive.

• **Dirac ket and bra notations:**

$$\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \Rightarrow |u\rangle, \quad \text{then} \quad \mathbf{u}^\dagger = [u_1^*, \dots, u_n^*] \Rightarrow \langle u| \quad (1.2.1a)$$

$$\mathbf{A}\mathbf{v} \Rightarrow \hat{A}|v\rangle \quad (\text{Matrix } \mathbf{A} \Rightarrow \text{Operator } \hat{A}) \quad (1.2.1b)$$

• **Inner product** of two vectors is a c-number, defined with

$$\langle u|v\rangle = [u_1^*, \dots, u_n^*] \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = u_1^* v_1 + \dots + u_n^* v_n \quad (1.2.2)$$

– *Normalized* vector refers to its length $\langle v|v\rangle^{\frac{1}{2}} = 1$ (in given unit)

– *Orthogonal* vectors refer to $\langle u|v\rangle = 0$

• **Outer product** of two vectors is an operator (a matrix)

$$|v\rangle\langle u| = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} [u_1^*, \dots, u_n^*] = \begin{bmatrix} v_1 u_1^* & \dots & v_1 u_n^* \\ \dots & \dots & \dots \\ v_n u_1^* & \dots & v_n u_n^* \end{bmatrix} \quad (1.2.3)$$

Exercise E1.1: (i) Show that

$$\text{Tr}[|v\rangle\langle u|] = \langle u|v\rangle = \langle v|u\rangle^* \quad (1.2.4)$$

(ii) Does the value of $\langle u|v\rangle$ depend on the representation?

(iii) Consider $|u\rangle = u_\alpha|\alpha\rangle + u_\beta|\beta\rangle$, where $\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$ and $\langle\alpha|\beta\rangle = 0$. Show that

$$u_\alpha = \langle\alpha|u\rangle \quad \text{and} \quad \langle u|v\rangle = u_\alpha^* v_\alpha + u_\beta^* v_\beta$$

(iv) Prove further the *Schwartz inequality*,

$$\boxed{\langle u|v\rangle\langle v|u\rangle \leq \langle u|u\rangle\langle v|v\rangle} \quad (1.2.5)$$

(v) Discuss the geometric pictures of (ii)–(iv).

(vi) For a basis set, $\{|n\rangle\}$, express its *orthogonality* and *completeness*.

• **Hermitian matrix** (self-adjoint):

$$\mathbf{A} = \mathbf{A}^\dagger \quad (\text{Hermitian})$$

This amounts to $A_{jk} = A_{kj}^*$. For example, $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ is Hermitian; but $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ is not.

In parallel, **Hermitian operator** satisfies $\hat{A} = \hat{A}^\dagger$ that is equivalent to

$$\langle u | \hat{A} | v \rangle^* = \langle v | \hat{A}^\dagger | u \rangle = \langle v | \hat{A} | u \rangle \quad (\hat{A} = \hat{A}^\dagger) \quad (1.2.6)$$

The first identity above is valid for an arbitrary operator, not matter whether it is Hermitian or not.

Remark:

• **Classical-and-quantum correspondence:** Dynamical variables, such as the coordinate x , momentum p and the Hamiltonian function, $H(p, x)$, all becomes Hermitian operators, $\hat{x}$, $\hat{p}$ and $\hat{H}$, in quantum mechanics. As seen in Chapter 2, the identification of wave-particle-duality gives rise to $[\hat{x}, \hat{p}] = i\hbar/2$, where $[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A}$ denotes the *commutator*.

• Note that $\text{Tr}(\mathbf{AB}) = \text{Tr}(\mathbf{BA})$ for matrixes, since they are “finite supported” with finite trace values. This may not be the case, if both $\hat{A}$ and $\hat{B}$ are dynamical operators, such as $\text{Tr}(\hat{x}\hat{p}) \neq \text{Tr}(\hat{p}\hat{x})$. Nevertheless, *all traces of physical interest are finite supported*. The aforementioned “operator-to-matrix isomorphism” is practically valid.

Exercise E1.2

(i) Show that (a) the commutator between two Hermitian operators is anti-Hermitian, satisfying $[\hat{A}, \hat{B}]^\dagger = -[\hat{A}, \hat{B}] = [\hat{B}, \hat{A}]$; (b) $e^{i\lambda\hat{A}}$ is unitary, where λ is a real c-number.

(ii) Show that

$$(a) [\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}; \quad (b) [\hat{A}, \hat{B}\hat{C}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C}$$

$$(c) \left[\frac{\partial^n}{\partial x^n}, x \right] = n \frac{\partial^{n-1}}{\partial x^{n-1}}; \quad (d) \left[\frac{\partial}{\partial x}, f(x) \right] = f'(x)$$

1.2.3 Theorems of Hermitian operators

Statements of the theorems:

- (i) The eigenvalues of a Hermitian operator are real;
- (ii) The eigenvectors associated with different eigenvalues are orthogonal.

• Terminologies:

- (a) The vectors $|u\rangle$ and $|v\rangle$ are orthogonal, if $\langle u|v\rangle = 0$;
- (b) The vector $|u\rangle$ is normalized if $\langle u|u\rangle = 1$;
- (c) The *eigenequation* of operator $\hat{A}$ refers to

$$\boxed{\hat{A}|v\rangle = \lambda|v\rangle} \quad (1.2.7)$$

with $|v\rangle$ and λ being the *eigenvector* and the associated *eigenvalue*, respectively. Note that there may be multiple eigenvectors, $|v_1\rangle, \dots, |v_m\rangle$, having a same eigenvalue $\lambda = \lambda_1 = \dots = \lambda_m$, with the multiplicity of m .

• Proofs of the Theorems:

Theorem-(i): The eigenvalues of a Hermitian operator are real.

Proof: From the eigenequation, $\hat{A}|v\rangle = \lambda|v\rangle$, we have

$$\langle v|\hat{A}|v\rangle = \lambda\langle v|v\rangle$$

Taking complex conjugate, following by using $\hat{A}^\dagger = \hat{A}$, we have further

$$\langle v|\hat{A}|v\rangle = \lambda^*\langle v|v\rangle$$

The above two equations result in $(\lambda - \lambda^*)\langle v|v\rangle = 0$, with noting that $\langle v|v\rangle > 0$. Therefore, $\lambda = \lambda^*$ that is real.

Theorem-(ii): The eigenvectors of a Hermite operator, $\hat{A} = \hat{A}^\dagger$, associated with different eigenvalues are orthogonal.

Proof: Let us start with recasting Eq. (1.2.7) as $\hat{A}|v'\rangle = \lambda'|v'\rangle$. Taking the Hermitian conjugate, followed by using the properties of $\hat{A}^\dagger = \hat{A}$ and λ' being real, we have

$$\langle v'|\hat{A} = \lambda'\langle v'| \quad \implies \quad \langle v'|\hat{A}|v\rangle = \lambda'\langle v'|v\rangle$$

Together with $\langle v'|\hat{A}|v\rangle = \lambda\langle v'|v\rangle$ via Eq. (1.2.7), we obtain

$$(\lambda - \lambda')\langle v'|v\rangle = 0$$

Thus, $\langle v'|v\rangle = 0$, if $\lambda' \neq \lambda$. This proves the **Theorem-(ii)** above.

Remarks and Exercises

- A Hermitian operator, $\hat{A} = \hat{A}^\dagger$, can always be diagonalized with a unitary operator, $\hat{U}$ that satisfies $\hat{U}^\dagger \hat{U} = \hat{I}$, such that

$$\hat{U}^\dagger \hat{A} \hat{U} = \text{diag}\{\lambda_1, \lambda_2, \dots\} \quad (1.2.8)$$

Consider now a Hermitian operator $\hat{A} = \hat{A}^\dagger$ in an arbitrary unitary transformation, $\hat{A}' = \hat{U}^\dagger \hat{A} \hat{U}$. Show that (i) $\hat{A}'$ remains Hermitian and (ii) the eigenvalues are invariance with respect to unitary transformation.

- The system's Hamiltonian operator $\hat{H}$ is Hermitian and its eigenequation is the time-independent Schrödinger equation,

$$\hat{H}|\psi\rangle = E|\psi\rangle \quad (1.2.9)$$

The solutions are now the system's eigenenergies and eigenvectors that are often given with normalization, satisfying $\langle\psi|\psi\rangle = 1$.

Exercise: Let $\hat{H} = \begin{bmatrix} 0 & \beta \\ \beta & 0 \end{bmatrix}$. Solve Eq. (1.2.9) and discuss the solutions, assuming $\beta < 0$.

- Schrödinger equation (time-dependent) reads

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle \quad (1.2.10)$$

Verify the following results:

- (i) If $\Psi_1(t)$ and $\Psi_2(t)$ are the solutions, $c_1\Psi_1(t) + c_2\Psi_2(t)$ must also be.
- (ii) Under given $\hat{H}$, the initial $\Psi(t=0)$ completely determines $\Psi(t)$.
- (iii) The formal solution reads (assuming $\hat{H}$ is time-independent)

$$|\Psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\Psi(0)\rangle \quad (1.2.11)$$

which is equivalent to

$$|\Psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |\psi_n\rangle \quad (1.2.11')$$

Here, $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$, the eigenequation of the Hamiltonian operator.

(iv) As seen Eq. (1.2.11), $\hat{U}(t) \equiv e^{-i\hat{H}t/\hbar}$ governs the wave vector evolution and is called the *propagator* that is unitary.

- For $f(\hat{A})$, a function of the Hermite operator, there are two equivalent expressions, in terms of the Taylor's expansion and the eigen solutions, respectively. The latter goes by (*Prove it if you can*)

$$f(\hat{A}) = \sum_n f(\lambda_n) |\phi_n\rangle \langle \phi_n|, \quad \text{where } \hat{A}|\phi_n\rangle = \lambda_n |\phi_n\rangle \quad (1.2.12)$$

1.3 Classical Mechanics and Remarks

1.3.1 Newton's Laws and remarks

Classical mechanics is the limiting scenario of quantum physics whenever the Planck constant ($\hbar$) can effectively be neglected. It is worth reviewing classical mechanism in this section, together with the conceptual level remarks on the quantum correspondences and/or analogues, by simple considering the wave-particle duality. These remarks are mainly for your reference when you study quantum physics in later chapters.

The First Law: “*A body at rest will remain at rest, and a body in motion will remain in motion unless it is acted upon by an external force*”. The Newton's First Law naturally implies the Galileo's law of inertial: (i) The property of a massive body, in terms of the momentum $\mathbf{p} = m\dot{\mathbf{x}}$, to resist changes in its state of motion, is called **inertia**, giving rise to the concept of **inertial reference frames**; (ii) An inertial reference frame is a 3-dimensional space that can be in uniform linear motion with respect to some other inertial reference frame; (iii) All inertial reference frames are physically identical.

Remarks (Only the conceptual understanding is needed now, as you will learn the details in Chapter 2):

- Quantum physics is rooted at the wave-particle duality. In particular, a force free-moving particle, having the momentum $\mathbf{p}$, is a *plane wave*, with the wave vector of $\mathbf{k} = \mathbf{p}/\hbar$ (and vice versa, $\mathbf{p} = \hbar\mathbf{k}$, for the particle nature of a plane light wave). Here, $\hbar = h/(2\pi) = 1.05457 \times 10^{-34} \text{ J s}$ is the (reduced) Planck constant. The above de Broglies relation had been extensively verified and widely used in such as electronic microscopy and neutron scattering experiments.

- The principle of reference frames invariance, originally implied in the Newton's First Law, remains true in quantum theory, manifesting further in terms of both the representation invariance and the prescription invariance.

(i) **Representation invariance:** First of all, quantum mechanics states are described by the wave vectors that constitute a linear space (the *Hilbert-space*) algebra in mathematics. The wave functions, $\psi(\mathbf{x}, t)$, are just the (daily-life) real-space representation of wave vectors. In analogy to the geometric vectors, the representation invariance refers to the lengths of individual vectors and the relative orientation between two vectors are invariance in different reference frames.

(ii) **Prescription invariance:** In quantum mechanics, the *Schrödinger picture* refers to the time evolutions of state wave vectors that satisfy the time-dependent Schrödinger equation. The dynamic variables such as coordinates and momentums are fixed. The *Heisenberg picture* is the opposite prescription and the *interaction picture* is an intermediate in between. The prescription invariance refers to

the fact that experimentally observations formulated in these equivalent pictures are all identical.

The Second Law: The dynamics of classical particles follows

$$\mathbf{F} = m\mathbf{a} = m \frac{d^2 \mathbf{x}}{dt^2} = \frac{d\mathbf{p}}{dt} \equiv \dot{\mathbf{p}} \quad (1.3.1)$$

For a *closed* (external field-free) many-particle system, all forces are internal from the system's potential $V(\mathbf{x})$,

$$F_x = -\frac{\partial}{\partial x} V(\mathbf{x}) \quad \text{or} \quad \mathbf{F} = -\nabla V \quad (1.3.2)$$

Combining Eqs. (1.3.1) and (1.3.2) leads to

$$\dot{\mathbf{p}} = m\ddot{\mathbf{x}} = -\nabla V(\mathbf{x}) \quad (1.3.3)$$

The solution to Eq. (1.3.3), the classical *trajectory*, is completely determined by the given pair of initial position and momentum vectors, $\mathbf{x}(t=0)$ and $\mathbf{p}(t=0)$, which constitute a *point* in the classical phase space.

In contrast, the quantum state is a phase-space *wave packet*, due to the wave-particle duality. We will learn that the *minimum uncertainty* is of $\Delta x \Delta p_x \simeq \hbar$ in each dimension, where $\Delta A \equiv \langle A^2 - \bar{A}^2 \rangle^{\frac{1}{2}}$, is the standard deviation in statistics, with respect to the wave packet distribution function (*a la* the Born's probability postulation).

The Third Law (the law of action and reaction): When two bodies interact, they apply forces to one another that are equal in magnitude and opposite in direction. This law is valid in general and related to the *total linear momentum conservation*. This together with the total energy conservation are exploited in such as the Compton scattering experiment (1923) that verifies the particle nature of light wave [A. H. Compton, *Phys. Rev.* **21**, 483 (1923); **22**, 409 (1923)].

Beside the total linear momentum conservation and energy conservation, there is also the total angular momentum conservation. All these remain important in quantum mechanics. It is also noticed that the Newton's Second Law, Eq. (1.3.1) and the resulting Eq. (1.3.3), is written in term of force that is a vector in real (i.e. Euclidean) space. Alternately, Eq. (1.3.3) can be recast in terms of energy-type functions, discussed in Analytical Mechanics. As scalar properties, energy functions readily support the generalized coordinates and conjugate momenta description. In the coming subsection, we will present the related formulations and key results. Again, these are for the *cross-references* when you study quantum mechanics counterparts in later chapters.

1.3.2 Lagrange's and Hamilton's formalisms

A. The basic formulations

First of all, the *Hamiltonian function describes the total energy of system*, in terms of the *generalized coordinates and conjugate momentums*. The resulting Hamilton's formalism refers the canonical equation of classical mechanics [cf. Eq. (1.3.9)]. It is the Hamiltonian function that corresponds to the *Hamiltonian operator* in quantum mechanics.

On the other hand, the Lagrange's formalism is the most versatile approach to analytical mechanics, facilitating various problems under complex constraints, which are however largely limited to classical systems. In the following, we present only the key ingredients of the aforementioned two formalisms that are equivalent to the Newton's Second Law.

• **Lagrangian function**, $L(\mathbf{q}, \dot{\mathbf{q}}; t)$, in terms of generalized coordinates $\mathbf{q} \equiv \{q_\alpha\}$ and velocities $\dot{\mathbf{q}} = \{\dot{q}_\alpha\}$, describes the difference between the kinetic and potential energies:

$$L(\mathbf{q}, \dot{\mathbf{q}}; t) \equiv K(\dot{\mathbf{q}}, \mathbf{q}) - V(\mathbf{q}; t) \quad (1.3.4)$$

Potential energy $V(\mathbf{q}; t)$ involves the generalized coordinates only, including couplings with time-dependent external fields. Kinetic energy $K(\dot{\mathbf{q}}, \mathbf{q})$ depends on velocities *quadratically*, whereas non-Cartesian coordinates would also enter in general. The Lagrange's formalism reads

$$\boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_\alpha} \right) - \frac{\partial L}{\partial q_\alpha} = 0} \quad (1.3.5)$$

(It is obtained in general via the *principle of least action* or *Hamilton's principle*)

• **Hamiltonian function**, $H(\mathbf{p}, \mathbf{q}; t) \equiv K + V$, describes the total energy, in terms of generalized coordinates and conjugate momentums. It is related to the Lagrangian via the Legendra transformation (changing variables):

$$H(\mathbf{p}, \mathbf{q}; t) = \sum_{\alpha} p_{\alpha} \dot{q}_{\alpha} - L(\mathbf{q}, \dot{\mathbf{q}}; t) \quad (1.3.6)$$

with the conjugate momentum the universal expression,

$$\boxed{p_{\alpha} = \frac{\partial L(\mathbf{q}, \dot{\mathbf{q}}; t)}{\partial \dot{q}_{\alpha}}} \quad (1.3.7)$$

The Hamilton's formalism via Eqs. (1.3.5)–(1.3.7) has the canonical form,

$$\boxed{\dot{q}_{\alpha} = \frac{\partial H}{\partial p_{\alpha}}}, \quad \boxed{\dot{p}_{\alpha} = -\frac{\partial H}{\partial q_{\alpha}}} \quad (1.3.8)$$

engaging the generalized coordinate and momentum in a similar manner.

B. Angular momentum conservation, which reads

$$\boxed{\frac{d}{dt} \sum_n (\mathbf{r}_n \times \mathbf{p}_n) = 0} \quad (1.3.9)$$

Physically, this is concerned with a rigid many-particles system rotates as a whole in real space. The sum runs over all composing particles, each contributes the angular momentum of $\mathbf{r}_n \times \mathbf{p}_n$, where $\mathbf{r}_n = x_n \vec{i} + y_n \vec{j} + z_n \vec{k}$ in the Cartesian space. While Eq. (1.3.9) holds in general, the value of the angular momentum itself depends on the point with respect to which it is defined. *The convention choice of this point is the center of mass.*

The proof of Eq. (1.3.9): It is noticed that the Lagrangian function is unchanged against *any* rigid-body rotations. That is

$$\begin{aligned} 0 = \delta L &= \sum_n \left(\frac{\partial L}{\partial \mathbf{r}_n} \delta \mathbf{r}_n + \frac{\partial L}{\partial \dot{\mathbf{r}}_n} \delta \dot{\mathbf{r}}_n \right) \\ &= \sum_n \left[\frac{\partial L}{\partial \mathbf{r}_n} \cdot (\delta \boldsymbol{\Omega} \times \mathbf{r}_n) + \frac{\partial L}{\partial \dot{\mathbf{r}}_n} \cdot (\delta \boldsymbol{\Omega} \times \dot{\mathbf{r}}_n) \right] \end{aligned} \quad (1.3.10)$$

with an infinitesimal rotation vector, $\delta \boldsymbol{\Omega}$, so that $\delta \mathbf{r}_n = \delta \boldsymbol{\Omega} \times \mathbf{r}_n$ and $\delta \dot{\mathbf{r}}_n = \delta \boldsymbol{\Omega} \times \dot{\mathbf{r}}_n$. Moreover, $\partial L / \partial \dot{\mathbf{r}}_n = \mathbf{p}_n$ via Eq. (1.3.7) and $\partial L / \partial \mathbf{r}_n = -\partial V / \partial \mathbf{r}_n = \dot{\mathbf{p}}_n$ via Eq. (1.3.3), since the kinetic energy does not depend on the Cartesian coordinates. Substituting the above two identities into Eq. (1.3.10), followed by using the determinant identity of $\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A})$, we obtain

$$\begin{aligned} 0 &= \sum_n [\dot{\mathbf{p}}_n \cdot (\delta \boldsymbol{\Omega} \times \mathbf{r}_n) + \mathbf{p}_n \cdot (\delta \boldsymbol{\Omega} \times \dot{\mathbf{r}}_n)] \\ &= \delta \boldsymbol{\Omega} \cdot \sum_n (\mathbf{r}_n \times \dot{\mathbf{p}}_n + \dot{\mathbf{r}}_n \times \mathbf{p}_n) \\ &= \delta \boldsymbol{\Omega} \cdot \sum_n \frac{d}{dt} (\mathbf{r}_n \times \mathbf{p}_n) \end{aligned}$$

This completes the proof of Eq. (1.3.9).

(Q.E.D.)

C. Momentums via the free-particle scenario

As known, generalized momentums appear only in the kinetic term. It is sufficient to consider them only in the potential-free particle setup, with the Lagrangian,

$$L = \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{dl}{dt} \right)^2 \quad (\text{free-particle}) \quad (1.3.11)$$

Here, dl is the *differential distance*, for the free particle traveled in $[t, t+dt]$. The evaluation starts with the Cartesian coordinates; i.e.,

$$(dl)^2 = (dx)^2 + (dy)^2 + (dz)^2 \quad (1.3.12)$$

Exemplified below will be two generalized coordinates scenarios.

• **Polar Coordinates:**

$$x = \varrho \cos \phi, \quad y = \varrho \sin \phi, \quad z = z$$

We obtain

$$dx = \cos \phi d\varrho - \varrho \sin \phi d\phi \quad \text{and} \quad dy = \sin \phi d\varrho + \varrho \cos \phi d\phi$$

Hence,

$$(dl)^2 = (dx)^2 + (dy)^2 + (dz)^2 = (d\varrho)^2 + \varrho^2 (d\phi)^2 + (dz)^2$$

The resulting Eq. (1.3.11) is then

$$L = \frac{1}{2}m(\dot{\varrho}^2 + \varrho^2 \dot{\phi}^2 + \dot{z}^2) \quad (\text{free-particle}) \quad (1.3.13)$$

The conjugate momentums are [cf. Eq. (1.3.7)]

$$p_\varrho = m\dot{\varrho}, \quad p_\phi = m\varrho^2 \dot{\phi}, \quad p_z = m\dot{z} \quad (1.3.14)$$

The Hamiltonian counterpart to Eq. (1.3.13) is

$$H = \frac{1}{2m}(p_\varrho^2 + p_z^2) + \frac{p_\phi^2}{2m\varrho^2} \quad (1.3.15)$$

• **Spherical Coordinates:**

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

and the corresponding

$$\begin{aligned} dx &= \sin \theta \cos \phi dr - r \cos \theta \cos \phi d\theta - r \sin \theta \sin \phi d\phi \\ dy &= \sin \theta \sin \phi dr + r \cos \theta \sin \phi d\theta + r \sin \theta \cos \phi d\phi \\ dz &= \cos \theta dr - r \sin \theta d\theta \end{aligned}$$

Hence [cf. Eq. (1.3.12)],

$$(dl)^2 = (dr)^2 + r^2 (d\theta)^2 + r^2 \sin^2 \theta (d\phi)^2$$

The resulting Eq. (1.3.11) reads

$$L = \frac{1}{2}m(\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) \quad (\text{free-particle}) \quad (1.3.16)$$

The conjugate momentums are [cf. Eq. (1.3.7)]

$$p_r = m\dot{r}, \quad p_\theta = mr^2\dot{\theta}, \quad p_\phi = mr^2 \sin^2 \theta \dot{\phi} \quad (1.3.17)$$

The Hamiltonian counterpart to Eq. (1.3.16) is

$$H = \frac{1}{2m}p_r^2 + \frac{1}{2mr^2} \left(p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta} \right) = \frac{1}{2m}p_r^2 + \frac{J^2}{2mr^2} \quad (1.3.18)$$

– **Angular momentum:**

$$\mathbf{J} = \mathbf{r} \times \mathbf{p} = (yp_z - zp_y)\mathbf{i} + (zp_x - xp_z)\mathbf{j} + (xp_y - yp_x)\mathbf{k} \quad (1.3.19)$$

Alternatively, angular momentum around any axis, $\mathbf{n}$, can be found via

$$J_{\mathbf{n}} = \frac{\partial L}{\partial \dot{\Omega}_{\mathbf{n}}} \quad (1.3.20)$$

In particular, Eq. (1.3.16) results in immediately

$$J_z = \frac{\partial L}{\partial \dot{\phi}} = mr^2 \sin^2 \theta \dot{\phi} = p_\phi \quad (1.3.21)$$

We can also obtain (*for reference only*)

$$\begin{bmatrix} J_x \\ J_y \end{bmatrix} = - \begin{bmatrix} \sin \phi & \cot \theta \cos \phi \\ -\cos \phi & \cot \theta \sin \phi \end{bmatrix} \begin{bmatrix} p_\theta \\ p_\phi \end{bmatrix} \quad (1.3.22)$$

The above expressions lead to immediately

$$J^2 = p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta} \quad (1.3.23)$$

This leads to Eq. (1.3.18) the last identity.

1.3.3 Poisson equation of motion and remarks

- Let $A(\mathbf{p}, \mathbf{q})$ be an arbitrary dynamic function of a mechanics system in the absence of time-dependent external field. We have

$$\frac{dA}{dt} = \sum_{\alpha} \left[\frac{\partial A}{\partial q_{\alpha}} \dot{q}_{\alpha} + \frac{\partial A}{\partial p_{\alpha}} \dot{p}_{\alpha} \right]$$

Together with the Hamilton's formalism, Eq. (1.3.9), we obtain the Poisson equation, in terms of *generalized coordinates and momentums*:

$$\boxed{\frac{dA}{dt} = \sum_{\alpha} \left[\frac{\partial A}{\partial q_{\alpha}} \frac{\partial H}{\partial p_{\alpha}} - \frac{\partial A}{\partial p_{\alpha}} \frac{\partial H}{\partial q_{\alpha}} \right] \equiv [A, H]_{\text{Poisson}}} \quad (1.3.24)$$

The last identity defines the Poisson brackets, $[A, B]_{\text{Poisson}}$, for two dynamics properties. Evidently,

$$[B, A]_{\text{Poisson}} = -[A, B]_{\text{Poisson}} \quad (1.3.25)$$

- The conservation property A is about $[A, H]_{\text{Poisson}} = 0$, since it amounts to $dA/dt = 0$. Evidently, $[H, H]_{\text{Poisson}} = 0$ (the energy conservation). In parallel, we can recast Eq. (1.3.9), the conservation of total angular momentum as

$$[\mathbf{J}, H]_{\text{Poisson}} = 0, \quad \text{where } \mathbf{J} = \sum_n (\mathbf{r}_n \times \mathbf{p}_n) \quad (1.3.26)$$

whereas that of the total linear momentum with

$$[\mathbf{P}, H]_{\text{Poisson}} = 0, \quad \text{where } \mathbf{P} = \sum_n \mathbf{p}_n \quad (1.3.27)$$

Each vector equation above amounts to three scalar equations.

Remarks (*Brainwash for the time being*)

- As mentioned earlier, there is a postulation, called the *classical-and-quantum correspondence* principle: Each individual dynamic variable $A(\mathbf{p}, \mathbf{q})$ corresponds to a Hermitian *operator* $\hat{A}$ (behaving like a *matrix*).
- (a) Define the *commutator* between two operators,

$$[\hat{A}, \hat{B}] \equiv \hat{A}\hat{B} - \hat{B}\hat{A} \quad (1.3.28)$$

This is the quantum correspondence of the Poisson brackets, $[A, B]_{\text{Poisson}}$, Eq. (1.3.24), with the phase-space partial derivatives.

- (b) In particular, that $[x, p_x]_{\text{Poisson}} = 1$ corresponds to

$$[\hat{x}, \hat{p}_x] = i\hbar \quad (1.3.29)$$

This is a direct consequence of the de Broglies' wave-particle-duality relation, to be elaborated soon in the coming chapter.

- (c) In parallel, the quantum correspondence to Eq. (1.3.24) is

$$\frac{d\hat{A}}{dt} = -\frac{i}{\hbar} [\hat{A}, \hat{H}] \quad (1.3.30)$$

This is the Heisenberg equation of motion in quantum mechanics. While the energy conservation is about $[\hat{H}, \hat{H}] = 0$, the total angular momentum conservation is related to $[\hat{\mathbf{J}}, \hat{H}] = 0$ that is true in quantum mechanics.

- Moreover, the classical statistical dynamics deals with the mean values measured in experiments,

$$\bar{A}(t) \equiv \iint d\mathbf{p} d\mathbf{q} A(\mathbf{p}, \mathbf{q}) \rho(\mathbf{p}, \mathbf{q}, t) \quad (1.3.31)$$

While the dynamic variable $A(\mathbf{p}, \mathbf{q})$ is considered to be time-independent, the classical phase-space distribution function, $\rho(\mathbf{p}, \mathbf{q}, t)$, is time-dependent. It satisfies

$$\frac{\partial}{\partial t} \rho(\mathbf{p}, \mathbf{q}, t) = [H(\mathbf{p}, \mathbf{q}), \rho(\mathbf{p}, \mathbf{q}, t)]_{\text{Poisson}} \quad (1.3.32)$$

(d) The classical-and-quantum correspondence, $\rho(\mathbf{p}, \mathbf{q}, t) \Rightarrow \hat{\rho}(t)$, is the system density operator that satisfies [cf. Eq. (1.3.30) and also Eq. (2.2.11) later]

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}] \quad (1.3.33)$$

This actually is a form of the Schrödinger equation, which governs the dynamics of system wave functions that go by the Born's probability density interpretation. *The wave-particle duality plays the most fundamental role in quantum physics.*

Chapter 2

Quantum Mechanics: Basic Principles

2.1 Wave–Particle–Duality versus Momentum Operator

It witnessed the severe and fundamental limitations of classical physics. Classical particles such as electrons and nuclei obey the Newton's laws of dynamics, whereas classical light or electromagnetic waves are described with the Maxwell equations. From the epistemological point of view, particles and waves are two distinct types of entities in classical physics. There is no such distinction in quantum world. The wave–particle–duality is an *intrinsic* property of *any* entity in physics, no matter it is an electromagnetic (light) wave field or electron/atom/molecule.

The revolutionary change in understanding of microscopic phenomena that took place in 1900–1927 is unprecedented in the natural of sciences. A new parameter, the Planck's constant,

$$\hbar \equiv \frac{h}{2\pi} = 1.05457 \times 10^{-34} \text{ J s} \quad (2.1.1)$$

emerges ubiquitously in quantum physics.

2.1.1 Photons versus matter waves

A. Particle nature of light waves

- Electromagnetic (light) wave possesses electric and magnetic fields, $\mathbf{E}(\mathbf{r}, t)$ and $\mathbf{B}(\mathbf{r}, t)$, both are perpendicular to the direction the propagation. This reflects the *transverse* nature of a light wave.

For our purpose (or from the reductionism point of view), it is sufficient to consider only the case of *plane wave*, such as a laser field, with a monochromatic frequency ν , moving through vacuum space along $\mathbf{n} \equiv \mathbf{k}/|\mathbf{k}|$, the direction of a given wave vector $\mathbf{k}$. For plane waves,

$\mathbf{B} = \mathbf{n} \times \mathbf{E}$ and

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (2.1.2)$$

In general, $\mathbf{E}_0 \equiv (\mathbf{a}_1 + i\mathbf{a}_2)e^{-i\phi}$, with the real vectors $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{k}$ are mutually perpendicular. *Linearly polarized* light field goes with $\mathbf{a}_1 \neq 0$ and $\mathbf{a}_2 = 0$ or vice versa. *Elliptically polarized* light field has $\mathbf{a}_1^2 \neq 0$ and $\mathbf{a}_2^2 \neq 0$, with $\mathbf{a}_1^2 = \mathbf{a}_2^2$ for the circularly polarized case.

• The wave nature of Eq. (2.1.2) is described by $\boxed{e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}}$, as follows.

- 1) Circular frequency: $\omega = 2\pi\nu$
- 2) Wavelength: $\lambda = c/\nu$, where c is the speed of light in vacuum.
- 3) Wavenumber:

$$\tilde{\nu} \equiv \frac{1}{\lambda} = \frac{\nu}{c} \quad (2.1.3)$$

- 4) Wavevector $\mathbf{k}$ (direction the wave propagation), with the magnitude,

$$k = \sqrt{k_x^2 + k_y^2 + k_z^2} = \frac{2\pi}{\lambda} = \frac{\omega}{c} \quad (2.1.4)$$

• **Photons** (*the particle nature of light*)

(a) Electromagnetic (light) wave consists of *photons*. Each individual photon of frequency ω carries a quantized energy of

$$\boxed{E = \hbar\omega} \quad (2.1.5a)$$

(b) The momentum $\mathbf{p}$ of each photon. Consider first the amplitude, p . From Eq. (2.1.5a) and the Einstein's formula, $E = mc^2 = pc$, we have $p = \hbar\omega/c = h\nu/c = h/\lambda$. This reads in combining with Eq. (2.1.4) as

$$p = \hbar k = \frac{h}{\lambda} \quad (2.1.5b)$$

The first identity report the amplitude of the vector relation,

$$\boxed{\mathbf{p} = \hbar\mathbf{k}} \quad (2.1.5c)$$

B. Wave property of particles

Similarly, the issue of wave-particle-duality on matter (such as electron, nuclei and so on) is rooted also at Eq. (2.1.5) that results in

$$\boxed{e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} = e^{i(\mathbf{p} \cdot \mathbf{r} - Et)/\hbar}}$$

This is a plane-matter-wave, corresponding to a zero-force free-moving particle, with the specified momentum, $\mathbf{p} = m\mathbf{v}$, and the resultant kinetic energy $E = mv^2/2$. In 1923, Louis de Broglie (PhD in 1924 and Nobel

Laureate in 1929) proposed the wave nature of a particle be consistent with Eqs. (2.1.5b) and (2.1.5c):

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad \text{and} \quad \mathbf{k} = \frac{\mathbf{p}}{\hbar} = \frac{m\mathbf{v}}{\hbar} \quad (2.1.6)$$

- In 1927, Davisson and Germer experimentally confirmed de Broglie's hypothesis by reflecting electrons from metals and observing diffraction effects. In the same year, Thomson also carried out a diffraction-like experiment, by using photo-electrons as the source of diffraction wave. It is worth noting that the diffraction limit of spacial resolution is about a half-wavelength. Consider electrons of speed $1.0 \times 10^6 \text{ m} \cdot \text{s}$, having the de Broglie's wavelength of about $7\text{\AA}$. The wavelength of matter is extensively confirmed, with various particles, including fullerene C_{60} .

- **Photoelectric effect** (Einstein, 1905): When a metal exposed to a light, electrons can be ejected only when its frequency ν is sufficiently large. These ejected electrons are called photoelectrons. Each photoelectron has the kinetic energy, $E_{\text{kinetic}} = h\nu - \Phi$, where $\Phi > 0$ is the so-called "work function", an intrinsic property of the metal, for ejecting an electron from it.

- **Compton effect** (1922): As known, the collision of two particles, A and B, preserves the total energy, $E_0(A) + E_0(B) = E(A) + E(B)$, and the momentum, $\mathbf{p}_0(A) + \mathbf{p}_0(B) = \mathbf{p}(A) + \mathbf{p}(B)$. All momentum vectors are in a same plane.

(a) To account for his *photon-electron scattering* experiments, Compton exploited not only the wave-particle-duality, but also the Einstein's energy-momentum relation, $E^2 = (mc^2)^2 + (pc)^2$, for a free-moving particle.

(b) Focus on the X-ray photon scattering with a *zero-kinetic energy photoelectron*. The Compton equations are then

$$\mathbf{p}_0 = \mathbf{p} + m_e \mathbf{v} \quad \text{and} \quad h\nu_0 + m_e c^2 = h\nu + [(m_e c^2)^2 + (pc)^2]^{\frac{1}{2}}$$

with noting that $p = h\nu/c = h/\lambda < p_0 = h/\lambda_0$. Let θ be the scattering angle between $\mathbf{p}$ and $\mathbf{p}_0$. The resulting photon wavelength change reads

$$\lambda - \lambda_0 = 2\lambda_C \sin^2(\theta/2), \quad \text{with} \quad \lambda_C \equiv \frac{h}{m_e c} = \frac{hc}{m_e c^2} \approx 0.0241\text{\AA} \quad (2.1.7)$$

referred to the Compton wavelength via the static energy.

- Electron microscopy (EM) is another example of using the wavelength property of particles. The cyro-EM is a *revolutionary* invention, which

detects basically the structures of proteins *in vivo*, with about 1.25Å resolution nowadays. Before that, people were using the X-ray diffraction method, which is applicable only for crystal forms of proteins that are extremely painful to have. The inventors of cyro-EM are Jacques Dubochet, Joachim Frank and Richard Henderson – Physicists won the Nobel Prize in Chemistry (2017) for the invention, critically important to Biology!

2.1.2 Momentum operator in coordinate representation

- Without loss of generality, we consider a particle moving in one-dimension space. Its coordinate and conjugate momentum, $\hat{x}$ and $\hat{p}$, are both Hermitian operators. Their eigenequations are

$$\hat{x}|x\rangle = x|x\rangle \quad \text{and} \quad \hat{p}|p\rangle = p|p\rangle \quad (2.1.8)$$

- (a) Each individual $\{|x\rangle\}$ or $\{|p\rangle\}$ constitutes a representation basis set, satisfying

$$\begin{aligned} \langle x|x'\rangle &= \delta(x-x') \\ \langle p|p'\rangle &= \delta(p-p') \end{aligned} \quad (\text{Orthonormality}) \quad (2.1.9)$$

and

$$\int dx |x\rangle\langle x| = \int dp |p\rangle\langle p| = \hat{I} \quad (\text{Completeness}) \quad (2.1.10)$$

- (b) In comparison, the discrete representation basis set, $\{|n\rangle\}$, goes by

$$\langle m|n\rangle = \delta_{mn} \quad \text{and} \quad \sum_n |n\rangle\langle n| = \hat{I}$$

- The task now is to find the momentum operator $\hat{p}$ in the coordinate x -representation. This is concerned with the wave function, $\varphi_p(x)$, for the plane wave of a particle, with a given momentum p , moving along the x direction. The related representation transformation is given by

$$\varphi_p(x) = \langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} \quad (2.1.11)$$

This is the coordinate part of the plane wave. The coefficient, $1/\sqrt{2\pi\hbar}$, is concerned with the unitary requirement on the transformation between $\{|x\rangle\}$ and $\{|p\rangle\}$ representations. The established correspondence reads

$$\hat{p}|p\rangle = p|p\rangle \xrightarrow{x\text{-rep.}} \frac{\hbar}{i} \frac{\partial}{\partial x} \varphi_p(x) = p\varphi_p(x)$$

This concludes that (**memorized it!**)

$$\boxed{\hat{p} \doteq \frac{\hbar}{i} \frac{\partial}{\partial x}} \quad (2.1.12)$$

with $\doteq$ denoting a specified representation, which in the present case is the coordinate representation.

Exercises: Derive the following important results (**memorize them**).

- Heisenberg's (canonical) commutator relation in quantum mechanics:

$$[\hat{x}, \hat{p}] = i\hbar \quad (2.1.13)$$

- Equation (2.1.9) satisfies the basis set requirement of

$$\langle x|x'\rangle = \int dp \langle x|p\rangle \langle p|x'\rangle = \delta(x - x') \quad (2.1.14)$$

- Adopted in quantum mechanics are five **postulations**, with their consequences being subject to experimental verifications. One postulation is the Schrödinger equation,

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

Verify the following formal solution,

$$\boxed{|\Psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\Psi(0)\rangle} \quad (2.1.15)$$

Show further

$$\langle \Psi(t) | \Psi(t) \rangle = \langle \Psi(0) | \Psi(0) \rangle \quad (2.1.16)$$

- By using Eq. (2.1.12), show that in the x -representation Hamiltonian operator becomes

$$\hat{H} = \frac{1}{2m} \hat{p}^2 + V(\hat{x}) \doteq -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \quad (2.1.17)$$

Write down the Schrödinger equation in the x -representation, in which the wave vector $|\Psi(t)\rangle$ becomes the wave function,

$$\Psi(x, t) \equiv \langle x | \Psi(t) \rangle \quad (2.1.18)$$

- Show that [*Hint: using Eq. (2.1.9)*]

$$\tilde{\Psi}(p, t) \equiv \langle p | \Psi(t) \rangle = \frac{1}{\sqrt{2\pi\hbar}} \int dx e^{-ipx/\hbar} \Psi(x, t) \quad (2.1.19a)$$

and

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int dp e^{ipx/\hbar} \tilde{\Psi}(p, t) \quad (2.1.19b)$$

- Write down the bra-ket notation for

$$\int dx \Psi^*(x, t) \Psi(x, t) = \int dp \tilde{\Psi}^*(p, t) \tilde{\Psi}(p, t)$$

Remarks

- The scaled momentum-space, with the de Broglie relation, $\mathbf{k} = \mathbf{p}/\hbar$, is the wavevector $\mathbf{k}$ -space. The wave functions in the coordinate- $\mathbf{r}$ and wavevector- $\mathbf{k}$ spaces, $\psi(\mathbf{r})$ and $\tilde{\psi}(\mathbf{k})$, as inferred from Eq. (2.1.19), are related each other via the Fourier transformations. The pre-integral factor is $(2\pi)^{-N/2}$ for an N -dimension systems. Some literature may use the real-space and the reciprocal space for the $\mathbf{r}$ -space and $\mathbf{k}$ -space, respectively.
- The wave function $\psi(\mathbf{r})$ is the choice for atoms and (small) molecules. However, in solid-state physics, $\psi(\mathbf{r})$ spreads over systems of macroscopic size. In this case, the $\mathbf{k}$ -space description via $\tilde{\psi}(\mathbf{k})$ is preferable. For conducting polymers, the charge carriers are conjugated π -electrons and/or holes that are extensively *delocalized*. The $\mathbf{k}$ -space wave function description would also be the choice.
- Turn to optical spectroscopy of an atom. The transition between a pair of high and low energy levels of the atom,

$$E_m - E_n = \hbar\omega_{mn} \quad (2.1.20)$$

gives rise to the emission line at frequency $\omega = \omega_{mn}$. The transition dipole oscillation field engages the time-dependent component via Eq. (2.1.2),

$$\mu_{mn}(t) = \mu_{mn}e^{-i\omega_{mn}t} \quad (2.1.21)$$

The corresponding spectrum line is given by

$$S(\omega; n \leftarrow m) = \int_{-\infty}^{\infty} dt e^{i(\omega - \omega_{mn})t} e^{-\gamma_{mn}|t|} = \frac{2|\mu_{mn}|^2\gamma_{mn}}{(\omega - \omega_{mn})^2 + \gamma_{mn}^2} \quad (2.1.22)$$

which is centered at $\omega = \omega_{mn}$ of Eq. (2.1.20). Included here is also the line-broadening parameters, $\gamma_{mn} \approx \gamma_m + \gamma_n$, with $\gamma_m = 1/\tau_m$ being the inverse life-time of the m^{th} -level.

- Crystals are lattice structures, with translation symmetry through macroscopic three-dimension real space. Diffraction from a crystal forms the pattern described by the Bragg's formula. This is the reciprocal space image, the Fourier transformation of the lattice structure in real space.

- Energy reported in terms of wave property:

(a) Frequency (ν): $1 \text{ Hz} \Leftrightarrow 6.62610^{-34} \text{ J}$

(b) **Wavenumber** ($\nu/c = 1/\lambda$): $1 \text{ cm}^{-1} \Leftrightarrow 1.986^{-23} \text{ J}$

(c) Thermal energy $k_B T$ reported in such as $300 \text{ K} \approx 208 \text{ cm}^{-1}$

2.1.3 Implications of matter waves

People could inferred from the wave-particle-duality alone the some basic quantum-particle characteristics. These include (1) the self-interference property; (2) Uncertainty principle; (3) the zero-point energy; (4) quantum confinement principle, and (5) tunneling phenomena. The first three are more or less direct consequences of the wave-particle-duality. The last two general features of quantum particles are to be illustrated with simple examples.

(1) Self-interference:

Interference is a basic wave phenomenon. Classical waves, such as water wave or sound wave, describe a collective motion of many particles. On the other hand, the wave-particle-duality is concerned with the fact that each individual particle possesses the wave property. This is the origin of the **self-interference** in quantum physics.

(2) Uncertainty principle: $\Delta x \Delta p > 0$.

This is a qualitative result, prior to the identification of the lower-bound uncertainty value ($\hbar/2$). The uncertainty principle arises immediately from the wave-particle-duality.

(3) Zero-point energy:

$$E_0 \equiv E_{\text{lowest}}^{\text{quan}} - E_{\text{lowest}}^{\text{classical}} > 0 \quad (2.1.23)$$

(a) The lowest allowed quantum energy, $E_{\text{lowest}}^{\text{quan}} = E_g$, is the ground state energy. The classical counterpart is just the global-minimum potential energy, denoted as $E_{\text{lowest}}^{\text{classical}} = V_{\text{min}}$, together with the zero-kinetic energy (or momentum $p = 0$) at the same time. However, the uncertainty principle rules out the simultaneous precision in both x and p . This gives rise to the aforementioned zero-point energy principle which can be recast as $E_0 \equiv E_g - V_{\text{min}} > 0$, for any mechanic systems, with the global minimum potential energy V_{min} being finite.

(b) For Columbic systems, such as hydrogen atom whose potential energy is given by $V(r) \propto -1/r$ that goes with $V_{\text{min}} \rightarrow -\infty$. The resultant finite ground state energy $E_g = -13.6$ eV, from the Bohr's atomic orbit theory, explains the stability of atoms, rather than the zero-point energy.

(c) The name of "orbit", used in Bohr's theory, is completely abandoned now, due to its implication of trajectory in nature. Adopted in quantum mechanics is "**orbital**", defined to be the wave function of a *single-electron* system. Atomic orbitals versus molecular orbitals (including MO-LCAO)

are then self-defined. *Orbitals are the building blocks of many-electron wave functions.*

(4) Quantum confinement:

To elaborate the captioned picture, we may consider the simplest case, the particle standing wave scenario. This is concerned with a particle of mass m inside a one-dimension box of length L , with the potential energy,

$$V(0 < x < L) = 0 \text{ and } V(x) \rightarrow \infty \text{ if } x \notin (0, L).$$

The solutions to this model are the form of $\sin(kx) = \sin(2\pi x/\lambda)$, subject to the standing waves constraint,

$$n\lambda = 2L, \text{ where } n = 1, 2, \dots \quad (2.1.24)$$

This determines the de Broglie momentum amplitude of

$$p = \frac{2\pi\hbar}{\lambda} = \frac{n\pi\hbar}{L} \quad (2.1.25)$$

The resultant standing wave function and energy are, respectively,

$$\psi_n(x) \propto \sin(n\pi x/L) \quad (2.1.26)$$

and

$$E_n = \frac{1}{2m} \left(\frac{n\pi\hbar}{L} \right)^2, \text{ where } n = 1, 2, \dots \quad (2.1.27)$$

(a) As $V_{\min} = 0$ here, the zero-point energy, E_0 , is the ground state value,

$$E_0 = E_1 = \frac{\pi^2\hbar^2}{2mL^2}$$

The transition energy between two neighboring states is

$$E_{n+1} - E_n = (2n+1)E_0; \text{ where } n = 1, 2, \dots$$

(b) For a given particle mass m , the zero-point energy and also the transition energy increase as the box length L decrease. This is the so-called **quantum confinement**, a *universal phenomenon* for a broad range of systems.

(c) Black body radiation is a thermal equilibrium system. For the question that whether such a system behaves quantum or not, the relevant quantity is $\Delta E/(k_B T)$, where ΔE denotes the transition energy between the first excited state and the ground state of each particle.

(5) Tunneling: *Universal phenomena of particles appearing in classical forbidden regions.* Tunneling occurs in a variety of systems and scenarios.

(a) The simplest case may be the particle in square barrier/well potential systems. These are analytical solvable model systems to be elaborated in Chapter §3.1. We will see that tunneling phenomena appear even in the reflection waves above the potential barriers and wells.

(b) Harmonic oscillator is another analytical solvable model system; see the details in Chapter §3.2. This provides rich phenomena, including such as energy quantization, zero-point energy, and tunneling. Physically, harmonic oscillator is closely related to the electromagnetic field radiation and the heat capacity of solid problems.

(i) It is well established in general physics that radiation arises from the electric dipole oscillations and can be treated as harmonic systems. In the Planck's theory of blackbody radiation, the field energy quantization goes by $E_n = n\hbar\omega$; $n = 0, 1, \dots$. The zero-point energy is missing, as it is irrelevant to blackbody radiation.

(ii) Harmonic oscillator systems go by the potential energy the form of

$$V(x) = \frac{1}{2}m\omega^2 x^2 \quad (2.1.28)$$

The resulting eigen-energies are

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right); \quad n = 0, 1, 2, \dots \quad (2.1.29)$$

The ground state wave function reads

$$\psi_0(x) \propto \exp\left(-\frac{m\omega x^2}{2\hbar}\right) \quad (2.1.30)$$

Exercises:

E2.1.1 Are there **tunneling** phenomenon in $\psi_0(x)$? What are the classical turning points about?

E2.1.2 Write down the time-independent Schrödinger equation for the harmonic oscillator's wave function $\psi(x)$.

E2.1.3 Show that $\psi_0(x)$, Eq. (2.1.30), is a solution to the Schrödinger equation. Identify the associate eigen-energy and justify the fact that $\psi_0(x)$ is the ground state wave function. (*Hint: In relation to the number of nodes*)

E2.1.4 Write down the normalized $\psi_0(x)$.

Two basic observations:

(i) **Isotope effect:** Isotope with lighter mass behaves more quantum, in terms of both zero-point energy and transition energies, which are larger than the heavier isotope counterparts.

(ii) **Node** is defined at which $\psi(x_{\text{node}}) = 0$. General speaking, for one-dimension systems, the ground state wave function has no nodes and the k^{th} -excited state has k -number of nodes. **Self-interference** is responsible for this observation.

2.2 Postulations and Basic Principles

2.2.1 Schrödinger equation and postulations

Postulation-1: Wave Function Description

Quantum state of a system at time t is completely described by a given Hilbert-space vector $|\Psi(t)\rangle$. In the real-space (“human’s eye”) representation, this is the wave function, $\Psi(x, t) \equiv \langle x|\Psi(t)\rangle$. Its particle property goes with the Born’s interpretation (for normalized wave vector):

$$|\Psi(x, t)|^2 dx \propto \text{Probability of the system locate inside } [x, x + dx] \quad (2.2.1)$$

Postulation-2: Correspondence Principle

Dynamic variables, such as coordinate, momentum, angular momentum and Hamiltonian functions, are *Hermitian operators*.

Postulation-3: Measurement

- (a) The event of a single measurement on dynamic variable $\hat{A}$ gives rise a single outcome, say a_n , that should be an eigenvalue via $\hat{A}|a_n\rangle = a_n|a_n\rangle$.
- (b) The system state, which was $|\Psi(t^-)\rangle$ right before the measurement *collapses immediately* into $|\Psi(t)\rangle = |a_n\rangle$ at the time of measurement. (After the measurement, the system evolves following the Schrödinger equation)

Postulation-4: Schrödinger’s Equation

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle \quad (2.2.2)$$

This is the law of nonrelativistic quantum mechanics.

Postulation-5: Fermions versus Bosons

Particles possess *internal spin* degrees of freedom, which are no classical correspondence at all. The spin angular momentum is quantized with either half-integer (for fermions) or integer (for bosons).

- There are several ways of presenting the Postulations. The above version consists of five postulations. While Postulation-1 highlights the wave-particle-duality of physical object, Postulation-3 on measurement is given in two parts, with Postulation-3(b) for the measurement-induced “*instantaneous collapse*” of wave function. This has caused many philosophical debates; i.e., the “Schrödinger Cat” or the Einstein-Podolsky-Rosen (EPR) Paradox. This kind of debates is very fruitful and constructive to the modern understanding of quantum physics, giving rise to in particular the fields of quantum entanglement and quantum measurement [<https://www.thoughtco.com/epr-paradox-in-physics-2699186>].

- From the reductionism point of view, postulations are required only for a single degree of freedom. Consequences on multiple degrees of freedom will arise from deductions.

2.2.2 Basic principles with wave function description

A. Concepts and principles (by logic deductions, as §2.1.3)

- Only the direction of state wave vector $|\Psi(t)\rangle$ matters. The overall constant phase and the length of vector are physically irrelevant. Moreover, as inferred from Eq. (2.1.16), $\langle\Psi(t)|\Psi(t)\rangle = \langle\Psi(0)|\Psi(0)\rangle$, the wave function normalization needs to be carried out once, with the initial $|\Psi(0)\rangle$. Hence, the normalization constant to Eq. (2.2.1), the Born's probability description on the amplitude square of wave function, is a time-independent parameter.
- **Quantum superposition principle and self-interference** – It is worth emphasizing that *Born's probability* $\neq$ *Classical probability*, with remarkably differences by nature. The latter involves classical statistical ensemble, where each particle acquires a deterministic Newton's trajectory. In contrast, quantum particles are described by wave functions, with the Bohn's interpretation that goes by quantum superposition principle. This further predicts the remarkable **tunneling** and the **self-interference** properties, as we described earlier, for individual quantum particle!
- **Quantum entanglement** refers to the **mutual interference** between two particles, A and B, described with such as $|\Psi_{AB}\rangle = |\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle$. Now we revisit Postulation-3(b), the “*instantaneous collapse*” of wave function induced by measurement. Let A and B be distantly far apart and consider a local spin measurement on A. The single outcome, either $|\uparrow\rangle_A$ or $|\downarrow\rangle_A$, tells immediately (faster than the speed of light) that the distant B must be in the spin-opposite state. Quantum entanglement has now been extensively verified in recent experiments, including those USTC teams.
- “**Energy operator**” in quantum mechanics: The time-dependent part of the plane-matter-wave reads $e^{-iEt/\hbar}$. The fact,

$$i\hbar \frac{d}{dt}(e^{-iEt/\hbar}) = E \cdot (e^{-iEt/\hbar})$$

that would lead to

$$\boxed{\text{Energy operator} \equiv i\hbar \frac{d}{dt}} \quad (2.2.3)$$

Hence, the Schrödinger equation (2.2.2) assumes the form of

$$(\text{Energy operator}) \cdot |\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$$

- **Representation invariance principle** – In relation to the fact that the laws of physics are the same in all reference frames (Newton, Lorentz, Einstein), representations can be viewed as the inertial reference frames in quantum physics. The underlying laws are representation invariance.

B. Expectation values via wave function versus density matrix

- In relation to quantities in experimental measurements, the density operator/matrix description is usually more convenient than the wave vector/function counterpart. Let us start with the classical mean-value of

$$\langle A(t) \rangle = \iint dp dq A(p, q) \rho(p, q; t) \quad (2.2.4)$$

Physically, this amounts to the experimental measurement on a dynamic variable $A(p, q)$, with averaging over the time-dependent distribution $\rho(p, q; t)$ of the classical system.

- In parallel to $A(p, q) \Rightarrow \hat{A}$, we have also the correspondence principle of

$$\boxed{\rho(p, q; t) \Rightarrow \hat{\rho}(t)} = |\Psi(t)\rangle \langle \Psi(t)| \quad (2.2.5)$$

Moreover, $\text{Tr} \doteq \iint dp dq$, with noting that trace is representation invariance. The quantum description of Eq. (2.2.4) is called the **expectation value**:

$$\boxed{\langle \hat{A}(t) \rangle = \text{Tr}[\hat{A} \hat{\rho}(t)]} = \langle \Psi(t) | \hat{A} | \Psi(t) \rangle \quad (2.2.6)$$

Remarks on Eq. (2.2.5) and Eq. (2.2.6)

- In Eq. (2.2.5), $\hat{\rho}(t)$ is called the **system density operator**, the quantum correspondence of classical distribution function; see also Eq. (2.2.7). Density operator satisfies three basic properties: (i) *Hermitian*; (ii) *Positivity*; (iii) *Normalizable*.

(a) As a Hermitian operator, $\hat{\rho}$ can be diagonalized at any time t . Let $\{|\Psi_j\rangle\}$ be the orthonormal basis set that diagonalize $\hat{\rho}$ at local time. Together with the positivity and normalization, we have

$$\hat{\rho} = \sum_j P_j |\Psi_j\rangle \langle \Psi_j|, \text{ with } P_j = \langle \Psi_j | \hat{\rho} | \Psi_j \rangle \geq 0 \text{ and } \text{Tr} \hat{\rho} = \sum_j P_j = 1 \quad (2.2.7a)$$

This amounts to

$$\hat{\rho} = \sum_j P_j \hat{\rho}_j^{\text{pure}}, \text{ where } \hat{\rho}_j^{\text{pure}} \equiv |\Psi_j\rangle \langle \Psi_j| = (\hat{\rho}_j^{\text{pure}})^2 \quad (2.2.7b)$$

In other words, a *pure state* can be described by a single wave function.

(b) Generally speaking, $\hat{\rho}$ comprises a weighted (incoherent) sum over pure-states, with the eigenvalues $\{P_j\}$ being the *probabilities*. This differs from the coherent superposition on $|\Psi\rangle = \sum_a c_a |\varphi_a\rangle$ that is a pure state.

(c) From Eq. (2.2.7), we have further $\hat{\rho}^2 = \sum_j P_j^2 \hat{\rho}_j^{\text{pure}}$; thus

$$\text{Tr} \hat{\rho}^2 \leq \text{Tr} \hat{\rho} \quad (2.2.8)$$

with “=” and “<” for the pure-state and mixed-state cases, respectively.

(d) Thermal equilibrium is a mixed state, $\hat{\rho}^{\text{eq}}(T) = e^{-\beta\hat{H}}/Z$, with $Z = \text{Tre}^{-\beta\hat{H}}$ being the partition function and $\beta = 1/(k_B T)$, where k_B is the Boltzmann constant and T is the temperature.

$$P_j^{\text{eq}}(T) = \langle \psi_j | \hat{\rho}^{\text{eq}}(T) | \psi_j \rangle = \frac{e^{-\beta E_j}}{Z} = \frac{e^{-\beta E_j}}{\text{Tre}^{-\beta\hat{H}}} \quad (2.2.9)$$

depends on the solutions to the Schrödinger equation, $\hat{H}|\psi_j\rangle = E_j|\psi_j\rangle$.

(e) As also noticed, the density operator supports the description of entropy, in terms of

$$S = -k_B \text{Tr}(\hat{\rho} \ln \hat{\rho}) = -k_B \sum_j (P_j \ln P_j) \geq 0 \quad (2.2.10)$$

with the “=” sign for pure states.

(f) Moreover, the Schrödinger equation (2.2.2) immediately gives rise

$$\frac{d\hat{\rho}(t)}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}(t)] \quad (2.2.11)$$

This is the Liouville–von Neumann equation to be elaborated further in §3.2.

• **Expectation value, Eq. (2.2.6)**

As mentioned earlier, $\hat{\rho}$, Eq. (2.2.7), is the probability weighted composite of pure-states. Thus, we can verify Eq. (2.2.6), with the pure-state case only. Let it be

$$|\psi\rangle = \sum_n c_n |a_n\rangle, \quad \text{with} \quad \hat{A}|a_n\rangle = a_n|a_n\rangle \quad (2.2.12)$$

According to the Born's interpretation, $|c_n|^2$ denotes the probability of find the particle of $|\psi\rangle$ in the specified eigenstate $|a_n\rangle$ of $\hat{A}$, with the eigenvalue a_n . Note also that $c_n = \langle a_n | \psi \rangle$. As results,

$$\langle A \rangle = \sum_n a_n |c_n|^2 = \sum_n a_n \langle \psi | a_n \rangle \langle a_n | \psi \rangle = \sum_n \langle \psi | \hat{A} | a_n \rangle \langle a_n | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle$$

The last two identity are obtained by using Eq. (2.2.12) and the basis set completeness, respectively. The local time dependence is implied. Thus, we complete Eq. (2.2.6), with individual pure states. The mixed state is just a certain distribution of pure states.

2.2.3 Uncertainty relations

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle| \quad (2.2.13)$$

Both $\hat{A} = \hat{A}^\dagger$ and $\hat{B} = \hat{B}^\dagger$. Note also that $\Delta A = \sqrt{\langle (\hat{A} - \langle A \rangle)^2 \rangle}$.

Proof of Eq. (2.2.13): Consider the Schwartz inequality, $\langle u|u \rangle \langle v|v \rangle \geq |\langle u|v \rangle|^2$ [Eq. (1.2.5)], with setting $|u\rangle = \hat{A}|\psi\rangle$ and $|v\rangle = \hat{B}|\psi\rangle$. We have

$$\langle \psi | \hat{A}^\dagger \hat{A} | \psi \rangle \langle \psi | \hat{B}^\dagger \hat{B} | \psi \rangle \geq |\langle \psi | \hat{A}^\dagger \hat{B} | \psi \rangle|^2$$

Consider now $\hat{A}^\dagger = \hat{A}$ and $\hat{B}^\dagger = \hat{B}$, with noting that $\langle \hat{O} \rangle \equiv \langle \psi | \hat{O} | \psi \rangle$. We obtain

$$\langle \hat{A}^2 \rangle \langle \hat{B}^2 \rangle \geq |\langle \hat{A} \hat{B} \rangle|^2 = \langle \hat{A} \hat{B} \rangle_r^2 + \langle \hat{A} \hat{B} \rangle_i^2 \geq \langle \hat{A} \hat{B} \rangle_i^2 \quad (2.2.14)$$

The real and imaginary parts of $\langle \hat{A} \hat{B} \rangle = \langle \hat{A} \hat{B} \rangle_r + i \langle \hat{A} \hat{B} \rangle_i$ are respectively

$$\langle \hat{A} \hat{B} \rangle_r = \frac{1}{2} \langle \{\hat{A}, \hat{B}\} \rangle \quad \text{and} \quad \langle \hat{A} \hat{B} \rangle_i = -\frac{i}{2} \langle [\hat{A}, \hat{B}] \rangle \quad (2.2.14')$$

The commutator $[\hat{A}, \hat{B}] \equiv \hat{A} \hat{B} - \hat{B} \hat{A}$ is purely quantum. The **anticommutator**,

$$\{\hat{A}, \hat{B}\} \equiv \hat{A} \hat{B} + \hat{B} \hat{A} \quad (2.2.15)$$

which symmetrizes the product of two Hermitian operators, has the classical analogue; cf. Eq. (2.2.19). The resulting $\langle \{\hat{A}, \hat{B}\} \rangle$ could be zero in general [see Eq. (2.3.13) for an example]. The above observations give rise to Eq. (2.2.14) the form of

$$\langle \hat{A}^2 \rangle \langle \hat{B}^2 \rangle \geq \frac{1}{4} |\langle [\hat{A}, \hat{B}] \rangle|^2 \quad (2.2.16)$$

Applying it for $\hat{A} - \langle A \rangle$ and $\hat{B} - \langle B \rangle$, with noticing that $[\hat{A} - \langle A \rangle, \hat{B} - \langle B \rangle] = [\hat{A}, \hat{B}]$, we obtain Eq. (2.2.13). (Q.E.D.)

- **The Heisenberg's canonical uncertainty relations:**

$$\Delta x \Delta p_x \geq \hbar/2 \quad \text{and} \quad \Delta E \Delta t \geq \hbar/2 \quad (2.2.17)$$

- **Compatible** observables if $[\hat{A}, \hat{B}] = 0$. There are a set of common eigenstates for *compatible* observables, such that

$$\hat{A}|a, b\rangle = a|a, b\rangle \quad \text{and} \quad \hat{B}|a, b\rangle = b|a, b\rangle \quad (\text{if } [\hat{A}, \hat{B}] = 0) \quad (2.2.18)$$

- **Complementary** observables: They are compatible and their spanning spaces satisfy $S_B \subset S_A$, with the subspace S_B defined by $\hat{B}$ resolving the degeneracy of $\hat{A}$ in S_A , so that $\langle a, b|a', b'\rangle = \delta_{aa'} \delta_{bb'}$.

Remarks on Postulation-2, the correspondence principle: “*Classical dynamic variables be Hermitian operators in quantum mechanics*”. This would have implied

$$(AB)_{\text{classical}} \Rightarrow \frac{1}{2}\{\hat{A}, \hat{B}\} \equiv \frac{1}{2}(\hat{A}\hat{B} + \hat{B}\hat{A}) \quad (2.2.19)$$

While the anticommutator $\{\hat{A}, \hat{B}\}$ of two Hermitian operators is Hermitian, the commutator $[\hat{A}, \hat{B}]$ is anti-Hermitian and related to the uncertainty principle that is a pure quantum property.

Useful relations (*verify and memorize them*):

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} = \hat{A}\{\hat{B}, \hat{C}\} - \{\hat{A}, \hat{C}\}\hat{B} \quad (2.2.20)$$

2.2.4 Requirements of wave functions

A. Three basic requirements on wave functions

It is noticed that, by the postulations, quantum physics poses certain criterions on the acceptable Hilbert-space wavekets $\{|\psi\rangle\}$. The constraints imposed by the postulations 1–4 can in general be elaborated with one-degree of system. Consider the time-independent Schrödinger equation,

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x) = E\psi(x) \quad (2.2.21)$$

Together with the Born's interpretation, Eq. (2.2.1), we have the wave function $\psi(x)$ the following three basic requirements:

- (i) **Continuous functions**, $\psi(x)$ and $\psi'(x)$, as for Eq. (2.2.21);
- (ii) **Single-valued** $\psi(x)$, for its probability amplitude interpretation. Recall the fact that laws of physics are all representation invariance. The captioned requirement is the single-valued $|\psi\rangle$ in any chosen presentation.
- (iii) **Square-integrable** (Normalizable $|\psi\rangle$): The normalized wave function satisfies $\langle\psi|\psi\rangle = 1$. Moreover, that $\langle p'|p\rangle = \delta(p-p')$ should also be acceptable. The matter plane wave $\varphi_p(x) = \langle x|p\rangle = e^{ipx/\hbar}/\sqrt{2\pi\hbar}$, the normalized eigenfunction of momentum operator, satisfies

$$\langle p'|p\rangle = \int \varphi_{p'}^*(x) \varphi_p(x) dx = \frac{1}{2\pi\hbar} \int e^{i(p-p')x/\hbar} dx = \delta(p-p') \quad (2.2.22)$$

Exercise: Solve the Schrödinger equation (2.2.21) for the particle-in-box setup, with the normalize wave functions; cf. Eqs. (2.1.26) and (2.1.27).

B. Eigenequation in quantum mechanics

Schrödinger equation is the eigenequation of system Hamiltonian operator and dictates the steady-states (i.e., the ground and excited structures) of system in study. Quantum physics supports also $\hat{A}|a_n\rangle = a_n|a_n\rangle$, the eigenequations of other dynamic measurable operators, as stipulated in Postulations 2 and 3. These wave functions should also satisfy the above requirements.

C. Indistinguishable particles permutation symmetry

Postulation-5, fermions versus bosons, poses the different permutation symmetry for the many-particle wave functions, including both orbital and spin degrees of freedom. Permutation of two fermions (such as electrons) results in a minus (“−”) sign to the total N -particle wave functions, whereas bosons are permutable, without the sign change. We will elaborate the many-particle wave functions in Chapter 8.

2.3 Schrödinger Picture versus Heisenberg Picture

2.3.1 The Heisenberg's equation of motion

It is worth recalling the fact that classical mechanics goes largely with time-dependent dynamic variables such as $\dot{x}(t) = p(t)/m$. In quantum mechanics, this concerns the Heisenberg's picture,

$$\hat{A}(t) \equiv e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} \quad (2.3.1)$$

In this prescription, Eq. (2.2.6) is recast as [cf. Eq. (2.1.15)]

$$\langle \hat{A}(t) \rangle = \langle \Psi(t) | \hat{A} | \Psi(t) \rangle = \langle \Psi(0) | \hat{A}(t) | \Psi(0) \rangle \quad (2.3.2)$$

Comments:

- 1) Expectation values are measurable quantities in experiments. For this type of quantities, the resultant theorems and expressions are all **prescription invariance**, no matter with the Schrödinger's picture or the Heisenberg picture.
- 2) The Schrödinger's picture goes by time-dependent state $|\Psi(t)\rangle$ and time-independent dynamic variable $\hat{A} = \hat{A}(t=0)$.
- 3) The Heisenberg's picture goes by time-independent $|\Psi(t=0)\rangle$, but time-dependent $\hat{A}(t)$, Eq. (2.3.1). The effect of Hilbert-space propagator (time-evolution operator) on measurable quantities is completely absorbed into dynamic variables.
- 4) In practice, the Heisenberg's picture is limited only to time-independent Hamiltonian case.
- 5) Time-dependent $\hat{H}(t)$ occurs whenever the experimental measurements involved time-dependent external pulsed fields. In this case, the Schrödinger's picture will be the choice. On the other hand, when the time-dependent external field-matter coupling is weak comparing to the matter system Hamiltonian, the *interaction picture* would be preferable. I will discuss this together with the *time-dependent perturbation theory*.
- 6) **Heisenberg's equation of motion**, arising from Eq. (2.3.1):

$$\dot{\hat{A}} = -\frac{i}{\hbar} [\hat{A}, \hat{H}] \quad (2.3.3)$$

This is valid for $\hat{A}(t)$ at any given time, including $\hat{A}(t=0)$. For the comparison, it is worth recalling the classical Poisson equation (1.3.24) of motion and the remarks there.

7) **Ehrenfest's theorem:** Exemplified with one-dimension systems,

$$\hat{H} = \hat{p}^2/(2m) + V(\hat{x}) \quad (2.3.4)$$

we have

$$\boxed{\dot{\hat{x}} = \hat{p}/m} \quad \text{and} \quad \boxed{\dot{\hat{p}} = -V'(\hat{x})} \quad (2.3.5)$$

for the operators correspondence to classical mechanics. Moreover,

$$\langle \dot{\hat{x}} \rangle = \langle \hat{p} \rangle / m \quad \text{and} \quad \langle \dot{\hat{p}} \rangle = -\langle V'(\hat{x}) \rangle \quad (2.3.6)$$

The involving the average $\langle \cdots \rangle$ differs in nature from the classical ensemble one, as discussed earlier, Eq. (2.3.2) or (2.2.6) versus Eq. (2.2.4). The quantum average is rooted at matter wave function description and related to the Born's probability, where the quantum superposition principle supports the self-interference.

Exercises

E2.3.1 Consider a one-dimension harmonic oscillator system [cf. Eq. (2.1.28)],

$$\hat{H} = \frac{1}{2m} \hat{p}^2 + \frac{1}{2} m \omega^2 \hat{x}^2 \quad (2.3.7)$$

Derive the following formulations:

(a) The equations of motion read

$$\dot{\hat{x}} = \hat{p}/m, \quad \dot{\hat{p}} = -m\omega^2 \hat{x} \quad (2.3.8a)$$

(b) The solutions are

$$\begin{bmatrix} \hat{x}(t) \\ \hat{p}(t) \end{bmatrix} = \begin{bmatrix} \cos(\omega t) & \frac{1}{m\omega} \sin(\omega t) \\ -m\omega \sin(\omega t) & \cos(\omega t) \end{bmatrix} \begin{bmatrix} \hat{x}(0) \\ \hat{p}(0) \end{bmatrix} \quad (2.3.8b)$$

(c) Set the following operator,

$$\hat{a} \equiv \sqrt{\frac{m\omega}{\hbar}} \hat{x} + i\sqrt{\frac{1}{\hbar m\omega}} \hat{p} \quad (2.3.9)$$

This reduces Eq. (2.3.8) to

$$\dot{\hat{a}}(t) = -i\omega \hat{a}(t) \quad \text{and} \quad \hat{a}(t) = e^{-i\omega t} \hat{a}(0) \quad (2.3.10)$$

(d) Show $[\hat{a}, \hat{a}^\dagger] = 1$ and also that Eq. (2.3.7) can be recast as

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (2.3.11)$$

2.3.2 Steady-state theorems

A. Steady-state description

- Generally speaking, the steady states remain unchange under the influence of Hilbert-space propagator. Apparently, the eigenstates to time-independent Schrödinger equation constitute a set of steady-states, $\{|n\rangle\}$, where

$$\hat{H}|n\rangle = E_n|n\rangle \quad (2.3.12)$$

It follows that

$$[|n\rangle\langle n|, \hat{H}] = 0 \quad (2.3.13)$$

and the immediate result of

$$e^{-i\hat{H}t/\hbar}(|n\rangle\langle n|)e^{i\hat{H}t/\hbar} = |n\rangle\langle n| \quad (2.3.14)$$

- It is more general to discuss the steady-state physics with the *steady-state density operator*, $\hat{\rho}^{\text{st}}$, satisfying

$$e^{-i\hat{H}t/\hbar} \hat{\rho}^{\text{st}} e^{i\hat{H}t/\hbar} = \hat{\rho}^{\text{st}} \quad (2.3.15)$$

This is equivalent to

$$[\hat{\rho}^{\text{st}}, \hat{H}] = 0 = \dot{\hat{\rho}}^{\text{st}}; \quad (2.3.16)$$

In other words, the steady-state density operator takes the form of

$$\hat{\rho}^{\text{st}} = \hat{\rho}^{\text{st}}(\hat{H}) = \sum_n P_n |n\rangle\langle n|, \quad \text{with } \sum_n P_n = 1 \quad \text{and } P_n \geq 0 \quad (2.3.17)$$

Each $|n\rangle$ via the Schrödinger equation (2.3.12) is a pure-state case of steady states. Recall that $\langle \hat{O} \rangle^{\text{st}} \equiv \text{Tr}(\hat{O} \hat{\rho}^{\text{st}})$ [cf. Eq. (2.2.6)].

B. Canonical steady-state theorems

$$\boxed{\langle \hat{p} \rangle^{\text{st}} = 0} \quad \text{and} \quad \boxed{\langle \{\hat{x}, \hat{p}\} \rangle^{\text{st}} = 0} \quad (2.3.18)$$

While the above equation is concerned with $\hat{p} \propto \dot{\hat{x}}$, Eq. (2.3.5), we should in general have

$$\boxed{0 = \langle \dot{\hat{A}} \rangle^{\text{st}} = \langle \{\hat{A}, \dot{\hat{A}}\} \rangle^{\text{st}}} \quad (2.3.19)$$

(a) By using the Heisenberg's equation of motion (2.3.3), the first identity in Eq. (2.3.19) is equivalent to

$$\langle \dot{\hat{A}} \rangle^{\text{st}} \propto \langle [\hat{H}, \hat{A}] \rangle^{\text{st}} = 0 \quad (2.3.20a)$$

(b) In parallel,

$$\langle \{\hat{A}, \dot{\hat{A}}\} \rangle^{\text{st}} \propto \langle \{\hat{A}, [\hat{H}, \hat{A}]\} \rangle^{\text{st}} = i \langle [\hat{H}, \hat{A}^2] \rangle^{\text{st}} = 0 \quad (2.3.20b)$$

(b) Together with $[\hat{x}, \hat{p}_x] = i\hbar$, we obtain further

$$\langle \hat{x}\hat{p} \rangle^{\text{st}} = i\hbar/2 \quad \text{and} \quad \langle \hat{p}\hat{x} \rangle^{\text{st}} = -i\hbar/2 \quad (2.3.21)$$

C. The virial theorem

Consider now [cf. Eq. (2.3.5)]

$$0 = \frac{d}{dt} \langle \hat{x}\hat{p} \rangle^{\text{st}} = \langle \dot{\hat{x}}\hat{p} \rangle^{\text{st}} + \langle \hat{x}\dot{\hat{p}} \rangle^{\text{st}} = \frac{1}{m} \langle \hat{p}^2 \rangle^{\text{st}} - \langle \hat{x}V'(\hat{x}) \rangle^{\text{st}} \quad (2.3.22)$$

Write this for many-particle systems in the form of

$$\langle \hat{\mathbf{r}} \cdot \nabla V(\hat{\mathbf{r}}) \rangle^{\text{st}} = 2 \langle \hat{K} \rangle^{\text{st}} \quad (\text{Virial theorem}) \quad (2.3.23)$$

In classical mechanics, potential energy gradient is the negative force, whereas “*vire*” is “*force*” in Latin. The virial theorem, Eq. (2.3.23), is the quantum correspondence to the one in classical mechanics. As emphasized earlier, $\langle \hat{O} \rangle^{\text{st}}$ reduces to $\langle \hat{O} \rangle_n = \langle n | \hat{O} | n \rangle$, whenever the pure-state via the Schrödinger equation (2.3.12) is in study.

D. The virial theorem with homogeneous potential functions

- Homogeneous potentials refer to the scenario of

$$V(\lambda \hat{\mathbf{r}}) = \lambda^\nu V(\hat{\mathbf{r}}) \quad (2.3.24)$$

that goes with

$$\hat{\mathbf{r}} \cdot \nabla V(\hat{\mathbf{r}}) = \nu V(\hat{\mathbf{r}}) \quad (2.3.25)$$

Two typical cases: (a) The harmonic oscillator, $V(x) \propto x^2$, where $\nu = 2$;
(b) The Coulomb potential, $V(r) \propto -1/r$, where $\nu = -1$.

- For the system of Eq. (2.3.25), the virial theorem, Eq. (2.3.23), reads

$$\nu \langle \hat{V} \rangle^{\text{st}} = 2 \langle \hat{K} \rangle^{\text{st}} \quad (2.3.26)$$

This together with $\langle \hat{H} \rangle^{\text{st}} = \langle \hat{V} \rangle^{\text{st}} + \langle \hat{K} \rangle^{\text{st}}$ give rise to

$$\langle \hat{V} \rangle^{\text{st}} = \frac{2}{\nu + 2} \langle \hat{H} \rangle^{\text{st}} \quad \text{and} \quad \langle \hat{K} \rangle^{\text{st}} = \frac{\nu}{\nu + 2} \langle \hat{H} \rangle^{\text{st}} \quad (2.3.27)$$

For an individual eigen-state via $\hat{H}|n\rangle = E_n|n\rangle$, Eq. (2.3.27) becomes

$$\langle \hat{V} \rangle_n = \frac{2}{\nu + 2} E_n \quad \text{and} \quad \langle \hat{K} \rangle_n = \frac{\nu}{\nu + 2} E_n \quad (2.3.28)$$

E. The Hellmann–Feynman theorem

In general, time-independent Hamiltonian $\hat{H}$ involves some parameters, such as the particle mass m , harmonic frequency ω and nuclear charges, etc. As results,

$$\hat{H}(\lambda)|\psi_n(\lambda)\rangle = E_n(\lambda)|\psi_n(\lambda)\rangle; \quad \text{with} \quad \langle\psi_n(\lambda)|\psi_n(\lambda)\rangle = 1 \quad (2.3.29)$$

As derived soon below, we have

$$\left\langle\psi_n(\lambda)\left|\frac{\partial\hat{H}(\lambda)}{\partial\lambda}\right|\psi_n(\lambda)\right\rangle = \frac{\partial E_n(\lambda)}{\partial\lambda} \quad (2.3.30)$$

The choice of the λ -parameter is rather flexible, as long as it explicitly appears in E_n . Applications of various steady-state theorems will largely be presented in §5.4.4.

Derivations of Eq. (2.3.30): The λ -derivative on Eq. (2.3.29) reads

$$\hat{H}'(\lambda)|\psi_n(\lambda)\rangle + \hat{H}(\lambda)|\psi_n'(\lambda)\rangle = E_n'(\lambda)|\psi_n(\lambda)\rangle + E_n(\lambda)|\psi_n'(\lambda)\rangle$$

Applying $\langle\psi_n(\lambda)|$ to the above expression results in

$$\langle\psi_n(\lambda)|\hat{H}'(\lambda)|\psi_n(\lambda)\rangle + \langle\psi_n(\lambda)|\hat{H}(\lambda)|\psi_n'(\lambda)\rangle = E_n'(\lambda) + E_n\langle\psi_n(\lambda)|\psi_n'(\lambda)\rangle$$

The second terms in the two sides are equal, since $\langle\psi_n(\lambda)|\hat{H}(\lambda) = E_n\langle\psi_n(\lambda)|$. Thus,

$$\langle\psi_n(\lambda)|\hat{H}'(\lambda)|\psi_n(\lambda)\rangle = E_n'(\lambda)$$

This is Eq. (2.3.30).

(Q.E.D.)

Exercises

E2.3.2 Use the Hellmann–Feynman theorem, Eq. (2.3.30), to reproduce Eqs. (2.3.24)–(2.3.28).

E2.3.3 The electron in hydrogen atom's 1s orbital has the energy $E_{1s} = -13.6\text{ eV}$. Show that

$$\left\langle\frac{1}{r}\right\rangle_{1s} = \frac{1}{a_0}, \quad \text{with } a_0 = 0.529\text{ \AA} \quad (\text{Bohr's radius}) \quad (2.3.31)$$

E2.3.4 Consider the harmonic oscillator system, Eq. (2.3.7), with the eigen-energies of Eq. (2.1.29); i.e., $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$; $n = 0, 1, 2, \dots$. Show that

$$\Delta x \Delta p = \hbar\left(n + \frac{1}{2}\right) \quad (\text{harmonic oscillator}) \quad (2.3.32)$$

E2.3.5 Show the following general identities,

$$\frac{1}{\hbar^2} \langle\varphi|\hat{p}^2|\psi\rangle = -\int \varphi^*(\mathbf{r}) \nabla^2 \psi(\mathbf{r}) d\mathbf{r} = \int [\nabla \varphi(\mathbf{r})]^* \cdot \nabla \psi(\mathbf{r}) d\mathbf{r} \quad (2.3.33)$$

where $\hat{p}^2 = \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2$ and

$$\nabla \equiv \frac{\partial}{\partial x} \vec{\mathbf{i}} + \frac{\partial}{\partial y} \vec{\mathbf{j}} + \frac{\partial}{\partial z} \vec{\mathbf{k}} \quad (2.3.34)$$

Moreover,

$$\int \nabla^2 \rho(\mathbf{r}) d\mathbf{r} = 0, \quad \text{where } \rho(\mathbf{r}) \equiv |\psi(\mathbf{r})|^2 \quad (2.3.35)$$

Chapter 3

Theories of Quantum Dynamics

3.1 Hilbert-Space Dynamics Prescriptions

3.1.1 Causality and time-order exponential

Causality refers to the time ordering of cause-and-consequence events. This principle is related to the time-order description in quantum mechanics, especially in the case of $\hat{H}(t)$, with an explicit time-dependence arising from the external field actions. The resulting propagator turns out to be a *time-ordered exponential*, which reduces the ordinary exponential, $e^{-i\hat{H}t/\hbar}$, when the Hamiltonian is independent of time.

The Hilbert-space propagator $\hat{U}(t, t_0)$ is defined via

$$|\psi(t)\rangle \equiv \hat{U}(t, t_0)|\psi(t_0)\rangle \quad (3.1.1)$$

As it satisfies the Schrödinger equation with arbitrary initial $|\psi(t_0)\rangle$, we obtain

$$\frac{\partial}{\partial t} \hat{U}(t, t_0) = -\frac{i}{\hbar} \hat{H}(t) \hat{U}(t, t_0); \quad \text{with} \quad \hat{U}(t_0, t_0) = \hat{I} \quad (3.1.2)$$

Its formal integration gives rise to

$$\hat{U}(t, t_0) = \hat{I} - \frac{i}{\hbar} \int_{t_0}^t d\tau \hat{H}(\tau) \hat{U}(\tau, t_0) \quad (3.1.3)$$

Using this identity recursively results in

$$\hat{U}(t, t_0) = \hat{I} - \frac{i}{\hbar} \int_{t_0}^t d\tau \hat{H}(\tau) + \left(\frac{-i}{\hbar}\right)^2 \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \hat{H}(\tau_2) \hat{H}(\tau_1) + \cdots \quad (3.1.4)$$

In mathematics this is the *positively time-ordered* exponential,

$$\hat{U}(t, t_0) = \mathcal{T}_p \exp \left[-\frac{i}{\hbar} \int_{t_0}^t d\tau \hat{H}(\tau) \right] \equiv \exp_+ \left[-\frac{i}{\hbar} \int_{t_0}^t d\tau \hat{H}(\tau) \right] \quad (3.1.5)$$

Here, $\mathcal{T}_p$ is the so-called *positive time-ordering operator*, such that

$$\mathcal{T}_p\{\hat{H}(\tau)\hat{H}(\tau')\} \equiv \begin{cases} \hat{H}(\tau)\hat{H}(\tau'), & \text{if } \tau \geq \tau' \\ \hat{H}(\tau')\hat{H}(\tau), & \text{if } \tau < \tau' \end{cases}$$

Thus,

$$\mathcal{T}_p \left[\int_{t_0}^t d\tau \hat{H}(\tau) \right]^2 = 2! \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \hat{H}(\tau_2) \hat{H}(\tau_1)$$

Similarly (see the proof soon below),

$$\mathcal{T}_p \left[\int_{t_0}^t d\tau \hat{H}(\tau) \right]^n = n! \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau_{n-1} \cdots \int_{t_0}^{\tau_2} d\tau_1 \hat{H}(\tau_n) \cdots \hat{H}(\tau_1) \quad (3.1.6)$$

This proves the equivalence between Eq. (3.1.4) and Eq. (3.1.5).

Remarks

- **About the time-ordered actions:** It is noticed that Eq. (3.1.6) involves n time-ordered actions of $\hat{H}(\tau_j)$, with

$$t_0 \leq \tau_1 \leq \tau_2 \leq \cdots \leq \tau_n \leq t \quad (3.1.7)$$

In other words, the action of $\hat{H}(\tau_j)$ on $|\psi\rangle$ takes place before any $\hat{H}(\tau_k > \tau_j)$.

- **Causality:** As inferred from Eq. (3.1.6), the time-ordered exponential reduces to the ordinary exponential whenever $\hat{H}$ does not explicitly depend on time. In other words, the time-order exponential, Eq. (3.1.4) or (3.1.5), represents the universal expression for the Hilbert-space propagator $\hat{U}(t, t_0)$. It highlights the causality for the effect of Hamiltonian on the wave function, Eq. (3.1.1), at the final time t .

- **Proof of Eq. (3.1.6) (for your information only):** Evidently, Eq. (3.1.6) holds for $n = 1$. In the following, we present the proof via the induction method. For bookkeeping, we denote the n -fold time-ordered integral as

$$\mathcal{I}_n(t) \equiv \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau_{n-1} \cdots \int_{t_0}^{\tau_2} d\tau_1$$

The proof on Eq. (3.1.6) by induction is concerned with the following relation,

$$\int_{t_0}^t d\tau \int_{t_0}^{\tau} d\tau_n \mathcal{I}_{n-1}(\tau_n) = (n+1) \mathcal{I}_{n+1}(t) \quad (3.1.8)$$

assuming it had been valid when $n+1 = m$. The required proof exploits the identity,

$$\int_{t_0}^t d\tau \int_{t_0}^{\tau} d\tau_n = \int_{t_0}^t d\tau \int_{t_0}^{\tau} d\tau_n + \int_{t_0}^t d\tau_n \int_{t_0}^{\tau_n} d\tau \quad (3.1.9)$$

This results in

$$\int_{t_0}^t d\tau \int_{t_0}^{\tau} d\tau_n \mathcal{I}_{n-1}(\tau_n) = \mathcal{I}_{n+1}(t) + \int_{t_0}^t d\tau_n \left[\int_{t_0}^{\tau_n} d\tau \int_{t_0}^{\tau} d\tau_{n-1} \mathcal{I}_{n-2}(\tau_{n-1}) \right] \quad (3.1.10)$$

As inferred from Eq. (eq3.1.8), the square-bracketed term amounts to $n\mathcal{I}_n(\tau_n)$ that is set to be correct for the proof by induction. Hence, Eq. (3.1.10) reads

$$\int_{t_0}^t d\tau \int_{t_0}^{\tau} d\tau_n \mathcal{I}_{n-1}(\tau_n) = \mathcal{I}_{n+1}(t) + n \int_{t_0}^t d\tau_n \mathcal{I}_n(\tau_n) = (n+1)\mathcal{I}_{n+1}(t)$$

This proves Eq. (3.1.8) and consequently Eq. (3.1.6). (Q.E.D.)

Basic properties of propagator

(a) Unitary, $\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0) = \hat{U}(t, t_0)\hat{U}^\dagger(t, t_0) = \hat{I}$. Moreover,

$$\boxed{\hat{U}(t, t_0)\hat{U}^\dagger(\tau, t_0) = \hat{U}(t, \tau)} \quad \text{and} \quad \boxed{\hat{U}(t, \tau)\hat{U}(\tau, t_0) = \hat{U}(t, t_0)} \quad (3.1.11)$$

(b) Hermitian conjugate via Eqs. (3.1.4) and (3.1.5):

$$\hat{U}^\dagger(t, t_0) = \hat{I} + \frac{i}{\hbar} \int_{t_0}^t d\tau \hat{H}(\tau) + \left(\frac{i}{\hbar}\right)^2 \int_{t_0}^t d\tau_2 \int_{t_0}^{\tau_2} d\tau_1 \hat{H}(\tau_1) \hat{H}(\tau_2) + \cdots$$

that is equivalent to

$$\left\{ \exp_+ \left[-\frac{i}{\hbar} \int_{t_0}^t d\tau \hat{H}(\tau) \right] \right\}^\dagger = \exp_- \left[\frac{i}{\hbar} \int_{t_0}^t d\tau \hat{H}(\tau) \right] \quad (3.1.12)$$

The causality here is concerned with the Hermitian conjugate to Eq. (3.1.1); i.e.,

$$\langle \psi(t) | = \langle \psi(t_0) | \hat{U}^\dagger(t, t_0)$$

(c) As inferred from Eq. (3.1.11), we have

$$\hat{U}^\dagger(t, t_0) = \hat{U}^{-1}(t, t_0) = \hat{U}(t_0, t) \quad (3.1.13)$$

While they are mathematically equivalent, $\hat{U}^\dagger(t, t_0)$ is the choice of use in the quantum mechanics, for the reason of causality description.

3.1.2 Perturbative dynamics versus interaction picture

A. Time-dependent perturbation theory

- The interaction picture arises naturally as an intermediate step in the construction of the time-dependent perturbation theory. The total Hamiltonian consists of the unperturbed $\hat{H}(t)$ and a perturbed $\hat{V}(t)$ parts,

$$\hat{H}_T(t) = \hat{H}(t) + \hat{V}(t) \quad (3.1.14)$$

We investigate the total propagator in the form of

$$\hat{U}_T(t, t_0) \equiv \hat{U}(t, t_0) \hat{U}_I(t, t_0) \quad (3.1.15)$$

assuming that the unperturbed $\hat{U}(t, t_0)$ via $\hat{H}(t)$ had been known.

- The perturbed component, $\hat{U}_I(t, t_0)$ as defined via Eq. (3.1.15), is called the propagator in the interaction picture. This is the *intermediate quantity* of formulation, leading to the Schrödinger picture $\hat{U}_T(t, t_0)$ of Eq. (3.1.15),

$$\hat{U}_T(t, t_0) = \hat{U}(t, t_0) - \frac{i}{\hbar} \int_{t_0}^t d\tau \hat{U}(t, \tau) \hat{V}(\tau) \hat{U}_T(\tau, t_0) \quad (3.1.16)$$

This is similar to Eq. (3.1.4), but for the desired Dyson's by-orders expansion of the time-dependent perturbation $\hat{V}(\tau)$.

Derivations: By using Eqs. (3.1.14) and (3.1.15), together with

$$\frac{\partial \hat{U}_T}{\partial t} = \frac{\partial \hat{U}}{\partial t} \hat{U}_I + \hat{U} \frac{\partial \hat{U}_I}{\partial t} = -\frac{i}{\hbar} \hat{H}_T \hat{U}_T \quad \text{and} \quad \frac{\partial \hat{U}}{\partial t} = -\frac{i}{\hbar} \hat{H} \hat{U} \quad (3.1.17)$$

we obtain $\hat{U} \frac{\partial \hat{U}_I}{\partial t} = -\frac{i}{\hbar} \hat{V} \hat{U} \hat{U}_I$. This leads to

$$\frac{\partial}{\partial t} \hat{U}_I(t, t_0) = -\frac{i}{\hbar} \hat{V}_I(t) \hat{U}_I(t, t_0), \quad \text{with} \quad \hat{V}_I(t) \equiv \hat{U}^\dagger(t, t_0) \hat{V}(t) \hat{U}(t, t_0) \quad (3.1.18)$$

This is the interaction picture. We obtain [cf. Eq. (3.1.3)],

$$\hat{U}_I(t, t_0) = \hat{I} - \frac{i}{\hbar} \int_{t_0}^t d\tau [\hat{U}^\dagger(\tau, t_0) \hat{V}(\tau) \hat{U}(\tau, t_0)] \hat{U}_I(\tau, t_0) \quad (3.1.19)$$

Multiplying it from the left with $\hat{U}(t, t_0)$, followed by using Eq. (3.1.11), we obtain

$$\hat{U}(t, t_0) \hat{U}_I(t, t_0) = \hat{U}(t, t_0) - \frac{i}{\hbar} \int_{t_0}^t d\tau \hat{U}(t, \tau) \hat{V}(\tau) \hat{U}(\tau, t_0) \hat{U}_I(\tau, t_0) \quad (3.1.20)$$

This together with Eq. (3.1.15) give rise to Eq. (3.1.16). (Q.E.D.)

- For the (Schrödinger picture) wave function, we obtain

$$\psi_{\text{T}}(t) = \sum_{n=0}^{\infty} \psi_{\text{T}}^{(n)}(t), \quad (3.1.21)$$

where (setting $\tau_{n+1} \equiv t$)

$$|\psi_{\text{T}}^{(n)}(t)\rangle = \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t d\tau_n \cdots \int_{t_0}^{\tau_2} d\tau_1 \left[\prod_{j=1}^n \hat{U}(\tau_{j+1}, \tau_j) \hat{V}(\tau_j) \right] |\psi(\tau_1)\rangle \quad (3.1.22)$$

with $|\psi(\tau_1)\rangle = \hat{U}(\tau_1, t_0)|\psi(t_0)\rangle$ via the unperturbed propagator.

- (a) The underlying n -fold time-ordered integrand reads explicitly

$$\hat{U}(t, \tau_n) \hat{V}(\tau_n) \hat{U}(\tau_n, \tau_{n-1}) \hat{V}(\tau_{n-1}) \cdots \hat{U}(\tau_2, \tau_1) \hat{V}(\tau_1) |\psi(\tau_1)\rangle \quad (3.1.23)$$

The n actions of perturbation, $\{\hat{V}(\tau_j)\}$, occur at $\tau_1 \leq \tau_2 \leq \cdots \leq \tau_n$. Meanwhile, in between each time interval of actions, $[\tau_j, \tau_{j+1}]$ and also $[\tau_n, t \equiv \tau_{n+1}]$, the system propagates with the *unperturbed* $\hat{U}(\tau_{j+1}, \tau_j)$.

- (b) The n^{th} -order perturbed $|\psi_{\text{T}}^{(n)}(t)\rangle$ involves further the n -fold integrations over all action-time moments, as described by Eq. (3.1.22) that results from Eq. (3.1.16).

- Moreover, as inferred from Eq. (3.1.15), we have

$$|\psi_{\text{T}}^{(n)}(t)\rangle = \hat{U}(t, t_0) |\psi_I^{(n)}(t)\rangle \quad (3.1.24)$$

where

$$|\psi_I^{(n)}(t)\rangle \equiv \hat{U}_I(t, t_0) |\psi(t_0)\rangle \quad (3.1.25)$$

This is the *interaction picture* wave function, satisfying [cf. Eq. (3.1.18)]

$$\left[\frac{\partial}{\partial t} \psi_I(t) = -\frac{i}{\hbar} \hat{V}_I(t) \psi_I(t) \right], \quad \text{with} \quad \left[\hat{V}_I(t) \equiv \hat{U}^\dagger(t, t_0) \hat{V}(t) \hat{U}(t, t_0) \right] \quad (3.1.26)$$

B. Quantum prescription invariance principle

We have learned three prescriptions of quantum mechanics; i.e., the Schrödinger picture, the Heisenberg picture and the interaction picture. The initial wave function is set to be common in all these three pictures:

$$|\psi(t_0)\rangle = |\psi_{\text{T}}(t_0)\rangle = |\psi_I(t_0)\rangle = |\psi_{\text{H}}(t_0)\rangle \quad (3.1.27)$$

The prescription invariance principle reads

$$\begin{aligned} \bar{A}(t) &= \langle \psi_{\text{T}}(t) | \hat{A} | \psi_{\text{T}}(t) \rangle = \langle \psi(t_0) | \hat{A}_{\text{H}}(t) | \psi(t_0) \rangle \\ &= \langle \psi_I(t) | \hat{A}_I(t) | \psi_I(t) \rangle \end{aligned} \quad (3.1.28)$$

with the following detailed specifications.

$$\text{Schrödinger picture : } |\psi_T(t)\rangle = \hat{U}_T(t, t_0)|\psi(t_0)\rangle \quad (3.1.29a)$$

$$\text{Heisenberg picture : } \hat{A}_H(t) = \hat{U}_T^\dagger(t, t_0)\hat{A}\hat{U}_T(t, t_0) \quad (3.1.29b)$$

$$\text{Interaction picture : } \begin{cases} |\psi_I(t)\rangle = \hat{U}_I(t, t_0)|\psi(t_0)\rangle \\ \hat{A}_I(t) = \hat{U}^\dagger(t, t_0)\hat{A}\hat{U}(t, t_0) \end{cases} \quad (3.1.29c)$$

where $\hat{U}_T(t, t_0) = \hat{U}(t, t_0)\hat{U}_I(t, t_0)$ [cf. Eq. (3.1.15)]. The initial values of wave vectors are the same $|\psi(t_0)\rangle$, since all propagators at $t = t_0$ are the unit operator.

Remarks. Onset of interaction picture in open quantum systems

- Consider a *thermodynamic* system, such as a solute-solvent mixture, at *temperature* T . Let $\hat{H}_S$ and $\hat{H}_B$ be the isolated system (solute) and bath (solvent) Hamiltonians, respectively, before mixing. The total composite mixture Hamiltonian is time-independent and reads

$$\hat{H}_T = \hat{H}_S + \hat{H}_B + \hat{Q}^S \hat{F}^B \quad (3.1.30)$$

with $\hat{Q}^S \equiv \{\hat{Q}_j^S\}$ and $\hat{F}^B \equiv \{\hat{F}_j^B\}$ being operators in the system subspace and the bath subspace, respectively. Note that $[\hat{A}_S, \hat{H}_B] = 0$, where $\hat{A}_S$ is an arbitrary operator in the system subspace, including $\hat{H}_S$ and $\hat{Q}^S$.

(a) As the system is of the primary interest, one often adopts the bath $\hat{H}_B$ -interaction description, in which Eq. (3.1.30), exemplified with single system-bath coupling mode, becomes [cf. Eq. (3.1.26)]

$$\tilde{H}_T(t) = \hat{H}_S + \hat{Q}^S \hat{F}^B(t), \quad \text{with } \hat{F}^B(t) = e^{i\hat{H}_B t/\hbar} \hat{F}^B e^{-i\hat{H}_B t/\hbar} \quad (3.1.31)$$

(b) The bath-subspace has practically infinite degrees of freedom. The *statistical thermodynamics description* is applicable for the bath environment influence. The above interaction picture goes effectively with much *reduced* degrees of freedom.

(c) Physically, each individual $\hat{F}_j^B(t)$ is the *stochastic random force* acting on the system mode $\hat{Q}_j^S$. In the *statistical thermodynamics*, the influence of bath is described by the stochastic force-force correlation functions matrix, $\langle \hat{F}_j^B(t) \hat{F}_k^B(0) \rangle_B$, where $\langle (\cdot) \rangle_B \equiv \text{tr}_B[(\cdot)e^{-\beta \hat{H}_B}]/\text{tr}_B e^{-\beta \hat{H}_B}$, with $\beta \equiv 1/(k_B T)$.

(d) The aforementioned *reduced description* constitutes the starting point to the theory of open quantum systems, involving relaxation and decoherence processes. *Thermodynamic irreversibility* occurs, resulting in the total composite density operator toward $\hat{\rho}_T(t \rightarrow \infty) = \hat{\rho}_T^{\text{eq}}(T) = e^{-\beta \hat{H}_T}/\text{Tr} e^{-\beta \hat{H}_T}$, at thermal equilibrium.

- It is worth noting that quantum measurement apparatus is actually a type of environment. The readout-time is not immediate but finite, and so is the wave function collapse time.

3.2 Density Operators and Liouville–Space Algebra

3.2.1 General properties of density operator

A. Opening remarks

- Density operator (matrix), $\hat{\rho}$, comprises a *probability* type of linear combination of Hilbert–space outer products, $\{|\psi\rangle\langle\varphi|\}$. As results, the total composite system density operator satisfies

$$\dot{\hat{\rho}}_{\text{T}} = -\frac{i}{\hbar}[\hat{H}_{\text{T}}(t), \hat{\rho}_{\text{T}}] \equiv -i\mathcal{L}_{\text{T}}(t)\hat{\rho}_{\text{T}} \quad (3.2.1)$$

This is the Liouville–von Neumann equation. The last identity defines the total composite system Liouvillian,

$$\mathcal{L}_{\text{T}}\hat{O} \equiv [\hat{H}_{\text{T}}, \hat{O}]/\hbar \quad (3.2.2)$$

- The most distinct feature of the density operator approach is that it supports a *reduced* or system–bath description, which is closely related to the thermodynamic *irreversibility*. The reduced system density operator is defined as $\hat{\rho}_{\text{S}} \equiv \text{tr}_{\text{B}}\hat{\rho}_{\text{T}}$; where $\hat{\rho}_{\text{T}}$ is the total system–plus–bath composite density operator and tr_{B} is the environment bath subspace trace. Macroscopic or phenomenological variables such as temperature can be naturally included in the reduced description. The reduced system density operator satisfies the so-called the quantum master equation, in the framework of quantum dissipation theories that are beyond this course.
- This section focuses on the the Liouville–von Neumann equation (3.2.1), referred as the *complete Liouville–space description*. In mathematics, it is isomorphism to the Schrödinger equation. However, the Liouville–space dynamics provides a versatile vehicle in formulating various nonlinear response functions or spectrums. We will elaborate some basic results via Eq. (3.2.1) in due course.

B. Basic properties of density operators

- Density operator $\hat{\rho}$, either the total composite or reduced one, is required to be *Hermitian*, *positively defined*, and *integrable*. These requirements arise from the probability interpretation of density operator, as follows.

(a) As known, the Hilbert space supports only the *pure–states* description, where the system is characterized by a *single* wave function ψ , including the coherent superposition, $|\psi\rangle = \sum_n c_n |\phi_n\rangle$. The probability density of finding a particle in the vicinity of x is $|\psi(x)|^2$. On the other hand, the density operator for the pure–state system is given by $\hat{\rho} = |\psi\rangle\langle\psi|$.

(b) In the coordinate representation, it is $\rho(x, x') = \langle x | \hat{\rho} | x' \rangle$. The aforementioned pure-state probability density amounts to $\rho(x, x) = |\psi(x)|^2$. In a discrete basic-set representation, while $|\psi\rangle = \sum_n c_n |n\rangle$ assumes a vector, the density operator is a matrix, with the elements of $\rho_{mn} = \langle m | \hat{\rho} | n \rangle$. For the pure-state case it reads $\rho_{mn} = c_m c_n^*$ and satisfies $\rho_{mn} \rho_{nm} = \rho_{mm} \rho_{nn}$.

(c) A realistic system is in general a *mixed state* that cannot be described by a single wave function. However, the density operator description remains valid. It contains the information on the probabilities of individual pure states, i.e.,

$$\hat{\rho} = \sum_j P_j |\psi_j\rangle \langle \psi_j|, \quad \text{where } P_j \geq 0 \quad \text{and} \quad \sum_j P_j = 1 \quad (3.2.3)$$

These expressions lead immediately to

$$\text{Tr} \hat{\rho}^2 \leq \text{Tr} \hat{\rho} \quad (3.2.4)$$

The “=” sign hold only for pure states since $(\hat{\rho}^2 = \hat{\rho})_{\text{pure state}}$. In the thermal equilibrium, $\hat{\rho}^{\text{eq}}(T) = e^{-\hat{H}/(k_B T)} / \text{Tr} e^{-\hat{H}/(k_B T)}$, with $\{P_j\}$ being the Boltzmann distribution over the eigenstates $\{|\psi_j\rangle\}$.

- As noticed, Eq. (3.2.3) represents the diagonal representation of density operator, with the local-time dependence on $\hat{\rho}$, ψ_j and P_j . In general, one can exploit time-independent basic set and recast Eq. (3.2.3) as

$$\hat{\rho}(t) = \sum_{mn} \rho_{mn}(t) |m\rangle \langle n|, \quad \text{where } \rho_{mn}(t) = \langle m | \hat{\rho}(t) | n \rangle \quad (3.2.5)$$

satisfying $\rho_{mn} = \rho_{nm}^*$, since $\hat{\rho} = \hat{\rho}^\dagger$.

(d) Diagonal elements, $\{\rho_{mm}\}$, describes populations, with $\rho_{mm} \geq 0$ in any representation. This is the positivity requirement, in consistent with Eq. (3.2.3), on any density operator, satisfying the Schwartz inequality, $|\rho_{mn}|^2 \leq \rho_{mm} \rho_{nn}$, as implied in Eq. (3.2.4).

(e) Off-diagonal elements, $\{\rho_{mn}; m \neq n\}$, are named coherences between specified states. Apparently, the above classification between population and coherence depends on representation. Therefore, one should avoid using off-diagonal $\{\rho_{mn}(t)\}$ to the measure of dynamical decoherence. The latter should be an intrinsic property, depending only on the choice of reduced system description, in relation to the total system-bath matter composite Hamiltonian, $\hat{H}_T$ of Eq. (3.1.30).

C. Advantage of density operator description

- Density operator description is universal for *experimental measurements in response to external fields*.

(i) First of all, we can generalize the expectation value as

$$\bar{A}(t) = \text{Tr}[\hat{A}\hat{\rho}(t)] \quad (3.2.6)$$

One may argue that is just $\bar{A} = \sum_j P_j \langle \psi_j | \hat{A} | \psi_j \rangle$, if the density operator is written in its diagonal representation, Eq. (3.2.3), at local time; thus, the advantage of Eq. (3.2.6) over the Hilbert-space wave function description would rather be insignificant.

(ii) Experiments often engage certain external fields to interrogate, manipulate and detect systems in study. Measured quantities are certain spectrums or nonlinear response of system with respect to the external fields. The related perturbation expansion on expectation value via Eq. (3.2.6) is concerned with

$$\bar{A}(t) = \sum_{n=0}^{\infty} \bar{A}^{(n)}(t) = \sum_{n=0}^{\infty} \text{Tr}[\hat{A}\hat{\rho}^{(n)}(t)] \quad (3.2.7)$$

This is much simpler than the wave function description.

3.2.2 The Liouville-space descriptions

A. Conventions and tetradic notations

- First of all, a linear space in mathematics is concerned with the following three features: (i) Inner product of two elements in a linear space is defined to be a c-number; (ii) Outer product of two elements constitutes a linear operator in the given space; (iii) Linear space goes with linear operators that map one element to another element in the linear space.

(a) Both the Hilbert space and Liouville space are linear. However, the elements of Liouville-space are those Hilbert-space operators. In this contact, the Liouville-space operators are also called superoperators, which are labeled by script letters, such as $\mathcal{O}$, throughout this book.

(b) There are three types of Hilbert-space operators: (1) Dynamical variables, $\hat{A}$; (2) Density operators, $\hat{\rho}$; (3) The outer products of the Hilbert-space vectors, $|m\rangle\langle n|$, as needed for representations; see Eqs. (3.2.10) and (3.2.13). Evidently, the Liouville-space construction is a natural extension (isomorphism) of the Hilbert-space algebra.

- For convenience of manipulation, we adopt the Liouville-space *tetradic* bra-and-ket notation with the following **conventions**:

$$(i) \quad \boxed{\langle\langle \hat{A} | \iff \hat{A} \quad \text{and} \quad \hat{\rho} \iff |\hat{\rho}\rangle\rangle} \quad (3.2.8)$$

$$(ii) \quad \boxed{\langle\langle \hat{A} | \hat{\rho}(t) \rangle\rangle \equiv \text{Tr}[\hat{A}\hat{\rho}(t)] = \bar{A}(t)} \quad (3.2.9)$$

$$(iii) \quad \boxed{|mn\rangle\rangle \implies |m\rangle\langle n| \quad \text{and} \quad \langle\langle mn| \implies |n\rangle\langle m|} \quad (3.2.10)$$

Comments:

(a) Convention-(i) defines dynamic operators and state operators to be the Liouville-space ket and bra components, respectively. This convention will also be used later, in relation to the Heisenberg picture versus the Schrödinger picture; see Eq. (3.3.9).

(b) Convention-(ii) is set in line with Convention-(i), and defines further the Liouville-space inner product, $\langle\langle \hat{A} | \hat{\rho}(t) \rangle\rangle$. This also suggests that $\text{Tr}[\hat{A}\hat{\rho}(t)]$ be formally preferred over the equal-valued $\text{Tr}[\hat{\rho}(t)\hat{A}]$.

(c) Convention-(iii) is used only for relating between Liouville-space and Hilbert-space representations, whenever applicable. This sole usage is specified by the “one-way map” sign, $\implies$, in Eq. (3.2.10) that goes by $\langle\langle mn| = (|mn\rangle\rangle)^\dagger$.

(d) Combining (i) and (iii) results in

$$\boxed{\begin{aligned} \langle\langle \hat{A} | mn \rangle\rangle &= \langle n | \hat{A} | m \rangle \equiv A_{nm} \\ \langle\langle mn | \hat{\rho} \rangle\rangle &= \langle m | \hat{\rho} | n \rangle \equiv \rho_{mn} \end{aligned}} \quad (3.2.11)$$

- By using Eqs. (3.2.9) and (3.2.11), we have

$$\bar{A} \equiv \text{Tr}(\hat{A}\hat{\rho}) = \langle\langle \hat{A} | \hat{\rho} \rangle\rangle = \sum_{mn} \langle\langle \hat{A} | mn \rangle\rangle \langle\langle mn | \hat{\rho} \rangle\rangle \quad (3.2.12)$$

This is identical to the Hilbert-space based evaluation, but goes by

$$\boxed{\begin{aligned} \langle\langle mn | m' n' \rangle\rangle &= \delta_{mm'} \delta_{nn'} & (\text{Orthonormality}) \\ \sum_{mn} |mn\rangle\rangle \langle\langle mn| &= \mathcal{I} & (\text{Completeness}) \end{aligned}} \quad (3.2.13)$$

The coordinate or momentum representation is similar. The Liouville-space unit operator, $\mathcal{I}$, refers also to the unit superoperator or tensor.

- Consider then the Liouville-space representation identity,

$$\langle\langle \hat{A} | \mathcal{O} | \hat{\rho} \rangle\rangle = \sum_{mn, m' n'} \langle\langle \hat{A} | mn \rangle\rangle \langle\langle mn | \mathcal{O} | m' n' \rangle\rangle \langle\langle m' n' | \hat{\rho} \rangle\rangle \quad (3.2.14)$$

This is the *tetradic bra-and-ket notation*. While $\langle\langle \hat{A}|mn\rangle\rangle = A_{nm}$ and $\langle\langle m'n'|\hat{\rho}\rangle\rangle = \rho_{m'n'}$ are matrixes [cf. Eq. (3.2.11)], the superoperator $\mathcal{O}$ is a tensor, with the elements,

$$\mathcal{O}_{mn,m'n'} \equiv \langle\langle mn|\mathcal{O}|m'n'\rangle\rangle \quad (3.2.15)$$

The diagonal elements refer to $\mathcal{O}_{mn,mn}$.

B. Hermitian superoperator

- A *Hermitian superoperator*, symbolically denoted by $\mathcal{O}^\dagger = \mathcal{O}$, is defined via its action on an arbitrary state operator $\hat{\rho}$ as

$$\mathcal{O}^\dagger \hat{\rho} \equiv (\mathcal{O} \hat{\rho}^\dagger)^\dagger = \mathcal{O} \hat{\rho} \quad (\text{Hermitian } \mathcal{O}) \quad (3.2.16)$$

The following properties arise immediately: (i) A superoperator is Hermitian, only if it *maps a Hermitian operator to another Hermitian operator*; (ii) Product of Hermitian superoperators remains a Hermitian superoperator; (iii) The tensor elements of a Hermitian superoperator are of

$$(\mathcal{O}^\dagger)_{mn,m'n'} \equiv \mathcal{O}_{nm,n'm'}^* = \mathcal{O}_{mn,m'n'} \quad (3.2.17)$$

The first identity, which is equivalent to that in Eq. (3.2.17), defines $\mathcal{O}^\dagger$.

- As noticed, diagonal tensor elements refer to $\mathcal{O}_{mn,mn}$, which for a Hermitian superoperator could be complex. Consider, for example, the Liouvillian $\mathcal{L}$, defined via the Hamiltonian commutator action; i.e.,

$$\mathcal{L} \cdot \equiv [\hat{H}, \cdot] / \hbar \quad (3.2.18)$$

(a) Apparently, $i\mathcal{L}$ is a Hermitian superoperator, as it acts on any Hermitian operator resulting in another Hermitian operator. It is easy to obtain

$$(i\mathcal{L})_{mn,m'n'} = (i/\hbar)(\delta_{nn'}H_{mm'} - H_{n'n}\delta_{mm'}) \quad (3.2.19a)$$

Its diagonal elements are

$$(i\mathcal{L})_{mn,mn} = (i/\hbar)(H_{mm} - H_{nn}) \quad (3.2.19b)$$

that is purely imaginary.

(b) In the H -representation,

$$H_{mm'} = \langle m|\hat{H}|m'\rangle = \delta_{mm'}E_m \quad (3.2.20)$$

The corresponding Eq. (3.2.19) is diagonal; i.e.,

$$(i\mathcal{L})_{mn,m'n'} = i\omega_{mn}\delta_{mm'}\delta_{nn'} \quad (3.2.21)$$

with [cf. Eq. (2.1.20)]

$$\omega_{mn} \equiv \frac{E_m - E_n}{\hbar} \quad (3.2.22)$$

being the transition frequency between two eigen-states.

3.3 Liouville–Space Dynamics Prescriptions

3.3.1 Schrödinger picture and Heisenberg picture

- A Liouville–space propagator, $\mathcal{U}_T(t, t_0)$, is defined in the total system–and–bath composite space via

$$\hat{\rho}_T(t) \equiv \mathcal{U}_T(t, t_0) \hat{\rho}_T(t_0) \quad (3.3.1)$$

Evidently, $\mathcal{U}_T(t, t_0)$ preserves the Hermitian and positivity.

Remarks:

- In a *reduced* description, Eq. (3.3.1) would formally remain valid for $\hat{\rho}_S(t) \equiv \text{tr}_B \hat{\rho}_T(t) \equiv \mathcal{U}(t, t_0) \hat{\rho}_S(t_0)$. This is the so-called *reduced system density operator*. The reduced Liouville–space propagator, $\mathcal{U}(t, t_0)$, preserves the Hermitian and positivity, but engages the thermodynamic *irreversibility*. Neither $\mathcal{U}^{-1}(t, t_0)$ nor $\mathcal{U}^\dagger(t, t_0)$ would be of physical relevance, as long as the reduced description is engaged. We will not treat this case in this book.
- In a *complete* description, all degrees of freedom, including solute system and solvent bath, are explicitly treated. The total non-reduced Liouville–space propagator, $\mathcal{U}_T(t, t_0)$, relates to the Hilbert-space propagator as

$$\boxed{\mathcal{U}_T(t, t_0) \hat{\rho}_T(t_0) = \hat{U}_T(t, t_0) \hat{\rho}_T(t_0) \hat{U}_T^\dagger(t, t_0)} \quad (3.3.2)$$

with $\hat{U}_T(t, t_0)$ of Eq. (3.1.5) being unitary and dictated by the total Hamiltonian $\hat{H}_T(t)$ that can be time-dependent. The *microscopic reversibility* holds; i.e., $\mathcal{U}_T^{-1}(t, t_0) = \mathcal{U}_T(t_0, t)$. The corresponding $\mathcal{U}_T(t, t_0)$ is a unitary superoperator and satisfies $\mathcal{U}_T(t, t_0) \mathcal{U}_T^\dagger(t, t_0) = \mathcal{U}_T^\dagger(t, t_0) \mathcal{U}_T(t, t_0) = \mathcal{I}$.

- Quantum mechanics in a complete description which includes all degrees of freedom of composite system is non-dissipative, involving no entropy change. Therefore, the probabilities of system in different states are time-independent; i.e., $\hat{\rho}_T(t) = \sum_j P_j |\psi_j(t)\rangle \langle \psi_j(t)|$, with $\dot{P}_j = 0$; thus

$$\dot{\hat{\rho}}_T(t) = -\frac{i}{\hbar} [\hat{H}_T(t), \hat{\rho}_T(t)] \equiv -i \mathcal{L}_T(t) \hat{\rho}_T(t) \quad (3.3.3)$$

This is the Liouville–von Neumann equation. The Liouville-space propagator, $\mathcal{U}_T(t, t_0)$, defined via $\hat{\rho}_T(t) \equiv \mathcal{U}_T(t, t_0) \hat{\rho}_T(t_0)$, is then

$$\mathcal{U}_T(t, t_0) = \exp_+ \left[-i \int_{t_0}^t d\tau \mathcal{L}_T(\tau) \right] \quad (3.3.4)$$

This is the mathematic isomorphism of Eq. (3.1.5), but with the Liouvillian, $\mathcal{L}_T(t) \cdot \equiv \hbar^{-1} [\hat{H}_T(t), \cdot]$, in Eq. (3.3.3).

- The Liouville-space algebra unifies the Schrödinger picture and the Heisenberg picture. While the former goes by Eq. (3.3.3), the latter is concerned with the time evolution of dynamical variable $\hat{A}(t)$, considered in particular with time-independent Hamiltonian. In this case,

$$\mathcal{U}(t, t_0) = e^{-i\mathcal{L}(t-t_0)} \equiv \mathcal{G}(t - t_0) \quad (3.3.5)$$

in relation to the Hilbert-space counterpart, $\hat{G}(t) = e^{-i\hat{H}t/\hbar}$. The unification follows the convention of Eq. (3.2.12) and recast it as

$$\bar{A}(t) = \langle\langle \hat{A} | \hat{\rho}(t) \rangle\rangle = \langle\langle \hat{A} | \mathcal{G}(t) | \hat{\rho}(0) \rangle\rangle = \langle\langle \hat{A}(t) | \hat{\rho}(0) \rangle\rangle \quad (3.3.6)$$

While $\hat{\rho}(t) = \mathcal{G}(t)\hat{\rho}(0)$, the Heisenberg picture goes by

$$\hat{A}(t) \equiv \hat{A}\mathcal{G}(t) = \hat{A}e^{-i\mathcal{L}t} \quad (3.3.7)$$

that satisfies Eq. (2.3.3); i.e.,

$$\dot{\hat{A}} = -i\hat{A}\mathcal{L} = -\frac{i}{\hbar}[\hat{A}, \hat{H}] \quad (3.3.8)$$

The unification in line with Eq. (3.3.2) goes by

$$\boxed{\begin{array}{ll} \mathcal{B}\hat{\rho} \equiv [\hat{B}, \hat{\rho}] & \longleftrightarrow e^{-\alpha\mathcal{B}}\hat{\rho} = e^{-\alpha\hat{B}}\hat{\rho}e^{\alpha\hat{B}} \quad (\text{Schrödinger}) \\ \updownarrow & \\ \hat{A}\mathcal{B} \equiv [\hat{A}, \hat{B}] & \longleftrightarrow \hat{A}e^{-\alpha\mathcal{B}} = e^{\alpha\hat{B}}\hat{A}e^{-\alpha\hat{B}} \quad (\text{Heisenberg}) \end{array}} \quad (3.3.9)$$

Remarks: On the other hand, the mathematical equivalence includes

$$\hat{A}e^{-\alpha\mathcal{B}} = e^{\alpha\hat{B}}\hat{A}e^{-\alpha\hat{B}} = e^{\alpha\mathcal{B}}\hat{A} = \sum_{n=0}^{\infty} \frac{\alpha^n}{n!} \mathcal{B}^n \hat{A}$$

so that

$$e^{\hat{B}}\hat{A}e^{-\hat{B}} = \hat{A} + [\hat{B}, \hat{A}] + \frac{1}{2!}[\hat{B}, [\hat{B}, \hat{A}]] + \frac{1}{3!}[\hat{B}, [\hat{B}, [\hat{B}, \hat{A}]]] + \dots$$

3.3.2 Nonlinear response functions

A. General remarks

• The response theory is concerned with a matter system under external fields interrogations. Let us start with the Liouville-space interaction picture, the mathematical isomorphism to that in Hilbert space in §3.1.2. The former, as inferred from Eqs. (3.2.6) and (3.2.7), is more convenient in formulating experimental quantities. The total system consists of the unperturbed and the perturbed parts; i.e., $\hat{H}_T(t) = \hat{H}(t) + \hat{V}(t)$ [Eq. (3.1.14)] or the equivalent

$$\mathcal{L}_T(t) = \mathcal{L}(t) + \mathcal{V}(t) \quad (3.3.10)$$

with $\mathcal{V}(t)$ being treated perturbatively. Here,

$$\mathcal{L}(t)\hat{O} \equiv \hbar^{-1}[\hat{H}(t), \hat{O}] \quad \text{and} \quad \mathcal{V}(t)\hat{O} \equiv \hbar^{-1}[\hat{V}(t), \hat{O}] \quad (3.3.11)$$

Consider the density operator in perturbative expansion [cf. Eq. (3.1.21)],

$$\hat{\rho}_T(t) = \sum_{n=0}^{\infty} \hat{\rho}_T^{(n)}(t) \quad (3.3.12)$$

Here, $\hat{\rho}_T^{(n)}(t)$, which has the n^{th} -order dependence on $\mathcal{V}(t)$, is given by the Liouville-space isomorphism of Eq. (3.1.22). That is (setting $\tau_{n+1} \equiv t$)

$$\hat{\rho}_T^{(n)}(t) = (-i)^n \int_{t_0}^t d\tau_n \cdots \int_{t_0}^{\tau_2} d\tau_1 \left[\prod_{j=1}^n \mathcal{U}(\tau_{j+1}, \tau_j) \mathcal{V}(\tau_j) \right] \hat{\rho}(\tau_1) \quad (3.3.13)$$

with the time-order product of Liouville-space actions on the unperturbed $\hat{\rho}(\tau_1) \equiv \mathcal{U}(\tau_1, t_0)\hat{\rho}(t_0)$, at time τ_1 of the first action taking place.

The physical meaning of Eq. (3.3.13) is straightforward. The n perturbative actions of $\mathcal{V}(\tau_j)$, $j = 1, \dots, n$, occur sequently at $\tau_1 \leq \cdots \leq \tau_n$, between $t_0 \equiv \tau_0$ and $\tau_{n+1} \equiv t$. In each interval, $[\tau_j, \tau_{j+1}]$, the system propagates unperturbatively, as described by the perturbation-free propagator, $\mathcal{U}(\tau_{j+1}, \tau_j)$, governed by the unperturbed $\mathcal{L}(t)$. The final result on the n^{th} -order density operator contribution, $\hat{\rho}_T^{(n)}(t)$, takes the integrations over those n interrogation times, $\{\tau_j; j = 1, \dots, n\}$.

B. Nonlinear response functions

• We will be interested in an initially equilibrium or steady-state system, in response to a classical external field $\epsilon(t)$. The total matter-and-field composite Hamiltonian and Liouvillian read [cf. Eq. (3.3.9)]

$$\hat{H}_T(t) = \hat{H} - \hbar \hat{B}\epsilon(t) \quad \text{and} \quad \mathcal{L}_T(t) = \mathcal{L} - \mathcal{B}\epsilon(t) \quad (3.3.14)$$

with time-independent $\hat{H}$ and its associated $\mathcal{L}$ for the pure matter part. In this case, the unperturbed propagator refers also to the Green's function,

$$\mathcal{U}(t, \tau) = e^{-i\mathcal{L}(t-\tau)} \equiv \mathcal{G}(t - \tau) \quad (3.3.15)$$

(a) Moreover, before applying the external field, the system was in a steady state. This fact is denoted as $\hat{\rho}(t_0 \rightarrow -\infty) = \hat{\rho}^{\text{st}}$ that satisfies $[\hat{H}, \hat{\rho}^{\text{st}}] = 0$. Thus, the unperturbed density operator is

$$\hat{\rho}(\tau_1) = \mathcal{G}(\tau_1 - t_0)\hat{\rho}(t_0) = \hat{\rho}^{\text{st}} \quad (3.3.16)$$

Equation (3.3.13) reads now

$$\hat{\rho}_{\text{T}}^{(n)}(t) = i^n \int_{t_0}^t d\tau_n \cdots \int_{t_0}^{\tau_2} d\tau_1 \left[\prod_{j=1}^n \mathcal{G}(\tau_{j+1} - \tau_j) \mathcal{B} \epsilon(\tau_j) \right] \hat{\rho}^{\text{st}} \quad (3.3.17)$$

This leads to the expectation value of a dynamical variable the expression,

$$\bar{A}(t) = \text{Tr}[\hat{A}\hat{\rho}_{\text{T}}(t)] = \sum_{n=0}^{\infty} \langle\langle \hat{A} | \hat{\rho}_{\text{T}}^{(n)}(t) \rangle\rangle = \sum_{n=0}^{\infty} \bar{A}^{(n)}(t) \quad (3.3.18a)$$

where

$$\bar{A}^{(n)}(t) \equiv \text{Tr}[\hat{A}\hat{\rho}_{\text{T}}^{(n)}(t)] \equiv \langle\langle \hat{A} | \hat{\rho}_{\text{T}}^{(n)}(t) \rangle\rangle \quad (3.3.18b)$$

(b) Applying Eq. (3.3.17), we obtain

$$\bar{A}^{(n)}(t) = \int_{-\infty}^t d\tau_n \cdots \int_{-\infty}^{\tau_2} d\tau_1 \chi_{AB}^{(n)}(t - \tau_n, \cdots, \tau_2 - \tau_1) \epsilon(\tau_n) \cdots \epsilon(\tau_1) \quad (3.3.19)$$

with the n^{th} -order response function,

$$\chi_{AB}^{(n)}(t_n, \cdots, t_1) = i^n \langle\langle \hat{A} | \mathcal{G}(t_n) \mathcal{B} \mathcal{G}(t_{n-1}) \mathcal{B} \cdots \mathcal{G}(t_1) \mathcal{B} | \hat{\rho}^{\text{st}} \rangle\rangle \quad (3.3.20)$$

This must be *real* whenever both $\hat{A}$ and $\hat{B}$ are Hermite. The involving time variables are the *durations* of field-free propagations,

$$t_n \equiv t - \tau_n, \quad t_{n-1} \equiv \tau_n - \tau_{n-1}, \quad \cdots, \quad t_1 = \tau_2 - \tau_1 \quad (3.3.21a)$$

while the classical external field takes actions at

$$\tau_j = t - \sum_{k=j}^n t_k; \quad j = 1, \cdots, n \quad (3.3.21b)$$

Therefore, Eq. (3.3.19) reads also

$$\begin{aligned} \bar{A}^{(n)}(t) &= \int_0^\infty dt_n \cdots \int_0^\infty dt_1 \chi_{AB}^{(n)}(t_n, \cdots, t_1) \epsilon(t - t_n) \\ &\quad \times \epsilon(t - t_n - t_{n-1}) \cdots \epsilon(t - t_n - t_{n-1} - \cdots - t_1) \end{aligned} \quad (3.3.22)$$

(c) As known, the Heisenberg picture reads [cf. Eq. (3.3.7)]

$$\hat{O}(t) \equiv \hat{O}\mathcal{G}(t) = \hat{O}e^{-i\mathcal{L}t} = e^{iHt/\hbar}\hat{O}e^{-iHt/\hbar} \quad (3.3.23)$$

One can readily recast Eq. (3.3.20) in terms of n -fold commutators,

$$\begin{aligned} & \chi_{AB}^{(n)}(t_n, \dots, t_1) \\ &= i^n \left\langle \underbrace{[\dots]}_n [\hat{A}(t_n + \dots + t_1), \hat{B}(t_{n-1} + \dots + t_1)], \dots, \hat{B}(t_1), \hat{B}(0)] \right\rangle \end{aligned} \quad (3.3.24)$$

with $\langle(\cdot)\rangle \equiv \text{Tr}[(\cdot)\hat{\rho}^{\text{st}}]$ denoting the steady-state ensemble average.

C. Mixed Heisenberg–Schrödinger prescriptions

Let us illustrate the implementation of Eq. (3.3.20), with the generalized second-order response function,

$$\chi^{(2)}(t_2, t_1) = i^2 \langle \langle \hat{A} | \mathcal{G}(t_2) \mathcal{B} \mathcal{G}(t_1) \mathcal{C} | \hat{\rho}^{\text{st}} \rangle \rangle \quad (3.3.25)$$

Here, $\mathcal{B}\hat{O} = [\hat{B}, \hat{O}]$ and $\mathcal{C}\hat{O} = [\hat{C}, \hat{O}]$. The direct evaluation via either the Schrödinger or Heisenberg prescription involves two-dimension time propagations. Alternatively, one can exploit the mixed prescriptions, as follows. Let

$$\hat{\rho}(t_1; \mathcal{C}) \equiv i\mathcal{G}(t_1)\mathcal{C}|\hat{\rho}^{\text{st}}\rangle \equiv \mathcal{G}(t_1)|\hat{\rho}(t_1 = 0; \mathcal{C})\rangle \quad (3.3.26)$$

This describes the Schrödinger dynamics of the density operator, propagated from the initial value of

$$\hat{\rho}(t_1 = 0; \mathcal{C}) = i\mathcal{C}\hat{\rho}^{\text{st}} = i[\hat{C}, \hat{\rho}^{\text{st}}] \quad (3.3.27)$$

On the other hand, the Heisenberg picture, as defined in Eq. (3.3.7), reads

$$\hat{A}(t_2) = \hat{A}\mathcal{G}(t_2) = e^{i\hat{H}t_2/\hbar}\hat{A}e^{-i\hat{H}t_2/\hbar} \quad (3.3.28)$$

Therefore, we have Eq. (3.3.25) the expression,

$$\chi^{(2)}(t_2, t_1) = i \langle \langle \hat{A}(t_2) | \mathcal{B} | \hat{\rho}(t_1; \mathcal{C}) \rangle \rangle \quad (3.3.29)$$

Its evaluation requires only two one-dimension time propagations.

The above strategy of mixed Heisenberg–Schrödinger dynamics evaluations can readily be extended to higher-order nonlinear response or correlation functions. For example, the numerical cost on a third-order response function amounts to a one-dimension plus a two-dimension time propagations, whereas that on a fourth-order one goes by twice of two-dimension propagations.

3.4 Correlation Function Descriptions

This section deals with some basics of linear response theory, with focus on correlation functions. These are the imaginary parts of response functions, Eq. (3.3.24) at $n = 1$; i.e.,

$$\chi_{AB}(t) \equiv i\langle[\hat{A}(t), \hat{B}(0)]\rangle = -2\text{Im}\langle\hat{A}(t)\hat{B}(0)\rangle \quad (3.4.1)$$

where $\hat{A}$ and $\hat{B}$ are both Hermitian. The last brackets quantity describes the correlation functions,

$$\langle\hat{A}(t)\hat{B}(0)\rangle = \text{Tr}[e^{iHt/\hbar}\hat{A}e^{-iHt/\hbar}\hat{B}\rho^{\text{st}}(\hat{H})] \quad (3.4.2)$$

The steady-state density operator satisfies $[\hat{\rho}^{\text{st}}, \hat{H}] = 0$.

3.4.1 Correlation functions and spectrums

A. Basic properties of correlation functions

- As statistical thermodynamic properties, correlations take effects only with finite time lapses. In other words, correlation functions satisfy

$$\lim_{t \rightarrow \infty} \langle\hat{A}(t)\hat{B}(0)\rangle = \langle\hat{A}\rangle\langle\hat{B}\rangle \quad (3.4.3)$$

This ansatz holds in general, as it is in line with the basic principles of thermodynamics and also the energy-time uncertainty principle.

- Equation (3.4.2) results immediately in the following properties:

(a) Time-translation invariance:

$$\langle\hat{A}(t)\hat{B}(\tau)\rangle = \langle\hat{A}(t-\tau)\hat{B}(0)\rangle \quad (3.4.4)$$

Only the time lapse matters. Similarly, $\langle\hat{A}(t)\rangle = \langle\hat{A}(0)\rangle$ in any steady state.

(b) Time-reversal relation,

$$\langle\hat{B}(0)\hat{A}(t)\rangle = \langle\hat{A}(t)\hat{B}(0)\rangle^* \quad (3.4.5)$$

(c) Time-derivative relation:

$$\langle\dot{\hat{A}}(t)\hat{B}(0)\rangle = -\langle\hat{A}(t)\dot{\hat{B}}(0)\rangle \quad (3.4.6)$$

Exercises

E3.3.1 Verify Eqs. (3.4.4)–(3.4.6) and also $\langle\hat{B}(-t)\hat{A}(0)\rangle = \langle\hat{B}(0)\hat{A}(t)\rangle$.

E3.3.2 Show that on the basis of $\hat{H}|n\rangle = E_n|n\rangle$, Eq. (3.4.2) reads

$$\langle\hat{A}(t)\hat{B}(0)\rangle = \sum_{m,n} P_n A_{nm} B_{mn} e^{-i\omega_{mn}t} \quad (3.4.7a)$$

where

$$P_n = \langle n|\hat{\rho}^{\text{st}}|n\rangle, \quad A_{nm} \equiv \langle n|\hat{A}|m\rangle \quad \text{and} \quad \omega_{mn} = (E_m - E_n)/\hbar \quad (3.4.7b)$$

B. Spectrum functions and dispersion functions

- To proceed, we set $\delta\hat{A} \equiv \hat{A} - \langle\hat{A}\rangle$, so that [cf. Eq. (3.4.3)]

$$\lim_{t \rightarrow \infty} \langle \delta\hat{A}(t) \delta\hat{B}(0) \rangle = 0 \quad (3.4.8)$$

Define then

$$C_{AB}(\omega) \equiv \int_0^\infty dt e^{i\omega t} \langle \delta\hat{A}(t) \delta\hat{B}(0) \rangle \equiv S_{AB}(\omega) + iD_{AB}(\omega) \quad (3.4.9)$$

Here, $\{C_{AB}(\omega)\}$ is treated as a (2×2) -matrix, decomposed into its Hermitian and anti-Hermitian parts. Physically, these two components describes the spectrum and dispersion, respectively; i.e.,

$$\begin{aligned} S_{AB}(\omega) &= \frac{1}{2} [C_{AB}(\omega) + C_{BA}^*(\omega)] & (Spectrum) \\ D_{AB}(\omega) &= \frac{1}{2i} [C_{AB}(\omega) - C_{BA}^*(\omega)] & (Dispersion) \end{aligned} \quad (3.4.10)$$

Addition information: The fact that $\langle \delta\hat{A}(t) \delta\hat{B}(0) \rangle$ is finite-time supported, Eq. (3.4.8), enables the extension of $C_{AB}(\omega)$, Eq. (3.4.9), to an *analytical function*, $C_{AB}(z)$, in the upper-half plane, where $\text{Im}z > 0$. Its Cauchy's closed-contour integration takes the value of

$$\oint_C dz C_{AB}(z) = 0; \quad \forall C \in \text{upper-half plane}$$

As results, the spectrum and dispersion functions are related each other via

$$\begin{aligned} D_{AB}(\omega) &= \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{S_{AB}(\omega')}{\omega - \omega'} \\ S_{AB}(\omega) &= -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{D_{AB}(\omega')}{\omega - \omega'} \end{aligned} \quad (3.4.11)$$

These are the Kramers-Kronig relations, with $\mathcal{P}$ denoting the principle part.

- By using Eq. (3.4.5), the spectrum function in Eq. (3.4.10) reads

$$S_{AB}(\omega) = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \delta\hat{A}(t) \delta\hat{B}(0) \rangle \quad (3.4.12)$$

The sum-over-states equivalence reads [Eq. (3.4.7)]

$$S_{AB}(\omega) = \pi \sum_{m,n} P_n A_{nm} B_{mn} \delta(\omega - \omega_{mn})$$

with $\hat{H}|n\rangle = E_n|n\rangle$, by which $\hbar\omega_{mn} = E_m - E_n$ and $A_{nm} = \langle n | \delta\hat{A} | m \rangle = A_{mn}^*$, whereas $P_n = \langle n | \hat{\rho}^{\text{st}} | n \rangle \geq 0$.

(a) Spectrum functions satisfies the positivity properties,

$$S_{AA}(\omega) \geq 0 \quad \text{and} \quad |S_{AB}(\omega)|^2 \leq S_{AA}(\omega)S_{BB}(\omega) \quad (3.4.13)$$

(b) **Optical spectroscopy** arises from **electric dipole correlation**,

$$S(\omega) = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \hat{\mu}(t) \delta \hat{\mu}(0) \rangle = \pi \sum_{m,n} P_n |\mu_{mn}|^2 \delta(\omega - \omega_{mn}) \quad (3.4.14)$$

Involved is the matter system *dipole vector component*, $\hat{\mu}$, along the direction of external electric field.

• Let us revisit Eq. (3.4.12) in terms of

$$S_{BA}(-\omega) = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{-i\omega t} \langle \delta \hat{B}(t) \delta \hat{A}(0) \rangle = \frac{1}{2} \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \delta \hat{B}(0) \delta \hat{A}(t) \rangle$$

Consider then

$$\chi_{AB}(t) = i \langle [\hat{A}(t), \hat{B}(0)] \rangle = i \langle [\delta \hat{A}(t), \delta \hat{B}(0)] \rangle$$

We obtain

$$\chi_{AB}^{(-)}(\omega) \equiv \frac{1}{2i} \int_{-\infty}^{\infty} dt e^{i\omega t} \chi_{AB}(t) = S_{AB}(\omega) - S_{BA}(-\omega) \quad (3.4.15)$$

satisfying $\chi_{BA}^{(-)}(-\omega) = -\chi_{AB}^{(-)}(\omega)$. In parallel to Eq. (3.4.13), we have

$$\chi_{AA}^{(-)}(\omega)/\omega \geq 0 \quad \text{and} \quad |\chi_{AB}^{(-)}(\omega)|^2 \leq \chi_{AA}^{(-)}(\omega) \chi_{BB}^{(-)}(\omega) \quad (3.4.16)$$

3.4.2 Fluctuation–dissipation theorem

Turn to the equilibrium scenario, in which

$$\hat{\rho}^{\text{st}}(\hat{H}) \Rightarrow \hat{\rho}^{\text{eq}}(T) \propto e^{-\hat{H}/(k_B T)} \quad (3.4.17a)$$

Let $\beta \equiv \hbar/(k_B T)$ be the imaginary time. We have

$$\hat{A}(t - i\beta) = e^{i(t-i\beta)\hat{H}/\hbar} \hat{A} e^{-i(t-i\beta)\hat{H}/\hbar} = e^{\beta\hat{H}/\hbar} \hat{A}(t) e^{-\beta\hat{H}/\hbar} \quad (3.4.17b)$$

Hence,

$$\langle \hat{B}(0) \hat{A}(t) \rangle^{\text{eq}} = \langle \hat{A}(t - i\beta) \hat{B}(0) \rangle^{\text{eq}} \quad (3.4.18)$$

The frequency–domain counterpart is given by [cf. Eq. (3.4.12)]

$$S_{BA}(-\omega) = e^{-\beta\omega} S_{AB}(\omega) \quad (3.4.19)$$

Together with Eq. (3.4.15), we obtain

$$S_{AB}(\omega) = \frac{\chi_{AB}^{(-)}(\omega)}{1 - e^{-\beta\omega}} \quad (3.4.20)$$

Its inverse–Fourier transformation, with respect to Eq. (3.4.12), is then

$$\langle \delta \hat{A}(t) \delta \hat{B}(0) \rangle^{\text{eq}} = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega e^{-i\omega t} \frac{\chi_{AB}^{(-)}(\omega)}{1 - e^{-\beta\omega}} \quad (3.4.21)$$

This is called the fluctuation–dissipation theorem.

3.5 Wigner Phase-Space Representation

3.5.1 Basic properties

- Consider the quantum phase-space correspondence to Eq. (2.2.4),

$$\langle A(t) \rangle = \text{Tr}[\hat{A}\hat{\rho}(t)] = \iint dp dq A(p, q) \rho(p, q; t) \quad (3.5.1)$$

with [cf. Eq. (3.2.12)]

$$A(p, q) \equiv \langle \hat{A} | pq \rangle \quad \text{and} \quad \rho(p, q) \equiv \langle pq | \hat{\rho} \rangle \quad (3.5.2)$$

These are the Wigner phase-space $\{|pq\rangle\}$ -representations of dynamic variable and density operator, respectively.

- In parallel to Eq. (3.2.13), we set

$$\boxed{\begin{aligned} \langle pq | p'q' \rangle &= \delta(p - p')\delta(q - q') && \text{(Orthonormality)} \\ \iint dp dq |pq\rangle \langle pq| &= \mathcal{I} && \text{(Completeness)} \end{aligned}} \quad (3.5.3)$$

Set further

$$\text{Tr} \hat{\rho} \equiv \langle \hat{I} | \hat{\rho} \rangle = \iint dp dq \rho(p, q) = 1 \quad (3.5.4)$$

and $\langle \hat{I} | pq \rangle \equiv 1$, with $\hat{I}$ being the Hilbert-space unit operator. Moreover,

$$\langle f(\hat{x}) | pq \rangle = f(q) \quad \text{and} \quad \langle g(\hat{p}) | pq \rangle = g(p) \quad (3.5.5)$$

for any functions of either coordinate or momentum operator only.

- Turn to the formal expression of $\rho(p, q) \equiv \langle pq | \hat{\rho} \rangle$, in relation to its coordinate-representation counterpart,

$$\rho(x, x') = \langle xx' | \hat{\rho} \rangle = \langle x | \hat{\rho} | x' \rangle \quad (3.5.6)$$

Set the coordinates center and difference,

$$q \equiv \frac{1}{2}(x + x') \quad \text{and} \quad s \equiv x - x' \quad (3.5.7)$$

The Wigner phase-space transformation reads then

$$\rho(p, q) \equiv \frac{1}{2\pi\hbar} \int ds e^{-ips/\hbar} \langle q + s/2 | \hat{\rho} | q - s/2 \rangle \quad (3.5.8)$$

that satisfies Eq. (3.5.4). Together with Eqs. (3.5.1)–(3.5.3), we obtain

$$\boxed{\begin{aligned} |pq\rangle &= \int ds e^{ips/\hbar} |q + s/2, q - s/2\rangle \\ \langle pq| &= \frac{1}{2\pi\hbar} \int ds e^{-ips/\hbar} \langle q + s/2, q - s/2| \end{aligned}} \quad (3.5.9)$$

The equivalent counterparts via momentum-representation are

$$\begin{aligned} |pq\rangle &\equiv \int du e^{-iqu/\hbar} |p + u/2, p - u/2\rangle \\ \langle pq| &\equiv \frac{1}{2\pi\hbar} \int du e^{iqu/\hbar} \langle p + u/2, p - u/2| \end{aligned} \quad (3.5.10)$$

Both expressions in Eq. (3.5.5) are followed. However,

$$(\hat{A}\hat{B})_w \neq A(p, q)B(p, q) \quad \text{and} \quad (\hat{A}\hat{p})_w \neq A(p, q)\rho(p, q) \quad (3.5.11)$$

See Eqs. (3.5.19)–(3.5.21) for the exact expressions.

Remarks:

(a) In contrast to $|xx'\rangle \Rightarrow |x\rangle\langle x'|$ and $\langle xx'| = |xx'\rangle^\dagger$ via Eq. (3.2.10), the Wigner representation, with $|pq\rangle \not\Rightarrow |p\rangle\langle q|$ and $|pq\rangle^\dagger \neq \langle pq|$, does not have the simple Hilbert-space correspondence. As also known, Eq. (3.5.9) states the fact that the momentum p does not conjugate to $q = (x + x')/2$, but to $s = x - x'$.

(b) In the three-dimension space,

$$\rho(\mathbf{r}, \mathbf{r}') = \rho(\mathbf{q} + \mathbf{s}/2, \mathbf{q} - \mathbf{s}/2) \equiv \rho(\mathbf{q}, \mathbf{s}) \quad (3.5.12)$$

Consider then [cf. Eq. (2.3.30)]

$$\frac{1}{\hbar^2} \langle \hat{p}^2 \rangle = - \int \nabla_{\mathbf{r}}^2 \rho(\mathbf{r}, \mathbf{r}') \big|_{\mathbf{r}'=\mathbf{r}} d\mathbf{r} = \int \nabla_{\mathbf{r}} \cdot \nabla_{\mathbf{r}'} \rho(\mathbf{r}, \mathbf{r}') \big|_{\mathbf{r}'=\mathbf{r}} d\mathbf{r} \quad (3.5.13)$$

Note that $\mathbf{q} = \frac{1}{2}(\mathbf{r} + \mathbf{r}')$ and $\mathbf{s} = \mathbf{r} - \mathbf{r}'$. We have

$$\frac{\partial}{\partial r_\alpha} = \frac{\partial}{\partial q_\alpha} \frac{\partial q_\alpha}{\partial r_\alpha} + \frac{\partial}{\partial s_\alpha} \frac{\partial s_\alpha}{\partial r_\alpha} = \frac{1}{2} \frac{\partial}{\partial q_\alpha} + \frac{\partial}{\partial s_\alpha}; \quad \alpha = x, y, z \quad (3.5.14)$$

In other words,

$$\nabla_{\mathbf{r}} = \frac{1}{2} \nabla_{\mathbf{q}} + \nabla_{\mathbf{s}} \quad \text{and} \quad \nabla_{\mathbf{r}'} = \frac{1}{2} \nabla_{\mathbf{q}} - \nabla_{\mathbf{s}} \quad (3.5.15)$$

Thus,

$$\nabla_{\mathbf{r}}^2 = \frac{1}{4} \nabla_{\mathbf{q}}^2 + \nabla_{\mathbf{s}}^2 + \nabla_{\mathbf{q}} \cdot \nabla_{\mathbf{s}} \quad \text{and} \quad \nabla_{\mathbf{r}} \cdot \nabla_{\mathbf{r}'} = \frac{1}{4} \nabla_{\mathbf{q}}^2 - \nabla_{\mathbf{s}}^2 \quad (3.5.16)$$

Substituting these into Eq. (3.5.13), followed by using Eq. (2.3.32), we obtain

$$\int \nabla_{\mathbf{q}} \cdot \nabla_{\mathbf{s}} \rho(\mathbf{q}, \mathbf{s}) \big|_{\mathbf{s}=0} d\mathbf{q} = 0 \quad (3.5.17)$$

and

$$\langle \hat{p}^2 \rangle = -\hbar^2 \int \nabla_{\mathbf{s}}^2 \rho(\mathbf{q}, \mathbf{s}) \big|_{\mathbf{s}=0} d\mathbf{q} \quad (3.5.18)$$

This agrees with $\hat{\mathbf{p}} \doteq -i\hbar \nabla_{\mathbf{s}}$, the momentum conjugate to $\mathbf{s} = \mathbf{r} - \mathbf{r}'$.

3.5.2 Wigner representation of operators product

- The Wigner representation for operators product, as shown soon below [cf. Eqs. (3.5.22)–(3.5.25)], is given by

$$(\hat{A}\hat{B})_{\text{w}} = A(p, q) \exp\left(\frac{i\hbar}{2} \overleftrightarrow{T}\right) B(p, q) \quad (3.5.19)$$

or equivalently

$$\{\hat{A}, \hat{B}\}_{\text{w}} = 2A(p, q) \cos\left(\frac{\hbar}{2} \overleftrightarrow{T}\right) B(p, q) \quad (3.5.20a)$$

$$[\hat{A}, \hat{B}]_{\text{w}} = i 2A(p, q) \sin\left(\frac{\hbar}{2} \overleftrightarrow{T}\right) B(p, q) \quad (3.5.20b)$$

Here,

$$\overleftrightarrow{T} \equiv \frac{\overleftarrow{\partial}}{\partial q} \frac{\overrightarrow{\partial}}{\partial p} - \frac{\overleftarrow{\partial}}{\partial p} \frac{\overrightarrow{\partial}}{\partial q} \quad (3.5.21)$$

This involves in the *classical Poisson brackets*, Eq. (1.3.24), with the arrows specifying the directions of derivatives.

- The above formalism holds no matter whether both $\hat{A}$ and $\hat{B}$ are dynamical variables, or one of them is system density operator. In particular,

$$-\frac{i}{\hbar} [\hat{H}, \hat{\rho}]_{\text{w}} = \frac{2}{\hbar} H(p, q) \sin\left(\frac{\hbar}{2} \overleftrightarrow{T}\right) \rho(p, q) \approx \frac{\partial H}{\partial q} \frac{\partial \rho}{\partial p} - \frac{\partial H}{\partial p} \frac{\partial \rho}{\partial q} + \mathcal{O}(\hbar^2)$$

The Wigner representation of $\dot{\hat{\rho}} = -(i/\hbar)[\hat{H}, \hat{\rho}]$, Eq. (3.2.1), does reduce to the classical Poisson equation (1.3.32), in the limit of $\hbar \rightarrow 0$.

• Some basic results

Operator Form	Wigner Representation	Operator Form	Wigner Representation
$\{\hat{q}, \hat{\rho}\}$	$2q\rho(p, q)$	$\{\hat{p}, \hat{\rho}\}$	$2p\rho(p, q)$
$\{\hat{q}, \{\hat{q}, \hat{\rho}\}\}$	$4q^2\rho(p, q)$	$\{\hat{p}, \{\hat{p}, \hat{\rho}\}\}$	$4p^2\rho(p, q)$
$[\hat{q}, \hat{\rho}]$	$i\hbar \frac{\partial}{\partial p} \rho(p, q)$	$[\hat{p}, \hat{\rho}]$	$\frac{\hbar}{i} \frac{\partial}{\partial q} \rho(p, q)$
$[\hat{q}, \{\hat{p}, \hat{\rho}\}]$	$2i\hbar \frac{\partial}{\partial p} [p\rho(p, q)]$	$[\hat{p}, \{\hat{q}, \hat{\rho}\}]$	$2\frac{\hbar}{i} \frac{\partial}{\partial q} [q\rho(p, q)]$
a) $[\hat{q}, [\hat{p}, \hat{\rho}]]$	$\hbar^2 \frac{\partial^2}{\partial p \partial q} \rho(p, q)$	b) $\{\hat{p}, \{\hat{q}, \hat{\rho}\}\}$	$4pq\rho(p, q)$
c) $[\hat{q}^2, \hat{\rho}]$	$2i\hbar \frac{\partial}{\partial p} [q\rho(p, q)]$	d) $[\hat{p}^2, \hat{\rho}]$	$2\frac{\hbar}{i} \frac{\partial}{\partial q} [p\rho(p, q)]$

The bottom four operations go respectively with the following identities:

$$\begin{aligned} \text{a) } [\hat{q}, [\hat{p}, \hat{\rho}]] &= [\hat{p}, [\hat{q}, \hat{\rho}]] & \text{b) } \{\hat{p}, \{\hat{q}, \hat{\rho}\}\} &= \{\hat{q}, \{\hat{p}, \hat{\rho}\}\} \\ \text{c) } [\hat{q}^2, \hat{\rho}] &= \{\hat{q}, [\hat{q}, \hat{\rho}]\} = [\hat{q}, \{\hat{q}, \hat{\rho}\}] & \text{d) } [\hat{p}^2, \hat{\rho}] &= \{\hat{p}, [\hat{p}, \hat{\rho}]\} = [\hat{p}, \{\hat{p}, \hat{\rho}\}] \end{aligned}$$

Proof of Eq. (3.5.19): From Eq. (3.5.21) we have

$$\exp\left(\frac{i\hbar}{2}T\right) = \sum_{n=0}^{\infty} \frac{(i\hbar/2)^n}{n!} \sum_{j=0}^n (-1)^{n-j} \frac{n!}{j!(n-j)!} \frac{\overleftarrow{\partial}^n}{\partial q^j \partial p^{n-j}} \frac{\overrightarrow{\partial}^n}{\partial p^j \partial q^{n-j}} \quad (3.5.22)$$

On the other hand, from Eqs. (3.5.9) and (3.5.10) we have

$$\begin{aligned} A_W &= \langle\langle \hat{A}|pq \rangle\rangle = \iint dq' ds \delta(q - q') e^{ips/\hbar} \langle\langle \hat{A}|q' + s/2, q' - s/2 \rangle\rangle \\ &= \iiint dq' \frac{ds du}{2\pi\hbar} e^{i(q-q')u/\hbar} e^{ips/\hbar} \langle q' - \frac{1}{2}s | \hat{A} | q' + \frac{1}{2}s \rangle \end{aligned} \quad (3.5.23)$$

Thus,

$$\frac{\partial^n A_W}{\partial q^j \partial p^{n-j}} = \iiint dq' \frac{ds du}{2\pi\hbar} \left(\frac{i}{\hbar}\right)^n u^j s^{n-j} e^{\frac{i}{\hbar}[(q-q')u + ps]} \langle q' - \frac{1}{2}s | \hat{A} | q' + \frac{1}{2}s \rangle$$

Similarly,

$$\frac{\partial^n B_W}{\partial p^j \partial q^{n-j}} = \iiint d\bar{q} \frac{d\bar{s} d\bar{u}}{2\pi\hbar} \left(\frac{i}{\hbar}\right)^n \bar{s}^j \bar{u}^{n-j} e^{\frac{i}{\hbar}[(q-\bar{q})\bar{u} + p\bar{s}]} \langle \bar{q} - \frac{1}{2}\bar{s} | \hat{B} | \bar{q} + \frac{1}{2}\bar{s} \rangle$$

Therefore,

$$\begin{aligned} A_W \exp\left(\frac{i\hbar}{2}T\right) B_W &= \iiint \iiint dq' d\bar{q} \frac{ds du}{2\pi\hbar} \frac{d\bar{s} d\bar{u}}{2\pi\hbar} e^{\frac{i}{\hbar}[(q-q')u + (q-\bar{q})\bar{u} + p(s+\bar{s})]} \\ &\quad \times \left[\sum_{n=0}^{\infty} \frac{(i\hbar/2)^n}{n!} \left(\frac{i}{\hbar}\right)^{2n} \sum_{j=0}^n \frac{(-1)^{n-j} n!}{j!(n-j)!} (u\bar{s})^j (\bar{u}s)^{n-j} \right] \\ &\quad \times \langle q' - \frac{1}{2}s | \hat{A} | q' + \frac{1}{2}s \rangle \langle \bar{q} - \frac{1}{2}\bar{s} | \hat{B} | \bar{q} + \frac{1}{2}\bar{s} \rangle \\ &= \iiint \iiint dq' d\bar{q} \frac{ds du}{2\pi\hbar} \frac{d\bar{s} d\bar{u}}{2\pi\hbar} e^{\frac{i}{\hbar}[(q-q')u + (q-\bar{q})\bar{u} + p(s+\bar{s})]} \\ &\quad \times e^{-\frac{i}{2\hbar}(u\bar{s} - \bar{u}s)} \langle q' - \frac{1}{2}s | \hat{A} | q' + \frac{1}{2}s \rangle \langle \bar{q} - \frac{1}{2}\bar{s} | \hat{B} | \bar{q} + \frac{1}{2}\bar{s} \rangle \end{aligned}$$

In writing the second identity, we have used

$$e^{-\frac{i}{2\hbar}(u\bar{s} - \bar{u}s)} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{-i}{2\hbar}\right)^n \sum_{j=0}^n \frac{(-1)^{n-j} n!}{j!(n-j)!} (u\bar{s})^j (\bar{u}s)^{n-j}$$

Carrying out the integrations over u and $\bar{u}$, and then q' and $\bar{q}$, we obtain

$$\begin{aligned} A_W \exp\left(\frac{i\hbar}{2}T\right) B_W &= \iiint dq' d\bar{q} ds d\bar{s} e^{\frac{i}{\hbar}p(s+\bar{s})} \delta(q - q' - \frac{1}{2}\bar{s}) \delta(q - \bar{q} + \frac{1}{2}s) \\ &\quad \times \langle q' - \frac{1}{2}s | \hat{A} | q' + \frac{1}{2}s \rangle \langle \bar{q} - \frac{1}{2}\bar{s} | \hat{B} | \bar{q} + \frac{1}{2}\bar{s} \rangle \\ &= \iint ds d\bar{s} e^{\frac{i}{\hbar}p(s+\bar{s})} \langle q - \frac{1}{2}(s + \bar{s}) | \hat{A} | q + \frac{1}{2}(s - \bar{s}) \rangle \\ &\quad \times \langle q + \frac{1}{2}(s - \bar{s}) | \hat{B} | q + \frac{1}{2}(s + \bar{s}) \rangle \\ &= \iint ds' dx e^{\frac{i}{\hbar}ps'} \langle q - \frac{1}{2}s' | \hat{A} | x \rangle \langle x | \hat{B} | q + \frac{1}{2}s' \rangle \end{aligned} \quad (3.5.24)$$

The last expression is obtained by changing the integration variables to $s' = s + \bar{s}$ and $x = q + \frac{1}{2}(s - \bar{s})$. Consequently,

$$A_W \exp\left(\frac{i\hbar}{2}T\right) B_W = \int ds' e^{\frac{i}{\hbar}ps'} \langle q - \frac{1}{2}s' | \hat{A} \hat{B} | q + \frac{1}{2}s' \rangle = [\hat{A} \hat{B}]_W \quad (3.5.25)$$

This is Eq. (3.5.19).

(Q.E.D.)

Chapter 4

Elementary Systems

Hereafter, we deal only with the time-independent Schrödinger equation. The present chapter consists of four sections, with each representing one elementary but important class of systems: (§4.1) Piecewise constant potential systems, with some representing scenarios; (§4.2) One-dimension harmonic oscillator systems, solved via an algebraic approach that is mathematically elegant and physically insightful; (§4.3) Angular momentum along the z -direction, $\hat{L}_z$, followed by the canonical commutator relations. Included are also the angular momentum operators in the spherical coordinates, so that the later chapters can go smoother; (§4.4) Two-dimension harmonic oscillator, subject to the angular momentum requirement.

The basic skills, covering from §4.1 to §4.3 are required, whereas §4.4 is supplemental, *but worth to read with fun*. Moreover, during studies it is important to revisit those general features, presented in §2.1.3, “*Implications of matter waves*”. All these features will be quantitatively exemplified, with rich elaborations in the coming sections. We will also see the fact that when systems carry a certain *symmetry*, the resulting eigen wave functions must possess the *parities*, with respect to the symmetry. In particular, for one-dimension systems with symmetric potential, $V(-x) = V(x)$, the wave functions must possess either even or odd parity. Symmetry broken would occur, for example, in the scattering setup, where the incident wave comes only from one direction; see §4.1.2.

4.1 Piecewise Potential Systems

4.1.1 Quantum well: Bound-state solutions

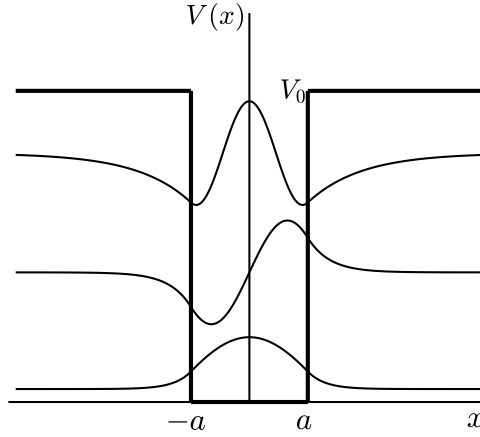


Fig. 4.1 Quantum well with three wave functions ($E < V_0$)

Depicted above is a quantum well system,

$$V(x) = \begin{cases} 0 & \text{for } |x| < a \\ V_0 & \text{for } |x| > a \quad (V_0 > 0) \end{cases} \quad (4.1.1)$$

with the depth V_0 and width $L = 2a$. The bound-state solutions are concerned with $E < V_0$. Consider further the symmetry, $V(-x) = V(x)$ of Eq. (4.1.1), the corresponding $\psi(x)$ must be either even or odd functions. Thus, we can focus on the $x > 0$ region, as $\psi_{\text{even/odd}}(-x) = \pm \psi_{\text{even/odd}}(x)$.

• Step (i): Solve the Schrödinger equation in each region and retain only those square-integrable terms. For $0 < E < V_0$, we obtain

$$\psi_{\text{even/odd}}(x > 0) = \begin{cases} \cos(kx) \text{ or } \sin(kx) & \text{for } 0 < x < a \\ Ae^{-\kappa x} \text{ or } Be^{-\kappa x} & \text{for } x > a \end{cases} \quad (4.1.2)$$

where

$$k = (2mE/\hbar^2)^{\frac{1}{2}} \quad \text{and} \quad \kappa = [(2m(V_0 - E)/\hbar^2)]^{\frac{1}{2}} \quad (4.1.3)$$

• Step (ii): The boundary-values equations,

$$\begin{aligned} \psi_{\text{even}}(x = a) : \quad & Ae^{-\kappa a} = \cos(ka) \quad \text{and} \quad \kappa Ae^{-\kappa a} = k \sin(ka) \\ \psi_{\text{odd}}(x = a) : \quad & Be^{-\kappa a} = \sin(ka) \quad \text{and} \quad \kappa Be^{-\kappa a} = -k \cos(ka) \end{aligned} \quad (4.1.4)$$

Apply the second column to the last column, resulting in

$$\boxed{\text{Even : } \kappa \cos(ka) = k \sin(ka)} \quad \boxed{\text{Odd : } \kappa \sin(ka) = -k \cos(ka)} \quad (4.1.5)$$

- Step (iii): Eigenenergies arising from Eq. (4.1.5). The final results read

$$\boxed{\begin{array}{l} \text{Even parity : } \xi = \chi |\cos \xi| \\ \text{Odd parity : } \xi = \chi |\sin \xi| \end{array}} \quad (4.1.6)$$

in terms of

$$\boxed{\xi \equiv ka = (2ma^2 E/\hbar^2)^{\frac{1}{2}}} \quad \text{and} \quad \chi = (2ma^2 V_0/\hbar^2)^{\frac{1}{2}} \quad (4.1.7)$$

In other words,

$$\boxed{E = \frac{\hbar^2 \xi^2}{2ma^2}} \quad (4.1.8)$$

with ξ being evaluated via Eq. (4.1.6). This is a function of χ that is the input parameter of system in study; cf. Eq. (4.1.7). It is also noticed that

$$(\kappa a)^2 = \chi^2 - \xi^2 \quad (4.1.9)$$

This gives rise to Eq. (4.1.6) from Eq. (4.1.5).

The solutions to eigenenergies, Eq. (4.1.8) with Eq. (4.1.6), can be obtained either graphically or via numerical iterations. Moreover, Eq. (4.1.4) leads to

$$A = e^{\kappa a} \cos(ka) \quad \text{and} \quad B = e^{\kappa a} \sin(ka) \quad (4.1.10)$$

The wave functions, Eq. (4.1.2), are now also determined, but up to the normalizations constants.

- Step (iv): As a normalized wave function satisfies $\int |\psi(x)|^2 dx = 1$, the normalization constants associated with Eq. (4.1.2) are given by

$$\begin{aligned} N_{\text{even}} &= \frac{1}{\sqrt{2}} \left[\int_0^a \cos^2(kx) dx + |A|^2 \int_a^\infty e^{-2\kappa x} dx \right]^{-\frac{1}{2}} \\ N_{\text{odd}} &= \frac{1}{\sqrt{2}} \left[\int_0^a \sin^2(kx) dx + |B|^2 \int_a^\infty e^{-2\kappa x} dx \right]^{-\frac{1}{2}} \end{aligned} \quad (4.1.11)$$

Exercises:

- Discuss the effects of V_0 and $L = 2a$ on the number of bound-states.
- Reproduce Eqs. (2.1.24)–(2.1.27), the particle-in-box limit ($V_0 \rightarrow \infty$). Show that Eq. (2.1.25), with Eq. (2.1.24), can be recast as

$$k = \sqrt{2mE}/\hbar = n\pi/L; \quad n = 1, 2, \dots \quad (4.1.12)$$

(Standing waves; Particle-in-box)

4.1.2 Transmission and reflection problems

Opening remarks and the setup

- For simplicity, we consider only the one-dimension scattering process. It involves an incident wave of *any given energy* from $x \rightarrow -\infty$, under the influence of potential $V(x)$ that takes effect within a finite region. A partial of incident wave transmits forward to $x \rightarrow \infty$ and the remainder reflects backward to $x \rightarrow -\infty$. The incident ψ_{in} and transmitted ψ_{tran} are forward waves, whereas the reflected ψ_{refl} is a backward wave. Their amplitude squares are proportional to the fluxes that also scale linearly with the speeds of particles. The transmission and reflection coefficients, T and R , are the ratios of the corresponding fluxes, with respect to the incident flux.
- In transmission-reflection problems, the incident wave, with any given energy, comes only from one side. Hence, the scattering wave functions are **asymmetric** and **unbounded** in general.
- The study of the simplest, analytically solvable one-dimension scattering systems, with single square-well (§4.1.2A and B) or stepwise (§4.1.2C) potentials, enables to “visualize” the basic picture of quantum characteristics. These include tunnellings in the unbounded region to both forward and backward directions. *Transmission resonances* ($T = 1$) could occur in a “standing-wave-like” quantization manner, as if a particle inside a box.
- The setup for linear transmission-reflection problems follows

$$\boxed{\begin{aligned} T &= \frac{k_{\text{tran}}}{k_{\text{in}}} |\psi_{\text{tran}}|^2 = \frac{k_{\text{tran}}}{k_{\text{in}}} |t|^2 \\ R &= |\psi_{\text{refl}}|^2 = |r|^2 = 1 - T \end{aligned}} \quad \boxed{\begin{aligned} (\psi_{\text{in}} &\equiv e^{ik_{\text{in}}x}) \\ (k_{\text{refl}} &= k_{\text{in}}) \end{aligned}} \quad (4.1.13)$$

- By convention, the incident wave is $\psi_{\text{in}} \equiv e^{ik_{\text{in}}x}$, for the normalization, with setting the incident flux to be 1. The incident energy can be any given value.
- The transmission wave is $te^{ik_{\text{tran}}x}$ and the reflection wave is $re^{-ik_{\text{refl}}x}$, with $k_{\text{refl}} = k_{\text{in}}$, since the reflection is elastic in study.
- The final results must satisfy $T + R = 1$ (*flux conservation*).

A. Transmission-reflection with a square-well potential

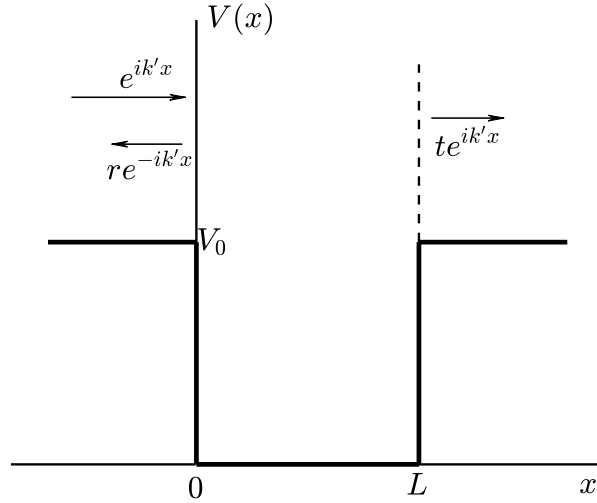


Fig. 4.2 square-well potential ($E > V_0$)

Consider the scattering against the square-well potential $V(x)$, Eq. (4.1.1) but shifted to the asymmetric region of $x \in [0, L]$, since the involving waves are asymmetric and unbounded. That is

$$V(x) = \begin{cases} 0 & x \in (0, L) \\ V_0 > 0 & \text{elsewhere} \end{cases} \quad (4.1.14)$$

The incident wave energy is given above the well depth; i.e., $E > V_0$. Denote

$$k' \equiv \sqrt{2m(E - V_0)}/\hbar \quad \text{and} \quad k \equiv \sqrt{2mE}/\hbar \quad (4.1.15)$$

• Solving the Schrödinger equation in the piecewise manner, followed by the setup convention, Eq. (4.1.13), we have

$$\psi(x) = \begin{cases} e^{ik'x} + re^{-ik'x}, & x < 0 \\ A \cos(kx) + B \sin(kx), & x \in (0, L) \\ te^{ik'x}, & x > L \end{cases} \quad (4.1.16)$$

Remarks: (a) In the region of $x < 0$ (before potential well), there are the incident wave and the reflected wave. The incident one is described by $\psi_{\text{in}}(x) = e^{ik'x}$,

coming from $x \rightarrow -\infty$ and moving in the forward direction. The reflected one, $\psi_{\text{refl}}(x) = re^{-ik'x}$, is a backward wave. (b) In the region of $x > L$, there is only the forward transmitted wave, $\psi_{\text{tran}}(x) = te^{ik'x}$, with $k_{\text{tran}} = k_{\text{in}} = k'$, since $V(x > L) = V(x < 0)$, as by Eq. (4.1.14).

- To proceed we exploit the continuity conditions to identify the unknowns in Eq. (4.1.16). Applying then Eq. (4.1.13), we obtain [cf. Eq. (4.1.23)]

$$\boxed{\frac{R}{T} = \frac{(k^2 - k'^2)^2}{(2kk')^2} \sin^2(kL) = \frac{V_0^2}{4E(E - V_0)} \sin^2(kL)} \quad (4.1.17)$$

- **Transmission resonances** ($T = 1$) occur when $\sin(kL) = 0$, the same as the “standing-wave-like” quantization, as if particle-in-box [cf. Eq. (4.1.12)],

$$\boxed{k \equiv \sqrt{2mE}/\hbar = n\pi/L; \quad n = 1, 2, \dots} \quad (T = 1) \quad (4.1.18)$$

(Above square-well bottom)

The boundary values of transmission resonance waves are found to be $\psi(0) = 1$ and $\psi(L) = (-1)^n$; see the remarks after Eq. (4.1.23).

Derivations of Eq. (4.1.17): To complete the scattering wave function, Eq. (4.1.16), we apply the continuity requirements, which read

$$\begin{aligned} 1 + r &= A, & A \cos(kL) + B \sin(kL) &= te^{ik'L} \\ ik'(1 - r) &= kB, & -k[A \sin(kL) - B \cos(kL)] &= ik'te^{ik'L} \end{aligned} \quad (4.1.19)$$

Introduce for bookkeeping

$$\tilde{t} \equiv te^{ik'L} \quad \text{and} \quad \alpha \equiv kL \quad (4.1.20)$$

We obtain from the right column of Eq. (4.1.19) the relations,

$$A = \frac{te^{ik'L}}{k} (k \cos \alpha - ik' \sin \alpha) \quad (4.1.21a)$$

$$B = \frac{te^{ik'L}}{k} (k \sin \alpha + ik' \cos \alpha) \quad (4.1.21b)$$

Combining the left column of Eq. (4.1.19), we complete the solutions further with

$$t = \frac{2ikk'e^{-ik'L}}{(k^2 + k'^2) \sin \alpha + 2ikk' \cos \alpha} \quad (4.1.22a)$$

$$r = -\frac{(k^2 - k'^2) \sin \alpha}{(k^2 + k'^2) \sin \alpha + 2ikk' \cos \alpha} \quad (4.1.22b)$$

The transmission and reflection coefficients are

$$T = \frac{(2kk')^2}{(2kk')^2 + (k^2 - k'^2)^2 \sin^2 \alpha} \quad (4.1.23a)$$

and

$$R = \frac{(k^2 - k'^2)^2 \sin^2 \alpha}{(2kk')^2 + (k^2 - k'^2)^2 \sin^2 \alpha} \quad (4.1.23b)$$

satisfying $T + R = 1$ and also Eq. (4.1.17), since $\alpha = kL$ [cf. Eq. (4.1.20)].

Transmission resonances ($T = 1$) occur at [Eq. (4.1.18)]

$$\alpha = kL = n\pi; \quad n = 1, 2, \dots$$

While this appears like Eq. (4.1.12), as if a particle inside box, the resonant transmission waves are of $\tilde{t} \equiv te^{ik'L} = (-1)^n$, according to Eq. (4.1.22), with $\sin \alpha = 1$, resulting in further $\psi(x=0) = 1$ and $\psi(x=L) = (-1)^n$.

B. Transmission–reflection with a square–barrier potential

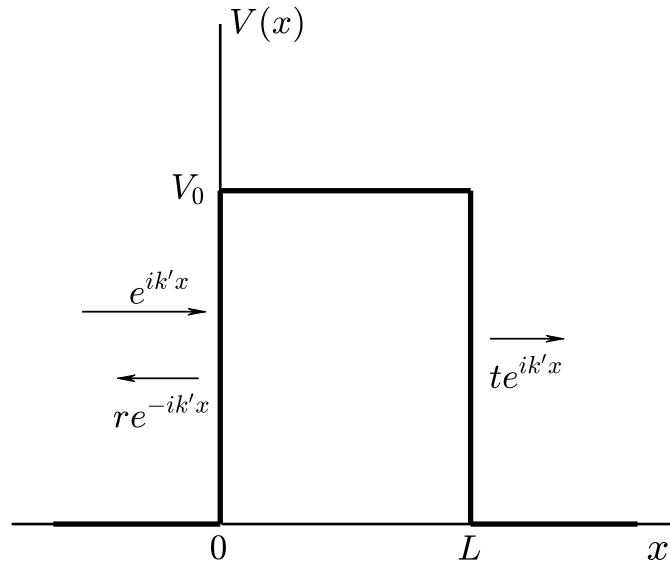


Fig. 4.3 square-barrier potential ($E < V_0$)

$$V(x) = \begin{cases} V_0 > 0, & x \in (0, L) \\ 0, & \text{elsewhere} \end{cases} \quad (4.1.24)$$

Let the steady-state scattering wave be formally identical to Eq. (4.1.16),

$$\psi(x) = \begin{cases} e^{ik'x} + re^{-ik'x}, & x < 0 \\ A \cos(kx) + B \sin(kx), & x \in (0, L) \\ te^{ik'x}, & x > L \end{cases} \quad (4.1.25)$$

This is achieved by setting [cf. Eq. (4.1.15) with k and k' swapped]

$$k \equiv \sqrt{2m(E - V_0)}/\hbar \quad \text{and} \quad k' \equiv \sqrt{2mE}/\hbar \quad (4.1.26)$$

Note that $k = i\kappa$ when $E < V_0$, with $\kappa > 0$. As Eq. (4.1.21) and (4.1.22) remain unchanged, we have [cf. Eq. (4.1.17)]

$$\frac{R}{T} = \begin{cases} b \sinh^2(\kappa L) & \text{when } E < V_0 \\ (k'L/2)^2 & \text{when } E = V_0 \\ b \sin^2(kL) & \text{when } E > V_0 \end{cases} \quad (4.1.27)$$

where

$$b \equiv \frac{(k^2 - k'^2)^2}{4|kk'|^2} = \frac{V_0}{2E|E - V_0|} \quad (4.1.28)$$

- **Transmission resonances** occur at [cf. Eq. (4.1.12) or (4.1.18)]

$$\boxed{k \equiv \sqrt{2m(E - V_0)}/\hbar = n\pi/L; \quad n = 1, 2, \dots} \quad (T = 1) \quad (4.1.29)$$

(Above square-barrier top)

Here, k via Eq. (4.1.26) is the wave vector amplitude above the square-barrier top, whereas that in Eq. (4.1.18) is above the square-well bottom.

Exercise: (a) Determine the unknowns in Eq. (4.1.25). (b) Verify Eqs. (4.1.27)–(4.1.29), by using $\cos(i\alpha) = \cosh \alpha$ and $\sin(i\alpha) = \sinh \alpha$. See also Eqs. (4.1.19)–(4.1.23).

C. Transmission–reflection with a stepwise potential

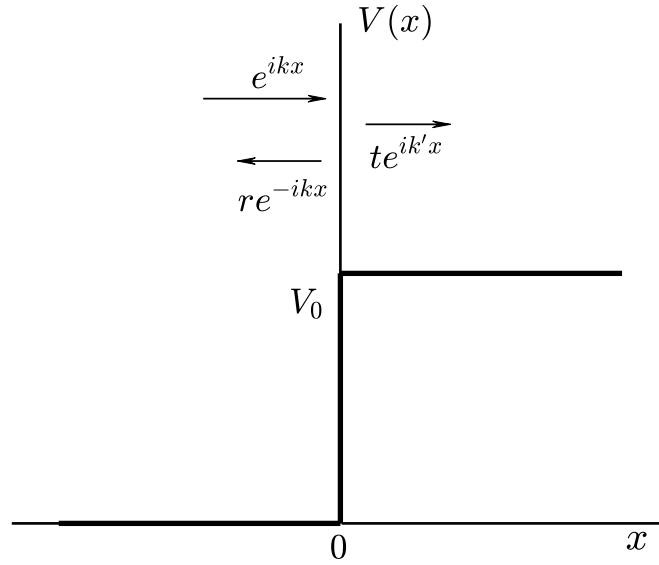


Fig. 4.4 stepwise potential($E > V_0$)

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x > 0 \end{cases} \quad (4.1.30)$$

Case (i): $E > \max(0, V_0)$

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0; \quad k = \sqrt{2mE}/\hbar \\ te^{ik'x}, & x > 0; \quad k' = \sqrt{2m(E - V_0)}/\hbar \end{cases} \quad (4.1.31)$$

The boundary conditions lead to

$$r = \frac{k - k'}{k + k'} \quad \text{and} \quad t = \frac{2k}{k + k'} \quad (4.1.32)$$

The resultant reflection and transmission coefficients via Eq. (4.1.13) are

$$R = r^2 = \frac{(k - k')^2}{(k + k')^2} \quad \text{and} \quad T = \frac{k'}{k} t^2 = \frac{4kk'}{(k + k')^2} \quad (4.1.33)$$

No transmission resonance occurs in the stepwise case.

Case (ii): $0 < E < V_0$

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0; \quad k = \sqrt{2mE}/\hbar \\ Ae^{-\kappa x}, & x > 0; \quad \kappa = \sqrt{2m(V_0 - E)}/\hbar \end{cases} \quad (4.1.34)$$

There is no transmission but pure tunnelling in the region of $x > 0$. We have [cf. Eq. (4.1.32)]

$$r = \frac{k - i\kappa}{k + i\kappa} \quad \text{and} \quad A = \frac{2k}{k + i\kappa} \quad (4.1.35)$$

Here, $R = |r|^2 = 1$; i.e., $T = 0$, as anticipated.

4.1.3 Delta-function potential systems

It is noticed that inside a square-well, the number of wave functions increases with the depth and width of well. It would be interesting to see the limiting scenario of delta-function potential,

$$V(x) = -\alpha\delta(x) \quad (4.1.36)$$

with $\alpha > 0$ (well) and $\alpha < 0$ (barrier). Note that $V(x=0) \rightarrow \pm\infty$. This results in the discontinuity of wave function derivative, $\psi'(x) \equiv d\psi(x)/dx$, at $x = 0$. To proceed, we shall first address this boundary value problem.

• **Boundary value, $\psi'(0+) - \psi'(0-)$, for $\delta(x)$ -potential systems**

Consider the Schrödinger equation the limiting expression,

$$\lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{\epsilon} \left[-\frac{\hbar^2}{2m} \psi''(x) + V(x)\psi(x) \right] dx = E \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^{\epsilon} \psi(x) dx = 0 \quad (4.1.37)$$

with noting that

$$\begin{aligned} \int_{-\epsilon}^{\epsilon} \psi''(x) dx &= \psi'(\epsilon) - \psi'(-\epsilon) \\ \int_{-\epsilon}^{\epsilon} V(x)\psi(x) dx &= -\alpha \int_{-\epsilon}^{\epsilon} \delta(x)\psi(x) dx = -\alpha\psi(0) \end{aligned} \quad (4.1.38)$$

Substitute them into Eq. (4.1.37) resulting in

$$\boxed{\psi'(0+) - \psi'(0-) = -\frac{2m\alpha}{\hbar^2} \psi(0)} \quad (4.1.39)$$

This is *boundary condition* for wave functions of $\delta(x)$ -potential systems.

A. The normalized bound state ($E < 0$) in the $\delta(x)$ -potential well case, where $\alpha > 0$. The wave function has the form of (normalized)

$$\boxed{\psi(x) = \sqrt{\kappa} e^{-\kappa|x|}}; \quad \kappa = \sqrt{-2mE}/\hbar \quad (4.1.40)$$

with the boundary values satisfying

$$\psi(0) = \sqrt{\kappa} \quad \text{and} \quad \psi'(0+) - \psi'(0-) = -2\kappa\sqrt{\kappa} \quad (4.1.40')$$

Apply then Eq. (4.1.39), resulting in

$$\boxed{\psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}} \quad \text{with} \quad \boxed{E = -\frac{m\alpha^2}{2\hbar^2}} \quad (4.1.41)$$

Only one bound state exists inside a $\delta(x)$ -well potential.

B. Scattering ($E > 0$) setup for $\delta(x)$ -potential:

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}; & x < 0 \\ te^{ikx}; & x > 0 \end{cases} \quad k = \sqrt{2mE}/\hbar \quad (4.1.42)$$

with the boundary values satisfying

$$\begin{aligned} \psi(x=0) &= 1 + r = t \\ \psi'(x=0+) - \psi'(x=0-) &= ik(t - 1 + r) = -\frac{2m\alpha}{\hbar^2}\psi(0) \end{aligned} \quad (4.1.43)$$

The last identity arises from Eq. (4.1.39). We obtain

$$r = \frac{i\beta}{1 - i\beta} \quad \text{and} \quad t = \frac{1}{1 - i\beta}, \quad \text{where} \quad \beta \equiv \frac{m\alpha}{k\hbar^2} = \frac{\alpha}{\hbar} \sqrt{\frac{m}{2E}} \quad (4.1.44)$$

4.2 Simple Harmonic Oscillator Systems

4.2.1 Algebraic solutions

A. Onsets of dimensionless coordinate and momentum

The Hamiltonian of a one-dimensional harmonic oscillator of mass m and frequency ω reads

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = \frac{\hbar\omega}{2}(\hat{P}^2 + \hat{Q}^2) \quad (4.2.1)$$

The second identity is expressed in terms of the dimensionless coordinate and momentum, defined as

$$\hat{Q} \equiv \sqrt{\frac{m\omega}{\hbar}}\hat{x} \quad \text{and} \quad \hat{P} \equiv \frac{1}{\sqrt{m\hbar\omega}}\hat{p} \quad (4.2.2)$$

which satisfy $[\hat{Q}, \hat{P}] = i$. Introduce now

$$\begin{aligned} \hat{a} &= \frac{1}{\sqrt{2}}(\hat{Q} + i\hat{P}) = \left(\frac{m\omega}{2\hbar}\right)^{\frac{1}{2}} \left(\hat{x} + \frac{i\hat{p}}{m\omega}\right) \\ \hat{a}^\dagger &= \frac{1}{\sqrt{2}}(\hat{Q} - i\hat{P}) = \left(\frac{m\omega}{2\hbar}\right)^{\frac{1}{2}} \left(\hat{x} - \frac{i\hat{p}}{m\omega}\right) \end{aligned} \quad (4.2.3)$$

We can then recast Eq. (4.2.1) in terms of (**Exercise**)

$$\hat{H} = \hbar\omega\left(\hat{n} + \frac{1}{2}\right), \quad \text{where } \hat{n} \equiv \hat{a}^\dagger\hat{a} \quad (4.2.4)$$

This defines the *number operator*, $\hat{n} = \hat{a}^\dagger\hat{a}$, whose eigenfunctions are just those of harmonic oscillator.

B. Ladder operators

As defined in Eq. (4.2.3), we have (**Exercise**)

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1} \Rightarrow \boxed{[\hat{a}^\dagger\hat{a}, \hat{a}] = -\hat{a} \quad \text{and} \quad [\hat{a}^\dagger\hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger} \quad (4.2.5)$$

resulting in further

$$\hat{a}^\dagger\hat{a}|n\rangle = n|n\rangle; \quad n = 0, 1, 2, \dots \quad (4.2.6)$$

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle \quad \text{and} \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \quad (4.2.7)$$

Derivations: Consider the eigenequation of $\hat{a}^\dagger\hat{a}$ that is Hermitian and *nonnegative*:

$$\hat{a}^\dagger\hat{a}|n\rangle = \lambda_n|n\rangle; \quad \lambda_n \geq 0 \quad (4.2.8)$$

Together with Eq. (4.2.5), we obtain

$$\begin{aligned} \hat{a}^\dagger\hat{a}(\hat{a}|n\rangle) &= \hat{a}(\hat{a}^\dagger\hat{a} - 1)|n\rangle = (\lambda_n - 1)(\hat{a}|n\rangle) \\ \hat{a}^\dagger\hat{a}(\hat{a}^\dagger|n\rangle) &= \hat{a}^\dagger(\hat{a}^\dagger\hat{a} + 1)|n\rangle = (\lambda_n + 1)(\hat{a}^\dagger|n\rangle) \end{aligned} \quad (4.2.9)$$

These conclude that

$$\hat{a}|n\rangle \propto |n-1\rangle \text{ and } \hat{a}^\dagger|n\rangle \propto |n+1\rangle, \text{ whereas } \lambda_{n\pm 1} = \lambda_n \pm 1 \quad (4.2.10)$$

In other words, $\hat{a}$ and $\hat{a}^\dagger$ are ladder operators, climbing down and up from $|n\rangle$ to its neighboring $|n-1\rangle$ and $|n+1\rangle$ states, respectively. Applying the ladder-down operator on the ground state should go by $\hat{a}|0\rangle = 0$. The resulting $\hat{a}^\dagger\hat{a}|0\rangle = 0$, together with the last identity of Eq. (4.2.10) give rise to

$$\lambda_n = n; \quad n = 0, 1, 2, \dots \quad (4.2.11)$$

The resulting Eq. (4.2.8) is Eq. (4.2.6). It is now easy to obtain $\langle n|\hat{a}^\dagger\rangle\langle\hat{a}|n\rangle = n$ and $\langle n|\hat{a}\rangle\langle\hat{a}^\dagger|n\rangle = n+1$, which together with Eq. (4.2.10), give rise to Eq. (4.2.7). (Q.E.D.)

C. Annihilation and creation operators versus ladder operators

First of all, naming $\hat{a}$ and $\hat{a}^\dagger$ the ladder operators refers to Eq. (4.2.7), for shifting $|n\rangle$ down and up to the neighboring $|n-1\rangle$ and $|n+1\rangle$ states, respectively. However, as the physical implications underline $\{|n\rangle\}$ are concerned, $\hat{a}$ and $\hat{a}^\dagger$ would rather be the *annihilation* and *creation* operators.

Annihilation operator, $\hat{a}$; Creation operator, $\hat{a}^\dagger$; Number operator, $\hat{n} \equiv \hat{a}^\dagger\hat{a}$.

The harmonic oscillator of frequency ω , Eq. (4.2.1), is equivalent to Eq. (4.2.4) and further Eq. (4.2.6), with the eigen-energies,

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right); \quad n = 0, 1, 2, \dots \quad (4.2.12)$$

In contact with the Planck–Einstein's radiation field quantization, each photon has energy of $\hbar\omega$, whereas the vacuum field $|0\rangle$ acquires the zero-point energy, $E_0 = \hbar\omega/2$. Hence, Eq. (4.2.12), associated with $|n\rangle$, comprises n number of photons. This leads to the terminology of *annihilation* ($\hat{a}$) and *creation* ($\hat{a}^\dagger$) operators, as implied in Eq. (4.2.7). Moreover, Eq. (4.2.6) reads $\hat{n}|n\rangle = n|n\rangle$, with $\hat{n} \equiv \hat{a}^\dagger\hat{a}$ [cf. Eq. (4.2.4)] named to be the *number operator*.

It is worth noting that $[\hat{a}, \hat{a}^\dagger] = 1$ [cf. Eq. (4.2.5)] is for *bosons* (including photons), giving rise to other identities [cf. Eqs. (4.2.5)–(4.2.7)]. The algebraic approach to harmonic system is highly recommended.

Exercises:

(a) Does Eq. (4.2.3) hold in general or for harmonic oscillator only?

(b) Let $\{|n\rangle; n = 0, 1, \dots\}$ be the eigenvectors of the number operator, $\hat{n} = \hat{a}^\dagger\hat{a}$. Write down the matrix form of $\hat{a}$ and $\hat{a}^\dagger$. [Hint: $A_{mn} = \langle m|\hat{A}|n\rangle$]

(c) Consider the expectation values, $\langle\hat{A}\rangle = \langle n|\hat{A}|n\rangle$. Show that

$$\langle\hat{x}\rangle = \langle\hat{p}\rangle = 0 \text{ and } \Delta x \Delta p = \hbar\left(n + \frac{1}{2}\right) \quad (4.2.13)$$

Are they general or for harmonic only? [Hint: Using Eq. (4.2.3)]

(d) For harmonic oscillator, the kinetic energy and potential energy are of

$$\langle\hat{K}\rangle = \langle\hat{V}\rangle = \frac{1}{2}\hbar\omega\left(n + \frac{1}{2}\right) \quad (\text{harmonic}) \quad (4.2.14)$$

(e) Show that harmonic oscillator satisfies [Hint: Using Eq. (2.3.3)]

$$\hat{a}(t) = \hat{a}(0)e^{-i\omega t}; \quad \hat{a}^\dagger(t) = \hat{a}^\dagger(0)e^{i\omega t} \quad (4.2.15)$$

Derive $\hat{x}(t)$ and $\hat{p}(t)$ and compare them with the *classical results*.

4.2.2 Eigenfunctions

As implied in Eq. (4.2.7), we have

$$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n|0\rangle; \quad n = 0, 1, 2, \dots \quad (4.2.16)$$

Consider now the wave functions,

$$\psi_n(q) = \langle q|n\rangle \quad (4.2.16')$$

Adopted hereafter is the dimensionless coordinate representation, in which

$$\hat{Q} \doteq q \quad \text{and} \quad \hat{P} \doteq -i\frac{\partial}{\partial q}, \quad \text{where} \quad q = \sqrt{\frac{m\omega}{\hbar}}x \quad (4.2.17)$$

The annihilation and creation operators [Eq. (4.2.3)] are then

$$\hat{a} \doteq \frac{1}{\sqrt{2}}\left(q + \frac{\partial}{\partial q}\right) \quad \text{and} \quad \hat{a}^\dagger \doteq \frac{1}{\sqrt{2}}\left(q - \frac{\partial}{\partial q}\right) \quad (4.2.18)$$

1) Consider first the ground state via $\hat{a}|0\rangle = 0$. That is

$$\left(q + \frac{\partial}{\partial q}\right)\psi_0(q) = 0 \quad (4.2.19)$$

This gives rise to the normalized ground-state wave function,

$$\psi_0(q) = \pi^{-\frac{1}{4}}e^{-q^2/2} \quad (4.2.20)$$

2) Equation (4.2.16) reads now

$$\psi_n(q) = \frac{1}{(2^n n! \sqrt{\pi})^{\frac{1}{2}}} \left(q - \frac{\partial}{\partial q}\right)^n e^{-q^2/2}; \quad n = 0, 1, 2, \dots \quad (4.2.21)$$

Supplementary materials (for information only): Alternative expression reads

$$\psi_n(q) = \frac{1}{(2^n n! \sqrt{\pi})^{\frac{1}{2}}} H_n(q) e^{-q^2/2}, \quad (4.2.22)$$

Involved here are the **Hermite polynomials**, $\{H_n(q); n = 0, 1, 2, \dots\}$, satisfying

$$H_n''(q) = 2qH_n'(q) - 2nH_n(q) \quad \text{and} \quad H_n'(q) = 2nH_{n-1}(q) \quad (4.2.23)$$

The second one follows $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$. Moreover [in line with Eq. (4.2.22)],

$$\int_{-\infty}^{\infty} dq H_m(q) H_n(q) e^{-q^2} = 2^n n! \sqrt{\pi} \delta_{mn} \quad (4.2.24)$$

The first two Hermite polynomials are

$$H_0(q) = 1 \quad \text{and} \quad H_1(q) = 2q \quad (4.2.25)$$

The higher-order Hermite polynomials can be evaluated via the recurrence relation,

$$H_{n+1}(q) = 2qH_n(q) - 2nH_{n-1}(q) \quad (4.2.26)$$

with the parity, $H_n(-q) = (-1)^n H_n(q)$. Moreover,

$$H_n(q) = (-1)^n e^{q^2} \frac{d^n}{dq^n} e^{-q^2} \quad \text{and} \quad e^{-z^2+2zq} = \sum_{n=0}^{\infty} \frac{z^n}{n!} H_n(q) \quad (4.2.27)$$

• **Direct derivations of Eq. (4.2.22) with the number operator:**

Consider the Schrödinger equation, in terms of the eigenequation for the occupation number operator [cf. Eq. (4.2.8)]:

$$\hat{n}|\psi\rangle = \lambda|\psi\rangle \quad (4.2.28)$$

In the dimensionless coordinate representation of Eq. (4.2.18), we have

$$\hat{n} \equiv \hat{a}^\dagger \hat{a} \doteq \frac{1}{2} \left(q^2 - \frac{\partial^2}{\partial q^2} - 1 \right) \quad (4.2.29)$$

This leads to Eq. (4.2.28) the expression,

$$\frac{1}{2} \left(q^2 - \frac{\partial^2}{\partial q^2} - 1 \right) \psi(q) = \lambda \psi(q) \quad (4.2.30)$$

To proceed, we first recognize that $e^{-q^2/2}$ is a solution. This in fact is the ground eigenstate, since $e^{-q^2/2}$ contains no node. The general solutions are of the same asymptotic behavior. Let $f(q)$ be the *undetermined polynomial* in

$$\psi(q) = f(q)e^{-q^2/2} \quad (4.2.31)$$

that satisfies Eq. (4.2.30). Evaluate then

$$\begin{aligned} \frac{\partial^2}{\partial q^2} [f(q)e^{-q^2/2}] &= f''(q)e^{-q^2/2} + 2f'(q)(e^{-q^2/2})' + f(q)(e^{-q^2/2})'' \\ &= [f''(q) - 2qf'(q) + (q^2 - 1)f(q)]e^{-q^2/2} \end{aligned}$$

Rearrange it as

$$\left(q^2 - \frac{\partial^2}{\partial q^2} - 1 \right) [f(q)e^{-q^2/2}] = [2qf'(q) - f''(q)]e^{-q^2/2}$$

Substituting it into Eq. (4.2.30) with Eq. (4.2.31) results in

$$2qf'(q) - f''(q) = 2\lambda f(q) \quad (4.2.32)$$

Compare it with Eq. (4.2.23), resulting in

$$f(q) \propto H_n(q) \quad \text{and} \quad \lambda_n = n \quad (4.2.33)$$

This determines the wave function, Eq. (4.2.31), as $\psi_n(q) \propto H_n(q)e^{-q^2/2}$. Applying then the normalization, Eq. (4.2.24), we obtain Eq. (4.2.22). (Q.E.D.)

4.3 Angular Momentum: Preliminary

4.3.1 Angular momentum along the z -direction

- Consider the angular momentum along the z -direction,

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = \frac{\hbar}{i} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \quad (4.3.1)$$

In the polar coordinate frame, $x = \varrho \cos \phi$ and $y = \varrho \sin \phi$, we have

$$\hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \phi} \quad (4.3.2)$$

Proof of Eq. (4.3.2): The involving Jacobian transformation is evaluated as follow.

$$\begin{bmatrix} \frac{\partial}{\partial \varrho} \\ \frac{\partial}{\partial \phi} \end{bmatrix} \equiv \mathbf{A} \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \varrho} & \frac{\partial y}{\partial \varrho} \\ \frac{\partial x}{\partial \phi} & \frac{\partial y}{\partial \phi} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi \\ -\varrho \sin \phi & \varrho \cos \phi \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} \quad (4.3.3a)$$

$$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} \frac{\partial}{\partial \varrho} \\ \frac{\partial}{\partial \phi} \end{bmatrix} = \varrho^{-1} \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial \varrho} \\ \frac{\partial}{\partial \phi} \end{bmatrix} \quad (4.3.3b)$$

Evaluate then

$$x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} = \begin{bmatrix} -y \\ x \end{bmatrix}^T \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} = \begin{bmatrix} -\sin \phi \\ \cos \phi \end{bmatrix}^T \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} \varrho \frac{\partial}{\partial \varrho} \\ \frac{\partial}{\partial \phi} \end{bmatrix} \quad (4.3.4)$$

and obtain

$$x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} = \frac{\partial}{\partial \phi} \quad (4.3.4')$$

Substituting it into Eq. (4.3.1) results in Eq. (4.3.2).

Q.E.D.

• Quantization of angular momentum along z -direction

This is concerned with the eigenequation, $\hat{L}_z |\Phi\rangle = \lambda |\Phi\rangle$. That is

$$\frac{\hbar}{i} \frac{\partial}{\partial \phi} \Phi(\phi) = \lambda \Phi(\phi) \Rightarrow \Phi(\phi) = C e^{i\lambda\phi/\hbar} \quad (4.3.5)$$

Now consider the single-valued constraint,

$$\Phi(\phi + 2\pi) = \Phi(\phi) \Rightarrow e^{i2\pi\lambda/\hbar} = 1$$

The resultant quantization of angular momentum along z -direction reads

$$\lambda = m\hbar; \quad m = 0, \pm 1, \pm 2, \dots \quad (4.3.6)$$

It is easy to obtain the normalization constant in Eq. (4.3.5), so that

$$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad (4.3.7)$$

Note also that the above quantization is often reported as

$$\hat{L}_z |\Phi\rangle = m\hbar |\Phi\rangle; \quad m = 0, \pm 1, \pm 2, \dots \quad (4.3.8)$$

4.3.2 Angular momentum in three dimension space

A. The basic relations

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \hat{x} & \hat{y} & \hat{z} \\ \hat{p}_x & \hat{p}_y & \hat{p}_z \end{vmatrix} \quad (4.3.9)$$

The canonical commutator relations give rise immediately to (**Exercises**)

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z; \quad (\text{also the two cyclic counterparts}) \quad (4.3.10a)$$

and further

$$[\hat{L}^2, \hat{L}_z] = [\hat{L}^2, \hat{L}_x] = [\hat{L}^2, \hat{L}_y] = 0 \quad (4.3.10b)$$

These fundamental relations are the starting points to the very important angular momentums theory that will be discussed in Chapter §4.

B. Angular momentum and Laplacian operators

• Angular momentum in spherical coordinates

$$\hat{L}_x \equiv \frac{\hbar}{i}\hat{\Lambda}_x, \quad \hat{L}_y \equiv \frac{\hbar}{i}\hat{\Lambda}_y, \quad \boxed{\hat{L}_z \equiv \frac{\hbar}{i}\hat{\Lambda}_z = \frac{\hbar}{i}\frac{\partial}{\partial\phi}} \quad (4.3.11)$$

We can also obtain (*Derivations will be provided for information only*)

$$\hat{\Lambda}_x = y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} = -\sin\phi\frac{\partial}{\partial\theta} - \cot\theta\cos\phi\frac{\partial}{\partial\phi} \quad (4.3.12a)$$

$$\hat{\Lambda}_y = z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z} = \cos\phi\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi} \quad (4.3.12b)$$

Equations (4.3.11)–(4.3.12) give rise to [cf. Eqs. (4.3.18)–(4.3.21)]

$$\boxed{\hat{L}^2 = -\hbar^2\hat{\Lambda}^2} \quad \text{and} \quad \boxed{\hat{\Lambda}^2 = \frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}} \quad (4.3.13)$$

• Laplacian operator in Cartesian versus spherical coordinates

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \frac{1}{r}\frac{\partial^2}{\partial r^2}r + \frac{1}{r^2}\hat{\Lambda}^2 \quad (4.3.14)$$

Correspondences to classical counterparts, Eqs. (1.3.21) and (1.3.22):

$$L_z = p_\phi \quad \text{and} \quad \begin{bmatrix} L_x \\ L_y \end{bmatrix} = - \begin{bmatrix} \sin\phi & \cot\theta\cos\phi \\ -\cos\phi & \cot\theta\sin\phi \end{bmatrix} \begin{bmatrix} p_\theta \\ p_\phi \end{bmatrix}$$

where p_ϕ and p_θ are the generalized momentums conjugate to the specified angles. In comparing with Eqs. (4.3.11) and (4.3.12), we would have the following correspondences,

$$p_\alpha \Rightarrow \boxed{\hat{p}_\alpha \doteq \frac{\hbar}{i}\frac{\partial}{\partial\alpha}} \quad (\text{for } \alpha = \theta, \phi)$$

The canonical commutator relation, $[\hat{\alpha}, \hat{p}_\alpha] = i\hbar$, holds here!

C. Derivations of Eqs. (4.3.12)–(4.3.14) (for information only)

First of all, the derivations are rather tedious but elementary, involving the Jacobian transformation, similar to that in Eqs. (4.3.3)–(4.3.4'). The details are as follows.

- **The Jacobian matrix** for $(x, y, z) = r(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ [cf. Eq. (4.3.3)]:

$$\mathbf{A} \equiv \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} & \frac{\partial z}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \theta} \\ \frac{\partial x}{\partial \phi} & \frac{\partial y}{\partial \phi} & \frac{\partial z}{\partial \phi} \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ r \cos \theta \cos \phi & r \cos \theta \sin \phi & -r \sin \theta \\ -r \sin \theta \sin \phi & r \sin \theta \cos \phi & 0 \end{bmatrix}$$

Evaluate then

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}} \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} \quad (4.3.15)$$

where

$$A_{11} = \begin{vmatrix} r \cos \theta \sin \phi & -r \sin \theta \\ r \sin \theta \cos \phi & 0 \end{vmatrix}, \quad A_{21} = - \begin{vmatrix} \sin \theta \sin \phi & \cos \theta \\ r \sin \theta \cos \phi & 0 \end{vmatrix}, \quad A_{31} = \begin{vmatrix} \sin \theta \sin \phi & \cos \theta \\ r \cos \theta \sin \phi & -r \sin \theta \end{vmatrix}$$

and so on. We obtain

$$\begin{aligned} A_{11} &= r^2 \sin^2 \theta \cos \phi, & A_{21} &= r \sin \theta \cos \theta \cos \phi, & A_{31} &= -r \sin \phi, \\ A_{12} &= r^2 \sin^2 \theta \sin \phi, & A_{22} &= r \sin \theta \cos \theta \sin \phi, & A_{32} &= r \cos \phi, \\ A_{13} &= r^2 \sin \theta \cos \theta, & A_{23} &= -r \sin^2 \theta, & A_{33} &= 0 \end{aligned}$$

and

$$\det \mathbf{A} = \cos \theta A_{13} - r \sin \theta A_{23} = r^2 \sin \theta$$

The resultant Eq. (4.3.15) reads

$$\mathbf{A}^{-1} = \begin{bmatrix} \sin \theta \cos \phi & \frac{1}{r} \cos \theta \cos \phi & -\frac{1}{r \sin \theta} \sin \phi \\ \sin \theta \sin \phi & \frac{1}{r} \cos \theta \sin \phi & \frac{1}{r \sin \theta} \cos \phi \\ \cos \theta & -\frac{1}{r} \sin \theta & 0 \end{bmatrix} \quad (4.3.15')$$

This determines

$$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial \phi} \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix} \quad (4.3.16)$$

- **Derivations of Eq. (4.3.12):** Consider for example,

$$y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} = \begin{bmatrix} 0 \\ -z \\ y \end{bmatrix}^T \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = -r \begin{bmatrix} 0 \\ \cos \theta \\ -\sin \theta \sin \phi \end{bmatrix}^T \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}$$

Applying Eq. (4.3.16) results in

$$\begin{aligned} y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} &= -r \begin{bmatrix} 0 \\ \cos \theta \\ -\sin \theta \sin \phi \end{bmatrix}^T \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix} \\ &= -r \begin{bmatrix} 0 \\ \sin \phi \\ \cos \theta \cos \phi \end{bmatrix}^T \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix} \end{aligned}$$

We obtain Eq. (4.3.12a); i.e.,

$$\hat{\Lambda}_x = y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} = - \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) \quad (4.3.17)$$

The derivations of Eq. (4.3.12b) are similar. (Q.E.D.)

• **Derivations on Eq. (4.3.13):** By using Eq. (4.3.12), we have

$$\begin{aligned} \hat{\Lambda}_x^2 &= \sin \phi \frac{\partial}{\partial \theta} \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right) \\ \hat{\Lambda}_y^2 &= \cos \phi \frac{\partial}{\partial \theta} \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right) \end{aligned}$$

Consequently,

$$\begin{aligned} \hat{\Lambda}_x^2 + \hat{\Lambda}_y^2 &= \frac{\partial^2}{\partial \theta^2} + \cot \theta \left(\cos \phi \frac{\partial}{\partial \phi} \sin \phi - \sin \phi \frac{\partial}{\partial \phi} \cos \phi \right) \frac{\partial}{\partial \theta} \\ &\quad + \cot^2 \theta \left(\cos \phi \frac{\partial}{\partial \phi} \cos \phi + \sin \phi \frac{\partial}{\partial \phi} \sin \phi \right) \frac{\partial}{\partial \phi} \end{aligned} \quad (4.3.18)$$

Note that

$$\frac{\partial}{\partial \phi} \sin \phi = \cos \phi + \sin \phi \frac{\partial}{\partial \phi} \quad \text{and} \quad \frac{\partial}{\partial \phi} \cos \phi = -\sin \phi + \cos \phi \frac{\partial}{\partial \phi} \quad (4.3.19)$$

Hence,

$$\begin{aligned} \cos \phi \frac{\partial}{\partial \phi} \sin \phi - \sin \phi \frac{\partial}{\partial \phi} \cos \phi &= 1 \\ \cos \phi \frac{\partial}{\partial \phi} \cos \phi + \sin \phi \frac{\partial}{\partial \phi} \sin \phi &= \frac{\partial}{\partial \phi} \end{aligned} \quad (4.3.20)$$

We obtain Eq. (4.3.18) the expression,

$$\hat{\Lambda}_x^2 + \hat{\Lambda}_y^2 = \frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \cot^2 \theta \frac{\partial^2}{\partial \phi^2} \quad (4.3.21)$$

This together with $\hat{\Lambda}_z^2 = \frac{\partial^2}{\partial \phi^2}$ result in

$$\hat{\Lambda}^2 = \hat{\Lambda}_x^2 + \hat{\Lambda}_y^2 + \hat{\Lambda}_z^2 = \frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \quad (4.3.22)$$

This is Eq. (4.3.13). (Q.E.D.)

• **Derivations on Eq. (4.3.14):** Let us start with

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}^T \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \quad (4.3.23)$$

The transpose to Eq. (4.3.16) goes through its individual elements as follows.

$$\begin{aligned}\frac{\partial}{\partial x} &= \sin \theta \cos \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \cos \phi \frac{\partial}{\partial \theta} - \frac{1}{r \sin \theta} \sin \phi \frac{\partial}{\partial \phi} \\ &= \left(\frac{\partial}{\partial r} \sin \theta \cos \phi + \frac{1}{r} \frac{\partial}{\partial \theta} \cos \theta \cos \phi - \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \sin \phi \right) \\ &\quad + \frac{1}{r} \sin \theta \cos \phi + \frac{1}{r \sin \theta} \cos \phi\end{aligned}\quad (4.3.24a)$$

$$\begin{aligned}\frac{\partial}{\partial y} &= \sin \theta \sin \phi \frac{\partial}{\partial r} + \frac{1}{r} \cos \theta \sin \phi \frac{\partial}{\partial \theta} + \frac{1}{r \sin \theta} \cos \phi \frac{\partial}{\partial \phi} \\ &= \left(\frac{\partial}{\partial r} \sin \theta \sin \phi + \frac{1}{r} \frac{\partial}{\partial \theta} \cos \theta \sin \phi + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \cos \phi \right) \\ &\quad + \frac{1}{r} \sin \theta \sin \phi + \frac{1}{r \sin \theta} \sin \phi\end{aligned}\quad (4.3.24b)$$

$$\frac{\partial}{\partial z} = \cos \theta \frac{\partial}{\partial r} - \frac{1}{r} \sin \theta \frac{\partial}{\partial \theta} = \left(\frac{\partial}{\partial r} \cos \theta - \frac{1}{r} \frac{\partial}{\partial \theta} \sin \theta \right) + \frac{1}{r} \cos \theta \quad (4.3.24c)$$

We obtain

$$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}^T = \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix}^T \mathbf{U}^T + \frac{1}{r} \begin{bmatrix} \sin \theta \cos \phi + \frac{1}{\sin \theta} \cos \phi \\ \sin \theta \sin \phi + \frac{1}{\sin \theta} \sin \phi \\ \cos \theta \end{bmatrix}^T \quad (4.3.25)$$

where

$$\mathbf{U} \equiv \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \quad (\mathbf{U}^T \mathbf{U} = \mathbf{1}) \quad (4.3.26)$$

with which Eq. (4.3.16) reads

$$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \mathbf{U} \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix} \quad (4.3.27)$$

By substituting Eqs. (4.3.25)–(4.3.27) into Eq. (4.3.23), we have

$$\begin{aligned}\nabla^2 &= \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}^T \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix} + \frac{1}{r} \begin{bmatrix} \sin \theta \cos \phi + \frac{1}{\sin \theta} \cos \phi \\ \sin \theta \sin \phi + \frac{1}{\sin \theta} \sin \phi \\ \cos \theta \end{bmatrix}^T \mathbf{U} \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix} \\ &= \frac{\partial^2}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} + \frac{1}{r} \begin{bmatrix} 2 \\ \cot \theta \\ 0 \end{bmatrix}^T \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \end{bmatrix} \\ &= \frac{\partial^2}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{\cot \theta}{r^2} \frac{\partial}{\partial \theta}\end{aligned}\quad (4.3.28)$$

Together with Eq. (4.3.13), we obtain

$$\boxed{\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \hat{\Lambda}^2} \quad (4.3.29)$$

Moreover,

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} r = \frac{1}{r} \left(r \frac{\partial^2}{\partial r^2} + 2 \frac{\partial}{\partial r} \right) = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \quad (4.3.30)$$

Substituting this into Eq. (4.3.29) results in Eq. (4.3.14). (Q.E.D.)

4.4 Generate Two-Dimension Harmonic Oscillators

4.4.1 General considerations

Consider a harmonic oscillator of frequency ω and reduced mass μ , moving in the xy -plane. While it can be readily treated in mathematics, by using the creation-annihilation operators technique as §4.2.1, the physically important angular momentum of this system would not stand out. Combining the eigen solutions of $\hat{L}_z$ in §4.3.1 would be preferred.

- Let us start with the established operators approach of §4.2.1, in which the two-dimension harmonic oscillator Hamiltonian reads [cf. Eq. (4.2.4)]

$$\hat{H} = \hat{H}_x + \hat{H}_y = \hbar\omega\left(\hat{a}_x^\dagger\hat{a}_x + \frac{1}{2}\right) + \hbar\omega\left(\hat{a}_y^\dagger\hat{a}_y + \frac{1}{2}\right) \quad (4.4.1)$$

The annihilation operators are [cf. Eq. (4.2.3)]

$$\hat{a}_x = \left(\frac{\mu\omega}{2\hbar}\right)^{\frac{1}{2}}\left(\hat{x} + \frac{i\hat{p}_x}{\mu\omega}\right) \quad \text{and} \quad \hat{a}_y = \left(\frac{\mu\omega}{2\hbar}\right)^{\frac{1}{2}}\left(\hat{y} + \frac{i\hat{p}_y}{\mu\omega}\right) \quad (4.4.2)$$

satisfying [cf. Eq. (4.2.5)]

$$[\hat{a}_u, \hat{a}_v^\dagger] = \delta_{uv} \quad \text{and} \quad [\hat{a}_u, \hat{a}_v] = 0; \quad u, v = x, y \quad (4.4.3)$$

The eigenvectors and energies are [cf. Eqs. (4.2.4)–(4.2.7)]

$$|n_x, n_y\rangle = |n_x\rangle \otimes |n_y\rangle = \frac{(\hat{a}_x^\dagger)^{n_x} (\hat{a}_y^\dagger)^{n_y}}{\sqrt{(n_x)!(n_y)!}} |0, 0\rangle \quad (4.4.4a)$$

$$E_{n_x, n_y} = E_{n_x} + E_{n_y} = \hbar\omega(n_x + n_y + 1); \quad n_x, n_y = 0, 1, 2, \dots \quad (4.4.4b)$$

The ground state is non-degenerate, satisfying

$$\hat{a}_x \hat{a}_y |0, 0\rangle = 0 \quad (4.4.5)$$

This determines the ground state wave function $\langle x, y | 0, 0 \rangle$, as demonstrated in §4.2.2, with $\hat{a}_x$ and $\hat{a}_y$ being expressed in the coordinate representation. The excited $\langle x, y | n_x, n_y \rangle$ via Eq. (4.4.4a) are then followed.

- Discussions and comments:

(i) For a multiple-dimension system, $\hat{H}|\Psi_n\rangle = E_n|\Psi_n\rangle$, is degenerate in general. For the illustrated case above, associated with a given

$$E_n = \hbar\omega(n + 1); \quad n = 0, 1, 2, \dots,$$

there are a total of $(n + 1)$ degenerated states,

$$|\Psi_n\rangle = |\psi_{n_x=n-k}\rangle \otimes |\psi_{n_y=n+k}\rangle; \quad k = 0, 1, \dots, n$$

where $n = n_x + n_y$.

(ii) The total Hamiltonian $\hat{H}$, as a whole or the resulting primary quantum number, n , is insufficient in the descriptions. a *complete set of commuting observables* (CSCO) is required.

(iii) Mathematically, either $(\hat{H}, \hat{H}_x)$ or $(\hat{H}, \hat{H}_y)$ constitutes a CSCO, which is however only valid for the present non-interacting two-dimension system in study. Physically, $\hat{H}_x$ and $\hat{H}_y$ describe the same type of properties. A CSCO with distinct properties would be preferred.

(vi) The $(\hat{H}, \hat{L}_z)$ is the choice for two-dimension systems, in relation to the underlying the angular momentum conservation.

• For the two-dimension harmonic oscillator system, Eq. (4.4.1), we revisit $\hat{L}_z$, Eq. (4.3.1), as follows. From Eq. (4.4.2), we have

$$\begin{aligned}\hat{x} &= \left(\frac{\hbar}{\mu\omega}\right)^{\frac{1}{2}} \frac{\hat{a}_x + \hat{a}_x^\dagger}{\sqrt{2}}, & \hat{p}_x &= (\hbar\mu\omega)^{\frac{1}{2}} \frac{\hat{a}_x - \hat{a}_x^\dagger}{i\sqrt{2}} \\ \hat{y} &= \left(\frac{\hbar}{\mu\omega}\right)^{\frac{1}{2}} \frac{\hat{a}_y + \hat{a}_y^\dagger}{\sqrt{2}}, & \hat{p}_y &= (\hbar\mu\omega)^{\frac{1}{2}} \frac{\hat{a}_y - \hat{a}_y^\dagger}{i\sqrt{2}}\end{aligned}\quad (4.4.6)$$

These give rise Eq. (4.4.2) the expression,

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = \frac{\hbar}{2i} [(\hat{a}_x + \hat{a}_x^\dagger)(\hat{a}_y - \hat{a}_y^\dagger) - (\hat{a}_y + \hat{a}_y^\dagger)(\hat{a}_x - \hat{a}_x^\dagger)]$$

We obtain

$$\hat{L}_z = -i\hbar(\hat{a}_x^\dagger\hat{a}_y - \hat{a}_y^\dagger\hat{a}_x) \quad (4.4.7)$$

• Let us revisit $\hat{a}_x$ and $\hat{a}_y$, Eq. (4.4.2), in the following representations.

$$\begin{aligned}\hat{a}_x &\doteq \left(\frac{\mu\omega}{2\hbar}\right)^{\frac{1}{2}} \left(x + \frac{\hbar}{\mu\omega} \frac{\partial}{\partial x}\right) = \frac{1}{\sqrt{2}} \left[\varrho \cos \phi + \left(\cos \phi \frac{\partial}{\partial \varrho} - \frac{\sin \phi}{\varrho} \frac{\partial}{\partial \phi}\right)\right] \\ \hat{a}_y &\doteq \left(\frac{\mu\omega}{2\hbar}\right)^{\frac{1}{2}} \left(y + \frac{\hbar}{\mu\omega} \frac{\partial}{\partial y}\right) = \frac{1}{\sqrt{2}} \left[\varrho \sin \phi + \left(\sin \phi \frac{\partial}{\partial \varrho} + \frac{\cos \phi}{\varrho} \frac{\partial}{\partial \phi}\right)\right]\end{aligned}\quad (4.4.8)$$

The second identities are the polar-coordinate representation, with the dimensionless polar radius, $\varrho = \sqrt{\mu\omega/\hbar} \rho$, and Eq. (4.3.3b) being exploited. Similarly,

$$\begin{aligned}\hat{a}_x^\dagger &\doteq \left(\frac{\mu\omega}{2\hbar}\right)^{\frac{1}{2}} \left(x - \frac{\hbar}{\mu\omega} \frac{\partial}{\partial x}\right) = \frac{1}{\sqrt{2}} \left[\varrho \cos \phi - \left(\cos \phi \frac{\partial}{\partial \varrho} - \frac{\sin \phi}{\varrho} \frac{\partial}{\partial \phi}\right)\right] \\ \hat{a}_y^\dagger &\doteq \left(\frac{\mu\omega}{2\hbar}\right)^{\frac{1}{2}} \left(y - \frac{\hbar}{\mu\omega} \frac{\partial}{\partial y}\right) = \frac{1}{\sqrt{2}} \left[\varrho \sin \phi - \left(\sin \phi \frac{\partial}{\partial \varrho} + \frac{\cos \phi}{\varrho} \frac{\partial}{\partial \phi}\right)\right]\end{aligned}\quad (4.4.9)$$

The above expressions will be used later; see Eqs. (4.4.23).

4.4.2 Bosonic algebra with angular momentum

A. Circular harmonic operators

- To proceed, we transform Eq. (4.4.7) into

$$\hat{L}_z = \hbar(\hat{n}_R - \hat{n}_L) \quad (4.4.10)$$

with

$$\hat{n}_R \equiv \hat{c}_R^\dagger \hat{c}_R \quad \text{and} \quad \hat{n}_L \equiv \hat{c}_L^\dagger \hat{c}_L \quad (4.4.11)$$

being the *right-circular* and *left-circular* number operators, respectively.

The required transformation reads

$$\hat{c}_R = \frac{1}{\sqrt{2}}(\hat{a}_x - i\hat{a}_y) \quad \text{and} \quad \hat{c}_L = \frac{1}{\sqrt{2}}(\hat{a}_x + i\hat{a}_y) \quad (4.4.12)$$

Together with Eq. (4.4.7), we obtain

$$\hat{n}_R = \frac{1}{2}(\hat{n}_x + \hat{n}_y) + \frac{\hat{L}_z}{2\hbar} \quad \text{and} \quad \hat{n}_L = \frac{1}{2}(\hat{n}_x + \hat{n}_y) - \frac{\hat{L}_z}{2\hbar} \quad (4.4.13)$$

where $\hat{n}_x \equiv \hat{a}_x^\dagger \hat{a}_x$ and $\hat{n}_y \equiv \hat{a}_y^\dagger \hat{a}_y$. While $\hat{n}_R - \hat{n}_L = \hat{L}_z/\hbar$ as Eq. (4.4.10), $\hat{n} = \hat{n}_L + \hat{n}_R = \hat{n}_x + \hat{n}_y$, resulting in Eq. (4.4.1) invariance with the form of

$$\hat{H} = \hbar\omega(\hat{n} + 1), \quad \text{where} \quad \hat{n} = \hat{n}_R + \hat{n}_L \quad (4.4.14)$$

Evidently, $[\hat{H}, \hat{L}_z] = 0$.

- Moreover, the transformation, Eq. (4.4.12), retains the canonical commutation relations of

$$[\hat{c}_\alpha, \hat{c}_{\alpha'}^\dagger] = \delta_{\alpha\alpha'} \quad \text{and} \quad [\hat{c}_\alpha, \hat{c}_{\alpha'}] = 0; \quad \alpha, \alpha' = \text{L, R} \quad (4.4.15)$$

The same harmonic oscillator algebra, as depicted in Eqs. (4.2.5)–(4.2.7), is applicable in the individual pair of transformed annihilation and creation operators. Both $\hat{H}$ and $\hat{L}_z$, Eq. (4.4.14) and Eq. (4.4.10), are now treated with the same operator algebra. We obtain

$$\hat{H}|n_R, n_L\rangle = \hbar\omega(n_R + n_L + 1)|n_R, n_L\rangle \quad (4.4.16a)$$

$$\hat{L}_z|n_R, n_L\rangle = \hbar(n_R - n_L)|n_R, n_L\rangle \quad (4.4.16b)$$

The form of Eq. (4.4.4a) remains the same, but changes $(\hat{a}_x^\dagger, \hat{a}_y^\dagger; n_x, n_y)$ to $(\hat{c}_R^\dagger, \hat{c}_L^\dagger; n_R, n_L)$. In terms of the recursive relation, it reads [cf. Eq. (4.2.7)]

$$|n_R, n_L\rangle = \frac{1}{\sqrt{n_R}}\hat{c}_R^\dagger|n_R - 1, n_L\rangle = \frac{1}{\sqrt{n_L}}\hat{c}_L^\dagger|n_R, n_L - 1\rangle \quad (4.4.17)$$

The ground state remains as Eq. (4.4.5).

- Recast Eq. (4.4.16) as

$$\hat{H}|n, m\rangle = \hbar\omega(n + 1)|n, m\rangle \quad \text{and} \quad \hat{L}_z|n, m\rangle = \hbar m|n, m\rangle \quad (4.4.18a)$$

with

$$n = 0, 1, 2, \dots \quad \text{and} \quad m = -n, -n + 2, \dots, n - 2, n \quad (4.4.18b)$$

The degeneracy of eigenenergy is $(n + 1)$. We will elaborate this representation further with the spherical symmetry counterpart; see §5.4.3.

B. Wave functions in the polar coordinates

Consider the polar-coordinate representation in line with Eq. (4.4.16),

$$\Psi_{n_R, n_L}(\varrho, \phi) \equiv \langle \varrho, \phi | n_R, n_L \rangle = \frac{1}{\sqrt{2\pi}} \psi_{n_R, n_L}(\varrho) e^{i(n_R - n_L)\phi} \quad (4.4.19)$$

Here $\varrho = \sqrt{\mu\omega/\hbar} \rho$ is the dimensionless polar radius. The angle part has been normalized. The radius part $\psi_{n_R, n_L}(\varrho)$ is real and follows

$$\int_0^\infty \varrho \psi_{n_R, n_L}^2(\varrho) d\varrho = 1 \quad (4.4.20)$$

The ground state is given by $\Psi_{0,0}(\varrho, \phi) = e^{-\varrho^2/2}/\sqrt{\pi}$ [cf. Eq. (4.2.20)] The corresponding normalized radius function is

$$\psi_{0,0}(\varrho) = \sqrt{2} e^{-\varrho^2/2} \quad (4.4.21)$$

As shown soon later, the recursion relations in Eq. (4.4.17) read now

$$\begin{aligned} \psi_{n_R, n_L}(\varrho) &= \frac{1}{2\sqrt{n_R}} \left(\varrho - \frac{d}{d\varrho} + \frac{n_R - n_L - 1}{\varrho} \right) \psi_{n_R-1, n_L}(\varrho) \\ &= \frac{1}{2\sqrt{n_L}} \left(\varrho - \frac{d}{d\varrho} + \frac{n_L - n_R - 1}{\varrho} \right) \psi_{n_R, n_L-1}(\varrho) \end{aligned} \quad (4.4.22)$$

The symmetry of $\psi_{n_\alpha, n_{\alpha'}}(\varrho) = \psi_{n_{\alpha'}, n_\alpha}(\varrho)$ is highlighted.

Derivations of Eq. (4.4.22): To complete Eq. (4.4.19) via Eq. (4.4.17), we exploit Eq. (4.4.9) and Eq. (4.4.12) to have $\hat{c}_R^\dagger$ and $\hat{c}_L^\dagger$ in the polar coordinate representation.

$$\hat{c}_R^\dagger = \frac{e^{i\phi}}{2} \left(\varrho - \frac{\partial}{\partial \varrho} - \frac{i}{\varrho} \frac{\partial}{\partial \phi} \right) \quad \text{and} \quad \hat{c}_L^\dagger = \frac{e^{-i\phi}}{2} \left(\varrho - \frac{\partial}{\partial \varrho} + \frac{i}{\varrho} \frac{\partial}{\partial \phi} \right) \quad (4.4.23)$$

Their actions on the wave function, Eq. (4.2.18) with Eq. (4.4.19), read

$$\begin{aligned} \hat{c}_R^\dagger [\psi_{n_R-1, n_L}(\varrho) e^{i(m-1)\phi}] &= \sqrt{n_R} \psi_{n_R, n_L}(\varrho) e^{im\phi} \\ \hat{c}_L^\dagger [\psi_{n_R, n_L-1}(\varrho) e^{i(m+1)\phi}] &= \sqrt{n_L} \psi_{n_R, n_L}(\varrho) e^{im\phi} \end{aligned} \quad (4.4.24)$$

where $m = n_R - n_L$. Substitute Eq. (4.4.23) into Eq. (4.4.24), resulting in

$$\begin{aligned} \left(\varrho - \frac{d}{d\varrho} + \frac{n_R - n_L - 1}{\varrho} \right) \psi_{n_R-1, n_L}(\varrho) &= 2\sqrt{n_R} \psi_{n_R, n_L}(\varrho) \\ \left(\varrho - \frac{d}{d\varrho} + \frac{n_L - n_R - 1}{\varrho} \right) \psi_{n_R, n_L-1}(\varrho) &= 2\sqrt{n_L} \psi_{n_R, n_L}(\varrho) \end{aligned} \quad (4.4.25)$$

They are equivalent to Eq. (4.4.22). (Q.E.D.)

Some basic results of Eq. (4.4.22):

(a) The first few excited states from Eq. (4.4.22) are as follows.

$$\psi_{1,0}(\varrho) = \psi_{0,1}(\varrho) = \frac{1}{2} \left(\varrho - \frac{d}{d\varrho} \right) \psi_{0,0}(\varrho) = \sqrt{2} \varrho e^{-\varrho^2/2} \quad (4.4.26)$$

$$\begin{aligned} \psi_{2,0}(\varrho) = \psi_{0,2}(\varrho) &= \frac{1}{2\sqrt{2}} \left(\varrho - \frac{d}{d\varrho} + \frac{1}{\varrho} \right) \psi_{1,0}(\varrho) = \varrho^2 e^{-\varrho^2/2} \\ \psi_{1,1}(\varrho) &= \frac{1}{2} \left(\varrho - \frac{d}{d\varrho} - \frac{1}{\varrho} \right) \psi_{0,1}(\varrho) = \sqrt{2}(\varrho^2 - 1)e^{-\varrho^2/2} \end{aligned} \quad (4.4.27)$$

and

$$\begin{aligned} \psi_{3,0}(\varrho) = \psi_{0,3}(\varrho) &= \frac{1}{2\sqrt{3}} \left(\varrho - \frac{d}{d\varrho} + \frac{2}{\varrho} \right) \psi_{2,0}(\varrho) = \frac{1}{\sqrt{3}} \varrho^3 e^{-\varrho^2/2} \\ \psi_{2,1}(\varrho) = \psi_{1,2}(\varrho) &= \frac{1}{2\sqrt{2}} \left(\varrho - \frac{d}{d\varrho} \right) \psi_{1,1}(\varrho) = \varrho(\varrho^2 - 2)e^{-\varrho^2/2} \end{aligned} \quad (4.4.28)$$

Each individual Eqs. (4.4.26)–(4.4.28) describes a set of degenerate wave functions, having the common energy, as specified by Eq. (4.4.16a).

(b) It is easy to show (by deduction) that

$$\psi_{n,0}(\varrho) = \psi_{0,n}(\varrho) = \sqrt{\frac{2}{n!}} \varrho^n e^{-\varrho^2/2} \quad (4.4.29)$$

We can then exploit Eq. (4.4.22) to recursively evaluate this degenerate series of common energy, $\psi_{n-1,1}(\varrho) = \psi_{1,n-1}(\varrho)$, $\psi_{n-2,2}(\varrho) = \psi_{2,n-2}(\varrho)$, and so on.

Chapter 5

Angular Momentum Theory and Central Force Systems

Angular momentum theory is extremely important, infiltrating many fields of physics, such as atomic and molecular spectroscopy, spin and magnetism properties, just name a few. The total angular momentum is a conserved property; see Eq. (1.3.9). This is related to rotations in real-space. Rotation matrixes are orthogonal, but noncommutable. All these have the standard quantum correspondences. In particular, we will present in §5.4 the Hilbert-space description of rotations via the angular momentum operator. In addition, quantum angular momentum involves further internal *spins that do not have classical correspondence*.

5.1 Angular Momentum Theorem

5.1.1 Basic results

Quantum particles posses also internal *spin*, $\hat{\mathbf{S}}$, identified in Stern–Gerlach experiments. This angular momentum does not have the classical analogue. On the other hand, the aforementioned $\hat{\mathbf{L}}$ has classical correspondences and is called the orbital angular momentum. The generic angular momentum, either $\hat{\mathbf{L}}$ or $\hat{\mathbf{S}}$ or $(\hat{\mathbf{L}} + \hat{\mathbf{S}})$, is

$$\hat{\mathbf{J}} = \hat{J}_x \mathbf{i} + \hat{J}_y \mathbf{j} + \hat{J}_z \mathbf{k} \quad (5.1.1)$$

The angular momentum theorem presented below is universal. It starts with the formal extension of Eq. (3.3.10a) as

$$\boxed{[\hat{J}_x, \hat{J}_y] = i\hbar \hat{J}_z \quad (\text{and the cyclic two})} \quad (5.1.2)$$

These constitute a set of canonical commutators for angular momentum. For bookkeeping, we present the results first and the derivations after.

A. Summary of the results

While Eq. (5.1.2) concludes that the angular momentum in different directions do not have common set of eigen functions, it also gives rise to

$$[\hat{J}^2, \hat{J}_z] = 0 \quad (\text{and the cyclic two}) \quad (5.1.3)$$

Let $\{|j, m\rangle\}$ be the eigenkets for both $\hat{J}^2$ and $\hat{J}_z$, which are found to be

$$\begin{aligned} \hat{J}^2|j, m\rangle &= \hbar^2 j(j+1)|j, m\rangle & [(2j+1) \text{ be integer}] \\ \hat{J}_z|j, m\rangle &= \hbar m|j, m\rangle; & m = -j, -j+1, \dots, j-1, j \end{aligned} \quad (5.1.4)$$

• Remarks:

- 1) The first identity specifies that the eigenvalue of $|\hat{J}|$ is $\hbar\sqrt{j(j+1)}$, with the degeneracy of $2j+1$. This degeneracy is resolved with the $(2j+1)$ values of quantum number m for $\hat{J}_z$, as specified in the second identity.
- 2) As $\hat{J}$ stands for both $\hat{L}$ and $\hat{S}$, the quantum number j in these two cases refer to l and s , respectively. The former is always an integer. This is evident from Eq. (3.3.8) that $\hat{L}_z$ has the eigenvalues of $m\hbar$; $m = 0, \pm 1, \dots, \pm l$, with an integer l . The atomic s, p, d and f orbitals are of $l = 0, 1, 2$ and 3, respectively. The molecular rotation is another example.
- 3) The spin quantum number s of an elementary particle can be either an *integer* or a *half-integer*. This differs between *bosons* and *fermions*.
- 4) The eigenvalues of $\hat{J}_x$ and $\hat{J}_y$ are the same as those of $\hat{J}_z$.

B. Algebraic approach to Eq. (5.1.4) (Self-Exercises)

- **Ladder operators** (for mapping $|j, m\rangle$ onto $|j, m \pm 1\rangle$):

$$\hat{J}_{\pm} \equiv \hat{J}_x \pm i\hat{J}_y \quad (5.1.5)$$

- 4) The ladder operators satisfy

$$[\hat{J}_z, \hat{J}_{\pm}] = \pm\hbar\hat{J}_{\pm} \quad (5.1.6)$$

This results in

$$\hat{J}_{\pm}|j, m\rangle \propto |j, m \pm 1\rangle \Rightarrow \hat{J}_+|j, m_{\max}\rangle = \hat{J}_-|j, m_{\min}\rangle = 0 \quad (5.1.7)$$

- 5) The ladder operators satisfy also

$$\hat{J}_+\hat{J}_- = \hat{J}^2 - \hat{J}_z^2 + \hbar\hat{J}_z \quad \text{and} \quad \hat{J}_-\hat{J}_+ = \hat{J}^2 - \hat{J}_z^2 - \hbar\hat{J}_z \quad (5.1.8)$$

These give rise to the normalized expression,

$$\hat{J}_{\pm}|j, m\rangle = \hbar[j(j+1) - m(m \pm 1)]^{\frac{1}{2}}|j, m \pm 1\rangle \quad (5.1.9)$$

This satisfies Eq. (5.1.7) and further results in Eq. (5.1.4). (Q.E.D.)

5.1.2 Spin angular momentum matrices

This section is concerned with the matrix representation of the spin angular momentum, with the basis set of $\{|s, m\rangle\}$, by using Eq. (5.1.9) and also the second expression of Eq. (5.1.4), where $j = s$. Adopt the notation, $^{2s+1}\hat{\mathbf{S}}$, so that the spin- $\frac{1}{2}$ and spin-1 angular momentums are $^2\hat{\mathbf{S}}$ and $^3\hat{\mathbf{S}}$, respectively. Note that $^1\hat{\mathbf{S}} = 0$ for spin-0.

- For spin- $\frac{1}{2}$, we have

$$^2\hat{S}_z = \frac{\hbar}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad ^2\hat{S}_x = \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad ^2\hat{S}_y = \frac{\hbar}{2} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (5.1.10)$$

The first identity arises from the second expression of Eq. (5.1.4). The last two via Eq. (5.1.5) are related to Eq. (5.1.9) that gives rise to

$$^2\hat{S}_+ = \hbar \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad ^2\hat{S}_- = ^2\hat{S}_+^\dagger = \hbar \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad (5.1.11)$$

Exercises:

E5.1.1 Show that

$$^3\hat{S}_z = \hbar \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad ^3\hat{S}_x = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad ^3\hat{S}_y = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{bmatrix} \quad (5.1.12)$$

E5.1.2 Derive the matrices for $^4\hat{\mathbf{S}}$

E5.1.3 Given the state $|j, m\rangle$, show that $\langle \hat{J}_z \rangle = \hbar m$ and $\Delta J_z = 0$, whereas

$$\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0 \quad \text{and} \quad \Delta J_x = \Delta J_y = \frac{\hbar}{\sqrt{2}} [j(j+1) - m^2]^{\frac{1}{2}} \quad (5.1.13)$$

E5.1.4 Discuss Eq. (5.1.2) from the uncertainty principle aspect. For illustrations, consider the spin- $\frac{1}{2}$ particle case, in the single versus sequential Stern–Gerlach measurements setup; see for example, the Figures 1.1–1.3 of Sakurai's book [2nd ed. (2017)].

5.2 Descriptions of Rotations and Spins

5.2.1 Rotation and spin operators

A. Displacement Operators

We will show below that the x -displacement operator is given by

$$\hat{D}(a) \equiv e^{-i\hat{p}a/\hbar} \quad (5.2.1)$$

where a is a real length parameter. While it is unitary, the real-space displacement operator takes the action of

$$\hat{D}(a)|x\rangle = |x+a\rangle \quad (5.2.2)$$

Its “Heisenberg” equivalence reads

$$\hat{D}^\dagger(a)F(\hat{x})\hat{D}(a) = F(\hat{x}+a) \quad (5.2.3)$$

(a) In fact, by using $[\hat{x}, \hat{p}] = i\hbar$, which results in $[\hat{x}, f(\hat{p})] = i\hbar f'(\hat{p})$, we have

$$[\hat{x}, \hat{D}(a)] = a\hat{D}(a) \Rightarrow \hat{x}\hat{D}(a) = \hat{D}(a)(\hat{x}+a) \Rightarrow \hat{D}^\dagger(a)\hat{x}\hat{D}(a) = \hat{x}+a$$

Hence, $\hat{D}^\dagger(a)\hat{x}\hat{D}(a) = \hat{x}+a$, since $\hat{D}(a)$ is unitary, leading further to Eq. (5.2.3).

(b) Consider then the identity, $\hat{x}|x\rangle = x|x\rangle$. We have

$$\hat{x}[\hat{D}(a)|x\rangle] = \hat{D}(a)(\hat{x}+a)|x\rangle = \hat{D}(a)(x+a)|x\rangle = (x+a)[\hat{D}(a)|x\rangle]$$

Comparing it with the identity, $\hat{x}|x+a\rangle = (x+a)|x+a\rangle$, we obtain Eq. (5.2.2).

(c) Turn to wave function, $\psi(x) \equiv \langle x|\psi\rangle$. Note that

$$\langle x|\hat{D}(a) = [\hat{D}^\dagger(a)|x\rangle]^\dagger = \langle x-a| \Rightarrow \langle x|\hat{D}(a)|\psi\rangle = \psi(x-a) \quad (5.2.4)$$

In other words, the displacement operator, $\hat{D}(a)$ of Eq. (5.2.1), moves the wave function in x -representation from $\psi(x)$ to $\psi(x-a)$. In geometry, it shifts the coordinate reference origin from $x=0$ to $x=a$.

(d) The generalization of Eqs. (5.2.1)–(5.2.4) to three-dimension real space are

$$\begin{aligned} \hat{D}(\mathbf{a}) &\equiv e^{-i\hat{\mathbf{p}}\cdot\mathbf{a}/\hbar}, & \hat{D}(\mathbf{a})|\mathbf{r}\rangle &= |\mathbf{r}+\mathbf{a}\rangle \\ \hat{D}^\dagger(\mathbf{a})F(\hat{\mathbf{r}})\hat{D}(\mathbf{a}) &= F(\hat{\mathbf{r}}+\mathbf{a}), & \langle\mathbf{r}|\hat{D}(\mathbf{a})|\psi\rangle &= \psi(\mathbf{r}-\mathbf{a}) \end{aligned} \quad (5.2.5)$$

(e) It is worth noting that the Hilbert-space propagator, $e^{-i\hat{H}t/\hbar}$. This is the type of time-displacement operator, governing [cf. Eqs. (5.2.2) and (5.2.3)]

$$e^{-i\hat{H}t/\hbar}|\psi(0)\rangle = |\psi(t)\rangle \quad \text{and} \quad e^{i\hat{H}t/\hbar}\hat{A}(0)e^{-i\hat{H}t/\hbar} = \hat{A}(t) \quad (5.2.6)$$

with $\hat{A}(0) \equiv \hat{A}$.

B. Rotations and Spins

Consider a geometric rotation, $\mathcal{R}_{\mathbf{n}}(\Omega)$, around the unit direction $\mathbf{n}$ by angle Ω . The rotation $\mathcal{R}_{\mathbf{n}}(\Omega)$, which brings $\mathbf{r} = (x, y, z)$ to $\mathbf{r}' = (x', y', z')$, in a fixed coordinate-frame space, is given by

$$\mathcal{R}_{\mathbf{n}}(\Omega) \cdot \mathbf{r} = \mathbf{r}' \quad \Rightarrow \quad \mathcal{R}_{\mathbf{n}}(\Omega)|\mathbf{r}\rangle = |\mathbf{r}'\rangle \quad (5.2.7)$$

While the geometric rotation matrix $\mathcal{R}_{\mathbf{n}}(\Omega)$ is orthogonal, its Hilbert-space correspondence is unitary and reads

$$\boxed{\mathcal{R}_{\mathbf{n}}(\Omega) = e^{-i\Omega\hat{L}_{\mathbf{n}}/\hbar}} \quad (\hat{L}_{\mathbf{n}} \equiv \hat{\mathbf{L}} \cdot \mathbf{n}) \quad (5.2.8)$$

As Eq. (5.2.7) is concerned, Eq. (5.2.8) refers to the rotation *displacement* operator, in parallel to Eq. (5.2.1) versus Eq. (5.2.2). Naturally and also truly, the spin counterpart to Eq. (5.2.8) is

$$\boxed{\mathcal{S}_{\mathbf{n}}(\Omega) = e^{-i\Omega\hat{S}_{\mathbf{n}}/\hbar}} \quad (\hat{S}_{\mathbf{n}} \equiv \hat{\mathbf{S}} \cdot \mathbf{n}) \quad (5.2.9)$$

Observations:

- (a) The rotation operator $\mathcal{R}_{\mathbf{n}}(\Omega)$, Eq. (5.2.8), is generated by orbital angular momentum, $\hat{\mathbf{L}}$. The spin one, $\mathcal{S}_{\mathbf{n}}(\Omega)$, Eq. (5.2.9), goes by $\hat{\mathbf{S}}$.
- (b) $\mathcal{R}(2\pi) = 1$ (unit operator), as the $\Omega = 2\pi$ rotations around *any* direction vector $\mathbf{n}$ restores all effects. The physical origin is, however, due to the quantum numbers of orbital angular momentum are *integer*. This gives rise also $\mathcal{S}_{\text{boson}}(\Omega = 2\pi) = 1$.
- (c) $\mathcal{S}_{\text{fermion}}(\Omega = 2\pi) = -1$, which is also independent of the direction $\mathbf{n}$.

5.2.2 Spinors and Pauli matrixes

A. Spin- $\frac{1}{2}$ system and Pauli matrixes

Let us focus on the spin- $\frac{1}{2}$ system and recast Eq. (5.1.10) in terms of

$$\hat{S}_j = \frac{\hbar}{2} \hat{\sigma}_j; \quad j = x, y, z \quad (5.2.10)$$

where

$$\hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \hat{\sigma}_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{\sigma}_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (5.2.11)$$

These are so-called the *Pauli matrixes*.

• Properties of Pauli matrixes (*Self-exercises*)

$$\begin{aligned} [\hat{\sigma}_x, \hat{\sigma}_y] &= 2i\hat{\sigma}_z \\ \hat{\sigma}_x \hat{\sigma}_y &= i\hat{\sigma}_z \end{aligned} \quad (\text{also cyclic ones}) \quad (5.2.12)$$

$$\{\hat{\sigma}_j, \hat{\sigma}_k\} = 2\delta_{jk}, \quad \text{thus} \quad \hat{\sigma}_j^2 = \hat{\sigma}_j^\dagger \hat{\sigma}_j = 1 \quad (5.2.13)$$

Each Pauli's matrix is *both Hermitian and unitary*. Moreover,

$$\text{Tr} \hat{\sigma}_j = 0, \quad \det \hat{\sigma}_j = -1 \quad (5.2.14)$$

B. Spinors, $\hat{\sigma}_{\mathbf{n}} \equiv \hat{\boldsymbol{\sigma}} \cdot \mathbf{n}$, the generic Pauli's matrix.

For spin- $\frac{1}{2}$ system, Eq. (5.2.9) with Eq. (5.2.10), reads

$$\mathcal{S}_{\mathbf{n}}(\Omega) = e^{-i\Omega \hat{S}_{\mathbf{n}}/\hbar} = e^{-i\Omega \hat{\sigma}_{\mathbf{n}}/2} \quad (\text{spin-}\frac{1}{2}) \quad (5.2.15)$$

By using the properties of Pauli matrixes, we obtain

$$\mathcal{S}_{\mathbf{n}}(\Omega) = \cos(\Omega/2) - i\hat{\sigma}_{\mathbf{n}} \sin(\Omega/2) \quad (5.2.16)$$

Derivations: As known, a function of operator, $f(\hat{A})$, can be expanded in Taylor's series. Involved in Eq. (5.2.15) are the Pauli matrixes. By using Eq. (5.2.13), we have

$$\hat{\sigma}_{\mathbf{n}}^2 = \sum_{j \in x, y, z} n_j^2 \hat{\sigma}_j^2 + \frac{1}{2} \sum_{j \neq k} n_j n_k \{\hat{\sigma}_j, \hat{\sigma}_k\} = n_x^2 + n_y^2 + n_z^2 = 1 \quad (5.2.17)$$

Hence, $\hat{\sigma}_{\mathbf{n}}^{2m} = 1$ and $\hat{\sigma}_{\mathbf{n}}^{2m+1} = \hat{\sigma}_{\mathbf{n}}$, where $m = 0, 1, 2, \dots$. These result in

$$e^{i\alpha \hat{\sigma}_{\mathbf{n}}} = \cos(\alpha \hat{\sigma}_{\mathbf{n}}) + i \sin(\alpha \hat{\sigma}_{\mathbf{n}}) = \cos \alpha + i \hat{\sigma}_{\mathbf{n}} \sin \alpha \quad (5.2.18)$$

giving rise to Eq. (5.2.16) from Eq. (5.2.15). (*Q.E.D.*)

- The *spinor* $\hat{\sigma}_{\mathbf{n}}$ in Eq. (5.2.16) reads

$$\hat{\sigma}_{\mathbf{n}} \equiv \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} = n_x \hat{\sigma}_x + n_y \hat{\sigma}_y + n_z \hat{\sigma}_z = \begin{bmatrix} n_z & n_x - in_y \\ n_x + in_y & -n_z \end{bmatrix} \quad (5.2.19)$$

As the vector of Pauli's matrixes [Eq. (5.2.11)] project onto the specified direction $\mathbf{n}$, spinors satisfy such as $\text{tr} \hat{\sigma}_{\mathbf{n}} = 0$ and $\det \hat{\sigma}_{\mathbf{n}} = -1$ [Eq. (5.2.14)]. Moreover [Eq. (5.2.13)],

$$\boxed{\hat{\sigma}_{\mathbf{n}} = \hat{\sigma}_{\mathbf{n}}^\dagger = \hat{\sigma}_{\mathbf{n}}^{-1}} \quad (\text{Hermitian and unitary}) \quad (5.2.20)$$

- In the spherical angles (the azimuth θ and the polar angle ϕ),

$$\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \quad (5.2.21)$$

Eq. (5.2.19) reads

$$\boxed{\hat{\sigma}_{\mathbf{n}} = \begin{bmatrix} \cos \theta & e^{-i\phi} \sin \theta \\ e^{i\phi} \sin \theta & -\cos \theta \end{bmatrix} \equiv \hat{\sigma}(\theta, \phi)} \quad (5.2.22)$$

The spinor diagonalization, $\hat{\sigma}_{\mathbf{n}} = \hat{U}^\dagger \hat{\sigma}_z \hat{U}$, can be expressed in the form of

$$\boxed{\hat{\sigma}(\theta, \phi) = \hat{\sigma}(\theta/2, \phi) \hat{\sigma}(\theta = 0) \hat{\sigma}(\theta/2, \phi)} \quad (5.2.23)$$

Here, $\hat{\sigma}(\theta = 0) = \text{diag}(1, -1) = \hat{\sigma}_z$. The other two Pauli's matrixes in Eq. (5.2.11) are $\hat{\sigma}_x = \hat{\sigma}(\theta = \pi/2, \phi = 0)$ and $\hat{\sigma}_y = \hat{\sigma}(\theta = \pi/2, \phi = \pi/2)$.

C. The spinor formalism of spin rotation operator

By using Eqs. (5.2.21)–(5.2.23), we obtain Eq. (5.2.15) the form of

$$\boxed{\mathcal{S}_{\mathbf{n}}(\Omega) = \hat{\sigma}(\theta/2, \phi) \mathcal{S}_z(\Omega) \hat{\sigma}(\theta/2, \phi)} \quad (\text{spin-}\frac{1}{2}) \quad (5.2.24)$$

with $\mathcal{S}_z(\Omega)$ being the spin rotation around the z -axis. That is

$$\mathcal{S}_z(\Omega) = \begin{bmatrix} e^{-i\Omega/2} & 0 \\ 0 & e^{i\Omega/2} \end{bmatrix} \quad (5.2.25)$$

Exercises:

E5.2.1 Prove Eq. (5.2.12) and write down also the cyclic ones.

E5.2.2 Prove Eqs. (5.2.13) and (5.2.14). Verify then Eq. (5.2.20).

E5.2.3 (a) Does $\hat{\sigma}_{\mathbf{n}}$ of Eq. (5.2.22) agree with Eq. (5.2.20)? (b) Verify the result of $\hat{\sigma}_{\mathbf{n}}$ in the form of Eq. (5.2.23); (c) How is this derived? Present the derivations in details.

E5.2.4 Derive Eq. (5.2.24) [Hint: For a diagonal matrix $\hat{A}$, express the matrix of $f(\hat{A})$].

E5.2.5 Discussions on Eq. (5.2.24): (a) Unitary? (b) Any geometric rotation analogue?

E5.2.6 Consider $\cos \theta$ and $e^{\pm i\phi} \sin \theta$ in Eq. (5.2.22). Are they c-numbers or operators? [See Eq. (5.3.31) for the answer].

5.2.3 Total angular momentum conservation

It is known that in classical mechanics, the total angular momentum of isolated systems is conserved, in the absence of external fields. Rotations alone do not change the laws of physics. Measurements on any scalar (nondirectional) quantities are independent of the change of reference rotation framework. The potential energy, $V(\mathbf{r})$, *regardless it is spherically symmetric or not*, is a such invariant property, with respect to rotations.

The above observation, as inferred from Eqs. (5.2.8) and (5.2.9) in quantum mechanics, reads

$$\mathcal{D}_{\mathbf{n}}^\dagger(\Omega)\hat{V}(\mathbf{r})\mathcal{D}_{\mathbf{n}}(\Omega) = \hat{V}(\mathbf{r}) \quad \text{where} \quad \mathcal{D}_{\mathbf{n}}(\Omega) = e^{-i\Omega\hat{\mathbf{J}}\cdot\mathbf{n}/\hbar} \quad (5.2.26)$$

with $\hat{\mathbf{J}}$ being the *total orbital-plus-spin angular momentum vector operator*. The above equation is equivalent to

$$[\hat{V}(\mathbf{r}), \hat{\mathbf{J}}] = 0 \quad (5.2.27)$$

This remains valid if $\hat{V}(\mathbf{r})$ is replaced by any scalar operators. In particular,

$$[\hat{H}, \hat{\mathbf{J}}] = 0 \quad \Rightarrow \quad \hat{\mathbf{J}}(t) = e^{i\hat{H}t/\hbar} \hat{\mathbf{J}} e^{-i\hat{H}t/\hbar} = \hat{\mathbf{J}}(0) \quad (5.2.28)$$

This is the law of total angular momentum conservation.

Remarks: In presence of *spin-orbit coupling* (SOC), neither the orbital $\hat{\mathbf{L}}$ nor the spin $\hat{\mathbf{S}}$ alone is a **conserved** property. The total composite angular momentum, $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$, **is always conserved, if there are no time-dependent external fields**. Hence, the *least* set of commutable observables comprises $(\hat{H}, \hat{J}^2, \hat{J}_z)$. In the absence of SOC, the commutable set includes further those from individual $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$.

5.3 Orbital Angular Momentum Wave Functions

5.3.1 Spherical harmonics

The orbital angular momentum wave functions, $Y_{l,m}(\theta, \phi) \equiv \langle \theta, \phi | l, m \rangle$, are the *spherical harmonics*, with the parities arising from the spherical symmetry; see Eq. (5.3.5). The final normalized results reads

$$Y_{l,m}(\theta, \phi) = \frac{1}{\sqrt{2\pi}} \Theta_{l,m}(\theta) e^{im\phi}; \quad m = 0, \pm 1, \dots, \pm l \quad (5.3.1)$$

with

$$\Theta_{l,m}(\theta) = \frac{(-1)^l}{2^l l!} \left[\frac{2l+1}{2} \frac{(l+m)!}{(l-m)!} \right]^{\frac{1}{2}} \frac{1}{(\sin \theta)^m} \frac{d^{l-m}(\sin \theta)^{2l}}{d(\cos \theta)^{l-m}} \quad (5.3.2)$$

While the properties of spherical harmonics are detailed from Part A to Part C, the derivations of Eq. (5.3.2) will be given in Part D, as follows.

A. Basic Properties

• Orthonormality and completeness

$$\langle l', m' | l, m \rangle = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta Y_{l',m'}^*(\theta, \phi) Y_{l,m}(\theta, \phi) = \delta_{ll'} \delta_{mm'} \quad (5.3.3a)$$

$$\sum_{l=0}^{\infty} \sum_{m=-l}^l Y_{l,m}^*(\theta, \phi) Y_{l,m}(\theta', \phi') = \delta(\cos \theta - \cos \theta') \delta(\phi - \phi') \quad (5.3.3b)$$

Note that $\delta(\cos \theta - \cos \theta') = \delta(\theta - \theta') / \sin \theta$. The above second expression arises from applying the completeness to $\langle \mathbf{r} | \mathbf{r}' \rangle = \delta(\mathbf{r} - \mathbf{r}')$.

• The sum theorem of spherical harmonics

Involved in Eq. (5.3.3b) are two direction vectors, $\mathbf{r}(\theta, \phi)$ and $\mathbf{r}'(\theta', \phi')$. Let α be the angle between them. The inside sum over m is actually related to the rotation of one vector around the other. In terms of spherical harmonics, this reads

$$\sum_{m=-l}^l Y_{l,m}^*(\theta, \phi) Y_{l,m}(\theta', \phi') = \frac{2l+1}{4\pi} P_l(\cos \alpha) \propto \Theta_{l,0}(\alpha) \quad (5.3.4)$$

Each Legendre polynomial $P_l(\cos \alpha) \propto \Theta_{l,0}(\alpha)$, differing only at the normalizations.

• **Parities:** Let $\mathbf{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ a unit sphere vector. The spherical symmetry is concerned with $\mathbf{r} \rightarrow \mathbf{r}' = -\mathbf{r}$, by which $\theta' = \pi - \theta$ and $\phi' = \pi + \phi$, so that $\cos \theta' = -\cos \theta$ and $\sin \theta' = \sin \theta$, while $e^{im\phi'} = (-1)^m e^{im\phi}$. The resulting spherical harmonics are of

$$\begin{aligned} Y_{l,-m}^*(\theta, \phi) &= (-1)^m Y_{l,m}(\theta, \phi) \\ Y_{l,m}(\pi - \theta, \pi + \phi) &= (-1)^l Y_{l,m}(\theta, \phi) \end{aligned} \quad (5.3.5)$$

The resulting $\Theta_{l,m}(\theta)$ -components acquire

$$\Theta_{l,-m}(\theta) = (-1)^m \Theta_{l,m}(\theta) = (-1)^l \Theta_{l,m}(\pi - \theta) \quad (5.3.6)$$

See also Eq. (5.3.14) and the remarks therein.

B. Conventions

- The overall phase of $\{Y_{l,m}(\theta, \phi); m = 0, \pm 1, \dots, \pm l\}$ is set to have

$$Y_{l,0}(\theta = 0, \phi) > 0 \quad [\text{Convention, same by } \Theta_{l,0}(0) > 0] \quad (5.3.7)$$

- For atomic orbitals, $l = 0, 1, 2, \dots$, are also labeled by s, p, d, $\dots$, and so on.

Exercises. The first few spherical harmonics via Eqs. (5.3.1) and (5.3.2) read

$Y_{0,0} = \frac{1}{\sqrt{4\pi}}$	$Y_{1,0}(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos \theta$ $Y_{1,\pm 1}(\theta, \phi) = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}$	$Y_{2,0}(\theta, \phi) = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$ $Y_{2,\pm 1}(\theta, \phi) = \mp \sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{\pm i\phi}$ $Y_{2,\pm 2}(\theta, \phi) = \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm i2\phi}$
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E5.3.1 (a) Verify the parities, Eq. (5.3.6); (b) Rationalize the convention, Eq. (5.3.7).

E5.3.2 The real-space orbitals: (i) $s = Y_{0,0}$; (ii) $p_z = Y_{1,0}$, $p_x = Y_{1,-1} - Y_{1,1}$, $p_y = i(Y_{1,-1} + Y_{1,1})$; (iii) $d_{z^2} = Y_{2,0}$, $d_{zx} = Y_{2,-1} - Y_{2,1}$, $d_{yz} = i(Y_{2,-1} + Y_{2,1})$, $d_{x^2-y^2} = Y_{2,-2} + Y_{2,2}$, $d_{xy} = i(Y_{2,-2} - Y_{2,2})$. Discuss the results of angular momentum measurements on d_{xy} .

C. The Use of Angular Momentum Theorem, Eqs. (5.1.4)–(5.1.9)

- Let us start with [cf. Eqs. (4.3.11)–(4.3.13)]

$$\hat{L}^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} \right) - \frac{\hbar^2}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \quad \text{and} \quad \hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \phi} \quad (5.3.8)$$

The ladder operators, Eq. (5.1.5), are given by [Eqs. (4.3.11) and (4.3.12)]

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y = \hbar e^{\pm i\phi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) \quad (5.3.9)$$

The corresponding Eq. (5.1.9) reads

$$\hat{L}_{\pm} Y_{l,m}(\theta, \phi) = \hbar c_{l,m}^{\pm} Y_{l,m\pm 1}(\theta, \phi) \quad (5.3.10)$$

where

$$c_{l,m}^{\pm} \equiv [l(l+1) - m(m\pm 1)]^{\frac{1}{2}} = [(l \mp m)(l \pm m + 1)]^{\frac{1}{2}} \quad (5.3.11)$$

The terminal values are evaluated via $\hat{L}_+ Y_{l,l} = \hat{L}_- Y_{l,-l} = 0$. Moreover,

$$Y_{l,m}(\theta, \phi) = \left[\frac{(l \mp m)!}{(2l)!(l \pm m)!} \right]^{\frac{1}{2}} \left(\frac{\hat{L}_{\pm}}{\hbar} \right)^{l \pm m} Y_{l,\mp l}(\theta, \phi) \quad (5.3.12)$$

The coefficient arises from $(c_{l,m}^{\pm} c_{l,m \mp 1}^{\pm} \cdots c_{l,\mp l}^{\pm})^{-1}$ of Eq. (5.3.11).

- The first expression of Eq. (5.1.4) reads

$$\hat{L}^2 [\Theta_{l,m}(\theta) e^{im\phi}] = \hbar^2 l(l+1) [\Theta_{l,m}(\theta) e^{im\phi}] \quad (5.3.13)$$

in line with $Y_{l,m}(\theta, \phi) \propto \Theta_{l,m}(\theta) e^{im\phi}$ [Eq. (5.3.1)]. We obtain [cf. Eq. (5.3.8)]

$$\left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + l(l+1) - \frac{m^2}{\sin^2 \theta} \right] \Theta_{l,m}(\theta) = 0 \quad (5.3.14)$$

As the operator (the square-bracketed term) is symmetric, with respect to

$$m \rightarrow -m \quad \text{and} \quad \theta \rightarrow \pi - \theta$$

the wave functions acquire the parities of Eq. (5.3.6).

D. Derivations of Eq. (5.3.2), $\Theta_{l,m}(\theta)$, by inductions

- To show that Eq. (5.3.2) satisfies Eq. (5.3.14), we revisit Eq. (5.3.10) the equivalent form of

$$e^{\pm i\phi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right) [\Theta_{l,m}(\theta) e^{im\phi}] = c_{l,m}^{\pm} [\Theta_{l,m \pm 1}(\theta) e^{i(m \pm 1)\phi}]$$

The resulting $\Theta_{l,m}$ -counterpart of Eq. (5.3.10) reads

$$\hat{\Gamma}_m^{\pm} \Theta_{l,m}(\theta) = c_{l,m}^{\pm} \Theta_{l,m \pm 1}(\theta) \quad (5.3.15)$$

with the m -associated ladder operators,

$$\hat{\Gamma}_m^{\pm} \equiv \pm \frac{d}{d\theta} - m \cot \theta, \quad \text{satisfying} \quad \hat{\Gamma}_m^{-} = -\hat{\Gamma}_{-m}^{+} \quad (5.3.16)$$

(a) The terminal equations, $\hat{\Gamma}_{\pm l}^{\pm} \Theta_{l,\pm l}(\theta) = 0$, read then

$$\frac{d\Theta_{l,\pm l}(\theta)}{d\theta} = l \cot \theta \Theta_{l,\pm l}(\theta) \quad \Rightarrow \quad \Theta_{l,\pm l}(\theta) \propto (\sin \theta)^l \quad (5.3.17)$$

Note that the overall phase of normalization constant should be determined in line with $\Theta_{l,0}(\theta = 0) > 0$ [Eq. (5.3.7)]. We obtain

$$\Theta_{l,l}(\theta) = \frac{(-1)^l}{2^l l!} \sqrt{\frac{(2l+1)!}{2}} (\sin \theta)^l \quad \text{and} \quad \Theta_{l,-l}(\theta) = (-1)^l \Theta_{l,l}(\theta) \quad (5.3.18)$$

These are the terminal values of $\Theta_{l,m}(\theta)$, Eq. (5.3.2) at $m = \pm l$.

(b) The second step of *inductions* to $\Theta_{l,m}(\theta)$ starts with assuming Eq. (5.3.2) be true, at any given $m \in (m_{\min}, l)$, and then evaluates $\Theta_{l,m-1}(\theta)$, via the ladder-down version of Eq. (5.3.15), with Eq. (5.3.16); i.e.,

$$c_{l,m}^- \Theta_{l,m-1}(\theta) = \hat{\Gamma}_m^- \Theta_{l,m}(\theta) = -\left(\frac{d}{d\theta} + m \cot \theta\right) \Theta_{l,m}(\theta) \quad (5.3.19)$$

Exploited will be the identity (*Self-exercise*),

$$\left(\frac{d}{d\theta} + m \cot \theta\right) \left[\frac{1}{(\sin \theta)^m} \frac{d^{l-m}(\sin \theta)^{2l}}{d(\cos \theta)^{l-m}} \right] = -\frac{1}{(\sin \theta)^{m-1}} \frac{d^{l-m+1}(\sin \theta)^{2l}}{d(\cos \theta)^{l-m+1}} \quad (5.3.20)$$

(c) To complete the derivation of Eq. (5.3.2), we recast it as

$$\Theta_{l,m}(\theta) = N_{l,m} \frac{(-1)^l}{2^l l!} \left[\frac{1}{(\sin \theta)^m} \frac{d^{l-m}(\sin \theta)^{2l}}{d(\cos \theta)^{l-m}} \right] \quad (5.3.21a)$$

with

$$N_{l,m} \equiv \left[\frac{2l+1}{2} \frac{(l+m)!}{(l-m)!} \right]^{\frac{1}{2}} \quad (5.3.21b)$$

Moreover [Eq. (5.3.11)],

$$\frac{N_{l,m}}{c_{l,m}^-} = \frac{N_{l,m}}{[(l+m)(l-m+1)]^{\frac{1}{2}}} = \left[\frac{2l+1}{2} \frac{(l+m-1)!}{(l-m+1)!} \right]^{\frac{1}{2}} = N_{l,m-1} \quad (5.3.21c)$$

As results, Eq. (5.3.19) amounts to have

$$\Theta_{l,m-1}(\theta) = N_{l,m-1} \frac{(-1)^l}{2^l l!} \left[\frac{1}{(\sin \theta)^{m-1}} \frac{d^{l-m+1}(\sin \theta)^{2l}}{d(\cos \theta)^{l-m+1}} \right] \quad (5.3.21d)$$

the form of Eq. (5.3.21a); i.e. $\Theta_{l,m}(\theta)$ of Eq. (5.3.2).

(*Q.E.D.*)

5.3.2 Relation to Legendre functions

A. The setup: Changing variable

Let us revisit Eq. (5.3.14) as follows.

$$\left[\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + l(l+1) - \frac{m^2}{\sin^2 \theta} \right] \Theta_{l,m}(\theta) = 0 \quad [\text{Eq. (5.3.14)}] \quad (5.3.22)$$

Set then $x \equiv \cos \theta \in [-1, 1]$. It follows

$$\begin{aligned} \frac{d}{d\theta} &= \frac{dx}{d\theta} \frac{d}{dx} = -\sin \theta \frac{d}{dx} \quad (x = \cos \theta) \\ \frac{d^2}{d\theta^2} &= (1-x^2) \frac{d^2}{dx^2} - x \frac{d}{dx} \quad \text{and} \quad \cot \theta \frac{\partial}{\partial \theta} = -x \frac{\partial}{\partial x} \end{aligned}$$

The transformed Eq. (5.3.22) reads

$$(1-x^2)\Theta_{l,m}''(x) - 2x\Theta_{l,m}'(x) + \left[l(l+1) - \frac{m^2}{1-x^2} \right] \Theta_{l,m}(x) = 0 \quad (5.3.23)$$

This is a standard differential equation,¹ with solutions being the associated Legendre functions; see Eqs. (5.3.28) and (5.3.29).

B. The $\Theta_{l,0}(\theta)$ function verses Legendre polynomial $P_l(x)$:

$$\Theta_{l,0}(\theta) = \sqrt{\frac{2l+1}{2}} P_l(\cos \theta) \quad (5.3.24)$$

The solutions to Eq. (5.3.23), with $m = 0$, are the Legendre polynomials, satisfying

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn} \quad (5.3.25)$$

This goes by the setting of $P_n(x=1) \equiv 1$, resulting in

$$\Theta_{l,0}(\theta=0) = \sqrt{(2l+1)/2}$$

Basic properties:

- (i) $(1-x^2)P_n''(x) - 2xP_n'(x) + n(n+1)P_n(x) = 0$ (Differential equation)
- (ii) $(1-x^2)P_n'(x) = -nP_n(x) + nP_{n-1}(x)$ (Derivative relation)
- (iii) $P_n(-x) = (-1)^n P_n(x)$ (Symmetry)
- (iv) $P_0(x) = 1$ and $P_1(x) = x$ (Initial values) (5.3.26)
- (v) $(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$ (Recursion)
- (vi) $\frac{1}{\sqrt{1-2xz+z^2}} = \sum_{n=0}^{\infty} P_n(x)z^n; \quad |z| < 1$ (Generating function)

¹Handbook of Mathematical Functions, Milton Abramowitz and Irene A. Stegun (ed.); Chapter 8

The direct relevance to Eq. (5.3.2) with $m = 0$ is

$$P_n(x) = \frac{(-1)^n}{2^n n!} \frac{d^n}{dx^n} (1-x^2)^n \quad (\text{Rodrigues' formula}) \quad (5.3.27)$$

C. The $\Theta_{l,m}(\theta)$ function versus the associated Legendre function:

$$P_n^m(x) \equiv (1-x^2)^{\frac{m}{2}} \frac{d^m P_n(x)}{dx^m}; \quad m = 0, 1, \dots, n \quad (5.3.28)$$

with the notion here of $P_n^m(x) \neq [P_n(x)]^m$, as the associated Legendre functions are concerned with. Applying Eq. (5.3.27) results in

$$P_n^m(x) = \frac{(-1)^n}{2^n n!} (1-x^2)^{\frac{m}{2}} \frac{d^{n+m}}{dx^{n+m}} (1-x^2)^{m+n} \quad (5.3.28')$$

Together with Eqs. (5.3.24), we obtain Eq. (5.3.2) the alternative expression,

$$\Theta_{l,m}(\theta) = (-1)^m \left[\frac{2l+1}{2} \frac{(l-m)!}{(l+m)!} \right]^{\frac{1}{2}} P_l^m(\cos \theta) \quad (5.3.29)$$

where $m = 0, 1, \dots, l$. We have also [cf. Eq. (5.3.6)]

$$\Theta_{l,-m}(\theta) = (-1)^m \Theta_{l,m}(\theta)$$

Exercises

E5.3.3 Verify (a) Eq. (5.3.12); (b) Eq. (5.3.18); (c) Eq. (5.3.20); (d) Eq. (5.3.29).

E5.3.4 Obtain the parity of $P_n^m(x)$, where $m = 0, \pm 1, \dots, \pm n$.

E5.3.5 By using Eqs. (5.3.28)–(5.3.30) and the last two expressions in Eq. (5.3.26), construct d -orbitals [The answers were given after Eq. (5.3.7)].

E5.3.6 (a) Show the ladder operators in Eq. (5.3.9) satisfying

$$[\hat{L}_{\pm}, \cos \theta] = \hbar e^{\pm i\phi} \sin \theta \quad \text{and} \quad [\hat{L}_{\pm}, e^{\pm i\phi} \sin \theta] = 0 \quad (5.3.30)$$

(b) What would be the implication to the spinor of Eq. (5.2.22)?

E5.3.7 As mentioned earlier, Eq. (5.3.5) describes the parity of $Y_{l,m}(\theta, \phi)$, with respect to the spherical symmetry of $\mathbf{r} \rightarrow \mathbf{r}' = -\mathbf{r}$, by which

$$\theta' = \pi - \theta \quad \text{and} \quad \phi' = \pi + \phi$$

so that $\cos \theta' = -\cos \theta$ and $\sin \theta' = \sin \theta$, while $e^{im\phi'} = (-1)^m e^{im\phi}$. Consider

$$\begin{aligned} \alpha_{l,m} &\equiv \langle l-1, m-1 | e^{-i\phi} \sin \theta | l, m \rangle \\ \beta_{l,m} &\equiv \langle l+1, m-1 | e^{-i\phi} \sin \theta | l, m \rangle \end{aligned} \quad (5.3.31a)$$

where $|l, m\rangle \doteq Y_{l,m}(\theta, \phi)$. Show that

$$\begin{aligned} \langle l-1, m+1 | e^{i\phi} \sin \theta | l, m \rangle &= -\alpha_{l,-m} \\ \langle l+1, m+1 | e^{i\phi} \sin \theta | l, m \rangle &= -\beta_{l,-m} \end{aligned} \quad (5.3.31b)$$

5.3.3 Selection rules on optical transitions

A. Other important properties of spherical harmonics

- Spherical harmonics, $Y_{l,m}(\theta, \phi) = |l, m\rangle$, satisfy

$$\cos \theta |l, m\rangle = \gamma_{l,m} |l-1, m\rangle + \gamma_{l+1,m} |l+1, m\rangle \quad (5.3.32a)$$

and [cf. Eq. (5.3.31)]

$$\begin{aligned} e^{-i\phi} \sin \theta |l, m\rangle &= \alpha_{l,m} |l-1, m-1\rangle + \beta_{l,m} |l+1, m-1\rangle \\ e^{i\phi} \sin \theta |l, m\rangle &= -\alpha_{l,-m} |l-1, m+1\rangle - \beta_{l,-m} |l+1, m+1\rangle \end{aligned} \quad (5.3.32b)$$

- The most important implications above are the *selection rules*,

$$\boxed{\langle l', m' | \hat{\mathbf{r}} | l, m \rangle \neq 0} \quad \text{only if} \quad \boxed{\Delta l = \pm 1; \Delta m = 0, \pm 1} \quad (5.3.33)$$

Here, $\hat{\mathbf{r}} = (x, y, z)$, with $z/r = \cos \theta$ and $(x \pm iy)/r = e^{\pm i\phi} \sin \theta$. Electric dipole operator has the form of $\hat{\boldsymbol{\mu}} \propto \hat{\mathbf{r}}$, with $\hat{\mathbf{r}}$ pointing from the negative charge center to the positive center. As results, Eq. (5.3.33) equally describes when the *transition dipoles* are nonzero. Optical transitions occur only with specified values of $\Delta l = l' - l$ and $\Delta m = m' - m$.

(a) The transitions between atomic orbitals, $s \leftrightarrow p$ and $p \leftrightarrow d$ are allowed, but $s \leftrightarrow d$ is forbidden, according to the selection rule of $\Delta l = \pm 1$.

(b) Moreover, diatomic molecular rotation wave functions are also described by spherical harmonics. The related spectrums will be detailed in §5.3.4.

- Final expressions in Eq. (5.3.32) include

$$\gamma_{l,m} = \left(\frac{l^2 - m^2}{4l^2 - 1} \right)^{\frac{1}{2}} \quad (5.3.34a)$$

and

$$\begin{aligned} \alpha_{l,m} &= - \left[\frac{(l+m)(l+m-1)}{(2l-1)(2l+1)} \right]^{\frac{1}{2}} \\ \beta_{l,m} &= \left[\frac{(l-m+2)(l-m+1)}{(2l+1)(2l+3)} \right]^{\frac{1}{2}} \end{aligned} \quad (5.3.34b)$$

B. Derivations of $\cos \theta |l, m\rangle$, Eq. (5.3.32a) with Eq. (5.3.34a)

- For spherical harmonics, $\cos \theta |l, m\rangle$ does not change the quantum number m and must take the form of

$$\cos \theta \Theta_{l,m}(\theta) = \gamma_{l,m} \Theta_{l-1,m}(\theta) + \gamma'_{l,m} \Theta_{l+1,m}(\theta) \quad (5.3.35)$$

(a) As inferred from Eq. (5.3.6), the undetermined parameters satisfy

$$\gamma_{l,-m} = \gamma_{l,m} \quad \text{and} \quad \gamma'_{l,-m} = \gamma'_{l,m} \quad (5.3.36)$$

(b) The boundary values, in line with Eq. (5.3.35), are

$$\gamma_{l,l} = 0 \quad \text{and} \quad \gamma'_{l,l+1} = 0 \quad (5.3.37)$$

(c) Consider now $\Theta_{l,0}(\theta) \propto P_l(\cos \theta)$, Eq. (5.3.24), followed by using the recursion relation of Legendre polynomials, the last line in Eq. (5.3.26). We have

$$\cos \theta \Theta_{l,0}(\theta) = \frac{l}{2l+1} \sqrt{\frac{2l+1}{2l-1}} \Theta_{l-1,0}(\theta) + \frac{l+1}{2l+1} \sqrt{\frac{2l+1}{2l+3}} \Theta_{l+1,0}(\theta) \quad (5.3.38)$$

This gives rise to

$$\gamma_{l,0} = \frac{l}{\sqrt{4l^2-1}} \quad \text{and} \quad \gamma'_{l,0} = \frac{l+1}{\sqrt{4(l+1)^2-1}} = \gamma_{l+1,0} \quad (5.3.39)$$

Together with Eqs. (5.3.36) and (5.3.37), we obtain

$$\gamma_{l,m} = \left(\frac{l^2 - m^2}{4l^2 - 1} \right)^{\frac{1}{2}} \quad \text{and} \quad \gamma'_{l,m} = \gamma_{l+1,m} \quad (5.3.40)$$

The derivations of Eq. (5.3.32a) with Eq. (5.3.34a) are now completed.

C. Derivations of $e^{-i\phi} \sin \theta |l, m\rangle$, Eq. (5.3.32b) with Eq. (5.3.34b)

• By using $[\hat{L}_-, e^{-i\phi} \sin \theta] = 0$ [cf. Eq. (5.3.30)], $e^{-i\phi} \sin \theta |l, m\rangle$ can be constructed with the ladder operator algebra. Recast the related Eq. (5.3.12) as

$$|l, m\rangle = C_{l,m} \left(\frac{\hat{L}_-}{\hbar} \right)^{l-m} |l, l\rangle, \quad \text{where} \quad C_{l,m} \equiv \left[\frac{(l+m)!}{(2l)!(l-m)!} \right]^{\frac{1}{2}} \quad (5.3.41)$$

that is $C_{l,m} = 1/(c_{l,m}^- c_{l,m+1}^- \cdots c_{l,l}^-)$, after Eq. (5.3.11). Moreover [cf. Eq. (5.3.18)],

$$|l, l\rangle = A_l e^{il\phi} (\sin \theta)^l, \quad \text{with} \quad A_l \equiv \frac{(-1)^l}{2^l l!} \left[\frac{(2l+1)!}{4\pi} \right]^{\frac{1}{2}} \quad (5.3.42)$$

Thus,

$$e^{-i\phi} \sin \theta |l, l\rangle = A_l e^{i(l-1)\phi} (\sin \theta)^{l+1} = \frac{A_l}{A_{l-1}} \sin^2 \theta |l-1, l-1\rangle \quad (5.3.43)$$

(a) Consider the boundary values in Eq. (5.3.32b). From Eqs. (5.3.1) and (5.3.2), we have

$$\begin{aligned} |l+1, l-1\rangle &= A_{l+1} C_{l+1,l-1} \frac{e^{i(l-1)\phi}}{(\sin \theta)^{l-1}} \frac{d^2[(\sin \theta)^{2(l+1)}]}{d(\cos \theta)^2} \\ &= -2(l+1) A_{l+1} C_{l+1,l-1} [(2l+1) \sin^2 \theta - 2l] [e^{i(l-1)\phi} (\sin \theta)^{l-1}] \\ &= -2(l+1) \frac{A_{l+1}}{A_{l-1}} C_{l+1,l-1} [(2l+1) \sin^2 \theta - 2l] |l-1, l-1\rangle \end{aligned} \quad (5.3.44)$$

The resulting final expression of Eq. (5.3.43) reads

$$e^{-i\phi} \sin \theta |l, l\rangle = \alpha_{l,l} |l-1, l-1\rangle + \beta_{l,l} |l+1, l-1\rangle \quad (5.3.45)$$

This is the form of Eq. (5.3.32b), at $m = l$, with [cf. Eq. (5.3.42)]

$$\alpha_{l,l} = \frac{2l A_l}{(2l+1) A_{l-1}} = - \left(\frac{2l}{2l+1} \right)^{\frac{1}{2}} \quad (5.3.46a)$$

and [cf. Eq. (5.3.41)]

$$\beta_{l,l} = -\frac{1}{2(l+1)(2l+1)} \frac{A_l}{A_{l+1}C_{l+1,l-1}} = \left[\frac{2}{(2l+1)(2l+3)} \right]^{\frac{1}{2}} \quad (5.3.46b)$$

(b) To proceed, we exploit Eq. (5.3.10) and obtain

$$\begin{aligned} \left(\frac{\hat{L}_-}{\hbar} \right)^{l-m} |l-1, l-1\rangle &= \left[\frac{(2l-2)!(l-m)!}{(l+m-2)!} \right]^{\frac{1}{2}} |l-1, m-1\rangle \\ \left(\frac{\hat{L}_-}{\hbar} \right)^{l-m} |l+1, l-1\rangle &= \left[\frac{(2l)!(l-m+2)!}{2(l+m)!} \right]^{\frac{1}{2}} |l+1, m-1\rangle \end{aligned} \quad (5.3.47)$$

The involving coefficients amount to $\prod_{k=m}^{l-1} c_{l-1,k}^-$ and $\prod_{k=m}^{l-1} c_{l+1,k}^-$, respectively, where $c_{l,m}^- \equiv [(l+m)(l-m+1)]^{\frac{1}{2}}$ [cf. Eq. (5.3.11)]. Applying then Eqs. (5.3.45)–(5.3.47) for Eq. (5.3.41), followed by using $[\hat{L}_-, e^{-i\phi} \sin \theta] = 0$ [cf. Eq. (5.3.30)], we obtain

$$e^{-i\phi} \sin \theta |l, m\rangle = \alpha_{l,m} |l-1, m-1\rangle + \beta_{l,m} |l+1, m-1\rangle \quad (5.3.48)$$

with [cf. Eqs. (5.3.41) and (5.3.46)]

$$\begin{aligned} \alpha_{l,m} &= \alpha_{l,l} C_{l,m} \left[\frac{(2l-2)!(l-m)!}{(l+m-2)!} \right]^{\frac{1}{2}} = - \left[\frac{(l+m)(l+m-1)}{(2l-1)(2l+1)} \right]^{\frac{1}{2}} \\ \beta_{l,m} &= \beta_{l,l} C_{l,m} \left[\frac{(2l)!(l-m+2)!}{2(l+m)!} \right]^{\frac{1}{2}} = \left[\frac{(l-m+2)(l-m+1)}{(2l+1)(2l+3)} \right]^{\frac{1}{2}} \end{aligned} \quad (5.3.49)$$

The derivations of Eq. (5.3.32b) with Eq. (5.3.34b) are completed.

5.3.4 Rotation spectroscopy of diatomic molecules

A. Opening remarks

• Recall the general formulation of spectrum presented in §3.4. In particular, we recast the optical spectroscopy, Eq. (3.4.14), in terms of

$$S(\omega) \propto \sum_{f,i} e^{-\beta E_i} |\boldsymbol{\mu}_{fi}|^2 \delta(\omega - \omega_{fi}) \equiv \sum_{f,i} e^{-\beta E_i} S(\omega; f \leftarrow i) \quad (5.3.50)$$

where

$$S(\omega; f \leftarrow i) = |\boldsymbol{\mu}_{fi}|^2 \delta(\omega - \omega_{fi}) \quad (5.3.51)$$

This is centered at $\omega_{fi} \equiv \hbar^{-1}(E_f - E_i)$, for the specified transition between two molecular *eigenstates*; i.e., $|\psi_f\rangle \leftarrow |\psi_i\rangle$ (or $f \leftarrow i$ for short). The optical transition rule is about when the transition dipole,

$$\boldsymbol{\mu}_{fi} \equiv \langle \psi_f | \hat{\boldsymbol{\mu}} | \psi_i \rangle \propto \langle \psi_f | \hat{\mathbf{r}} | \psi_i \rangle \neq 0 \quad (5.3.52)$$

since the matter system dipole operator takes the form of $\hat{\boldsymbol{\mu}} \propto \hat{\mathbf{r}}$.

B. Pure rotation motion of diatomic molecules

Molecular rotations and vibrations response to *microwave* and *infrared* light fields, respectively. The latter engages actually ro-vibration information, since the participated eigenstates are described by both vibration and rotation quantum numbers. Moreover, molecules in gas phase are randomly orientated. Average over the orientations, with respect to the direction of electric field, is also needed, on top of the Boltzmann distribution.

- Let us focus on pure rotation spectrum, with

$$\hat{H}_{\text{rot}} = \frac{\hat{L}^2}{2\mu R^2} \quad (5.3.53)$$

Here, μ and R are the reduced mass and the *bond length*, respectively. As known for diatomic molecules, the moment of inertia is $I \equiv \mu R^2$.

- (a) The rotation wave functions are the spherical harmonics, $Y_{l,m}(\theta, \phi)$.

$$\hat{H}_{\text{rot}}|Y_{l,m}\rangle = E_l^{\text{rot}}|Y_{l,m}\rangle; \quad m = 0, \pm 1, \dots, \pm l$$

with the degeneracy of $2l + 1$, and

$$E_l^{\text{rot}} = \hbar \tilde{B} l(l+1), \quad \text{where} \quad \tilde{B} \equiv \frac{\hbar}{2\mu R^2} \quad (5.3.54)$$

The *rotation constant*, $\tilde{B}$, goes with the unit of frequency.

- (b) The microwave active transitions are

$$\omega_{l+1,l} = (E_{l+1} - E_l)/\hbar = \tilde{B}[(l+1)(l+2) - l(l+1)] = 2\tilde{B}(l+1) \quad (5.3.55)$$

Thus, pure-rotation spectrum for each diatomic molecule comprises *a series of lines, with equal distance* of $2\tilde{B}$ in between.

- (c) As mentioned earlier, experimental spectrum acquires the average over molecular orientations, which can be expressed in terms of

$$\overline{(\mu_{l+1,l})^2} = \left(\frac{l+1}{2l+1} \right) \mu_0^2 \quad (5.3.56)$$

where μ_0 is the permanent dipole of the molecule. Together with the Boltzmann thermal distribution, the resulting pure-rotation spectrum reads

$$S_{\text{rot}}(\omega) = \sum_{l=0}^{\infty} (2l+1) \exp[-\beta \tilde{B} l(l+1)] \overline{(\mu_{l+1,l})^2} \delta(\omega - \omega_{l+1,l}) \quad (5.3.57)$$

Its maximum value is around

$$l^* \approx \left(\frac{k_B T}{2\hbar \tilde{B}} \right)^{\frac{1}{2}} \quad (5.3.58)$$

5.4 Central Force Systems

5.4.1 Radial wave functions: General remarks

A. Systems with spherical symmetry

• For an application of angular momentum theory, consider systems with spherical symmetry, where the potential $V(r)$ depends only on radial. The spherical coordinates frame is the choice. Together with the Laplacian operator, Eq. (4.3.14) with Eq. (4.3.13), the central force problems (without spins) are concerned with

$$\hat{H} = -\frac{\hbar^2}{2\mu} \left(\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right) + \frac{\hat{L}^2(\theta, \phi)}{2\mu r^2} + V(r) \quad (5.4.1)$$

We will be interest especially in the bound-states solutions to

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right) + \frac{\hat{L}^2(\theta, \phi)}{2\mu r^2} + V(r) \right] \Psi(\mathbf{r}) = E \Psi(\mathbf{r}) \quad (5.4.2)$$

The zero-energy must be included; otherwise $V(r)$ would no longer be central symmetric. The central force system Hamiltonian and the angular momentum satisfy

$$[\hat{H}, \hat{L}^2] = 0, \quad [\hat{H}, \hat{L}_z] = 0 \quad \text{and} \quad [\hat{L}^2, \hat{L}_z] = 0 \quad (5.4.3)$$

These constitute the *complete set of commuting observables* (CSCO) for central force systems.

B. Variable separation solutions

The standard set of wave functions via Eq. (5.4.2) will be the form of

$$\Psi_{k,l,m}(\mathbf{r}) = \langle \mathbf{r} | k, l, m \rangle = \langle r, \theta, \phi | k, l, m \rangle$$

The orthonormality is

$$\int_0^\infty dr \int_0^\pi d\theta \int_0^{2\pi} d\phi r^2 \sin\theta \Psi_{k',l',m'}^*(\mathbf{r}) \Psi_{k,l,m}(\mathbf{r}) = \delta_{kk'} \delta_{ll'} \delta_{mm'} \quad (5.4.4)$$

To proceed, we adopt the variables separation form of

$$\Psi_{k,l,m}(\mathbf{r}) = R_{k;l}(r) Y_{l,m}(\theta, \phi) \quad (5.4.5)$$

The angular part is the spherical harmonics that satisfies

$$\hat{L}^2 Y_{l,m}(\theta, \phi) = \hbar^2 l(l+1) Y_{l,m}(\theta, \phi); \quad l = 0, 1, \dots, l_{\max} \quad (5.4.6a)$$

$$\hat{L}_z Y_{l,m}(\theta, \phi) = \hbar m Y_{l,m}(\theta, \phi); \quad m = 0, \pm 1, \dots, \pm l \quad (5.4.6b)$$

The angular momentum quantum number acquires the upper bound, $l_{\max}$, due to the additional quantum number k that is nonnegative.

C. Radial wave function analysis

Substituting Eq. (5.4.6a) into Eq. (5.4.2) results in (Self-exercise)

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right) + \frac{\hbar^2 l(l+1)}{2\mu r^2} + V(r) \right] R_{k;l}(r) = E_{k;l} R_{k;l}(r) \quad (5.4.7)$$

The second term in the square-brackets is the centrifugal energy. Let the effective potential and the associated radial wave function be

$$V_l^{\text{eff}}(r) \equiv V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2} \quad \text{and} \quad \boxed{\psi_{k;l}(r) \equiv r R_{k;l}(r)} \quad (5.4.8)$$

respectively. Equation (5.4.7) becomes (Self-exercise)

$$\left[-\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial r^2} + V_l^{\text{eff}}(r) \right] \psi_{k;l}(r) = E_{k;l} \psi_{k;l}(r); \quad r \in [0, \infty) \quad (5.4.9)$$

Acceptable solutions are subject to

$$\psi_{k;l}(r=0) = \psi_{k;l}(r \rightarrow \infty) = 0 \quad (5.4.10)$$

The normalization reads

$$\int_0^\infty dr |\psi_{k;l}(r)|^2 = \int_0^\infty dr r^2 |R_{k;l}(r)|^2 = 1 \quad (5.4.11)$$

General features:

- Radial wave functions, $\psi_{k;l}(r) \equiv r R_{k;l}(r)$, are always real. Equation (5.4.9) reads in formal a one-dimension Schrödinger equation. The angular momentum quantum number l is a parameter and related to mean centrifugal energy via the Hellmann–Feynman theorem; see §5.4.4.

(i) The value of $\psi_{k;l}(r=0) = 0$ is universal, as in line with $\psi_{k;l}(r < 0) \equiv 0$.

(ii) Bound-states, $\psi_{k;l}(r)$, with $k = 0, 1, 2, \dots$, being the *number of nodes*. Acceptable bound-states are all of $\psi_{k;l}(r \rightarrow \infty) = 0$.

(iii) Unbound $\psi_{k;l}(\infty) \neq 0$, but $R_{k;l}(\infty) = 0$. This is the free-particle feature when $r \rightarrow \infty$. As usual, the unbounded wave functions go by the delta-function normalization.

Intrinsic versus accidental degeneracy

- Both $\psi_{k;l}(r)$ and eigenenergy $E_{k;l}$ depend l . Associated with each given l there are $m = 0, \pm 1, \dots, \pm l$. This $(2l+1)$ -folds degeneracy is *intrinsic*, since the radial Eq. (5.4.7) or Eq. (5.4.9) contains no quantum number m .
- Beside the intrinsic $(2l+1)$ -degeneracy, there are also the system-dependent ones, referred to *accidental* degeneracies. The best known example is the hydrogen-like atoms system, where the energy $E_n = E_{k;l}$ does not depend of l at all, but only on the principal shell-quantum number n , which goes also with the subshell of $l = 0, 1, \dots, n-1$.
- Another well-known accidental degeneracy scenario is the spherical harmonic oscillators system, with distinct subshell structures between an even or odd n ; see Eq. (5.4.41) and the Table below.

5.4.2 Spherical free-particle waves

A. Basic properties

Free-particles are plane waves. Spherical waves are concerned with

$$\Psi_{\mathbf{k}}(\mathbf{r}) \equiv \langle \mathbf{r} | \mathbf{k} \rangle = (2\pi)^{-\frac{3}{2}} e^{i\mathbf{k} \cdot \mathbf{r}} \equiv R_{k;l}(r) Y_{l,m}(\theta, \phi) \quad (5.4.12)$$

where $|\mathbf{r}\rangle = |r, \theta, \phi\rangle$,

$$\mathbf{k} = \mathbf{p}/\hbar \quad \text{and} \quad k = \sqrt{2\mu E}/\hbar \geq 0 \quad (5.4.13)$$

As an unbound-state scenario, the energy E is continuous, where μ is the mass of free-particle itself, with noting that m had appeared in $Y_{l,m}$. For the specified representation by Eq. (5.4.12), its radial part follows

$$R_{k;l}(r) = (2\pi)^{-\frac{3}{2}} \sum_m \int_0^\pi d\theta \int_0^{2\pi} d\phi e^{i\mathbf{k} \cdot \mathbf{r}} \sin \theta Y_{l,m}^*(\theta, \phi) \quad (5.4.14)$$

This is the formal definition of spherical free-particle waves.

- Spherical waves satisfy Eq. (5.4.7), with $V(r) = 0$, at a given kinetic energy, $E_{k;l} = (\hbar k)^2/(2\mu) = E$, Eq. (5.4.13). The corresponding Eq. (5.4.9), with eliminating the common factor of $\hbar^2/(2\mu)$, reads

$$\psi''_{k;l}(r) + \left[k^2 - \frac{l(l+1)}{r^2} \right] \psi_{k;l}(r) = 0; \quad \psi_{k;l}(r) \equiv r R_{k;l}(r) \quad (5.4.15)$$

with $\psi_{k;l}(r=0) = 0$ the boundary value. The normalization goes by

$$\int_0^\infty r^2 R_{k;l}(r) R_{k';l}(r) dr = \int_0^\infty \psi_{k;l}(r) \psi_{k';l}(r) dr = \delta(k - k') \quad (5.4.16)$$

Consider first the case of $l = 0$, with which Eq. (5.4.15) directly results in

$$\psi_{k,0}(r) = \sqrt{\frac{2}{\pi}} \sin(kr) \quad (5.4.17a)$$

We obtain further

$$\psi_{k;l+1}(r) = -\frac{r^{l+1}}{k} \frac{d}{dr} \frac{\psi_{k;l}(r)}{r^{l+1}} \quad (5.4.17b)$$

Derivations: Let $\psi_{k;l}(r) = r^{l+1} \chi_{k;l}(r)$, so that Eq. (5.4.15) becomes

$$r^{l+1} \left[\chi''_{k;l} + \frac{2(l+1)}{r} \chi'_{k;l} + \frac{l(l+1)}{r^2} \chi_{k;l} \right] + \left[k^2 - \frac{l(l+1)}{r^2} \right] r^{l+1} \chi_{k;l} = 0$$

That is

$$\chi''_{k;l}(r) + \frac{2(l+1)}{r} \chi'_{k;l}(r) + k^2 \chi_{k;l}(r) = 0 \quad (5.4.18)$$

The solutions are found satisfying

$$\chi'_{k;l}(r) = r \chi_{k;l+1} \quad (5.4.19)$$

as follows. Consider the derivative of Eq. (5.4.18),

$$\chi_{k;l}''' + \frac{2(l+1)}{r} \chi_{k;l}''(r) + \left[k^2 - \frac{2(l+1)}{r^2} \right] \chi_{k;l}'(r) = 0$$

This together with Eq. (5.4.19) give rise to

$$\begin{aligned} 0 &= (r\chi_{k;l+1}'' + 2\chi_{k;l+1}') + \frac{2(l+1)}{r} (r\chi_{k;l+1}' + \chi_{k;l+1}) + \left[k^2 - \frac{2(l+1)}{r^2} \right] (r\chi_{k;l+1}) \\ &= r\chi_{k;l+1}'' + 2(l+2)\chi_{k;l+1}' + k^2 r\chi_{k;l+1} \end{aligned}$$

The resulting

$$\chi_{k;l+1}'' + \frac{2(l+2)}{r} \chi_{k;l+1}' + k^2 \chi_{k;l+1} = 0$$

does reproduce Eq. (5.4.18), with l being replaced by $(l+1)$.

We have thus proved $\chi_{k;l}'(r) = r\chi_{k;l+1}$, Eq. (5.4.19), up to a scaling constant, which amounts to the use of dimensionless radial, kr , as implied in $\psi_{k,0}(r)$ of Eq. (5.4.17). Therefore,

$$\psi_{k;l+1}(r) \propto r^{l+2} \chi_{k;l+1}(r) = r^{l+1} \chi_{k;l}'(r)$$

resulting in

$$\psi_{k;l+1}(r) = -\frac{r^{l+1}}{k} \frac{d}{dr} \frac{\psi_{k;l}(r)}{r^{l+1}}$$

This is Eq. (5.4.17b), with the “ $-$ ” sign arising from Eq. (5.4.14) that defines spherical wave via $Y_{l,m}(\theta, \phi) = \Theta_{l,m}(\theta) e^{im\phi} / \sqrt{2\pi}$ and that $\Theta_{l,m}(\theta)$ of Eq. (5.3.2) carries $(-)^l$. (Q.E.D.)

B. Spherical waves versus Bessel functions

- Mathematically, $R_{k;l}(r)$ of Eq. (5.4.14), up to the normalization function, amounts to $j_l(kr)$, the *spherical Bessel function of the first kind*, defined as

$$j_l(x) \equiv \sqrt{\frac{\pi}{2x}} J_{l+\frac{1}{2}}(x) \quad (5.4.20)$$

with $J_\nu(x)$ being the Bessel function. The normalized spherical waves are

$$R_{k;l}(\rho \equiv kr) = \sqrt{\frac{2}{\pi}} j_l(\rho) \quad (5.4.21)$$

It is true that Eq. (5.4.15), expressed in terms of $R_{k;l}(\rho \equiv kr)$, does give rise the standard equation for the spherical Bessel functions in Handbook.

- Basic properties [cf. Eq. (5.4.17b)]:

(a) In relation to the Legendre polynomials [cf. Eqs. (5.3.24)–(5.3.27)],

$$j_n(x) = \frac{(-i)^n}{2} \int_0^\pi e^{ix \cos \theta} P_n(\cos \theta) \sin \theta d\theta \quad (5.4.22a)$$

(b) Compared with Eq. (5.4.17b),

$$j_n(x) = -x^n \frac{d}{dx} \frac{j_{n-1}(x)}{x^n} \quad (5.4.22b)$$

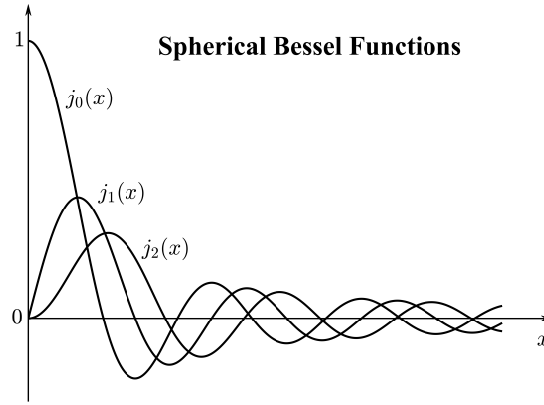
and

$$j_n(x) = \frac{x^n}{(2n+1)!!} \left\{ 1 - \frac{\frac{1}{2}x^2}{1!(2n+3)} + \frac{(\frac{1}{2}x^2)^2}{2!(2n+3)(2n+5)} - \dots \right\} \quad (5.4.22c)$$

The first three spherical Bessel functions are

$$j_0(x) = \frac{\sin x}{x}, \quad j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}, \quad j_2(x) = \left(\frac{3}{x^3} - \frac{1}{x} \right) \sin x - \frac{3}{x^2} \cos x \quad (5.4.22d)$$

The corresponding $R_{k;l}(\rho)$, Eq. (5.4.21), agrees with $\psi_{k;l}(r) = rR_{k;l}(r)$ of Eq. (5.4.17).



Plotted here, as inferred from Eq. (5.4.21), are the s, p and d waves of free-particle, with the x -axis standing for kr , where $k = |\mathbf{p}|/\hbar$ [cf. Eq. (5.4.13)]. The radial wave function, $R_l(\rho) = R_{k;l}(\rho) \propto j_l(\rho)$, Eq. (5.4.21), with the dependence on k through $\rho \equiv kr$.

C. Radial kinetics energy versus centrifugal energy

It is worth noting that k is the wavevector amplitude, as specified in Eq. (5.4.13), and related to the kinetic energy. This in general consists of the radial and the centrifugal parts and both are nonnegative. For free-particles, $V(r) = 0$, the centrifugal part is just $\hat{V}_l^{\text{eff}}(r) = \hat{L}^2/(2\mu r^2)$, whereas [cf. Eq. (5.4.13)]

$$E = \frac{\mathbf{p}^2}{2\mu} = \frac{\hbar^2 k^2}{2\mu} = \langle \hat{K}_r \rangle_{k;l} + \langle \hat{V}_l^{\text{eff}} \rangle_{k;l} \quad (5.4.23)$$

We would obtain

$$\frac{\langle \hat{V}_l^{\text{eff}} \rangle_{k;l}}{\langle \hat{K}_r \rangle_{k;l}} = \frac{l(l+1)}{l^2 + 4l + 2} \quad (5.4.24)$$

The derivations are as follows.

Derivations of Eq. (5.4.24):

(a) Let us start with Eq. (5.4.20) and recast Eq. (5.4.21) in terms of the Bessel function,

$$\psi_l(\rho) \equiv \rho R_{k;l}(\rho = kr) = \sqrt{\rho} J_{l+1/2}(\rho) \quad (5.4.25)$$

As known, the free-particle waves go by delta-function normalization, Eq. (5.4.16). Similar to the transmission-and-reflection problems in §3.1.2, the delta-function normalization serves the common scaling measure for any given $\psi_l(\rho)$. The standard expression of *expectation values*, via normalized wave functions, would remain valid for *their ratio*, as demonstrated below.

(b) The evaluations below go in formal via

$$\frac{\langle \hat{V}_l^{\text{eff}} \rangle_{k;l}}{\langle \hat{K}_r \rangle_{k;l}} = \frac{\bar{V}_l^{\text{eff}}}{\bar{K}_l^r} \quad (5.4.26)$$

with

$$\bar{V}_l^{\text{eff}} = l(l+1) \int_0^\infty \frac{1}{\rho^2} \psi_l^2(\rho) d\rho = \frac{l(l+1)}{2l+1} \quad (5.4.27a)$$

$$\bar{K}_l^r = - \int_0^\infty \psi_l(\rho) \frac{d^2}{d\rho^2} \psi_l(\rho) d\rho = \int_0^\infty [\psi_l'(\rho)]^2 d\rho \quad (5.4.27b)$$

where [cf. Eq. (5.4.17b)]

$$\psi_l'(\rho) \equiv \frac{d\psi_l(\rho)}{d\rho} = \frac{l+1}{\rho} \psi_l(\rho) - \psi_{l+1}(\rho) \quad (5.4.28)$$

The last identity of Eq. (5.4.27a) exploits Eq. (5.4.25), followed by using

$$\int_0^\infty \frac{1}{\rho} J_{l+1/2}^2(t) dt = \frac{1}{2l+1} \quad (5.4.29a)$$

This is a well-established property of Bessel functions.

(c) On the other hand, Eq. (5.4.27b), with Eq. (5.4.28), will engage

$$\int_0^\infty J_{l+3/2}(a\rho) J_{l+1/2}(b\rho) d\rho = \begin{cases} 0; & a < b \\ 1/(2a); & a = b \\ a^{-1}(b/a)^{l+1/2}; & a > b \end{cases} \quad (5.4.29b)$$

where $a, b > 0$. Together with Eq. (5.4.25), we obtain in formal

$$\bar{K}_l^r = \int_0^\infty \left[\frac{l+1}{\rho} \psi_l(\rho) - \psi_{l+1}(\rho) \right]^2 d\rho = \left[\frac{(l+1)^2}{2l+1} + 1 \right] - 2(l+1)C \quad (5.4.30)$$

The first term follows Eq. (5.4.29a) and the standard normalization, with respect to the aforementioned common scaling measure. The second term goes with

$$C \equiv \int_0^\infty \frac{1}{\rho} \psi_{l+1}(\rho) \psi_l(\rho) d\rho = \int_0^\infty J_{l+3/2}(\rho) J_{l+1/2}(\rho) d\rho = 0 \quad (5.4.31)$$

The zero-value case of Eq. (5.4.29b) is the only choice, since physically Eq. (5.4.30) must always be *nonnegative*. Together with Eq. (5.4.27a), the final result of Eq. (5.4.26) reads

$$\frac{\langle \hat{V}_l^{\text{eff}} \rangle_{k;l}}{\langle \hat{K}_r \rangle_{k;l}} = \frac{l(l+1)}{(l+1)^2 + 2l+1} \quad (5.4.32)$$

This is Eq. (5.4.24).

(Q.E.D.)

5.4.3 Spherical harmonic oscillators

A. General considerations

- Let us start with the general feature of *all systems*:

Eigenenergies and their degeneracies are representation invariance

- Spherical harmonic oscillator is a special case, with the potential,

$$V(r) = \frac{1}{2}\mu\omega^2(x^2 + y^2 + z^2) = \frac{1}{2}\mu\omega^2r^2 \quad (5.4.33)$$

We had learn one-dimension harmonic oscillator systems in §4.2. The total composite wave functions in three-dimension Cartesian coordinates take the direct product form,

$$\Psi_{\mathbf{n}}(\mathbf{r}) = \psi_{n_x}(x)\psi_{n_y}(y)\psi_{n_z}(z) \quad (5.4.34a)$$

The total composite energies are

$$E_n = n\hbar\omega + E_0; \quad n = n_x + n_y + n_z = 0, 1, 2, \dots \quad (5.4.34b)$$

with $E_0 = 3\hbar\omega/2$, the zero-point energy. The degeneracy is found to be

$$g_n = \frac{1}{2}(n+1)(n+2) \quad (5.4.35)$$

While this is a simple combinatorial problem, we may also obtain the results as follows: The ground-state is non-degenerate ($g_0 = 1$). The first two excite-states are of $g_1 = 3$ and $g_2 = 6$, respectively. These three values are sufficient in dictating the form of g_n , as the degeneracy of three-dimension systems is quadratic.

- As had learnt, the CSCO, $(\hat{H}, \hat{L}^2, \hat{L}_z)$, is preferred, for its distinct physical properties. This is concerned with

$$\hat{H}|k, l, m\rangle = E_{k;l}|k, l, m\rangle \quad (5.4.36a)$$

$$\hat{L}^2|k, l, m\rangle = \hbar^2l(l+1)|k, l, m\rangle \quad (5.4.36b)$$

$$\hat{L}_z|k, l, m\rangle = \hbar m|k, l, m\rangle \quad (5.4.36c)$$

In the spherical coordinates representation,

$$\Psi_{\mathbf{n}}(\mathbf{r}) \equiv \langle r, \theta, \phi | k, l, m \rangle = R_{k;l}(r)Y_{l,m}(\theta, \phi) \quad (5.4.37)$$

The angle part, $Y_{l,m}(\theta, \phi)$, had been detailed in §5.3. The radial part, $R_{k;l}(r)$ of Eq. (5.4.7), goes now with $V(r)$ of Eq. (5.4.33). That is

$$\frac{\hbar^2}{2\mu} \left[-\left(\frac{1}{r} \frac{\partial^2}{\partial r^2} r \right) + \frac{l(l+1)}{r^2} + \left(\frac{\mu\omega r}{\hbar} \right)^2 \right] R_{k;l}(r) = E_{k;l} R_{k;l}(r) \quad (5.4.38)$$

with $k = 0, 1, \dots$. This index specifies also the number of nodes of $R_{k;l}(r)$.

B. Spherical harmonic oscillators: $n = 2k + l$

Before verifying that $n = 2k + l$, consider first

$$E_{k;l} = E_n = \hbar\omega\left(n + \frac{3}{2}\right) = \hbar\omega\left(n_{k;l} + \frac{3}{2}\right); \quad n = 0, 1, 2, \dots \quad (5.4.39)$$

the invariance of eigenenergy, $E_{k;l}$ in Eq. (5.4.38) versus E_n in Eq. (5.4.34b). In terms of quantum numbers, the above expressions read

$$n = n_x + n_y + n_z = n_{k;l} = 2k + l \quad (5.4.40)$$

The last identity is to be identified, along the solutions to Eq. (5.4.38); see Eqs. (5.4.42)–(5.4.47) and discussions therein.

The consequence of $n = 2k + l$ is as follows. Recall the fact that k stands for the number of nodes associated with $\psi_{k;l}(r) = rR_{k;l}(r)$. Thus, at a given energy level, E_n , there are $k = 0, 1, \dots, \text{Int}(n/2)$, and

$$l = \begin{cases} 0, 2, \dots, n - 2, n; & \text{even } n \\ 1, 3, \dots, n - 2, n; & \text{odd } n \end{cases} \quad (5.4.41)$$

Each l is of the intrinsic $(2l+1)$ -degeneracy from $m = 0, \pm 1, \dots, \pm l$. The resulting total degeneracy g_n to E_n , via Eq. (5.4.41), does satisfy Eq. (5.4.35). Further elaborations and clarifications are tabulated as follows.

n	g_n	$(l; 2k)$	$2l + 1$	Orbitals (n, l)
0	1	(0; 0)	1	0s
1	3	(1; 0)	3	1p
2	6	(0; 2)	1	2s
		(2; 0)	5	2d
3	10	(1; 2)	3	3p
		(3; 0)	7	3f
4	15	(0; 4)	1	4s
		(2; 2)	5	4d
		(4; 0)	9	4g

C. Radial wave functions via Eq. (5.4.38)

Consider Eq. (5.4.38) in terms of its corresponding $\psi_{k;l} \equiv rR_{k;l}$, Eq. (5.4.9), with setting $E_{k;l} = \hbar\omega(n_{k;l} + 3/2)$. In the dimensionless unit, the equation reads

$$\psi_{k;l}''(\rho) + \left[(2n_{k;l} + 3) - \frac{l(l+1)}{\rho^2} - \rho^2 \right] \psi_{k;l}(\rho) = 0 \quad (5.4.42)$$

where

$$\psi_{k;l}(\rho) \equiv \rho R_{k;l}(\rho) \quad \text{and} \quad \rho \equiv \left(\frac{\mu\omega}{\hbar} \right)^{\frac{1}{2}} r \quad (5.4.43)$$

Consider the solutions the form of

$$\psi_{k;l}(\rho) = e^{-\rho^2/2} \rho^{l+1} \chi_{k;l}(\rho) \quad (5.4.44)$$

which satisfies $\psi_{k;l}(\rho=0) = 0$ and the ground state $\propto e^{-\rho^2/2}$ [Eq. (4.2.20)]. Substitute the above into Eq. (5.4.42), resulting in

$$\chi_{k;l}''(\rho) + \frac{2}{\rho} [(l+1) - \rho^2] \chi_{k;l}'(\rho) + 2(n_{k;l} - l) \chi_{k;l}(\rho) = 0 \quad (5.4.45)$$

As mentioned earlier, $\chi_{k;l}(\rho)$ would have k number of nodes. Moreover, it is an even function as seen from Eq. (5.4.45) or Eq. (5.4.42). Consider then

$$\chi_{k;l}(\rho) = \sum_{j=0}^k c_j \rho^{2j} \quad (5.4.46)$$

Applying it to Eq. (5.4.45) gives rise to (see derivations below)

$$\frac{c_{j+1}}{c_j} = \frac{1}{j+1} \frac{j - \frac{1}{2}(n_{k;l} - l)}{j + l + \frac{3}{2}} = \frac{1}{j+1} \frac{j - k}{j + l + \frac{3}{2}} \quad (5.4.47)$$

Derivations: (a) The last identity above, $n_{k;l} = 2k + l$, Eq. (5.4.40), arises from the requirement of $c_{j_{\max}+1} = 0$; hence, $j_{\max} = \frac{1}{2}(n_{k;l} - l) = k$, which had been used in Eq. (5.4.46) and also Eq. (5.4.40). (b) Turn to Eq. (5.4.45), with changing the variable,

$$z \equiv \rho^2 \Rightarrow \frac{d}{d\rho} = 2\rho \frac{d}{dz} \quad \text{and} \quad \frac{d^2}{d\rho^2} = 2 \frac{d}{dz} + 4z \frac{d^2}{dz^2}$$

The transformed Eq. (5.4.45), where $n_{k;l} = 2k + l$, reads $[\chi(z) \equiv \chi_{k;l}(z)]$ below

$$z\chi''(z) + \left(l + \frac{3}{2} - z\right)\chi'(z) + k\chi(z) = 0 \quad (5.4.48)$$

This is the form of Kummer's equation. The solutions, *Kummer's functions*, are a kind of *confluent hypergeometric functions*. Nevertheless, the recursion solutions are rather straightforward, as follows.

To proceed, we adopt the *dummy-sum* convention and recast Eq. (5.4.46) as

$$\chi(z) = c_j z^j \quad (\text{dummy-sum-over } j) \quad (5.4.49a)$$

This highlights the coefficient to z^j in $\chi(z)$ and further those in

$$z\chi'' = j(j+1)c_{j+1}z^j, \quad \chi' = (j+1)c_{j+1}z^j, \quad z\chi' = jc_jz^j \quad (5.4.49b)$$

Applying them for Eq. (5.4.48) results in the recursion relation for coefficients,

$$(j+1)\left(j + l + \frac{3}{2}\right)c_{j+1} = (j-k)c_j \quad (5.4.50)$$

This is the last identity of Eq. (5.4.47). (Q.E.D.)

D. Normalized radial wave functions

- Recast Eq. (5.4.47) in terms of

$$\frac{c_j}{c_{j-1}} = -\frac{k+1-j}{j(\alpha+j)}; \quad \alpha \equiv l + \frac{1}{2} \quad (5.4.51)$$

By setting $c_0 \equiv 1$, we obtain

$$c_j = \frac{(-1)^j}{(\alpha)_j} \binom{k}{j} \quad (5.4.52)$$

Here,

$$(\alpha)_j = (\alpha+1) \cdots (\alpha+j); \quad (\alpha)_0 \equiv 1 \quad (5.4.53)$$

- For the normalization, consider first

$$\chi_{k;l}^2(\rho) = \left[\sum_{j=0}^k c_j \rho^{2j} \right]^2 = \sum_{m=0}^k (-1)^m A_m \rho^{2m} \quad (5.4.54)$$

where

$$A_m = \sum_{j=0}^k \frac{1}{(\alpha)_j (\alpha)_{m-j}} \binom{k}{j} \binom{k}{m-j} \quad (5.4.55)$$

with $\binom{k}{j'} \equiv 0$, if $j' < 0$. Consider now

$$\frac{1}{N_{k;l}^2} = \int_0^\infty \rho^{2(l+1)} \chi_{k;l}^2(\rho) e^{-\rho^2} d\rho = \sum_{m=0}^k (-1)^m A_m \int_0^\infty \rho^{2(l+1+m)} e^{-\rho^2} d\rho \quad (5.4.56)$$

Exploit the standard Gaussian integration identity,

$$\boxed{\frac{1}{\sqrt{2\pi\Delta^2}} \int_{-\infty}^\infty x^{2(n+1)} e^{-\frac{x^2}{2\Delta^2}} dx = (2n+1)!! \Delta^{2(n+1)}} \quad (5.4.57)$$

Finally, we obtain

$$N_{k;l} = \left(\frac{2^{l+2-k}}{\sqrt{\pi} k!} \right)^{\frac{1}{2}} \frac{[(2l+2k+1)!!]^{\frac{1}{2}}}{(2l+1)!!} \quad (5.4.58)$$

The normalized radial wave function reads

$$R_{k;l}(\rho) = N_{k;l} e^{-\rho^2/2} \rho^l \sum_{j=0}^k c_j \rho^{2j} \quad (5.4.59)$$

with c_j being specified in Eq. (5.4.52).

5.4.4 Onset of centrifugal potential energies

A. The use of Hellmann–Feynman theorem: General remarks

- The centrifugal potential energy, the second term in $V_l^{\text{eff}}(r)$ of Eq. (5.4.8), is a fraction of the total kinetic energy and always nonnegative; i.e.,

$$V_l^{\text{eff}}(r) \equiv V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2} \geq V(r) = V_{l=0}^{\text{eff}}(r) \quad (5.4.60)$$

- Moreover,

$$E_{k;l} > \langle \psi_{k;l}(r) | V_l^{\text{eff}}(r) | \psi_{k;l}(r) \rangle = \int_0^\infty dr V_l^{\text{eff}}(r) \psi_{k;l}^2(r) \quad (5.4.61)$$

It is worth re-emphasizing that $\psi_{k;l}(r) \equiv rR_{k;l}(r)$ is always real, with k being the number of nodes.

- On the other hand, often used are n and l , the shell and subshell quantum numbers, with which the energies and orbitals are reported. Therefore, the expression of $n = n_{k;l}$, by which $E_{n,l} = E_{k;l}$ and $\psi_{n,l}(r) = \psi_{k;l}(r)$ in values, is an essential ingredient in the solutions to the radial wave functions.

- Recall the virial theorem and its application to the case of homogeneous potentials, Eq. (2.3.21), and the resulting Eq. (2.3.25):

$$V(\lambda \hat{\mathbf{r}}) = \lambda^\nu V(\hat{\mathbf{r}}) \Rightarrow \frac{\langle \hat{V} \rangle_{n,l}}{\langle \hat{K} \rangle_{n,l}} = \frac{2}{\nu} \quad (5.4.62)$$

with $E_{n,l} = \langle \hat{K} \rangle_{n,l} + \langle \hat{V} \rangle_{n,l}$.

- Turn to the Hellmann–Feynman (HF) theorem, Eq. (2.3.27),

$$\frac{\partial E_{n,l}(\lambda)}{\partial \lambda} = \left\langle \frac{\partial \hat{H}(\lambda)}{\partial \lambda} \right\rangle_{n,l} \quad (5.4.63)$$

with understanding that $E_{n,l}$ stands for $E_{k;l}$ here. The subshell quantum number, l , can be a HF variation parameter. For systems with spherical symmetry, we have

$$\left. \frac{\partial E_{n,l}(\lambda)}{\partial \lambda} \right|_{\lambda=l} = \frac{\hbar^2}{2\mu} (2l+1) \left\langle \frac{1}{r^2} \right\rangle_{n,l}$$

Therefore,

$$\left\langle \frac{\hat{L}^2}{2\mu r^2} \right\rangle_{n,l} = \frac{l(l+1)}{2l+1} \cdot \left. \frac{\partial E_{n,l}(\lambda)}{\partial \lambda} \right|_{\lambda=l} \quad (5.4.64)$$

This is a viable method of evaluating the mean centrifugal potential energy, whenever both the formal expressions of $n = n_{k;l}$ and $E_{n,l} = E_{k;l}$ are identified. In the following illustrations, we exploit these pre-required results, for two representing systems. One is the spherical harmonic oscillator, where $n = n_{k;l} = 2k+l$ [cf. Eq. (5.4.40)]. Another is the hydrogenlike atoms, where $n = n_{k;l} = k+l+1$ [cf. Eq. (5.4.73)].

B. Spherical harmonic oscillators:

$$V(r) = \frac{1}{2}\mu\omega^2 r^2 \quad (5.4.65)$$

As identified in §5.4.3, Eqs. (5.4.39) and (5.4.40),

$$\begin{aligned} E_{n,l} = E_{k;l} &= \hbar\omega \left(n + \frac{3}{2}\right); & n &= 0, 1, 2, \dots \\ n = n_{k;l} &= 2k + l; & k &= 0, 1, 2, \dots \end{aligned} \quad (5.4.66)$$

The corresponding Eq. (5.4.64) reads

$$\left\langle \frac{\hat{L}^2}{2\mu r^2} \right\rangle_{n,l} = \frac{l(l+1)}{2l+1} \hbar\omega \quad (5.4.67)$$

resulting in

$$\left\langle \frac{1}{r^2} \right\rangle_{n,l} = \frac{2}{2l+1} \frac{\mu\omega}{\hbar} \quad (5.4.68)$$

On the other hand,

$$\langle r^2 \rangle_{n,l} = \frac{\hbar}{\mu\omega} \left(n + \frac{3}{2}\right) \quad (5.4.69)$$

arising from Eq. (5.4.62), which for spherical harmonic oscillators reads

$$\langle \hat{V} \rangle_{n,l} = \langle \hat{K} \rangle_{n,l} = \frac{1}{2} E_{n,l} = \frac{1}{2} \hbar\omega \left(n + \frac{3}{2}\right) \quad (5.4.70)$$

The mean value for the *radial kinetic energy*, $\hat{K}_r$, the difference between the total and the centrifugal ones, is given by

$$\langle \hat{K}_r \rangle_{n,l} = \frac{1}{2} \hbar\omega \left[n + \frac{3}{2} - \frac{2l(l+1)}{2l+1} \right] \quad (5.4.71)$$

Note that $l_{\max} = n$; see Eq. (5.4.66) or Eq. (5.4.41).

C. Hydrogenlike atomic systems:

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} = -2\mathcal{R} \frac{a_0}{r} \quad (5.4.72)$$

See Eq. (5.1.4) for both the Rydberg constant $\mathcal{R}$ and the Bohr's radius a_0 .

As identified later [cf. Eqs. (6.1.18) and (6.1.19)],

$$\begin{aligned} E_{n,l} = E_{k;l} &= -\frac{Z^2}{n^2} \mathcal{R}; & n &= 1, 2, 3, \dots \\ n = n_{k;l} &= k + l + 1; & k &= 0, 1, 2, \dots \end{aligned} \quad (5.4.73)$$

The corresponding Eq. (5.4.64), with $\mu \simeq m_e$, the electron mass, reads

$$\left\langle \frac{\hat{L}^2}{2m_e r^2} \right\rangle_{n,l} = \frac{l(l+1)}{2l+1} \cdot \frac{2Z^2}{n^3} \mathcal{R} \quad (5.4.74)$$

We obtain

$$\left\langle \frac{a_0^2}{r^2} \right\rangle_{n,l} = \frac{2Z^2}{n^3(2l+1)} \quad (\text{H-like atoms}) \quad (5.4.75)$$

As shown below, we can also obtain

$$\left\langle \frac{a_0^3}{r^3} \right\rangle_{n,l} = \frac{2Z^3}{n^3 l(l+1)(2l+1)} \quad (5.4.76)$$

Note that $\langle a_0/r \rangle_{n,l} = Z/n^2$ via the virial theorem, Eq. (5.4.62). All these three identities will be used in evaluating the fine structure of hydrogenlike atomic system; see §5.3.3. Note also that Eq. (5.4.76) diverges for s-orbitals.

Equation (5.4.76) via the steady-state theorem:

$$\langle [\hat{A}, \hat{H}] \rangle = 0 \quad (5.4.77)$$

As mentioned earlier, Eq. (5.4.9) can be treated as a regular Schrödinger equation; i.e.,

$$\hat{H}^{\text{eff}}|\psi\rangle = E|\psi\rangle$$

For hydrogenlike atoms, we adopt the *atomic unit*, by which

$$\hat{H}^{\text{eff}} = -\frac{1}{2} \frac{\partial^2}{\partial r^2} + \frac{l(l+1)}{2r^2} - \frac{Z}{r} \quad (5.4.78)$$

Consider now the following identity,

$$\left[\frac{\partial}{\partial r}, \hat{H}^{\text{eff}} \right] = \frac{l(l+1)}{2} \left[\frac{\partial}{\partial r}, \frac{1}{r^2} \right] - Z \left[\frac{\partial}{\partial r}, \frac{1}{r} \right] = -\frac{l(l+1)}{r^3} + \frac{Z}{r^2}$$

Applying the steady-state theorem, Eq. (5.4.77), followed by using Eq. (5.4.75), we obtain

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{Z}{l(l+1)} \left\langle \frac{1}{r^2} \right\rangle = \frac{Z}{l(l+1)} \frac{2Z^2}{n^3(2l+1)}$$

This is the result of Eq. (5.4.76).

(Q.E.D.)

Exercises

E5.4.1 Derive $\langle a_0^4/r^4 \rangle_{n,l}$ for the hydrogenlike atom systems.

E5.4.2 Derive Eq. (5.4.74) and the corresponding $\langle \hat{K}_r \rangle_{n,l}$.

E5.4.3 Show that Eq. (5.4.71), the mean radial kinetic energy for spherical harmonic oscillator, does satisfy $\langle \hat{K}_r \rangle_{n,l} > 0$.

Chapter 6

Hydrogenlike Atomic Systems

6.1 Atomic Orbitals and General Remarks

6.1.1 *Energies and angular momentums*

Orbitals refer to wave functions of single-electron systems. Atomic orbitals (AOs) are the solutions to hydrogen and/or hydrogen-like ions, consisting of only one electron and one nucleus, with the Coulomb potential,

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} \quad (6.1.1)$$

Here, Z is the atomic number, ϵ_0 the vacuum permittivity and e the electron charge. AOs are analytically solvable and serve as the building blocks for many-electron atoms and further for molecules.

In this section, we focus on the bound-state AOs. In the absence of spin degree of freedom, the CSCO comprises $(\hat{H}, \hat{L}^2, \hat{L}_z)$, with

$$\hat{H}|n, l, m\rangle = E_n|n, l, m\rangle; \quad n = 1, 2, \dots \quad (6.1.2a)$$

$$\hat{L}^2|n, l, m\rangle = \hbar^2 l(l+1)|n, l, m\rangle; \quad l = 0, 1, \dots, n-1 \quad (6.1.2b)$$

$$\hat{L}_z|n, l, m\rangle = \hbar m|n, l, m\rangle; \quad m = 0, \pm 1, \dots, \pm l$$

As shown later, the bound-state energies are [cf. Eq. (6.1.19)]

$$E_n = -\frac{Z^2}{n^2} \mathcal{R}; \quad n = 1, 2, \dots \quad (6.1.3)$$

where

$$\mathcal{R} = \frac{1}{2} \cdot \frac{e^2}{4\pi\epsilon_0 a_0} \quad \text{and} \quad a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} \quad (6.1.4)$$

The Rydberg constant $\mathcal{R} \approx 13.6 \text{ eV}$ and the Bohr's radius $a_0 \approx 0.529 \text{ \AA}$. More precisely, we shall use the reduced mass, μ , rather than the electron

mass, m_e . While electron's $\mathcal{R} \approx 1.09737 \times 10^5 \text{cm}^{-1}$, the hydrogen's Rydberg constant is $\mathcal{R}_H \approx 1.09677 \times 10^5 \text{cm}^{-1}$ (energy in terms of wavenumber). This is the ionization energy of hydrogen atom, from its ground (1s) state.

Bohr's orbit theory

- As well known, Eq. (6.1.3) was first obtained by Neil Bohr in 1913. The Bohr's orbit theory violates the basic quantum principles, but it does offer some valuable insights on atomic structure. For the later reference, it is worth to present the Bohr's theory below.
- In his theory, the electron moves around the nuclei in the circular orbit of radius r , subject to not only the centrifugal force, but also the standing-wave quantization; i.e.,

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m_e \frac{v^2}{r} \quad \text{and} \quad m_e r v = n\hbar; \quad n = 1, 2, \dots \quad (6.1.5)$$

The resulting velocity and radius read [see a_0 in Eq. (6.1.4)]

$$v_n = \frac{Ze^2}{4\pi\epsilon_0} \cdot \frac{1}{n\hbar} \quad \text{and} \quad r_n = \frac{n\hbar}{m_e v_n} = n^2 a_0 / Z \quad (6.1.6)$$

The corresponding kinetic and potential energies are [see $\mathcal{R}$ in Eq. (6.1.4)]

$$K(v_n) = \frac{m_e}{2} v_n^2 = \frac{Z^2}{n^2} \mathcal{R} \quad \text{and} \quad V(r_n) = -\frac{e^2}{4\pi\epsilon_0 r_n} = -2 \frac{Z^2}{n^2} \mathcal{R} \quad (6.1.7)$$

The total energy is then

$$E_n = K(v_n) + V(r_n) = -\frac{Z^2}{n^2} \mathcal{R}; \quad n = 1, 2, \dots \quad (6.1.8)$$

- Not only Eq. (6.1.8) agrees with the exact Eq. (6.1.3), the kinetic and potential energies, Eq. (6.1.7), satisfy also the virial theorem; see Eq. (5.4.62). The Bohr's theory offers some valuable insights. It will further be referred in this Chapter whenever it fits; see, for example, Eq. (6.1.6) versus Eq. (6.1.20).

6.1.2 Radial wave functions

A. General features for systems with $V(r=0) \rightarrow -\infty$

- As a central force system, the wave functions go by the generic form,

$$\Psi_{k;l,m}(\mathbf{r}) = R_{k;l}(r)Y_{l,m}(\theta, \phi) \quad (6.1.9)$$

with $Y_{l,m}(\theta, \phi)$, the spherical harmonics, being detailed in §5.3.1. The radial part, $R_{k;l}(r)$, Eq.(5.4.7), expressed in terms of $\psi_{k;l}(r) \equiv rR_{k;l}(r)$, reads

$$\left[-\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial r^2} + \frac{\hbar^2 l(l+1)}{2\mu r^2} + V(r) \right] \psi_{k;l}(r) = E \psi_{k;l}(r) \quad (6.1.10)$$

with $\psi_{k;l}(r=0) = \psi_{k;l}(r \rightarrow \infty) = 0$.

- General features for systems with $V(r=0) \rightarrow -\infty$ and $V(r \rightarrow \infty) = 0$:

$$\psi_{k;l}(r) \equiv rR_{k;l}(r) \sim e^{-\kappa r} \cdot r^{l+1} \cdot (k^{\text{th}}\text{-order polynomial}) \quad (6.1.11)$$

(i) The cusp $e^{-\kappa r}$, where $\kappa = \sqrt{2\mu|E|}/\hbar$, with the bound-state energies ($E < 0$) yet to be solved. The cusp component is universal for systems with $V(r=0) \rightarrow -\infty$, whereas $V(r \rightarrow \infty) = 0$. These include such as the Coulomb potential and the delta-potential well studied in §4.1.3; see Eq. (4.1.40). Formally, $e^{-\kappa r}$ arises from $\psi'' \simeq \kappa^2 \psi$, the asymptotic Eq. (6.1.10) in the $r \rightarrow \infty$ regime.

(ii) The second component, r^l , associated with $R_{k;l}(r)$ is the generic feature, arising from the centrifugal potential in Eq. (6.1.10). This had been demonstrated in §5.4.2 and §5.4.3, with two remarkably distinct scenarios.

(iii) In Eq. (6.1.11) the k^{th} -order polynomial component is responsible for the fact of $\psi_{k;l}(r) \equiv rR_{k;l}(r)$ has the specified number of nodes.

B. Solutions to hydrogenlike atomic systems

- Exploit the internal dimensionless radius and energy [cf. Eq. (6.1.4)],

$$\rho \equiv \frac{Zr}{a_0} \quad \text{and} \quad \varepsilon \equiv -\frac{E}{Z^2 \mathcal{R}} \quad (6.1.12)$$

The resulting Eq. (6.1.10), with $\mu \approx m_e$ and the Coulomb potential, $V(r)$ of Eq. (6.1.1), reads

$$\psi_{k;l}''(\rho) = \left[\frac{l(l+1)}{\rho^2} - \frac{2}{\rho} + \varepsilon \right] \psi_{k;l}(\rho) \quad (6.1.13)$$

Consider the solutions the form of Eq. (6.1.11),

$$\psi_{k;l}(\rho) = e^{-\sqrt{\varepsilon}\rho} \rho^{l+1} \chi_{k;l}(\rho) \quad (6.1.14)$$

Substituting it into Eq. (6.1.13) results in

$$\rho \chi_{k;l}''(\rho) + 2(l+1 - \sqrt{\varepsilon}\rho) \chi_{k;l}'(\rho) + 2[1 - \sqrt{\varepsilon}(l+1)] \chi_{k;l}(\rho) = 0 \quad (6.1.15)$$

for the k^{th} -order polynomial, as discussed in Eq. (6.1.11); i.e.,

$$\chi_{k;l}(\rho) = \sum_{j=0}^k c_j \rho^j \quad (6.1.16)$$

It is easy to obtain [cf. Eqs. (5.4.48)–(5.4.52) for the standard method]

$$\frac{c_{j+1}}{c_j} = 2 \frac{\sqrt{\varepsilon}(j+l+1)-1}{(j+1)(j+2l+2)} \quad (6.1.17)$$

The resulting $c_{j_{\max}+1} = c_{k+1} = 0$ gives rise to

$$\boxed{\sqrt{\varepsilon} = \frac{1}{k+l+1} = \frac{1}{n}}; \quad (n = k+l+1) \quad (6.1.18)$$

The corresponding eigenenergy in Eq. (6.1.12) reads

$$E_n = E_{k;l} = -\frac{Z^2}{n^2} \mathcal{R}; \quad n = 1, 2, \dots \quad (6.1.19)$$

This is Eq. (6.1.3), with noting that $k = 0, 1, \dots$, as it represents the number of nodes. The resulting $n = k+l+1$ gives rise to not only $n = 1, 2, \dots$, but also $l_{\max} = n-1$, as specified in Eq. (6.1.2).

Remarks: Evidently, $R_{n,l}(r)$ has $(n-l-1)$ number of poles. Moreover, from Eq. (6.1.14) with Eq. (6.1.18), it is readily to obtain

$$\max [|r R_{n,n-1}(r)|^2] \quad \text{at} \quad r_n^{\max} = n^2 a_0 / Z \quad (6.1.20)$$

This is the truth about the Bohr's result, Eq. (6.1.6).

• To proceed, we denote

$$\lambda \equiv \frac{2}{n} \quad \text{and} \quad \alpha \equiv 2l+1 \quad (6.1.21)$$

and recast Eq. (6.1.17) as

$$\frac{c_j}{c_{j-1}} = -\frac{k+1-j}{(\alpha+j)j} \lambda \quad \Rightarrow \quad c_j = (-1)^j \frac{\alpha!}{(\alpha+j)!} \binom{k}{j} \lambda^j \quad (6.1.22)$$

with $c_0 \equiv 1$. Let us recast Eq. (6.1.16) as $\chi_k^{(2l+1)}(\rho) \equiv \chi_{k;l}(\rho)$. We obtain

$$\chi_k^{(\alpha)}(\rho) = \sum_{j=0}^k (-1)^j \frac{\alpha!}{(\alpha+j)!} \binom{k}{j} (\lambda \rho)^j \quad (6.1.23)$$

Note that $\chi_k^{(\alpha)}(0) = 1$. The normalized radial wave function reads

$$R_{n,l}(\rho) = N_{n,l} e^{-\rho/n} \rho^l \chi_{n-l-1}^{(2l+1)}(\rho) \quad (6.1.24a)$$

with

$$N_{n,l} = \frac{(2/n)^{l+1}}{n(2l+1)!} \left[\frac{(n+l)!}{(n-l-1)!} \right]^{\frac{1}{2}} \quad (6.1.24b)$$

(i) The normalization goes by

$$\int_0^\infty \rho^2 R_{n,l}^2(\rho) d\rho = 1 \quad (6.1.25)$$

(ii) Radical wave functions, Eq. (6.1.24), with $n \leq 3$.

$$R_{1s}(\rho) = 2e^{-\rho}; \quad \left(\rho = \frac{Zr}{a_0}\right) \quad (6.1.26a)$$

$$R_{2s}(\rho) = \frac{e^{-\rho/2}}{\sqrt{2}} \left(1 - \frac{1}{2}\rho\right), \quad R_{2p}(\rho) = \frac{e^{-\rho/2}}{2\sqrt{3!}} \rho \quad (6.1.26b)$$

$$\begin{aligned} R_{3s}(\rho) &= \frac{2e^{-\rho/3}}{3\sqrt{3}} \left(1 - \frac{2}{3}\rho + \frac{2}{27}\rho^2\right) \\ R_{3p}(\rho) &= \frac{8e^{-\rho/3}}{27\sqrt{3!}} \rho \left(1 - \frac{1}{6}\rho\right), \quad R_{3d}(\rho) = \frac{8e^{-\rho/3}}{81\sqrt{5!}} \rho^2 \end{aligned} \quad (6.1.26c)$$

(iii) The radical wave functions can be recast in terms of the Laguerre polynomials. That is $\chi_k^{(\alpha)}(\rho) = L_k^{(\alpha)}(\lambda\rho)/L_k^{(\alpha)}(0)$; see Eq. (6.1.29), with Eqs. (6.1.30)–(6.1.32), in particular, the orthonormality and recursion relations. Their usage, as highlighted in Eqs. (6.1.35)–(6.1.40), will in general give rise to

$$\frac{\nu+1}{n^2} \langle \rho^\nu \rangle_{n,l} - (2\nu+1) \langle \rho^{\nu-1} \rangle_{n,l} + \frac{\nu}{4} [(2l+1)^2 - \nu^2] \langle \rho^{\nu-2} \rangle_{n,l} = 0 \quad (6.1.27)$$

where $\nu = -2l, -2l+1, \dots$. At $\nu = 0$, it reproduces $\langle 1/\rho \rangle_{n,l} = 1/n^2$, a result of the virial theorem.

Exercises:

E6.1.1 Confirm that $R_{3p}(\rho)$ is from Eq. (6.1.24), with the normalization of Eq. (6.1.25).

E6.1.2 Confirm that Eq. (6.1.27) reproduces Eqs. (5.4.75) and (5.4.76), also

$$\left\langle \frac{a_0}{r} \right\rangle_{n,l} = \frac{Z}{n^2} \quad \text{and} \quad \left\langle \frac{r}{a_0} \right\rangle_{n,l} = \frac{1}{2Z} [3n^2 - l(l+1)] \quad (6.1.28)$$

E6.1.3 (a) Plot $R_{n,l}(\rho)$, for all those with $n \leq 3$. Arrange these six curves in individual frames of a same figure; (b) $|rR_{n,l}(\rho)|^2$, for comparison between 3s, 3p and 3d orbitals; (c) Discuss the related Eq. (6.1.42) and the penetration-and-shielding mechanism underlying many-electron atoms.

E6.1.4 Evaluate $\langle R_{1s}|R_{3p} \rangle$ and $\langle R_{3d}|R_{3p} \rangle$. Comment on the results.

E6.1.5 Obtain the value of $dR_{nl}(r=0)/dr$.

[‡]**C. Radical wave functions via Laguerre polynomials** (*For information*)

- Let us recast Eq. (6.1.23) as

$$\chi_k^{(\alpha)}(\rho) = \frac{\alpha! k!}{(\alpha + k)!} \sum_{j=0}^k \frac{(-1)^j}{j!} \binom{\alpha + k}{\alpha + j} (\lambda \rho)^j = \frac{\alpha! k!}{(\alpha + k)!} L_k^{(\alpha)}(\lambda \rho) \quad (6.1.29)$$

Here, $L_k^{(\alpha)}(x)$ is the Laguerre polynomial, with $x = \lambda \rho = 2\rho/n$

- The Laguerre polynomials (*Handbook of Mathematical Functions*):

$$L_k^{(\alpha)}(0) = \frac{(\alpha + k)!}{\alpha! k!}, \quad \text{with setting } \frac{\partial^k}{\partial x^k} L_k^{(\alpha)}(x) \equiv (-1)^k \quad (6.1.30)$$

$$\int_0^\infty e^{-x} x^\alpha L_k^{(\alpha)}(x) L_{k'}^{(\alpha)}(x) dx = \frac{(\alpha + k)!}{k!} \delta_{kk'} \quad (6.1.31)$$

$$\begin{aligned} L_0^{(\alpha)}(x) &= 1; & L_1^{(\alpha)}(x) &= \alpha + 1 - x \\ (k+1)L_{k+1}^{(\alpha)}(x) &= (\alpha + 2k + 1 - x)L_k^{(\alpha)}(x) - (\alpha + k)L_{k-1}^{(\alpha)}(x) \end{aligned} \quad (6.1.32)$$

- The new variable of presentation would be [cf. Eqs. (6.1.12) and (6.1.21)]

$$\varrho \equiv \lambda \rho = \frac{2\rho}{n} = \frac{2Zr}{na_0} \quad (6.1.33)$$

The transformed Eq. (6.1.28) reads

$$\left\langle \frac{1}{\varrho} \right\rangle_{n,l} = \frac{1}{2n} \quad \text{and} \quad \langle \varrho \rangle_{n,l} = \frac{1}{n} [3n^2 - l(l+1)] \quad (6.1.34)$$

The first one is a well known expression and the second one is to be derived. The transformed Eq. (6.1.24), expressed in terms of $\psi(\varrho) \equiv \varrho R(\varrho)$, reads

$$\psi_k(\varrho) = \mathcal{N}_k (e^{-\varrho} \varrho^{\alpha+1})^{\frac{1}{2}} L_k^{(\alpha)}(\varrho); \quad \alpha = 2l + 1 \quad (6.1.35)$$

Individual wave functions, with the variable $\varrho = 2\rho/n$ depending on $k = n - l - 1$, would be *nonorthogonal* to others. The normalization is given by

$$\int_0^\infty \psi_k^2(\varrho) d\varrho = \mathcal{N}_k^2 \int_0^\infty e^{-\varrho} \varrho^{\alpha+1} [L_k^{(\alpha)}(\varrho)]^2 d\varrho = 1 \quad (6.1.36)$$

Consider now [cf. Eq. (6.1.31)]

$$\left\langle \frac{1}{\varrho} \right\rangle_{n,l} = \mathcal{N}_k^2 \int_0^\infty e^{-\varrho} \varrho^\alpha [L_k^{(\alpha)}(\varrho)]^2 d\varrho = \mathcal{N}_k^2 \frac{(\alpha + k)!}{k!} = \frac{1}{\alpha + 2k + 1} \quad (6.1.37)$$

The last identity follows that of Eq. (6.1.34), with noting that $2n = \alpha + 2k + 1$.

- To evaluate $\langle \varrho \rangle_{n,l}$ in Eq. (6.1.34), we exploit the recursion Eq. (6.1.32) and obtain

$$\varrho^2 L_k^{(\alpha)}(\varrho) = A L_k^{(\alpha)}(\varrho) + \left\{ \text{Terms of } L_{k' \neq k}^{(\alpha)}(\varrho) \right\} \quad (6.1.38)$$

where [noting that $\alpha = 2l + 1$ and $n = k + l + 1 = (\alpha + 2k + 1)/2$]

$$A = (\alpha + 2k + 1)^2 + (\alpha + k)k + (k + 1)(\alpha + k + 1) = 2[3n^2 - l(l + 1)] \quad (6.1.39)$$

Together with Eq. (6.1.31), followed by using Eq. (6.1.37), we obtain

$$\langle \varrho \rangle_{n,l} = A \mathcal{N}_k^2 \frac{(\alpha + k)!}{k!} = \frac{A}{\alpha + 2k + 1} = \frac{A}{2n} = \frac{1}{n} [3n^2 - l(l + 1)] \quad (6.1.40)$$

This is the second identity in Eq. (6.1.34), the equivalence to Eq. (6.1.28), which represents two special values arising from Eq. (6.1.27). The latter can in general be obtained via the recursion relation of Laguerre polynomials. (*Q.E.D.*)

6.1.3 Empirical rules of many-electron atoms

A. Penetration and shielding

• Many-electrons systems involve Coulomb interactions between electrons. In a given atom, electrons are *indistinguishable* from each other. The total N -electrons wave functions, including spins, must be antisymmetric with respect to the permutation of any two electrons, as they are *fermions*, with spin quantum number $s = \frac{1}{2}$. This is concerned with the *Pauli exclusion principle* that will be discussed in §8.1. Exact solutions to many-particles systems can only be obtained numerically.

• On the other hand, effective one-electron models would be useful, in particular, for a *qualitative* understanding the basic properties of atoms, with respect to their locations in the Mendeleev Periodic Table. Physically, the effective models treat individual electron moving in the mean-field of other electrons, in addition to the local Coulomb interaction with the nucleus. The resulting mean-field orbital energy, $E_{n,l}$, would read in the form like Eq. (6.1.10), but with an effective core charge of Z_{eff} ,

$$E_{n,l} \approx -\frac{Z_{\text{eff}}^2}{n^2} \mathcal{R} \quad (\text{Coulomb mean-field}) \quad (6.1.41)$$

The degeneracy removal in different subshell index values of l can be understood with the *penetration and shielding* picture, as follows.

(i) Consider the centrifugal repulsion potential, $V_l(r) = l(l+1)/(2\mu r^2)$. The smaller value of l , the less repulsive $V_l(r)$ will be. In terms of the hydrogenlike radial functions, $R_{n,l}(r)$ is less stretched than $R_{n,l+1}(r)$.

(ii) As results, for example, the 3s electron is more *penetrative*, closer to the nuclei, and effectively shields the nuclei charge, so that the 3p electron feels smaller Z_{eff} , and so on. The penetration-and-shielding mechanism is a mean-field picture, resulting in

$$Z_{\text{eff}}(3s) > Z_{\text{eff}}(3p) > Z_{\text{eff}}(3d) \quad (6.1.42)$$

and the consequence of

$$E_{3s} < E_{3p} < E_{3d} \quad (\text{many-electron atoms}) \quad (6.1.43)$$

(iii) Note also that $E_{1s} < E_{2s} < E_{2p} < E_{3s}$. Moreover,

$$\underbrace{E_{3p} < E_{4s}}_{n+l=4} < \underbrace{E_{3d} < E_{4p} < E_{5s}}_5 < \underbrace{E_{4d} < E_{5p} < E_{6s}}_6 < \underbrace{E_{4f} < E_{5d} < E_{6p} < E_{7s}}_7$$

This is the empirical order of electrons filling in shells and subshells, for the ground-state *configurations* of most atoms (up to $Z = 88$). In convention, one also uses capital letters for the principal quantum numbers, such as the K-shell ($n = 1$), the L-shell ($n = 2$), the M-shell ($n = 3$), and so on.

B. Configuration versus term symbols of atoms

• Electrons in an atom are indistinguishable, but orbitals are. According to the Pauli exclusion principle, each spin-orbital, $|n, l, m, m_s = \pm \frac{1}{2}\rangle$, can at most be occupied by one electron. **Configuration** describes the occupation numbers at the (n, l) -level, such that $1s^2 2s^2 2p^2$ for carbon atom, with the order of orbital energies described earlier. *Configuration is a single-electron (orbital-level) approximation.*

• Many-electron effects are rooted at Coulomb interaction, spin correlation and spin-orbit coupling. *For light atoms, spin-orbit coupling is relatively weak.* The corresponding many-particle corrections enter through the total orbital, spin and their composite angular momentums,

$$\hat{\mathbf{L}} = \hat{\mathbf{l}}_1 + \cdots + \hat{\mathbf{l}}_N, \quad \hat{\mathbf{S}} = \hat{\mathbf{s}}_1 + \cdots + \hat{\mathbf{s}}_N \quad \text{and} \quad \hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}} \quad (6.1.44)$$

via their quantum numbers, L, S and

$$J = L + S, L + S - 1, \cdots, |L - S| \quad (6.1.45)$$

the standard *angular momentum series* (Clebsch-Gordan series).

• Equation (6.1.44) is concerned with the $(L + S)$ -coupling scheme, in relation to a succinct **term symbol** [cf. Eq. (6.1.47)], for many-electron effects in light atoms. For heavy atoms, where the spin-orbit coupling is strong, the jj -coupling scheme becomes preferred, with first $\hat{\mathbf{j}}_k = \hat{\mathbf{l}}_k + \hat{\mathbf{s}}_k$ and then $\hat{\mathbf{J}} = \hat{\mathbf{j}}_1 + \cdots + \hat{\mathbf{j}}_N$. Moreover,

$$M_J = -J, -J + 1, \cdots, J - 1, J \quad (6.1.46)$$

for the magnetic degeneracy that can be lifted by external magnetic fields.

• In the $(L + S)$ -coupling scheme, each configuration splits into several terms at three individual hierarchical steps,

$$(\text{Configuration}) \Rightarrow \boxed{{}^{2S+1}L} \Rightarrow \boxed{{}^{2S+1}L_J} \Rightarrow \boxed{{}^{2S+1}L_{J;M_J}} \quad (6.1.47)$$

(a) In a given ${}^{2S+1}L$, the total number of states is $(2S+1)(2L+1)$, whereas the number of states in a given ${}^{2S+1}L_J$ is $2J+1$.

(b) The orbital $L = 0, 1, 2, \cdots$ go by the letters, S, P, D and so on. Others are reported by their values, including $2S+1$ the spin-multiplicity.

(c) As the term symbol is concerned, closed sub-shells, s^2, p^6, d^{10} and so on, contribute only with 1S . Moreover, the *particle-hole symmetry* gives rise to the same set of terms, such that

Configuration	Terms	Configuration	Terms
p^1 and p^5	2P	p^2 and p^4	${}^3P, {}^1D, {}^1S$
p^3	${}^4S, {}^2D, {}^2P$	d^2 and d^8	${}^3F, {}^3P, {}^1G, {}^1D, {}^1S$

(d) *Two-electron configurations*, illustrated with $l_1 = l_2 = 1$, where $L = 2, 1, 0$ and $S = 1, 0$ [cf. Eq. (6.1.45)]; thus $\{^3D, ^3P, ^3S, ^1D, ^1P, ^1S\}$. All these emerge from $2p^1 3p^1$ and there are $6^2 = 36$ states in total. However, $2p^2$ has only $\binom{6}{2} = 15$ states, for two equivalent electrons distributed over six spin-specified p-orbitals. The survival terms are only 3P , 1D and 1S , due to the Pauli exclusion principle. *The rule about the value of $L + S$ being even is general for the equivalent two-electron case.*

(e) As mentioned earlier, heavy atoms possess strong spin-orbit coupling. In this case, the jj -scheme becomes more appropriate. The use of ^{2S+1}L is then just for labelling, with little relation to the physical reality.

• **Hund's rules** (1927): For those $\{^{2S+1}L_J\}$ arising from a given configuration, the **lower energies** go by (i) **first larger S** and **then larger L** , followed **further** by (ii) **smaller or larger J** , **before or after** the subshell configuration is half-filled, respectively.

Illustrations: Both p^2 and p^4 give rise to $\{^3P, ^1D, ^1S\}$, with the energy increasing order, according to the rule-(i) above. The rule-(ii) suggests further the energy order of $^3P_0 < ^3P_1 < ^3P_2$ from p^2 , whereas the reversed order, $^3P_2 < ^3P_1 < ^3P_0$, for those from p^4 .

• **Optical selection rules** (for single-photon processes):

$$\begin{cases} \Delta l = \pm 1 & (\text{as only one electron involved}) \\ \Delta S = 0 & (\text{may fail for heavy atoms}) \\ \Delta L = 0, \pm 1 \\ \Delta J = 0, \pm 1 & (\text{but } J = 0 \leftrightarrow J = 0) \end{cases} \quad (6.1.48)$$

Remarks: In spectroscopy, term symbols often go also with the principal quantum number. For example, the valent electron of Na atom in $3s^1$ goes by $3^2S_{\frac{1}{2}}$, whereas $3p^1$ splits into $3^2P_{\frac{1}{2}}$ and $3^2P_{\frac{3}{2}}$. The underlying optical transitions are

$$3^2S_{\frac{1}{2}} \xleftarrow{589.6 \text{ nm}} 3^2P_{\frac{1}{2}} \quad \text{and} \quad 3^2S_{\frac{1}{2}} \xleftarrow{589.0 \text{ nm}} 3^2P_{\frac{3}{2}} \quad (6.1.49)$$

These are the famous *yellow double-lines* of Na emission spectrum.

C. Hybrid atomic orbitals

In chemistry, encountered often are hybrid atomic orbitals, such as sp^3 and sp^2 , in molecular environments. The set of hybrid orbitals satisfy the three requirements: **(a)** *Orthonormality*, $\langle h_j | h_k \rangle = \delta_{jk}$; **(b)** *Completeness* – the total contribution from each s or p orbital is 1; **(c)** *Directionality* as dictated by the p-orbital components in individual h_j -orbitals. While **(a)** and **(b)** go by the standard basis set transformation, **(c)** is chemistry related, toward a suitable bonding geometry.

- The standard sp^3 hybrid of C atom, which promote 2s, $2p_x$, $2p_y$ and $2p_z$ to four equivalent atomic orbitals:

$$\begin{aligned} h_1(sp^3) &= \frac{1}{2}(s + p_x + p_y + p_z) \\ h_2(sp^3) &= \frac{1}{2}(s - p_x - p_y + p_z) \\ h_3(sp^3) &= \frac{1}{2}(s - p_x + p_y - p_z) \\ h_4(sp^3) &= \frac{1}{2}(s + p_x - p_y - p_z) \end{aligned} \quad (6.1.50)$$

These four equivalent sp^3 orbitals orientate along the tetrahedral axes, with the angle of 109.47° between them.

- To demonstrate the *nonequivalent* case, consider the three sp^2 orbitals of atom C in formaldehyde, CH_2O . The one pointing to O differs from those pointing to individual H. Let CH_2O define the xy -plane, $C \rightarrow O$ be the x -direction, and ϑ the angle between two O-H bonds. The three nonequivalent sp^2 orbitals of atom C are determined by setting

$$\begin{aligned} h_1 &= \frac{\varphi_{2s} + a\varphi_{2p_x}}{\sqrt{1+a^2}}, \quad h_2 = \frac{\varphi_{2s} - b\varphi_{2p_x} + c\varphi_{2p_y}}{\sqrt{1+b^2+c^2}} \\ h_3 &= \frac{\varphi_{2s} - b\varphi_{2p_x} - c\varphi_{2p_y}}{\sqrt{1+b^2+c^2}} \end{aligned} \quad (6.1.51)$$

The orthogonality gives rise to $1 - ab = 0$ and $1 - b^2 - c^2 = 0$, whereas the angle ϑ between h_2 and h_3 amounts to $b^2 - c^2 = (b^2 + c^2) \cos \vartheta$. These three equations lead to the positive parameters in Eq. (6.1.51)

$$a = \left(\frac{-2 \cos \vartheta}{1 + \cos \vartheta} \right)^{\frac{1}{2}}, \quad b = \left(\frac{1 + \cos \vartheta}{-2 \cos \vartheta} \right)^{\frac{1}{2}}, \quad c = \left(\frac{1 - \cos \vartheta}{-2 \cos \vartheta} \right)^{\frac{1}{2}} \quad (6.1.52)$$

The standard (equivalent) sp^2 orbitals, where $\vartheta = 120^\circ$, go by $a = \sqrt{2}$, $b = 1/\sqrt{2}$ and $c = \sqrt{3/2}$.

6.2 Electromagnetic Properties of Atoms

6.2.1 Electronic current density in hydrogen atom

A. Time-dependent Schrödinger equation and probability flux

Let us revisit the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial \Psi(\mathbf{r}, t)}{\partial t} = \left[-\frac{\hbar^2}{2\mu} \nabla^2 + V(\mathbf{r}) \right] \Psi(\mathbf{r}, t) \quad (6.2.1)$$

Consider the wave function the form of

$$\Psi(\mathbf{r}, t) \equiv a(\mathbf{r}, t) e^{i\varphi(\mathbf{r}, t)} \quad (6.2.2a)$$

with the Born's *probability density*,

$$\rho(\mathbf{r}, t) \equiv |\Psi(\mathbf{r}, t)|^2 = a^2(\mathbf{r}, t) \quad (6.2.2b)$$

Both $a(\mathbf{r}, t)$ and $\varphi(\mathbf{r}, t)$ are real scalar field function of space $\mathbf{r}$, with time t being a parameter. Physically, the above form of $\Psi(\mathbf{r}, t)$ is closely related to the Bohn's *probability flux*, $\mathcal{J}(\mathbf{r}, t)$, to be given in terms of the probability density $\rho(\mathbf{r}, t) = a^2(\mathbf{r}, t)$ and phase $\varphi(\mathbf{r}, t)$. The details are as follows.

First of all, the Schrödinger equation (6.2.1) gives rise to

$$\frac{\partial \rho}{\partial t} = \frac{i\hbar}{2\mu} (\Psi^* \nabla^2 \Psi - \Psi \nabla^2 \Psi^*) = \frac{i\hbar}{2\mu} \nabla \cdot (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*) \quad (6.2.3)$$

Recast it in the form of continuity equation,

$$\frac{\partial \rho(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathcal{J}(\mathbf{r}, t) = 0 \quad (6.2.4)$$

with the Bohn's probability flux,

$$\mathcal{J}(\mathbf{r}, t) \equiv -\frac{i\hbar}{2\mu} (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*) = \frac{\hbar}{\mu} \text{Im}(\Psi^* \nabla \Psi) \quad (6.2.5)$$

On the other hand, from Eq. (6.2.2), we have

$$\Psi^*(\mathbf{r}, t) \nabla \Psi(\mathbf{r}, t) = a(\mathbf{r}, t) \nabla a(\mathbf{r}, t) + i\rho(\mathbf{r}, t) \nabla \varphi(\mathbf{r}, t) \quad (6.2.6)$$

Comparing it with Eq. (6.2.5) results in the probability flux the final expression,

$$\mathcal{J}(\mathbf{r}, t) = \frac{\hbar}{\mu} \rho(\mathbf{r}, t) \nabla \varphi(\mathbf{r}, t) \quad (6.2.7)$$

arising from the wave function the form of Eq. (6.2.2).

As well-known, the gradient of a scale field is of zero-curl. Hence,

$$\nabla \times \frac{\mathcal{J}(\mathbf{r}, t)}{\rho(\mathbf{r}, t)} = \frac{\hbar}{\mu} \nabla \times \nabla \varphi(\mathbf{r}, t) = 0 \quad (6.2.8)$$

The multiple solutions of $\varphi(\mathbf{r}, t)$ differ only by real constants that are the physically irrelevant overall phases to $\Psi(\mathbf{r}, t)$.

B. Electronic probability flux in hydrogen atom

In the following, we exploit Eq. (6.2.7) to examine the electronic probability flux, the Bohn's electron probability current, underlying the steady-states of hydrogen-like atoms,

$$\Psi_{n,l,m}(\mathbf{r}) = R_{n,l}(r)Y_{l,m}(\theta, \phi) = \frac{1}{\sqrt{2\pi}}R_{n,l}(r)\Theta_{l,m}(\theta)e^{im\phi} \quad (6.2.9)$$

The phase function is $\varphi(\mathbf{r}) = m\phi$. On the other hand [cf. . Eq (3.3.16)],

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} = \begin{bmatrix} \sin\theta \cos\phi & \cos\theta \cos\phi & -\sin\phi \\ \sin\theta \sin\phi & \cos\theta \sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{1}{r} \frac{\partial}{\partial \theta} \\ \frac{1}{r \sin\theta} \frac{\partial}{\partial \phi} \end{bmatrix} \quad (6.2.10)$$

We obtain

$$\nabla\varphi(\mathbf{r}) = \frac{m}{r \sin\theta} \mathbf{n}_\phi \quad (6.2.11)$$

where

$$\mathbf{n}_\phi \equiv -\mathbf{i} \cdot \sin\phi + \mathbf{j} \cdot \cos\phi = (\mathbf{k} \times \mathbf{r})/(r \sin\theta) \quad (6.2.12)$$

The resulting Bohn's electronic current, Eq. (6.2.7), reads

$$\mathcal{J}_{n,l,m}(\mathbf{r}) = \hbar m \frac{\rho_{n,l,m}(\mathbf{r})}{m_e r \sin\theta} \mathbf{n}_\phi = \hbar m \frac{R_{n,l}^2(r)\Theta_{l,m}^2(\theta)}{2\pi m_e r \sin\theta} \mathbf{n}_\phi \quad (6.2.13)$$

The unit direction vector, $\mathbf{n}_\phi$, Eq. (6.2.12) for the current, lies in the xy -plane and is also perpendicular to $\mathbf{r}$. There is no current when $m = 0$.

Remarks:

- The circulating current of charge particle carries an *orbit* magnetic moment, $\mathbf{M}_L$, along the direction of angular momentum. This concludes that $\mathbf{M}_L \propto \mathbf{L}$ and further $\mathbf{M}_S \propto \mathbf{S}$, for the *spin magnetic moment*; see Eqs. (6.2.16)–(6.2.20).
- Spin-orbit coupling would arise and contributes to atoms the fine structure that will be elaborated in §6.3. The total composite magnetic moment is

$$\hat{\mathbf{M}} = \hat{\mathbf{M}}_L + \hat{\mathbf{M}}_S \quad (6.2.14)$$

This is a vector sum, with two components arising from the total orbital and spin angular momentums, respectively.

- The interaction between system and external (classical) magnetic field $\mathbf{B}$ reads

$$\hat{H}_{\text{int}} = -\hat{\mathbf{M}} \cdot \mathbf{B} \xrightarrow{\mathbf{B}=\mathbf{k}B_z} -\hat{M}_z B_z \quad (6.2.15)$$

Here, the direction of $\mathbf{B}$ is set to be the z -axis. This is in line with the Bohn's electronic current, specified in Eq. (6.2.12) and Eq. (6.2.13). In other words, we can refer the quantum number m via $\hat{L}_z$ to be the *magnetic quantum number*.

- As known, $\mathbf{B}(\mathbf{r}) = \nabla \times \mathbf{A}(\mathbf{r})$ and $\mathbf{E}(\mathbf{r}) = -\partial\mathbf{A}(\mathbf{r})/\partial t - \nabla\phi(\mathbf{r})$, where $\mathbf{A}(\mathbf{r})$ and $\phi(\mathbf{r})$ are the vector and scalar fields, respectively. We will focus on the weak external field limit. The zero-field wave function description remains valid.

6.2.2 Magnetic moment quantizations

A. Magnetic properties

- As just discussed, the orbital angular momentum in hydrogen atom, $\hat{\mathbf{L}}$, gives rise to the magnetic moment [cf. Eq. (6.2.22)],

$$\hat{\mathbf{M}}_L = -\frac{e\hat{\mathbf{L}}}{2m_e} \equiv -\frac{\mu_B\hat{\mathbf{L}}}{\hbar} \quad (6.2.16)$$

The *Bohr's magneton*, the *quantized magnetic moment unit*, is given by

$$\mu_B \equiv \frac{e\hbar}{2m_e} \quad (\sim 9.274 \times 10^{-24} \text{ Joule/Tesla}) \quad (6.2.17)$$

- In parallel, the electron spin angular momentum, $\mathbf{S}$, carries the magnetic moment of

$$\hat{\mathbf{M}}_S = -\frac{\mu_S\hat{\mathbf{S}}}{\hbar} \equiv -g_s\frac{\mu_B\hat{\mathbf{S}}}{\hbar} \quad (6.2.18)$$

The involving Landé g -factor reads (via the quantum field theory)

$$g_s = 2 + (\alpha/\pi) - 0.657(\alpha/\pi)^2 + \cdots \quad (6.2.19)$$

with the fine structure constant,

$$\alpha \equiv \frac{e^2}{4\pi\epsilon_0\hbar c} \simeq \frac{1}{137} \quad (6.2.20)$$

The electron spin magneton μ_S is about twice of the orbital μ_B .

Remarks:

(i) The classical analogue of Eq. (6.2.16) is related to a particle of charge Q and mass m , moving in a circle of the radius $\varrho = (x^2 + y^2)^{\frac{1}{2}}$ in the xy -plane. The current circulating around the z -axis has the value of

$$\mathcal{J} = Q\frac{\dot{\phi}}{2\pi} = \frac{Q}{2\pi m\varrho^2}p_\phi, \quad \text{where } p_\phi = m\varrho^2\dot{\phi} \quad (6.2.21)$$

See Eq. (1.3.14) for the generalized momentum, $p_\phi = m\varrho^2\dot{\phi}$, in the polar coordinates, with noting that $p_\phi = L_z$ remains valid in the spherical coordinates. The resulting magnetic moment goes along the z -direction, with the value of

$$M_z = \pi\varrho^2\mathcal{J} = \frac{Q}{2m}p_\phi = \frac{Q}{2m}L_z \quad (6.2.22)$$

Its quantum correspondence is Eq. (6.2.16), specified with the charge $Q = -e$ and mass $m = m_e$. The name of Bohr's magneton, μ_B of Eq. (6.2.17), is related to his standing wave quantization on the circular orbit motion of electron.

(ii) The nuclear magneton is given by $\mu_N = Ze\hbar/(2m_N) \approx Z\mu_B/2000$, in line with Eq. (6.2.17). However, the nuclear spin magneton, μ_I , is rather complicated, due to its internal structure, with contributions from *both protons and neutrons*. At any rate, $\mu_I/\mu_S \sim 10^{-3}$. The additional energy shift due to the total nuclear magnetic moment contributes only to the hyperfine structure. On the other hand, nucleus spin is response for nuclear magnetic resonance (NMR) spectroscopy, an extremely important technique in molecular structure characterizations.

B. The total spin-and-orbital composite magnetic moment

- The total magnetic moment, Eq. (6.2.14), is given by

$$\hat{\mathbf{M}} = \hat{\mathbf{M}}_L + \hat{\mathbf{M}}_S = -\frac{\mu_B}{\hbar}(\hat{\mathbf{L}} + g_S\hat{\mathbf{S}}) \quad (6.2.23)$$

The resulting system and magnetic field interaction, Eq. (6.2.15), reads

$$\hat{H}_{\text{int}} = -M_z B_z = \frac{\mu_B}{\hbar}(\hat{L}_z + g_S\hat{S}_z)B_z = \Omega_L(\hat{L}_z + g_S\hat{S}_z) \quad (6.2.24)$$

with the Larmor frequency,

$$\Omega_L \equiv \frac{\mu_B B_z}{\hbar} = \frac{eB_z}{2m_e} \quad (6.2.25)$$

We obtain

$$\hat{H}_{\text{int}}|l, m, m_S\rangle = \hbar(m + g_S m_S)\Omega_L|l, m, m_S\rangle \quad (6.2.26)$$

[The Larmor frequency is also called cyclotron frequency. Its relation to the charge-mass ratio and magnetic field strength, Eq. (6.2.25), is used in various contexts, such as discovering particles and measuring magnetic fields of stars in galaxy.]

- Turn to the total angular momentum,

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}} \quad (6.2.27)$$

This is a conserved property, for general systems involving spin-orbit couplings. The basis set $\{|l, j, m_j\rangle\}$ would be preferred.

(a) Note that $\hat{\mathbf{L}} + g_S\hat{\mathbf{S}}$ does not align with $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$. However, there is the so-called *projection* identity,

$$\hat{\mathbf{L}} + g_S\hat{\mathbf{S}} = \left[(\hat{\mathbf{L}} + g_S\hat{\mathbf{S}}) \frac{\hat{\mathbf{J}}}{\hat{J}^2} \right] \hat{\mathbf{J}} \equiv \hat{\Lambda}_{\text{eff}} \hat{\mathbf{J}} \quad (6.2.28)$$

for the projection of the left-hand-side vector onto the right-hand-side direction of $\hat{\mathbf{J}}$. As results [cf. Eq. (6.2.26)],

$$\hat{H}_{\text{int}}|l, j, m_j\rangle = \hbar m_j \Omega_L \hat{\Lambda}_{\text{eff}}|l, j, m_j\rangle \quad (6.2.29)$$

(b) We will show soon below that [cf. Eqs. (6.2.33)–(6.2.35)]

$$\hat{\Lambda}_{\text{eff}}|l, j, m_j\rangle = g_J|l, j, m_j\rangle \quad (6.2.30)$$

with the eigenvalue,

$$g_J = \frac{g_s + 1}{2} - \frac{g_s - 1}{2j(j+1)} \left[l(l+1) - \frac{3}{4} \right] \quad (6.2.31)$$

The resulting Eq. (6.2.29) reads

$$\hat{H}_{\text{int}}|l, j, m_j\rangle = \hbar m_j g_J \Omega_L |l, j, m_j\rangle \quad (6.2.32)$$

Derivations of Eq. (6.2.30) with Eq. (6.2.31): Let us start with

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}} \quad \text{and} \quad \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2) \quad (6.2.33)$$

Hence [cf. Eq. (6.2.28)],

$$\hat{\Lambda}_{\text{eff}} \equiv (\hat{\mathbf{L}} + g_s \hat{\mathbf{S}}) \frac{\hat{\mathbf{J}}}{\hat{J}^2} = \frac{g_s + 1}{2} - \frac{g_s - 1}{2} (\hat{L}^2 - \hat{S}^2) \frac{1}{\hat{J}^2} \quad (6.2.34)$$

Note also that

$$\begin{aligned} \hat{J}^2|l, j, m_j\rangle &= \hbar^2 j(j+1)|l, j, m_j\rangle \\ \hat{L}^2|l, j, m_j\rangle &= \hbar^2 l(l+1)|l, j, m_j\rangle \\ \hat{S}^2|l, j, m_j\rangle &= \hbar^2 s(s+1)|l, j, m_j\rangle \end{aligned}$$

Applying the above expressions for Eq. (6.2.30), we obtain

$$g_J = \frac{g_s + 1}{2} - \frac{g_s - 1}{2j(j+1)} \left[l(l+1) - s(s+1) \right] \quad (6.2.35)$$

and the resulting Eq. (6.2.31), where $s = \frac{1}{2}$ is used. (Q.E.D.)

Remarks:

(a) As known, $m_j = -j, -j+1, \dots, j-1, j$. For hydrogenlike atoms,

$$j = \begin{cases} \frac{1}{2}; & l = 0 \\ l + \frac{1}{2}, l - \frac{1}{2}; & l \neq 0 \end{cases} \quad (6.2.36)$$

This completes Eq. (6.2.31) the g -factor of hydrogenlike atoms. The total number of angular momentum states is $(2l+1)(2s+1)$. This is an invariance property, with respect to different presentations, such as $\{|l, j, m_j\rangle\}$ used in Eq. (6.2.32) and $\{|l, m, m_s\rangle\}$ in Eq. (6.2.26).

(b) Equation (6.2.36) is concerned with $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$. Its amplitude quantum number, which must be nonnegative, goes by Eq. (6.1.45); i.e.,

$$J = L + S, L + S - 1, \dots, |L - S| \quad (6.2.37)$$

The terminal values, $J_{\max} = L + S$ and $J_{\min} = |L - S|$, correspond to that $\hat{L}_z$ and $\hat{S}_z$ are aligned in parallel and antiparallel, respectively. For hydrogenlike atoms, $L = l$ and $S = s = \frac{1}{2}$, resulting in Eq. (6.2.36). The capital letters follow the convention of many-electron atomic systems; see Eq. (6.1.47).

C. System and magnetic B -field interaction

• Consider now $\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$. The field-free atomic system Hamiltonian goes with energies of $E_{n,l}^{(0)} = \langle n, l, j, m_j | \hat{H}_0 | n, l, j, m_j \rangle$. The system-field interaction follows Eq. (6.2.32), with Eq. (6.2.25), having

$$\langle n, l, j, m_j | \hat{H}_{\text{int}} | n, l, j, m_j \rangle = m_j g_J \mu_B B_z \quad (6.2.38)$$

The Landé factor g_J and the Bohr's magneton μ_B are given by Eq. (6.2.35) and Eq. (6.2.17), respectively. The external magnetic field B_z , along the z -axis, leads to the unperturbed energy $E_{n,l}^{(0)}$ splitting into $(2j + 1)$ different values, in relation to the magnetic angular momentum,

$$m_j = -j, -j + 1, \dots, j - 1, j \quad (6.2.39)$$

associated with each individual j from Eq. (6.2.36). The above formalism provides the principle underlying the experimental observations of magnetic angular momentum quantization.

• **Stern–Gerlach experiments:** A beam of atoms passing through a *gradient magnetic field* experiences the force, along the gradient z -direction,

$$F_z \simeq -\frac{\partial H_{\text{int}}}{\partial z} = -m_j g_J \mu_B \frac{\partial B_z}{\partial z} \quad (6.2.40)$$

This is the physical picture of the experiments, in relation to Eq. (6.2.38). As results, the atomic beam splits into symmetric $(2j + 1)$ -bands, hitting the detection plate at distance, assuming $\partial B_z / \partial z \approx \text{constant}$. The atoms with $m_j > 0$ appear in the lower side and $m_j < 0$ in the upper side, with respect to the magnetic field gradient, whereas those of $m_j = 0$ experience no force and reach at the original beam center direction. However, the scenario of $m_j = 0$ exists only when j is integer.

The most important observation in the Stern–Gerlach experiments is the number of the band, the value of $2j + 1$. The celebrated experiment in 1922 on silver atoms results in a double-arcs pattern in the detection plate. This concludes that electron is a spin- $\frac{1}{2}$ particle. More specific, silver atom

ground-state configuration is $[\text{Kr}](4d)^{10}5s$. Its angular momentum goes by $J = \frac{1}{2}$, since $L = 0$ and $S = \frac{1}{2}$. The *atomic core*, $[\text{Kr}](4d)^{10}$, is of overall zero-angular momentum. In this sense, silver atom is a hydrogenlike atom, with the ground-state of $^2S_{\frac{1}{2}}$. It is the electron spin responsible for the observed double-splitting pattern.

• **Zeeman effect** refers to that under a *constant magnetic field*, each optical transition between configurations, such as $ns^2 \leftrightarrow n(s^1p^1)$, splits into three lines. This is the normal Zeeman effect (1896). Anomalous Zeeman effect is about the individual optical level splitting into two or other even numbers of sub-lines.

(a) The Zeeman splitting can be understood in terms of the field *dressed-state* energy [cf. Eq. (6.2.38), where $\mu_B B_z = \hbar\Omega_L$, Eq. (6.2.25)],

$$E_{n,l,j,m_j} = E_{n,l}^{(0)} + \hbar m_j g_J \Omega_L \quad (6.2.41)$$

The optical transition occurs now between two dressed states. Equation (5.3.35) for the spectrum remains valid in general and the electric dipole operator is commutable with spin. The resulting selection rule goes by $\Delta l = \pm 1$ and $\Delta s = 0$ [cf. Eq. (6.1.48)].

(b) In contact with the term symbol in many-electron atoms [cf. Eq. (6.1.47)], the Zeeman effect of magnetic field splits each $^{2S+1}L_J$ into $(2J+1)$ number of sub-lines. Observed now are those of $^{2S+1}L_{J,M_J}$, with Eq. (6.1.46), $M_J = -J, -J+1, \dots, J-1, J$.

6.3 Fine Structures of Hydrogenlike Atoms

6.3.1 Origins of fine structures

- To facilitate the later analysis, we specify the fine structure constant, α of Eq. (6.2.20), the ratio between Compton wavelength and Bohr's radius,

$$\alpha \equiv \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{\lambda_e}{a_0} \simeq \frac{1}{137} \quad (\text{Fine structure constant}) \quad (6.3.1)$$

$$\lambda_e \equiv \frac{\hbar}{m_e c} \simeq 3.86 \times 10^{-3} \text{Å} \quad (\text{Compton wavelength}) \quad (6.3.2)$$

(a) *Einstein's $E = mc^2$ is for spectroscopists.* As known, Compton wavelength, arising from his photon–electron scattering experiments, refers to $\lambda_C \equiv 2\pi\lambda_e$. This wavelength corresponds to the static energy of electron, since Eq. (6.3.2) satisfies [cf. Eq. (2.1.7)]

$$\frac{\hbar c}{\lambda_C} = \frac{\hbar c}{\lambda_e} = m_e c^2$$

(b) The Rydberg constant and Bohr's radius, Eq. (6.1.4), can be recast as

$$\boxed{\mathcal{R} = \frac{\alpha^2}{2} m_e c^2} \quad \text{and} \quad \boxed{a_0 = \frac{\lambda_e}{\alpha}} \quad (6.3.3)$$

(c) Relativity quantum mechanics is described by the Dirac equation. This by nature includes spin degrees of freedom. Hydrogen atom comprises one electron and one proton, both are spin- $\frac{1}{2}$ fermions. This is an analytical solvable system, with rich fine structure and hyperfine structure. The latter is related to the nuclear spin interactions. All these detailed structures can be accurately measured with high-precision spectroscopy.

- As known, hydrogen atom is a weak relativity system. The reference is then the non-relativistic Hamiltonian, $\hat{H}_0$, with [cf. Eqs. (6.1.2) and (6.1.3)]

$$\hat{H}_0 |n, l, m\rangle = E_n^{(0)} |n, l, m\rangle, \quad \text{with} \quad E_n^{(0)} = -\frac{Z^2}{n^2} \mathcal{R}; \quad n = 1, 2, \dots \quad (6.3.4)$$

The fine structure perturbed Hamiltonian is given by

$$\hat{H} = \hat{H}_0 + \hat{W}_{\text{RKE}} + \hat{W}_{\text{SOC}} \quad (\text{Relativistic}) \quad (6.3.5)$$

with $\hat{W}_{\text{RKE}}$ and $\hat{W}_{\text{SOC}}$ denoting the relativity kinetic energy correction and the spin-orbit coupling, respectively. The former goes by a well-defined classical correspondence, but the latter involving spin does not. As detailed in §6.3.2 and §6.3.3, both $\langle \hat{W}_{\text{RKE}} \rangle$ and $\langle \hat{W}_{\text{SOC}} \rangle$ are of the relative α^2 -order, with respect to Eq. (6.3.4) the non-relativistic energies.

6.3.2 Relativity kinetic energy correction

A. Basic picture

In Eq. (6.3.5), the relativity kinetic energy correction is found to be

$$\hat{W}_{\text{RKE}} = \hat{W}_{\text{KEC}} + \hat{W}_{\text{Darwin}} = -\frac{m_e c^2}{8} \left(\frac{\lambda_e \hat{\mathbf{p}}}{\hbar} \right)^4 + \frac{\lambda_e^2}{8} \nabla^2 V(\mathbf{r}) \quad (6.3.6)$$

The correction term, $\hat{W}_{\text{KEC}}$, arises from the Einstein energy-momentum relation, but is treated at the lowest-order level; see Eq. (6.3.8). The Darwin term, $\hat{W}_{\text{Darwin}}$, describes the *self-energy* that compensates the *nonlocality error*, implied in the perturbative $\hat{W}_{\text{KEC}}$. The details are as follows.

• Relativity kinetic energy correlation, $\hat{W}_{\text{KEC}}$

(i) Relativity effects take place when the speed ratio v/c becomes comparable or larger than the fine structure constant, $\alpha \approx 1/137$. In terms of relative kinetic energy, with respect to $m_e c^2$, the fine structure constant enters as α^2 . To that end, it requires spectroscopy a higher than 10^{-5} relative precision, which had been well achieved in electronic spectroscopy experiments for decades. Ultrahigh precision spectroscopy is a foundation of the modern metrology, including the high precision atomic clock used in the Global Position System covering the Earth and also space stations.

(ii) Recall Eq. (6.1.6), the velocity of an electron moving around the Bohr's atomic orbit. Recast the expression in terms of

$$v_n^{\text{Bohr}} = Z\alpha c/n \quad (6.3.7)$$

This suggests that the relativity effect would be important for the inner-core electrons in heavy atoms. On the other hand, hydrogen atom is a weak relativity system. The first-order perturbation treatment on the relativity effects would be enough.

(iii) For a free-moving particle, $E^2 = (mc^2)^2 + (pc)^2$. This is the well-known energy-momentum relation. In the lower-order expansion, it reads

$$E(\text{free-moving}) = mc^2 [1 + (v/c)^2]^{\frac{1}{2}} = mc^2 [1 + \frac{1}{2}(v/c)^2 - \frac{1}{8}(v/c)^4 + \dots]$$

In particular, $v/c = \lambda_e p/\hbar$, for electron, with $p = m_e v$ being the non-relativistic momentum, and λ_e , Eq. (6.3.2), the Compton wavelength. We can then recast the relativity kinetic energy correction [cf. Eq. (6.3.2)],

$$W_{\text{KEC}} = E(\text{free-moving}) - m_e c^2 - \frac{m_e v^2}{2} \simeq -\frac{m_e c^2}{8} \frac{v^4}{c^4} = -\frac{m_e c^2}{8} \left(\frac{\lambda_e p}{\hbar} \right)^4 \quad (6.3.8)$$

This gives rise to the first term in Eq. (6.3.6).

• **Onset of locality versus nonlocality correction, $\hat{W}_{\text{Darwin}}$**

(1) Let us revisit the Coulomb potential [Eq. (6.1.1) with Eq. (6.1.4)],

$$V(\mathbf{r}) = -\frac{Ze^2}{4\pi\epsilon_0 r} = -2Z\mathcal{R}\frac{a_0}{r}, \quad \text{with } r = |\mathbf{r}_e - \mathbf{R}_N| \quad (6.3.9)$$

This is a *point-charge* description, in which an electron, located at $\mathbf{r}_e = \mathbf{r}$, interacting with the electrostatic field, $\phi(\mathbf{r}) = Ze/(4\pi\epsilon_0 r)$, of the nucleus at $\mathbf{R}_N = \mathbf{0}$.

- a) The physical size of nuclei is the order of $\alpha\lambda_e$ and that of electron is even much smaller. Both are irrelevant to the fine structure. In fact, Dirac equation goes also with $V(\mathbf{r})$ of Eq. (6.3.9), the *local* point-charge description for hydrogenlike atoms.
- b) However, exploited in W_{KEC} , Eq. (6.3.8), is the low-order expansion *approximation*. This gives rise to an effective nonlocal $V_{\text{NL}}(\mathbf{r}) \neq V(\mathbf{r})$, no longer the truly required locality type. Express $V_{\text{NL}}(\mathbf{r})$ in terms of a certain electron charge distribution, $f(\mathbf{r}) \neq \delta(\mathbf{r})$, by which

$$V_{\text{NL}}(\mathbf{r}) \equiv \int f(\mathbf{r}')V(\mathbf{r} + \mathbf{r}')d\mathbf{r}' \quad (6.3.10a)$$

Here, $f(\mathbf{r})$ is a spherical distribution function, centered at $f(\mathbf{r} = 0)$, with the broadening width of $\Delta r \sim \lambda_e$, *the order of Compton wavelength* [Eq. (6.3.2)]. On the other hand, $\hat{W}_{\text{KEC}}$, the first term of Eq. (6.3.6), goes by the minimum uncertainty, $\Delta r \Delta p = \hbar/2$. Thus,

$$f(\mathbf{r}) \simeq \delta(\mathbf{r}) + \frac{\lambda_e^2}{8}\delta''(\mathbf{r}) \quad (6.3.10b)$$

(2) The above nonlocality error is then compensated with

$$\hat{W}_{\text{Darwin}}(\mathbf{r}) \equiv V_{\text{NL}}(\mathbf{r}) - V(\mathbf{r}) \simeq \frac{\lambda_e^2}{8} \int \frac{\partial^2 \delta(\boldsymbol{\lambda})}{\partial \lambda^2} V(\mathbf{r} + \boldsymbol{\lambda}) d\boldsymbol{\lambda} = \frac{\lambda_e^2}{8} \nabla^2 V(\mathbf{r}) \quad (6.3.11)$$

This together with Eq. (6.3.8) complete Eq. (6.3.6). (Q.E.D.)

(3) To proceed, we exploit the spherical coordinates delta-function identity,

$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi\delta(\mathbf{r}) \quad (6.3.12)$$

In relation to $V(\mathbf{r})$ of Eq. (6.3.9), we obtain Eq. (6.3.11) the final expression,

$$\boxed{\hat{W}_{\text{Darwin}}(\mathbf{r}) = Z\pi\alpha^2\mathcal{R}a_0^3\delta(\mathbf{r})} \quad (6.3.13)$$

This will be used later evaluation; see Eqs. (6.3.21)–(6.3.23).

B. The fine structure via $\hat{W}_{\text{RKE}}$, Eq. (6.3.6)

The fine structure via $\hat{W}_{\text{RKE}}$, Eq. (6.3.6), removes the subshell degeneracy in Eq. (6.3.4), with (see the derivations soon below)

$$\langle \hat{W}_{\text{RKE}} \rangle_{n,l} = -Z^4 \alpha^2 \mathcal{R} \left[\frac{2}{n^3(2l+1)} - \frac{3}{4n^4} \right] + \delta_{l0} \frac{Z^4 \alpha^2 \mathcal{R}}{n^3} \quad (6.3.14)$$

While the first term arising from $\hat{W}_{\text{KEC}}$ depends explicitly on individual subshell quantum number $l = 0, 1, \dots, n-1$, the second term via $\hat{W}_{\text{Darwin}}$ contributes to the s-orbitals only.

Derivations of Eq. (6.3.14): Denote for later use [cf. Eqs. (6.3.6) and (6.3.13)],

$$\Xi_{n,l} \equiv \frac{\langle \hat{W}_{\text{KEC}} \rangle_{n,l}}{\alpha^2 \mathcal{R}} = -\frac{a_0^4}{4\hbar^4} \langle \hat{\mathbf{p}}^4 \rangle_{n,l} \quad (6.3.15)$$

$$\tilde{\Xi}_{n,l} \equiv \frac{\langle \hat{W}_{\text{Darwin}} \rangle_{n,l}}{\alpha^2 \mathcal{R}} = Z\pi a_0^3 \langle \delta(\mathbf{r}) \rangle_{n,l} \quad (6.3.16)$$

(a) By using $\hat{\mathbf{p}}^4 = [2m_e(\hat{H}_0 - \hat{V})]^2 = 4m_e^2(\hat{H}_0^2 - \hat{H}_0\hat{V} - \hat{V}\hat{H}_0 + \hat{V}^2)$ and Eq. (6.3.4), we obtain

$$\langle \hat{\mathbf{p}}^4 \rangle_{n,l} = 4m_e^2 \left(\frac{Z^4 \mathcal{R}^2}{n^4} + 2 \frac{Z^2 \mathcal{R}}{n^2} \langle \hat{V} \rangle_{n,l} + \langle \hat{V}^2 \rangle_{n,l} \right) \quad (6.3.17)$$

On the other hand, the virial theorem gives rise to

$$\langle \hat{V} \rangle_{n,l} = 2E_n^{(0)} = -2 \frac{Z^2 \mathcal{R}}{n^2} \quad (6.3.18a)$$

whereas Eq. (5.4.75) results in [cf. Eq. (6.1.4)]

$$\langle \hat{V}^2 \rangle_{n,l} = \left(\frac{Ze^2}{4\pi\epsilon_0 a_0} \right)^2 \left\langle \frac{a_0^2}{r^2} \right\rangle_{n,l} = 4Z^2 \mathcal{R}^2 \frac{2Z^2}{n^3(2l+1)} \quad (6.3.18b)$$

We obtain Eq. (6.3.15) the expression,

$$\Xi_{n,l} = -Z^4 \frac{a_0^4 m_e^2}{\hbar^4} \mathcal{R}^2 \left[\frac{8}{n^3(2l+1)} - \frac{3}{n^4} \right] = -Z^4 \left[\frac{2}{n^3(2l+1)} - \frac{3}{4n^4} \right] \quad (6.3.19)$$

where Eqs. (6.3.1)–(6.3.3) are used. This together with Eq. (6.3.15) give rise to

$$\langle \hat{W}_{\text{KEC}} \rangle_{n,l} = -Z^4 \alpha^2 \mathcal{R} \left[\frac{2}{n^3(2l+1)} - \frac{3}{4n^4} \right] \quad (6.3.20)$$

This is the first term of Eq. (6.3.14).

(b) Turn to Eq. (6.3.16),

$$\tilde{\Xi}_{n,l} \equiv \frac{\langle \hat{W}_{\text{Darwin}} \rangle_{n,l}}{\alpha^2 \mathcal{R}} = Z\pi a_0^3 \langle n, l, m | \delta(\mathbf{r}) | n, l, m \rangle = Z\pi a_0^3 |\Psi_{n,l,m}(\mathbf{0})|^2 \quad (6.3.21)$$

For hydrogenlike atoms [cf. Eq. (6.1.24) and Eq. (5.3.1)],

$$|\Psi_{n,l,m}(\mathbf{0})|^2 = \frac{1}{4\pi} R_{n,l}^2(0) = \frac{1}{\pi} \frac{Z^3}{n^3 a_0^3} \delta_{l0} \quad (6.3.22)$$

with $Y_{0,0} = 1/\sqrt{4\pi}$ being applied. We obtain

$$\langle \hat{W}_{\text{Darwin}} \rangle_{n,l} = \frac{Z^4 \alpha^2 \mathcal{R}}{n^3} \delta_{l0} \quad (6.3.23)$$

This is the second term of Eq. (6.3.14).

6.3.3 Spin-orbit coupling and Thomas precession

The spin-orbit coupling, $\hat{W}_{\text{soc}}$ in Eq. (6.3.5), is given by [see Eqs. (6.3.28)–(6.3.32) for the derivations]

$$\hat{W}_{\text{soc}} = \xi(r) \frac{\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}}{\hbar^2}, \quad \text{where } \xi(r) = \frac{\lambda_e^2}{2r} \frac{dV(r)}{dr} \quad (6.3.24)$$

By using Eq. (5.4.76), we have

$$\langle \xi(r) \rangle_{n,l} = Z\alpha^2 \mathcal{R} \left\langle \frac{a_0^3}{r^3} \right\rangle_{n,l} = \frac{2Z^4 \alpha^2 \mathcal{R}}{n^3 l(l+1)(2l+1)} \quad (6.3.25)$$

whereas Eqs. (6.2.33) and (6.2.34) give rise to

$$\left\langle \frac{\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}}{\hbar^2} \right\rangle_{l,j} = \frac{1}{2} \left[j(j+1) - l(l+1) - \frac{3}{4} \right] \quad (6.3.26)$$

We obtain

$$\langle \hat{W}_{\text{soc}} \rangle_{n,l,j} = (1 - \delta_{l0}) Z^4 \alpha^2 \mathcal{R} \frac{j(j+1) - l(l+1) - \frac{3}{4}}{n^3 l(l+1)(2l+1)} \quad (6.3.27)$$

This is nonzero only for $l \neq 0$; see Eq. (6.2.36).

Derivations of Eq. (6.3.24): The spin-orbit coupling arises from the interaction between the electron spin magnetic momentum and the magnetic field, generated by the electron moving around the nucleus, as follows.

(i) Let us revisit the Coulomb potential, Eq. (6.3.4), in terms of $V(\mathbf{r}) = -e\phi(\mathbf{r})$, with $\phi(\mathbf{r}) = Ze/(4\pi\epsilon_0 r)$ denoting the electric scalar potential of nucleus (positive charge of Ze) at $\mathbf{r} = \mathbf{0}$. The corresponding electrostatic field of nucleus reads

$$\mathbf{E}_0(\mathbf{r}) = -\nabla\phi(\mathbf{r}) = \frac{1}{e} \nabla V(\mathbf{r}) = \frac{\mathbf{r}}{er} \frac{dV(\mathbf{r})}{dr} \quad (6.3.28)$$

(ii) The electron (charge of $-e$ and mass m_e), moving at velocity, $\mathbf{v} = \mathbf{p}/m_e$, around the electric $\mathbf{E}_0(\mathbf{r})$, gives rise to the magnetic induction field,

$$\mathbf{B}_{\text{ind}}(\mathbf{r}) = -\frac{\mathbf{p}}{m_e} \times \frac{\mathbf{E}_0(\mathbf{r})}{c^2} = -\frac{1}{m_e c^2} \frac{dV(\mathbf{r})}{dr} \frac{\mathbf{p} \times \mathbf{r}}{er} = \frac{1}{m_e c^2} \frac{dV(\mathbf{r})}{dr} \frac{\mathbf{L}}{er} \quad (6.3.29)$$

Its interaction with the electron spin magnetic momentum, $\mathbf{M}_s$ of Eq. (6.2.18), is responsible for the spin-orbit coupling.

(iii) However, differing from the external magnetic field, $\mathbf{B}_{\text{ind}}(\mathbf{r})$ of Eq. (6.3.29) is induced by the orbital rotation of electron around the nucleus. For the spin-orbit coupling, $W_{\text{soc}}(\mathbf{r})$, the *Coriolis force* should be taken into account, since the spin of electron goes along with its orbital rotation. The resulting spin-orbit coupling energy takes the standard form,

$$W_{\text{soc}}(\mathbf{r}) = -\mathbf{M}_s^{\text{eff}} \cdot \mathbf{B}_{\text{ind}}(\mathbf{r}) \equiv \mu_s^{\text{eff}} \mathbf{S} \cdot \mathbf{B}_{\text{ind}}(\mathbf{r})/\hbar \quad (6.3.30)$$

with the effective or Thomas spin precession magneton,

$$\mu_S^{\text{eff}} = \mu_S - \mu_B = (g_S - 1)\mu_B \quad (6.3.31)$$

See Eqs. (6.3.33)–(6.3.36) for the derivations.

(iv) We obtain Eq. (6.3.30) the expression,

$$W_{\text{soc}}(\mathbf{r}) = \frac{g_S - 1}{2(m_e c)^2} \frac{dV(\mathbf{r})}{dr} \frac{\mathbf{S} \cdot \mathbf{L}}{r} = (g_S - 1) \frac{\lambda_e^2}{2r} \frac{dV(\mathbf{r})}{dr} \frac{\mathbf{S} \cdot \mathbf{L}}{\hbar^2} \quad (6.3.32)$$

At the relative α^2 -order, it is sufficient to take $g_S \approx 2$, the value from the Dirac equation. On the other hand, g_S of Eq. (6.2.19), is the result of quantum field theory, with the α -dependence being referred to the anomalous electron magneton contribution, for μ_S being not exact twice of μ_B . In other words, for the purpose of fine structure, Eq. (6.3.32) is identical to Eq. (6.3.24). (Q.E.D.)

Derivations of Eq. (6.3.30) and Eq. (6.3.31): The electron spin motion is given by

$$\frac{\partial \hat{\mathbf{S}}}{\partial t} = \left(\frac{\mu_S}{\hbar} \mathbf{B}_{\text{ind}} + \boldsymbol{\Omega}_T \right) \times \hat{\mathbf{S}} \quad (\text{Thomas precession}) \quad (6.3.33)$$

with the spin magneton μ_S in Eq. (6.2.18). The Thomas precession frequency reads

$$\boldsymbol{\Omega}_T = \frac{\mathbf{v} \times \nabla V(\mathbf{r})}{2m_e c^2} \quad (6.3.34)$$

with the Coriolis force, $\nabla V(\mathbf{r})$, acting on the electron (charge of $-e$) in orbital motion. Evaluate then [cf. Eq. (6.2.17) and Eq. (6.3.29)]

$$\frac{\hbar}{\mu_B} \boldsymbol{\Omega}_T = \frac{2m_e}{e} \boldsymbol{\Omega}_T = \frac{m_e \mathbf{v} \times \nabla V(\mathbf{r})}{em_e c^2} = -\frac{1}{m_e c^2} \frac{dV(\mathbf{r})}{dr} \frac{\hat{\mathbf{L}}}{er} = -\mathbf{B}_{\text{ind}} \quad (6.3.35)$$

The spin-orbit coupling via Eqs. (6.3.33)–(6.3.35) is given by

$$\hat{W}_{\text{soc}}(\mathbf{r}) = \left(\frac{\mu_S}{\hbar} \mathbf{B}_{\text{ind}} + \boldsymbol{\Omega}_T \right) \cdot \hat{\mathbf{S}} = \frac{1}{\hbar} (\mu_S - \mu_B) \mathbf{B}_{\text{ind}} \cdot \hat{\mathbf{S}} \quad (6.3.36)$$

This is Eq. (6.3.30) with Eq. (6.3.31). (Q.E.D.)

Remarks: In literature and textbooks, reported often is $\mu_S^{\text{eff}} \simeq \mu_S/2$, for its participating in the spin-orbit coupling. Nevertheless, the true value is $\mu_S^{\text{eff}} = \mu_S - \mu_B$, which highlights the *prescription invariance principle*. This resembles to the interaction between the positive charge of nucleus (Ze) and the negative charge of electron ($-e$). The resulting Coulomb potential depends on $r = |\mathbf{r}_e - \mathbf{R}_N|$, no matter whether it is from the aspect that the electron interacts with the electrostatic field of nucleus or the opposite aspect. The underlying prescription invariance principle holds for any mutual interactions in general, including the spin-orbit coupling W_{soc} . It is anticipated that the final result be invariance against swapping the roles of spin and orbital in Eqs. (6.3.28)–(6.3.30).

6.3.4 Final results and remarks

• As mentioned earlier in Eq. (6.3.5), the fine structure comprises the sum of $\langle \hat{W}_{\text{RKE}} \rangle_{n,l}$, Eq. (6.3.14) and $\langle \hat{W}_{\text{SOC}} \rangle_{n,l,j}$, Eq. (6.3.27). Note also that the atomic unit for energy is given by $2\mathcal{R} = \alpha^2 m_e c^2$ [cf. Eq. (6.3.3)]. Report the fine structure to hydrogenlike atom, in terms of

$$\begin{aligned} \frac{\langle \hat{W}_{\text{FS}} \rangle_{n,l,j}}{(Z\alpha)^4 m_e c^2} = & - \left[\frac{1}{n^3(2l+1)} - \frac{3}{8n^4} \right] + \frac{\delta_{l0}}{2n^3} \\ & + (1 - \delta_{l0}) \frac{j(j+1) - l(l+1) - \frac{3}{4}}{2n^3 l(l+1)(2l+1)} \end{aligned} \quad (6.3.37)$$

The second term describes the Darwin reorganization, Eq. (6.3.23). The last term arises from the spin-orbit coupling, Eq. (6.3.27) that is nonzero only for $l \neq 0$. Note also that Eq. (6.2.36),

$$j = \begin{cases} \frac{1}{2}; & l = 0 \\ l \pm \frac{1}{2}; & l \neq 0 \end{cases} \quad (6.3.38)$$

The resulting Eq. (6.3.37), the fine structure of hydrogenlike atoms, is then

$$\boxed{\langle \hat{W}_{\text{FS}} \rangle_{n,l,j} = -m_e c^2 \frac{(Z\alpha)^4}{n^3} \left(\frac{1}{2j+1} - \frac{3}{8n} \right)} \quad (6.3.39)$$

The l -dependence is precisely canceled out at the fine structure level.

Result of the Dirac equation for hydrogenlike atom:

$$\begin{aligned} E_{n,j} &= m_e c^2 \left[1 + \left(\frac{Z\alpha}{n - j - \frac{1}{2} + [(j + \frac{1}{2})^2 - (Z\alpha)^2]^{\frac{1}{2}}} \right)^2 \right]^{-\frac{1}{2}} \\ &= m_e c^2 \left[1 - \frac{(Z\alpha)^2}{2n^2} - \frac{(Z\alpha)^4}{n^3} \left(\frac{1}{2j+1} - \frac{3}{8n} \right) + \cdots \right] \end{aligned} \quad (6.3.40)$$

In the above expansion, the first two terms are the static energy and the nonrelativistic binding energy, respectively. The third term is identical to Eq. (6.3.39), the leading contribution to the fine structure. For hydrogenlike atom, the exact energies at the fine-structure level depend only on n and j . Physically, this is related to the fact that $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$ is a universal conserved property, as discussed in §5.2.3.

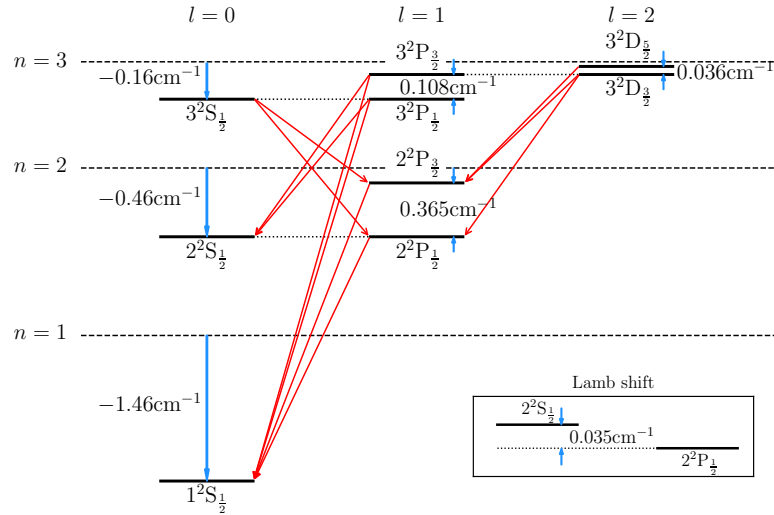


Fig. 6.1 The fine structure of hydrogen atom via Eq. (6.3.39) that agrees well with the Dirac result, Eq. (6.3.40), that does not depend on the orbital angular momentum quantum number l . The additional Lamb shift, arising from the coupling with the universal electromagnetic fluctuating field, primarily affects on the s -orbitals.

Reported in Fig. 6.1 are the hydrogen atom spin-orbital states, labeled by the spectroscopic term symbols, $n^{2S+1}L_J$; see Eq. (6.1.46) and remarks there. The principal quantum number n is placed in the front. For hydrogenlike atom, $S = s$, $L = l$ and $J = j$. Energies depend only on n and j , where $j = \frac{1}{2}$ for s orbitals and $j = l \pm \frac{1}{2}$ for others [cf. Eqs. (6.3.38)–(6.3.40)].

The Lamb shift is a type of system energy reorganization, due to the *universal environment of fluctuating electromagnetic field, even at vacuum*. This stochastic field is *spherically symmetric* and primarily influences on atoms through $|\Psi(\mathbf{r} = 0)|^2$, the same algebra to the Darwin term, Eq. (6.3.23), but with an additional order in α . At this level of treatment, only s -orbitals are subject to the Lamb shifts.

Exercises:

E6.3.1 (a) How many states in the individual terms of $2^2S_{1/2}$, $2^2P_{1/2}$ and $2^2P_{3/2}$? Are they related to the configurations of $2s^1$ and $2p^1$ and how? (b) Calculate the fine structure energies of hydrogen atom, associated with the above three terms;

E6.3.2 List all optical transitions between $n = 2$ and $n = 3$ for hydrogen atom. Specify the positions of individual optical lines, satisfying the selection rules, Eq. (6.1.48).

E6.3.3 Choose an optical line to illustrate the Zeeman effects in the presence of external constant magnetic field [Hint: Eq. (6.2.41)].

Chapter 7

Approximation Theories and Methods

This chapter is concerned with the some approximation methods and the underlying principles. In §7.1, we present the steady-state perturbation theory, in which the system Hamiltonian is partitioned into the unperturbed referent part ($\hat{H}_0$) and the perturbation part ($\hat{V}$). The latter is responsible for small deviation from the reference energies and wave functions. This is the case of time-dependent perturbation theory, described in §3.1.2 and §3.3.2, for wave functions and density operators, respectively. Moreover, the fine structure of hydrogen atom studied in §6.3 arises from the first-order relativity energy perturbations.

In §7.2, we discuss the variation principle, in which trial wave functions are optimized against the minimal energy expectation values. Differing from the perturbation approximations, variational energies are lower-bounded by the exact ground state energy. Various mean field approximations are the functional variation methods. On the other hand, the linear variation represents an equivalent theory of basis set transformation, including the use of nonorthogonal basis set wave functions. This is the case of the LCAO-MO theory: Molecular orbitals are formed via linear combination of atomic orbitals that from different atoms are nonorthogonal in general. Moreover, the truncated basis set expansion constitutes a variation type approximation. Increasing the basis set size results in lowering the evaluated ground-state energy.

In §7.3, we discuss the Born-Oppenheimer approximation that separates the fast (electron) and slow (nucleus) motions, followed by the related adiabatic steady-state theorems and nonadiabatic corrections.

7.1 Rayleigh–Schrödinger Perturbation Theory

7.1.1 Basic setup and perturbation energies

A. The intermediate normalization setup

In perturbation theories, the total Hamiltonian, denoted as $\hat{H} \equiv \hat{H}_0 + \hat{V}$, consists of the unperturbed and perturbed parts. The solutions to Schrödinger equation,

$$\hat{H}|\Psi_n\rangle = (\hat{H}_0 + \hat{V})|\Psi_n\rangle = E_n|\Psi_n\rangle, \quad n = 0, 1, \dots \quad (7.1.1)$$

are to be expressed in the Taylor's-like expansion, in the order of $\hat{V}$, the perturbative part of the total Hamiltonian. The zeroth-order solutions,

$$\hat{H}_0|n\rangle = E_n^{(0)}|n\rangle, \quad \langle m|n\rangle = \delta_{mn}, \quad \sum_m |m\rangle\langle m| = 1 \quad (7.1.2)$$

are the *references* and also the *basis set* that spans over the entire space.

To proceed, we set the solutions to Eq. (7.1.1) the forms of

$$E_n \equiv E_n^{(0)} + \mathcal{E}_n \quad \text{and} \quad |\Psi_n\rangle \equiv |n\rangle + |\Phi_n\rangle \quad \text{with} \quad \boxed{\langle n|\Phi_n\rangle = 0} \quad (7.1.3)$$

that amounts to

$$\langle n|\Psi_n\rangle = 1 \quad (\text{intermediate normalization}) \quad (7.1.4)$$

This convention facilitates the theoretical development. In particular,

$$\langle n|\hat{H}|\Psi_n\rangle = E_n \quad \text{and} \quad \langle n|\hat{H}_0|\Psi_n\rangle = E_n^{(0)} \quad (7.1.5)$$

We obtain

$$\mathcal{E}_n = \langle n|\hat{V}|\Psi_n\rangle = E_n - E_n^{(0)} \quad (7.1.6)$$

Moreover, we can recast Eq. (7.1.1) as $(\hat{H} - E_n)|\Phi_n\rangle = (E_n - \hat{H}_0 - \hat{V})|n\rangle$ and obtain

$$(\hat{H} - E_n)|\Phi_n\rangle = (\mathcal{E}_n - \hat{V})|n\rangle \quad (7.1.7)$$

This is an inhomogeneous equation for $|\Phi_n\rangle$, in relation to the specified $|n\rangle$. As known, the steady-state wave function $|\Psi_n\rangle$ via the Schrödinger equation (7.1.1) is determined up to a trivial phase constant, which is common in $|n\rangle$ and $|\Phi_n\rangle$. The resulting $\mathcal{E}_n$, Eq. (7.1.6), is real, as it should be.

B. The key intermediate results

The Rayleigh–Schrödinger perturbation theory (RSPT) is a standard perturbation-order expansion method. It starts in formal with

$$\hat{H} = \hat{H}_0 + \lambda \hat{V}$$

The corresponding Eqs. (7.1.6) and (7.1.7) are now

$$\mathcal{E}_n = \langle n | \lambda \hat{V} | \Psi_n \rangle \quad \text{and} \quad (\hat{H}_0 + \lambda \hat{V} - E_n) | \Phi_n \rangle = (\mathcal{E}_n - \lambda \hat{V}) | n \rangle \quad (7.1.8)$$

The working parameter λ is related to the eigenenergies and wave functions in Eq. (7.1.3), with the perturbation-order expansion of

$$\mathcal{E}_n = \sum_{k=1}^{\infty} \lambda^k E_n^{(k)} \quad \text{and} \quad | \Psi_n \rangle = | n \rangle + \sum_{k=1}^{\infty} \lambda^k | \Phi_n^{(k)} \rangle \equiv \sum_{k=0}^{\infty} \lambda^k | \Phi_n^{(k)} \rangle \quad (7.1.9)$$

The unperturbed energy and wave function are $E_n^{(0)} = \langle n | \hat{H}_0 | n \rangle$ and $| \Phi_n^{(0)} \rangle \equiv | n \rangle$, respectively, satisfying $\langle \Phi_n^{(0)} | \Phi_n^{(k)} \rangle = \delta_{k0}$ [cf. Eq. (7.1.3)]. The two expressions of Eq. (7.1.8) become

$$E_n^{(k)} = \langle n | \hat{V} | \Phi_n^{(k-1)} \rangle; \quad k \geq 1 \quad (7.1.10)$$

and

$$\sum_{k=1}^{\infty} \lambda^k (\hat{H}_0 + \lambda \hat{V} - \sum_{j=0}^{\infty} \lambda^j E_n^{(j)}) | \Phi_n^{(k)} \rangle = \left(\sum_{k=1}^{\infty} \lambda^k E_n^{(k)} - \lambda \hat{V} \right) | \Phi_n^{(0)} \rangle \quad (7.1.11)$$

respective, with noting that

$$\sum_{k=1}^{\infty} \sum_{j=0}^{\infty} \lambda^{k+j} E_n^{(j)} | \Phi_n^{(k)} \rangle = \sum_{k=1}^{\infty} \sum_{j=0}^{k-1} \lambda^k E_n^{(j)} | \Phi_n^{(k-j)} \rangle$$

Now, by equating Eq. (7.1.11) at the λ^k -order terms, we obtain

$$(E_n^{(0)} - \hat{H}_0) | \Phi_n^{(k)} \rangle = \hat{V} | \Phi_n^{(k-1)} \rangle - \sum_{j=0}^{k-1} E_n^{(k-j)} | \Phi_n^{(j)} \rangle \quad (7.1.12)$$

This is the key intermediate result to be resolved into the final recursive RSPT solutions in §7.1.3. On the other hand, we exploit Eq. (7.1.12) to construct an advanced perturbation energy formalism in the coming section.

7.1.2 The Wigner's $(2k+1)$ -rule

As shown, $E_n^{(k)} = \langle n | \hat{V} | \Phi_n^{(k-1)} \rangle$, Eq. (7.1.10), is the basic expression of perturbation energies. The generalized expression, see Eq. (7.1.14) later, can be obtained by using Eq. (7.1.12). In particular, as shown by E. Wigner in 1935, $\{\Phi_n^{(0)}, \dots, \Phi_n^{(k)}\}$ dictate the eigenenergies up to the order of $E_n^{(2k+1)}$:

$$\begin{aligned} E_n^{(2k)} &= \langle \Phi_n^{(k-1)} | \hat{V} | \Phi_n^{(k)} \rangle - \sum_{i=1}^{k-1} \sum_{j=1}^k E_n^{(2k-i-j)} \langle \Phi_n^{(i)} | \Phi_n^{(j)} \rangle \\ E_n^{(2k+1)} &= \langle \Phi_n^{(k)} | \hat{V} | \Phi_n^{(k)} \rangle - \sum_{i=1}^k \sum_{j=1}^k E_n^{(2k+1-i-j)} \langle \Phi_n^{(i)} | \Phi_n^{(j)} \rangle \end{aligned} \quad (7.1.13)$$

- More precisely, what is to be proved is that Eq. (7.1.12) enables Eq. (7.1.10) the generalized expression,

$$E_n^{(K)} = \langle \Phi_n^{(m-1)} | \hat{V} | \Phi_n^{(K-m)} \rangle - \sum_{i=1}^{m-1} \sum_{j=1}^{K-m} E_n^{(K-i-j)} \langle \Phi_n^{(i)} | \Phi_n^{(j)} \rangle \quad (7.1.14)$$

that holds for any value of integer $m \in [1, K]$.

- (i) When $m = 1$, it recovers the basic expression, $E_n^{(K)} = \langle n | \hat{V} | \Phi_n^{(K-1)} \rangle$, Eq. (7.1.10) with $|\Phi_n^{(0)}\rangle \equiv |n\rangle$;
- (ii) The Wigner's $(2k+1)$ -rule, Eq. (7.1.13), goes by $m = K - k$, with $K = 2k$ and $K = 2k + 1$ for the two expressions, respectively.

Prove of Eq. (7.1.14) by induction; i.e.,

$$E_{n;\text{I}}^{(K)}(m+1) - E_{n;\text{II}}^{(K)}(m+1) = E_{n;\text{I}}^{(K)}(m) - E_{n;\text{II}}^{(K)}(m) = E_n^{(K)} \quad (7.1.15)$$

- (a) First of all, $E_{n;\text{II}}^{(K)}(m+1)$ via the second term in Eq. (7.1.14) is given by

$$\begin{aligned} E_{n;\text{II}}^{(K)}(m+1) &= \sum_{i=1}^m \sum_{j=1}^{K-m-1} E_n^{(K-i-j)} \langle \Phi_n^{(i)} | \Phi_n^{(j)} \rangle \\ &= E_{n;\text{II}}^{(K)}(m) + \Delta_n^{(K)}(m) \end{aligned} \quad (7.1.16)$$

where

$$\Delta_n^{(K)}(m) \equiv \sum_{j=1}^{K-m-1} E_n^{(K-m-j)} \langle \Phi_n^{(m)} | \Phi_n^{(j)} \rangle - \sum_{i=1}^{m-1} E_n^{(m-i)} \langle \Phi_n^{(i)} | \Phi_n^{(K-m)} \rangle \quad (7.1.17)$$

- (b) Turn to the first term of Eq. (7.1.14), with the corresponding

$$E_{n;\text{I}}^{(K)}(m+1) = \langle \Phi_n^{(m)} | \hat{V} | \Phi_n^{(K-m-1)} \rangle \quad (7.1.18)$$

The task now is to obtain its counterpart to Eq. (7.1.17). The key idea is to exploit the following properties behind Eq. (7.1.12):

$$\hat{V}|\Phi_n^{(K-m-1)}\rangle = (E_n^{(0)} - \hat{H}_0)|\Phi_n^{(K-m)}\rangle + \sum_{j=0}^{K-m-1} E_n^{(K-m-j)}|\Phi_n^{(j)}\rangle \quad (7.1.19)$$

and

$$\langle \Phi_n^{(m)} | (E_n^{(0)} - \hat{H}_0) = \langle \Phi_n^{(m-1)} | \hat{V} - \sum_{i=0}^{m-1} E_n^{(m-i)} \langle \Phi_n^{(i)} | \quad (7.1.20)$$

Together with $\langle \Phi_n^{(m>0)} | \Phi_n^{(0)} \rangle = 0$ and $\Delta_n^{(K)}(m)$ of Eq. (7.1.18), we obtain

$$E_{n;1}^{(K)}(m+1) = \langle \Phi_n^{(m)} | \hat{V} | \Phi_n^{(K-m-1)} \rangle = E_{n;1}^{(K)}(m) + \Delta_n^{(K)}(m) \quad (7.1.21)$$

Comparing this to Eq. (7.1.17) gives rise to Eq. (7.1.15) that completes the by-induction proof of Eq. (7.1.14). (Q.E.D.)

Remarks

- As emphasized earlier, all $\{\Phi_n^{(k)}; k = 0, 1, \dots\}$ carry a common phase; hence $\langle \Phi_n^{(i)} | \Phi_n^{(j)} \rangle$ is real, as implied Eq. (7.1.14).
- While $E_n^{(k)} = \langle n | \hat{V} | \Phi_n^{(k-1)} \rangle$, Eq. (7.1.10), is the standard, the Wigner's $(2k+1)$ -rule, Eq. (7.1.13), would be preferred for $E_n^{(k>2)}$. As also known, in Eq. (7.1.14), the first term and the $\langle \Phi_n^{(i=m-1)} | \Phi_n^{(j=K-m)} \rangle$ component of the second term can be combined. The final results are then often reported in terms of

$$\hat{W} \equiv \hat{V} - E_n^{(1)} = \hat{V} - \langle n | \hat{V} | n \rangle \equiv \hat{V} - \langle \hat{V} \rangle \quad (7.1.22)$$

Together with Eq. (7.1.13), we have

$$E_n^{(3)} = \langle \Phi_n^{(1)} | \hat{W} | \Phi_n^{(1)} \rangle \quad (7.1.23a)$$

$$E_n^{(4)} = \langle \Phi_n^{(1)} | \hat{W} | \Phi_n^{(2)} \rangle - E_n^{(2)} \langle \Phi_n^{(1)} | \Phi_n^{(1)} \rangle \quad (7.1.23b)$$

$$E_n^{(5)} = \langle \Phi_n^{(2)} | \hat{W} | \Phi_n^{(2)} \rangle - E_n^{(3)} \langle \Phi_n^{(1)} | \Phi_n^{(1)} \rangle - 2E_n^{(2)} \langle \Phi_n^{(1)} | \Phi_n^{(2)} \rangle \quad (7.1.23c)$$

and so on.

7.1.3 Perturbation wave functions and remarks

A. Algebraic construction and the recursive solutions

- To revolve Eq. (7.1.12) for perturbation wave functions, we exploit the standard resolvent operator approach. Formally, it engages the *projection operator*, $\hat{P}$, to specify the subspace of primary interest. The complementary projection operator is then $\hat{Q} \equiv \hat{I} - \hat{P}$. All projection operators are *idempotent*; i.e., $\hat{P}^2 = \hat{P}$ and $\hat{Q}^2 = \hat{Q}$.
- The RSPT formalism goes by the universal algebraic construction, involving the aforementioned operators for both the degenerate and non-degenerate scenarios. The terminology here is concerned only with the unperturbed target $|n\rangle$, whether its associated $E_n^{(0)}$ is degenerate or non-degenerate, regardless the nature of $E_{m \neq n}^{(0)}$. We will elaborate the scenario of degenerate $E_n^{(0)}$ in §7.1.4.
- The projection operator associated with a non-degenerate $|n\rangle$ is

$$\hat{P}_n = |n\rangle\langle n| \quad (7.1.24)$$

The complementary projection operator is then

$$\hat{Q}_n \equiv \hat{I} - \hat{P}_n = \sum_m (1 - \delta_{mn}) |m\rangle\langle m| = \sum_{m \neq n} |m\rangle\langle m| \quad (7.1.25)$$

with $\hat{H}_0|m\rangle = E_m^{(0)}|m\rangle$, satisfying

$$[\hat{Q}_n, \hat{H}_0] = 0 \quad \text{and} \quad \hat{Q}_n|\Phi_n^{(k)}\rangle = |\Phi_n^{(k)}\rangle; \quad k > 0 \quad (7.1.26)$$

The perturbation wave functions are the $\hat{Q}_n$ -subspace properties.

- The RS resolvent operator to Eq. (7.1.12) is identified with

$$\hat{R} \equiv \frac{\hat{Q}_n}{E_n^{(0)} - \hat{H}_0} = \sum_{m \neq n} \frac{|m\rangle\langle m|}{E_n^{(0)} - E_m^{(0)}} \quad (7.1.27)$$

Together with $\hat{W} = \hat{V} - E_n^{(1)}$ [Eq. (7.1.22)] and $\hat{R}|n\rangle = 0$ [Eq. (7.1.25)], we obtain Eq. (7.1.12) the *recursive* solution,

$$|\Phi_n^{(k)}\rangle = \hat{R}\hat{W}|\Phi_n^{(k-1)}\rangle - \sum_{j=1}^{k-2} E_n^{(k-j)} \hat{R}|\Phi_n^{(j)}\rangle \quad (7.1.28)$$

Remarks: The summation term contributions only for $|\Phi_n^{(k \geq 3)}\rangle$. We obtain

$$|\Phi_n^{(1)}\rangle = \hat{R}\hat{W}|\Phi_n^{(0)}\rangle = \hat{R}\hat{V}|n\rangle \quad (7.1.29a)$$

$$|\Phi_n^{(2)}\rangle = \hat{R}\hat{W}|\Phi_n^{(1)}\rangle = \hat{R}\hat{W}\hat{R}\hat{V}|n\rangle \quad (7.1.29b)$$

with noting that $\hat{W} = \hat{V} - E_n^{(1)}$ and $\hat{R}|n\rangle = 0$. Moreover,

$$|\Phi_n^{(3)}\rangle = \hat{R}\hat{W}|\Phi_n^{(2)}\rangle - E_n^{(2)}\hat{R}|\Phi_n^{(1)}\rangle = (\hat{R}\hat{W}\hat{R}\hat{W}\hat{R}\hat{V} - E_n^{(2)}\hat{R}^2\hat{V})|n\rangle \quad (7.1.30)$$

The involving $\hat{R}^j$ in $|\Phi_n^{(k \geq 3)}\rangle$ via Eq. (7.1.27), with $\hat{Q}_n^2 = \hat{Q}_n$, reads

$$\hat{R}^j = \frac{\hat{Q}_n}{(E_n^{(0)} - \hat{H}_0)^j} = \sum'_m \frac{|m\rangle\langle m|}{(E_n^{(0)} - E_m^{(0)})^j} \quad (7.1.31)$$

The primed sum runs over all $m \neq n$.

B. The first-order RSPT results with error estimations

In the unperturbed $\hat{H}_0$ -representation, Eq. (7.1.29a) reads

$$|\Phi_n^{(1)}\rangle = \hat{R}\hat{V}|n\rangle = \sum'_m c_{m;n}|m\rangle \quad (7.1.32)$$

where

$$c_{m;n} = \langle m|\hat{R}\hat{V}|n\rangle = \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}}; \quad m \neq n \quad (7.1.33)$$

On the other hand, often evaluated are only

$$E_n^{(1)} = \langle n|\hat{V}|n\rangle \equiv \langle \hat{V} \rangle \quad (7.1.34)$$

It would then require for the error estimation. This task is related to the second-order energy,

$$E_n^{(2)} = \langle n|\hat{V}|\Phi_n^{(1)}\rangle = \sum'_m \frac{V_{nm}V_{mn}}{E_n^{(0)} - E_m^{(0)}} \quad (7.1.35)$$

whose upper-bound is to be formulated. To that end, we exploit the identity,

$$\langle \hat{W}^2 \rangle = \langle n|\hat{V}^2|n\rangle - \langle n|\hat{V}|n\rangle^2 = \sum'_m |V_{nm}|^2 \quad (7.1.36)$$

and obtain Eq. (7.1.35) the upper-bound,

$$|E_n^{(2)}| \leq |E_n^{(2)}|_{\max} = \langle \hat{W}^2 \rangle / \Omega_n, \quad \text{with } \Omega_n \equiv \min_{\{m \neq n\}} |E_n^{(0)} - E_m^{(0)}| \quad (7.1.37)$$

The resulting $|E_n^{(2)}|_{\max}/|E_n^{(1)}|$ estimates the relative error by the first-order energy correction.

7.1.4 Unified formalism and illustrations

A. The RSPT for degenerate states

- In general, an unperturbed $E_n^{(0)}$ goes with a defined degeneracy, g_n , and the target states in the perturbation theory are

$$|\mathbf{n}\rangle \equiv \{|n_p\rangle; p = 1, \dots, g_n\} \quad (7.1.38)$$

that satisfy $\langle n_p | n_{p'} \rangle = \delta_{pp'}$ and

$$\hat{H}_0 |n_p\rangle = E_n^{(0)} |n_p\rangle \quad (7.1.39)$$

As known, the degenerate subspace is itself closure and any linear combination of the vectors satisfies Eq. (7.1.39). In cope with the unified RSPT, the question arises then about the working set of unperturbed wave vectors, $\{|n_p\rangle; p = 1, \dots, g_n\}$. We will address this issue in Part B.

- All formulations of RSPT presented in §7.1.1–§7.1.3 remain valid for the working set of degenerate $\{|n_p\rangle; p = 1, \dots, g_n\}$, as follows.

(a) First of all, $|\Phi_{n_p}^{(0)}\rangle \equiv |n_p\rangle$, as before, and Eq. (7.1.10) reads

$$E_{n_p}^{(k)} = \langle n_p | \hat{V} | \Phi_{n_p}^{(k-1)} \rangle; \quad k \geq 1 \quad (7.1.40)$$

(b) As known, Eq. (7.1.12), the key intermediate result of RSPT, holds for individual n_p . The resulting Wigner's $(2k+1)$ -rule, Eq. (7.1.13), remains true, such as [cf. Eq. (7.1.23a) with Eq. (7.1.22)]

$$E_{n_p}^{(3)} = \langle \Phi_{n_p}^{(1)} | \hat{W}_{n_p} | \Phi_{n_p}^{(1)} \rangle; \quad \hat{W}_{n_p} \equiv \hat{V} - E_{n_p}^{(1)} \quad (7.1.41)$$

(c) The projection operators are [cf. Eq. (7.1.24)]

$$\hat{P}_n = |\mathbf{n}\rangle \langle \mathbf{n}| \equiv \sum_{p=1}^{g_n} |n_p\rangle \langle n_p| \quad \text{and} \quad \hat{Q}_n \equiv \hat{I} - \hat{P}_n = \sum_{m \notin \mathbf{n}} |m\rangle \langle m| \quad (7.1.42)$$

satisfying $[\hat{Q}_n, \hat{H}_0] = 0$ and $\hat{Q}_n \hat{H}_0 \hat{Q}_n |m\rangle = E_m^{(0)} |m\rangle$, with $E_m^{(0)} \neq E_n^{(0)}$. The RS resolvent operator, Eq. (7.1.27), reads then

$$\hat{R} = \frac{\hat{Q}_n}{E_n^{(0)} - \hat{H}_0} = \sum_{m \notin \mathbf{n}} \frac{|m\rangle \langle m|}{E_n^{(0)} - E_m^{(0)}} \quad (7.1.43)$$

This gives rise to Eq. (7.1.12) the solution of [cf. Eq. (7.1.28)]

$$|\Phi_{n_p}^{(k)}\rangle = \hat{R} \hat{W}_{n_p} |\Phi_{n_p}^{(k-1)}\rangle - \sum_{j=1}^{k-2} E_n^{(k-j)} \hat{R} |\Phi_{n_p}^{(j)}\rangle \quad (7.1.44)$$

Note that $\hat{Q}_n |\Phi_{n_p}^{(k)}\rangle = |\Phi_{n_p}^{(k)}\rangle$ for $k \geq 1$, whereas $\langle \Phi_{n_p}^{(0)} | \Phi_{n_p}^{(k)} \rangle = \delta_{0k}$.

B. Working set of unperturbed degenerate wave vectors

- To clarify the issue, let us recast the unperturbed Eq. (7.1.39) as

$$E_n^{(0)} = \langle n_p | \hat{P}_n \hat{H}_0 \hat{P}_n | n_p \rangle \quad (7.1.45)$$

and the first-order perturbation energy via Eq. (7.1.40) as

$$E_{n_p}^{(1)} = \langle n_p | \hat{P}_n \hat{V} \hat{P}_n | n_p \rangle \quad (7.1.46)$$

The involving projection operator, Eq. (7.1.42), satisfies

$$\hat{P}_n = \sum_{p=1}^{g_n} |n_p\rangle \langle n_p| = \sum_{p=1}^{g_n} |\psi_p\rangle \langle \psi_p| \equiv \hat{I}_P \quad (7.1.47)$$

with $\langle n_p | n_{p'} \rangle = \delta_{pp'} = \langle \psi_p | \psi_{p'} \rangle$ and the fact that the projection operator is itself the unit matrix of the degenerate-subspace. Both $\{n_p; p = 1, \dots, g_n\}$ and $\{\psi_p; p = 1, \dots, g_n\}$ constitute the basis sets that are formally equivalent. However, they would be different in physics, such as localized/atomic versus delocalized/molecular orbitals.

In line with Eq. (7.1.42), $\{|n_p\rangle\}$ is the working set exploited in the degenerate RSPT, to be specified with the $\{|\psi_p\rangle\}$ -representation. In particular,

$$\hat{P}_n \hat{V} \hat{P}_n = \begin{bmatrix} V_{11} & \cdots & V_{1g_n} \\ \cdots & \cdots & \cdots \\ V_{n_g 1} & \cdots & V_{n_g g_n} \end{bmatrix} \quad \text{and} \quad |n_p\rangle = \begin{bmatrix} c_{1p} \\ \cdots \\ c_{g_n p} \end{bmatrix} \quad (7.1.48)$$

where

$$V_{ij} \equiv \langle \psi_i | \hat{V} | \psi_j \rangle \quad \text{and} \quad c_{jp} = \langle \psi_j | n_p \rangle \quad (7.1.49)$$

We can then relate Eq. (7.1.46) with

$$(\hat{P}_n \hat{V} \hat{P}_n) | n_p \rangle = E_{n_p}^{(1)} | n_p \rangle; \quad p = 1, \dots, g_n \quad (7.1.50)$$

This is the eigenequation of the $g_n \times g_n$ matrix, specified in Eq. (7.1.48).

- We can now conclude the formal RSPT, as follows: The targeted unperturbed degenerate $\{|n_p\rangle; p = 1, \dots, g_n\}$ satisfy both Eq. (7.1.45) and Eq. (7.1.50). While Eq. (7.1.45) goes by $\hat{P}_n \hat{H}_0 \hat{P}_n = E_n^{(0)} \hat{I}_P$, with $\hat{I}_P$ being the unit matrix in the degenerate subspace, Eq. (7.1.50) is the eigenequation, with the $(g_n \times g_n)$ matrix specified by Eq. (7.1.48). The nondegenerate RSPT described in §7.1.3 would become applicable, if those $\{E_{n_p}^{(1)}\}$ via Eq. (7.1.50) were all different.

- It is worth re-emphasizing that the unperturbed Hamiltonian, $\hat{H}_0$, is often readily solvable. The applications of RSPT, where $\hat{P}_n \hat{H}_0 \hat{P}_n = E_n^{(0)} \hat{I}_P$ is degenerate, would then exploit a chosen reference, $\hat{P}_n \hat{V}_{\text{ref}} \hat{P}_n$, with analytical models that remove the degeneracy in physics, as exemplified below.

C. Illustrations

(C1) Consider the simplest N -site tight-binding model system,

$$\hat{H} = \epsilon \mathbf{I} + \lambda \mathbf{V} \quad (7.1.51)$$

with the linear and cyclic coupling scenarios, respectively,

$$\mathbf{V} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 1 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 1 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \quad (7.1.52)$$

As $\mathbf{I}$ is an $N \times N$ unit matrix, the wave vectors $\{|\Psi_n\rangle\}$ are completely determined by matrix $\mathbf{V}$. In Eq. (7.1.51), ϵ and λ are the on-site energy and the tight-binding coupling parameters, respectively.

Linear coupling case: The (normalized) solutions are

$$\begin{aligned} |\Psi_n\rangle &= \left(\frac{2}{N+1}\right)^{\frac{1}{2}} \sum_{j=1}^N \sin\left(\frac{jn\pi}{N+1}\right) |\psi_j\rangle \\ E_n &= \epsilon + 2\lambda \cos\left(\frac{n\pi}{N+1}\right); \quad n = 1, \dots, N \end{aligned} \quad \text{(Linear)} \quad (7.1.53)$$

Cyclic coupling case: The (normalized) solutions are

$$\begin{aligned} |\Psi_n\rangle &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \exp\left(i\frac{2jn\pi}{N}\right) |\psi_j\rangle \\ E_n &= \epsilon + 2\lambda \cos\left(\frac{2n\pi}{N}\right); \quad n = 0, \dots, N-1 \end{aligned} \quad \text{(Cyclic)} \quad (7.1.54)$$

Exercises

E7.1.1 (a) Verify Eqs. (7.1.53) and (7.1.54); (b) Discuss their wave functions with respect to those of a particle in box and in circle, respectively.

E7.1.2 Let $\hat{H}_0 = \epsilon \mathbf{I}$ be the unperturbed part of Eq. (7.1.51). Whether the first-order perturbation is exact or not? Discuss also the degeneracy removals of the above two scenarios.

E7.1.3 Consider the non-degenerate case ($\epsilon_2 > \epsilon_1$),

$$\hat{H} = \begin{bmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{bmatrix} + \lambda \begin{bmatrix} 0 & e^{-i\varphi} \\ e^{i\varphi} & 0 \end{bmatrix} \equiv \hat{H}_0 + \hat{V} \quad (7.1.55)$$

(a) It is easy to obtain $E_1^{(k)} + E_2^{(k)} = 0$ and $E_n^{(1)} = E_n^{(3)} = 0$. What are the general properties used here? (b) Derive $|\Phi_1^{(1)}\rangle$ and show $E_1^{(2)} = -\lambda^2/(\epsilon_2 - \epsilon_1)$; (c) Verify that $E_1^{(2)} + E_2^{(2)} = 0$.

(C2) Stark effect on hydrogen atom

- As known in physics, wave functions are formulated with the symmetry-adapted CSCO construction. For hydrogen atom in the absence of external field, we have [cf. Eq. (6.2.9)]

$$\Psi_{n,l,m}(\mathbf{r}) = R_{n,l}(r)Y_{l,m}(\theta, \phi) \equiv |n, l, m\rangle \quad (7.1.56)$$

with $l = 0, \dots, n-1$ and $m = 0, \pm 1, \dots, \pm l$. The unperturbed eigenenergy depends only the principle quantum number n ; i.e.,

$$E_n^{(0)} = \langle n, l, m | \hat{H}_0 | n, l, m \rangle = -\frac{1}{2n^2} \quad (\text{atomic unit}) \quad (7.1.57)$$

with the degeneracy of $g_n = n^2$, excluding spins.

On top of the bare $\hat{H}_0$, external static electric field, $\vec{\mathcal{E}}$, gives rise to

$$\hat{V}_{\text{int}} = -\vec{\mu} \cdot \vec{\mathcal{E}} = -e r \mathcal{E} \cos \theta \quad (7.1.58)$$

In the original n^2 -degenerate subspace, the interaction matrix reads

$$\langle n, l', m' | \hat{V}_{\text{int}} | n, l, m \rangle = -e \mathcal{E} \langle n, l', m' | r \cos \theta | n, l, m \rangle \quad (7.1.59)$$

Note that [cf. Eq. (5.3.32) with Eq. 5.3.33)]

$$\cos \theta |l, m\rangle = \gamma_{l,m} |l-1, m\rangle + \gamma_{l+1,m} |l+1, m\rangle \quad (7.1.60)$$

with $\gamma_{l,m} = \sqrt{(l^2 - m^2)/(4l^2 - 1)}$. Denote further

$$\lambda_{n,l,m} \equiv -e \mathcal{E} \gamma_{l,m} \langle R_{n,l-1}(r) | r | R_{n,l}(r) \rangle \quad (7.1.61)$$

Those nonzero matrix elements in Eq. (7.1.59) are then

$$\begin{aligned} \langle n, l-1, m | \hat{V}_{\text{int}} | n, l, m \rangle &= \lambda_{n,l,m} \\ \langle n, l+1, m | \hat{V}_{\text{int}} | n, l, m \rangle &= \lambda_{n,l+1,m} \end{aligned} \quad (7.1.62)$$

The resulting energy shift is called the **Stark effect**.

- For illustration, consider the four degenerate orbitals with $n = 2$. The nonzero system-field coupling matrix elements, satisfying Eq. (7.1.62), occur only between $|2s\rangle \equiv |2, 0, 0\rangle$ and $|2p_z\rangle \equiv |2, 1, 0\rangle$; i.e.,

$$\langle 2p_z | \hat{V}_{\text{int}} | 2s \rangle = \lambda_{2,1,0} = 3ea_0\mathcal{E} \quad (7.1.63)$$

with the value being obtained via Eq. (7.1.61) and Eq. (6.1.26b). On the other hand, $\langle 2p_{+1} | \hat{V}_{\text{int}} | 2p_{-1} \rangle = 0$, where $|2p_{\pm 1}\rangle \equiv |2, 1, \pm 1\rangle$. Hence, the working set of vectors used in the RSPT now consists of $2^{-\frac{1}{2}}(|2s\rangle \pm |2p_z\rangle)$ and $|2p_{\pm 1}\rangle$. While the latter pair remain intact, the external electric field splits the sp_z -hybrid pair into $E_2^{(0)} \pm 3ea_0\mathcal{E}$, which are no longer degenerate.

7.2 Variation Principle Methods

7.2.1 Variation principle

The variation principle states that the energy of any acceptable trial wave function, $|\Phi\rangle$, never be lower than the exact ground state energy; i.e.,

$$\mathcal{E}(\text{trial}) = \frac{\langle \Phi | \hat{H} | \Phi \rangle}{\langle \Phi | \Phi \rangle} \geq E_0(\text{exact}) \quad (7.2.1)$$

Proof of Eq. (7.2.1): Note that $E_0 = \min\{E_n\}$ and $\hat{H}|\Psi_n\rangle = E_n|\Psi_n\rangle$. We have

$$\langle \Phi | \hat{H} | \Phi \rangle = \sum_n \langle \Phi | \hat{H} | \Psi_n \rangle \langle \Psi_n | \Phi \rangle = \sum_n E_n \langle \Phi | \Psi_n \rangle \langle \Psi_n | \Phi \rangle \geq E_0 \sum_n \langle \Phi | \Psi_n \rangle \langle \Psi_n | \Phi \rangle$$

That is $\langle \Phi | \hat{H} | \Phi \rangle \geq E_0 \langle \Phi | \Phi \rangle$; i.e., $\mathcal{E} \geq E_0$, Eq. (7.2.1). (Q.E.D.)

• **Inequality:** Variation energies, as implied in Eq. (7.2.1), are lower bounded by the exact ground state energy. This observation will be exploited in various contexts of many-particle physics, including the Hartree–Fock mean field theory and density functional theory to be discussed in Chapter §8 and Chapter §9, respectively. Both involve the functional variations.

In §7.2.2, we focus on the minimum basis He atom, with both of its two electrons occupying the 1s spatial orbital, involving a single variation parameter. The resulting effective 1s orbital for individual electron shows the shielding characteristics, described by the effective $Z_{\text{eff}} < Z$, where $Z = 2$ is the nuclei charge of He atom; see Eq. (7.2.5).

• **Uniqueness on the equal sign:**

- (i) In Eq. (7.2.1), the “=” sign holds *only* $\Phi = \Psi_0$, the exact ground state wave function (or the degenerate one that may be selected by considering CSCO).
- (ii) If the trial $|\Phi\rangle$ is orthogonal to the ground-state subspace, Eq. (7.2.1) remains valid, but with E_0 being the first excited state energy. The equal sign holds when $|\Phi\rangle$ is the first excited wave function.
- (iii) Higher excited states follow the same strategy.

The above properties emerge naturally in the linear variation theory, concerning in particular the issues of representation, with nonorthogonal basis. This will be exemplified in §7.2.3, with the linear combination of atomic orbitals to molecular orbitals (LCAO–MO) construction. Evidently, the eigenenergies via a finite basis, according to the variation principle, are individually lower-bounded by their exact values with infinite basis.

7.2.2 Variation evaluation of He atom

The Hamiltonian of He atom reads (in atomic unit)

$$\hat{H} = -\left(\frac{1}{2}\nabla_{\mathbf{r}_1}^2 + \frac{Z}{r_1} + \frac{1}{2}\nabla_{\mathbf{r}_2}^2 + \frac{Z}{r_2}\right) + \frac{1}{r_{12}} = \hat{h}(1) + \hat{h}(2) + \frac{1}{r_{12}} \quad (7.2.2)$$

The ground-state configuration is $1s^2$, with the $1s$ orbital being doubly occupied. Here, we focus on the penetration and shielding picture, discussed in §6.1.3A, with the effective Z_{eff} to be determined.

- To that end, consider the trial wave function (normalized),

$$\Phi(1, 2) = \varphi_{1s}(\mathbf{r}_1)\varphi_{1s}(\mathbf{r}_2) \quad (7.2.3a)$$

with

$$\varphi_{1s}(\mathbf{r}) = 2\zeta^{3/2}e^{-\zeta r}Y_{0,0}(\theta, \phi) = (\zeta^3/\pi)^{1/2}e^{-\zeta r} \quad (7.2.3b)$$

The exponent ζ is the variation parameter. The resulting expectation energy, as derived soon later, is found to be

$$\mathcal{E}(\zeta) = \langle \Phi | \hat{H} | \Phi \rangle = \zeta^2 - 2Z\zeta + \frac{5}{8}\zeta \quad (7.2.4)$$

The variation method, $d\mathcal{E}(\zeta)/d\zeta = 0$, results in

$$\boxed{\zeta = Z - \frac{5}{16} \equiv Z_{\text{eff}}} \Rightarrow \mathcal{E} = -Z_{\text{eff}}^2 \quad (7.2.5)$$

The screening effect is evident. The resulting variation energy $\mathcal{E}$ is 1.88% higher than the experimental ground-state energy of He atom.

Derivations of Eq. (7.2.4):

The Hamiltonian, Eq. (7.2.2), consists of one-electron and two-electron components. The resulting energy expectation value reads

$$\mathcal{E} = \langle \Phi | \hat{H} | \Phi \rangle = \mathcal{E}_I + \mathcal{E}_{II} \quad (7.2.6)$$

- (a) The one-electron component contributes to

$$\mathcal{E}_I = -2\left\langle \varphi_{1s}(\mathbf{r}) \left| \frac{1}{2}\nabla_{\mathbf{r}}^2 + \frac{Z}{r} \right| \varphi_{1s}(\mathbf{r}) \right\rangle = \zeta^2 - 2Z\zeta \quad (7.2.7)$$

This is the hydrogenlike formulas.

- (b) The two-electron component contributes to Eq. (7.2.6) with

$$\mathcal{E}_{II} = \iint \frac{d\mathbf{r}_1 d\mathbf{r}_2}{r_{12}} |\varphi_{1s}(\mathbf{r}_1)|^2 |\varphi_{1s}(\mathbf{r}_2)|^2 \quad (7.2.8)$$

Here, $\mathbf{r}_j \equiv (r_j, \theta_j, \phi_j)$ and $r_{12} \equiv |\mathbf{r}_1 - \mathbf{r}_2|$, with $\mathbf{r}_1 \mathbf{r}_2 = r_1 r_2 \cos \theta$, where

$$\cos \theta = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 \cos(\phi_1 - \phi_2)$$

To proceed, we exploit the Legendre polynomial's generating function formulas, the last expression of Eq. (5.3.26),

$$\frac{1}{r_{12}} = \frac{1}{\sqrt{r_1^2 - 2r_1 r_2 \cos \theta + r_2^2}} = \frac{1}{r_>} \sum_{k=0}^{\infty} \frac{r_<^k}{r_>^k} P_k(\cos \theta) \quad (7.2.9)$$

with

$$r_> = \max(r_1, r_2) \quad \text{and} \quad r_< = \min(r_1, r_2) \quad (7.2.10)$$

Applying then Eq. (5.3.4), we recast Eq. (7.2.9) in terms of spherical harmonics,

$$\frac{1}{r_{12}} = \sum_{k=0}^{\infty} \frac{r_<^k}{r_>^{k+1}} \frac{4\pi}{2k+1} \sum_{m=-k}^k Y_{k,m}^*(\theta_1, \phi_1) Y_{k,m}(\theta_2, \phi_2) \quad (7.2.11)$$

Substituting it into Eq. (7.2.8) with Eq. (7.2.3), followed by using

$$\int_0^{2\pi} d\phi_j \int_0^\pi d\theta_j \sin \theta_j Y_{k,m}^*(\theta_j, \phi_j) Y_{0,0}(\theta_j, \phi_j) = \delta_{k0} \delta_{m0}$$

and $Y_{0,0}(\theta_j, \phi_j) = (4\pi)^{-\frac{1}{2}}$, we obtain

$$\mathcal{E}_{\text{II}} = (4\zeta^3)^2 \int_0^\infty dr_1 \int_0^\infty dr_2 \frac{r_1^2 r_2^2}{r_>} e^{-2\zeta(r_1+r_2)} = \zeta \int_0^\infty x^2 e^{-x} dx \int_x^\infty x' e^{-x'} dx'$$

The last identity is obtained by using Eq. (7.2.10), followed by changing the integration variables. We obtain

$$\mathcal{E}_{\text{II}} = \zeta \int_0^\infty x^2 (x+1) e^{-2x} dx = \zeta \left(\frac{3!}{2^4} + \frac{2!}{2^3} \right) = \frac{5}{16} \zeta \quad (7.2.12)$$

Together with Eq. (7.2.7) for Eq. (7.2.6), we obtain Eq. (7.2.4).

Exercises

E7.2.1 Verify Eq. (7.2.7) and Eq. (7.2.12). Note that one can also treat $\hat{V}_{ee} = 1/r_{12}$, the last term of two-electron Hamiltonian, $\hat{H}$ of Eq. (7.2.2), as perturbation. Make the general remarks on the variation principle methods versus the perturbation approximations.

E7.2.2 For many-electron atoms, the wave functions comprise the atomic orbitals, assumed the form of $\varphi_{n,l,m}(r, \theta, \phi) = R_{n,l}(r) Y_{l,m}(\theta, \phi)$. As in line with Eq. (7.2.3), only some specified parameters involved in $R_{n,l}(r)$ are subject to the variation determination. Discuss the underlying physical consideration.

7.2.3 Molecular orbitals via atomic orbitals

A. Opening remarks

- Linear variation method, as discussed to the end of §7.2.1, constitutes a representation, with an arbitrary basis wave functions that can be *nonorthogonal*. The linear expansion parameters are subject to the variation determination. The resulting secular equation would be formally identical to the Schrödinger equation, if the basis set is completed; however, any truncation corresponds to a type variation approximation, with the *individual* $\mathcal{E}_n$ being lower-bounded by the exact value:

$$E_n \leq \mathcal{E}_n; \quad n = 0, 1, \dots, K$$

Equal sign holds when the basis size $K \rightarrow \infty$.

- The most important application of linear variation method is the construction of molecular orbitals (MOs) from the atomic orbitals (AOs). Before presenting the formal theory, we would like to emphasize the two observations as follows.
 - (a) Molecular structure theory is based on the Born–Oppenheimer approximation, where nuclear coordinates are parameters; see §7.3. This is the same for MOs, the wave functions of single-electron molecular systems. Unlike AOs arising from hydrogenlike atoms, MOs are in general not analytically solvable. As the building blocks of molecular structure theory, MOs are rooted at their atomic origin. It desires MOs be linear combination of AOs (LCAOs), arising from their constituent atoms.
 - (b) The AOs in molecular environments are generally not identical to those in isolated atoms. Related issues will be discussed in §10.3.3, with respect to some practically used AOs in computational chemistry. As results, AOs in molecules are generally nonorthogonal, no matter they are associated with different nuclear centers or same nucleus.

B. The LCAO–MO construction

- The MOs are solutions to one-electron molecular system,

$$\hat{H}|\psi_j(\mathbf{r})\rangle = \mathcal{E}_j|\psi_j(\mathbf{r})\rangle \quad (7.2.13)$$

where

$$\hat{H} = -\frac{1}{2}\nabla_{\mathbf{r}}^2 - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \quad (7.2.14)$$

The LCAO–MO expression reads

$$|\psi_j\rangle = \sum_{\mu} C_{\mu j} |\varphi_{\mu}\rangle \quad (7.2.15)$$

The basis set AOs, $\{|\varphi_\mu\rangle\}$, arise from all constituent atoms and are nonorthogonal in general. The Hamiltonian matrix and overlap matrix, $\mathbf{H}$ and $\mathbf{S}$, respectively, are

$$\begin{aligned}\mathbf{H} &\equiv \{H_{\mu\nu}\}, \text{ with } H_{\mu\nu} = \langle\varphi_\mu|\hat{H}|\varphi_\nu\rangle \\ \mathbf{S} &\equiv \{S_{\mu\nu}\}, \text{ with } S_{\mu\nu} = \langle\varphi_\mu|\varphi_\nu\rangle\end{aligned}\quad (7.2.16)$$

with the matrix positivity of $\mathbf{S}$ and also $|S_{\mu\nu}| \leq 1$, whereas $S_{\mu\mu} = 1$. In line with the BO approximation, nuclear coordinates enter as parameters, as implied in Eq. (7.2.14) and the resulting $\mathbf{H}(\{\mathbf{R}_A\})$ and $\mathbf{S}(\{\mathbf{R}_A\})$.

• The LCAO–MO matrix, $\mathbf{C} \equiv \{C_{\mu j}\}$, comprises the coefficients in Eq. (7.2.15). Another unknown is $\mathcal{E} = \text{diag}\{\mathcal{E}_j\}$, which is diagonal, involving the eigenvalues of Eq. (7.2.13). As derived soon below, we obtain

$$\boxed{\mathbf{H}\mathbf{C} = \mathbf{S}\mathbf{C}\mathcal{E}}; \quad (\text{secular equation}) \quad (7.2.17)$$

Derivations: First of all, this is a *linear variation problem*, targeting at the linear expansion coefficients in Eq. (7.2.15). It is to minimize

$$\mathcal{E}_j = \langle\psi_j|\hat{H}|\psi_j\rangle = \sum_{\mu,\nu} C_{\mu j}^* C_{\nu j} H_{\mu\nu} \quad (7.2.18)$$

under the constraint,

$$\langle\psi_j|\psi_j\rangle = \sum_{\mu,\nu} C_{\mu j}^* C_{\nu j} S_{\mu\nu} = 1 \quad (7.2.19)$$

The required Lagrange's multiplier method is then

$$\begin{aligned}0 &= \frac{\partial}{\partial C_{\mu j}^*} \left\{ \langle\psi_j|\hat{H}|\psi_j\rangle - \mathcal{E}_j (\langle\psi_j|\psi_j\rangle - 1) \right\} \\ &= \frac{\partial}{\partial C_{\mu j}^*} \left\{ \sum_{\mu,\nu} [C_{\mu j}^* C_{\nu j} H_{\mu\nu} - \mathcal{E}_j (C_{\mu j}^* C_{\nu j} S_{\mu\nu} - 1)] \right\}\end{aligned}\quad (7.2.20)$$

We obtain

$$\sum_{\nu} C_{\nu j} H_{\mu\nu} = \mathcal{E}_j \sum_{\nu} C_{\nu j} S_{\mu\nu} \quad (7.2.21)$$

Its matrix form is given by Eq. (7.2.17). (Q.E.D.)

• General speaking, the secular equation (7.2.17) is just the Schrödinger equation in a finite basis set representation, applicable to both the orthogonal and nonorthogonal scenarios. In Part C soon below, we will transform Eq. (7.2.17) into its canonical orthogonalization equivalence. In Part D, we illustrate the secular equation itself with two-level molecular systems and comments on the general features of bonding and anti-bonding MOs.

C. Canonical orthogonalization of Eq. (7.2.17)

- For the numerical solution to Eq. (7.2.17), we like to convert it into the canonical form. This is concerned with the AOs basis set orthogonalization,

$$|\tilde{\varphi}_m\rangle = \sum_{\mu} A_{\mu m} |\varphi_{\mu}\rangle, \quad \text{with } \langle \tilde{\varphi}_m | \tilde{\varphi}_n \rangle = \delta_{mn} \quad (7.2.22)$$

whereas $\langle \varphi_{\mu} | \varphi_{\nu} \rangle = S_{\mu\nu}$. The task is then equivalent to have

$$\boxed{\mathbf{A}^{\dagger} \mathbf{S} \mathbf{A} = \mathbf{1}} \quad (7.2.23)$$

The LCAO–MO expression (7.2.15) becomes

$$|\psi_j\rangle = \sum_{\mu} C_{\mu j} |\varphi_{\mu}\rangle = \sum_m \tilde{C}_{mj} |\tilde{\varphi}_m\rangle, \quad \text{with } \boxed{\mathbf{C} = \mathbf{A} \tilde{\mathbf{C}}} \quad (7.2.24)$$

Together with Eq. (7.2.23), we transform Eq. (7.2.17) into

$$\tilde{\mathbf{H}} \tilde{\mathbf{C}} = \tilde{\mathbf{C}} \mathbf{E}, \quad \text{where } \boxed{\tilde{\mathbf{H}} \equiv \mathbf{A}^{\dagger} \mathbf{H} \mathbf{A}} \quad (7.2.25)$$

- To obtain the required transformation matrix $\mathbf{A}$, we exploit the properties that $\mathbf{S}$ is Hermitian and positive, satisfying

$$\boxed{\mathbf{s} \equiv \mathbf{U}^{\dagger} \mathbf{S} \mathbf{U} = \text{diag}\{s_{\mu}\}} \quad (7.2.26)$$

The involving unitary matrix $\mathbf{U}$ is determined, together with

$$0 < s_1 \leq s_2 \leq \cdots \leq s_k \leq 1$$

Set then

$$\boxed{\mathbf{A} = \mathbf{U} \mathbf{s}^{-\frac{1}{2}} \mathbf{U}^{\dagger} = \mathbf{A}^{\dagger}} \quad (7.2.27)$$

This is the canonical (or **Löwdin**) orthogonalization. The required Eq. (7.2.23) is not only satisfied, but also assumes to be *symmetric*, in terms of $\mathbf{A} \mathbf{S} \mathbf{A} = \mathbf{1}$ and also $\tilde{\mathbf{H}} \equiv \mathbf{A} \mathbf{H} \mathbf{A}$.

Exercises

E7.2.3 Verify the consistency of Eqs. (7.2.22)–(7.2.27).

E7.2.4 Show that $\mathbf{S}^{-1} = \mathbf{U} \mathbf{s}^{-1} \mathbf{U}^{\dagger}$,

$$\text{tr } \mathbf{E} = \text{tr } \tilde{\mathbf{H}} = \text{tr}(\mathbf{S}^{-1} \mathbf{H}) \quad \text{and} \quad \det \mathbf{E} = \det \tilde{\mathbf{H}} = \frac{\det \mathbf{H}}{\det \mathbf{S}} \quad (7.2.28)$$

in relation to the matrix traces and determinants, respectively.

E7.2.5 Let $\tilde{\mathbf{U}}$ be the following unitary transformation,

$$\tilde{\mathbf{U}}^{\dagger} \tilde{\mathbf{H}} \tilde{\mathbf{U}} = \text{diag}\{\mathbf{E}_j\} \quad (7.2.29)$$

Denote then

$$\mathbf{W} \equiv \mathbf{U} \mathbf{s}^{-\frac{1}{2}} \mathbf{U}^{\dagger} \tilde{\mathbf{U}} \quad (7.2.30)$$

Show that

$$\mathbf{W}^{\dagger} \mathbf{S} \mathbf{W} = \mathbf{1} \quad \text{and} \quad \mathbf{W}^{\dagger} \mathbf{H} \mathbf{W} = \text{diag}\{\mathbf{E}_j\} \quad (7.2.31)$$

Compare the above results, with the normal modes analysis in classical mechanics.

D. Illustrations and remarks

B.1 Degenerated scenario

- Consider first the simplest model of homonuclear diatomic molecules,

$$\mathbf{H} = \begin{bmatrix} \epsilon & \lambda \\ \lambda & \epsilon \end{bmatrix} \quad \text{and} \quad \mathbf{S} = \begin{bmatrix} 1 & S \\ S & 1 \end{bmatrix}, \quad \text{where } S = \langle \varphi_A | \varphi_B \rangle \quad (7.2.32)$$

The LCAO–MO is given by

$$|\psi\rangle = c_A |\varphi_A\rangle + c_B |\varphi_B\rangle \quad (7.2.33)$$

with setting the coefficients be real. The secular equation (7.2.17) reads now

$$\begin{bmatrix} \epsilon - \mathcal{E} & \lambda - S\mathcal{E} \\ \lambda - S\mathcal{E} & \epsilon - \mathcal{E} \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = 0 \quad (7.2.34)$$

The solutions are

$$\mathcal{E}_{\pm} = \frac{\epsilon \pm \lambda}{1 \pm S} \quad \text{and} \quad |\psi_{\pm}\rangle = \frac{|\varphi_A\rangle \pm |\varphi_B\rangle}{\sqrt{2(1 \pm S)}} \quad (7.2.35)$$

Both the coupling parameter λ and overlap S are functions of distance between two atoms, whereas ϵ denotes the local AO energy.

- As known, two AOs from different atoms, form a pair of *binding* and *anti-binding* MOs, distinct by whether there is a node or not. Each binding/anti-binding MO contribute to a positive/negative half covalence bond. For those MOs in Eq. (7.2.35),

$$\boxed{\psi_{\text{bind}} = \psi_+} \quad \text{and} \quad \boxed{\psi_{\text{anti}} = \psi_-} \quad (7.2.36)$$

and $\mathcal{E}_+ \leq \epsilon \leq \mathcal{E}_-$; i.e., $\lambda < 0$. This is a rather general case, for diatomic molecules in the binding region. This would also be true for the nearest-neighboring bonds in general molecules.

B.2 Nondegenerated scenario

- Consider now a nondegenerate two-level binding scenario,

$$\mathbf{H} = \begin{bmatrix} \Delta & \lambda \\ \lambda & -\Delta \end{bmatrix} \quad \text{and} \quad \mathbf{S} = \begin{bmatrix} 1 & S \\ S & 1 \end{bmatrix} \quad (7.2.37)$$

The center of two-level energy is set to be zero and $\epsilon_A - \epsilon_B = 2\Delta > 0$. The secular equation reads

$$\begin{bmatrix} \Delta - \mathcal{E} & \lambda - S\mathcal{E} \\ \lambda - S\mathcal{E} & -\Delta - \mathcal{E} \end{bmatrix} \begin{bmatrix} c_A \\ c_B \end{bmatrix} = 0 \quad (7.2.38)$$

The eigenvalues via the determinant equation are given by

$$\mathcal{E}_{\pm} = -\frac{\lambda S}{1-S^2} \pm \frac{\lambda}{1-S^2} \sqrt{S^2 + (\eta^2 + 1)(1-S^2)} \quad (7.2.39)$$

where

$$\eta \equiv \frac{\Delta}{|\lambda|} > 0; \quad (\lambda < 0) \quad (7.2.40)$$

Note that $\mathcal{E}_+ < \mathcal{E}_-$. Denote further

$$\kappa_{\pm} = \frac{S\mathcal{E}_{\pm} - \lambda}{\Delta \mp \mathcal{E}_{\pm}} \quad (7.2.41)$$

We can then report the MOs in terms of (**Exercise**)

$$\begin{aligned} |\psi\rangle_{\text{bind}} &= \frac{\kappa_+ |\varphi_A\rangle + |\varphi_B\rangle}{(1 + 2\kappa_+ S + \kappa_+^2)^{\frac{1}{2}}} \\ |\psi\rangle_{\text{anti}} &= \frac{|\varphi_A\rangle - \kappa_- |\varphi_B\rangle}{(1 - 2\kappa_- S + \kappa_-^2)^{\frac{1}{2}}} \end{aligned} \quad (7.2.42)$$

Remarks: To highlight the underlying picture above, we focus on the Hückel's MO (HMO) theory, with setting the overlap integral $S = 0$. In this limit,

$$\kappa = \kappa_+ = \kappa_- = \sqrt{\eta^2 + 1} - \eta; \quad \kappa \in (0, 1] \quad (7.2.43)$$

with η being given in Eq. (7.2.40). The corresponding energies and MOs are

$$\begin{aligned} \mathcal{E}_+^{\text{HMO}} &= -\Delta + \kappa\lambda \\ |\psi\rangle_{\text{bind}} &= \frac{\kappa|\varphi_A\rangle + |\varphi_B\rangle}{\sqrt{1 + \kappa^2}} \end{aligned} \quad \text{and} \quad \begin{aligned} \mathcal{E}_-^{\text{HMO}} &= \Delta - \kappa\lambda \\ |\psi\rangle_{\text{anti}} &= \frac{|\varphi_A\rangle - \kappa|\varphi_B\rangle}{\sqrt{1 + \kappa^2}} \end{aligned} \quad (7.2.44)$$

- General speaking, in the bonding regime of molecular systems, the coupling parameter $\lambda < 0$ between neighboring AOs. Note also that λ depends the nuclear coordinates as parameters
- The binding MO $|\psi\rangle_{\text{bind}}$ has no node between two AOs, whereas the anti-binding $|\psi\rangle_{\text{anti}}$ has one node. Evidently, $\mathcal{E}_- < \epsilon_B < \epsilon_A < \mathcal{E}_+$, since $\lambda < 0$ and $\epsilon_A - \epsilon_B = 2\Delta > 0$ that is the AOs energies difference.
- The parameter $\kappa \in (0, 1]$, Eq. (7.2.43), measures the polarization, the fact that $|\psi\rangle_{\text{bind}}$ has more contribution from the lower energy AO $|\varphi_B\rangle$, whereas $|\psi\rangle_{\text{anti}}$ does the opposite.

The above basic pictures remain valid for Eqs. (7.2.39)–(7.2.42).

Exercise E7.2.6 Verify Eq. (7.2.42) with Eq. (7.2.41).

7.3 Born–Oppenheimer Approximation Formalism

7.3.1 Born–Oppenheimer approximation

Born–Oppenheimer (BO) approximation plays the pivotal roles in the study of molecular or solid systems. BO approximation is about the least error among various practically used approximations, for its root at the electron–and–nucleus masses difference and the separation of fast–and–slow motion.

The electronic part of Hamiltonian reads (in atomic unit)

$$\hat{H}_e = \sum_{i=1}^N \hat{h}_i(\mathbf{r}_i; \{\mathbf{R}_A\}) + \frac{1}{2} \sum_{i,j=1}^N \frac{1}{r_{ij}} \quad (7.3.1)$$

The single–electron component involves

$$\hat{h}_i(\mathbf{r}_i; \{\mathbf{R}_A\}) = -\frac{1}{2} \nabla_i^2 + v(\mathbf{r}_i), \text{ with } v(\mathbf{r}) = - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \quad (7.3.2)$$

A. The BO approximation and three basic implications

(i) When electronic motion is in study, nuclei coordinates are not dynamics variables, but fixed parameters. That is to solve

$$\hat{H}_e |\Psi_n(\{\mathbf{R}_A\})\rangle = E_n(\{\mathbf{R}_A\}) |\Psi_n(\{\mathbf{R}_A\})\rangle \quad (7.3.3)$$

at each fixed nuclear configurations, $\{\mathbf{R}_A\}$. Equation (7.3.3) is the subject of computational quantum physics.

(ii) The nuclear potential associated with each electronic state is given by

$$U_n(\{\mathbf{R}_A\}) \equiv E_n(\{\mathbf{R}_A\}) + \sum_{A>B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|} \quad (7.3.4)$$

This is the so–called *adiabatic* potential surface, for its arising from the separation of fast–and–slow motion. Included here is also the Coulomb interaction between nuclei. In each individual electronic state, the Schrödinger equation for nuclear motion reads

$$[\hat{T}_{\text{nuc}} + U_n(\{\mathbf{R}_A\})] \Xi_n(\{\mathbf{R}_A\}) = \mathcal{E} \Xi_n(\{\mathbf{R}_A\}) \quad (\text{adiabatic}) \quad (7.3.5)$$

The nonadiabatic influence is neglected; see §7.3.3 for the details.

(iii) Optical transition between adiabatic potential surfaces is essentially *vertical*, without the nuclear positions change. Subsequent nuclear motion occurs on individual adiabatic potentials, whereas surface crossings between different electronic states are nonadiabatic events.

B. Electronic energies versus adiabatic nuclear potentials

- Let us combine Eq. (7.3.1) and the last term of Eq. (7.3.4) into

$$\hat{H}_{\text{elec}} \equiv \hat{H}_e + \sum_{A>B} \frac{Z_A Z_B}{|\mathbf{R}_A - \mathbf{R}_B|} \quad (7.3.6)$$

We can then recast Eq. (7.3.3) as

$$\hat{H}_{\text{elec}}|\Psi(\{\mathbf{R}_A\})\rangle = U(\{\mathbf{R}_A\})|\Psi(\{\mathbf{R}_A\})\rangle \quad (7.3.7)$$

The individual electronic energy, $U \in \{U_n(\{\mathbf{R}_A\}); n = 0, 1, \dots\}$, directly serves the adiabatic potential surface for nuclear motion; cf. Eq. (7.3.5).

- In terms of expectation values, we have

$$U(\{\mathbf{R}_A\}) = \langle \hat{H}_{\text{elec}} \rangle = \langle \hat{T}_e \rangle + \langle \hat{V} \rangle \quad (7.3.8)$$

where $\langle \hat{O} \rangle \equiv \langle \Psi | \hat{O} | \Psi \rangle$. While $\hat{T}_e$ denotes the electronic kinetic energy operator, the electronic potential reads

$$\hat{V} = \hat{V}_{\text{en}} + \hat{V}_{\text{ee}} + V_{\text{nn}} = - \sum_{i,A} \frac{Z_A}{r_{iA}} + \sum_{i>j} \frac{1}{r_{ij}} + \sum_{A>B} \frac{Z_A Z_B}{R_{AB}} \quad (7.3.9)$$

with $r_{iA} \equiv |\mathbf{r}_{iA}| \equiv |\mathbf{r}_i - \mathbf{R}_A|$ and so on.

- Let us focus on the electron–nuclei interaction,

$$\langle \hat{V}_{\text{en}} \rangle = - \left\langle \sum_{i,A} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} \right\rangle = \left\langle \sum_{i=1}^N v(\mathbf{r}_i) \right\rangle = \int v(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} \quad (7.3.10)$$

with $v(\mathbf{r})$ being defined in Eq. (7.3.2) and

$$\rho(\mathbf{r}) \equiv \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \quad (7.3.11)$$

that satisfies

$$\int \rho(\mathbf{r}) d\mathbf{r} = N \quad (7.3.12)$$

As known, Eq. (7.3.11) is the *classical* definition of electron density function. Its quantum correspondence will be given later; see §12.1.1.

C. Electrostatic force of adiabatic potentials

Consider U , Eq. (7.3.8) with Eq. (7.3.9), and set $\lambda = \mathbf{R}_A$ to be the variable for the Hellmann–Feynman theorem, Eq. (2.3.27). We obtain

$$\nabla_A U(\{\mathbf{R}_A\}) = -Z_A \int \rho(\mathbf{r}) \frac{\mathbf{r} - \mathbf{R}_A}{|\mathbf{r} - \mathbf{R}_A|^3} d\mathbf{r} + \frac{1}{2} Z_A \sum_B' Z_B \frac{\mathbf{R}_{AB}}{R_{AB}^3} \quad (7.3.13)$$

This is the *classical electrostatic force theorem*: The electrostatic force on each nucleus arises from the Coulomb fields of other nuclei and all electrons. Equation (7.3.13) holds in general, for any individual adiabatic nuclear potential surface, $U_n(\{\mathbf{R}_A\})$. The electrostatic forces on nuclei are the functional of the corresponding electron density, $\rho(\mathbf{r})$, associated with individual N –electron eigenfunction; see §12.1.1.

7.3.2 Adiabatic electronic steady-state theorems

A. The virial theorem revisit

- First of all, the virial theorem, Eq. (2.3.20), reads

$$2\langle\hat{T}\rangle_{\text{total}} = \sum_i \mathbf{r}_i \cdot \nabla_i V + \sum_A \mathbf{R}_A \cdot \nabla_A V \quad (7.3.14)$$

On the other hand, homogeneous functions,

$$V(\{\lambda \mathbf{r}_i\}, \{\lambda \mathbf{R}_A\}) = \lambda^\nu V(\{\mathbf{r}_i\}, \{\mathbf{R}_A\}) \quad (7.3.15a)$$

satisfy

$$\nu V = \sum_i \mathbf{r}_i \cdot \nabla_i V + \sum_A \mathbf{R}_A \cdot \nabla_A V \quad (7.3.15b)$$

Coulomb potential, $\hat{V}$ of Eq. (7.3.9), is such a function, with the homogeneity order of $\nu = -1$.

- However, the virial theorem, Eq. (7.3.14), is not directly related to the BO approximation. This is concerned with the electronic $U = \langle\hat{T}_e\rangle + \langle\hat{V}\rangle$, where nuclear positions are fixed. The nuclear kinetic energy should be excluded. The resulting adiabatic counterpart to Eq. (7.3.14) reads

$$2\langle\hat{T}_e\rangle = -\langle\hat{V}\rangle - \sum_A \mathbf{R}_A \cdot \langle\nabla_A \hat{V}\rangle = -\langle\hat{V}\rangle - \sum_A \mathbf{R}_A \cdot \nabla_A U \quad (7.3.16)$$

Together with $U = \langle\hat{T}_e\rangle + \langle\hat{V}\rangle$, we obtain

$$\langle\hat{T}_e\rangle = -U(\{\mathbf{R}_A\}) - \sum_A \mathbf{R}_A \cdot \nabla_A U(\{\mathbf{R}_A\}) \quad (7.3.17a)$$

$$\langle\hat{V}\rangle = 2U(\{\mathbf{R}_A\}) + \sum_A \mathbf{R}_A \cdot \nabla_A U(\{\mathbf{R}_A\}) \quad (7.3.17b)$$

Evidently, the second terms account for the influences of adiabatic electrostatic force, as in line with the BO approximation.

Equation (7.3.17) stands for the adiabatic virial theorem. It holds for any individual electronic states, either molecules or solids.

- (i) At the nuclear potential local extrema, Eq. (7.3.17) reduces to

$$\langle\hat{V}\rangle = 2U(\{\mathbf{R}_A\}) \text{ and } \langle\hat{T}_e\rangle = -U(\{\mathbf{R}_A\}), \text{ where } \nabla_A U = 0 \quad (7.3.18)$$

These appear like the standard virial theorem and hold for either minimums or maximums of any adiabatic surfaces.

- (ii) In the atomic limit, with *an arbitrary number of electrons*, we obtain

$$\langle\hat{V}\rangle_{\text{atoms}} = 2E \text{ and } \langle\hat{T}\rangle_{\text{atoms}} = -E \quad (7.3.19)$$

The single nucleus position is set to be the coordinate origin; i.e., $\mathbf{R} = \mathbf{0}$. The standard virial theorem is directly applicable.

B. Ground electronic state energy expression

On the other hand, by setting all individual nuclear Z_a to λZ_a , consider the Hellmann–Feynman theorem integrated over $\lambda = 0$ to $\lambda = 1$. Note that $\lambda = 0$ represents *the nuclei-free limit*. In this limit, the electronic energy, $\langle \hat{H}_{\text{elec}} \rangle_{\lambda=0} = U(\lambda=0)$ via Eq. (7.3.8), consists of only the electron kinetic energy, $\langle \hat{T}_e \rangle$, and electron–electron Coulomb repulsion energy, $\langle \hat{V}_{ee} \rangle$. This is an unbound case, with potential approaches to zero, resulting in $U_0(\lambda=0) = 0$, for the nuclei-free ground state.

As results, for the electronic ground state $|\Psi_0\rangle$, the Hellmann–Feynman theorem gives rise to the expression,

$$U_0(\{\mathbf{R}_A\}) = \int_0^1 d\lambda \left\langle \Psi_0(\lambda) \left| \frac{\partial \hat{H}_{\text{elec}}(\lambda)}{\partial \lambda} \right| \Psi_0(\lambda) \right\rangle \quad (7.3.20)$$

where [cf. Eq. (7.3.9)]

$$\frac{\partial \hat{H}_{\text{elec}}(\lambda)}{\partial \lambda} = \sum_{i=1}^N v(\mathbf{r}_i) + 2\lambda \sum_{A>B} \frac{Z_A Z_B}{R_{AB}} \quad (7.3.21)$$

We obtain the *ground state energy an exact expression*,

$$U_0(\{\mathbf{R}_A\}) = \int_0^1 d\lambda \int d\mathbf{r} v(\mathbf{r}) \rho_0(\mathbf{r}; \lambda) + \sum_{A>B} \frac{Z_A Z_B}{R_{AB}} \quad (7.3.22)$$

The λ -augmented electron density function, $\rho_0(\mathbf{r}; \lambda)$, arises from the N -electron ground-state wave function, $\Psi_0(\lambda)$, with the scaled nucleic charges $\{\lambda Z_A\}$. Equation (7.3.22) holds for systems with any number of electrons. This observation is closely related to the density functional theory: The ground state $\rho(\mathbf{r})$ alone determines the ground state properties of many-electron systems; see §12.4.

7.3.3 Nonadiabatic corrections

Nonadiabatic processes are *ubiquitous* in the surface crossing regime, where electrons and nuclei move at comparable time scales. Thus, it is anticipated that nonadiabaticity be *purely dynamic* in nature. The nuclear momentums will participate in the nonadiabatic couplings.

To proceed, we exploit the electronic wave functions via Eq. (7.3.3),

$$|n\rangle \equiv |\Psi_n(\{\mathbf{R}_A\})\rangle; \quad n = 0, 1, \dots \quad (7.3.23)$$

This refers also to the adiabatic representation basis set, satisfying the orthonormality and completeness, at each given nuclear configuration, $\{\mathbf{R}_A\}$. The inner product runs over the electronic subspace only, such as

$$\langle n|\hat{O}|m\rangle \equiv \int d\mathbf{r}^N \Psi_n^*(\{\mathbf{r}_i\}; \{\mathbf{R}_A\}) [\hat{O} \cdot \Psi_m(\{\mathbf{r}_i\}; \{\mathbf{R}_A\})] \quad (7.3.24)$$

This is a generic *vertical transition* scenario: The nuclear coordinates, associated with different electronic states ($n \neq m$), are fixed in common.

A. The adiabatic interaction formalism

As known, the total composite Hamiltonian reads $\hat{H}_{\text{total}} = \hat{T}_{\text{nuc}} + \hat{H}_{\text{elec}}$, with $\hat{H}_{\text{elec}}$ being specified by Eq. (7.3.6) and $\hat{T}_{\text{nuc}}$ the nuclear kinetic energy operator. In place of Eq. (7.3.5), consider now

$$(\hat{T}_{\text{nuc}} + \hat{H}_{\text{elec}})|\Psi_{\text{total}}\rangle = \mathcal{E}|\Psi_{\text{total}}\rangle \quad (7.3.25)$$

With the adiabatic representation, described after Eq. (7.3.23), the total composite wave function is given by

$$|\Psi_{\text{total}}\rangle = \sum_n \Xi_n(\{\mathbf{R}_A\})|n\rangle; \quad |n\rangle \equiv |\Psi_n(\{\mathbf{R}_A\})\rangle \quad (7.3.26)$$

The standard basis set expansion algebra gives rise to [cf. Eq. (7.3.5)]

$$(\hat{T}_{\text{nuc}} + U_n - \mathcal{E})\Xi_n(\{\mathbf{R}_A\}) = \sum_m \hat{\Lambda}_{nm}\Xi_m(\{\mathbf{R}_A\}) \quad (7.3.27)$$

where

$$\hat{\Lambda}_{nm} = \langle n|\hat{T}_{\text{nuc}}|m\rangle + (1 - \delta_{nm}) \sum_A \frac{1}{M_A} \langle n|\hat{\mathbf{p}}_A|m\rangle \cdot \hat{\mathbf{p}}_A \quad (7.3.28)$$

These are operators in the adiabatic nuclear subspace. The detailed derivations are as follows.

Derivations of Eqs. (7.3.27) and (7.3.28):

(a) Recast Eq. (7.3.25), with Eq. (7.3.26) the adiabatic $\{|n\rangle\}$ -representation, as

$$\langle n|\hat{T}_{\text{nuc}}[\Xi_m|m]\rangle = \langle n|(\mathcal{E} - \hat{H}_{\text{elec}})[\Xi_m|m]\rangle \quad (7.3.29)$$

where $[\Xi_n|n] \equiv |\Xi_n\rangle \cdot |n\rangle$. The right-hand-side via Eq. (7.3.26) goes by

$$\langle n|(\mathcal{E} - \hat{H}_{\text{elec}})[\Xi_m|m]\rangle = \delta_{nm}(\mathcal{E} - U_n)|\Xi_n\rangle \quad (7.3.30)$$

(b) On the other hand,

$$\hat{T}_{\text{nuc}} = \sum_A \frac{\hat{\mathbf{p}}_A^2}{2M_A} \quad (7.3.31)$$

with noting that $\hat{\mathbf{p}}_A = -i\nabla_A$, and

$$\hat{\mathbf{p}}_A^2[\Xi_m|m] = |m\rangle \cdot (\hat{\mathbf{p}}_A^2|\Xi_m\rangle) + (\hat{\mathbf{p}}_A^2|m\rangle) \cdot |\Xi_m\rangle + 2(\hat{\mathbf{p}}_A|m\rangle) \cdot (\hat{\mathbf{p}}_A|\Xi_m\rangle) \quad (7.3.32)$$

We obtain the left-hand-side of Eq. (7.3.29) the expression,

$$\langle n|\hat{T}_{\text{nuc}}[\Xi_m|m]\rangle = \delta_{nm}\hat{T}_{\text{nuc}}|\Xi_n\rangle + \hat{\Lambda}_{nm}|\Xi_m\rangle \quad (7.3.33)$$

with

$$\hat{\Lambda}_{nm} = \langle n|\hat{T}_{\text{nuc}}|m\rangle + \sum_A \frac{1}{M_A} \langle n|\hat{\mathbf{p}}_A|m\rangle \cdot \hat{\mathbf{p}}_A \quad (7.3.34)$$

(c) Apply then the steady-state theorem, $\langle n|\hat{\mathbf{p}}_A|n\rangle = 0$ [cf. Eq. (2.3.13)], and obtain

$$\hat{\Lambda}_{nm} = \langle n|\hat{T}_{\text{nuc}}|m\rangle + (1 - \delta_{nm}) \sum_A \frac{1}{M_A} \langle n|\hat{\mathbf{p}}_A|m\rangle \cdot \hat{\mathbf{p}}_A \quad [Eq. (7.3.28)]$$

Moreover, substitute Eqs. (7.3.33) and (7.3.30) into Eq. (7.3.29), resulting in

$$(\hat{T}_{\text{nuc}} + U_n - \mathcal{E})|\Xi_n\rangle = \sum_m \hat{\Lambda}_{nm}|\Xi_m\rangle \quad [Eq. (7.3.27)]$$

The derivations of Eqs. (7.3.27) and (7.3.28) are now completed. (Q.E.D.)

B. Analysis: Nonadiabaticity versus curve-crossings

- First of all, there are the effective adiabatic potential surfaces,

$$\hat{U}_n^{\text{eff}} \equiv U_n + \hat{\Lambda}_{nn} = U_n + \langle n | \hat{T}_{\text{nucl}} | n \rangle \quad (7.3.35)$$

resulting in Eq. (7.3.27) the expression,

$$(\hat{T}_{\text{nucl}} + \hat{U}_n^{\text{eff}} - \mathcal{E}) \Xi_n(\{\mathbf{R}_A\}) = \sum_m' \hat{\Lambda}_{nm} \Xi_m(\{\mathbf{R}_A\}) \quad (7.3.36)$$

The right-hand-side involves only off-diagonal $\hat{\Lambda}_{nm}$, Eq. (7.3.28); i.e.,

$$\hat{\Lambda}_{nm} = \langle n | \hat{T}_{\text{nucl}} | m \rangle + \sum_A \frac{1}{M_A} \langle n | \hat{\mathbf{p}}_A | m \rangle \cdot \hat{\mathbf{p}}_A \quad (7.3.37)$$

that dictate the nonadiabatic transitions, as comparing between Eq. (7.3.36) and Eq. (7.3.5). Consider first

$$\langle n | \hat{T}_{\text{nucl}} | m \rangle = \sum_A \frac{1}{2M_A} \int d\mathbf{r}^N (\nabla_A \Psi_n^*) \cdot (\nabla_A \Psi_m) \quad (7.3.38)$$

where $\Psi_n \equiv \Psi_n(\{\mathbf{r}_i\}; \{\mathbf{R}_A\})$. The diagonal $\langle n | \hat{T}_{\text{nucl}} | n \rangle$ is the nonadiabatic correction and a part the effective potential U_n^{eff} ; see Eqs. (7.3.35) and (7.3.36) versus Eq. (7.3.5).

- To evaluate the second term in Eq. (7.3.36), we exploit the identity,

$$\langle n | \hat{O} | m \rangle = \frac{\langle n | [\hat{H}_{\text{elec}}, \hat{O}] | m \rangle}{E_n(\{\mathbf{R}_A\}) - E_m(\{\mathbf{R}_A\})}; \quad (n \neq m) \quad (7.3.39)$$

Note that $E_n - E_m = U_n - U_m$ [cf. Eq. (7.3.4)]. Together with

$$\langle n | [\hat{H}_{\text{elec}}, \mathbf{p}_A] | m \rangle = -Z_A \sum_i \left\langle n \left| \left[\frac{1}{r_{iA}}, \mathbf{p}_A \right] \right| m \right\rangle \quad (7.3.40)$$

where $r_{iA} \equiv |\mathbf{r}_{iA}|$, and $\mathbf{r}_{iA} \equiv \mathbf{r}_i - \mathbf{R}_A$, we obtain

$$\langle n | \hat{\mathbf{p}}_A | m \rangle = \frac{-iZ_A}{U_n(\{\mathbf{R}_A\}) - U_m(\{\mathbf{R}_A\})} \sum_i \left\langle n \left| \frac{\mathbf{r}_{iA}}{r_{iA}^3} \right| m \right\rangle \quad (7.3.41)$$

Note also that [cf. Eq. (7.3.24)]

$$\left\langle n \left| \frac{\mathbf{r}_{iA}}{r_{iA}^3} \right| m \right\rangle = \int d\mathbf{r}^N \frac{\mathbf{r}_{iA}}{r_{iA}^3} \Psi_n^*(\{\mathbf{r}_i\}; \{\mathbf{R}_A\}) \Psi_m(\{\mathbf{r}_i\}; \{\mathbf{R}_A\}) \quad (7.3.42)$$

Equations (7.3.35)–(7.3.42) constitute the basic nonadiabatic formalism for nuclear dynamics. The electronic wave functions, Eq. (7.3.23), are used as the expansion basis set.

C. Remarks

- The kinetic energy operator matrix, $\{\hat{T}_{nm}^{\text{nucl}} \equiv \langle n | \hat{T}_{\text{nucl}} | m \rangle\}$, is positive. The diagonal elements contribute to the effective adiabatic potential surfaces; see Eq. (7.3.35). General speaking, $|\hat{T}_{nn}^{\text{nucl}}| \ll |U_n|$. This would remain true when the diagonalization on $\{\hat{T}_{nm}^{\text{nucl}}\}$ is performed.
- The aforementioned diagonalization stands for the transformation from *adiabatic* to *diabatic* representations, engaging *effective nuclear coordinates couplings*. Together with the effective potential surfaces description, Eq. (7.3.35), survived in Eq. (7.3.37) will only be the second term. It involves the total composite of $\langle n | \hat{\mathbf{p}}_A | m \rangle \cdot \hat{\mathbf{p}}_A$, where $n \neq m$; see Eq. (7.3.28).
- Turn to $\langle n | \hat{\mathbf{p}}_A | m \rangle$, Eq. (7.3.41). This is nonradiative but nonadiabatic vertical transition. The primary contributions arise from the curve-crossing regime, where nuclei and electrons possess comparable speeds, with the *singularity* crossing point, as seen from Eq. (7.3.41). Nevertheless, it does not cause divergency to Eq. (7.3.36).
- It is worth emphasizing that Eq. (7.3.36) is rooted at the BO approximation. The backactions of moving nuclei on electrons are neglected. On the other hand, the crossing surface exhibits the *conical intersection* geometry and nonadiabatic dynamics takes place. The aforementioned backactions give rise to the characteristic Berry's geometric phases [David Vanderbilt (2018), *Berry Phases in Electronic Structure Theory*, University Press, Cambridge].

Chapter 8

Many-Particle Systems: Principles and Descriptions

8.1 Indistinguishable Particles Systems

Fundamental particles are either fermions or bosons. The *indistinguishability* of many-particles in a system is concerned with the total wave function, *including spins*. The required particle-permutation property reads

$$\hat{\mathcal{P}}_{ij}\Psi(\cdots, i, \cdots, j, \cdots) \equiv \Psi(\cdots, j, \cdots, i, \cdots) = \pm \Psi(\cdots, i, \cdots, j, \cdots) \quad (8.1.1)$$

with the “+” sign for bosons and “−” for fermions, respectively. The indistinguishable particles permutation operator has the eigenvalues ± 1 , since $\hat{\mathcal{P}}_{ij}^2 = 1$.

- (a) Fermions: Electrons, neutrons, protons, ^{13}C , ^3He , $\cdots$.
- (b) Bosons: Photons, ^{12}C , ^4He , gravitational wave, $\cdots$.

8.1.1 Pauli exclusion principle

A. General statements

- The Pauli's exclusion principle is rooted at the aforementioned anti-symmetry requirement on many-electron wave functions, including spins. The most basic statement of the principle reads as follows.

No more than one electron can occupy a single spin orbital state

- In the real $\mathbf{r}$ -space representation, the Pauli's exclusion principle refers to the nature of **Fermi hole**: “*No more than one electron with same spin can simultaneously appear at a single real space point*”; see Eq. (8.1.6) for the further elaborations. This scenario of exclusion is particularly important in unbound-state cases, in such as electron scattering problems.

- The statement of Pauli's exclusion principle in the reciprocal $\mathbf{k}$ -space (or momentum space) is also self-evident now. General speaking, electrons with same spin cannot meet each other. This is true in either the spin-orbital space or the real space or the reciprocal space.
- Pauli's exclusion principle is universal for indistinguishable fermions. The implications are profound, in the electronic structure of atoms, molecules and solids, the functionality of electronic or spintronic devices, and so on.

B. Slater determinants representation

Slater determinants refer also to the *configurations*: Each specifies a set of N spin-orbitals occupied by N electrons. As orbitals are single-electron wave functions, each Slater determinant, $|\Phi\rangle \in \{|\Phi_{\mathbf{j}}\rangle\}$, describes an *noninteracting N -electron wave function*. The exact N -electron wave functions can then be expanded in terms of

$$|\Psi\rangle = \sum_{\mathbf{j}} c_{\mathbf{j}} |\Phi_{\mathbf{j}}\rangle \quad (8.1.2)$$

This is also called the configuration interaction wave function, to be elaborated in §11.1.1.

Without loss of generality, we specify the two-electron Slater determinant, consisting of the spin-orbitals, χ_a and χ_b , such as $|\chi_a\rangle = |n, l, m, m_s\rangle$. Denote further $\chi_a(1)$ and $\chi_b(2)$ for labeling electrons, but they are indistinguishable. The Fock's product, $\Phi_{\text{Fock}}(1, 2) = \chi_a(1)\chi_b(2)$, does not satisfy the Pauli exclusion principle; i.e., $\Phi_{\text{Fock}}(2, 1) \neq -\Phi_{\text{Fock}}(1, 2)$. The appropriate noninteracting products that serve as basis many-particle wave functions would be the Slater determinants,

$$\Phi(1, 2) = |\chi_a(1)\chi_b(2)\rangle \equiv \frac{1}{\sqrt{2!}} \begin{vmatrix} \chi_a(1) & \chi_b(1) \\ \chi_a(2) & \chi_b(2) \end{vmatrix} \quad (8.1.3)$$

with $\Phi(2, 1) = -\Phi(1, 2)$ for individual Slater determinant. The configuration interaction wave function, $\Psi(1, 2)$ of Eq. (8.1.2), does satisfy the anti-symmetric requirement. Evidently, $|\chi_a(1)\chi_a(2)\rangle = 0$. This is the Pauli exclusion principle that no two electrons can occupy a same spin-orbital state. Each spin orbital, $|\chi_a\rangle$, is of the occupation number either 0 or 1. The related fermionic algebra, with creation/annihilation and occupation number operators, will be elaborated in §8.2.

8.1.2 Slater determinant: Noninteracting nature

A. Noninteracting nature of individual determinants

• Slater determinants are *noninteracting* many-electron wave functions, satisfying the Pauli's exclusion principle, but *no correlations between electrons with opposite spins*.

(i) For illustrations, we examine closely two-electron Slater determinants, Eq. (8.1.3), the equivalent form of

$$\Phi(\mathbf{x}_1, \mathbf{x}_2) = |\chi_a(\mathbf{x}_1)\chi_b(\mathbf{x}_2)\rangle, \quad \text{where } \chi(\mathbf{x}) \equiv \psi(\mathbf{r})\sigma(s) \quad (8.1.4)$$

with the spin $\sigma(s) = \alpha$ or β .

(ii) Consider the joint probability density in real space,

$$\begin{aligned} |\Phi(\mathbf{r}_1, \mathbf{r}_2)|^2 &= \frac{1}{2} \left[\psi_a^*(\mathbf{r}_1)\psi_b^*(\mathbf{r}_2)\langle\sigma_a(s_1)\sigma_b(s_2)| - \psi_b^*(\mathbf{r}_1)\psi_a^*(\mathbf{r}_2)\langle\sigma_b(s_1)\sigma_a(s_2)| \right] \\ &\quad \times \left[\psi_a(\mathbf{r}_1)\psi_b(\mathbf{r}_2)|\sigma_a(s_1)\sigma_b(s_2)\rangle - \psi_b(\mathbf{r}_1)\psi_a(\mathbf{r}_2)|\sigma_b(s_1)\sigma_a(s_2)\rangle \right] \\ &= \frac{1}{2} [|\psi_a(\mathbf{r}_1)|^2|\psi_b(\mathbf{r}_2)|^2 + |\psi_b(\mathbf{r}_1)|^2|\psi_a(\mathbf{r}_2)|^2] \\ &\quad - \frac{1}{2} \delta_{\sigma_a\sigma_b} \underbrace{[\psi_a^*(\mathbf{r}_1)\psi_b(\mathbf{r}_1)\psi_a(\mathbf{r}_2)\psi_b^*(\mathbf{r}_2) + \text{c.c.}]}_{\text{Exchange term}} \end{aligned} \quad (8.1.5)$$

As highlighted above, the exchange term is nonclassical, associated with $\langle\sigma_a|\sigma_b\rangle = \delta_{\sigma_a\sigma_b}$, occurring only when two electrons are of the same spin.

(iii) At $\mathbf{r}_1 = \mathbf{r}_2$, Eq. (8.1.5) becomes

$$|\Phi(\mathbf{r}_1, \mathbf{r}_1)|^2 = \begin{cases} |\psi_a(\mathbf{r}_1)|^2|\psi_b(\mathbf{r}_1)|^2; & \sigma_a \neq \sigma_b \quad (\text{no correlation}) \\ 0; & \sigma_a = \sigma_b \quad (\text{Fermi hole}) \end{cases} \quad (8.1.6)$$

The observed Fermi hole here agrees with the Pauli exclusion principle: “No more than one electron with same spin can simultaneously appear at a single real space point”, as mentioned earlier.

• Individual determinantal wave function contains no correlations between electrons with opposite spins. The existence of *Coulomb hole* is completely missing. *Any single determinant overestimates the electron-electron interaction.*

B. Slater determinants: Notations and remarks

The Slater determinant wave function for N -electron systems reads

$$|\Phi\rangle = |\chi_1\chi_2\cdots\chi_N\rangle \equiv \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(1) & \chi_2(1) & \cdots & \chi_N(1) \\ \chi_1(2) & \chi_2(2) & \cdots & \chi_N(2) \\ \cdots & \cdots & \cdots & \cdots \\ \chi_1(N) & \chi_2(N) & \cdots & \chi_N(N) \end{vmatrix} \quad (8.1.7)$$

satisfying $\langle\Phi|\Phi\rangle = 1$, with

$$\langle\chi_a|\chi_b\rangle = \delta_{ab}; \quad a, b = 1, 2, \cdots, N \quad (8.1.8)$$

(a) Each Slater determinant, $|\Phi\rangle$ of Eq. (8.1.7), describes an N -electron *configuration*, in terms of those N number of *occupied spin-orbitals*. Moreover, spin-orbitals can be either the spin-unrestricted or the spin-restricted; see Eqs. (8.1.11)–(8.1.14) and the discussions therein.

(b) As detailed in Chapter 10, the Hartree–Fock (HF) theory that minimizes the determinantal energy, $\langle\Phi|\hat{H}|\Phi\rangle$, gives rise to all related spin-orbitals and the associated energies, $\{\chi_j\}$ and $\{\epsilon_j\}$. The convention order of arranging $\{\chi_j\}$ to the columns of determinant follows that of $\epsilon_1 \leq \epsilon_2 \leq \cdots \leq \epsilon_N$.

(c) It is sufficient to specify only the diagonal elements; i.e., the first expression of Eq. (8.1.7). The antisymmetry, with respect to the permutation on two electrons, is represented by swapping two rows of the Slater determinant. However, the labeling of indistinguishable particles is physically irrelevant. This is about the notation, such as $|\chi_a\chi_b\rangle \equiv |\chi_a(1)\chi_b(2)\rangle$.

Exercises

E8.1.1 In line with Eq. (8.1.7), the inner product of determinants is given by

$$\langle\tilde{\chi}_1\cdots\tilde{\chi}_N|\chi_1\cdots\chi_N\rangle = \frac{1}{N!} \int d\mathbf{x}^N \begin{vmatrix} \tilde{\chi}_1^*(\mathbf{x}_1) & \cdots & \tilde{\chi}_1^*(\mathbf{x}_N) \\ \cdots & \cdots & \cdots \\ \tilde{\chi}_N^*(\mathbf{x}_1) & \cdots & \tilde{\chi}_N^*(\mathbf{x}_N) \end{vmatrix} \begin{vmatrix} \chi_1(\mathbf{x}_1) & \cdots & \chi_N(\mathbf{x}_1) \\ \cdots & \cdots & \cdots \\ \chi_1(\mathbf{x}_N) & \cdots & \chi_N(\mathbf{x}_N) \end{vmatrix}$$

Show that

$$\langle\tilde{\chi}_1\cdots\tilde{\chi}_N|\chi_1\cdots\chi_N\rangle = \begin{vmatrix} \langle\tilde{\chi}_1|\chi_1\rangle & \cdots & \langle\tilde{\chi}_1|\chi_N\rangle \\ \cdots & \cdots & \cdots \\ \langle\tilde{\chi}_N|\chi_1\rangle & \cdots & \langle\tilde{\chi}_N|\chi_N\rangle \end{vmatrix} \quad (8.1.9)$$

E8.1.2 Show that the determinants basis set satisfy

$$\langle\chi_{j_1}\cdots\chi_{j_N}|\chi_{k_1}\cdots\chi_{k_N}\rangle = \det|\delta_{j_1k_1}\cdots\delta_{j_Nk_N}| \quad (\text{Orthonormality}) \quad (8.1.10a)$$

$$\frac{1}{N!} \sum_{j_1\cdots j_N} |\chi_{j_1}\cdots\chi_{j_N}\rangle \langle\chi_{j_1}\cdots\chi_{j_N}| = \hat{I}_N \quad (\text{Completeness}) \quad (8.1.10b)$$

The factor $N!$ arises from the permutations of N -electrons.

E8.1.3 Elaborate the above expressions with the two-electron case.

C. Spin unrestricted versus restricted determinants

- Each spin-orbital, $\chi_j(\mathbf{x}) \equiv \langle \mathbf{x} | \chi_j \rangle \equiv \langle \mathbf{r} s | \chi_j \rangle$, participating in the determinant, is given by

$$\chi_j(\mathbf{x}) \equiv \chi_j(\mathbf{r} s) = \psi_j^\sigma(\mathbf{r}) \sigma_j(s); \quad \sigma_j(s) = \alpha, \beta \quad (8.1.11)$$

where α and β specify the spin $\uparrow$ and $\downarrow$, with $m_s = \frac{1}{2}$ and $-\frac{1}{2}$, respectively.

(a) In general, $\psi_j^\alpha(\mathbf{r}) \neq \psi_j^\beta(\mathbf{r})$. This is the type of spin *unrestricted* spatial orbitals, determined via the unrestricted Hartree-Fock (UHF) theory that is to be detailed in §10.3.1. The spin *restricted* counterpart or the RHF theory goes by $\psi_j^\alpha(\mathbf{r}) = \psi_j^\beta(\mathbf{r}) = \psi_j(\mathbf{r})$; see §10.3.2 for the details.

(b) Usually the spin restricted description is adopted for close-shell systems. However, in open-shell systems, there are *effective spin-orbital couplings*, arising from the spin polarization. The spin unrestricted description, with $\psi_j^\alpha(\mathbf{r}) \neq \psi_j^\beta(\mathbf{r})$, would be preferred.

(c) For illustration, consider Li atom the ground state configuration, $1s^2 2s$. The spin unrestricted and restricted determinants are given by

$$\begin{aligned} |\Phi^{\text{UHF}}\rangle &= |\psi_{1s}^\alpha(1) \bar{\psi}_{1s}^\beta(2) \psi_{2s}^\alpha(3)\rangle \\ |\Phi^{\text{RHF}}\rangle &= |\psi_{1s}(1) \bar{\psi}_{1s}(2) \psi_{2s}(3)\rangle \end{aligned} \quad (8.1.12)$$

with the unrestricted and restricted spin-orbitals being rewritten as

$$\chi(\mathbf{x}) = \begin{cases} \psi^\alpha(\mathbf{x}) \equiv \psi^\alpha(\mathbf{r}) \alpha(s) \\ \bar{\psi}^\beta(\mathbf{x}) \equiv \psi^\beta(\mathbf{r}) \beta(s) \end{cases} \quad \text{and} \quad \begin{cases} \psi(\mathbf{x}) \equiv \psi(\mathbf{r}) \alpha(s) \\ \bar{\psi}(\mathbf{x}) \equiv \psi(\mathbf{r}) \beta(s) \end{cases} \quad (8.1.13)$$

respectively. For Eq. (8.1.12), it is anticipated that

$$\epsilon(\psi_{1s}^\alpha) < \epsilon(\bar{\psi}_{1s}^\beta) < \epsilon(\psi_{2s}^\alpha), \quad \text{whereas} \quad \epsilon(\psi_{1s}) = \epsilon(\bar{\psi}_{1s}) < \epsilon(\psi_{2s}) \quad (8.1.14)$$

While the restricted ψ_{1s} and $\bar{\psi}_{1s}$, differ only in the spin parts, the unrestricted ψ_{1s}^α and $\bar{\psi}_{1s}^\beta$ are distinct also by $\psi_{1s}^\alpha(\mathbf{r}) \neq \bar{\psi}_{1s}^\beta(\mathbf{r})$, since the α -spin electron in $\psi_{2s}^\alpha(\mathbf{r})$ polarizes these two $1s$ orbitals.

(d) It is worth noting that for molecules and solid, spatial orbitals go by the Bohn-Oppenheimer approximation, with the parametric dependence on nuclear coordinates being implied.

- General speaking, $\langle \Phi^{\text{UHF}} | \hat{H} | \Phi^{\text{UHF}} \rangle \leq \langle \Phi^{\text{RHF}} | \hat{H} | \Phi^{\text{RHF}} \rangle$, for open-shell systems. The spin unrestricted determinant is energetically favored, but in the expense of *spin contamination* that will be discussed soon below, as if the effective mean field Hamiltonian in the presence of spin polarization no longer commute with $\hat{S}^2$.

8.1.3 Spin properties of Slater determinants

A. Opening remarks

- For an N -electron wave function, $|\Psi\rangle$, the spin adaptivity refers to the properties of (atomic unit)

$$\hat{S}^2|\Psi\rangle = S(S+1)|\Psi\rangle \quad \text{and} \quad \hat{S}_z|\Psi\rangle = M_S|\Psi\rangle \quad (8.1.15)$$

where $M_S = -S, -S+1, \dots, S-1, S$. The total composite spin angular momentum vector comprises those of individual electrons:

$$\hat{\mathbf{S}} = \hat{\mathbf{s}}_1 + \dots + \hat{\mathbf{s}}_N \quad (8.1.16)$$

- Turn to the single determinant approximant,

$$|\Psi\rangle \approx |\Phi\rangle = |\chi_1(1) \cdots \chi_N(N)\rangle \quad (8.1.17)$$

(a) It is easy to show that

$$\hat{S}_z|\Phi\rangle = M_S|\Phi\rangle, \quad \text{with} \quad M_S = (N_\alpha - N_\beta)/2 \quad (8.1.18)$$

Here, N_α and $N_\beta \equiv N - N_\alpha$ are the numbers of spin-specified electrons. The eigenvalue of $\hat{S}_z$ is half the number of unpaired electrons, $N_\alpha - N_\beta$.

(b) Individual determinant would not be an eigenstate of $\hat{S}^2$, the first expression of Eq. (8.1.15), but rather be of [cf. Eqs. (8.1.28)–(8.1.31)]

$$S(S+1) \geq \langle \Phi | \hat{S}^2 | \Phi \rangle \geq M_S(M_S+1) \quad (8.1.19)$$

The spin-adaptivity can be achieved with one or few spin restricted determinant(s); see Part C later.

B. The Fock–Dirac spin operator formalism

- Unrestricted determinants are by nature spin contaminated. The quantification goes by the Fock–Dirac spin operator, $\hat{S}_{\text{FD}}^2 \neq S^2$, but setting $S_{\text{FD}}^2|\Phi\rangle \equiv \hat{S}^2|\Phi\rangle$ [cf. Eq. (8.1.28)], for any unrestricted determinant. Without loss of generality, we set $N_\alpha \geq N_\beta$, with $N = N_\alpha + N_\beta$.

Let us start with individual electron the following identities,

$$\hat{s}_z|\alpha\rangle = \frac{1}{2}|\alpha\rangle \quad \text{and} \quad \hat{s}_z|\beta\rangle = -\frac{1}{2}|\beta\rangle \quad (8.1.20)$$

The ladder operators, $\hat{s}_\pm \equiv \hat{s}_x \pm i\hat{s}_y$, satisfy Eqs. (5.1.5)–(5.1.7); i.e.,

$$\hat{s}_+|\alpha\rangle = \hat{s}_-|\beta\rangle = 0, \quad \hat{s}_+|\beta\rangle = |\alpha\rangle \quad \text{and} \quad \hat{s}_-|\alpha\rangle = |\beta\rangle \quad (8.1.21)$$

Turn to Eq. (5.1.8) for the total N -electron spin operator,

$$\hat{S}^2 = \hat{S}_z^2 + \hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+ \quad (8.1.22)$$

From Eq. (8.1.18), we have

$$(\hat{S}_z^2 + \hat{S}_z)|\Phi\rangle = M_S(M_S + 1)|\Phi\rangle \quad (8.1.23)$$

Consider then

$$\hat{S}_- \hat{S}_+ |\Phi\rangle = (\text{I}) + (\text{II}) \quad (8.1.24)$$

to be evaluated via [cf. Eq. (8.1.16)]

$$\hat{S}_- \hat{S}_+ = \sum_{j,k=1}^N \hat{s}_-(j) \hat{s}_+(k) = \sum_{j=1}^N \hat{s}_-(j) \hat{s}_+(j) + \sum_{j,k=1}^{N'} \hat{s}_-(j) \hat{s}_+(k) \quad (8.1.25)$$

Together with Eq. (8.1.21) and $N_\alpha \geq N_\beta$, we obtain

$$(\text{I}) \equiv \sum_{j=1}^N \hat{s}_-(j) \hat{s}_+(j) |\Phi\rangle = N_\beta |\Phi\rangle \quad (8.1.26a)$$

$$(\text{II}) \equiv \sum_{j,k=1}^{N'} \hat{s}_-(j) \hat{s}_+(k) |\Phi\rangle = \sum_{j=1}^{N_\alpha} \sum_{k=1}^{N_\beta} \hat{\mathcal{P}}_{jk}^S |\Phi\rangle \quad (8.1.26b)$$

Here, $\hat{\mathcal{P}}_{jk}^S$, the spin permutation operator, acts only on the spin-subspace and can be described in terms of [cf. Eq. (8.1.13)]

$$\begin{cases} \psi_j^\alpha(\mathbf{x}_j) \equiv \psi_j^\alpha(\mathbf{r}_j) \alpha(s_j) \\ \bar{\psi}_k^\beta(\mathbf{x}_k) \equiv \psi_k^\beta(\mathbf{r}_k) \beta(s_k) \end{cases} \xrightarrow{\hat{\mathcal{P}}_{jk}^S} \begin{cases} \bar{\psi}_j^\alpha(\mathbf{x}_j) \equiv \psi_j^\alpha(\mathbf{r}_j) \beta(s_j) \\ \psi_k^\beta(\mathbf{x}_k) \equiv \psi_k^\beta(\mathbf{r}_k) \alpha(s_k) \end{cases} \quad (8.1.27)$$

which in short reads

$$\hat{\mathcal{P}}_{jk}^S [\alpha(s_j) \beta(s_k)] = \beta(s_j) \alpha(s_k) \quad (8.1.27')$$

Neither the *particle-indexes* j and k inside the parentheses, nor the spatial wave functions, $\psi_j^\alpha(\mathbf{r}_j)$ and $\psi_k^\beta(\mathbf{r}_k)$, are altered. The implication here will be evident later; see Eqs. (8.1.33)–(8.1.37).

Now combining Eqs. (8.1.22)–(8.1.26), we obtain

$$\hat{S}^2 |\Phi\rangle = [M_S(M_S + 1) + N_\beta] |\Phi\rangle + \sum_{j=1}^{N_\alpha} \sum_{k=1}^{N_\beta} \hat{\mathcal{P}}_{jk}^S |\Phi\rangle \equiv \hat{S}_{\text{FD}}^2 |\Phi\rangle \quad (8.1.28)$$

The last identity defines

$$\hat{S}_{\text{FD}}^2 \equiv M_S(M_S + 1) + N_\beta + \sum_{j=1}^{N_\alpha} \sum_{k=1}^{N_\beta} \hat{\mathcal{P}}_{jk}^S \quad (8.1.29)$$

This is so-called the Fock–Dirac spin angular momentum operator.

- Consider then

$$\langle \Phi | \hat{S}^2 | \Phi \rangle = M_S(M_S + 1) + N_\beta + \sum_{j=1}^{N_\alpha} \sum_{k=1}^{N_\beta} \langle \Phi | \hat{\mathcal{P}}_{jk}^S | \Phi \rangle \quad (8.1.30)$$

As shown below, we can obtain

$$\langle \Phi | \hat{\mathcal{P}}_{jk}^S | \Phi \rangle = -|\langle \psi_j^\alpha(\mathbf{r}) | \psi_k^\beta(\mathbf{r}) \rangle|^2 \quad (8.1.31)$$

Derivations: For general purpose, let $|\Phi\rangle$ be an unrestricted determinant,

$$|\Phi\rangle = |\cdots \psi_j^\alpha(\mathbf{x}_j) \cdots \bar{\psi}_k^\beta(\mathbf{x}_k) \cdots\rangle \quad (8.1.32)$$

The action of particle permutation is given by

$$\hat{\mathcal{P}}_{jk} |\Phi\rangle = |\cdots \bar{\psi}_k^\beta(\mathbf{x}_k) \cdots \psi_j^\alpha(\mathbf{x}_j) \cdots\rangle = -|\Phi\rangle \quad (8.1.33)$$

Together with the action of spin permutation operator, Eq. (8.1.27), we have

$$\hat{\mathcal{P}}_{jk}^S |\Phi\rangle = -\hat{\mathcal{P}}_{jk}^S (\hat{\mathcal{P}}_{jk} |\Phi\rangle) = -|\cdots \psi_k^\beta(\mathbf{x}_k) \cdots \bar{\psi}_j^\alpha(\mathbf{x}_j) \cdots\rangle \quad (8.1.34)$$

Thus,

$$\langle \Phi | \hat{\mathcal{P}}_{jk}^S | \Phi \rangle = -\langle \Phi | \hat{\mathcal{P}}_{jk}^S \hat{\mathcal{P}}_{jk} | \Phi \rangle = -\langle \psi_j^\alpha(\mathbf{x}_1) \bar{\psi}_k^\beta(\mathbf{x}_2) | \psi_k^\beta(\mathbf{x}_1) \bar{\psi}_j^\alpha(\mathbf{x}_2) \rangle \quad (8.1.35)$$

The involving determinantal inner product both the spatial orbitals and spin degrees of freedom. The latter can be integrated out. We obtain

$$\langle \psi_j^\alpha(\mathbf{x}_1) \bar{\psi}_k^\beta(\mathbf{x}_2) | \psi_k^\beta(\mathbf{x}_1) \bar{\psi}_j^\alpha(\mathbf{x}_2) \rangle = \langle \psi_j^\alpha(\mathbf{r}_1) | \psi_k^\beta(\mathbf{r}_1) \rangle \langle \psi_k^\beta(\mathbf{r}_2) | \psi_j^\alpha(\mathbf{r}_2) \rangle \quad (8.1.36)$$

Substitute it into Eq. (8.1.35), resulting in

$$\langle \Phi | \hat{\mathcal{P}}_{jk}^S | \Phi \rangle = -|\langle \psi_j^\alpha(\mathbf{r}) | \psi_k^\beta(\mathbf{r}) \rangle|^2 \quad [Eq. (8.1.31)]$$

The derivations of Eq. (8.1.31) are completed.

(Q.E.D.)

Remarks on Eq. (8.1.30) and alternative expression:

(i) The unrestricted spatial wave functions, $\{\psi_j^\alpha(\mathbf{r})\}$ and $\{\psi_k^\beta(\mathbf{r})\}$, are eigenvectors of two distinct single-particle operators, in the presence of spin polarization. As results, $\langle \psi_j^\alpha | \psi_k^\alpha \rangle = \langle \psi_j^\beta | \psi_k^\beta \rangle = \delta_{jk}$, whereas

$$S_{jk}^{\alpha\beta} \equiv \langle \psi_j^\alpha(\mathbf{r}) | \psi_k^\beta(\mathbf{r}) \rangle \neq \delta_{jk} \quad (8.1.37)$$

(ii) The final expression for Eq. (8.1.30) reads

$$\langle \Phi | \hat{S}^2 | \Phi \rangle = M_S(M_S + 1) + N_\beta - \sum_{j=1}^{N_\alpha} \sum_{k=1}^{N_\beta} |S_{jk}^{\alpha\beta}|^2 \quad (8.1.38)$$

where $M_S = (N_\alpha - N_\beta)/2 \geq 0$; cf. Eq. (8.1.18). The first term represents an eigenvalue of $\hat{S}^2$, with the angular momentum quantum number, $S = M_S$. This is the scenario that all unpaired electrons are of same spins.

(iii) In Eq. (8.1.38), the last term satisfies (see the derivations soon later)

$$\sum_{j=1}^{N_\alpha} \sum_{k=1}^{N_\beta} |S_{jk}^{\alpha\beta}|^2 \leq \min(N_\alpha, N_\beta) = N_\beta \quad (8.1.39)$$

The equal sign holds only for restricted determinants; i.e., $\psi_j^\alpha = \psi_j^\beta = \psi_j$. In this case, Eq. (8.1.38) becomes $\langle \Phi | \hat{S}^2 | \Phi \rangle = M_S(M_S + 1)$, without spin contamination. The inequalities of Eq. (8.1.19) are now evident.

(iv) General speaking, unrestricted determinants are neither eigenfunctions of spin operator $\hat{S}^2$, nor spin-adapted by combining a small number of them. Individual unrestricted determinant consists of two sets spin polarized spatial orbitals, with $\langle \Phi | \hat{S}^2 | \Phi \rangle > M_S^{\max}(M_S^{\max} + 1)$ being the spin contamination. *Effective spin purification* is possible in formal, especially for *half-filled* open systems, with all unpaired electrons in parallel spins.

Derivations of Eq. (8.1.39): As known, the Slater determinant is invariance, if the individual j^{th} -column is added with the linear combination of other columns. This is about the linear transformation on the orbitals, without altering the indistinguishability of electrons. For $\{\psi_j^\alpha(\mathbf{r})\}$ and $\{\psi_k^\beta(\mathbf{r})\}$, there exist the so-called correspondence pair of linear transformations, with the transformed orbitals satisfying

$$\langle \tilde{\psi}_j^\alpha(\mathbf{r}) | \tilde{\psi}_k^\alpha(\mathbf{r}) \rangle = \langle \tilde{\psi}_j^\beta(\mathbf{r}) | \tilde{\psi}_k^\beta(\mathbf{r}) \rangle = \delta_{jk} \quad (8.1.40a)$$

$$\langle \tilde{\psi}_j^\alpha(\mathbf{r}) | \tilde{\psi}_k^\beta(\mathbf{r}) \rangle = \delta_{jk} \langle \tilde{\psi}_k^\alpha(\mathbf{r}) | \tilde{\psi}_k^\beta(\mathbf{r}) \rangle \quad (8.1.40b)$$

The Schwartz inequality can now be applied, resulting in

$$|\langle \tilde{\psi}_k^\alpha(\mathbf{r}) | \tilde{\psi}_k^\beta(\mathbf{r}) \rangle|^2 \leq \langle \tilde{\psi}_k^\alpha(\mathbf{r}) | \tilde{\psi}_k^\alpha(\mathbf{r}) \rangle \langle \tilde{\psi}_k^\beta(\mathbf{r}) | \tilde{\psi}_k^\beta(\mathbf{r}) \rangle = 1 \quad (8.1.41)$$

The above expressions give rise to Eq. (8.1.39) the expressions,

$$\sum_{j=1}^{N_\alpha} \sum_{k=1}^{N_\beta} |S_{jk}^{\alpha\beta}|^2 = \sum_{k=1}^{N_\beta} |\langle \tilde{\psi}_k^\alpha(\mathbf{r}) | \tilde{\psi}_k^\beta(\mathbf{r}) \rangle|^2 \leq N_\beta \quad [Eq. (8.1.39)]$$

C. Spin-adapted configurations

- First of all, spin adaptivity is concerned with $[\hat{H}, \hat{S}^2] = [\hat{H}, \hat{S}_z] = 0$, underlying Eq. (8.1.15), in the absence of spin-orbit couplings. It requires the total electronic wave functions be *factorizable* into the spatial and spin components, in relation to the term symbols at level of $^{2S+1}\Lambda_{M_S}$.

Unlike atoms, molecular electronic structures are considered under the Born–Oppenheimer approximation. The total spatial angular momentum of electrons in a molecule is not conserved, since the electric field of multiple nuclei is not centrally symmetric. The letter Λ in electronic term symbol of molecules arises from the symmetry of nuclei at a fixed geometric configuration; see Chapter 9.

- Turn to the determinantal form of wave functions. As mentioned earlier, unrestricted determinants are spin contaminated in general. Spin adaptivity (i.e., the spatial-and-spin factorization) is achievable with one or *few* spin *restricted* Slater determinant(s).

For illustration, we focus on the configurations of two electron occupying two spatial orbitals. Consider first

$$|a\bar{b}\rangle = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_a(1)\alpha(1) & \psi_b(1)\beta(1) \\ \psi_a(2)\alpha(2) & \psi_b(2)\beta(2) \end{vmatrix} \quad (8.1.42a)$$

and

$$|\bar{a}b\rangle = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_a(1)\beta(1) & \psi_b(1)\alpha(1) \\ \psi_a(2)\beta(2) & \psi_b(2)\alpha(2) \end{vmatrix} \quad (8.1.42b)$$

Instead of these individual determinants, the spin adaptivity goes with

$$^1\Phi_{ab}(1, 2) = 2^{-\frac{1}{2}} (|a\bar{b}\rangle - |\bar{a}b\rangle) = \Psi_{ab}^+(1, 2)S(1, 2) \quad (8.1.43a)$$

$$^3\Phi_{ab}(1, 2) = 2^{-\frac{1}{2}} (|a\bar{b}\rangle + |\bar{a}b\rangle) = \Psi_{ab}^-(1, 2)T_0(1, 2) \quad (8.1.43b)$$

The spatial components are

$$\Psi_{ab}^{\pm}(1, 2) \equiv \frac{1}{\sqrt{2}} [\psi_a(1)\psi_b(2) \pm \psi_a(2)\psi_b(1)] = \pm \Psi_{ab}^{\pm}(2, 1) \quad (8.1.44)$$

with the last identity specifying the parity of particle permutation. In Eq. (8.1.43), the spin components are

$$S(1, 2) \equiv 2^{-\frac{1}{2}} [\alpha(1)\beta(2) - \alpha(2)\beta(1)] \quad (8.1.45a)$$

$$T_0(1, 2) \equiv 2^{-\frac{1}{2}} [\alpha(1)\beta(2) + \alpha(2)\beta(1)] \quad (8.1.45b)$$

The latter is the $M_S = 0$ in the spin-triplet set of

$$T(1, 2) \equiv \{\alpha(1)\alpha(2), T_0(1, 2), \beta(1)\beta(2)\} \quad (8.1.46)$$

The $M_S = 1$ and $M_S = -1$ of the spin triplet are associated with

$$\begin{aligned} ^3_1\Phi_{ab}(1, 2) &= |ab\rangle = \Psi_{ab}^-(1, 2)\alpha(1)\alpha(2) \\ ^3_{-1}\Phi_{ab}(1, 2) &= |\bar{a}\bar{b}\rangle = \Psi_{ab}^-(1, 2)\beta(1)\beta(2) \end{aligned} \quad (8.1.47)$$

Each Slater determinant is now spin adaptive. Together with Eqs. (8.1.43b) and (8.1.46), we have

$$^3\Phi_{ab}(1, 2) = \Psi_{ab}^-(1, 2)T(1, 2) \quad (8.1.48)$$

8.2 Second Quantization Formalism

8.2.1 Fermionic Fock-space descriptions

A. General remarks

• In quantum mechanics the first quantization refers to the operator correspondences to dynamical observables such as coordinates and momentum. The second quantization is concerned with the fermionic and bosonic operators descriptions of many-particle wave functions. It involves annihilation and creation operators, $\{\hat{a}_j\}$ and $\{\hat{a}_j^\dagger\}$, associated with spin orbitals that are physically specifiable, whereas particles are indistinguishable.

- (a) The *number operator* is given by $\hat{n}_j \equiv \hat{a}_j^\dagger \hat{a}_j$. Its eigenvalues are $n_j = 0$ and 1 for fermions, whereas $n_j = 0, 1, 2, \dots$ for bosonic systems.
- (b) The Fock space is a many-particle version of the Hilbert space, with the elements being configuration wave vectors, either fermionic or bosonic. The number of particles occupied in a configuration can be $N = 0, 1, 2, \dots$.
- (c) The full Fock space comprises subspaces of different number of particles. *These subspaces are mutually orthogonal from each others.*

• **Many-boson systems:** Wave functions are symmetric with respect to particle permutations. Bose-Einstein condensation refers to a macroscopic number of particles occupying a single microscopic state.

(i) Denote $|\chi_1^{n_1} \cdots \chi_K^{n_K}\rangle$, an N -bosons configuration, where $n_1 + \cdots + n_K = N$, with $n_j = 0, 1, 2, \dots$ being the occupation number, such that

$$|\chi_1^{n_1}\rangle \equiv |\chi_1(1) \cdots \chi_1(n_1)\rangle \quad \text{and} \quad |\chi_2^{n_2}\rangle \equiv |\chi_2(n_1 + 1) \cdots \chi_2(n_1 + n_2)\rangle$$

Bosonic configurations are *permanental* wave functions, satisfying the symmetric relation with respect to particle permutations; i.e.,

$$|\chi_j(1)\chi_k(2)\rangle = |\chi_j(2)\chi_k(1)\rangle$$

(ii) Electromagnetic field is bosonic in nature and can be described in terms of a collection of plane wave modes. Each mode is a harmonic oscillator, with the occupation number of $n_j = 0, 1, 2, \dots$. This is the celebrated Planck-Einstein descriptions of black body radiation and photons. The many-bosons plane wave function amounts to a permanental configuration, $|\chi_1^{n_1} \chi_2^{n_2} \cdots\rangle$.

(iii) The vacuum electromagnetic field is $|\mathbf{0}\rangle = |00 \cdots\rangle$. Its fluctuation is rooted at the *intrinsic degeneracy* of photons that are spin-1 particles. Each photon has the spin angular momentum $^3\mathbf{S}$ of Eq. (5.1.12), with the uncertainties, $\Delta S_x = \Delta S_y = \hbar\sqrt{1 - m^2/2}$, where $m = 0, \pm 1$; see Eq. (5.1.13).

B. Fermionic Fock-space formalism

- For fermions, any spin orbital χ_j is either unoccupied or occupied. The occupation numbers can only be $n_j = 0$ or 1. For N electrons distributed over the chosen $\mathcal{N}$ spin orbitals, there are a total $\binom{\mathcal{N}}{N}$ number of distinct determinants. Their collection constitutes the full configurations space, with the chosen $\mathcal{N}$ being fixed. The full Fock space size is

$$\sum_{N=0}^{\mathcal{N}} \binom{\mathcal{N}}{N} = 2^{\mathcal{N}} \quad (8.2.1)$$

As just mentioned earlier, subspaces of different particle numbers are mutually orthogonal from each others.

- Basis set wave functions in N -electron Fock subspace are the Slater determinants, $\{|\chi_{j_1} \cdots \chi_{j_N}\rangle\}$, Eq. (8.1.7). To have the *one-to-one* correspondence to physical configurations, these occupied spin orbitals in every individual Slater determinant are arranged in an overall *ordered* sequence.
- The N -particle subspace orthonormality is given by Eq. (8.1.10a),

$$\langle \chi_{j_1} \cdots \chi_{j_N} | \chi_{j'_1} \cdots \chi_{j'_N} \rangle = \det |\delta_{j_1 j'_1} \cdots \delta_{j_N j'_N}| \quad (8.2.2)$$

In particular,

$$\langle \chi_{j_1} \chi_{j_2} | \chi_{j'_1} \chi_{j'_2} \rangle = \delta_{j_1 j'_1} \delta_{j_2 j'_2} - \delta_{j_2 j'_1} \delta_{j_1 j'_2} \quad (8.2.3)$$

with the last term specifying the *exchange*, in relation to the antisymmetry requirement. The completeness is given by Eq. (8.1.10b),

$$\frac{1}{N!} \sum_{j_1 \cdots j_N} |\chi_{j_1} \cdots \chi_{j_N}\rangle \langle \chi_{j_1} \cdots \chi_{j_N}| = \hat{I}_N \quad (8.2.4)$$

with $\hat{I}_N$ being the N -particle subspace unit operator.

- The Fock space by nature includes also the *empty* ($N = 0$) or *vacuum* state, $|0\rangle$, satisfying

$$\langle 0|0\rangle = 1 \quad \text{and} \quad |0\rangle\langle 0| = \hat{I}_0 \quad (\hat{I}_0 \equiv 1) \quad (8.2.5)$$

This contains only one state, with $\hat{I}_0 \equiv 1$, a mathematical property of *empty* subspace. On the other hand, a given reference, often chosen to be the ground electronic state, may be called the *vacuum hole state*, such as the *excitation-free* scenario.

C. Fermionic creation and annihilation operators

Associating with each spin orbital, $|\chi_j\rangle$, is an *annihilation operators* $\hat{a}_j$. Its Hermitian conjugate is the *creating operator* $\hat{a}_j^\dagger$. These operators are the key ingredients in the second quantization, along with the Fock states that specify the particle occupations among different orbitals.

We will specify the Fock-space matrix representation of annihilation and creation operators; see Eqs. (8.2.17)–(8.2.20) and remarks therein. The basic principles are as follows. First of all,

$$\boxed{\hat{a}_j^\dagger|0\rangle = |\chi_j\rangle} \quad \text{and} \quad \boxed{\hat{a}_j|\chi_j\rangle = |0\rangle} \quad (8.2.6)$$

with $|0\rangle \equiv |\mathbf{0}\rangle$ being the vacuum state [cf. Eq. (8.2.5)]. Moreover,

$$\hat{a}_j^\dagger|\chi_j\rangle = 0 = \hat{a}_j|0\rangle \quad (8.2.7)$$

and

$$\hat{a}_j|\chi_j\chi_k\rangle = |\chi_k\rangle, \quad \text{but} \quad \hat{a}_k|\chi_j\chi_k\rangle = -|\chi_j\rangle \quad (8.2.8a)$$

$$\hat{a}_k\hat{a}_j|\chi_j\chi_k\rangle = |0\rangle, \quad \text{but} \quad \hat{a}_j\hat{a}_k|\chi_j\chi_k\rangle = -|0\rangle \quad (8.2.8b)$$

since $|\chi_j\chi_k\rangle = -|\chi_k\chi_j\rangle$. The two-electron Slater determinant reads

$$|\chi_j\chi_k\rangle = \hat{a}_j^\dagger\hat{a}_k^\dagger|0\rangle \quad \text{and} \quad \langle\chi_j\chi_k| = \langle 0|\hat{a}_k\hat{a}_j \quad (8.2.9)$$

Remarks on Eqs. (8.2.6)–(8.2.9):

(a) Equation (8.2.6): As implied by the first identity there, we have $\langle 0|\hat{a}_j = \langle\chi_j|$ and further $\langle\chi_j|\chi_j\rangle = \langle 0|\hat{a}_j|\chi_j\rangle = 1 = \langle 0|0\rangle$; thus $\hat{a}_j|j\rangle = |0\rangle$, the second identity.

(b) Equation (8.2.7): The first identity arises from the Pauli exclusion principle. The last one, $\hat{a}_j|0\rangle = 0$, since $0 = \langle\chi_j|0\rangle = \langle 0|\hat{a}_j|0\rangle$.

(c) Equation (8.2.8): This exemplifies the rule of permutation in relation to removing (or adding) individual fermion from (or into) the existing sequence of orbitals. Furthermore, the rule of permutation here leads to the canonical anticommutator relations for fermion operators; see Eq. (8.2.10).

(d) Equation (8.2.9): The first expression goes by $\hat{a}_j^\dagger\hat{a}_k^\dagger|\mathbf{0}\rangle = \hat{a}_j^\dagger|0_j\chi_k\rangle = |\chi_j\chi_k\rangle$. Its Hermitian conjugate gives rise to the second expression. Similarly, $|\chi_j\chi_k\chi_l\rangle = \hat{a}_j^\dagger\hat{a}_k^\dagger\hat{a}_l^\dagger|0\rangle$ and $\langle\chi_j\chi_k\chi_l| = \langle 0|\hat{a}_l\hat{a}_k\hat{a}_j$, and so on. In relation to the corresponding Slater determinant, the order of creation operators sequence and that of annihilation operators are both self-explained. *The latter is via the Hermitian conjugate to the former.*

D. Canonical fermionic relations

$$\boxed{\{\hat{a}_k, \hat{a}_j^\dagger\} = \delta_{kj}} \quad \text{and} \quad \boxed{\{\hat{a}_k, \hat{a}_j\} = 0} \quad (8.2.10)$$

as inferred from Eqs. (8.2.6)–(8.2.9). Evidently,

$$\{\hat{a}_k^\dagger, \hat{a}_j^\dagger\} = \hat{a}_k^2 = 0 \quad (8.2.11)$$

The *number operators* remain to be

$$\hat{n}_k \equiv \hat{a}_k^\dagger \hat{a}_k, \quad \text{satisfying} \quad \hat{n}_k |0\rangle = 0 \quad \text{and} \quad \hat{n}_k |1_k\rangle = |1_k\rangle \quad (8.2.12)$$

with the eigenvalues of 0 and 1, the occupation numbers, respectively. Here, $|1_k\rangle \equiv |\chi_k\rangle$ by the notation defined in Eq. (8.2.17). Moreover, the *hole* number operator is given by $\hat{a}_k \hat{a}_k^\dagger = 1 - \hat{n}_k$.

Fermionic field operators: The annihilation and creation operators associated with coordinate $\mathbf{x} \equiv \mathbf{rs}$ are called field operators, denoted with $\hat{\psi}(\mathbf{x})$ and $\hat{\psi}^\dagger(\mathbf{x})$, formally related to the basis set transformation; i.e.,

$$|\chi_j\rangle = \int d\mathbf{x} |\mathbf{x}\rangle \langle \mathbf{x} | \chi_j \rangle = \int d\mathbf{x} \chi_j(\mathbf{x}) |\mathbf{x}\rangle \quad (8.2.13)$$

giving rise to

$$\boxed{\hat{a}_j^\dagger = \int d\mathbf{x} \chi_j(\mathbf{x}) \hat{\psi}^\dagger(\mathbf{x})} \quad (8.2.14)$$

Together with Eq. (8.2.10), we readily obtain

$$\boxed{\{\hat{\psi}(\mathbf{x}), \hat{\psi}^\dagger(\mathbf{x}')\} = \delta(\mathbf{x} - \mathbf{x}')} \quad \text{and} \quad \boxed{\{\hat{\psi}(\mathbf{x}), \hat{\psi}(\mathbf{x}')\} = 0} \quad (8.2.15)$$

It is worth reemphasizing that $\mathbf{x} \equiv \mathbf{rs}$ comprises both the real-space $\mathbf{r}$ and spin s degrees of freedom. The momentum or $\mathbf{k}$ -space counterparts, often used in solid state physics, are similar.

Exercises

E8.2.1 Show the following relations,

$$[\hat{a}_k^\dagger \hat{a}_{k'}, \hat{a}_j] = -\delta_{kj} \hat{a}_{k'} \quad \text{and} \quad [\hat{a}_k^\dagger \hat{a}_{k'}, \hat{a}_j^\dagger] = \delta_{k'j} \hat{a}_k^\dagger \quad (8.2.16)$$

and compare them to the bosonic counterparts.

E8.2.2 Consider an N -particle state and its $(N-1)$ -particle associates:

$$\begin{aligned} |\mathbf{j}\rangle &\equiv |j_1 \cdots j_N\rangle \equiv |\chi_{j_1} \cdots \chi_{j_N}\rangle \\ |\mathbf{j}_r^-\rangle &\equiv |j_1 \cdots j_{r-1} j_{r+1} \cdots j_N\rangle \end{aligned} \quad (8.2.17)$$

Show that

$$|\mathbf{j}\rangle = \hat{a}_{j_1}^\dagger \cdots \hat{a}_{j_N}^\dagger |\mathbf{0}\rangle = (-1)^{r-1} \hat{a}_{j_r}^\dagger |\mathbf{j}_r^-\rangle \quad (8.2.18a)$$

$$\hat{a}_k |\mathbf{j}\rangle = \sum_{r=1}^N (-1)^{r-1} \delta_{kj_r} |\mathbf{j}_r^-\rangle \quad \text{and} \quad \langle \mathbf{j}_r^- | \hat{a}_{j_r} |\mathbf{j}\rangle = (-1)^{r-1} \quad (8.2.18b)$$

E8.2.3 As implied above, each row or column of the matrix $\hat{a}_j$ has *at most one* nonzero element that is either 1 or -1 . The total nonzero elements is $\mathcal{M}/2$, where $\mathcal{M} = 2^N$ is the Fock space size, so that $\hat{a}_j$ is an $\mathcal{M} \times \mathcal{M}$ matrix. Generate a computer code for all annihilation and creation operators for N -electron systems, with the minimum spin-orbital basis set size of $\mathcal{N} = 2N$.

E8.2.4 Consider a Fock space spanning over $(|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle)$; i.e., the vacuum state, two single-occupation states and the double occupation state, respectively. Only single-occupation states are spin orbitals, associated with creation and annihilation operators.

(a) Verify that

$$\hat{a}_\uparrow \doteq \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \hat{a}_\downarrow \doteq \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (8.2.19)$$

The nonzero matrix elements are only

$$\langle 0 | \hat{a}_\uparrow | \uparrow \rangle = 1 \quad \text{and} \quad \langle \downarrow | \hat{a}_\uparrow | \uparrow \downarrow \rangle = 1 \quad (8.2.20a)$$

$$\langle 0 | \hat{a}_\downarrow | \downarrow \rangle = 1 \quad \text{and} \quad \langle \uparrow | \hat{a}_\downarrow | \uparrow \downarrow \rangle = -1 \quad (8.2.20b)$$

(b) Verify further that $\{\hat{a}_s, \hat{a}_{s'}^\dagger\} = \delta_{ss'}$ and $\{\hat{a}_s, \hat{a}_{s'}\} = 0$ [cf. Eq. (8.2.10)].

(c) Show that

$$\hat{n}_\uparrow \equiv \hat{a}_\uparrow^\dagger \hat{a}_\uparrow = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \hat{n}_\downarrow \equiv \hat{a}_\downarrow^\dagger \hat{a}_\downarrow = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (8.2.21)$$

(d) Complete the Hamiltonian matrix for the *single impurity Anderson model*,

$$\hat{H} = \epsilon_\uparrow \hat{n}_\uparrow + \epsilon_\downarrow \hat{n}_\downarrow + U \hat{n}_\uparrow \hat{n}_\downarrow \quad (8.2.22)$$

where $U > 0$ is the Coulomb interaction. The Fermi energy is set to be the zero. Each ϵ_s , the spin polarized on-site energy, can be either positive or negative. Plot the energy scheme as implied in $\hat{H}$ of Eq. (8.2.22).

(Remark: Anderson model is often used for the active parts of total composite systems in such as electronic devices and catalysis reactions).

8.2.2 Dynamic operators in second quantization

A. Electronic Hamiltonian

- To complete the second quantization formalism, we shall also establish the fermionic algebra on dynamical operators. Consider first the nonrelativistic N -electron Hamiltonian (in atomic unit),

$$\hat{H} \equiv \hat{H}_I + \hat{H}_{II} = \sum_{i=1}^N \hat{h}(i) + \frac{1}{2} \sum'_{i,j=1}^N \frac{1}{r_{ij}} \quad (8.2.23)$$

The one-electron component, $\hat{H}_I$, describes the noninteracting electrons moving individual in the Coulomb field of the nucleus. The two-electron component, $\hat{H}_{II} \equiv \hat{V}_{ee}$, is the Coulomb interaction between electrons, with the primed sum over all $i \neq j$. The factor $1/2$ takes care of double counting for electrons in pair. The nucleus-nucleus interaction is not included, without loss of generality.

- Second quantization of the Hamiltonian, Eq. (8.2.23), reads

$$\hat{H} = \sum_{jk} h_{jk} \hat{a}_j^\dagger \hat{a}_k + \frac{1}{2} \sum_{ijkl} V_{ijkl} \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k \quad (8.2.24)$$

where

$$h_{jk} = \int d\mathbf{x} \chi_j^*(\mathbf{x}) \hat{h} \chi_k(\mathbf{x}) \quad (8.2.25a)$$

$$V_{ijkl} = \iint d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1) \chi_j^*(\mathbf{x}_2) \frac{1}{r_{12}} \chi_k(\mathbf{x}_1) \chi_l(\mathbf{x}_2) \quad (8.2.25b)$$

Derivations: The one-electron term of Eq. (8.2.24) can readily be obtained via the harmonic oscillator-like algebra. In the following, we focus on

$$\hat{O}(1, 2) = \sum'_{i,j=1}^N \frac{1}{r_{ij}} \quad (8.2.26)$$

to be formulated via Eq. (8.2.4). We obtain

$$\hat{O}(1, 2) = \sum_{ijkl} \langle ij | \hat{O}(1, 2) | kl \rangle |ij\rangle \langle kl| = \sum_{ijkl} \langle ij | \hat{O}(1, 2) | kl \rangle \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k \quad (8.2.27)$$

since $|ij\rangle \langle kl| = \hat{a}_i^\dagger \hat{a}_j^\dagger |0\rangle \langle 0| \hat{a}_l \hat{a}_k = \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k$ [cf. Eq. (8.2.9)]. The involving $\langle ij | \hat{O}(1, 2) | kl \rangle$ is just V_{ijkl} of Eq. (8.2.25). The canonical order of operators goes by $\hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k$. Its index series, $(ijkl)$, differs from that in V_{ijkl} . The Fock space size, $\mathcal{N}$, is that of spin orbitals basis set, implied in the summations in Eq. (8.2.24). (Q.E.D.)

Exercises

E8.2.5 Recast Eq. (8.2.25) in terms of the spatial integrals, by using Eq. (8.1.8).

B. Spin- $\frac{1}{2}$ operator in second quantization

• In strong correlation systems, the Hamiltonian contains often electronic spin operators. Without loss of generality, we focus on an individual electron spin operator,

$$\hat{S} = \frac{1}{2}\hat{\sigma} = \frac{1}{2} \sum_{s,s'=\uparrow,\downarrow} |s\rangle\langle s|\hat{\sigma}|s'\rangle\langle s'| = \frac{1}{2} \sum_{s,s'=\uparrow,\downarrow} \hat{a}_s^\dagger \boldsymbol{\sigma}_{ss'} \hat{a}_{s'} \quad (8.2.28)$$

where $\boldsymbol{\sigma}_{ss'} \equiv \langle s|\hat{\sigma}|s'\rangle$, with $\hat{\sigma} = \hat{\sigma}_x\mathbf{i} + \hat{\sigma}_y\mathbf{j} + \hat{\sigma}_z\mathbf{k}$ being the vector of Pauli matrices; i.e., Eq. (5.2.11),

$$\hat{\sigma}_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \hat{\sigma}_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \hat{\sigma}_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (8.2.29)$$

in the one-electron spin $\{|s\rangle\}$ -subspace, with $s = \uparrow, \downarrow$.

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Remarks on Eq. (8.2.28):

(a) In contact with the Fock-space representation, consider the spin-spatial description; i.e., $\mathbf{x} \equiv \mathbf{r}s$, where $s = \uparrow, \downarrow$. As results,

$$\begin{aligned} \hat{\sigma} &= \sum_{s,s'} \iint d\mathbf{r}d\mathbf{r}' |\mathbf{r}s\rangle\langle\mathbf{r}s|\hat{\sigma}|\mathbf{r}'s'\rangle\langle\mathbf{r}'s'| \\ &= \sum_{s,s'} \iint d\mathbf{r}d\mathbf{r}' \delta(\mathbf{r} - \mathbf{r}') |\mathbf{r}s\rangle\langle\mathbf{r}s|\hat{\sigma}|\mathbf{r}'s'\rangle\langle\mathbf{r}'s'| = \sum_{s,s'} |s\rangle\langle s|\hat{\sigma}|s'\rangle\langle s'| \end{aligned} \quad (8.2.30)$$

This gives rise to the second identity of Eq. (8.2.28).

(b) The last identity of Eq. (8.2.28) is the second quantization equivalence. Exploited are $|s\rangle = \hat{a}_s^\dagger|0\rangle$ and $\langle s'| = \langle 0|\hat{a}_{s'}$, followed by using $|0\rangle\langle 0| = 1$.

C. Angular momentum in second quantization

• Let $\hat{A}$ be an arbitrary one-electron operator in an N -electron system:

$$\hat{A} = \sum_{i=1}^N \hat{A}(i) \quad (8.2.31)$$

Its second quantization goes by

$$\hat{A} = \sum_{jk} |\chi_j\rangle\langle\chi_j|\hat{A}(1)|\chi_k\rangle\langle\chi_k| = \sum_{jk} A_{jk} \hat{a}_j^\dagger \hat{a}_k \quad (8.2.32)$$

where $A_{jk} \equiv \langle\chi_j|\hat{A}(1)|\chi_k\rangle$ [cf. Eq. (8.2.25a)] and $\chi_j(\mathbf{x}) = \psi_j(\mathbf{r})\sigma_j(s)$, with the spin part of $\sigma_j(s) = \uparrow$ or $\downarrow$.

• Angular momentum, $\hat{\mathbf{J}} = \hat{J}_x\mathbf{i} + \hat{J}_y\mathbf{j} + \hat{J}_z\mathbf{k}$, is a one-electron operator and follows the above second quantization. Exemplified in Eq. (8.2.28) is for the spin $\hat{\mathbf{S}}$, with the spatial one-particle unit operator $\hat{I}_1^{\text{spatial}}$ being implied. The orbital angular momentum, $\hat{\mathbf{L}}$, is similar. The corresponding Eq. (8.2.32) goes by $\hat{L}_{jk} \equiv \langle\psi_j(\mathbf{r})|\hat{\mathbf{L}}|\psi_k(\mathbf{r})\rangle$, with the spin one-particle unit operator $\hat{I}_1^{\text{spin}}$ being implied.

Chapter 9

Symmetry Adapted Wave Functions

In the Born–Oppenheimer approximation scheme, the electronic properties do not have the nuclear isotopic effect. The geometry of a molecule can differ in different electronic states. For example, HCN is linear in its ground electronic state, but nonlinear in certain excited states. Unless otherwise specified, we refer molecular symmetry with ground electronic states at local equilibriums. Molecular Hamiltonian, either electronic and nuclear, is invariance with respect to the permutations of identical nuclei. On the other hand, this type of permutations can often be achieved by the symmetry operations that are commutable with the molecular Hamiltonian. In other words, *molecular symmetry* is related to certain *constants of motion*.

Electronic wave functions should be *symmetry-adapted*, with respect to their parametric dependence on nuclear positions. This requirement can be addressed within the symmetry-adapted linear combination (SALC) framework, starting from the LCAO–MOs level. The symmetry adaptivity will be preserved in the consequent construction of many-electron wave functions. The constituent AOs possess own local orientations and symmetries. The resulting SALC MOs are crucial in such as the celebrated Woodward–Hoffmann rules of pericyclic reactions, satisfying the symmetry-preserving principle. Chemical bonds can be either localized or delocalized between two or more atoms through their valent AOs.

In §9.1, we consider diatomic molecules, with focus on the general principles of symmetry-adapted LCAO–MOs, chemical bonds and spectroscopic term symbols. While the symmetry group theory is presented in §9.2, its applications will be illustrated in §9.3 and §9.4, for analysing the normal vibrations and the electronic structures, respectively.

9.1 Diatomic Molecules and Spectroscopic Terms

9.1.1 Symmetry adapted molecular orbitals

• Diatomic molecules possess the *axial* symmetry, satisfying $[\hat{H}, \hat{L}_z] = 0$. The axial angular momentum is quantized. Each diatomic MO has the component of $e^{im\phi}$, with $m = 0, \pm 1, \pm 2, \dots$. Energy depends on m^2 .

(a) Diatomic MOs are classified with the values of $|m| = 0, 1, 2, 3, \dots$, labelled by $\sigma, \pi, \delta, \varphi$ and so on. Moreover, $|m|$ is the *number of nodal planes* through molecular axis. While σ -MO is non-degenerate and retains the axial symmetry, non- σ -MO is double degenerate and non-axial symmetric.

(b) To form a symmetry-adapted MO in diatomic molecule AB, the constituent AOs must be properly selected and oriented, as follows.

1) σ -MO is the “head-to-head” type, with (s, p_z, d_{z^2}) from each atom,

$$|\sigma_{\text{MO}}\rangle = C_A^{\mathfrak{s}}|\varphi_A^{\mathfrak{s}}\rangle + C_B^{\mathfrak{s}}|\varphi_B^{\mathfrak{s}}\rangle, \text{ with } \mathfrak{s} = s, p_z, d_{z^2} \quad (9.1.1)$$

where

$$|\varphi_A^{\mathfrak{s}}\rangle = c_{A1}|\varphi_s\rangle_A + c_{A2}|\varphi_{p_z}\rangle_A + c_{A3}|\varphi_{d_{z^2}}\rangle_A \quad (9.1.2)$$

This is a *generalized* AO, located at atom A.

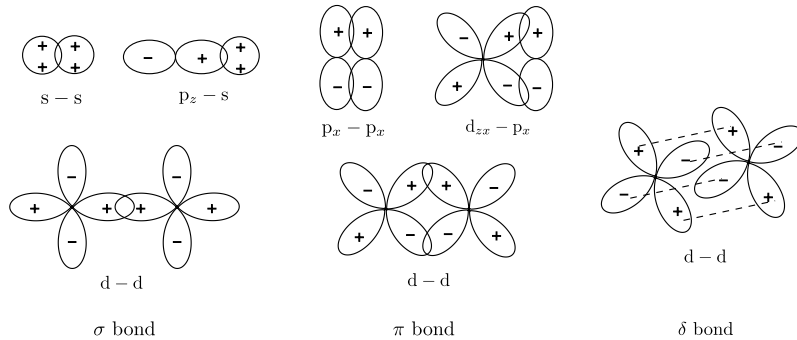
2) π -MO is the “shoulder-to-shoulder” type,

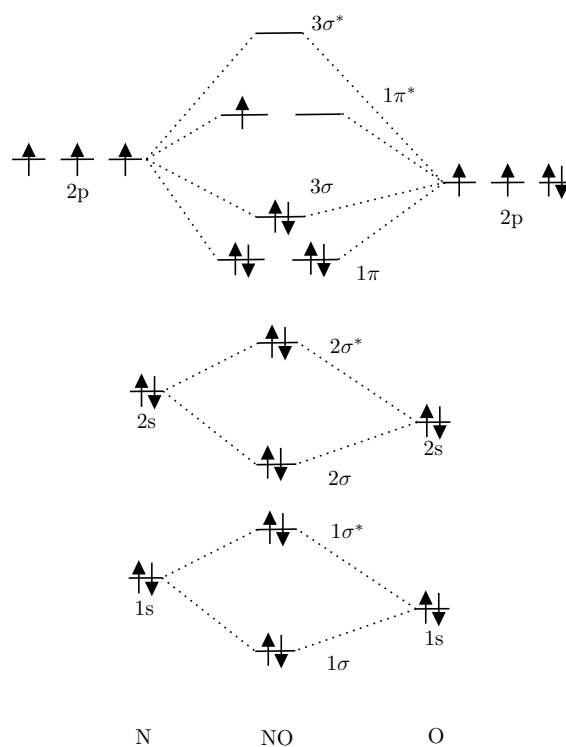
$$|\pi_{\text{MO}}\rangle = C_A^{\mathfrak{p}}|\varphi_A^{\mathfrak{p}}\rangle + C_B^{\mathfrak{p}}|\varphi_B^{\mathfrak{p}}\rangle, \text{ with } \mathfrak{p} = (p_x, p_y) \text{ and } (d_{xz}, d_{yz}) \quad (9.1.3)$$

3) δ -MO is the “face-to-face” type,

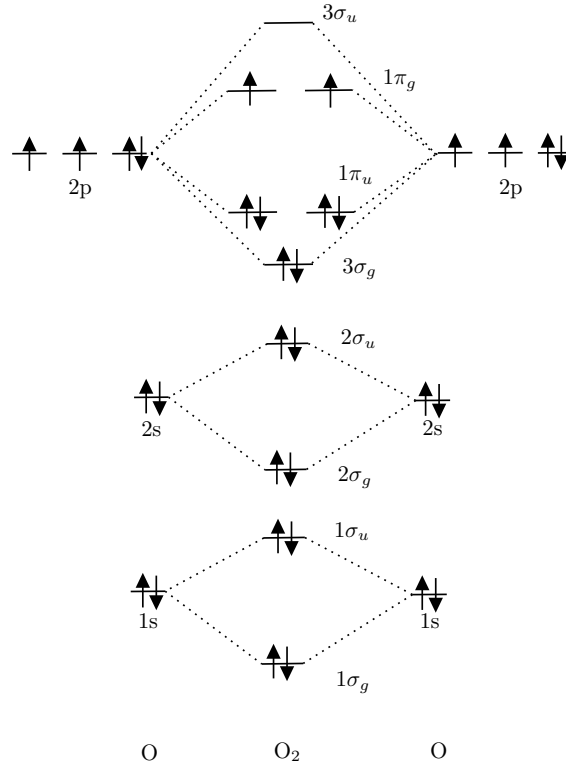
$$|\delta_{\text{MO}}\rangle = C_A^{\mathfrak{d}}|\varphi_A^{\mathfrak{d}}\rangle + C_B^{\mathfrak{d}}|\varphi_B^{\mathfrak{d}}\rangle, \text{ with } \mathfrak{d} = (d_{xy}, d_{x^2-y^2}) \quad (9.1.4)$$

forming in particular metal-metal bonds.





- LCAO–MOs are determined via a self-consistent field method that preserves symmetry (cf. §10.2). Empirical rules observed are as follows. (i) Chemical bonds are formed primarily from *valence* (or *frontier*) AOs, with *maximum overlap integral* and *similar energies*; (ii) In heterogeneous diatomic molecules, bonding MO carries more low-level AO components, whereas anti-bonding MO (labeled by $*$) does the opposite [cf. Eqs. (7.2.42)–(7.2.44) and the remarks therein].
- Homonuclear diatomic molecule possesses the inversion symmetry. The resulting MOs are of either the “ g ” (gerade) or “ u ” (ungerade) parity, such that $(\sigma, \sigma^*) \rightarrow (\sigma_g, \sigma_u)$ and $(\delta, \delta^*) \rightarrow (\delta_g, \delta_u)$, but $(\pi, \pi^*) \rightarrow (\pi_u, \pi_g)$, with respect to the orbital orientations in space.



9.1.2 Configurations and spectroscopic terms

For illustration, consider homonuclear diatomics, covering from H_2^+ to F_2 . Two ordered series of MO energies are found as follows:

(i) The short-bond scenario, covering from H_2^+ to N_2 , has

$$1\sigma_g < 1\sigma_u < 2\sigma_g < 2\sigma_u < 1\pi_u < \underbrace{3\sigma_g}_{2p_z} < 1\pi_g < \underbrace{3\sigma_u}_{2p_z} \quad (9.1.5)$$

(ii) The long-bond scenario, covering from O_2^+ to F_2 , has

$$\underbrace{1\sigma_g}_{1s} < \underbrace{1\sigma_u}_{1s} < \underbrace{2\sigma_g}_{2s} < \underbrace{2\sigma_u}_{2s} < \underbrace{3\sigma_g}_{2p_z} < \underbrace{1\pi_u}_{2p_x, 2p_y} < \underbrace{1\pi_g}_{2p_x, 2p_y} < \underbrace{3\sigma_u}_{2p_z} \quad (9.1.6)$$

The bottom line specifies the primary participating AOs. While σ_g and π_u are bonding, σ_u and π_g are anti-bonding orbitals, anti-bonding orbitals, respectively, as in line with the related geometries. The energy difference

between $3\sigma_g$ and $1\pi_u$ is rather small, subject to the observed swap in the above two series, Eq. (9.1.5) versus Eq. (9.1.6). The Aufbau Principle of filling electrons into orbitals remains valid, resulting in configurations.

- For the energies of those terms from a same configuration, consider the *generalization of Hund's rules*:

- (1) The *highest spin-multiplicity* term has the lowest energy.
- (2) For a given multiplicity, the *highest spatial degeneracy* term has the lowest energy.

- For O_2 (a 16-electron molecule) with Eq. (9.1.6), the ground-state configuration is $[Be_2(3\sigma_g)^2](1\pi_u)^4(1\pi_g)^2$, with the subshell configuration,

$$(1\pi_u)^4(1\pi_g)^2 \Rightarrow {}^3\Sigma_g^-, {}^1\Delta_g, {}^1\Sigma_g^+ \quad (9.1.7)$$

The resulting three terms goes by the following order of energies,

$$\mathcal{E}({}^3\Sigma_g^-) < \mathcal{E}({}^1\Delta_g) < \mathcal{E}({}^1\Sigma_g^+) \quad (9.1.8)$$

The ground-state of O_2 molecule is *paramagnetic*, with the of ${}^3\Sigma_g^-$.

- For any linear molecules, $[\hat{H}, \hat{L}_z] = 0$, the axial rotation is quantized and described by the wave function component of

$$C_z(\phi) = e^{iM\phi}; \quad M = 0, \pm 1, \pm 2, \dots \quad (9.1.9)$$

For many-electron molecules, $\hat{L}_z = \hat{l}_z(1) + \dots + \hat{l}_z(N)$. We report

$$\Lambda = |M| = |m(1) + \dots + m(N)| = 0, 1, 2, 3, \dots \quad (9.1.10)$$

with the symbols of $\Sigma, \Pi, \Delta, \Phi$ and so on. Any full-filled configurations contribute to ${}^1\Sigma^+$, where $M = 0$ and $S = 0$.

(a) The $(\pm)$ -parity, associated with $\Sigma^\pm$, arises from that linear molecules possess also the symmetry, with respect to the vertical axial-plane reflection, $\hat{\sigma}_v$, which satisfies

$$[\hat{H}, \hat{\sigma}_v] = 0, \text{ but } [\hat{L}_z, \hat{\sigma}_v] \neq 0 \quad (9.1.11)$$

The common wave functions of both $\hat{L}_z$ and $\hat{\sigma}_v$ exist only when $M = 0$, described by the $\Sigma^\pm$ -term, to be further specified with Eq. (9.1.16).

(b) Revisit now Eq. (9.1.7), for diatomic molecules without central inversion symmetry. It is the configuration $(\pi_{MO})^2$ responsible for the term symbols. The corresponding two-electron Slater determinants are

$$\pi_{MO}(\mathbf{r}_1)\pi_{MO}(\mathbf{r}_2) \Rightarrow \{|\psi_m\bar{\psi}_m\rangle, |\psi_m\bar{\psi}_{-m}\rangle, |\psi_m\psi_{-m}\rangle\}_{|m|=1} \quad (9.1.12)$$

As inferred from Eqs. (8.1.43)–(8.1.48), we have

$${}^1\Delta(\chi_1, \chi_2) = |\psi_1\bar{\psi}_1\rangle = \left\{ \begin{array}{c} \pi_1(\mathbf{r}_1)\pi_1(\mathbf{r}_2) \\ \pi_{-1}(\mathbf{r}_1)\pi_{-1}(\mathbf{r}_2) \end{array} \right\} \cdot S(1, 2) \quad (9.1.13)$$

with $\chi_j \equiv \mathbf{r}_j s_j$ and $S(1, 2) \equiv S(s_1, s_2)$, whereas

$$\begin{aligned} {}^1\Sigma^+(\chi_1, \chi_2) &= |\psi_1 \bar{\psi}_{-1}\rangle = \Psi_+(\mathbf{r}_1, \mathbf{r}_2) \cdot S(1, 2) \\ {}^3\Sigma^-(\chi_1, \chi_2) &= |\psi_1 \psi_{-1}\rangle = \Psi_-(\mathbf{r}_1, \mathbf{r}_2) \cdot T(1, 2) \end{aligned} \quad (9.1.14)$$

The spin singlet $S(1, 2)$ [Eq. (8.1.45a)] and triplet $T(1, 2)$ [Eq. (8.1.46)] are antisymmetric and symmetric, respectively, with respect to the particle-permutation. While the spacial components of Eq. (9.1.13) are symmetric, those of Eq. (9.1.14) are given explicitly by (unnormalized)

$$\Psi_{\pm}(\mathbf{r}_1, \mathbf{r}_2) \equiv \pi_1(\mathbf{r}_1)\pi_{-1}(\mathbf{r}_2) \pm \pi_1(\mathbf{r}_2)\pi_{-1}(\mathbf{r}_1) = \pm\Psi_{\pm}(\mathbf{r}_2, \mathbf{r}_1) \quad (9.1.15)$$

The resulting $\Sigma^{\pm}$, Eq. (9.1.14), obeys the Pauli principle.

• It is worth emphasizing that the *term symbols primarily report the molecular symmetry*. The related group theory and applications will be discussed in the coming sections of this chapter. In particular, linear molecules possess both the axial rotation and the axial-plane reflection symmetries; see Eq. (9.1.11) and the remarks therein. The $\pm$ -parity in $\Sigma^{\pm}$ rather reports the axial-plane reflection symmetry aspect, for the spacial components of Eq. (9.1.14) the required parity,

$$\hat{\sigma}_v(xz)\Psi_{\pm}(\mathbf{r}_1, \mathbf{r}_2) = \pm\Psi_{\pm}(\mathbf{r}_1, \mathbf{r}_2) \quad (9.1.16)$$

Derivations: As known, the π -orbital has the form of

$$\pi_{\pm 1}(\mathbf{r}_i) = A(r_i, \theta_i)e^{\pm i\phi_i} \quad (9.1.17)$$

On the other hand, the axial-plane reflection goes with

$$(r_i, \theta_i, \phi_i) \xrightarrow{\hat{\sigma}_v(xz)} (r_i, \theta_i, -\phi_i) \quad (9.1.18)$$

As results, the wave functions in Eq. (9.1.15) acquire the form of

$$\begin{aligned} \Psi_+(\mathbf{r}_1, \mathbf{r}_2) &= A(r_1, \theta_1)A(r_2, \theta_2)\cos(\phi_2 - \phi_1) \xrightarrow{\hat{\sigma}_v(xz)} \Psi_+(\mathbf{r}_1, \mathbf{r}_2) \\ \Psi_-(\mathbf{r}_1, \mathbf{r}_2) &= A(r_1, \theta_1)A(r_2, \theta_2)\sin(\phi_2 - \phi_1) \xrightarrow{\hat{\sigma}_v(xz)} -\Psi_-(\mathbf{r}_1, \mathbf{r}_2) \end{aligned} \quad (9.1.19)$$

This is Eq. (9.1.16). (Q.E.D.)

Exercises:

E9.1.1 For Be_2 , B_2 , C_2 , N_2^+ , N_2 , O_2^+ , O_2 and F_2 , write down their ground-state configurations and the resulting spectroscopic terms [Hint: Eqs. (9.1.5) and (9.1.6)].

E9.1.2 As known, bond order equals to the half of difference between the number of bonding electrons and that of anti-bonding electrons. The B_2 molecule has the **bond-order** of 1, with the ground configuration $[\text{Be}_2](\pi_u 2p)^2$, while Be_2 is of the bond-order of 0. Write down the spectroscopic terms for B_2 and specify the ground-state one. Compare the results of B_2 with those of O_2 .

9.2 Group Representation Theories

9.2.1 Mathematical group descriptions

In mathematics, group ($\mathbb{G}$) describes a collection of elements, together with the rule of group multiplication, which satisfy the following five criterions: (i) $AB \in \mathbb{G}$, if $A \in \mathbb{G}$ and $B \in \mathbb{G}$; (ii) Group multiplication is associative; i.e., $A(BC) = (AB)C$; (iii) Inverse $A^{-1} \in \mathbb{G}$; (iv) **Identity element**, denoted as E that satisfies $E = A^{-1}A = AA^{-1}$ for all $A \in \mathbb{G}$, is *universal* in all groups; (v) *Matrix-representability* of the group elements.

Remarks: In principle, the rule of group multiplication can be arbitrary. For example, the collection of integer numbers, with the rule of group multiplication being the summation (+), constitute the group $\mathbb{G} = (\cdots, -2, -1, 0, 1, 2, \cdots)$: The identity element is number zero 0 and inverse of n is $-n$. Consider also numbers as (1×1) -matrixes. The above five criterions are all satisfied. Evidently, the collection all even numbers, $\mathbb{G}' = (\cdots, -4, -2, 0, 2, 4, \cdots) \subset \mathbb{G}$, is a *subgroup* of the integer group, whereas the collection of odd numbers is not.

- Any practical use of the group theory concerns a collection of symmetry operations that correspond to matrixes in real-space and the group multiplication is just that of matrixes.
- A group can be specified via its *generating elements*. An *Abelian* group refers to the case if its members are *all commutable* each other. Evidently, for a group being Abelian or non-Abelian, it is sufficient to examine whether its generating elements are all commutable or not. The total number of the group elements refers also the group order (h).
- Group elements (matrixes) are partitioned into **classes**, each of them comprising those matrixes that are related each other via similarity transformations *within the given group*. The elements of a same class have a common *matrix trace* value, called the **character** of the class. The group characters describe **irreducible representations**. This is closely related to the symmetry-adapted linear combination (SALC) construction of LACO-MOs. The Slater determinantal wave functions go along the direct-product description.
- Collection of all symmetry operations of a molecule forms a **point group**, since there is a common point at which they intersect. Rigid molecular translation and rotation motions are not included in the point group.

[‡]Crystals engage also three-dimension translation symmetries and only 32 molecular point-groups can form crystals. These are topics of solid state physics, beyond the scope of this course. Nevertheless, the general principles discussed in the coming §9.2.2 and §9.2.3 remain valid.

9.2.2 Theory of irreducible representations

A. Onset of group representations

• *Group representation* is to have all its elements in matrixes. Denote hereafter $\mathbb{G} = \{\hat{\Gamma}_i; i = 1, \dots, h\}$, with all $\hat{\Gamma}_i$ being matrixes, including the identity $\hat{E}$ that is the unit matrix. The total number of elements (i.e., matrixes) is the group order (h).

(a) Molecular Hamiltonian, satisfying $[\hat{H}, \hat{\Gamma}_i] = 0$ with any $\hat{\Gamma}_i \in \mathbb{G}$, possesses the same symmetry, and $\hat{H}$ can be either electronic or nuclear in nature. For geometry symmetry operations on rigid molecules, the (x, y, z) -basis would be enough. On the other hands, electronic structure theory goes by the basis of single-electron wave functions (i.e., the LCAO-MOs).

(b) Various representations differ at their basis sets, but are related each other via *similarity transformations* that support the use of nonorthogonal basis, such as LCAO-MOs. Similarity transformations are *trace invariant* properties.

B. Irreducible representations and projection operators

• Irreducible representation of a group, $\mathbb{G} = (\hat{\Gamma}_1, \dots, \hat{\Gamma}_h)$, is the simultaneous *block-diagonalizations* description. Different block subspaces are orthogonal. Denote hereafter Λ_k be the k^{th} -irreducible representation subspace, with the dimensionality l_k , in which

$$\Lambda_k(\hat{\Gamma}_i) \equiv \{\Lambda_{mn}^k(\hat{\Gamma}_i)\}_{l_k \times l_k}; \quad i = 1, \dots, h \quad (9.2.1a)$$

where

$$\Lambda_{mn}^k(\hat{\Gamma}_i) \equiv \langle \varphi_m^k | \hat{\Gamma}_i | \varphi_n^k \rangle; \quad m, n = 1, \dots, l_k \quad (9.2.1b)$$

As usual,

$$\langle \varphi_m^k | \varphi_n^k \rangle = \delta_{mn} \quad \text{and} \quad \sum_{m=1}^{l_k} |\varphi_m^k\rangle \langle \varphi_m^k| = \mathbf{1}_{l_k \times l_k} \quad (9.2.2)$$

• Irreducible subspace *projection operator matrix* is given by

$$\hat{\mathbf{P}}_k = \{\hat{P}_{mn}^k\}_{l_k \times l_k}, \quad \text{with} \quad \hat{P}_{mn}^k \equiv |\varphi_m^k\rangle \langle \varphi_n^k| \quad (9.2.3a)$$

It satisfies [cf. Eq. (9.2.1)]

$$\hat{\mathbf{P}}_k^2 = \hat{\mathbf{P}}_k \quad \text{and} \quad \hat{\mathbf{P}}_k \Lambda_k(\hat{\Gamma}_i) \hat{\mathbf{P}}_k = \Lambda_k(\hat{\Gamma}_i) \quad (9.2.3b)$$

We obtain

$$\hat{P}_{mn}^k = |\varphi_m^k\rangle \langle \varphi_n^k| = \frac{l_k}{h} \sum_{i=1}^h \langle \varphi_m^k | \hat{\Gamma}_i | \varphi_n^k \rangle^* \hat{\Gamma}_i \quad (9.2.4)$$

This provides a universal means for the SALC constructions.

C. The great orthogonality theorem (GOT)

- Combining that different irreducible subspaces are orthogonal, we can recast Eq. (9.2.4) in the form of

$$\langle \varphi_m^k | \hat{P}_{m'n'}^{k'} | \varphi_n^k \rangle = \frac{l_k}{h} \sum_{i=1}^h \langle \varphi_m^k | \hat{\Gamma}_i | \varphi_n^k \rangle \langle \varphi_{m'}^{k'} | \hat{\Gamma}_i | \varphi_{n'}^{k'} \rangle^* = \delta_{kk'} \delta_{mm'} \delta_{nn'} \quad (9.2.5)$$

This is the great orthogonality theorem (GOT) for the irreducible representations of a group.

- In terms of $\Lambda_{mn}^k(\hat{\Gamma}_i) \equiv \langle \varphi_m^k | \hat{\Gamma}_i | \varphi_n^k \rangle$, Eq. (9.2.1b), the GOT reads

$$\sum_{i=1}^h \Lambda_{mn}^k(\hat{\Gamma}_i) [\Lambda_{m'n'}^{k'}(\hat{\Gamma}_i)]^* = \frac{h}{l_k} \delta_{kk'} \delta_{mm'} \delta_{nn'} \quad (9.2.6)$$

Some observations and immediate consequences are as follows.

- (a) Let $\{\Lambda_{mn}^k(\hat{\Gamma}_1), \dots, \Lambda_{mn}^k(\hat{\Gamma}_h)\}$ be a h -dimension vector, whose length square is h/l_k ; cf. Eq. (9.2.6). The total l_k^2 -number of such vectors, with different pairs of m and n indexes, are mutually orthogonal. As evident later [see Eq. (9.2.14) and Eq. (9.2.17) for example], the GOT implies that

$$\sum_{k=1}^K l_k^2 = h \quad (9.2.7)$$

- (b) Irreducible representation is *completely* described by the **characters**,

$$\mathfrak{X}_k(\hat{\Gamma}_i) \equiv \text{tr } \mathbf{\Lambda}_k(\hat{\Gamma}_i) = \sum_{m=1}^{l_k} \Lambda_{mm}^k(\hat{\Gamma}_i), \quad \text{where } \begin{cases} k = 1, \dots, K \\ i = 1, \dots, h \end{cases} \quad (9.2.8)$$

The identity element goes by $\mathbf{\Lambda}_k(\hat{E}) = \mathbf{1}_{l_k \times l_k}$, so that

$$\mathfrak{X}_k(\hat{E}) = l_k \quad (9.2.9)$$

For the characters, Eq. (9.2.8), the GOT, Eq. (9.2.6), reduces to

$$\sum_{i=1}^h \mathfrak{X}_k(\hat{\Gamma}_i) \mathfrak{X}_{k'}^*(\hat{\Gamma}_i) = h \delta_{kk'} \quad (9.2.10)$$

with noting that $\sum_{m,n} \delta_{mn} = l_k$, where $m, n = 1, \dots, l_k$.

- (c) Irreducible description is free of redundancy, having the property:

Number (K) of irreducible subspaces equals the number of classes

Verifications: (i) Consider first the Abelian group scenario: Each group element itself is in a class and each irreducible subspace has the dimensionality of 1.

Both the numbers of classes and irreducible subspaces equal to h ; see Eq. (9.2.7); (ii) Turn to the non-Abelian group case, with a given or representing class of g elements. The number of irreducible subspaces and that of classes both decrease by $g - 1$, with respect to the Abelian or reference scenario. Repeat the above inductions, giving rise to the captioned observation.

- The total symmetry (TS) representation has

$$\mathfrak{X}_{\text{TS}}(\hat{\Gamma}_i) = 1; \quad \text{for all } \hat{\Gamma}_i \in \mathbb{G} \quad (9.2.11)$$

which by nature is irreducible, with the dimensionality of 1, and exists in any groups, as like the identity $\hat{E}$ does.

- In terms of classes, Eq. (9.2.10) becomes

$$\sum_{\alpha=1}^K g_{\alpha} \mathfrak{X}_k(\hat{\Gamma}_{\alpha}) \mathfrak{X}_{k'}^*(\hat{\Gamma}_{\alpha}) = h \delta_{kk'} \quad (9.2.12)$$

with g_{α} specifying the number of group elements in the α^{th} -class. Characters in any group ($\mathbb{G}$) form a $(K \times K)$ -matrix, the so-called **character matrix**, denoted as

$$\mathbf{X}(\mathbb{G}) = \{\mathfrak{X}_k(\hat{\Gamma}_{\alpha}); \quad k, \alpha = 1, \dots, K\} \quad (9.2.13)$$

In convention, the first row ($k = 1$) is the TS representation, Eq. (9.2.11), and the first column ($\alpha = 1$) reports those $\{\mathfrak{X}_k(\hat{E}) = l_k\}$, Eq. (9.2.9), the dimensionality of individual irreducible subspace.

Exercises: Some basic features of group character matrixes

E9.2.1 Consider an n -fold rotation group, $\mathcal{C}_n = (\hat{E}, \hat{C}_n, \dots, \hat{C}_n^{n-1})$, to which cyclic tight-binding molecules [cf. Eq. (7.1.54)] belong. The generating element of the $\mathcal{C}_n$ group is only the n -fold rotation, $\hat{C}_n(z)$, around the z -axis by angle $2\pi/n$. The character table, exemplified with $\mathcal{C}_3$, reads

$\mathcal{C}_3$	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$
Λ_1	1	1	1
Λ_2	1	ϵ	ϵ^2
Λ_3	1	ϵ^*	$(\epsilon^*)^2$

$$(\epsilon = e^{i2\pi/n}, \text{ with } n = 3) \quad (9.2.14)$$

- Verify its satisfying all required properties, Eqs. (9.2.7)–(9.2.11).
- In the $\mathcal{C}_n$ group, complex conjugate characters appear in pair, such that

$$\left\{ \begin{matrix} 1 & \epsilon & \epsilon^2 \\ 1 & \epsilon^* & (\epsilon^*)^2 \end{matrix} \right\} = \left\{ \begin{matrix} 1 & \epsilon & \epsilon^* \\ 1 & \epsilon^* & \epsilon \end{matrix} \right\} \text{ in } \mathcal{C}_3, \text{ where } \epsilon = e^{i2\pi/3}$$

$$\left\{ \begin{matrix} 1 & i & -1 & -i \\ 1 & -i & -1 & i \end{matrix} \right\} \text{ in } \mathcal{C}_4, \text{ since } e^{i2\pi/4} = i \quad (9.2.15)$$

Complete the character table of $\mathcal{C}_4$.

(c) In contact with the wave functions of Eq. (7.1.54), discuss the physical picture for the C_n group irreducible representations. Complete the character tables for C_5 and C_6 , with complex conjugate characters pairs being specified in curly brackets.

E9.2.2 While C_n has a single generating element, $\hat{C}_n(z)$, adding to it with another generating element, the horizontal plane reflection, $\hat{\sigma}_h(xy)$, results in the C_{nh} group. Verify the following observations: The character matrix of C_{nh} is related to that of C_n [cf. Eq. (9.2.13)] and $C_s = (\hat{E}, \hat{\sigma}_h)$ the plane reflection group via

$$\mathbf{X}(C_{nh}) = \mathbf{X}(C_n) \otimes \mathbf{X}(C_s) = \mathbf{X}(C_n) \otimes \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad (9.2.16)$$

That is

$$\mathbf{X}(C_{nh}) = \begin{bmatrix} \mathbf{X}(C_n) & \mathbf{X}(C_n) \\ \mathbf{X}(C_n) & -\mathbf{X}(C_n) \end{bmatrix} \quad (9.2.16')$$

E9.2.3 The $C_{3v} = (\hat{E}, 2\hat{C}_3, 3\hat{\sigma}_v)$ group of order 6, for such as a pyramidal AB_3 molecule, has two generating elements: (i) $\hat{C}_3(z)$ the three-fold rotation by angle $2\pi/3$ around the z -axis; (ii) $\hat{\sigma}_v(xz)$ the axial vertical plane reflection. The $\hat{C}_3$ rotation will bring one vertical plane to another, resulting in three equivalent σ_v -planes.

- (a) Show that $(\hat{C}_3, \hat{C}_3^{-1})$ are in one class and $(\hat{\sigma}_v(1), \hat{\sigma}_v(2), \hat{\sigma}_v(3))$ in another class.
 (b) Write down the steps of using Eqs. (9.2.7)–(9.2.11) to obtain

C_{3v}	$\hat{E}$	$2\hat{C}_3$	$3\hat{\sigma}_v$
Λ_1	1	1	1
Λ_2	1	1	-1
Λ_3	2	-1	0

(9.2.17)

(c) Consider $D_3 = (\hat{E}, 2\hat{C}_3, 3\hat{C}_2)$, with three C_2 -axes perpendicular to the principle $C_3(z)$ -axis. Verify that D_3 is isomorphous to C_{3v} , having the same character matrix in Eq. (9.2.13), as in general,

$$\mathbf{X}(D_n) = \mathbf{X}(C_{nv}) \quad (\text{Isomorphism}) \quad (9.2.18)$$

Moreover, the D_{nh} group, arising from adding to D_n with the horizontal plane reflection, goes by [cf. Eq. (9.2.16)]

$$\mathbf{X}(D_{nh}) = \mathbf{X}(D_n) \otimes \mathbf{X}(C_s) = \begin{bmatrix} \mathbf{X}(D_n) & \mathbf{X}(D_n) \\ \mathbf{X}(D_n) & -\mathbf{X}(D_n) \end{bmatrix} \quad (9.2.19)$$

E9.2.4 As known, the collection of all symmetry operations of a molecule forms a point group. Character tables are well documented; [See for example, F. A. Cotton (1990), “*Chemical Applications of Group Theory*”, 3rd ed., Appendix IIA]. Search over the full list of character tables for (i) Those individual isomorphic sets of point groups, with each set having a same character matrix; (ii) Those group-subgroup pairs, related via the direct product via $\mathbf{X}(C_s)$ as Eq. (9.2.16).

9.2.3 Direct product and decompositions

• **Direct product representation:** Let $\{|\psi_m^a\rangle\}$ and $\{|\psi_n^b\rangle\}$ be two arbitrary basis functions used in the group representation. In other words,

$$\hat{\Gamma}|\psi_m^a\rangle = \sum_{m'} c_{mm'}^a |\psi_{m'}^a\rangle \quad \text{and} \quad \hat{\Gamma}|\psi_n^b\rangle = \sum_{n'} c_{nn'}^b |\psi_{n'}^b\rangle \quad (9.2.20)$$

Consider then

$$\hat{\Gamma}|\psi_m^a \psi_n^b\rangle = \sum_{m'n'} c_{mn,m'n'}^{ab} |\psi_{m'}^a \psi_{n'}^b\rangle \quad (9.2.21)$$

where

$$c_{mn,m'n'}^{ab} = \langle \psi_m^a \psi_n^b | \hat{\Gamma} | \psi_{m'}^a \psi_{n'}^b \rangle = c_{mm'}^a c_{nn'}^b \quad (9.2.22)$$

The direct products $\{|\psi_m^a \psi_n^b\rangle\}$ also form a basis set functions for the group representation. The corresponding character is given by

$$\chi_{ab}(\hat{\Gamma}) \equiv \sum_{mn} \langle \psi_m^a \psi_n^b | \hat{\Gamma} | \psi_m^a \psi_n^b \rangle = \sum_{mn} c_{mn,mn}^{ab} = \sum_{mn} c_{mm}^a c_{nn}^b$$

The last quantity is the product of two characters associated with those subspaces defined in Eq. (9.2.20). We obtain

$$\chi_{ab}(\hat{\Gamma}) = \chi_a(\hat{\Gamma}) \chi_b(\hat{\Gamma}) \quad (9.2.23)$$

• **Reducible representation decompositions:** Any reducible representation $\mathfrak{R}$ -space can be decomposed into irreducible representation $\{\Lambda_k\}$ -subspaces. This is about the existence of the aforementioned simultaneous block diagonalization. The group irreducible representation removes also the redundancy, so that each irreducible subspace is unique.

Let $n_k(\mathfrak{R})$ be the number of times the k^{th} -irreducible subspaces, contained within the reducible $\mathfrak{R}$ -space via the $\{|\psi_m\rangle\}$ -basis functions. In terms of the characters (matrix traces), we have

$$\mathfrak{R}(\hat{\Gamma}) \equiv \sum_m \langle \psi_m | \hat{\Gamma} | \psi_m \rangle = \sum_{k=1}^K n_k(\mathfrak{R}) \chi_k(\hat{\Gamma}); \quad \forall \Gamma \in \mathbb{G} \quad (9.2.24)$$

By using Eq. (9.2.10), we obtain

$$n_k(\mathfrak{R}) = \frac{1}{h} \sum_{i=1}^h \mathfrak{R}(\hat{\Gamma}_i) \chi_k^*(\hat{\Gamma}_i) \geq 0 \quad (\text{Integer}) \quad (9.2.25)$$

This provides a method for decomposing an arbitrary representation.

- In particular, the TS component, Eq. (9.2.11), contained a direct product representation, $\mathfrak{R} = \mathfrak{X}_{ab}$, Eq. (9.2.23), is given by

$$n_{\text{TS}}(\mathfrak{X}_{ab}) = \frac{1}{h} \sum_{i=1}^h \mathfrak{X}_a(\hat{\Gamma}_i) \mathfrak{X}_b(\hat{\Gamma}_i) \quad (9.2.26)$$

It is *nonzero* only if $\mathfrak{X}_a(\hat{\Gamma})$ and $\mathfrak{X}_b(\hat{\Gamma})$ have at least one *irreducible representation in common*. The proof is rather straightforward by combining Eq. (9.2.24) and the reduced GOT, Eq. (9.2.10), in terms of characters.

9.2.4 Molecular symmetry group descriptions

A. Symmetry elements versus symmetry operations

- Symmetry operations covering all molecules are

$\hat{C}_n$: An n -fold rotation by angle $2\pi/n$ (counterclockwise)

$\hat{\sigma}$: Plane reflection (σ_h or σ_v)

$\hat{i}$: Inversion (responsible for the “ g ” or “ u ” parity)

$\hat{S}_n$: An n -fold improper rotation; i.e., $\hat{S}_n \equiv \hat{\sigma}_h \hat{C}_n = \hat{C}_n \hat{\sigma}_h$

The collection of all symmetry operations of a molecule forms a point group, since there is a common point at which they intersect. There is a systematic procedure for determining what a symmetry group a molecule belongs to.¹

— By convention, we set the *principle* (highest) n -fold rotation-axis be the z -direction, σ_v be a vertical z -axial plane, and σ_h the horizontal xy -plane.

$$\hat{C}_n(z) \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos(2\pi/n) & -\sin(2\pi/n) & 0 \\ \sin(2\pi/n) & \cos(2\pi/n) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (9.2.27)$$

$$\begin{aligned} (x, y, z) &\xrightarrow{\hat{\sigma}_v(xz)} (x, \bar{y}, z), & (x, y, z) &\xrightarrow{\hat{\sigma}_h} (x, y, \bar{z}) \\ (x, y, z) &\xrightarrow{\hat{\sigma}_v(yz)} (\bar{x}, y, z), & (x, y, z) &\xrightarrow{\hat{i}} (\bar{x}, \bar{y}, \bar{z}) \end{aligned} \quad (9.2.28)$$

with $\bar{x} \equiv -x$ and so on. The matrix $\hat{S}_n \equiv \hat{C}_n \hat{\sigma}_h$ is then followed. Moreover,

$$\begin{aligned} \hat{\sigma}_h \hat{\sigma}_v(xz) &= \hat{C}_2(x), & \hat{\sigma}_v(xz) \hat{\sigma}_v(yz) &= \hat{C}_2(z) \\ \hat{\sigma}_h \hat{\sigma}_v(yz) &= \hat{C}_2(y), & \hat{\sigma}_h \hat{C}_2(z) &= \hat{i} \end{aligned} \quad (9.2.29)$$

Presented below are those commutable pairs of operations:

- (1) $[\hat{C}_2(x), \hat{C}_2(y)] = 0$, for two C_2 -rotations about perpendicular axes.

¹**Self-study** and doing **exercises** of the following references:

(a) Ira N. Levine (2014), *Quantum Chemistry*, 7th ed., Chapter 12.

(b) F. A. Cotton (1990), *Chemical Applications of Group Theory*, 3rd ed., Chapter 3.

- (2) $[\hat{C}_2(z), \hat{\sigma}_v] = 0$, but $[\hat{C}_{n>2}(z), \hat{\sigma}_v] \neq 0$.
- (3) $[\hat{C}_m(z), \hat{C}_n(z)] = 0$, for any two rotations about a same axis.
- (4) $[\hat{C}_n(z), \hat{\sigma}_h] = 0$, for the z -axis rotation and (xy) -plane reflection.
- (5) $[\hat{i}, \hat{\sigma}] = [\hat{i}, \hat{C}_n] = 0$, for the inversion and any reflection or rotation.
- (6) $[\hat{\sigma}_h, \hat{\sigma}_v] = [\hat{\sigma}_v(xz), \hat{\sigma}_v(yz)] = 0$, for the pairs of reflections through planes perpendicular to each other.

- As noticed, symmetry elements are *geometrical entities* (point or line or plane), with respect to which symmetry operations are performed, such that the geometric C_n -axis versus the rotation $\hat{C}_n$. Symmetry group symbols manifest geometric properties, such as $C_n \doteq [C_n]$. See also Eqs. (9.2.14)–(9.2.19) and remarks therein. The $\mathcal{T}_d$ group describes the tetrahedral symmetry, to which the CH_4 molecule belongs, where the carbon atom engages four equivalent sp^3 -hybrid orbitals.

(a) Different conformers of a same molecule can have different symmetries. For example, ferrocene $\text{Fe}(\text{C}_5\text{H}_5)_2$ molecule has the following three conformers: (i) Eclipsed ferrocene belongs to $\mathcal{D}_{5h}$; (ii) Staggered ferrocene goes by $\mathcal{D}_{5d}$; (iii) Any intermediate conformation of $\text{Fe}(\text{C}_5\text{H}_5)_2$ goes by $\mathcal{D}_5$.

(b) *Asymmetric* molecules (zero-symmetry) belong to the $\mathcal{C}_1$ group, involving only the identity operation.

(c) The $\mathcal{C}_s$ group is for molecules having only symmetry plane, while $\mathcal{C}_i$ having only inversion point. Moreover, $\mathcal{C}_s = \mathcal{S}_1$ and $\mathcal{C}_i = \mathcal{S}_2$.

(d) Moreover, $\mathcal{S}_n = \mathcal{C}_{nh}$ when n is odd, such that $\mathcal{S}_3 = \mathcal{C}_{3h}$. The latter notation is preferred, since $\hat{S}_3^3 = \hat{\sigma}_h$ and $\hat{S}_3^4 = \hat{C}_3$, which are the generating elements of $\mathcal{C}_{3h} = (\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_h)$.

- Improper rotation axis, $\mathcal{S}_n$, is by itself a geometrical property only when an **even** $n \geq 4$, in $\mathcal{S}_4 \doteq [S_4]$, $\mathcal{S}_6 \doteq [S_6]$ and so on. The involving single generating operation, $\hat{S}_n = \hat{\sigma}_h \hat{C}_n$, gives rise to

$$\mathcal{S}_4 = (\hat{E}, \hat{S}_4, \hat{C}_2, \hat{S}_4^3) \quad \text{and} \quad \mathcal{S}_6 = (\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{i}, \hat{S}_6^5, \hat{S}_6)$$

Improper rotation axis $\mathcal{S}_n$ exists, without the C_n -axis and the σ_h -plane. The inversion center point emerges in $\mathcal{S}_6$, since $\hat{S}_6^3 = (\hat{C}_6 \hat{\sigma}_h)^3 = \hat{C}_2 \hat{\sigma}_h = \hat{i}$.

- *Dissymmetry* refers to having no superimposable mirror image. This concerns the optical rotary isomer, called *enantiomer*, that exists only for dissymmetric molecules, containing neither symmetry plane nor inversion center nor improper rotation axis. We may refer dissymmetry to that having no any form of improper rotation axis, since $\mathcal{S}_1 = \sigma_h$ and $\mathcal{S}_2 = i$. Evidently, asymmetry must be dissymmetrical, but the converse is not true.

- *Symmetry versus dipole vectors*: If there is a C_n -axis, the molecular dipole must lie along this axis. Molecules having multiple *noncoincident* rotation axes or having central inversion symmetry are *nonpolar*.

B. General features of character tables

• Beside the character matrix that is shared by isomorphic groups, character tables report also the **basis functions**, associated with irreducible representations that are labeled by the Mulliken convention.² For examples,

$$\begin{array}{c|cc|c|c}
 C_2 & \hat{E} & \hat{C}_2 & & \\
 \hline
 A & 1 & 1 & z, R_z & x^2, y^2, z^2, xy \\
 B & 1 & -1 & x, y, R_x, R_y & xz, yz \\
 \hline
 \end{array} \quad (9.2.30a)$$

$$\begin{array}{c|cc|c|c}
 C_h & \hat{E} & \hat{\sigma}_h & & \\
 \hline
 A' & 1 & 1 & x, y, R_z & x^2, y^2, z^2, xy \\
 A'' & 1 & -1 & z, R_x, R_y & xz, yz \\
 \hline
 \end{array} \quad (9.2.30b)$$

$$\begin{array}{c|cc|c|c}
 C_i & \hat{E} & \hat{i} & & \\
 \hline
 A_g & 1 & 1 & R_x, R_y, R_z & x^2, y^2, z^2, xy, xz, yz \\
 A_u & 1 & -1 & x, y, z & \\
 \hline
 \end{array} \quad (9.2.30c)$$

The main differences in these isomorphic groups are the basis functions, the Cartesian (x, y, z) -coordinates and the rotation (R_x, R_y, R_z) -coordinates, associated with individual irreducible representation subspaces. The involving symmetry operations on these basis functions are

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \xrightarrow{\hat{C}_2} \begin{bmatrix} -x \\ -y \\ z \end{bmatrix}, \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \xrightarrow{\hat{\sigma}_h} \begin{bmatrix} x \\ y \\ -z \end{bmatrix}, \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \xrightarrow{\hat{i}} - \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad (9.2.31)$$

$$\begin{bmatrix} R_x \\ R_y \\ R_z \end{bmatrix} \xrightarrow{\hat{C}_2, \hat{\sigma}_h} \begin{bmatrix} -R_x \\ -R_y \\ R_z \end{bmatrix}, \quad \begin{bmatrix} R_x \\ R_y \\ R_z \end{bmatrix} \xrightarrow{\hat{i}} \begin{bmatrix} R_x \\ R_y \\ R_z \end{bmatrix} \quad (9.2.32)$$

The underlying parities specify the basis functions, associated with the irreducible subspaces of individual groups in Eq. (9.2.30). Their total symmetry representations, Eq. (9.2.7), go also by any products of the Cartesian coordinates, x^2 , xy and so on [cf. Eq. (9.2.23)]. As known, while x , y and z are the symmetries of three atomic p -orbitals, their products in general are related to d -orbitals. Applications to the spectroscopic activity issue will also be illustrated.

²**Irreducible subspace symbols** (originally suggested by R. S. Mulliken):

1D (one-dimension): A or B ; 2D: E ; 3D: T ; 4D: G ; 5D: H

- (i) One-dimension notations, A and B , go by $\mathfrak{X}_A(\hat{C}_n(z)) = 1$ and $\mathfrak{X}_B(\hat{C}_n(z)) = -1$.
- (ii) Subscripts 1 and 2 attached to A and B go by the parity signs, $+$ and $-$, respectively, with respect to the chosen $\hat{C}_2(x)$ or $\hat{\sigma}_v(xz)$, if the $C_2(x)$ axis is lacking. Numerical subscripts in two-dimension and three-dimension symbols, E and T , may go differently, but in a self-explained manner.
- (iii) Prime ($'$) and double-prime ($''$) attach all letters, for the parity with respect to the horizontal σ_h -plane symmetry.
- (iv) Subscripts g and u are used *whenever* the inversion i -symmetry exists.

9.3 Symmetry Analysis of Normal Vibrations

9.3.1 Symmetry aspects of vibrations

A. Preludes

• Symmetry group theory provides a unified description for molecular electronic structures *at equilibrium* and nuclear motions *near equilibrium*. For a polyatomic molecule the nuclear potential $V(\mathbf{r})$ is a function of $3N$ -Cartesian coordinates, $\mathbf{r} = (\mathbf{r}_1, \dots, \mathbf{r}_N)$, with $\mathbf{r}_i = (x_i, y_i, z_i)$ at each atom, which describe the *displacements* from equilibrium. The nuclear motions is considered at the Taylor expansion level; i.e., $V(\mathbf{r}) \simeq \frac{1}{2} \mathbf{r}^T \mathbf{F} \mathbf{r}$, with $\mathbf{F}$ being the force constant matrix. The nuclear Hamiltonian function is then

$$H_{\text{nucl}}(\mathbf{p}, \mathbf{q}) \simeq \frac{1}{2} (\mathbf{p}^T \mathbf{m}^{-1} \mathbf{p} + \mathbf{r}^T \mathbf{F} \mathbf{r}) = \frac{1}{2} (\tilde{\mathbf{p}}^2 + \tilde{\mathbf{r}}^T \tilde{\mathbf{F}} \tilde{\mathbf{r}}) \quad (9.3.1)$$

with $\tilde{\mathbf{p}} = \mathbf{m}^{-\frac{1}{2}} \mathbf{p}$, $\tilde{\mathbf{r}} = \mathbf{m}^{\frac{1}{2}} \mathbf{r}$ and $\tilde{\mathbf{F}} = \mathbf{m}^{-\frac{1}{2}} \mathbf{F} \mathbf{m}^{-\frac{1}{2}}$ being scaled by the mass matrix, $\mathbf{m} \equiv \text{diag}(m_1, \dots, m_N)$.

(a) The mass-scaled force constant matrix remains real and symmetric, diagonalizable via an orthogonal matrix that satisfies $\mathbf{U} \mathbf{U}^T = 1$. As results,

$$\mathbf{U} \tilde{\mathbf{F}} \mathbf{U}^T = \text{diag}\{f_1, \dots, f_{3N}\}, \text{ where } f_j \geq 0 \quad (9.3.2)$$

The zero-eigenvalues correspond to the rigid molecular motions.

(b) *Normal modes* refer to those $\mathbf{f}_{\text{vib}} = \text{diag}\{f_1, \dots, f_M\}$ of $f_j > 0$, with $M = 3N - 6$ or $3N - 5$ for nonlinear or linear molecules, respectively. The j^{th} -normal mode coordinate and momentum are

$$q_j = (\mathbf{U} \mathbf{m}^{\frac{1}{2}} \mathbf{r})_j \text{ and } p_j = (\mathbf{U} \mathbf{m}^{-\frac{1}{2}} \mathbf{p})_j \quad (9.3.3)$$

The normal vibration Hamiltonian function is

$$H_{\text{vib}}(\mathbf{p}', \mathbf{q}') = \frac{1}{2} \sum_{j=1}^M (p_j'^2 + f_j q_j'^2) \quad (9.3.4)$$

• *Normal modes are the irreducible representations* of the group, to which the molecule belongs. The symmetry group analysis is applicable to not only translation and rotation, but also normal vibration motions. The standard character table provides also rich information about the symmetries of normal modes (§9.3.1B) and their optical activities (§9.3.1C). (**Remarks:** Normal modes are *nonlocal* and harmonic, but true molecules are anharmonic, having overtone excitations that are often *local modes*, where the symmetry is broken).

• Moreover, the projection operator technique via Eq. (9.2.4) is a universal tool to the construction of symmetry adapted linear combination (SALC) wave functions. Illustrations with normal modes will be given in §9.3.2.

B. Group theory for molecular nuclear motion

• Nuclear motion of N -atomic molecules has $3N$ degrees of freedom and is representable by the Cartesian coordinates vectors, $\mathbf{r} = (\mathbf{r}_1, \dots, \mathbf{r}_N)$, with $\mathbf{r}_i = (x_i, y_i, z_i)$ at each atom. In this representation, the traces (i.e., characters) of symmetry operation matrixes, Eq. (9.2.24), are given by

$$\mathfrak{R}(\hat{\Gamma}_\alpha; \mathbf{r}) = \sum_{i=1}^N \mathfrak{R}(\hat{\Gamma}_\alpha; \mathbf{r}_i) = \sum_{k=1}^K n_k(\mathfrak{R}; \mathbf{r}) \mathfrak{X}_k(\hat{\Gamma}_\alpha); \quad \forall \hat{\Gamma}_\alpha \in \mathbb{G} \quad (9.3.5)$$

where [cf. Eqs. (9.2.25) and (9.2.12)]

$$n_k(\mathfrak{R}; \mathbf{r}) = \frac{1}{h} \sum_{\alpha=1}^K g_\alpha \mathfrak{R}(\hat{\Gamma}_\alpha; \mathbf{r}) \mathfrak{X}_k^*(\hat{\Gamma}_\alpha) \quad (9.3.6)$$

This is the number of times the specified k^{th} -irreducible representation, contained within the $\mathbf{r}$ -representation space. The number (K) of classes equals that of irreducible subspaces (§9.2.2C). As before, h is the group order and g_α is the number of group elements in the α^{th} -class. Symmetry operations in a same class have the same character, $\mathfrak{R}(\hat{\Gamma}_\alpha; \mathbf{r})$. Note that $\mathfrak{R}(\hat{E}; \mathbf{r}) = 3N$, which amounts to the dimensionality of N -atomic molecules, described with $\mathbf{r} = (\mathbf{r}_1, \dots, \mathbf{r}_N)$, the full Cartesian coordinates vectors on atoms.

(i) In general, the nuclear $\mathfrak{R}(\hat{\Gamma}_\alpha; \mathbf{r})$ of a molecule is the sum of all atomic contributions, as highlighted by the first identity of Eq. (9.3.5), that can be easily determined; see the illustrations soon later.

(ii) The resulting $\{\mathfrak{R}(\hat{\Gamma}; \mathbf{r})\}$ for all $\hat{\Gamma} \in \mathbb{G}$, denoted as

$$\mathfrak{R}_{\text{nucl}}(\mathbb{G}) \equiv \{\mathfrak{R}(\hat{\Gamma}; \mathbf{r}); \quad \forall \hat{\Gamma} \in \mathbb{G}\} \quad (9.3.7)$$

are then used in Eq. (9.3.6) to determine the second identity of Eq. (9.3.5), using the character table of the specified $\mathbb{G}$ group.

(iii) Turn to the authentic normal vibration contribution [cf. Eq. (9.3.3)],

$$\mathfrak{R}_{\text{vib}}(\mathbb{G}) \equiv \{\mathfrak{R}_{\text{vib}}(\hat{\Gamma}; \mathbf{q}'); \quad \forall \hat{\Gamma} \in \mathbb{G}\} \quad (9.3.8)$$

This can be achieved by using the character table to separate out the rigid molecular contributions *by symmetries* from $\mathfrak{R}_{\text{nucl}}(\mathbb{G})$; see Eq. (9.3.11) later. As mentioned earlier, character tables report also the translation and rotation motions, associated with irreducible representations.

• Moreover, the selection rules for normal modes transitions, involved in either infrared excitation or Raman scattering, can also be readily addressed by using the character tables; see Part C later. To proceed, we first elaborate the aforementioned (i)–(iii) steps with detailed illustrations.

• **Illustration:** Pyramidal molecules, AB_3 , belongs to the $\mathcal{C}_{3v} = (\hat{E}, 2\hat{C}_3, 3\hat{\sigma}_v)$ group, with $\mathfrak{R}(\hat{E}; \mathbf{r}) = 3N = 12$. The step-(i) deals with the first identity of Eq. (9.3.5); i.e., $\mathfrak{R}(\hat{\Gamma}_\alpha; \mathbf{r})$ via individual atoms, for all $\hat{\Gamma}_\alpha \in \mathcal{C}_{3v}$, as follows.

(a) The character $\mathfrak{R}(\hat{C}_3; \mathbf{r})$: The $\hat{C}_3$ operation brings one B-atom to another B-atom, resulting in only off-diagonal matrix elements; i.e., $\mathfrak{R}(\hat{C}_3; \mathbf{r}_B) = 0$. Only the A-atom contribution to $\mathfrak{R}(\hat{C}_3; \mathbf{r})$, Eq. (9.3.5), is subject to further evaluation, with the final value of [cf. Eq. (9.2.27)]

$$\mathfrak{R}(\hat{C}_3; \mathbf{r}) = \mathfrak{R}(\hat{C}_3; \mathbf{r}_A) = 2 \cos(2\pi/3) + 1 = 0 \quad (9.3.9)$$

(b) Turn to $\mathfrak{R}(\hat{\sigma}_v; \mathbf{r})$: Each σ_v -plane contains the A-atom and one B-atom. Other two B-atoms have no diagonal contributions. As results [cf. Eq. (9.2.28)],

$$\mathfrak{R}(\hat{\sigma}_v; \mathbf{r}) = \mathfrak{R}(\hat{\sigma}_v; \mathbf{r}_A) + \mathfrak{R}(\hat{\sigma}_v; \mathbf{r}_B) = 2(1 - 1 + 1) = 2 \quad (9.3.10)$$

The aforementioned step-(i), the determination of $\mathfrak{R}_{\text{nucl}}(\mathcal{C}_{3v})$, is completed.

The step-(ii) deals with the second identity of Eq. (9.3.5); i.e., decomposing the determined $\mathfrak{R}_{\text{nucl}}(\mathcal{C}_{3v})$ into irreducible subspaces, whereas the step-(iii) determines further $\mathfrak{R}_{\text{vib}}(\mathcal{C}_{3v})$, as the remarks after Eq. (9.3.8). Both these two steps make use of the character table,

$\mathcal{C}_{3v}$	$\hat{E}$	$2\hat{C}_3$	$3\hat{\sigma}_v$		
A_1	1	1	1	z	$x^2 + y^2, z^2$
A_2	1	1	-1	R_z	
E	2	-1	0	$(x, y), (R_x, R_y)$	$(xz, yz), (xy, x^2 - y^2)$
$\mathfrak{R}_{\text{nucl}}$	12	0	2	$3A_1 + A_2 + 4E$	
$\mathfrak{R}_{\text{vib}}$	6	0	2	$2A_1 + 2E$	

which shows also that the translation and rotation motions belong to $A_1 + E$ and $A_2 + E$, respectively. The added bottom region reports the final results from the aforementioned three steps, together with the characters of $\mathfrak{R}_{\text{vib}}(\mathcal{C}_{3v}) = 2A_1 + 2E$.

Exercises

E9.3.1 Verify Eqs. (9.3.9)–(9.3.11).

E9.3.2 Consider the $\mathcal{D}_{3h}$ group, with [cf. Eqs. (9.2.18) and (9.2.19)]

$\mathcal{D}_{3h}$	$\hat{E}$	$2\hat{C}_3$	$3\hat{C}_2$	$\hat{\sigma}_h$	$2\hat{S}_3$	$3\hat{\sigma}_v$		
A'_1	1	1	1	1	1	1		$x^2 + y^2, z^2$
A'_2	1	1	-1	1	1	-1	R_z	
E'	2	-1	0	2	-1	0	(x, y)	$(xy, x^2 - y^2)$
A''_1	1	1	1	-1	-1	-1		
A''_2	1	1	-1	-1	-1	1	z	
E''	2	-1	0	-2	1	0	(R_x, R_y)	(xz, yz)

Show that

$$\mathfrak{R}_{\text{vib}}(\mathcal{D}_{3h}) = A'_1 + 2E' + A''_2 \quad (\text{Planar AB}_3 \text{ molecules}) \quad (9.3.13)$$

C. Infrared and Raman activities of normal mode transitions

• Normal vibration of a molecule consists of $3N-5$ or $3N-6$ normal modes, which are harmonic oscillators, Eq. (9.3.4), with the eigenstates being detailed in §4.2. The selection rule of $\Delta n \equiv m - n = \pm 1$ is applicable, if the oscillatory motion alters the molecular dipole vector. As known, homogenous diatomic molecules are infrared (IR) inactive, but Raman active, whereas heterogenous diatomic molecules are both IR and Raman active, due to the underlying symmetry. The related issues for polyatomic molecules can in general be addressed via symmetry group analysis.

(a) First of all, different normal modes of a given molecule may go with different irreducible representations of the molecular group. For examples, the normal vibration of a pyramidal AB_3 molecule, $\mathfrak{R}_{\text{vib}}(\mathcal{C}_{3v}) = 2A_1 + 2E$, Eq. (9.3.11), involves the two specified types of symmetry in the $\mathcal{C}_{3v}$ group.

(b) Without loss of generality, we focus on the fundamental transition between the ground state and the first excited state. Adopt also the dimensionless coordinate, q_j , for the j^{th} -normal mode. For the normal vibration of a molecule, the group state wave function is [cf. Eq. (4.2.20)]

$$\Psi_0(\mathbf{q}) = e^{-(q_1^2 + \dots + q_M^2)/2} = e^{-\mathbf{q}^2/2} \quad (9.3.14)$$

This is a basis function in the total symmetry (TS) representation of the group, to which the molecule belongs to. On the other hand, the first excited state constitutes a manifold; i.e.,

$$\Psi_1(\mathbf{q}) \equiv \{\Psi_{1j}(\mathbf{q}); j = 1, \dots, M\}, \text{ with } \Psi_{1j}(\mathbf{q}) = q_j e^{-\mathbf{q}^2/2} \quad (9.3.15)$$

Only one normal mode is excited to $\psi_1(q_j) = q_j e^{-q_j^2/2}$ [cf. Eq. (4.2.22)]. Each Ψ_{1j} has the same symmetry of normal mode q_j in the group.

• **The IR active normal modes:** Recall Eq. (5.3.35) with Eq. (5.3.36), which for the fundamental excitation on the j^{th} -normal mode of frequency ω_j , is written as

$$S_{\text{IR}}(\omega; 1_j \leftarrow 0) \propto |\langle \Psi_{1j} | \hat{\mathbf{r}} | \Psi_0 \rangle|^2 \delta(\omega - \omega_j) \quad (9.3.16)$$

Here, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ describes the *direction* of molecular dipole vector in the three-dimension space; i.e., $\mu \propto \mathbf{r}$. The IR activity requires that

$$\langle \Psi_{1j} | \hat{\mathbf{r}} | \Psi_0 \rangle = \langle \Psi_0 | q_j \hat{\mathbf{r}} | \Psi_0 \rangle \neq 0 \quad (9.3.17)$$

This holds only if $q_j \mathbf{r}$ contains contributions from the total symmetry representation of the molecular group. This question is addressed by the direct product property, Eq. (9.2.26) and the remarks therein; i.e., the IR selection rule is that the **normal mode q_j be the same symmetry of any Cartesian coordinates**, either x or y or z , whose information is a part of the character table.

• **The Raman active normal modes:** As known, Raman scattering is a *coherent two-photon process*, in which the $|f\rangle \leftarrow |i\rangle$ transition is not direct, but

involves an intermediate state, considered to be virtual, without real occupation. Instead of the electric dipole vector, Raman spectrum responses to the *polarizability tensor*, depending on the quadratics of Cartesian coordinates, x^2 , y^2 , z^2 , xy , xz and yz . The resulting *Raman selection rule* is that the *normal mode* q_j be the *same symmetry of any Cartesian coordinates products*, which are often also reported in character tables.

• **Illustrations: Pyramidal versus planar AB_3 molecules**

(i) Pyramidal AB_3 molecules belong to the $\mathcal{C}_{3v}$ group, with

$$\mathfrak{R}_{\text{vib}}(\mathcal{C}_{3v}; \text{AB}_3) = 2A_1 + 2E \quad (9.3.18)$$

specified at the bottom of Eq. (9.3.11). These two types of symmetry adapted normal modes are *both IR and Raman active*.

(ii) Planar AB_3 molecules belong to the $\mathcal{D}_{3h}$ group. The resulting normal vibration follows Eq. (9.3.13); i.e.,

$$\mathfrak{R}_{\text{vib}}(\mathcal{D}_{3h}; \text{AB}_3) = A'_1 + 2E' + A''_2 \quad (9.3.19)$$

As seen from Eq. (9.3.12), these three types of symmetry adapted normal modes go with the following optical properties:

- (a) The A'_1 -type is IR inactive, but Raman active.
- (b) The E' -type is both IR and Raman active.
- (c) The A''_2 -type is IR active only.

Exercises

E9.3.3 Can a molecule of the $\mathcal{C}_{3v}$ group have the A_2 -symmetry normal modes? Discuss the IR and Raman activities of this case.

E9.3.4 As inferred from Eq. (9.3.11) and Eq. (9.3.12), each term symbol in $\mathcal{C}_{nv}$ splits into two in $\mathcal{D}_{nh}$. Summarize this type of splitting, in terms the Mulliken's irreducible subspace symbols, as specified earlier in §9.3.1B, and discuss further the related optical activities of normal modes.

E9.3.5 Consider the $\mathcal{C}_{\infty v}$ and $\mathcal{D}_{\infty h}$ groups, the $n \rightarrow \infty$ limits of $\mathcal{C}_{nv}$ and $\mathcal{D}_{nh}$, respectively, to which linear molecules belong, without and with central inversion symmetry. Together with the equivalent term symbols used in §9.1, we have

$\mathcal{C}_{\infty v}$	$\hat{E}$	$2\hat{C}(\phi)$	$\hat{\sigma}_v(\phi)$		
$A_1 = \Sigma^+$	1	1	1	z	$(x^2 + y^2), z^2$
$A_2 = \Sigma^-$	1	1	-1	R_z	
$E_1 = \Pi$	2	$2\cos\phi$	0	$(x, y), (R_x, R_y)$	(xz, yz)
$E_2 = \Delta$	2	$2\cos(2\phi)$	0		$(xy, x^2 - y^2)$
$E_3 = \Phi$	2	$2\cos(3\phi)$	0		
...	...	...	...		

(9.3.20)

Answer the same questions of *E9.3.4* for these two infinite-order groups, with further the illustrations on the molecules, (a) HBr and HCN; (b) H_2 and CO_2 .

9.3.2 Symmetry adapted normal modes

To obtain the symmetry adapted linear combination (SALC) form of normal modes, exploited will be the projection operator matrix, Eq. (9.2.4),

$$\hat{P}_{mn}^k(\mathbb{G}) = \frac{l_k}{h} \sum_{i=1}^h \langle \varphi_m^k | \hat{\Gamma}_i | \varphi_n^k \rangle^* \hat{\Gamma}_i; \quad \forall \hat{\Gamma}_i \in \mathbb{G} \quad (9.3.21)$$

with $\{|\varphi_n^k\rangle; n = 1, \dots, l_k\}$ being the k^{th} -irreducible basis set functions.

(i) The projection operator technique is *universal* and applicable for the constructions of SALC normal modes from a broad range of functions, including those involved in Eqs. (9.3.1)–(9.3.4).

(ii) Particularly interested will be the *internal functions*, such as the molecular *bond length and angle displacements*, without translation-and-rotation contaminations. The resulting SALC normal modes would often be either stretching or bending dominant, if the aforementioned two types of internal motions correlations can be neglected. This is the scenario to be illustrated in details below.

• **Illustrations:** Consider a pyramidal AB_3 molecule, whose the internal displacement functions are the bond lengths and angles, $\{|r_j\rangle$ and $|\theta_j\rangle; j = 1, 2, 3\}$. To have these two sets of functions behave in the same way [cf. Eq. (9.3.24)], the angles are indexed such that $|\theta_j\rangle$ is directly opposite to $|r_j\rangle$. As inferred from Eq. (9.3.11), the SALC construction of normal vibrations is concerned with

$$\mathfrak{R}_{\text{vib}}(\mathcal{C}_{3v}; \text{AB}_3) = 2A_1 + 2E \quad (9.3.22)$$

(a) Consider first the one-dimension A_1 -subspace normal mode construction. The related projection operator, Eq. (9.3.21), with the $\mathcal{C}_{3v}$ group, reads

$$\hat{P}_{A_1}(\mathcal{C}_{3v}) = \frac{1}{6} [\hat{E} + \hat{C}_3 + \hat{C}_3^2 + \hat{\sigma}_v(1) + \hat{\sigma}_v(2) + \hat{\sigma}_v(3)] \quad (9.3.23)$$

It is sufficient to have the actions of $\hat{P}_{A_1}(\mathcal{C}_{3v})$ on $|r_1\rangle$ and $|\theta_1\rangle$, since

$$\begin{aligned} |r_1\rangle &\xrightarrow{\hat{C}_{3v}, \hat{\sigma}_v(3)} |r_2\rangle \xrightarrow{\hat{C}_{3v}} |r_3\rangle, & |r_1\rangle &\xrightarrow{\hat{E}, \hat{\sigma}_v(1)} |r_1\rangle \xrightarrow{\hat{\sigma}_v(2)} |r_3\rangle \\ |\theta_1\rangle &\xrightarrow{\hat{C}_{3v}, \hat{\sigma}_v(3)} |\theta_2\rangle \xrightarrow{\hat{C}_{3v}} |\theta_3\rangle, & |\theta_1\rangle &\xrightarrow{\hat{E}, \hat{\sigma}_v(1)} |\theta_1\rangle \xrightarrow{\hat{\sigma}_v(2)} |\theta_3\rangle \end{aligned} \quad (9.3.24)$$

spanning all $\{|r_j\rangle$ and $|\theta_j\rangle; j = 1, 2, 3\}$. Thus, Eq. (9.3.23) goes by

$$\begin{aligned} \hat{P}_{A_1}|r_1\rangle &= \frac{1}{6} (|r_1\rangle + |r_2\rangle + |r_3\rangle + |r_1\rangle + |r_3\rangle + |r_2\rangle) \\ \hat{P}_{A_1}|\theta_1\rangle &= \frac{1}{6} (|\theta_1\rangle + |\theta_2\rangle + |\theta_3\rangle + |\theta_1\rangle + |\theta_2\rangle + |\theta_3\rangle) \end{aligned} \quad (9.3.25)$$

The corresponding SALC constructions are (unnormalized)

$$A_1(\mathcal{C}_{3v}) : \begin{cases} |r_1\rangle + |r_2\rangle + |r_3\rangle \\ |\theta_1\rangle + |\theta_2\rangle + |\theta_3\rangle \end{cases} \quad (9.3.26)$$

for the two total symmetry normal modes, described by $2A_1$ in Eq. (9.3.22).

(b) Turn to the two-dimension E -subspace, with the projection operator matrix, Eq. (9.3.21), the form of [cf. Eq. (9.3.23)]

$$\hat{P}_E(\mathcal{C}_{3v}) = \frac{1}{3}(\hat{E} + \mathbf{C}_3\hat{C}_3 + \mathbf{C}_3^2\hat{C}_3^2) + \frac{1}{3}\sum_{j=1}^3\sigma_v(j)\hat{\sigma}_v(j) \quad (9.3.27)$$

The E -symmetry counterpart to Eq. (9.3.25) reads (up to constant of $1/3$)

$$\begin{aligned} \hat{P}_E|r_1\rangle &= [\mathbf{1} + \sigma_v(1)]|r_1\rangle + [\mathbf{C}_3 + \sigma_v(3)]|r_2\rangle + [\mathbf{C}_3^2 + \sigma_v(2)]|r_3\rangle \\ \hat{P}_E|\theta_1\rangle &= [\mathbf{1} + \sigma_v(1)]|\theta_1\rangle + [\mathbf{C}_3 + \sigma_v(3)]|\theta_2\rangle + [\mathbf{C}_3^2 + \sigma_v(2)]|\theta_3\rangle \end{aligned} \quad (9.3.28)$$

Note also that $\mathbf{C}_3^{-1}\sigma_v(1)\mathbf{C}_3 = \sigma_v(3)$ and $\mathbf{C}_3^{-1}\sigma_v(3)\mathbf{C}_3 = \sigma_v(2)$.

To proceed, we exploit the Cartesian (x, y) -representation, which supports the E -symmetry, as seen from Eq. (9.3.11), so that [cf. Eqs. (9.2.27) and (9.2.28)]

$$\begin{aligned} \mathbf{C}_3 &= -\frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}, & \mathbf{C}_3^2 &= -\frac{1}{2} \begin{bmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} \\ \sigma_v(1) &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, & \sigma_v(2) &= -\frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{bmatrix}, & \sigma_v(3) &= \frac{1}{2} \begin{bmatrix} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{bmatrix} \end{aligned} \quad (9.3.29)$$

The resulting Eq. (9.3.28) can be expressed as

$$\hat{P}_E|r_1\rangle = (2|r_1\rangle - |r_2\rangle - |r_3\rangle) \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \sqrt{3}(|r_2\rangle - |r_3\rangle) \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad (9.3.30a)$$

$$\hat{P}_E|\theta_1\rangle = (2|\theta_1\rangle - |\theta_2\rangle - |\theta_3\rangle) \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & 0 \end{bmatrix} + \sqrt{3}(|\theta_2\rangle - |\theta_3\rangle) \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \quad (9.3.30b)$$

Hence,

$$E(\mathcal{C}_{3v}) : \begin{cases} 2|r_1\rangle - |r_2\rangle - |r_3\rangle & \text{and } |r_2\rangle - |r_3\rangle \\ 2|\theta_1\rangle - |\theta_2\rangle - |\theta_3\rangle & \text{and } |\theta_2\rangle - |\theta_3\rangle \end{cases} \quad (9.3.31)$$

for the $2E_1$ in Eq. (9.3.22). Together with Eq. (9.3.26), we complete the required SALC form for all six normal modes of a pyramidal AB_3 molecule, assuming that $\{|r_j\rangle, |\theta_j\rangle; j = 1, 2, 3\}$ are uncorrelated each other. Note that the force constants in Eq. (9.3.4) depend on the correlated stretches and bends via Eq. (9.3.3) that quantitatively defines the normal modes, whereas the projection operator technique remains applicable in general.

9.4 Symmetry Aspects of Electronic Structures

9.4.1 Revisits of diatomic molecules

- Consider heteronuclear diatomics that belong to the $\mathcal{C}_{\infty v}$ group, with the character table, Eq. (9.3.20), the form of

$\mathcal{C}_{\infty v}$	$\hat{E}$	$2\hat{C}(\phi)$	$\hat{\sigma}_v(\phi)$	
A_1	1	1	1	$(k = 1, 2, \dots)$
A_2	1	1	-1	
E_k	2	$2 \cos(k\phi)$	0	

(9.4.1)

(a) The $\mathcal{C}_{\infty v}$ group is the infinite limit of $\mathcal{C}_{nv}$, having the symmetry elements, $C_n(z)|_{n \rightarrow \infty} = C(\phi)$ and $\sigma_v(n)|_{n \rightarrow \infty} = \sigma_v(\phi)$, that are both continuous. While the pair of rotation operations, $\hat{C}^m(\phi) = \hat{C}(m\phi)$ and $\hat{C}(-m\phi)$, are in a same class, all vertical plane reflections are in a single class. The orthogonality requirement, Eq. (9.2.32), is satisfied, since

$$\int_0^\infty \frac{d\phi}{\pi} \cos(k\phi) \cos(k'\phi) = \delta_{kk'}; \quad k, k' = 0, 1, 2, \dots \quad (9.4.2)$$

(b) In comparison with the $\hat{L}_z$ -based description in §9.1, $\hat{C}(m\phi)$ takes the place of axial angular momentum $\hat{L}_z$, including that in $[\hat{L}_z, \hat{\sigma}_v] \neq 0$ [cf. Eq. (9.1.11)]. The irreducible representations in $\mathcal{C}_{\infty v}$ [cf. Eq. (9.4.1)] are equivalent to the spectroscopic term symbols used in §9.1; i.e.,

$$A_1 \leftrightarrow \Sigma^+, A_2 \leftrightarrow \Sigma^-, E_1 \leftrightarrow \Pi, E_2 \leftrightarrow \Delta, \dots \quad (9.4.3)$$

where $\pm$ describes the parity with respect to the axial-plane- σ_v reflection.

- The corresponding MOs are labeled by lower-case letters, such as

$$a_1: \sigma_{\text{MO}}(\mathbf{r}), \quad e_1: \pi_{\text{MO}}(\mathbf{r}) \quad \text{and} \quad e_2: \delta_{\text{MO}}(\mathbf{r}) \quad (9.4.4)$$

They are the three LCAO-MOs, as specified in Eqs. (9.1.1)–(9.1.3), and also the basis functions, associated with the specified irreducible subspaces, as implied in Eq. (9.3.20) for the $\mathcal{C}_{\infty v}$ group. The aforementioned Mulliken's convention remains valid, such that the a -subspace is one-dimensional and the e -subspace is two-dimensional.

(i) As described in §9.1.1, diatomic MOs are classified with the values of $|m| = 0, 1, 2, \dots$, and labeled by symbols $\sigma_{\text{MO}}, \pi_{\text{MO}}, \delta_{\text{MO}}, \dots$, respectively. While σ_{MO} having $m = 0$ is one-dimensional, others having $|m| \neq 0$ are two-dimensional; e.g., $\pi_{\text{MO}}(\mathbf{r}) = (\pi_1(\mathbf{r}), \pi_{-1}(\mathbf{r}))$ and $\delta_{\text{MO}}(\mathbf{r}) = (\delta_2(\mathbf{r}), \delta_{-2}(\mathbf{r}))$, associated with $|m| = 1$ and 2, respectively.

(ii) Moreover, only the symmetry preserved LCAO-MOs are required, as inferred from Eqs. (9.1.1)–(9.1.4). In other words, the symmetry of a given MO is rooted

at that of AOs, with respect to their orientations in space.

For example, Eq. (9.1.1),

$$|\sigma_{\text{MO}}\rangle = C_{\text{A}}^{\mathfrak{s}}|\varphi_{\text{A}}^{\mathfrak{s}}\rangle + C_{\text{B}}^{\mathfrak{s}}|\varphi_{\text{B}}^{\mathfrak{s}}\rangle, \text{ with } \mathfrak{s} = s, p_z \text{ and } d_{z^2}$$

are all the total symmetry A_1 -representation in the $C_{\infty v}$ group [cf. Eq. (9.2.26)]. Similarly,

$$|\pi_{\text{MO}}\rangle = C_{\text{A}}^{\mathfrak{p}}|\varphi_{\text{A}}^{\mathfrak{p}}\rangle + C_{\text{B}}^{\mathfrak{p}}|\varphi_{\text{B}}^{\mathfrak{p}}\rangle, \text{ with } \mathfrak{p} = (p_x, p_y) \text{ and } (d_{xz}, d_{yz})$$

having the E_1 -symmetry, while

$$|\delta_{\text{MO}}\rangle = C_{\text{A}}^{\mathfrak{d}}|\varphi_{\text{A}}^{\mathfrak{d}}\rangle + C_{\text{B}}^{\mathfrak{d}}|\varphi_{\text{B}}^{\mathfrak{d}}\rangle, \text{ with } \mathfrak{d} = (d_{xy}, d_{x^2-y^2})$$

having the E_2 -symmetry.

- As known, the construction of many-electron wave functions goes through configurations to final spectroscopic terms, such that [cf. Eq. (9.1.7)]

$$\pi_{\text{MO}}(\mathbf{r}_1)\pi_{\text{MO}}(\mathbf{r}_2) \Rightarrow {}^3\Sigma^-, {}^1\Delta, {}^1\Sigma^+ \quad (9.4.5)$$

This describes how the open-shell configuration of $(\pi_{\text{MO}})^2$ splits into three spectroscopic terms that are both symmetry and spin adapted. For the spin-adapted two-electron configurations, see Eqs. (8.1.42)–(8.1.48).

- The symmetry aspects of electronic wave functions deal with the spacial components. To proceed, we adopt the Mulliken's convention, which for the $C_{\infty v}$ group, Eq. (9.4.1), goes by $e_1 = \pi_{\text{MO}}$ in Eq. (9.4.4) and also those in Eq. (9.4.3). Applying then the direct product, Eq. (9.2.23), and the reducible representation decomposition, Eq. (9.2.24), we obtain

$$e_1^2 \doteq A_1 + A_2 + E_2 \quad (9.4.6)$$

This is equivalent to $\pi_{\text{MO}}^2 \doteq \Sigma^+ + \Sigma^- + \Delta$. The spacial components of Eq. (9.4.5) are reproduced, *without invoking the explicit wave function descriptions*.

As usual, the spectroscopic terms include also the spin $(2S+1)$ -multiplicity. Instead of the detailed construction in Eqs. (9.1.12)–(9.1.19), we can recover the multiplicities of ${}^1A_1 = {}^1\Sigma^+$, ${}^3A_2 = {}^3\Sigma^-$ and ${}^1E_2 = {}^1\Delta$, as follows.

(a) First of all, 1A_1 recasts the general result of ${}^1A_{\text{TS}}$, for total symmetry (TS) representation. While the symmetry adaptivity is rooted at molecular geometry, the *TS for many-electron wave function includes further that its spacial part is symmetric with respect to the particle-permutations*, whereas its spin part is antisymmetric. The participating MOs in the TS term are all spin-paired.

(b) Moreover, the configuration of e_1^2 has $\binom{4}{2} = 6$ number of spin-orbital states, so that the two-dimension E -subspace can only be 1E_2 that consists of two spin-orbital states. Thus, the remaining one-dimension subspace is 3A_2 .

9.4.2 Principles of term symbol description

• As $[\hat{H}_e, \hat{\Gamma}_i] = 0$, for any symmetry operation $\hat{\Gamma}_i \in \mathbb{G}$, the many-electron Hamiltonian $\hat{H}_e$ can always be block-diagonalized into sub-matrixes. In the group theory this is the irreducible representation for the symmetry adapted wave functions description. To this end, it is sufficient to start with the tight-binding model Hamiltonian, followed by SALC construction. *The obtained spectroscopic terms remain true for exact many-electron wave functions, without further details.*

• While the group method is universal, the symmetry adapted construction of electronic wave functions, discussed in parallel to Eqs. (9.3.7)–(9.3.10) for the nuclear motion, involves the following steps and remarks.

(a) Consider first the AOs basis representation, $\{|\varphi_\nu(n)\rangle\}$, located at individual atoms, labeled by n . The resulting reducible representation characters are

$$\mathfrak{R}_{\text{AOs}}(\mathbb{G}) \equiv \{\mathfrak{R}_{\text{AOs}}(\hat{\Gamma}_i); \forall \hat{\Gamma}_i \in \mathbb{G}\} \quad (9.4.7a)$$

with

$$\mathfrak{R}_{\text{AOs}}(\hat{\Gamma}_i) = \sum_{\nu, n} \mathfrak{R}(\hat{\Gamma}_i; |\varphi_\nu(n)\rangle) \quad (9.4.7b)$$

arising from individual AO contributions, identified to be

$$\mathfrak{R}(\hat{\Gamma}_i; |\varphi_\nu(n)\rangle) = \pm \delta_{mn}, \text{ where } \hat{\Gamma}_i |\varphi_\nu(n)\rangle = \pm |\varphi_\nu(m)\rangle \quad (9.4.7c)$$

This covers all scenarios, under individual symmetry operation, $\hat{\Gamma}_i \in \mathbb{G}$, on AOs. The scenario of $m \neq n$ describes that AO changes location, so that the diagonal elements are all zeros. The other two scenarios describe that the AO remains at the location ($m = n$), upon the Γ_i symmetry operation, having the same amplitude, but different signs, since the AO goes by its own parity; see §9.4.3C for comparison between s-orbitals and p_z-orbitals in a same molecular group.

– General speaking, for the SALC constructions, each participated AO $|\varphi_\nu(n)\rangle$ in the molecular environment, turns out to be *hybrid* and properly *orientated*, with a specified *parity*, such as those described by Eq. (9.1.1)–(9.1.4) for diatomic molecules. Similarly, engaged from the C atom participating in CH₄ molecule are four sp³-hybrid AOs, Eq. (6.1.50).

– As the symmetry aspect is concerned, it is sufficient to work on simplified models, such as the tight-binding Hamiltonian, using only the *frontier* AOs of *similar energies*. Moreover, we can also set that $S_{\mu\nu} \equiv \langle \mu | \nu \rangle \simeq \delta_{\mu\nu}$ for the overlap matrix between *all* AOs, no matter they belong to same atoms or not. The above symmetry aspect considerations are applied in obtaining the AOs-based reducible representation of Eq. (9.4.7).

(b) The decompositions of Eq. (9.4.7) into irreducible representations are then followed by using Eq. (9.2.24) of the group theory. Complete the symmetry adapted LCAO-MOs and relate them to the determined irreducible subspaces, in which the secular equation are block-diagonalized and solved. Note also that symmetry adapted MOs are labeled with the lower-case letters of Mulliken's symbols.

(c) Build the configurations via the Aufbau Principle of filling electrons into the obtained SALC MOs. As a direct-product presentation, a configuration-level many-electron wave function is reducible, whose irreducible decompositions are carried out, again by using Eq. (9.2.24).

(d) Each irreducible representation term corresponds to the *spacial symmetry* of the many-electron wave function. The obtained symmetry adapted many-electron wave functions via a given configuration are labeled with the upper-case letters of Mulliken's symbols.

- As usual, the spectroscopic terms include also the spin $(2S + 1)$ -multiplicity. In this relation, there are some simple and important observations, as follows.

(i) The total symmetry (TS) representation goes with $^1A_{TS}$. *The TS of many-electron wave function includes also that its spacial part is symmetric with respect to the particle-permutations*, whereas the associated spin part is antisymmetric. Moreover, for the term symbols, we can focus only on the open-subshell configuration, since the closed-subshell is totally symmetric, having no contribution to the symbols.

(ii) The total number of microscopic states is conserved. This property can be exploited in determining the spin multiplicities, associated with individual resulting spectroscopic terms from a single configuration; see Eq. (9.4.6) and the remarks therein, with $e_1^2 \Rightarrow ^1A_1 + ^3A_2 + ^1E_2$, for the $C_{\infty v}$ group.

(iii) The term symbols report both the symmetry and the spin properties of many-electron wave functions. The energy-order of those terms from a same configuration, is often estimated by the *Hund's rules*: (1) The *high-spin multiplicity* term has low energy; (2) For a given spin multiplicity, the *high-spacial dimensionality* (l_k) term has low energy. Further details would not be needed.

- **Optical selection rules** are concerned with [cf. Eq. (5.3.52)]

$$\mu_{fi} \equiv \langle \Psi_f | \hat{\mu} | \Psi_i \rangle \propto \langle \Psi_f | \hat{\mathbf{r}} | \Psi_i \rangle \neq 0 \quad (9.4.8)$$

for many-electron molecular systems, with $|\Psi_i\rangle$ and $|\Psi_f\rangle$ being described by the term symbols, $^{2S_i+1}\Lambda_i$ and $^{2S_f+1}\Lambda_f$, respectively. While the spin part of selection rule is $\Delta S \equiv S_f - S_i = 0$, the spacial part via Eq. (9.4.8) states that the direct product, $\Psi_i^*(\mathbf{r})[\mathbf{r}\Psi_f(\mathbf{r})]$, contains nonzero TS component.

(a) Note that the group character table reports also the symmetries for the Cartesian coordinates, $\mathbf{r} = (x, y, z)$. In parallel to the IR activity of normal modes discussed in §9.3.1C, the required spacial selection rules via Eq. (9.4.8) can be readily obtained by using the character table, together with Eqs. (9.2.23)–(9.2.26), which determine the direct product and its total symmetry components.

(b) In parallel to those of Eq. (6.1.48) for atomic systems, the selection rules in molecular systems engage both the configuration level and the term level, as in line with the single-photon and single-electron excitation processes. In other words, $|\Psi_i\rangle$ and $|\Psi_f\rangle$, with the Λ_i and Λ_f symmetries, actually arise from two configurations, differing in an optically allowed single-electron excitation. Equation (9.4.8) are used at both the configuration and spectroscopic term levels.

9.4.3 Conformers of H_3 molecule and remarks

The H_3 molecule can be either linear or bent or trigonal, belongs to $\mathcal{D}_{\infty h}$ or $\mathcal{C}_{2v}$ or $\mathcal{D}_{3h}$ groups, respectively. To that end, we consider the tight-binding model, using the 1s orbitals on each H atom as the bases, with $S_{\mu\nu} \equiv \langle \mu | \nu \rangle \simeq \delta_{\mu\nu}$ for the AOs overlap matrix, where $|\nu\rangle \equiv |\varphi_{1s}(\nu)\rangle$. Set the 1s orbital energy $\langle \nu | \hat{H} | \nu \rangle = 0$. The unified Hamiltonian matrix for the above three conformers reads

$$\hat{H} \doteq \begin{bmatrix} 0 & \lambda & \lambda' \\ \lambda & 0 & \lambda \\ \lambda' & \lambda & 0 \end{bmatrix} \quad (1s \text{ AOs bases}) \quad (9.4.9)$$

where $\lambda = \langle 1 | \hat{H} | 2 \rangle = \langle 2 | \hat{H} | 3 \rangle$ and $\lambda' = \langle 1 | \hat{H} | 3 \rangle = 2\lambda[1 - \sin(\theta/2)]$ that is an extended Hückel model, with $60^\circ \leq \theta \leq 180^\circ$ being the bond-angle of $\widehat{H_1H_2H_3}$.

A. Linear H_3 molecule: The $\mathcal{D}_{\infty h}$ symmetry

- Consider first $\mathcal{C}_{\infty v}$ for an arbitrary triatomic linear molecule, with noting that [cf. Eqs. (9.4.1) and (9.4.7)]

$\mathcal{C}_{\infty v}$	$\hat{E}$	$2\hat{C}(\phi)$	$\hat{\sigma}_v(\phi)$	
A_1	1	1	1	
A_2	1	1	-1	
E_k	2	$2\cos(k\phi)$	0	
$\mathfrak{R}_{s\text{-AOs}}$	3	3	3	$3A_1$

(9.4.10a)

We then focus on [cf. Eq. (9.2.19) with Eq. (9.2.18)]

$\mathcal{D}_{\infty h}$	$\hat{E}$	$2\hat{C}(\phi)$	$\hat{\sigma}_v(\phi)$	$\hat{i}$	$2\hat{S}(\phi)$	$\hat{C}_2(\phi)$
A_{1g}	1	1	1	1	1	1
A_{1u}	1	1	1	-1	-1	-1
$\mathfrak{R}_{s\text{-AOs}}$	3	3	3	1	1	1

(9.4.10b)

As results, that $3A_1$ of $\mathcal{C}_{\infty v}$ in Eq. (9.4.10a) splits into $2A_{1g} + A_{1u}$ of $\mathcal{D}_{\infty h}$; i.e.,

$$\mathfrak{R}_{s\text{-AOs}}(\mathcal{C}_{\infty v}) = 3A_1 \rightarrow \mathfrak{R}_{s\text{-AOs}}(\mathcal{D}_{\infty h}) = 2A_{1g} + A_{1u} \quad (9.4.11)$$

(a) Moreover, for the linear H_3 molecule, the tight-binding Hamiltonian, Eq. (9.4.9), goes with $\lambda' = 0$. The MOs and energies are [cf. Eq. (7.1.53)]

$$\begin{cases} |\psi_1\rangle = |1\rangle + 2^{\frac{1}{2}}|2\rangle + |3\rangle \\ |\psi_2\rangle = |1\rangle - |3\rangle \\ |\psi_3\rangle = |1\rangle - 2^{\frac{1}{2}}|2\rangle + |3\rangle \end{cases} \quad \text{and} \quad \begin{cases} \varepsilon_1 = 2^{\frac{1}{2}}\lambda < 0 \\ \varepsilon_2 = 0 \\ \varepsilon_3 = -2^{\frac{1}{2}}\lambda \end{cases} \quad (9.4.12)$$

where $|\nu\rangle \equiv |\varphi_{1s}(\nu)\rangle_{\text{AO}}$. These are the SALC MOs, specified in Eq. (9.4.11) as

$$a_{1g} : |\psi_1\rangle, |\psi_3\rangle \quad \text{and} \quad a_{1u} : |\psi_2\rangle \quad (9.4.13)$$

(b) The ground-state configuration of the linear H_3 molecule and the related term symbol are

$$\psi_1^2 \psi_2^1 = 1a_{1g}^2 a_{1u}^1 \rightarrow {}^2A_{1u} \quad (9.4.14)$$

(c) Consider an excited-state configuration of the linear H_3 molecule,

$$\psi_1^1 \psi_2^2 = 1a_{1g}^1 a_{1u}^2 \rightarrow {}^2A_{1g} \quad (9.4.15)$$

As known, $A_{1g} = \Sigma_g^+$ is the total symmetry in $\mathcal{D}_{\infty h}$, whereas $A_{1u} = \Sigma_u^+$ has the same symmetry of z . The optical transition ${}^2A_{1g} \leftrightarrow {}^2A_{1u}$ is allowed.

B. Bent H_3 : The C_{2v} symmetry and remarks on AOs parity

• Consider $\Re(\hat{\Gamma}_\alpha; s\text{-AOs})$, involving three 1s-AOs representation of the C_{2v} group. By setting the bent H_3 molecule in the xz -plane of the character table, we obtain

C_{2v}	$\hat{E}$	$\hat{C}_2$	$\hat{\sigma}_v(xz)$	$\hat{\sigma}_v(yz)$	
A_1	1	1	1	1	z
A_2	1	1	-1	-1	
B_1	1	-1	1	-1	x
B_2	1	-1	-1	1	y
$\Re_{s\text{-AOs}}$	3	1	3	1	$2A_1 + B_1$

(9.4.16)

• Let the resulting symmetry adapted MOs be [cf. Eq. (9.4.13)]

$$a_1 : |\psi_1\rangle, |\psi_3\rangle \quad \text{and} \quad b_1 : |\psi_2\rangle \quad (9.4.17)$$

(i) As the b_1 symmetry has only one MO, the projection operator, $\hat{P}_{b_1}$, gives rise immediately the wave function the (unnormalized) form of

$$b_1 : |\psi_2\rangle = |1\rangle - |3\rangle, \quad \text{with} \quad \varepsilon_2 = -\lambda' \quad (9.4.18)$$

satisfying $\hat{H}|\psi_2\rangle = \varepsilon_2|\psi_2\rangle$, with the Hamiltonian matrix of Eq. (9.4.9).

(ii) On the other hand, the a_1 symmetry engages two MOs. The single projection operator, $\hat{P}_{a_1}$, determines only up to

$$a_1 : |\psi_j\rangle = |1\rangle + c_j|2\rangle + |3\rangle; \quad j = 1, 3 \quad (9.4.19)$$

Applying then $\hat{H}|\psi_j\rangle = \varepsilon_j|\psi_j\rangle$ results in $\boxed{c_j = 2\lambda/\varepsilon_j}$, with $c_1 > 0$ and $c_3 < 0$ for the bonding $|\psi_1\rangle$ and antibonding $|\psi_3\rangle$, respectively.

(iii) To determine the unknown MO energies, consider the secular determinant of $\hat{H}$, Eq. (9.4.9), with the removal of the determined root of $\varepsilon_2 = -\lambda'$ of Eq. (9.4.18). Solve the resulting quadratic equation and obtain

$$\varepsilon_{\pm} = \frac{\lambda'}{2} \pm 2^{\frac{1}{2}} \lambda \left(1 + \frac{\lambda'^2}{8\lambda^2} \right)^{\frac{1}{2}} \quad (9.4.20)$$

that determines $\varepsilon_1 \equiv \varepsilon_+$ and $\varepsilon_3 \equiv \varepsilon_-$. Note that $\lambda < 0$.

- Let us report the MO energies in terms of $\varepsilon_{\text{MOs}} \equiv (\varepsilon_1, \varepsilon_2, \varepsilon_3) = (\varepsilon_+, -\lambda', \varepsilon_-)$, with $\varepsilon_{\pm}$ being specified by Eq. (9.4.20) and $\varepsilon_2 = -\lambda'$ from Eq. (9.4.18). Recall that in the extended Hückel model, $\lambda' = 2\lambda[1 - \sin(\theta/2)]$, with $60^\circ \leq \theta \leq 180^\circ$ being the bond-angle of $\text{H}_1\text{H}_2\text{H}_3$. For linear H_3 , where $\lambda' = 0$, the results $\varepsilon_{\text{MOs}} = (2^{\frac{1}{2}}\lambda, 0, -2^{\frac{1}{2}}\lambda)$ of Eq. (9.4.12) are recovered. Moreover, for trigonal H_3 , where $\lambda' = \lambda$, the MO energies can be obtained from Eq. (7.1.54), with $N = 3$, resulting in $\varepsilon_{\text{MOs}} = (2\lambda, -\lambda, -\lambda)$ that are also recovered.
- To elaborate the effect of AO's parity, consider a three-center $p\pi$ -conjugate system, the allyl ion, $[\text{H}_2\text{C}(\text{CH})\text{CH}_2]^-$ that belongs to the C_{2v} group. While set again the three carbon atoms in the xz -plane, with respect to $\hat{C}_2(z)$, the three $p\pi$ -AOs, for their participating in the three-center $p\pi$ bonding, are perpendicular to the xz -plane. We have $\Re_{p\pi\text{-AOs}}(\hat{C}_2) = -1$ and $\Re_{p\pi\text{-AOs}}(\hat{\sigma}_v(xz)) = -3$, according to the rules of characters, Eq. (9.4.7b), whereas $\Re_{p\pi\text{-AOs}}(\hat{\sigma}_v(yz)) = 1$ and $\Re_{p\pi}(\hat{E}) = 3$. In comparison to Eq. (9.4.16), we have now

C_{2v}	$\hat{E}$	$\hat{C}_2$	$\hat{\sigma}_v(xz)$	$\hat{\sigma}_v(yz)$	
A_1	1	1	1	1	z
A_2	1	1	-1	-1	
B_1	1	-1	1	-1	x
B_2	1	-1	-1	1	y
$\Re_{p\pi\text{-AOs}}$	3	-1	-3	1	$A_2 + 2B_2$

(9.4.21)

Note also that $p\pi$ -orbitals refer often the p_z -orbitals, with respect to the sp^2 -hybrid AOs considered in the xy -plane. Set then the principle $\hat{C}_2 = \hat{C}_2(x)$, for the property of cyclic permutations on x , y and z , in relation to the character table, so that $\Re_{p\pi\text{-AOs}} = A_2 + 2B_2$ remains valid.

C. Trigonal H_3 : Onsets of $C_n \subset \mathcal{D}_{nh}$ and remarks on AO's parity

- Let us examine trigonal H_3 from the aspect of $C_n \subset \mathcal{D}_{nh}$, for a generic n -atomic cyclic molecule the tight-binding Hamiltonian [cf. Eq. (7.1.52)],

$$\hat{H} = \lambda \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 1 & \cdots & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}_{n \times n} \quad (9.4.22)$$

in the real AO functions $\{|\nu\rangle\}$ -bases. Each equivalent $|\nu\rangle$ locates at its own atom and satisfies $\hat{C}_n|\nu\rangle = |\nu+1\rangle$ in cyclic molecules, so that $\Re_{\text{AOs}}(\hat{C}_n) = 0$, according to the rules of character in Eq. (9.4.7).

- As known, the solutions to Eq. (9.4.22) had been given in Eq. (7.1.54). The obtained LCAO-MOs are all SALCs in the Abelian C_n group, as follows.

(a) The ground MO, $|\psi_0\rangle$, together with its symmetry and energy, are given by

$$A : |\psi_0\rangle = |0\rangle + |1\rangle + \cdots + |n-1\rangle, \quad \text{with } \varepsilon_0 = 2\lambda \quad (9.4.23)$$

The excited MOs are

$$E_j : \begin{cases} |\psi_j\rangle = |0\rangle + \epsilon^j |1\rangle + \cdots + \epsilon^{(n-1)j} |n-1\rangle \\ |\psi_{n-j}\rangle = (|\psi_j\rangle)^*, \quad \text{where } \epsilon \equiv e^{i2\pi/n} \end{cases} \quad (9.4.24)$$

with the energies of

$$\varepsilon_j = \varepsilon_{n-j} = 2\lambda \cos\left(\frac{2j\pi}{n}\right); \quad k = 1, \dots, \text{Int}(n/2) \quad (9.4.25)$$

As known, if $n/2$ is integer, $\varepsilon_{n/2} = -2\lambda$ that is associated with

$$B : |\psi_{n/2}\rangle = |0\rangle - |1\rangle + \cdots + (-1)^{n-1} |n-1\rangle \quad (9.4.26)$$

(b) Moreover, the above LCAO–MO coefficients are the characters of $\mathcal{C}_n$ and satisfy

$$\sum_{k=0}^{n-1} \langle \nu | \psi_k \rangle = \sum_{k=0}^{n-1} e^{i \frac{2\pi}{n} \nu k} = n \delta_{0\nu} \quad (9.4.27)$$

Together with $\Re_{\text{AOs}}(\hat{E}) = n$ and $\Re_{\text{AOs}}(\hat{C}_n) = 0$, we obtain

$$\Re_{\text{AOs}}(\mathcal{C}_n) = \Lambda_1 + \cdots + \Lambda_K \quad (9.4.28)$$

the sum over *all irreducible representations* in $\mathcal{C}_n$, such that

$$\Re_{\text{AOs}}(\mathcal{C}_3) = A + E \quad \text{and} \quad \Re_{\text{AOs}}(\mathcal{C}_6) = A + B + E_1 + E_2 \quad (9.4.29)$$

• Now turn to $\Re_{\text{AOs}}(\mathcal{D}_{3h})$, with the results from $\mathcal{C}_3 \subset \mathcal{D}_{3h}$ as the input, following by considering the effect of AO's parity. We obtain

$$\begin{cases} \Re_{\text{AOs}}(\mathcal{C}_3) = A + E \\ \Re_{\text{s-AOs}}(\mathcal{D}_{3h}) = A'_1 + E' \\ \Re_{\text{p}\pi\text{-AOs}}(\mathcal{D}_{3h}) = A''_2 + E'' \end{cases} \quad (9.4.30)$$

While the letters arise from those of $\mathcal{C}_3$, the indexes depend on the parity of basis functions, in relation to the results [cf. Eq. (9.4.16) versus Eq. (9.4.21)],

$\mathcal{D}_{3h}$	$\hat{E}$	$2\hat{C}_3$	$3\hat{C}_2$	$\hat{\sigma}_h$	$2\hat{S}_3$	$3\hat{\sigma}_v$	(9.4.31)
$\Re_{\text{s-AOs}}$	3	0	1	3	0	1	
$\Re_{\text{p}\pi\text{-AOs}}$	3	0	-1	-3	0	1	

Recall the aforementioned Mulliken's convention: (i) Subscripts 1 and 2 attached to A and B go by the parity of the $\hat{C}_2(x)$ -operation; (ii) Prime and double-prime go by the parity of the $\hat{\sigma}_h$ -operation, whereas subscripts g and u are used *whenever* the inversion i -symmetry exists. The above observations conclude the symbols in Eq. (9.4.30). Note also that the MOs energies, Eqs. (9.4.23) and (9.4.25), remain valid, due to common Hamiltonian of Eq. (9.4.22).

Chapter 10

Determinantal Mean Field Theory

This chapter focuses on the energetic properties of a single Slater determinant. In §10.1, we describe the Coulomb and exchange contributions to determinantal energies, and also the correlations in many-electron Hamiltonian matrix. All these properties are reported in terms of two-electron integrals. In §10.2, we present the Hartree-Fock (HF) theory, which is a single-determinantal self-consistent field (SCF) description. The resulting HF spin-orbitals are the building blocks of various theories of many-electron systems. In §10.3, we discuss the HF theory for spatial LCAO-MOs. In §10.4, we exploit the minimal basis H_2 , an analytically solvable model, to illustrate the HF approximations and an exact method with correlations. The existence of nuclear-spin isomers is also discussed.

10.1 Energetics of Determinantal Wave Function

10.1.1 Coulomb integral and exchange integral

Consider the Slater determinant (referred also configuration),

$$|\Phi\rangle = |\chi_a(\mathbf{x}_1) \cdots \chi_b(\mathbf{x}_N)\rangle \equiv \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_a(1) & \cdots & \chi_b(1) \\ \cdots & \cdots & \cdots \\ \chi_a(N) & \cdots & \chi_b(N) \end{vmatrix} \quad (10.1.1)$$

The mean energy, to be derived soon later, reads

$$\mathcal{E} = \langle \Phi | \hat{H} | \Phi \rangle = \sum_{a=1}^N h_a + \frac{1}{2} \sum_{a,b=1}^N (J_{ab} - \delta_{\sigma_a \sigma_b} K_{ab}) \quad (10.1.2)$$

with

$$h_a = \langle \chi_a(\mathbf{x}) | \hat{h}(\mathbf{r}) | \chi_a(\mathbf{x}) \rangle = \langle \psi_a | \hat{h} | \psi_a \rangle = \int d\mathbf{r} [\psi_a^*(\mathbf{r}) \hat{h}(\mathbf{r}) \psi_a(\mathbf{r})] \quad (10.1.3)$$

The Coulomb integral and exchange integral are

$$J_{ab} = \iint d\mathbf{r}_1 d\mathbf{r}_2 \left[\psi_a^*(\mathbf{r}_1) \psi_b^*(\mathbf{r}_2) \frac{1}{r_{12}} \psi_a(\mathbf{r}_1) \psi_b(\mathbf{r}_2) \right] \quad (10.1.4a)$$

$$K_{ab} = \iint d\mathbf{r}_1 d\mathbf{r}_2 \left[\psi_a^*(\mathbf{r}_1) \psi_b^*(\mathbf{r}_2) \frac{1}{r_{12}} \psi_b(\mathbf{r}_1) \psi_a(\mathbf{r}_2) \right] \quad (10.1.4b)$$

respectively. In Eq. (10.1.2), K_{ab} goes also by $\delta_{\sigma_a \sigma_b}$. Only those parallel-spin electrons pairs contribute to the exchange energy. Moreover,

$$J_{aa} = K_{aa} \quad \text{and} \quad J_{ab} > K_{ab} > 0 \quad (10.1.5)$$

The first identity eliminates the self-energy contamination.

Derivations of Eqs. (10.1.2)–(10.1.4): To proceed, we decompose the total Hamiltonian, Eq. (10.1.1), in terms of one-electron and two-electron components:

$$\hat{H} = \sum_{i=1}^N \hat{h}(\mathbf{r}_i) + \frac{1}{2} \sum_{i,j=1}^N \frac{1}{r_{ij}} \equiv \hat{H}_I(1) + \hat{H}_{II}(1, 2) \quad (10.1.6)$$

(a) The one-electron component contribution to $|\Phi\rangle = |\chi_a(\mathbf{x}_1) \cdots \chi_b(\mathbf{x}_N)\rangle$ reads

$$\mathcal{E}_I = \langle \Phi | \hat{H}_I(1) | \Phi \rangle = \sum_{i=1}^N \langle \Phi | \hat{h}(\mathbf{r}_i) | \Phi \rangle = N \langle \Phi | \hat{h}(\mathbf{r}_1) | \Phi \rangle \quad (10.1.7)$$

where

$$\begin{aligned} \langle \Phi | \hat{h}(\mathbf{r}_1) | \Phi \rangle &= \frac{(N-1)!}{N!} \sum_a \int d\mathbf{x}_1 \chi_a^*(\mathbf{x}_1) \hat{h}(\mathbf{r}_1) \chi_a(\mathbf{x}_1) \\ &= \frac{1}{N} \sum_a \int d\mathbf{r} \psi_a^*(\mathbf{r}) \hat{h}(\mathbf{r}) \psi_a(\mathbf{r}) \end{aligned} \quad (10.1.8)$$

with the spin integral, $\langle \sigma_a | \sigma_a \rangle = 1$, being used. The resulting Eq. (10.1.7) reads

$$\mathcal{E}_I = \sum_a h_a = \sum_a \int d\mathbf{r} \psi_a^*(\mathbf{r}) \hat{h}(\mathbf{r}) \psi_a(\mathbf{r}) \quad (10.1.9)$$

(b) Consider then the two-electron component contribution,

$$\mathcal{E}_{II} = \langle \Phi | \hat{H}_{II} | \Phi \rangle = \frac{1}{2} \sum_{i,j=1}^N \left\langle \Phi \left| \frac{1}{r_{ij}} \right| \Phi \right\rangle = \binom{N}{2} \left\langle \Phi \left| \frac{1}{r_{12}} \right| \Phi \right\rangle \quad (10.1.10)$$

where

$$\begin{aligned} \left\langle \Phi \left| \frac{1}{r_{12}} \right| \Phi \right\rangle &= \frac{(N-2)!}{N!} \sum_{a,b} \left\langle \chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2) \left| \frac{1}{r_{12}} \right| \chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2) \right\rangle \\ &= \frac{(N-2)!}{N!} \sum_{a,b} (J_{ab} - \delta_{\sigma_a \sigma_b} K_{ab}) \end{aligned} \quad (10.1.11)$$

with J_{ab} and K_{ab} being given in Eqs. (10.1.4). The last identity is obtained in line with Eq. (10.1.2) and (10.1.3). The indexes a and b are effectively *always distinct*, due to their origins of spin orbitals. We obtain

$$\mathcal{E}_{II} = \frac{1}{2} \sum_{a,b} (J_{ab} - \delta_{\sigma_a \sigma_b} K_{ab}) \quad (10.1.12)$$

This together with Eq. (10.1.9) give rise to $\mathcal{E} = \mathcal{E}_I + \mathcal{E}_{II}$, Eqs. (10.1.2)–(10.1.4).

10.1.2 Correlation integral and Hamiltonian matrix

A. Two-electron integral notations

Denote the two-electron determinantal integral,

$$\begin{aligned} \langle ij||kl \rangle &\equiv \left\langle \chi_i(\mathbf{x}_1)\chi_j(\mathbf{x}_2) \left| \frac{1}{r_{12}} \right| \chi_k(\mathbf{x}_1)\chi_l(\mathbf{x}_2) \right\rangle \\ &= \iint d\mathbf{x}_1 d\mathbf{x}_2 \chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2) \frac{1}{r_{12}} \begin{vmatrix} \chi_k(\mathbf{x}_1) & \chi_l(\mathbf{x}_1) \\ \chi_k(\mathbf{x}_2) & \chi_l(\mathbf{x}_2) \end{vmatrix} \end{aligned} \quad (10.1.13)$$

While $\langle \chi_i(\mathbf{x}_1)\chi_j(\mathbf{x}_2) |$ is a ket of Slater determinant, $\chi_i^*(\mathbf{x}_1)\chi_j^*(\mathbf{x}_2)$ is the ordinary product. Denote further the spinless counterpart,

$$(ij||kl) \equiv \iint d\mathbf{r}_1 d\mathbf{r}_2 \psi_i^*(\mathbf{r}_1)\psi_j^*(\mathbf{r}_2) \frac{1}{r_{12}} \psi_k(\mathbf{r}_1)\psi_l(\mathbf{r}_2) \quad (10.1.14)$$

This gives rise to Eq. (10.1.13) the expression,

$$\langle ij||kl \rangle = \delta_{\sigma_i \sigma_k} \delta_{\sigma_j \sigma_l} (ij||kl) - \delta_{\sigma_i \sigma_l} \delta_{\sigma_j \sigma_k} (ij||lk) \quad (10.1.15)$$

with $\delta_{\sigma\sigma'}$ arising from the spin integration. In particular,

$$\langle ab||ab \rangle = (ab||ab) - \delta_{\sigma_a \sigma_b} (ab||ba) \equiv J_{ab} - \delta_{\sigma_a \sigma_b} K_{ab} \quad (10.1.16)$$

where $J_{ab} = (ab||ab)$ and $K_{ab} = (ab||ba)$ are identical to those of Eq. (10.1.4). We can then recast Eq. (10.1.2) in terms of

$$\mathcal{E} = \langle \chi_a \cdots \chi_b | \hat{H} | \chi_a \cdots \chi_b \rangle = \sum_{a=1}^N \langle a | \hat{h} | a \rangle + \frac{1}{2} \sum_{a,b=1}^N \langle ab||ab \rangle \quad (10.1.17)$$

B. Hamiltonian matrix

- As known, the Hamiltonian of an N -electron system is an $\binom{N}{N} \times \binom{N}{N}$ matrix, where N is the size of basis spin orbitals. Many elements of the Hamiltonian matrix are zeroes. The selection rule here is that the bra and ket Slater determinants differ by no more than two spin orbitals.

- Consider the elements between $|\Phi\rangle = |\chi_1 \cdots \chi_a \cdots \chi_b \cdots \chi_N\rangle$ and

$$|\Phi_a^r\rangle = |\chi_1 \cdots \chi_r \cdots \chi_b \cdots \chi_N\rangle = \hat{a}_r^\dagger \hat{a}_a |\Phi\rangle \quad (10.1.18a)$$

$$|\Phi_{ab}^{rs}\rangle = |\chi_1 \cdots \chi_r \cdots \chi_s \cdots \chi_N\rangle = \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a |\Phi\rangle \quad (10.1.18b)$$

It is easy to obtain

$$\langle \Phi | \hat{H} | \Phi_a^r \rangle = h_{ar} + \sum_{b=1}^N \langle ab||rb \rangle; \quad (\text{but zero}) \quad (10.1.19a)$$

$$\langle \Phi | \hat{H} | \Phi_{ab}^{rs} \rangle = \langle \Phi | \hat{V}_{ee} | \Phi_{ab}^{rs} \rangle = \langle ab||rs \rangle \quad (10.1.19b)$$

(a) Note that $\langle \Phi | \hat{H} | \Phi_a^r \rangle = 0$, Eq. (10.1.19a), arises from the Brillouin theorem [cf. Eq. (10.2.30)], associated with the canonical HF orbitals.

(b) Equation (10.1.19b) describes the double excitation. Participated is only the two-electron component of Hamiltonian,

$$\hat{H}_{II} = \hat{V}_{ee} = \frac{1}{2} \sum_{i,j=1}^N \frac{1}{r_{ij}} \quad (10.1.20)$$

(c) The above hierarchy, Eqs. (10.1.17)–(10.1.19), holds for an arbitrary reference state, $|\tilde{\Phi}\rangle$ and its two associates, $|\tilde{\Phi}_{\tilde{a}}^{\tilde{r}}\rangle$ and $|\tilde{\Phi}_{\tilde{a}\tilde{b}}^{\tilde{r}\tilde{s}}\rangle$, until all nonzero Hamiltonian matrix elements are completed; see §11.1.

(d) Consider

$$\mathcal{K}_{ab} \equiv (aa||bb) = \iint d\mathbf{r}_1 d\mathbf{r}_2 \psi_a^*(\mathbf{r}_1) \psi_a^*(\mathbf{r}_2) \frac{1}{r_{12}} \psi_b(\mathbf{r}_1) \psi_b(\mathbf{r}_2) \quad (10.1.21a)$$

This is neither Coulomb integral nor exchange, but related to

$$\langle a\bar{a} | \hat{H} | b\bar{b} \rangle = \langle a\bar{a} || b\bar{b} \rangle = (aa||bb) = \mathcal{K}_{ab} \equiv K_{ab}^{\text{corr}} \quad (10.1.21b)$$

While $|b\bar{b}\rangle = \hat{a}_b^\dagger \hat{a}_{\bar{b}}^\dagger \hat{a}_{\bar{a}} \hat{a}_a |a\bar{a}\rangle$ [cf. Eq. (10.1.18b)], the determinant $|a\bar{a}\rangle$ consists of the spin-specified $|a\rangle \equiv |\psi_a(\mathbf{r})\alpha(s)\rangle$ and $|\bar{a}\rangle = |\psi_a(\mathbf{r})\beta(s)\rangle$. In other words, $\mathcal{K}_{ab}$ describes the *correlation* between two configurations. The resulting exchange-correlation integral, which describes the nonclassical contribution, is given by

$$\mathcal{K}_{ab}^{\text{xc}} \equiv K_{ab} + \mathcal{K}_{ab} = (ab||ba) + (aa||bb) \quad (10.1.21c)$$

Exercises

E10.1.1 (a) Verify Eqs. (10.1.13)–(10.1.21); (b) Show that Eq. (10.1.14) satisfies

$$(ij||kl)^* = (kl||ij) = (lk||ji) \quad (10.1.22)$$

E10.1.2 Discuss the inequalities in Eq. (10.1.5), where $J_{ab} = (ab||ab)$ and $K_{ab} = (ab||ba)$, which are *positive* quantities.

E10.1.3 For the minimum basis H₂ molecule studied in §10.4, the relevant spin-singlet block of Hamiltonian matrix reads

$$\mathbf{H} = \begin{bmatrix} \langle \Phi | \hat{H} | \Phi \rangle & \langle \Phi | \hat{H} | \Phi_{11}^{22} \rangle \\ \langle \Phi_{11}^{22} | \hat{H} | \Phi \rangle & \langle \Phi_{11}^{22} | \hat{H} | \Phi_{11}^{22} \rangle \end{bmatrix} = \begin{bmatrix} 2h_{11} + J_{11} & \mathcal{K}_{12} \\ \mathcal{K}_{21} & 2h_{22} + J_{22} \end{bmatrix} \quad (10.1.23)$$

Discuss the results in terms of the spin-adaptivity and various integrals.

E10.1.4 By using Eq. (10.1.2), complete the energy expressions for $|1\bar{1}\rangle$, $|1\bar{2}\rangle$ and $|12\rangle$, together with the schematic plot of these *spin-restricted* configurations; see the last column of Eq. (8.1.13). Discuss also the fact that $|12\rangle$ represents a spin-triplet state.

E10.1.5 Show the determinantal energies, $\mathcal{E}(1\bar{1}2\bar{3}) = \mathcal{E}(1\bar{1}23) + K_{23}$ and

$$\mathcal{E}(1\bar{1}23) = 2h_{11} + h_{22} + h_{33} + J_{11} + 2J_{12} + 2J_{13} + J_{23} - K_{12} - K_{13} - K_{23}$$

10.2 The Hartree–Fock Theory

10.2.1 Single-particle mean field description

A. The determinantal energy functional derivatives

- The Hartree–Fock (HF) theory is concerned with the single Slater determinant, subject to the variational *self-consistent mean field* (SCF) treatment. The HF–SCF theory describes each individual electron moving in the Coulomb mean field of others.
- For a Slater determinant, $|\Phi\rangle = |\chi_a(1) \cdots \chi_b(N)\rangle$, its energy is a functional of participating spin orbitals, $\chi(\mathbf{x}) = \{\chi_a(\mathbf{x}), \cdots, \chi_b(\mathbf{x})\}$. That is

$$\mathcal{E}[\chi(\mathbf{x})] \equiv \langle \Phi | \hat{H} | \Phi \rangle = \mathcal{E}_I[\chi(\mathbf{x})] + \mathcal{E}_{II}[\chi(\mathbf{x})] \quad (10.2.1)$$

where

$$\mathcal{E}_I[\chi(\mathbf{x})] = \sum_{a=1}^N h_a[\chi(\mathbf{x})] = \sum_{a=1}^N \langle \chi_a(\mathbf{x}_1) | \hat{h}(1) | \chi_a(\mathbf{x}_1) \rangle \quad (10.2.2)$$

and the determinantal expression of [cf. Eq. (10.1.13)]

$$\mathcal{E}_{II}[\chi(\mathbf{x})] = \frac{1}{2} \sum_{a,b=1}^N W_{ab}[\chi(\mathbf{x})] \quad (10.2.3a)$$

$$W_{ab}[\chi(\mathbf{x})] = \left\langle \chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2) \left| \frac{1}{r_{12}} \right| \chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2) \right\rangle \quad (10.2.3b)$$

Here, $|\chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2)\rangle$ is a two-electron Slater determinant.

- The minimization is subject to the constraints of

$$\langle \chi_a(\mathbf{x}) | \chi_b(\mathbf{x}) \rangle = \int d\mathbf{x} \chi_a^*(\mathbf{x}) \chi_b(\mathbf{x}) = \delta_{ab}; \quad a, b = 1, 2, \cdots, N \quad (10.2.4)$$

The target functional for minimization via $\delta \mathcal{J}_{\min} / \delta \chi_a^* = 0$ is then

$$\mathcal{J}[\chi(\mathbf{x}); \{\varepsilon_{ab}\}] = \mathcal{E}[\chi(\mathbf{x})] - \sum_{ab} \varepsilon_{ab} [\langle \chi_a(\mathbf{x}) | \chi_b(\mathbf{x}) \rangle - \delta_{ab}] \quad (10.2.5)$$

The Lagrangian multipliers satisfy $\varepsilon_{ba} = \varepsilon_{ab}^*$. The required functional derivatives are elaborated as follows.

Evaluate $\delta \mathcal{J}[\chi(\mathbf{x})] / \delta \chi_a^*(\mathbf{x})$: The basic skill is to treat (i) $\chi_a^*(\mathbf{x})$ and $\chi_a(\mathbf{x})$ as different variables; (ii) integration as a sum of variables series.

(a) For the last term of Eq. (10.2.5), we have

$$\frac{\delta \langle \chi_{a'}(\mathbf{x}) | \chi_b(\mathbf{x}) \rangle}{\delta \chi_a^*(\mathbf{x})} = \frac{\delta}{\delta \chi_a^*(\mathbf{x})} \int d\mathbf{x}' \chi_{a'}^*(\mathbf{x}') \chi_b(\mathbf{x}') = \delta_{aa'} \int d\mathbf{x}' \delta(\mathbf{x} - \mathbf{x}') \chi_b(\mathbf{x}')$$

That is

$$\frac{\delta \langle \chi_{a'}(\mathbf{x}) | \chi_b(\mathbf{x}) \rangle}{\delta \chi_a^*(\mathbf{x})} = \delta_{aa'} \chi_b(\mathbf{x}) \quad (10.2.6)$$

As highlighted here, the functional derivative is in formal concerned only with the *bra-wave functions* involving in the inner-product integral. The resulting functional derivative on the last term of Eq. (10.2.5) reads

$$\sum_{a',b} \varepsilon_{ab} \frac{\delta \langle \chi_{a'}(\mathbf{x}) | \chi_b(\mathbf{x}) \rangle}{\delta \chi_a^*(\mathbf{x})} = \sum_b \varepsilon_{ab} \chi_b(\mathbf{x}) \quad (10.2.7)$$

(b) Similarly, for $h_a[\chi(\mathbf{x})]$ in Eq. (10.2.2), we have

$$\frac{\delta h_{a'}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} = \frac{\delta}{\delta \chi_a^*(\mathbf{x})} \langle \chi_{a'}(\mathbf{x}) | \hat{h}(1) | \chi_{a'}(\mathbf{x}) \rangle = \delta_{aa'} \hat{h}(1) \chi_{a'}(\mathbf{x}) \quad (10.2.8)$$

and the resulting [cf. Eq. (10.2.2)]

$$\frac{\delta \mathcal{E}_I[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} = \sum_{a'} \frac{\delta h_{a'}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} = \hat{h}(1) \chi_a(\mathbf{x}) \quad (10.2.9)$$

(c) Turn to the functional derivative on $W_{ab}[\chi(\mathbf{x})]$, Eq. (10.2.3b). By using Eq. (10.1.13), the standard notation of two-electron determinantal integral, we obtain immediately

$$\begin{aligned} \frac{\delta W_{a'b'}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} &= \frac{\delta}{\delta \chi_a^*(\mathbf{x})} \langle \chi_{a'}(\mathbf{x}_1) \chi_{b'}(\mathbf{x}_2) | | \chi_{a'}(\mathbf{x}_1) \chi_{b'}(\mathbf{x}_2) \rangle \\ &= \delta_{aa'} \langle \delta(\mathbf{x} - \mathbf{x}_1) \chi_{b'}(\mathbf{x}_2) | | \chi_{a'}(\mathbf{x}_1) \chi_{b'}(\mathbf{x}_2) \rangle \\ &\quad + \delta_{ab'} \langle \chi_{a'}(\mathbf{x}_1) \delta(\mathbf{x} - \mathbf{x}_2) | | \chi_{a'}(\mathbf{x}_1) \chi_{b'}(\mathbf{x}_2) \rangle \end{aligned} \quad (10.2.10)$$

This gives rise to [cf. Eq. (10.2.3a)]

$$\begin{aligned} \frac{\delta \mathcal{E}_{II}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} &= \frac{1}{2} \sum_{a'b'} \frac{\delta W_{a'b'}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} \\ &= \frac{1}{2} \sum_{b'} \langle \delta(\mathbf{x} - \mathbf{x}_1) \chi_{b'}(\mathbf{x}_2) | | \chi_a(\mathbf{x}_1) \chi_{b'}(\mathbf{x}_2) \rangle \\ &\quad + \frac{1}{2} \sum_{a'} \langle \chi_{a'}(\mathbf{x}_1) \delta(\mathbf{x} - \mathbf{x}_2) | | \chi_{a'}(\mathbf{x}_1) \chi_a(\mathbf{x}_2) \rangle \\ &= \sum_b \langle \delta(\mathbf{x} - \mathbf{x}_1) \chi_b(\mathbf{x}_2) | | \chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2) \rangle \end{aligned} \quad (10.2.11)$$

Applying then

$$\begin{aligned} |\delta(\mathbf{x} - \mathbf{x}_1) \chi_b(\mathbf{x}_2)\rangle &= 2^{-\frac{1}{2}} [\delta(\mathbf{x} - \mathbf{x}_1) \chi_b(\mathbf{x}_2) - \chi_b(\mathbf{x}_1) \delta(\mathbf{x} - \mathbf{x}_2)] \\ |\chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2)\rangle &= 2^{-\frac{1}{2}} [\chi_a(\mathbf{x}_1) \chi_b(\mathbf{x}_2) - \chi_b(\mathbf{x}_1) \chi_a(\mathbf{x}_2)] \end{aligned} \quad (10.2.12)$$

we obtain

$$\frac{\delta \mathcal{E}_{II}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} = \sum_b \int \frac{d\mathbf{x}'}{|\mathbf{r} - \mathbf{r}'|} \chi_b^*(\mathbf{x}') [\chi_b(\mathbf{x}') \chi_a(\mathbf{x}) - \chi_a(\mathbf{x}') \chi_b(\mathbf{x})] \quad (10.2.13)$$

(d) Finally, the minimal target functional, Eq. (10.2.5), satisfies

$$\frac{\delta \mathcal{J}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} = \frac{\delta \mathcal{E}_I[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} + \frac{\delta \mathcal{E}_{II}[\chi(\mathbf{x})]}{\delta \chi_a^*(\mathbf{x})} - \sum_b \varepsilon_{ab} \chi_b(\mathbf{x}) = 0 \quad (10.2.14)$$

with Eq. (10.2.7) being used. Together with Eqs. (10.2.9) and (10.2.13), we complete the formal derivations of the HF equation (10.2.14).

B. The Hartree–Fock equation

- Let us summarize Eq. (10.2.14) as follows.

$$\hat{F}(\mathbf{x})|\chi_a(\mathbf{x})\rangle = \sum_{b=1}^N \varepsilon_{ab}|\chi_b(\mathbf{x})\rangle; \quad a = 1, \dots, N \quad (10.2.15)$$

The Fock operator reads in general

$$\hat{F}(\mathbf{x}) = \hat{h} + \hat{v}^{\text{HF}}(\mathbf{x}) \equiv \hat{h} + \hat{J}(\mathbf{x}) - \hat{K}(\mathbf{x}) \quad (10.2.16)$$

While $\hat{h}$ is the one-electron component of the Hamiltonian in Eq. (10.1.6), the Hartree–Fock potential, $\hat{v}^{\text{HF}} \equiv \hat{J} - \hat{K}$, is an *effective* one-electron Hermitian operator. It treats the two-electron Hamiltonian contributions at the self-consistent determinantal mean field (SCF) approximation level.

(a) Both Coulomb $\hat{J}$ and exchange $\hat{K}$ are one-electron Hermitian operators, defined via their actions on an arbitrary function, $f(\mathbf{x})$:

$$\hat{J}(\mathbf{x})f(\mathbf{x}) \equiv \sum_{b=1}^N \int \frac{d\mathbf{x}'}{|\mathbf{r} - \mathbf{r}'|} [\chi_b^*(\mathbf{x}')\chi_b(\mathbf{x}')]f(\mathbf{x}) \quad (10.2.17a)$$

$$\hat{K}(\mathbf{x})f(\mathbf{x}) \equiv \sum_{b=1}^N \int \frac{d\mathbf{x}'}{|\mathbf{r} - \mathbf{r}'|} [\chi_b^*(\mathbf{x}')f(\mathbf{x}')] \chi_b(\mathbf{x}) \quad (10.2.17b)$$

(b) We can then recast $\hat{v}^{\text{HF}}(\mathbf{x}) = \hat{J}(\mathbf{x}) - \hat{K}(\mathbf{x})$ via

$$\hat{v}^{\text{HF}}(\mathbf{x})f(\mathbf{x}) \equiv \sum_{b=1}^N \int \frac{d\mathbf{x}'}{|\mathbf{r} - \mathbf{r}'|} \chi_b^*(\mathbf{x}') \begin{vmatrix} f(\mathbf{x}) & \chi_b(\mathbf{x}) \\ f(\mathbf{x}') & \chi_b(\mathbf{x}') \end{vmatrix} \quad (10.2.18)$$

This is equivalent to

$$\langle \chi_j | \hat{v}^{\text{HF}} | \chi_k \rangle = \sum_{b=1}^N \langle \chi_j \chi_b | | \chi_k \chi_b \rangle \quad (10.2.19)$$

The summation runs over all *occupied* $\{\chi_1(\mathbf{x}), \dots, \chi_N(\mathbf{x})\}$ within $|\Phi\rangle$, Eq. (10.1.1). As results, $\hat{v}^{\text{HF}}(\mathbf{x})$ is a one-electron SCF in Eq. (10.2.15) that should be iteratively solved [cf. Eq. (10.2.20) for the canonical equivalence].

(c) As a single determinant description, the HF–SCF determinantal energy, $\mathcal{E}$ of Eq. (10.1.17), engages no correlations. On the other hand, the Fock operator $\hat{F}$ dictates the full Fock space. The correlation $\mathcal{K}$ -type two-electron integrals do appear in the Hamiltonian matrix, as highlighted in Eqs. (10.1.21)–(10.1.23) and the remarks therein.

10.2.2 Canonical Hartree–Fock equation

A. Occupied orbitals versus virtual orbitals

- Let us start with the fact that Eq. (10.2.15) is *physically* equivalent to

$$\boxed{\hat{F}|\chi_a(\mathbf{x})\rangle = \epsilon_a|\chi_a(\mathbf{x})\rangle}; \quad a = 1, \dots, N \quad (10.2.20)$$

Justifications: Exploited will be two well-known observations, as follows.

(a) Physical properties are representation invariance. Different representations are related each other by unitary transformations. While $\{\chi_a(\mathbf{x})\}$ via Eq. (10.2.15) refer often the *local* orbitals, other types of orbitals can be carried out for various purposes, including the canonical equivalence, Eq. (10.2.20), that is used in the formal development.

(b) To proceed, we rewrite Eq. (10.2.15) with

$$\hat{F}'|\chi'_a\rangle = \sum_b \varepsilon_{ab}|\chi'_b\rangle \quad (9.2.15)$$

As mentioned after Eq. (10.2.5), $\varepsilon \equiv \{\varepsilon_{ab}\}$ is an $N \times N$ Hermitian matrix. Its diagonalization reads in formal

$$\mathbf{U}\varepsilon\mathbf{U}^\dagger = \text{diag}(\epsilon_1, \dots, \epsilon_N) \equiv \epsilon, \quad \text{with } \mathbf{U}\mathbf{U}^\dagger = \mathbf{I} \quad (10.2.21)$$

Let $|\chi'_a\rangle$ be the a^{th} -vector of the matrix χ' , whereas $|\chi_a\rangle$ be that of χ in Eq. (10.2.20). The unitary transformation from Eq. (10.2.15) to Eq. (10.2.20) reads

$$\hat{F}'\chi' = \varepsilon\chi' \xrightarrow[\mathbf{U}^\dagger \hat{F}' \mathbf{U} = \hat{F}]{\mathbf{U}\chi' = \chi} \hat{F}\chi = \epsilon\chi \quad (10.2.22)$$

(c) The overall phase of Slater determinant wave function is physically irrelevant. It is noticed that

$$|\chi'_a \cdots \chi'_b\rangle = \det \chi' \quad \text{and} \quad |\chi_a \cdots \chi_b\rangle = \det \chi \quad (10.2.23a)$$

Applying then $\chi = \mathbf{U}\chi'$, we obtain

$$\det \chi = \det(\mathbf{U}\chi') = (\det \mathbf{U})(\det \chi') = e^{i\phi}(\det \chi') \quad (10.2.23b)$$

since the determinant of a unitary matrix is a pure phase that is physically irrelevant. The equivalence between Eq. (10.2.20) and Eq. (10.2.15) is now proved. (*Q.E.D.*)

- The above unitary transformation is needed only for its *existence*, due to the nature of Eq. (10.2.20) that is a SCF method. The Fock operator, $\hat{F} = \hat{h} + \hat{v}^{\text{HF}}$, Eq. (10.2.16) with Eq. (10.2.18), itself requires the knowledge of *occupied spin orbitals*. In other words, Eq. (10.2.20) is to be solved in a self-consistent iteration

until convergence, including both the eigenvalues and spin orbitals, $\{\epsilon_a; |\chi_a\rangle; a = 1, \dots, N\}$.

- Upon the converged $\hat{F}$ being determined by those N -occupied spin orbitals, we have further

$$\boxed{\hat{F}|\chi_j(\mathbf{x})\rangle = \epsilon_j|\chi_j(\mathbf{x})\rangle}; \quad j = 1, \dots, \mathcal{N} \quad (10.2.24)$$

There are $(\mathcal{N} - N)$ number of *virtual* spin orbitals outside the SCF construction space. The Fock-space basis spin orbitals of size $\mathcal{N}$ are all determined.

B. Spin orbital energies versus the determinantal energy

- In Eq. (10.2.24), the spin orbital energy is [cf. Eq. (10.2.19)]

$$\epsilon_j = \langle j|\hat{F}|j\rangle = \langle j|\hat{h} + \hat{v}^{\text{HF}}|j\rangle = \langle j|\hat{h}|j\rangle + \sum_{b=1}^N \langle jb||jb\rangle \quad (10.2.25)$$

By convention, the occupied spin orbitals go with indexes of $j = a, b, c, \dots$, and the unoccupied virtual ones with $j = r, s, t, \dots$.

- Denote the SCF N -electron determinant,

$$|\Phi\rangle = |\chi_a(1) \cdots \chi_b(N)\rangle \quad (10.2.26)$$

All properties in Eq. (10.1.17) are now determined. In particular,

$$\boxed{\mathcal{E} = \langle \Phi|\hat{H}|\Phi\rangle = \sum_{a=1}^N \langle a|\hat{h}|a\rangle + \frac{1}{2} \sum_{a,b=1}^N \langle ab||ab\rangle < \sum_{a=1}^N \epsilon_a} \quad (10.2.27)$$

The last quantity is the sum of individual orbital energies from Eq. (10.2.25), which doubly counts the two-electron interactions.

- In reference to $|\Phi\rangle$, Eq. (10.2.26), the singly excited determinant, $|\Phi_a^r\rangle$, arises as the electron in the originally occupied $|\chi_a\rangle$ being moved to the virtual $|\chi_r\rangle$. The higher excited determinants, $|\Phi_{ab}^{rs}\rangle$, $|\Phi_{abc}^{rst}\rangle$, and so on, are also self-explained. All these associated Slater determinants are *unrelaxed*, without the SCF treatment. Together with the reference $|\Phi\rangle$, they constitute an *orthonormal* basis set of configurations, with $\{|\chi_1\rangle, \dots, |\chi_{\mathcal{N}}\rangle\}$, the HF spin orbitals via Eq. (10.2.24), being the building blocks.

• The Koopman's theorem:

- (a) The deviation on $\mathcal{E}_{\text{HF}}^{(N)} - E_{\text{exact}}^{(N)} > 0$ increases with the number N of electrons. Consider then

$$\mathcal{E}^{(N)} - \mathcal{E}_{c;\text{unrelaxed}}^{(N-1)} = \langle c|\hat{h}|c\rangle + \sum_{b=1}^N \langle cb||cb\rangle = \epsilon_c \approx -I_c \quad (10.2.28)$$

in relation between the occupied orbital energy and the vertical (unrelaxed) ion-

ization energy; i.e.,

$$I_c(\text{unrelaxed}) = \mathcal{E}_{c;\text{unrelaxed}}^{(N-1)} - \mathcal{E}^{(N)} = -\epsilon_c \approx I_c(\text{true}) \quad (10.2.29a)$$

This approximation is fairly accurate, due to the underlying error cancelation, as inferred from the aforementioned HF energy deviation behavior.

(b) On the other hand,

$$A_r(\text{unrelaxed}) = \mathcal{E}_{r;\text{unrelaxed}}^{(N+1)} - \mathcal{E}^{(N)} = \epsilon_r \not\approx A_r(\text{true}) \quad (10.2.29b)$$

This significantly deviates from the true electron affinity, due to the lack of error cancelation here.

• **The Brillouin's theorem:** Singly excited determinants, $|\Phi_a^r\rangle$, do not directly interact with the reference HF determinant, $|\Phi\rangle$; i.e.,

$$\boxed{\langle \Phi | \hat{H} | \Phi_a^r \rangle = 0} \quad (\text{Brillouin theorem}) \quad (10.2.30)$$

The derivation starts with Eq. (10.1.19a), followed by using the HF equation (10.2.24); see also Eq. (11.1.11) for the generalization.

Exercises

E10.2.1 Verify Eqs. (10.2.15)–(10.2.19). Show that both the Coulomb and exchange operators, involving in $\hat{v}^{\text{HF}} = \hat{J} - \hat{K}$, Eq. (10.2.18), are Hermitian.

E10.2.2 Write down the expressions for $\langle \chi_j | \hat{J} | \chi_k \rangle$ and $\langle \chi_j | \hat{K} | \chi_k \rangle$, followed by the underlying relations to those two-electron integrals of Eq. (10.1.4).

E10.2.3 Show that

$$\mathcal{E} = \sum_{a=1}^N \epsilon_a - \frac{1}{2} \sum_{a,b=1}^N \langle ab || ab \rangle \quad (10.2.31)$$

E10.2.4 Consider $|\Phi_a^r\rangle$ and $|\Phi_{ab}^{rs}\rangle$, Eq. (10.1.18). Evaluate the following properties:

- (a) $\langle \Phi_a^r | \hat{H} - \mathcal{E} | \Phi_a^r \rangle$; (b) $\langle \Phi_a^r | \hat{H} - \mathcal{E} | \Phi_{ab}^{rs} \rangle$; (c) $\langle \Phi_{ab}^{rs} | \hat{H} - \mathcal{E} | \Phi_{ab}^{rs} \rangle$; (d) $\langle \Phi_a^r | \hat{H} - \mathcal{E} | \Phi_b^s \rangle$.

10.3 The Spatial Orbitals Formalism

10.3.1 Spin-unrestricted Pople–Nesbet equation

The HF–SCF methods in practical use are required only for *spatial* orbital components, which exploit further the LCAO–MO construction of §7.2.3, so that the SCF concerns only the expansion coefficients. Spatial orbitals can be either spin restricted or spin unrestricted; cf. Eqs. (8.1.11)–(8.1.13) and the remarks there.

In the following, we start with the unrestricted HF (UHF) theory, which will readily be reduced to the restricted HF (RHF) formalism. To convert the formal spin–orbital HF theory into the spatial subspace formalism, we recast $|\Phi\rangle = |\chi_a(1) \cdots \chi_b(N)\rangle$ in the spin unrestricted form,

$$|\Phi^{\text{UHF}}\rangle = |\psi_1^\alpha(\mathbf{x}_1) \cdots \psi_{N_\alpha}^\alpha(\mathbf{x}_{N_\alpha}) \bar{\psi}_1^\beta(\mathbf{x}_{N_\alpha+1}) \cdots \bar{\psi}_{N_\beta}^\beta(\mathbf{x}_N)\rangle \quad (10.3.1)$$

with [cf. Eq. (8.1.13)]

$$\chi_j(\mathbf{x}) = \begin{cases} \psi_j^\alpha(\mathbf{x}) \equiv \psi_j^\alpha(\mathbf{r})\alpha(s) \\ \bar{\psi}_j^\beta(\mathbf{x}) \equiv \psi_j^\beta(\mathbf{r})\beta(s) \end{cases} \quad (10.3.2)$$

Denote also $\bar{\sigma}$ for the opposite spin to $\sigma = \alpha, \beta$. As before, we adopt the convention that $j = a, b, \dots$, for occupied molecular orbitals (MOs), whereas $j = r, s, \dots$ for virtual MOs. The LCAO–MO construction will be exploited, which converts the functional mean field to the parameters variational problem. The Green letters, $\mu, \nu, \dots$, are used for atomic orbitals.

A. Canonical UHF theory for spatial wave functions

- There are two sets of UHF–SCF equations (for $\sigma = \alpha$ and β),

$$[\hat{h}(\mathbf{r}) + \hat{v}^{\sigma;\text{HF}}(\mathbf{r})]|\psi_a^\sigma(\mathbf{r})\rangle = \epsilon_a^\sigma |\psi_a^\sigma(\mathbf{r})\rangle; \quad a = 1, \dots, N_\sigma \quad (10.3.3)$$

with N_σ being the number of spin- σ electrons and

$$\hat{v}^{\sigma;\text{HF}}(\mathbf{r}) = \sum_{b=1}^{N_\sigma} [\hat{J}_b^\sigma(\mathbf{r}) - \hat{K}_b^\sigma(\mathbf{r})] + \sum_{b=1}^{N_{\bar{\sigma}}} \hat{J}_b^{\bar{\sigma}}(\mathbf{r}) \quad (10.3.4)$$

The nonzero matrix elements are [cf. Eq. (10.2.17)]

$$\begin{aligned} J_{ab}^{\sigma\sigma'} &= \langle \psi_a^\sigma(\mathbf{r}) | \hat{J}_b^{\sigma'}(\mathbf{r}) | \psi_a^\sigma(\mathbf{r}) \rangle = (\psi_a^\sigma \psi_b^{\sigma'} | \psi_a^\sigma \psi_b^{\sigma'}) \\ K_{ab}^{\sigma\sigma} &= \langle \psi_a^\sigma(\mathbf{r}) | \hat{K}_b^\sigma(\mathbf{r}) | \psi_a^\sigma(\mathbf{r}) \rangle = (\psi_a^\sigma \psi_b^\sigma | \psi_b^\sigma \psi_a^\sigma) \end{aligned} \quad (10.3.5)$$

Coulomb interactions always exist, whereas exchange integrals appear only with same spin. Each set of $\{\psi_a^\sigma(\mathbf{r})\}$ are determined via Eq. (10.3.3). The involving HF potential, $\hat{v}^{\sigma;\text{HF}}(\mathbf{r})$, Eq. (10.3.4) with Eq. (10.3.5), is a functional of both $\{\psi_a^\alpha(\mathbf{x})\}$ and $\{\psi_a^\beta(\mathbf{x})\}$. The above expressions constitute the UHF–SCF formalism.

- The converged $\hat{v}^{\sigma;\text{HF}}(\mathbf{r})$ is then used to finalize [cf. Eq. (10.2.24)]

$$[\hat{h}(\mathbf{r}) + \hat{v}^{\sigma;\text{HF}}(\mathbf{r})]|\psi_j^\sigma(\mathbf{r})\rangle = \epsilon_j^\sigma |\psi_j^\sigma(\mathbf{r})\rangle; \quad \sigma = \alpha, \beta \quad (10.3.6)$$

covering also virtual orbitals, resulting in

$$\epsilon_j^\sigma = h_{jj}^\sigma + v_{jj}^{\sigma;\text{HF}} = \langle \psi_j^\sigma(\mathbf{r}) | \hat{h}(\mathbf{r}) | \psi_j^\sigma(\mathbf{r}) \rangle + v_{jj}^{\sigma;\text{HF}} \quad (10.3.7)$$

where [cf. Eq. (10.3.4)]

$$v_{jj}^{\sigma;\text{HF}} = \langle \psi_j^\sigma(\mathbf{r}) | \hat{v}^{\sigma;\text{HF}}(\mathbf{r}) | \psi_j^\sigma(\mathbf{r}) \rangle = \sum_{b=1}^{N_\sigma} (J_{jb}^{\sigma\sigma} - K_{jb}^{\sigma\sigma}) + \sum_{b=1}^{N_{\bar{\sigma}}} J_{jb}^{\sigma\bar{\sigma}} \quad (10.3.8)$$

Equation (10.3.5) covers now

$$\begin{aligned} J_{jb}^{\sigma\sigma'} &= \langle \psi_j^\sigma(\mathbf{r}) | \hat{J}_b^{\sigma'}(\mathbf{r}) | \psi_j^\sigma(\mathbf{r}) \rangle = (\psi_j^\sigma \psi_b^{\sigma'} | \psi_j^\sigma \psi_b^{\sigma'}) \\ K_{jb}^{\sigma\sigma} &= \langle \psi_j^\sigma(\mathbf{r}) | \hat{K}_b^\sigma(\mathbf{r}) | \psi_j^\sigma(\mathbf{r}) \rangle = (\psi_j^\sigma \psi_b^\sigma | \psi_b^\sigma \psi_j^\sigma) \end{aligned} \quad (10.3.9)$$

- As known, $\langle \psi_j^\sigma(\mathbf{r}) | \psi_k^\sigma(\mathbf{r}) \rangle = \delta_{jk}$, but [cf. Eq. (8.1.37)]

$$S_{jk}^{\alpha\beta} \equiv \langle \psi_j^\alpha(\mathbf{r}) | \psi_k^\beta(\mathbf{r}) \rangle \neq \delta_{jk} \quad (10.3.10)$$

which for those occupied orbitals satisfy [cf. Eq. (8.1.39)]

$$\sum_{a=1}^{N_\alpha} \sum_{b=1}^{N_\beta} |S_{ab}^{\alpha\beta}|^2 \leq \min(N_\alpha, N_\beta) \equiv N_\beta \quad (10.3.11)$$

The convention of setting $N_\alpha \geq N_\beta$ had also been used in §8.1.3.

B. The Pople–Nesbet formalism

In practice, the HF equation is implemented with the atomic orbitals (AOs) representation; cf. §7.2.3. The nuclear coordinates are parameters in the AOs basis, $\{\varphi_\mu(\mathbf{r})\}$, that are nonorthogonal, with

$$S_{\mu\nu} \equiv \langle \varphi_\mu(\mathbf{r}) | \varphi_\nu(\mathbf{r}) \rangle \quad (10.3.12)$$

The one-electron Hamiltonian matrix reads [cf. Eq. (7.3.2)]

$$h_{\mu\nu} \equiv \langle \varphi_\mu(\mathbf{r}) | \hat{h}(\mathbf{r}) | \varphi_\nu(\mathbf{r}) \rangle = \int d\mathbf{r} \varphi_\mu^*(\mathbf{r}) \left[-\frac{1}{2} \nabla_{\mathbf{r}}^2 + v(\mathbf{r}) \right] \varphi_\nu(\mathbf{r}) \quad (10.3.13)$$

The two-electron spatial integral, Eq. (10.1.14), reads

$$(\mu\kappa || \nu\eta) \equiv \iint d\mathbf{r}_1 d\mathbf{r}_2 \varphi_\mu^*(\mathbf{r}_1) \varphi_\kappa^*(\mathbf{r}_2) \frac{1}{r_{12}} \varphi_\nu(\mathbf{r}_1) \varphi_\eta(\mathbf{r}_2) \quad (10.3.14)$$

The above quantities are fixed in the HF–SCF equation, as follows.

- The LCAO-MOs are

$$|\psi_j^\sigma(\mathbf{r})\rangle = \sum_{\mu=1}^{\mathcal{N}} C_{\mu j}^\sigma |\varphi_\mu(\mathbf{r})\rangle; \quad \sigma = \alpha, \beta \quad (10.3.15)$$

We have

$$\langle \psi_j^\sigma(\mathbf{r}) | \psi_k^\sigma(\mathbf{r}) \rangle = \sum_{\mu, \nu} (C_{\mu j}^\sigma)^* C_{\nu k}^\sigma S_{\mu\nu} = (\mathbf{C}^{\sigma\dagger} \mathbf{S} \mathbf{C}^\sigma)_{jk} = \delta_{jk} \quad (10.3.16)$$

Evidently, only those $\{C_{\mu a}^\sigma; a = 1, \dots, N_\sigma\}$, with $\sigma = \alpha, \beta$, related to the occupied molecular orbitals, are subject to the SCF evaluation.

- Substitute Eq. (12.4.24) into Eq. (10.3.6), resulting in

$$\sum_{\nu} (h_{\mu\nu} + v_{\mu\nu}^{\sigma;\text{HF}}) C_{\nu j}^\sigma = \epsilon_j^\sigma \sum_{\nu} S_{\mu\nu} C_{\nu j}^\sigma \quad (10.3.17)$$

This is the *Pople–Nesbet equation*, with the details as follows.

(a) While $h_{\mu\nu}$ had been given in Eq. (10.3.13), the Hartree–Fock matrix, Eq. (10.3.4), with Eqs. (10.3.5), (10.3.14) and (12.4.24), reads

$$v_{\mu\nu}^{\sigma;\text{HF}} = \sum_{\kappa, \eta} (P_{\eta\kappa}^\sigma + P_{\eta\kappa}^{\bar{\sigma}}) (\mu\kappa || \nu\eta) - \sum_{\kappa, \eta} P_{\eta\kappa}^\sigma (\mu\kappa || \eta\nu) \quad (10.3.18)$$

The spin-polarized densities are

$$P_{\mu\nu}^\sigma \equiv \langle \varphi_\mu | \hat{\rho}_1^\sigma | \varphi_\nu \rangle \approx \sum_{a=1}^{N_\sigma} \langle \varphi_\mu | \psi_a^\sigma \rangle \langle \psi_a^\sigma | \varphi_\nu \rangle = \sum_{a=1}^{N_\sigma} C_{\mu a}^\sigma (C_{\nu a}^\sigma)^* \quad (10.3.19)$$

(b) The first identity is exact, formally defined via the reduced one-electron density operator, $\hat{\rho}_1^\sigma$, that is in principle a mixed state; see §12.1.1. The second identity exploits the HF theory, which is a pure-state approximation, with

$$\hat{\rho}_1^\sigma(\text{HF}) = \sum_{a=1}^{N_\sigma} |\psi_a^\sigma\rangle \langle \psi_a^\sigma| \quad (10.3.20)$$

resulting from the single-determinantal description. We will revisit the spin-polarized densities, Eq. (12.4.28), via the formally exact $\hat{\rho}_1^\sigma$ in §12.1.

10.3.2 Spin-restricted Roothann equation

- The spin restricted description goes by setting $\psi_j^\alpha = \psi_j^\beta = \psi_j$, arising from a single set of equations,

$$[\hat{h}(\mathbf{r}) + \hat{v}^{\text{HF}}(\mathbf{r})]|\psi_j(\mathbf{r})\rangle = \epsilon_j|\psi_j(\mathbf{r})\rangle \quad (10.3.21)$$

The LCAO–MO expression is given by [cf. Eq. (12.4.24)]

$$|\psi_j(\mathbf{r})\rangle = \sum_{\mu} C_{\mu j} |\varphi_{\mu}(\mathbf{r})\rangle \quad (10.3.22)$$

- In line with Eq. (12.4.26), we have now

$$\sum_{\nu} (h_{\mu\nu} + v_{\mu\nu}^{\text{HF}}) C_{\nu j} = \epsilon_j \sum_{\nu} S_{\mu\nu} C_{\nu j} \quad (10.3.23a)$$

This is the **Roothann equation**, with [cf. Eq. (12.4.27)]

$$v_{\mu\nu}^{\text{HF}} = \frac{1}{2} \sum_{\kappa, \eta} [2P_{\eta\kappa}(\mu\kappa|\nu\eta) - (P_{\eta\kappa} + P_{\eta\kappa}^{\text{S}})(\mu\kappa|\eta\nu)] \quad (10.3.23b)$$

The involving charge and spin density matrixes are [cf. Eq. (12.4.28)]

$$\mathbf{P} \equiv \mathbf{P}^{\alpha} + \mathbf{P}^{\beta} \quad \text{and} \quad \mathbf{P}^{\text{S}} \equiv \mathbf{P}^{\alpha} - \mathbf{P}^{\beta} \quad (10.3.24)$$

respectively, with setting $N_{\alpha} \geq N_{\beta}$, so that $\mathbf{P} + \mathbf{P}^{\text{S}} = 2\mathbf{P}^{\alpha}$ appearing in Eq. (10.3.23b).

Remarks: UHF versus RHF

- First of all, the HF theory is a *symmetry adapted method*, which supports the electronic term symbol descriptions for both atoms and molecules, together with the Hund's empirical rules. The conventional electronic terms adopt also the spin adaptivity that goes with the spin-restricted configurations, since the unrestricted determinants are spin contaminated, as detailed in §8.1.3.
- However, from the energy aspect, UHF is often the choice for open-shell systems, where the spin polarization takes in place. On the other hand, a closed-shell system is often exploited the RHF; but it may turn into an open-shell structure, such as the dissociation of H₂ molecule into two H atoms; see §10.4.
- To start the SCF evaluation, the initial value of Hartree–Fock potential can use $\hat{h}$ (or the *core-counterpart*) and setting both $\mathbf{P}^{\alpha}$ and $\mathbf{P}^{\beta}$ to zero. The unrestricted solution will merge after the first iteration, whenever $N_{\alpha} \neq N_{\beta}$. However, if $N_{\alpha} = N_{\beta}$, UHF may result in either the RHF result or the true UHF result, depending on the initial values of the SCF iteration.

10.3.3 Atomic orbitals in molecules and computations

- Atoms in molecules adopt the Slater-type orbitals (STOs, Slater, 1930),

$$\varphi_{nlm}^{\text{STO}}(\mathbf{r}; \zeta) = R_n(r; \zeta) Y_{lm}(\theta, \phi) \quad (10.3.25)$$

with $Y_{lm}(\theta, \phi)$ being the spherical harmonics and

$$R_n(r; \zeta) = (2\zeta)^n \left[\frac{2\zeta}{(2n)!} \right]^{\frac{1}{2}} r^{n-1} e^{-\zeta r} \quad (10.3.26)$$

(i) The radial functions are free of nodes [cf. Eq. (6.1.26)], assuming the form of $R_n(r; \zeta) = R_{n,n-1}(r)$, with $l = n - 1$. This is intimately related to the Bohr's orbit theory; Eq. (6.1.20) and the discussions there.

(ii) The Slater exponent, ζ in Eq. (10.3.26): Atomic orbitals in molecular environments go often with $\zeta > Z/n$. The latter is the value of atomic systems. For example, the 1s orbital of hydrogen atom is about $\zeta \simeq 1.24$, in a number of small molecules.

(iii) In principle, one can apply the variation principle to optimize the Slater exponents, as we did in §7.2.3 for He atom. On the other hand, H_2 has the equilibrium bond length of $R_{\text{eq}} = 1.4$ (atomic unit) that is smaller than the twice of H atom size. The optimal value of ζ is 1.19. However, computational physics and chemistry is mainly concerned with large systems. The values of ζ are rather pre-tested and stored, for individual AOs in averaged molecular environments at equilibrium.

(iv) For polyatomic molecules, the STO implementations are expensive. The numerical cost of HF theory is concerned with the large number of two-electron integrals, $(\mu\kappa|\nu\eta)$, which can engage up to four atoms centers.

- To facilitate the calculations, exploited often is the so-called STO-LG scheme, such as STO-3G, in which each STO is expanded a small L -number of Gaussian-type orbitals (GTOs). That is

$$\varphi_{\mu}^{\text{STO}}(\mathbf{r}; \zeta) \simeq \sum_{p=1}^L c_{\mu p} \varphi_p^{\text{GTO}}(\mathbf{r}; \xi_{\mu p}) \quad (10.3.27)$$

with the STO-LG parameters via nonlinear fitting. Each GTO is written in Cartesian coordinates,

$$\varphi_{n_x n_y n_z}^{\text{GTO}}(\mathbf{r}; \xi) = (2\xi/\pi)^{\frac{3}{4}} \mathcal{N}_{n_x} \mathcal{N}_{n_y} \mathcal{N}_{n_z} x^{n_x} y^{n_y} z^{n_z} e^{-\xi(x^2+y^2+z^2)} \quad (10.3.28)$$

where

$$\mathcal{N}_0 = 1 \quad \text{and} \quad \mathcal{N}_{n>0} = [(4\xi)^n / (2n-1)!!]^{\frac{1}{2}} \quad (10.3.29)$$

The aforementioned four-center two-electron integrals can be analytically reformulated as two-center integrals. However, GTOs fall off too fast comparing with the STOs and also can not reproduce the cusp behavior near nuclear centers.

Exercises

E10.3.1 Verify Eq. (10.3.3) that arises from Eq. (10.2.24) and verify also Eq. (12.4.27).

E10.3.2 Derive the Roothaan equation (10.3.23) and its canonical orthogonalization [Hint: See Eqs. (7.2.22)–(7.2.27)].

E10.3.3 Write down the RHF–SCF determinantal energy for close-shell systems. Verify also that [cf. Eq. (10.3.4)]

$$\hat{v}^{\text{HF}}(\mathbf{r}) = \sum_{b=1}^{N/2} [2\hat{J}_b(\mathbf{r}) - \hat{K}_b(\mathbf{r})] \quad (\text{close-shell system}) \quad (10.3.30)$$

In AOs basis it reads [cf. Eq. (10.3.23b)]

$$v_{\mu\nu}^{\text{HF}} = \frac{1}{2} \sum_{\kappa, \eta} P_{\eta\kappa} [2(\mu\kappa||\nu\eta) - (\mu\kappa||\eta\nu)] \quad (10.3.31)$$

where [cf. Eq. (12.4.28)]

$$P_{\mu\nu} = 2 \sum_{b=1}^{N/2} \langle \varphi_\mu | \psi_b \rangle \langle \psi_b | \varphi_\nu \rangle = 2 \sum_{b=1}^{N/2} C_{\mu b} C_{\nu b}^* \quad (10.3.32)$$

E10.3.4 Verify the following results:

(a) The UHF–SCF determinantal energy is given by

$$\mathcal{E}^{\text{UHF}} = \sum_{\sigma=\alpha, \beta} \sum_{a=1}^{N_\sigma} \left[h_{aa}^{\sigma\sigma} + \frac{1}{2} \sum_{b=1}^{N_\sigma} (J_{ab}^{\sigma\sigma} - K_{ab}^{\sigma\sigma}) \right] + \sum_{a=1}^{N_\alpha} \sum_{b=1}^{N_\beta} J_{ab}^{\alpha\beta} \quad (10.3.33)$$

The factor $\frac{1}{2}$ eliminates the double counting in the unconstraint summations. The self-interaction also disappears, since [cf. Eq. (10.3.9)]

$$J_{aa}^{\sigma\sigma} - K_{aa}^{\sigma\sigma} = 0 \quad (10.3.34)$$

(b) For $|\Phi\rangle = |\psi_1^\alpha \bar{\psi}_1^\beta \psi_2^\alpha\rangle$, we have

$$\mathcal{E}^{\text{UHF}} = h_{11}^{\alpha\alpha} + h_{11}^{\beta\beta} + h_{22}^{\alpha\alpha} + J_{12}^{\alpha\alpha} - K_{12}^{\alpha\alpha} + J_{11}^{\alpha\beta} + J_{21}^{\alpha\beta} \quad (10.3.35)$$

E10.3.5 Consider $|\Phi\rangle = |\psi_1 \bar{\psi}_1 \psi_2\rangle$, the restricted determinant. (a) Write down the charge density matrix and the spin density matrix; (b) Show that

$$\hat{v}^{\text{HF}} = 2\hat{J}_1 + \hat{J}_2 - \hat{K}_1 - \hat{K}_2 \quad (10.3.36)$$

E10.3.6 Construct (or at least get familiar with) the HF simulation codes. Self-study also the STO–LG evaluations of all involving one- and two-electron integrals.

10.4 Illustrations: The Minimal Basis H₂ Molecule

10.4.1 Restricted Hartree–Fock solutions

A. The model and the LCAO–MO solutions

- The minimal basis H₂ exploits the 1s orbital in each H atom,

$$\varphi_\mu(\mathbf{r}) \equiv \varphi_{1s}(\mathbf{r}; \mathbf{R}_\mu), \quad \text{with } \mu = A, B \quad (10.4.1)$$

where $S \equiv \langle \varphi_A | \varphi_B \rangle$. The bonding and anti-bonding spatial MOs are

$$\psi_1(\mathbf{r}) = [2(1 + S)]^{-\frac{1}{2}} [\varphi_A(\mathbf{r}) + \varphi_B(\mathbf{r})] \quad (10.4.2a)$$

$$\psi_2(\mathbf{r}) = [2(1 - S)]^{-\frac{1}{2}} [\varphi_A(\mathbf{r}) - \varphi_B(\mathbf{r})] \quad (10.4.2b)$$

In other words, $\psi = \mathbf{C}\varphi$, with

$$\mathbf{C} \equiv \begin{bmatrix} C_{A1} & C_{A2} \\ C_{B1} & C_{B2} \end{bmatrix} = \begin{bmatrix} [2(1 + S)]^{-\frac{1}{2}} & [2(1 - S)]^{-\frac{1}{2}} \\ [2(1 + S)]^{-\frac{1}{2}} & -[2(1 - S)]^{-\frac{1}{2}} \end{bmatrix} \quad (10.4.3)$$

- In contact with the HF formalism, we set the ground-state determinant that involves the bonding MO,

$$|\Phi\rangle = |\psi_1 \bar{\psi}_1\rangle \equiv |1\bar{1}\rangle \quad (10.4.4)$$

to be the reference. The corresponding Eq. (10.3.30) with $N = 2$ reads

$$\hat{v}^{\text{HF}} = 2\hat{J}_1 - \hat{K}_1 \quad (10.4.5)$$

- (a) The HF orbital eigenenergies are [cf. Eq. (10.3.7) with Eq. (10.3.8)]

$$\epsilon_1 = h_{11} + 2J_{11} - K_{11} = h_{11} + J_{11} \quad (10.4.6a)$$

$$\epsilon_2 = h_{22} + 2J_{21} - K_{21} \quad (10.4.6b)$$

where [cf. Eq. (10.3.9)]

$$\begin{aligned} J_{j1} &\equiv \langle \psi_j | \hat{J}_1 | \psi_j \rangle = (\psi_j \psi_1 | \psi_j \psi_1) \\ K_{j1} &\equiv \langle \psi_j | \hat{K}_1 | \psi_j \rangle = (\psi_j \psi_1 | \psi_1 \psi_j) \end{aligned} \quad (10.4.7)$$

The HF–SCF determinantal energy is

$$\mathcal{E}_0 = \langle \Phi | \hat{H} | \Phi \rangle = 2\epsilon_1 - J_{11} = 2h_{11} + J_{11} \quad (10.4.8)$$

- (b) Not only the Coulomb and exchange integrals [cf. Eq. (10.1.16)],

$$J_{jk} \equiv (\psi_j \psi_k | \psi_j \psi_k) > 0 \quad \text{and} \quad K_{jk} \equiv (\psi_j \psi_k | \psi_k \psi_j) > 0 \quad (10.4.9)$$

identified is also correlation integral [cf. Eq. (10.1.21)],

$$\mathcal{K}_{jk} \equiv \langle \psi_j \bar{\psi}_j | | \psi_k \bar{\psi}_k \rangle = (\psi_j \psi_j | | \psi_k \psi_k) = \mathcal{K}_{kj}^* \quad (10.4.10)$$

that participates in off-diagonal Hamiltonian matrix elements; see §10.4.3.

B. The Roothaan equation for close-shell systems

- Let us revisit Eq. (10.3.31) for close-shell systems,

$$v_{\mu\nu}^{\text{HF}} = \frac{1}{2} \sum_{\kappa, \eta} P_{\eta\kappa} [2(\mu\kappa | | \nu\eta) - (\mu\kappa | | \eta\nu)] \quad (10.4.11)$$

in the AOs representation, where [cf. Eq. (10.3.32)]

$$P_{\mu\nu} = 2 \sum_{b=1}^{N/2} \langle \varphi_\mu | \psi_b \rangle \langle \psi_b | \varphi_\nu \rangle = 2 \sum_{b=1}^{N/2} C_{\mu b} C_{\nu b}^* \quad (10.4.12)$$

- For $|\Phi\rangle = |\psi_1 \bar{\psi}_1\rangle$, the charge density matrix is given by [cf. Eq. (10.4.3)]

$$\mathbf{P} = 2|\mathbf{C}_1\rangle\langle\mathbf{C}_1| = \frac{1}{1+S} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad (10.4.13)$$

Equation (10.4.11) reads

$$v_{\mu\nu}^{\text{HF}} = \frac{1}{2(1+S)} \sum_{\kappa, \eta} [2(\mu\kappa | | \nu\eta) - (\mu\kappa | | \eta\nu)] \quad (10.4.14)$$

that is real. Together with the symmetry consideration, we obtain

$$\mathbf{h} + \mathbf{v}^{\text{HF}} \equiv \begin{bmatrix} h_{AA} + v_{AA}^{\text{HF}} & h_{AB} + v_{AB}^{\text{HF}} \\ h_{BA} + v_{BA}^{\text{HF}} & h_{BB} + v_{BB}^{\text{HF}} \end{bmatrix} = \begin{bmatrix} \xi & \lambda \\ \lambda & \xi \end{bmatrix} \quad (10.4.15)$$

The Roothaan equation (9.3.23) reads then [cf. Eq. (7.2.34)]

$$\begin{bmatrix} \xi & \lambda \\ \lambda & \xi \end{bmatrix} \begin{bmatrix} C_{Aj} \\ C_{Bj} \end{bmatrix} = \epsilon_j \begin{bmatrix} 1 & S \\ S & 1 \end{bmatrix} \begin{bmatrix} C_{Aj} \\ C_{Bj} \end{bmatrix} \quad (10.4.16)$$

The eigenenergies are

$$\epsilon_1 = \frac{\xi + \lambda}{1 + S} = h_{11} + J_{11} \quad (10.4.17a)$$

$$\epsilon_2 = \frac{\xi - \lambda}{1 - S} = h_{22} + 2J_{21} - K_{21} \quad (10.4.17b)$$

The last identities are the results of Eq. (10.4.6) and the MO wave functions are identical to those of Eq. (10.4.2). General speaking, $\lambda \equiv h_{AB} + v_{AB}^{\text{HF}} < 0$, in line with that $\epsilon_1 < \epsilon_2$, since $|\psi_1\rangle$ is bonding and $|\psi_2\rangle$ is anti-bonding. The

present results are all analytical without iteration. Moreover, while $\langle 1\bar{1}|\hat{H}|1\bar{1}\rangle = 2h_{11} + J_{11} < 2\epsilon_1$, Eq. (10.4.8), $\langle 2\bar{2}|\hat{H}|2\bar{2}\rangle = 2h_{22} + J_{22} < 2\epsilon_2$ would also hold.

10.4.2 Unrestricted Hartree–Fock solutions

A. Two sets of MOs via unitary transformation

- In closed-shell systems, where $N_\alpha = N_\beta = N/2$, the spin unrestricted and restricted MOs are related each other via the unitary transformation,

$$|\psi^\alpha\rangle \equiv \mathbf{U}|\psi\rangle \quad \text{and} \quad |\psi^\beta\rangle \equiv \mathbf{U}^\dagger|\psi\rangle \quad (10.4.18)$$

From $|\psi\rangle\langle\psi| = \mathbf{1} = \mathbf{U}\mathbf{U}^\dagger = \mathbf{U}^\dagger\mathbf{U}$, we have

$$|\psi^\sigma\rangle\langle\psi^\sigma| = \mathbf{1} \quad \text{and} \quad \mathbf{S}^{\alpha\beta} = |\psi^\alpha\rangle\langle\psi^\beta| = \mathbf{U}^2 \quad (10.4.19)$$

in relation to $\langle\psi_j^\sigma|\psi_k^\sigma\rangle = \delta_{jk}$ and $S_{jk}^{\alpha\beta} = \langle\psi_j^\alpha|\psi_k^\beta\rangle = (\mathbf{U}^2)_{jk}$, respectively.

- For the minimal basis H_2 , with real spatial MOs of Eq. (10.4.2), the transformation matrix is the rotation type,

$$\mathbf{U} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}; \quad 0 \leq \theta \leq \pi/4 \quad (10.4.20)$$

(i) $\theta = 0$: The restricted scenario, with $\psi_j^\sigma = \psi_j$.

(ii) $0 < \theta < \pi/4$: ψ_1^σ has more contribution from ψ_1 rather than ψ_2 .

(iii) $\theta = \pi/4$: Each ψ_j^σ comprises equally ψ_1 and ψ_2 , the bonding and the antibonding MOs, respectively. In other words, $\theta = \pi/4$ would be the solution to the UHF, when the internuclear distance $R \rightarrow \infty$. This is the dissociation limit, where MOs reduce to AOs; see Eqs. (10.4.35) and (10.4.36) with the discussions there.

B. Basic results and spin contamination

- The ground state unrestricted Slater determinant reads

$$|\psi_1^\alpha \bar{\psi}_1^\beta\rangle = \cos^2\theta |\psi_1 \bar{\psi}_1\rangle - \sin^2\theta |\psi_2 \bar{\psi}_2\rangle - \sqrt{2} \sin\theta \cos\theta |{}_0^3\Phi_{12}\rangle \quad (10.4.21)$$

Equations (10.4.18)–(10.4.20) were exploited. The last term is the spin triplet contamination, with [cf. Eq. (8.1.43b)]

$$|{}_0^3\Phi_{12}\rangle = \frac{1}{\sqrt{2}}(|\psi_1 \bar{\psi}_2\rangle + |\bar{\psi}_1 \psi_2\rangle) \quad (10.4.22)$$

- The determinantal energy is given by

$$\mathcal{E}(\theta) = \langle\psi_1^\alpha \bar{\psi}_1^\beta|\hat{H}|\psi_1^\alpha \bar{\psi}_1^\beta\rangle = h_{11}^{\alpha\alpha} + h_{11}^{\beta\beta} + J_{11}^{\alpha\beta}$$

where $h_{11}^{\sigma\sigma} = \langle \psi_1^\sigma | \hat{h} | \psi_1^\sigma \rangle$ and $J_{11}^{\alpha\beta} = (\psi_1^\alpha \psi_1^\beta | | \psi_1^\alpha \psi_1^\beta)$. Together with Eqs. (10.4.18)–(10.4.20) result in

$$\begin{aligned} \mathcal{E}(\theta) = & 2h_{11} \cos^2\theta + 2h_{22} \sin^2\theta + J_{11} \cos^4\theta + J_{22} \sin^4\theta \\ & + 2(J_{21} - 2K_{21}) \sin^2\theta \cos^2\theta \end{aligned} \quad (10.4.23)$$

The RHF determininal energy, Eq. (10.4.8), amounts to

$$\mathcal{E}(\theta = 0) = 2h_{11} + J_{11} = \mathcal{E}^{\text{RHF}} \quad (10.4.24)$$

- The value of θ can be obtained from the solution to the UHF–SCF equations. This amounts to the minimization of $\mathcal{E}(\theta)$, Eq. (10.4.23), with

$$\mathcal{E}'(\theta) \equiv \frac{d\mathcal{E}(\theta)}{d\theta} = 2 \sin(2\theta) \Theta(\theta) = 0 \quad (10.4.25)$$

where

$$\begin{aligned} \Theta(\theta) = & h_{22} - h_{11} + J_{22} \sin^2\theta - J_{11} \cos^2\theta \\ & + (J_{21} - 2K_{21})(\cos^2\theta - \sin^2\theta) \end{aligned} \quad (10.4.26)$$

- (a) The unrestricted solution of $\mathcal{E}'(\theta) = 0$ corresponds $\Theta(\theta) = 0$ in Eq. (10.4.25), resulting in

$$\cos^2\theta = \frac{h_{22} - h_{11} + J_{22} - J_{21} + 2K_{21}}{J_{11} + J_{22} - 2J_{21} + 4K_{21}} \equiv \eta(R) \quad (10.4.27)$$

While $\cos^2\theta$ is the unrestricted characteristics, the second term describes the specified RHF property, denoted as $\eta(R)$, with $R \equiv |\mathbf{R}_A - \mathbf{R}_B|$ the internuclear distance.

- (b) The UHF formalism, as mentioned below Eq. (10.4.20), is concerned with the region of $0 \leq \theta \leq \pi/4$ that corresponds to $1 \geq \eta(R) \geq 0.5$. The distinct UHF feature will be dominant in the dissociation ($R \rightarrow \infty$) limit, where $\theta = \pi/4$ emerges; see Eqs. (10.4.32) later.

- (c) The RHF scenario ($\theta = 0$) acquires $\mathcal{E}'(\theta) = 0$, as if setting $\sin(2\theta) = 0$ to Eq. (10.4.25). The corresponding Eq. (10.4.26) reads [cf. Eq. (10.4.6)]

$$\Theta(\theta = 0) = h_{22} - h_{11} - J_{11} + J_{21} - 2K_{21} = \epsilon_2 - \epsilon_1 - J_{21} - K_{21} \quad (10.4.28)$$

Together with Eq. (10.4.25), we obtain

$$\left. \frac{d^2\mathcal{E}(\theta)}{d\theta^2} \right|_{\theta=0} = 4\Theta(\theta = 0) \quad (10.4.29)$$

As results, $\Theta(\theta = 0) > 0$ or < 0 or $= 0$ determines whether the RHF energy, $\mathcal{E}(\theta = 0)$, Eq. (10.4.24), is a minimum or maximum or saddle point, respectively.

The latter satisfies $h_{22} - h_{11} = J_{11} - J_{21} + 2K_{21}$ via Eq. (10.4.28) and further

$$\eta_{\text{saddle}}(R) \equiv \frac{h_{22} - h_{11} + J_{22} - J_{21} + 2K_{21}}{J_{11} + J_{22} - 2J_{21} + 4K_{21}} \Big|_{\theta=0} = 1 \quad (10.4.30)$$

At the saddle point, $\theta = 0$, both components of Eq. (10.4.25) are zero. This scenario occurs when internuclear distance R is large, where the UHF takes in place, for its lower energy than the RHF result.

Exercises

E10.4.1 Verify Eqs. (10.4.21)–(10.4.30). Express $|\frac{3}{0}\Phi_{12}\rangle$ of Eq. (10.4.22) in terms of AOs.

E10.4.2 Evaluate $\langle \psi_1^\alpha \bar{\psi}_1^\beta | \hat{S}^2 | \psi_1^\alpha \bar{\psi}_1^\beta \rangle$, the spin contamination, as by Eq. (8.1.30).

C. RHF versus UHF wave functions analysis

To proceed, we denote

$$|\Phi_{\text{cov}}\rangle = \frac{1}{2} [|\varphi_A \varphi_B\rangle + |\varphi_B \varphi_A\rangle] \cdot |\alpha(1)\beta(2)\rangle \quad (10.4.31a)$$

$$|\Phi_{\text{ionic}}\rangle = \frac{1}{2} [|\varphi_A \varphi_A\rangle + |\varphi_B \varphi_B\rangle] \cdot |\alpha(1)\beta(2)\rangle \quad (10.4.31b)$$

for the covalence and ionic contributions, respectively, to the specified spin-singlet state, with $|\alpha(1)\beta(2)\rangle = 2^{-\frac{1}{2}} [\alpha(1)\beta(2) - \alpha(2)\beta(1)]$. The spatial components are given in terms of AOs: $|\varphi_\mu \varphi_\nu\rangle \equiv \varphi_\mu(\mathbf{r}_1) \varphi_\nu(\mathbf{r}_2)$.

- The RHF determinant, Eq. (10.4.4) with Eq. (10.4.2a), reads then

$$|\psi_1 \bar{\psi}_1\rangle = \frac{|\Phi_{\text{cov}}\rangle + |\Phi_{\text{ionic}}\rangle}{1 + S} \quad (10.4.32)$$

This overestimates the ionic contribution to the ground-state H_2 molecule, with the dissociation limit of



The energy of proton (H^+) is set to be zero, whereas the HF theory for hydrogen atom is exact. We obtain

$$\lim_{R \rightarrow \infty} [\mathcal{E}^{\text{RHF}}(\text{H}_2) - 2E(\text{H})] = \frac{1}{2} (\varphi_A \varphi_A | \varphi_A \varphi_A) > 0 \quad (10.4.34)$$

This is the Coulomb interaction of two electrons in H^- , set to be associated with the nuclei A. The contaminations by the ionic species are evident.

- On the other hand, the UHF determinant, Eq. (10.4.21), reads

$$\begin{aligned} |\psi_1^\alpha \bar{\psi}_1^\beta\rangle &= \left[\frac{\cos^2 \theta}{1+S} + \frac{\sin^2 \theta}{1-S} \right] |\Phi_{\text{cov}}\rangle + \left[\frac{\cos^2 \theta}{1+S} - \frac{\sin^2 \theta}{1-S} \right] |\Phi_{\text{ionic}}\rangle \\ &\quad + \frac{\sin 2\theta}{(1-S^2)^{\frac{1}{2}}} [|\varphi_A \varphi_B\rangle - |\varphi_B \varphi_A\rangle] \cdot T_0(1, 2) \end{aligned} \quad (10.4.35)$$

The last term is another covalence contribution, but spin contaminated as by $|\Phi_{12}^3\rangle$, with $T_0(1, 2) = 2^{-\frac{1}{2}}[\alpha(1)\beta(2) + \alpha(2)\beta(1)]$; cf. Eq. (8.1.45b).

The second term describes the ionic contribution, which vanishes in the dissociation regime, where $\theta = \pi/4$ and $S = 0$. The asymptotic Eq. (10.4.35) reads then $|\psi_1^\alpha \bar{\psi}_1^\beta\rangle \xrightarrow{R \rightarrow \infty} |\varphi_A \varphi_B\rangle$ that describes the correct limit of

$$\text{H}_2 \xrightarrow{R \rightarrow \infty} 2\text{H} \quad \text{and} \quad \lim_{R \rightarrow \infty} [\mathcal{E}^{\text{UHF}}(\text{H}_2) - 2E(\text{H})] = 0 \quad (10.4.36)$$

- In brief, the HF approximation often gives proper molecular equilibrium geometry, but too high energies, due to the lack of correlations. For the model H_2 , the RHF overestimates the ionic contribution, resulting in the wrong dissociation limit, Eq. (10.4.33). On the other hand, the UHF is in general spin contaminated, as by the last term of Eq. (10.4.21) or Eq. (10.4.35): $|\psi_1^\alpha \bar{\psi}_1^\beta\rangle$ is no longer factorizable into spatial and spin components, with different particle-permutation parities.

10.4.3 Exact solutions with correlation

- By considering the parities of MOs, Eq. (10.4.2), we obtain the block-matrix relevance to the exact ground state the form of

$$\mathbf{H} = \begin{bmatrix} \langle 1\bar{1}|\hat{H}|1\bar{1}\rangle & \langle 1\bar{1}|\hat{H}|2\bar{2}\rangle \\ \langle 2\bar{2}|\hat{H}|1\bar{1}\rangle & \langle 2\bar{2}|\hat{H}|2\bar{2}\rangle \end{bmatrix} = \begin{bmatrix} \mathcal{E}_0 & \mathcal{K}_{12} \\ \mathcal{K}_{12} & \mathcal{E}_0 + 2\Delta \end{bmatrix} \quad (10.4.37)$$

with $|j\bar{j}\rangle \equiv |\psi_j \bar{\psi}_j\rangle$ [cf. Eq. (10.4.4)]. Here,

$$\mathcal{E}_0 \equiv \langle 1\bar{1}|\hat{H}|1\bar{1}\rangle = 2h_{11} + J_{11} \quad (10.4.38a)$$

$$\mathcal{E}_0 + 2\Delta \equiv \langle 2\bar{2}|\hat{H}|2\bar{2}\rangle = 2h_{22} + J_{22} \quad (10.4.38b)$$

and the correlation integral [Eq. (9.1.21)],

$$\mathcal{K}_{12} = \langle 1\bar{1}|\hat{H}|2\bar{2}\rangle = \langle 2\bar{2}|\hat{H}|1\bar{1}\rangle = (11||22) > 0 \quad (10.4.39)$$

Equation (10.4.37) is spin-adapted for spin-singlet states.

- Solving $\mathbf{H}|\Psi\rangle = E|\Psi\rangle$, results in the ground state (unnormalized),

$$|\Psi_0\rangle = |1\bar{1}\rangle - c|2\bar{2}\rangle, \quad \text{with} \quad E_0 = \mathcal{E}_0 + \Delta - (\Delta^2 + \mathcal{K}_{12}^2)^{\frac{1}{2}} \quad (10.4.40)$$

and the excited state,

$$|\Psi_1\rangle = c|1\bar{1}\rangle + |2\bar{2}\rangle, \text{ with } E_1 = \mathcal{E}_0 + \Delta + (\Delta^2 + \mathcal{K}_{12}^2)^{\frac{1}{2}} \quad (10.4.41)$$

While $E_0 < \langle 1\bar{1}|\hat{H}|1\bar{1}\rangle$ and $E_1 > \langle 2\bar{2}|\hat{H}|2\bar{2}\rangle$,

$$c = \frac{\mathcal{K}_{12}}{\Delta + (\Delta^2 + \mathcal{K}_{12}^2)^{\frac{1}{2}}} \quad (10.4.42)$$

satisfying $0 \leq c \leq 1$. All results are exact in the given basis description.

- For analysis, we exploit Eq. (10.4.32),

$$|1\bar{1}\rangle = \frac{|\Phi_{\text{cov}}\rangle + |\Phi_{\text{ionic}}\rangle}{1 + S} \text{ and } |2\bar{2}\rangle = -\frac{|\Phi_{\text{cov}}\rangle - |\Phi_{\text{ionic}}\rangle}{1 - S} \quad (10.4.43)$$

with $|\Phi_{\text{cov}}\rangle$ and $|\Phi_{\text{ionic}}\rangle$ being specified by Eq. (10.4.31). The parametric dependence on the internuclear distance, $R = |\mathbf{R}_A - \mathbf{R}_B|$, is implied.

- (a) The ground-state wave function in Eq. (10.4.40) reads then

$$|\Psi_0\rangle = \eta_+ \Phi_{\text{cov}} + \eta_- \Phi_{\text{ionic}} \quad (10.4.44)$$

where

$$\eta_{\pm}(R) = \frac{1}{1 + S} \pm \frac{c}{1 - S}; \text{ with } R = |\mathbf{R}_A - \mathbf{R}_B| \quad (10.4.45)$$

In particular, $\eta_-(R \rightarrow \infty) = 0$, since $S \rightarrow 0$ and $c \rightarrow 1$ [cf. Eq. (10.4.42)]. Thus, $|\Psi_0\rangle$ is covalence-bond dominated, dissociating into $\text{H}_2 \rightarrow 2\text{H}$, as anticipated.

- (b) The excited-state wave function in Eq. (10.4.41) is

$$|\Psi_1\rangle = \eta'_+ \Phi_{\text{cov}} + \eta'_- \Phi_{\text{ionic}} \quad (10.4.46)$$

where

$$\eta'_{\pm}(R) = \frac{c}{1 + S} \mp \frac{1}{1 - S} \quad (10.4.47)$$

that satisfies $\eta'_+(R \rightarrow \infty) = 0$. The excited $|\Psi_1\rangle$ is an ionic-bond dominated state and dissociates into $\text{H}_2 \rightarrow \text{H}^+ + \text{H}^-$.

Exercises

E10.4.3 For the minimum basis H_2 , there are a total of $\binom{\mathcal{N}}{2} = \binom{4}{2} = 6$ configurations, where $\mathcal{N} = 2M$, with M being the number of spatial basis orbitals. With setting $|\Phi\rangle = |1\bar{1}\rangle$ the reference, $|2\bar{2}\rangle = |\Phi_{11}^{22}\rangle$ is the double excitation. Other four are single-excitation configurations, constituting a spin-singlet and a spin-triplet.

- (a) Recast $|\Phi_{12}^3\rangle$ of Eq. (10.4.22), in terms of excitations from the reference $|\Phi\rangle = |1\bar{1}\rangle$.

(b) While the full Hamiltonian is a (6×6) -matrix, Eq. (10.4.37) is the only non-diagonal block. Verify this observation with either the symmetry consideration or the Brillouin's theorem, Eq. (10.2.30), regardless of symmetry.

E10.4.4 Sketch the ground electronic-state potential, $U_0(R)$, for H_2 vibration, with comparisons between the RHF, UHF and FCI results, followed by qualitative discussions. Discuss also the basic properties of $U_1(R)$, as suggested by the related formula.

10.4.4 Nuclear-spin isomers: Ortho- H_2 and para- H_2

When considering the nuclear properties of H_2 , the fact that protons are spin- $\frac{1}{2}$ particles should be taken into account. In any stable electronic state, there are two nuclear-spin isomers: Ortho- H_2 and para- H_2 , referring to the two proton-spins being parallel and anti-parallel, respectively. In NMR spectroscopy, ortho- H_2 is triplet and para- H_2 is singlet.

(a) The total composite nuclear wave function must be antisymmetric with respect to the permutation of two protons. While it is irrelevant to vibration, nuclear spin wave function of ortho- H_2 is symmetric and that of para- H_2 is antisymmetric; see Eqs. (8.1.45) and (8.1.46). Rotation wave functions of diatomic molecules are the spherical harmonics, satisfying $Y_{l,m}(\pi - \theta, \pi + \phi) = (-1)^L Y_{l,m}(\theta, \phi)$, Eq. (5.3.5), that effectively permutes two protons. As results, ortho- H_2 goes by $l = 1, 3, \dots$ and para- H_2 goes by $l = 0, 2, \dots$.

(b) The equilibrium ensemble averaged energy of ortho- H_2 is higher than that of para- H_2 . The latter would be the dominant nuclear-spin isomer in the low-temperature regime.

Chapter 11

Many-Electron Wave Function Theories

Electronic structure theory and methods are the subjects of Computational Physics/Chemistry, covered in many specialized textbooks and courses. Quantum simulation softwares are accessible through universities and getting familiar of their use is recommended, while you learn the theories. Simulation codes are also all-symmetry adapted, including the molecular point-group geometric operations that are subject of other course.

This chapter is concerned with the basic principles and methods of many-electron systems. In §11.1, we start with the basic configuration-space relations in the Hartree-Fock orbital representation. We then present the configuration interaction (CI) formalism and the many-body perturbation theory (MBPT), followed by the issues of size extensivity requirement. In principle, the CI formalism is exact, but extremely expensive, whereas the truncated CI approximations violate the size extensivity requirement. In contrast, MBPT satisfies this requirement, but is often not accurate enough. It is worth noting that MBPT implements the Rayleigh-Schrödinger perturbation theory to many-electron systems, under the Moller-Plesset setup framework. Section §11.2 deals with the coupled cluster theory and related methods, which comprise the modern wave function approaches to practical many-electron systems, in the presence of strong correlations.

References

- 1 A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory* (Dover, 1982)
- 2 T. Helgaker, P. Jørgensen, and J. Olsen, *Molecular Electronic-Structure Theory* (Wiley, 2000)
- 3 http://en.wikipedia.org/wiki/Quantum_chemistry_computer_programs

11.1 Configuration Space Formulations

11.1.1 Basic relations

A. Correlation energy

- First of all, any *single* determinant theory describes an effective *noninteracting* electronic system; see Eqs. (8.1.4)–(8.1.6) and the remarks there. The Fermi hole has been taken into account, but the “*Coulomb hole*” is completely missing.
- (a) The aforementioned noninteracting scenario includes the HF–SCF approximant,

$$|\Phi\rangle = |\chi_1 \cdots \chi_a \cdots \chi_b \cdots \chi_N\rangle \approx |\Psi\rangle \quad (11.1.1)$$

The resulting HF determinantal energy, Eq. (10.2.27),

$$\mathcal{E} = \langle \Phi | \hat{H} | \Phi \rangle = \sum_{a=1}^N h_{aa} + \frac{1}{2} \sum_{a,b=1}^N \langle ab || ab \rangle = \sum_{a=1}^N \epsilon_a - \frac{1}{2} \sum_{a,b=1}^N \langle ab || ab \rangle \quad (11.1.2)$$

which is always overestimated over the true value.

(b) Formally, correlation energy is the difference between the exact ground state energy and the HF–SCF counterpart; i.e.,

$$E_{\text{corr}} \equiv E_0(\text{exact}) - \mathcal{E}_0(\text{HF}) < 0; \quad (\text{Correlation energy}) \quad (11.1.3)$$

with the converged $\mathcal{E}_0(\text{HF})$ via Eq. (11.1.2) against the basis set size.

B. The N –electron configuration space descriptions

- Turn to the canonical HF equation (10.2.24),

$$[\hat{h} + \hat{v}^{\text{HF}}(\mathbf{x})]|\chi_j\rangle = \epsilon_j|\chi_j\rangle; \quad \text{with} \quad \begin{cases} |\chi_j\rangle \in |\Phi\rangle; & j = a, b, \cdots \\ |\chi_j\rangle \notin |\Phi\rangle; & j = r, s, \cdots \end{cases} \quad (11.1.4)$$

Those relaxed N orbitals, labeled with $j = a, b, \cdots$, determine both $|\Phi\rangle$, Eq. (11.1.1), and the Hartree–Fock potential $\hat{v}^{\text{HF}}(\mathbf{x})$ via Eq. (10.2.18). Other spin–orbitals outside the SCF subspace, labeled with $j = r, s, \cdots$, are unrelaxed and called the virtual orbitals.

- In the second–quantization, Eq. (11.1.1) amounts to

$$|\Phi\rangle = |\chi_1 \cdots \chi_a \cdots \chi_N\rangle = (\hat{a}_1^\dagger \cdots \hat{a}_a^\dagger \cdots \hat{a}_N^\dagger)|0\rangle \quad (11.1.5)$$

The singly excited determinant is then

$$|\Phi_a^r\rangle \equiv |\chi_1 \cdots \chi_{a-1} \chi_r \chi_{a+1} \cdots \chi_N\rangle = \hat{a}_r^\dagger \hat{a}_a |\Phi\rangle \quad (11.1.6)$$

Higher excited ones are all self–explained, such as

$$|\Phi_{ab}^{rs}\rangle = \hat{a}_s^\dagger \hat{a}_b |\Phi_a^r\rangle = \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a |\Phi\rangle \quad (11.1.7)$$

$$\begin{aligned}
|\Phi_{abc}^{rst}\rangle &= \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_t^\dagger \hat{a}_c \hat{a}_b \hat{a}_a |\Phi\rangle \\
|\Phi_{abcd}^{rstu}\rangle &= \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_t^\dagger \hat{a}_u^\dagger \hat{a}_d \hat{a}_c \hat{a}_b \hat{a}_a |\Phi\rangle
\end{aligned} \tag{11.1.8}$$

and so on.

C. Hamiltonian matrix in the configuration space

• For N electrons distributed over the chosen $\mathcal{N}$ spin orbitals, there are a total number $\binom{N}{N}$ of distinct configurations, arising from the *canonical* HF equation (11.1.4). The N -electron Fock space basis elements are

$$|\tilde{\Phi}\rangle \in (|\Phi\rangle, \{|\Phi_a^r\rangle\}, \{|\Phi_{ab}^{rs}\rangle\}, \{|\Phi_{abc}^{rst}\rangle\}, \dots) \tag{11.1.9}$$

It is worth noting that the HF theory preserves the *molecular configuration symmetry*, under the Bohn–Oppenheimer description. As results, $\langle \tilde{\Phi}_A | \hat{H} | \tilde{\Phi}_B \rangle \neq 0$ only when both $|\tilde{\Phi}_A\rangle$ and $|\tilde{\Phi}_B\rangle$ belong to a same characteristic subspace of the symmetry groups. However, the related topics will not be discussed in this book.

• Any given $\tilde{\Phi}(\mathbf{x}^N) \equiv \tilde{\Phi}[\{\chi_j(\mathbf{x}_j)\}]$ can be a *local reference*. The Hamiltonian integrals higher than double excitations are all zero,

$$0 = \langle \tilde{\Phi} | \hat{H} | \tilde{\Phi}_{abc}^{rst} \rangle = \langle \tilde{\Phi} | \hat{H} | \tilde{\Phi}_{abcd}^{rstu} \rangle = \dots \tag{11.1.10}$$

(a) Turn to Eq. (10.1.19a), which is now extended to

$$\langle \tilde{\Phi} | \hat{H} | \tilde{\Phi}_c^t \rangle = h_{ct} + \sum_{b=1}^N \langle cb || tb \rangle = \langle c | \hat{h} + \hat{v}^{\text{HF}} | t \rangle = 0 \tag{11.1.11a}$$

In other words, the Brillouin theorem, Eq. (10.2.30), goes by

$$0 = \langle \Phi | \hat{H} | \Phi_a^r \rangle = \langle \Phi_a^r | \hat{H} | \Phi_{ab}^{rs} \rangle = \langle \Phi_{ab}^{rs} | \hat{H} | \Phi_{abc}^{rst} \rangle = \dots \tag{11.1.11b}$$

(b) Equation (10.1.17) becomes

$$\langle \tilde{\Phi} | \hat{H} | \tilde{\Phi} \rangle = \sum_{j=1}^N h_{jj} + \frac{1}{2} \sum_{j,b=1}^N \langle jb || jb \rangle = \sum_{j=1}^N \epsilon_j - \frac{1}{2} \sum_{j,b=1}^N \langle jb || jb \rangle \tag{11.1.12}$$

(c) Similarly, the double excitations, Eq. (10.1.19b), are

$$\langle \tilde{\Phi} | \hat{H} | \tilde{\Phi}_{ab}^{rs} \rangle = \langle \tilde{\Phi} | \hat{V}_{ee} | \tilde{\Phi}_{ab}^{rs} \rangle = \langle ab || rs \rangle \tag{11.1.13}$$

Moreover, double excitations include further

$$\langle \Phi_a^r | \hat{V}_{ee} | \Phi_b^s \rangle = \langle rb || as \rangle \tag{11.1.14a}$$

$$\langle \Phi_a^r | \hat{V}_{ee} | \Phi_{bc}^{rs} \rangle = \langle rb || st \rangle \tag{11.1.14b}$$

$$\langle \Phi_a^r | \hat{V}_{ee} | \Phi_{bc}^{rs} \rangle = \langle bc || sa \rangle \tag{11.1.14c}$$

and

$$\langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{ac}^{rt} \rangle = \langle cs || tb \rangle \quad (11.1.15a)$$

$$\langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{cd}^{rs} \rangle = \langle cd || ab \rangle \quad (11.1.15b)$$

$$\langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{ab}^{tu} \rangle = \langle rs || tu \rangle \quad (11.1.15c)$$

Remarks

- To proceed, we would like to elaborate the *local reference* states, exemplified with $|\tilde{\Phi}\rangle \equiv |\Phi_a^r\rangle$ used in Eq. (11.1.14). In this reference state, the specified spin orbital $|\chi_a\rangle$ becomes unoccupied, with the hole left being filled by $|\chi_r\rangle$. In other words, $\hat{a}_a^\dagger \hat{a}_a |\Phi_a^r\rangle = 0$, but $\hat{a}_r^\dagger \hat{a}_r |\Phi_a^r\rangle = |\Phi_a^r\rangle$. Moreover, we have

$$|\Phi_b^s\rangle = \hat{a}_s^\dagger \hat{a}_b |\Phi\rangle = \hat{a}_a^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_r |\Phi_a^r\rangle = |(\Phi_a^r)_{rb}^{as}\rangle \quad (11.1.16)$$

where Eq. (11.1.7) is used, but in relation to the reference $|\Phi_a^r\rangle$. Similarly,

$$\begin{aligned} |\Phi_{ab}^{st}\rangle &= \hat{a}_s^\dagger \hat{a}_t^\dagger \hat{a}_b \hat{a}_r |\Phi_a^r\rangle = |(\Phi_a^r)_{rb}^{st}\rangle \\ |\Phi_{bc}^{rs}\rangle &= \hat{a}_s^\dagger \hat{a}_a^\dagger \hat{a}_c \hat{a}_b |\Phi_a^r\rangle = |(\Phi_a^r)_{bc}^{sa}\rangle \end{aligned} \quad (11.1.17)$$

Let us summary the above results, together with Eq. (11.1.14), as

$$\begin{array}{|l} |\Phi_b^s\rangle = |(\Phi_a^r)_{rb}^{as}\rangle \\ |\Phi_{ab}^{st}\rangle = |(\Phi_a^r)_{rb}^{st}\rangle \\ |\Phi_{bc}^{rs}\rangle = |(\Phi_a^r)_{bc}^{sa}\rangle \end{array} \Rightarrow \begin{array}{|l} \langle \Phi_a^r | \hat{V}_{ee} | \Phi_b^s \rangle = \langle rb || as \rangle \\ \langle \Phi_a^r | \hat{V}_{ee} | \Phi_{ab}^{st} \rangle = \langle rb || st \rangle \\ \langle \Phi_a^r | \hat{V}_{ee} | \Phi_{bc}^{rs} \rangle = \langle bc || sa \rangle \end{array} \quad (11.1.18)$$

- For Eq. (11.1.15), the reference state is $|\tilde{\Phi}\rangle \equiv |\Phi_{ab}^{rs}\rangle$, with

$$\begin{array}{|l} |\Phi_{ac}^{rt}\rangle = |(\Phi_{ab}^{rs})_{cs}^{tb}\rangle \\ |\Phi_{cd}^{rs}\rangle = |(\Phi_{ab}^{rs})_{cd}^{ab}\rangle \\ |\Phi_{ab}^{tu}\rangle = |(\Phi_{ab}^{rs})_{rs}^{tu}\rangle \end{array} \Rightarrow \begin{array}{|l} \langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{ac}^{rt} \rangle = \langle cs || tb \rangle \\ \langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{cd}^{rs} \rangle = \langle cd || ab \rangle \\ \langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{ab}^{tu} \rangle = \langle rs || tu \rangle \end{array} \quad (11.1.19)$$

- The above technique is rather general for identifying the matrix elements involving double excitations, where $\hat{V}_{ee}$ can be replaced by $\hat{H}$, such as Eq. (11.1.13), when all indexes are distinct. This is the cast in the restricted-index convention, with $a < b < c < d$ and $r < s < t < u$; cf. Eq. (11.1.20) versus Eq. (11.1.21) soon below. See also §11.1.2B for further details.

11.1.2 Configuration interaction theory

A. Basic observations

- By using the basis determinants of Eq. (11.1.9) that are orthogonal, the N -electron wave functions read the configuration interaction (CI) form,

$$|\Psi\rangle = |\Phi\rangle + \sum_{a,r} g_a^r |\Phi_a^r\rangle + \sum_{\substack{a<b \\ r<s}} g_{ab}^{rs} |\Phi_{ab}^{rs}\rangle + \sum_{\substack{a<b<c \\ r<s<t}} g_{abc}^{rst} |\Phi_{abc}^{rst}\rangle + \cdots \quad (11.1.20)$$

This can be recast in terms of *unrestricted summations* as

$$|\Psi\rangle = |\Phi\rangle + \sum_{a,r} g_a^r |\Phi_a^r\rangle + \frac{1}{(2!)^2} \sum_{ab,rs} g_{ab}^{rs} |\Phi_{ab}^{rs}\rangle + \frac{1}{(3!)^2} \sum_{\substack{abc \\ rst}} g_{abc}^{rst} |\Phi_{abc}^{rst}\rangle + \cdots \quad (11.1.21)$$

for its in line with the fermionic permutation relations,

$$\boxed{|\Phi_{ab}^{rs}\rangle = -|\Phi_{ab}^{sr}\rangle = |\Phi_{ba}^{sr}\rangle = -|\Phi_{ba}^{rs}\rangle} \Rightarrow \boxed{g_{ab}^{rs} = -g_{ab}^{sr} = g_{ba}^{sr} = -g_{ba}^{rs}}$$

and so on, implying also such that $g_{ab}^{rr} = g_{aa}^{rs} = 0$. The intermediate normalization is adopted; i.e., $\langle\Phi|\Psi\rangle \equiv 1$, so that

$$\langle\Phi_a^r|\Psi\rangle = g_a^r, \quad \langle\Phi_{ab}^{rs}|\Psi\rangle = g_{ab}^{rs}, \quad \langle\Phi_{abc}^{rst}|\Psi\rangle = g_{abc}^{rst}, \quad \cdots \quad (11.1.22)$$

These go by Eq. (11.1.20), rather than Eq. (11.1.21).

- The CI theory is *formally exact*, satisfying

$$\hat{H}|\Psi\rangle = E|\Psi\rangle \quad (11.1.23)$$

which is actually the Hamiltonian matrix eigenequation; see Eqs. (11.1.5)–(11.1.14) for the matrix elements, including also Eq. (11.1.2) for the HF-SCF energy, $\mathcal{E} = \langle\Phi|\hat{H}|\Phi\rangle$. The individual eigenvector $|\Psi\rangle$ comprises the CI coefficients in Eq. (11.1.20).

(a) To obtain the correlation energy, $E_{\text{corr}} \equiv E - \mathcal{E}$, in terms of the CI coefficients, we recast Eq. (11.1.23) as

$$(\hat{H} - \mathcal{E})|\Psi\rangle = E_{\text{corr}}|\Psi\rangle \quad (11.1.24)$$

(b) Apply $\langle\Phi|\Psi\rangle \equiv 1$ resulting in

$$\boxed{E_{\text{corr}} = \langle\Phi|\hat{H} - \mathcal{E}|\Psi\rangle = \frac{1}{4} \sum_{ab,rs} g_{ab}^{rs} \langle ab||rs\rangle} \quad (11.1.25)$$

The second identity is obtained by using Eq. (11.1.21).

(c) On the other hand, the CI coefficients via Eq. (11.1.22) satisfy

$$\begin{aligned}\langle \Phi_a^r | \hat{H} - \mathcal{E} | \Psi \rangle &= g_a^r E_{\text{corr}} \\ \langle \Phi_{ab}^{rs} | \hat{H} - \mathcal{E} | \Psi \rangle &= g_{ab}^{rs} E_{\text{corr}}\end{aligned}\quad (11.1.26)$$

and so on. These together with Eq. (11.1.25) constitute a hierarchical set of equations for the CI coefficients.

- As inferred from Eq. (11.1.25) for E_{corr} , we would have

$$|D\rangle \gg |Q\rangle > |S\rangle > |T\rangle \quad (11.1.27)$$

for electronic ground states. The double $|D\rangle \equiv \{|\Phi_{ab}^{rs}\rangle\}$ excitation manifold are the dominant ($> 90\%$) contributions, followed by the quadruple $|Q\rangle \equiv \{|\Phi_{abcd}^{rstu}\rangle\}$, the single $|S\rangle \equiv \{|\Phi_a^r\rangle\}$ and triple $|T\rangle \equiv \{|\Phi_{abc}^{rst}\rangle\}$ excitation manifolds. Furthermore, within the $|Q\rangle$ -manifold, those connected (or linked) double-double pairs are the dominant contributions. Similarly, within the $|T\rangle$ -manifold, those linked single-double pairs are dominant. The above observations are closely related to the coupled cluster formulations; see §11.2.2.

B. Doubly excited configuration interaction formalism

- The lowest truncated CI approximation to Eq. (11.1.20) is the doubly excited CI (DCI); i.e.,

$$|\Psi_{\text{DCI}}\rangle = |\Phi\rangle + \sum_{a<b} \sum_{r<s} g_{ab}^{rs} |\Phi_{ab}^{rs}\rangle \equiv |0\rangle + \mathbf{g}_D |D\rangle \doteq \begin{bmatrix} 1 \\ \mathbf{g}_D \end{bmatrix} \quad (11.1.28)$$

with $\mathbf{g}_D \equiv \{g_{ab}^{rs}\}$ being a column vector. The corresponding Schrödinger equation (11.1.24), with the reference energy $\mathcal{E}$ -shifted Hamiltonian, reads

$$\begin{bmatrix} 0 & \mathbf{B}^\dagger \\ \mathbf{B} & \mathbf{D} \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{g}_D \end{bmatrix} = E_{\text{corr}}^{\text{DCI}} \begin{bmatrix} 1 \\ \mathbf{g}_D \end{bmatrix} \quad (11.1.29)$$

Here, $\mathbf{B} \equiv \{B_{ab}^{rs}\}$ is a column vector, with the elements of

$$B_{ab}^{rs} \equiv \langle \Phi | \hat{H} - \mathcal{E} | \Phi_{ab}^{rs} \rangle = \langle \Phi | \hat{V}_{ee} | \Phi_{ab}^{rs} \rangle = \langle ab || rs \rangle \quad (11.1.30)$$

and $\mathbf{D} \equiv \{D_{ab,cd}^{rs,tu}\}$ is a Hermitian matrix, with

$$D_{ab,cd}^{rs,tu} \equiv \langle \Phi_{ab}^{rs} | \hat{H} - \mathcal{E} | \Phi_{cd}^{tu} \rangle \quad (11.1.31)$$

- The DCI coefficients via Eqs. (11.1.29)–(11.1.31) go by

$$E_{\text{corr}}^{\text{DCI}} g_{ab}^{rs} = \langle ab || rs \rangle + \sum_{c<d} \sum_{t<u} \langle \Phi_{ab}^{rs} | \hat{H} - \mathcal{E} | \Phi_{cd}^{tu} \rangle g_{cd}^{tu} \quad (11.1.32)$$

As known,

$$\langle \Phi_{ab}^{rs} | \hat{H} - \mathcal{E} | \Phi_{ab}^{rs} \rangle = -\epsilon_{ab}^{rs} + \langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{ab}^{rs} \rangle \quad (11.1.33)$$

via Eq. (11.1.12) versus Eq. (11.1.2), with

$$\epsilon_{ab}^{rs} \equiv \epsilon_a + \epsilon_b - \epsilon_r - \epsilon_s \quad (11.1.34)$$

We can then evaluate Eq. (11.1.32) via

$$\begin{aligned} (E_{\text{corr}}^{\text{DCI}} + \epsilon_{ab}^{rs})g_{ab}^{rs} &= \langle ab||rs \rangle + \sum_{c<d} \sum_{t<u} \langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{cd}^{tu} \rangle g_{cd}^{tu} \\ &= \langle ab||rs \rangle + \sum_{c,t} \langle cs||tb \rangle g_{ac}^{rt} \\ &\quad + \sum_{c<d} \langle cd||ab \rangle g_{cd}^{rs} + \sum_{t<u} \langle rs||tu \rangle g_{ab}^{tu} \end{aligned} \quad (11.1.35)$$

where Eq. (11.1.19) is used. Together with [cf. Eq. (11.1.25)]

$$E_{\text{corr}}^{\text{DCI}} = \frac{1}{4} \sum_{ab,rs} g_{ab}^{rs} \langle ab||rs \rangle \quad (11.1.36)$$

we complete the DCI equations.

Exercises

E11.1.1 Show that

$$E_{\text{corr}}^{\text{DCI}} = \mathbf{B}^\dagger \mathbf{g}_D = \mathbf{B}^\dagger \frac{1}{E_{\text{corr}}^{\text{DCI}} - \mathbf{D}} \mathbf{B} \quad (11.1.37)$$

E11.1.2 Consider the *noninteracting* H₂ aggregate systems, comprising M number of H₂ molecules. Each H₂ is described by the minimal basis model; see Eqs. (10.4.37)–(10.4.39) for the monomer parameters,

$$\Delta \equiv \frac{1}{2} (\langle \Phi_{11}^{22} | \hat{H} | \Phi_{11}^{22} \rangle - \langle \Phi | \hat{H} | \Phi \rangle) \quad \text{and} \quad \mathcal{K} \equiv \langle \Phi | \hat{H} | \Phi_{11}^{22} \rangle \quad (11.1.38)$$

Together with Eq. (10.4.40) for the monomer correlation energy, the full CI result for the noninteracting aggregate is given by

$$E_{\text{corr}}^{\text{FCI}} = M [\Delta - (\Delta^2 + |\mathcal{K}|^2)^{\frac{1}{2}}] \quad (11.1.39)$$

In contrast, the DCI counterpart via Eq. (11.1.37) is rather

$$E_{\text{corr}}^{\text{DCI}} = \frac{M|\mathcal{K}|^2}{E_{\text{corr}}^{\text{DCI}} - 2\Delta} = \Delta - (\Delta^2 + M|\mathcal{K}|^2)^{\frac{1}{2}} \quad (11.1.40)$$

Derive and discuss the above results.

E11.1.3 (a) Verify Eq. (11.1.11) and (11.1.12); (b) Show that

$$\begin{aligned} g_a^r E_{\text{corr}} &= \langle \Phi_a^r | \hat{H} - \mathcal{E} | \Psi \rangle \\ &= \sum_{b,s} g_b^s \langle \Phi_a^r | \hat{H} - \mathcal{E} | \Phi_b^s \rangle + \frac{1}{2^2} \sum_{bc,st} g_{bc}^{st} \langle \Phi_a^r | \hat{H} - \mathcal{E} | \Phi_{bc}^{st} \rangle \\ &\quad + \frac{1}{2^2} \sum_{bc,st} g_{abc}^{rst} \langle \Phi_a^r | \hat{H} - \mathcal{E} | \Phi_{abc}^{rst} \rangle \end{aligned} \quad (11.1.41)$$

(c) Evaluate each individual matrix element in the canonical HF-orbital representation. [Hint: All Hamiltonian matrix elements in the canonical HF spin-orbital representation had been specified by Eqs. (11.1.10)–(11.1.15), together with Eq. (11.1.33)].

C. Size extensivity requirement

- As an extensive property, the energy of a total composite system, when it dissociates into uncorrelated segments, must be additive, such that

$$E(AB) \xrightarrow{\text{Dissociation}} E(A) + E(B) \quad (11.1.42)$$

for a dimer complex AB. Any exact theory satisfies the size extensivity (or size consistency) requirement.

- Exact theories implemented with *either finite basis or perturbation expansion series* remain the size consistency.

(a) A *full CI* refers to the implementation of the CI formalism in §11.1.2, with a given *finite basis*. While it preserves the size extensivity, full CI also a *variational principle-based* method, involving a total $\binom{N}{N}$ configurations, where N is the size of basis spin orbitals and N number of electrons. General speaking, CI methods are often too expansive and limited for small systems.

(b) The RSPT presented in §7.1.2 preserves the size extensivity, at each individual order of perturbation, which can be verified by simple inspection. As the many-body implementation of RSPT, the Moller-Plesset theory, presented in §11.1.3, is a size-consistency method. While the zeroth-order wave functions take the Hartree product form, the Hartree-Fock theory is just the first-order RSPT formalism; see Eq. (11.1.51) later. This concludes that the reference $|\Phi\rangle$ and $\mathcal{E}$ are size extensive.

- *Any truncated CI series methods violate the size extensivity requirement.* This had just been illustrated with the DCI approximation, with

$$|\Psi_{\text{DCI}}(AB)\rangle \neq |\Psi_{\text{DCI}}(A)\rangle \otimes |\Psi_{\text{DCI}}(B)\rangle \quad (11.1.43)$$

including the dissociation limit. This prohibits the correct dissociation limit, Eq. (11.1.42). The size extensivity requires each individual monomer be doubly excited, rather than $|\Psi_{\text{DCI}}(AB)\rangle$ whose double excitations occur just in the dimer complex. Moreover, as seen from Eq. (11.1.40), for an M -molecular aggregate system, the DCI approximation gives rise to

$$\lim_{M \rightarrow \infty} E_{\text{corr}}(\text{DCI}) \propto \lim_{M \rightarrow \infty} M^{\frac{1}{2}} \quad (\text{Noninteracting limit}) \quad (11.1.44)$$

The correlation energy per monomer, $E_{\text{corr}}(\text{DCI})/M \rightarrow 0$, when $M \rightarrow \infty$. In other words, the DCI approximation becomes *completely useless* for large molecular aggregate or solids systems; see Eq. (11.1.40) and remarks therein for further elaborations.

11.1.3 Many-body perturbation theory

A. The Moller–Plesset setup

- In the following, we apply the RSPT, Rayleigh–Schrödinger perturbation theory detailed in §7.1, to many-electron systems, with the target at

$$|0\rangle \equiv |\Phi\rangle = |\chi_1 \cdots \chi_a \cdots \chi_b \cdots \chi_N\rangle \quad (11.1.45)$$

The involving N occupied spin orbitals are self-consistently determined together with the HF potential that satisfies [Eq. (10.2.19)]

$$\langle \chi_j | \hat{v}^{\text{HF}} | \chi_k \rangle = \sum_{b=1}^N \langle jb || kb \rangle \quad (11.1.46)$$

where [Eq. (10.1.13)]

$$\langle ij || kl \rangle \equiv \left\langle \chi_i(\mathbf{x}_1) \chi_j(\mathbf{x}_2) \left| \frac{1}{r_{12}} \right| \chi_k(\mathbf{x}_1) \chi_l(\mathbf{x}_2) \right\rangle \quad (11.1.47)$$

The individual spin orbital satisfies Eq. (10.2.24), with $j = a, b, \dots$, for those relaxed orbitals, and $j = r, s, t, \dots$, for the unrelaxed virtual orbitals.

- Having the above background in hand, we adopt the Moller–Plesset setup,

$$\hat{H}_0 \equiv \sum_{i=1}^N [\hat{h}(i) + \hat{v}^{\text{HF}}(i)] \quad (11.1.48)$$

which corresponds to the Hartree product wave functions. The unperturbed energy is then

$$E_0^{(0)} = \sum_{a=1}^N \epsilon_a = \sum_{a=1}^N (h_{aa} + v_{aa}^{\text{HF}}) = \sum_{a=1}^N h_{aa} + \sum_{a,b=1}^N \langle ab || ab \rangle \quad (11.1.49)$$

The perturbation is concerned with

$$\hat{V} \equiv \hat{H} - \hat{H}_0 = \hat{V}_{ee} - \sum_{i=1}^N \hat{v}^{\text{HF}}(i) \quad (11.1.50)$$

The first-order perturbation energy is then [cf. Eqs. (11.1.45)–(11.1.47)]

$$E_0^{(1)} = \langle \Phi | \hat{V} | \Phi \rangle = -\frac{1}{2} \sum_{a,b=1}^N \langle ab || ab \rangle \quad (11.1.51a)$$

It together with Eq. (11.1.49) results in Eq. (10.1.27); i.e.,

$$\mathcal{E} = \langle \Phi | \hat{H} | \Phi \rangle = E_0^{(0)} + E_0^{(1)} = \sum_{a=1}^N h_{aa} + \frac{1}{2} \sum_{a,b=1}^N \langle ab || ab \rangle \quad (11.1.51b)$$

The correlation energy in the perturbation series reads

$$E_{\text{corr}} \equiv E_0 - \mathcal{E} = E_0^{(2)} + E_0^{(3)} + \dots \quad (11.1.52)$$

B. The use of the standard resolvent technique

- The key step is to identify the RSPT resolvent operator, Eq. (7.1.27),

$$\hat{R} = \sum_{n>0} \frac{|m\rangle\langle m|}{E_0^{(0)} - E_n^{(0)}} \quad (11.1.53)$$

based on the solutions of the unperturbed system,

$$\hat{H}_0 |n\rangle = E_n^{(0)} |n\rangle; \quad n = 0, 1, 2, \dots \quad (11.1.54)$$

For many-electron systems, we have

$$|0\rangle \equiv |\Phi\rangle, \quad |1\rangle \in \{|\Phi_a^r\rangle\}, \quad |2\rangle \in \{|\Phi_{ab}^{rs}\rangle\}, \quad |3\rangle \in \{|\Phi_{abc}^{rst}\rangle\}, \quad \dots \quad (11.1.55)$$

satisfying

$$\langle 0 | \hat{V} | 1 \rangle = 0 \quad \text{and} \quad \langle 0 | \hat{V} | n \rangle = 0, \quad \text{if } n > 2 \quad (11.1.56)$$

due to the Brillouin theorem, Eq. (11.1.11), and the rules underlying Eq. (8.4.19), respectively. On the other hand, Eq. (11.1.53) reads

$$\hat{R} = \sum_{ar} \frac{|\Phi_a^r\rangle\langle\Phi_a^r|}{\epsilon_a - \epsilon_r} + \frac{1}{4} \sum_{abrs} \frac{|\Phi_{ab}^{rs}\rangle\langle\Phi_{ab}^{rs}|}{\epsilon_a + \epsilon_b - \epsilon_r - \epsilon_s} + \dots \quad (11.1.57)$$

This gives rise to $|\Phi_0^{(k)}\rangle$ and further $E_0^{(k+1)} = \langle \Phi | \hat{V} | \Phi_0^{(k)} \rangle$, as Eq. (7.1.28) and Eq. (7.1.6), respectively. The many-body perturbation theory, also called the Moller–Plesset (MP) theory, is now formally completed. [The lowest-order is MP2, Eq. (11.1.59), which contributes 80% $\sim$ 90% of E_{corr} , whereas MP3, Eq. (11.1.62), contributes about 5%. Nevertheless, RSPT is not a variation principle method. Higher-order perturbations may not be better].

- Consider the first-order wave function [cf. Eq. (7.1.29a)],

$$|\Psi_0^{(1)}\rangle = \hat{R} \hat{V} |\Phi\rangle \quad (11.1.58)$$

which is related to [cf. Eq. (7.1.23a)]

$$E_0^{(2)} = \langle \Phi | \hat{V} | \Psi_0^{(1)} \rangle \quad \text{and} \quad E_0^{(3)} = \langle \Psi_0^{(1)} | \hat{V} - E_0^{(1)} | \Psi_0^{(1)} \rangle \quad (11.1.59)$$

Introduce for bookkeeping below

$$X_{ij}^{kl} \equiv \frac{\langle ij||kl \rangle}{\epsilon_i + \epsilon_j - \epsilon_k - \epsilon_l} \quad (11.1.60)$$

satisfying $X_{ij}^{kl} = X_{ji}^{lk} = -X_{ij}^{lk} = -X_{ij}^{kl}$. We obtain

$$E_0^{(2)} = \frac{1}{4} \sum_{abrs} X_{ab}^{rs} \langle rs||ab \rangle = \frac{1}{4} \sum_{abrs} \frac{|\langle ab||ra \rangle|^2}{\epsilon_a + \epsilon_b - \epsilon_r - \epsilon_s} \quad (11.1.61)$$

$$E_0^{(3)} = -\frac{1}{8} \sum_{ab,rs} X_{ab}^{rs} \left[\sum_{cd} \langle cd||ab \rangle X_{rs}^{cd} + \sum_{tu} \langle rs||tu \rangle X_{tu}^{ab} + 8 \sum_{ct} \langle cs||tb \rangle X_{rt}^{ac} \right] \quad (11.1.62)$$

The derivations are as follows.

Derivation of $E_0^{(2)}$, Eq. (11.1.61):

Let us start with $E_0^{(2)} = \langle 0|\hat{V}\hat{R}\hat{V}|0 \rangle$ [cf. Eq. (11.1.59) with Eq. (11.1.58) and $|0 \rangle \equiv |\Phi \rangle$]. For $\hat{R}$, Eq. (11.1.57), satisfying Eq. (11.1.56), the relevant block goes only by

$$|2 \rangle \langle 2|\hat{R}|2 \rangle \langle 2| = \frac{1}{2 \cdot 2!} \sum_{abrs} \frac{|\Phi_{ab}^{rs} \rangle \langle \Phi_{ab}^{rs}|}{\epsilon_a + \epsilon_b - \epsilon_r + \epsilon_s} \quad (11.1.63)$$

Together with $\langle 0|\hat{V}|\Phi_{ab}^{rs} \rangle \equiv \langle \Phi|\hat{V}|\Phi_{ab}^{rs} \rangle = \langle ab||rs \rangle$, Eq. (11.1.56), we obtain

$$E_0^{(2)} = \langle 0|\hat{V}|2 \rangle \langle 2|\hat{R}|2 \rangle \langle 2|\hat{V}|0 \rangle = \frac{1}{4} \sum_{abrs} \frac{\langle ab||rs \rangle \langle rs||ab \rangle}{\epsilon_a + \epsilon_b - \epsilon_r - \epsilon_s} \quad (11.1.64)$$

This is Eq. (11.1.61). (Q.E.D.)

Derivation of $E_0^{(3)}$, Eq. (11.1.62):

(a) From Eq. (11.1.59) with Eq. (11.1.58), we have

$$E_0^{(3)} = \langle 0|\hat{V}\hat{R}\hat{W}\hat{R}\hat{V}|0 \rangle = \langle \Phi|\hat{V}|2 \rangle \langle 2|\hat{R}|2 \rangle \langle 2|\hat{W}|2' \rangle \langle 2'|\hat{R}|2' \rangle \langle 2'|\hat{V}|\Phi \rangle \quad (11.1.65)$$

where [cf. Eqs. (11.1.50) and (11.1.51a)]

$$\hat{W} \equiv \hat{V} - E_0^{(1)} = \hat{V}_{ee} - \sum_{i=1}^N \hat{v}^{\text{HF}}(i) + \frac{1}{2} \sum_{a,b=1}^N \langle ab||ab \rangle \quad (11.1.66)$$

As known,

$$\langle \Phi|\hat{V}|\Phi_{ab}^{rs} \rangle = \langle ab||rs \rangle \quad \text{and} \quad \langle \Phi_{cd}^{tu}|\hat{V}|\Phi \rangle = \langle cd||tu \rangle^* = \langle tu||cd \rangle \quad (11.1.67)$$

(b) Together with X_{ij}^{kl} , Eq. (11.1.60), we obtain Eq. (11.1.65) the expression,

$$\begin{aligned} E_0^{(3)} &= \frac{1}{4 \cdot 2!} \sum_{\substack{abcd \\ rstu}} \frac{\langle \Phi | \hat{V} | \Phi_{ab}^{rs} \rangle \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{cd}^{tu} \rangle \langle \Phi_{cd}^{tu} | \hat{V} | \Phi \rangle}{(\epsilon_a + \epsilon_b - \epsilon_r - \epsilon_s)(\epsilon_c + \epsilon_d - \epsilon_t - \epsilon_u)} \\ &= -\frac{1}{8} \sum_{ab,rs} X_{ab}^{rs} \left[\sum_{cd,tu} \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{cd}^{tu} \rangle X_{tu}^{cd} \right] \end{aligned} \quad (11.1.68)$$

The nonzero elements inside the brackets are only those via Eq. (11.1.15); i.e.,

$$\sum_{cd,tu} \delta_{tr} \delta_{us} \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{cd}^{tu} \rangle X_{tu}^{cd} = \sum_{cd} \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{cd}^{rs} \rangle X_{rs}^{cd} = \sum_{cd} \langle cd || ab \rangle X_{rs}^{cd} \quad (11.1.69a)$$

$$\sum_{cd,tu} \delta_{ca} \delta_{db} \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{cd}^{tu} \rangle X_{tu}^{cd} = \sum_{tu} \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{ab}^{tu} \rangle X_{tu}^{ab} = \sum_{tu} \langle rs || tu \rangle X_{tu}^{ab} \quad (11.1.69b)$$

$$\sum_{cd,tu} \delta_{da} \delta_{ur} \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{cd}^{tu} \rangle X_{tu}^{cd} = \sum_{c,t} \langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{ac}^{rt} \rangle X_{rt}^{ac} = \sum_{c,t} \langle cs || tb \rangle X_{rt}^{ac} \quad (11.1.69c)$$

with noticing that

$$\langle \Phi_{ab}^{rs} | \hat{W} | \Phi_{cd}^{tu} \rangle = \langle \Phi_{ab}^{rs} | \hat{V}_{ee} | \Phi_{cd}^{tu} \rangle \quad (11.1.70)$$

Only the two-electron components of $\hat{W}$, Eq. (11.1.66), have contributions. Exploited in Eq. (11.1.69c) are also $|\Phi_{ca}^{tr}\rangle = |\Phi_{ac}^{rt}\rangle$ and $X_{tr}^{ca} = X_{rt}^{ac}$ [cf. Eq. (11.1.60)].

(c) **Weighting ratios of 1:1:8 to the three expressions of Eq. (11.1.69):** Each describes a *pair-mediated double excitation* that involves also a specified intermediate transition. This gives rise to a distinct weighting factor, associated with the individual degeneracy of diagrammatic path, as follows.

- (i) $\langle \Phi_{ab}^{rs} | \hat{H} | \Phi_{cd}^{rs} \rangle = \langle cd || ab \rangle$ goes by $(c, d) \leftarrow (r, s) \leftarrow (a, b)$, having $g = 1^3$, the degeneracy of the diagrammatic path. Each pair comprise *one type* of orbitals, either the occupied or the virtual pair.
- (ii) $\langle \Phi_{ab}^{rs} | \hat{H} | \Phi_{ab}^{tu} \rangle = \langle rs || tu \rangle$ goes by $(r, s) \leftarrow (a, b) \leftarrow (t, u)$, having $g = 1^3$.
- (iii) $\langle \Phi_{ab}^{rs} | \hat{H} | \Phi_{ac}^{rt} \rangle = \langle cs || tb \rangle$ goes by $(c, s) \leftarrow (r, a) \leftarrow (t, b)$, having $g = 2^3$, since the involving three pairs all comprise *two types* of orbitals.
- (d) Substituting Eq. (11.1.69) into Eq. (11.1.68), following by using the aforementioned weighting ratio of 1:1:8, we obtain

$$E_0^{(3)} = -\frac{1}{8} \sum_{ab,rs} X_{ab}^{rs} \left[\sum_{cd} \langle cd || ab \rangle X_{rs}^{cd} + \sum_{tu} \langle rs || tu \rangle X_{tu}^{ab} + 8 \sum_{ct} \langle cs || tb \rangle X_{rt}^{ac} \right]$$

This is Eq. (11.1.62).

(Q.E.D.)

Exercises

E11.1.4 Write down the first-order perturbative wave function, $\Psi_0^{(1)}$.

E11.1.5 Revisit the minimal basis H_2 , with the spin-restricted description; §10.4. Compare the exact results with respect to $\Psi_0^{(1)}$ and also the first three perturbation energies.

E11.1.6 It is worth noting that the aforementioned weighting ratios of 1:1:8, associated with the three expressions of Eq. (11.1.69) is often given via the diagrammatic representation; see for example, Attila Szabo and Neil S. Ostlund, *Modern Quantum Chemistry*, New York: Dover Publications, 1996; Chapter 6. Self-study the diagrams for Eq. (11.1.61) and Eq. (11.1.62).

11.2 Coupled Cluster Wave Function Theory

11.2.1 Coupled cluster formalism

A. Basic relations

• First of all, the coupled cluster (CC) is a size extensive theory. The original CC theory was proposed by Fritz Coester and Hermann Kümmel in the field of nuclear physics (Coester, 1958; Coester and Kümmel, 1960). Applications to atomic and molecular systems have been widely carried out since then. In principle, the full CC theory and the full CI method are equivalent; but in applications they are different, since practical truncations are unavoidable. The size extensivity of CC formalism makes it the choice of wave function theory for strongly correlated systems.

- For a close comparison, we recast the CI wave function, Eq. (11.1.20), as

$$|\Psi_{\text{CI}}\rangle \equiv (1 + \hat{\mathcal{T}}_1 + \hat{\mathcal{T}}_2 + \cdots + \hat{\mathcal{T}}_N)|\Phi\rangle \quad (11.2.1)$$

While $|\Phi\rangle$ is the HF reference wave function, $\hat{\mathcal{T}}_n$ denotes the n -tuple excitation operator, such as

$$\hat{\mathcal{T}}_1 \equiv \sum_{ar} g_a^r \hat{a}_r^\dagger \hat{a}_a \quad \text{and} \quad \hat{\mathcal{T}}_2 \equiv \frac{1}{(2!)^2} \sum_{abrs} g_{ab}^{rs} \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a \quad (11.2.2)$$

- The CC equivalence to Eq. (11.2.1) exploits the Magnus expansion,

$$|\Psi_{\text{CC}}\rangle = e^{\hat{T}_1 + \hat{T}_2 + \cdots + \hat{T}_N} |\Phi\rangle \equiv e^{\hat{T}} |\Phi\rangle \quad (11.2.3)$$

with noting that $\hat{T}_n \neq \hat{\mathcal{T}}_n$. The CC counterparts to Eq. (11.2.2) read

$$\hat{T}_1 = \sum_{ar} \eta_a^r \hat{a}_r^\dagger \hat{a}_a \quad \text{and} \quad \hat{T}_2 = \frac{1}{(2!)^2} \sum_{abrs} \eta_{ab}^{rs} \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a \quad (11.2.4)$$

Both the CI and CC wave functions adopt the convention of $\langle \Phi | \Psi \rangle \equiv 1$. As also known [cf. Eqs. (11.1.7) and (11.1.8)],

$$\hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a |\Phi\rangle = \hat{a}_s^\dagger \hat{a}_b |\Phi_a^r\rangle = |\Phi_{ab}^{rs}\rangle \quad (11.2.5a)$$

$$\langle \Phi_{ab}^{rs} | \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a = \langle \Phi_a^r | \hat{a}_r^\dagger \hat{a}_a = \langle \Phi | \quad (11.2.5b)$$

(a) In parallel to Eq. (11.1.22), we have

$$\langle \Phi_a^r | \hat{T}_1 | \Phi \rangle = \eta_a^r, \quad \langle \Phi_{ab}^{rs} | \hat{T}_2 | \Phi \rangle = \eta_{ab}^{rs}, \quad \langle \Phi_{abc}^{rst} | \hat{T}_3 | \Phi \rangle = \eta_{abc}^{rst} \quad (11.2.6)$$

and so on. The CC amplitudes are all real and satisfy the fermionic permutation relations, such that

$$\eta_{ab}^{rs} = -\eta_{ab}^{sr} = \eta_{ba}^{sr} = -\eta_{ba}^{rs} \quad (11.2.7)$$

(b) In the CC framework, truncation is concerned with the exponent $\hat{T}$ -series in $|\Psi_{\text{CC}}\rangle \equiv e^{\hat{T}}|\Phi\rangle$, Eq. (11.2.3). The truncated cluster excitation operator remains an exponential form that is size extensive. However, the CC approximation is not a variational method.

B. Canonical CC equations

- Let us start with $\hat{H}|\Psi\rangle = E|\Psi\rangle$, where $|\Psi\rangle = |\Psi_{\text{CC}}\rangle \equiv e^{\hat{T}}|\Phi\rangle$, in terms of

$$e^{-\hat{T}}\hat{H}e^{\hat{T}}|\Phi\rangle = E|\Phi\rangle \quad (11.2.8)$$

This is the canonical CC equation and can be recast as

$$e^{-\hat{T}}(\hat{H} - \mathcal{E})|\Psi\rangle = E_{\text{corr}}|\Phi\rangle \quad (11.2.8')$$

- (a) While $\langle\Phi|\Phi\rangle = 1$ and $\langle\Phi|e^{-\hat{T}} = \langle\Phi|$, any of those $\langle\tilde{\Phi}|\Phi\rangle = 0$ from

$$\tilde{\Phi} \in \{\Phi_a^r, \Phi_{ab}^{rs}, \Phi_{abc}^{rst}, \Phi_{abcd}^{rstu}, \dots\} \quad (11.2.9)$$

and $\langle\tilde{\Phi}|e^{-\hat{T}}$ introduces **de-excitations**, as inferred from Eq. (11.2.5b).

- (b) Equation (11.2.8') satisfies

$$E_{\text{corr}} = \langle\Phi|\hat{H} - \mathcal{E}|\Psi\rangle \quad (11.2.10)$$

$$\langle\tilde{\Phi}|e^{-\hat{T}}(\hat{H} - \mathcal{E})|\Psi\rangle = 0 \quad (11.2.11)$$

The resulting equations, which dictate the energy E and the CC amplitudes $\{\eta_a^r, \eta_{ab}^{rs}, \dots\}$, are *nonlinear* and *hierarchically coupled* with each other.

- As derived in details soon below, we have [cf. Eq. (11.1.25)]

$$E_{\text{corr}} = \frac{1}{4} \sum_{abrs} (\eta_{ab}^{rs} + 2\eta_a^r \eta_b^s) \langle ab || rs \rangle \quad (11.2.12)$$

where $\mathcal{E} = \langle\Phi|\hat{H}|\Phi\rangle$, Eq. (11.1.2). Moreover,

$$\langle\Phi_a^r|\hat{H} - \mathcal{E}|\Psi\rangle = \eta_a^r E_{\text{corr}} \quad (11.2.13a)$$

$$\langle\Phi_{ab}^{rs}|\hat{H} - \mathcal{E}|\Psi\rangle = (\eta_{ab}^{rs} + \eta_a^r \eta_b^s) E_{\text{corr}} \quad (11.2.13b)$$

and so on for all CC required coefficients.

Derivations of Eq. (11.2.12):

- As known, the Hamiltonian integrals involving more than two excitations are zeroes and so for those of single excitations, duo to the Brillouin's theorem, Eq. (11.1.11). Together with Eq. (11.2.4), the above observations give rise to

Eq. (11.2.10) the expression,

$$E_{\text{corr}} = \langle \Phi | \hat{H} \hat{T}_2 | \Phi \rangle + \frac{1}{2} \langle \Phi | \hat{H} \hat{T}_1^2 | \Phi \rangle \quad (11.2.14)$$

By using Eq. (11.1.7) and $\hat{a}_r^\dagger \hat{a}_a \hat{a}_s^\dagger \hat{a}_b = \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a$, we identity

$$\begin{aligned} \hat{T}_2 | \Phi \rangle &= \frac{1}{4} \sum_{abrs} \eta_{ab}^{rs} (\hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a) | \Phi \rangle = \frac{1}{4} \sum_{abrs} \eta_{ab}^{rs} | \Phi_{ab}^{rs} \rangle \\ \hat{T}_1^2 | \Phi \rangle &= \sum_{abrs} \eta_a^r \eta_b^s (\hat{a}_r^\dagger \hat{a}_a) (\hat{a}_s^\dagger \hat{a}_b) | \Phi \rangle = \sum_{abrs} \eta_a^r \eta_b^s | \Phi_{ab}^{rs} \rangle \end{aligned} \quad (11.2.15)$$

Together with $\langle \Phi | \hat{H} | \Phi_{ab}^{rs} \rangle = \langle ab || rs \rangle$, the resulting Eq. (11.2.14) reads

$$E_{\text{corr}} = \frac{1}{4} \sum_{abrs} (\eta_{ab}^{rs} + 2\eta_a^r \eta_b^s) \langle ab || rs \rangle \quad (11.2.16)$$

This is Eq. (11.2.12).

(Q.E.D.)

Derivations of Eqs. (11.2.13):

(a) Equation (11.2.13a) is related to Eq. (11.2.11), with $\tilde{\Phi} = \Phi_a^r$; i.e.,

$$\langle \Phi_a^r | e^{-\hat{T}} (\hat{H} - \mathcal{E}) | \Psi \rangle = 0 \quad (11.2.17)$$

As known [cf. Eqs. (11.2.4) and (11.2.5)],

$$\langle \Phi_a^r | e^{-\hat{T}} = \langle \Phi_a^r | (1 - \hat{T}_1) = \langle \Phi_a^r | - \eta_a^r \langle \Phi | \quad (11.2.18)$$

Substituting it into Eq. (11.2.17), followed by using Eq. (11.2.10), we obtain

$$\langle \Phi_a^r | \hat{H} - \mathcal{E} | \Psi \rangle = \eta_a^r \langle \Phi | \hat{H} - \mathcal{E} | \Psi \rangle = \eta_a^r E_{\text{corr}} \quad (11.2.19)$$

This is Eq. (11.2.13a).

(b) Similarly, Eq. (11.2.13b) is related to Eq. (11.2.11), with $\tilde{\Phi} = \Phi_{ab}^{rs}$; i.e.,

$$\langle \Phi_{ab}^{rs} | e^{-\hat{T}} (\hat{H} - \mathcal{E}) | \Psi \rangle = 0 \quad (11.2.20)$$

On the other hand, Eqs. (11.2.4)–(11.2.6) give rise to

$$\langle \Phi_{ab}^{rs} | (e^{-\hat{T}} - 1) = -\langle \Phi_{ab}^{rs} | \left(\hat{T} - \frac{1}{2} \hat{T}^2 \right) = -\langle \Phi_{ab}^{rs} | \left(\hat{T}_1 - \frac{1}{2} \hat{T}_1^2 + \hat{T}_2 \right) \quad (11.2.21)$$

$$\begin{aligned} \langle \Phi_{ab}^{rs} | \hat{T}_1 &= \eta_a^r \langle \Phi_b^s | + \eta_b^s \langle \Phi_a^r | \\ \langle \Phi_{ab}^{rs} | \hat{T}_1^2 &= 2\eta_a^r \eta_b^s \langle \Phi | \quad \text{and} \quad \langle \Phi_{ab}^{rs} | \hat{T}_2 = \eta_{ab}^{rs} \langle \Phi | \end{aligned} \quad (11.2.22)$$

The resulting Eq. (11.2.21) can be arranged as

$$\langle \Phi_{ab}^{rs} | e^{-\hat{T}} = \langle \Phi_{ab}^{rs} | - (\eta_a^r \langle \Phi_b^s | + \eta_b^s \langle \Phi_a^r |) - (\eta_{ab}^{rs} - \eta_a^r \eta_b^s) \langle \Phi | \quad (11.2.23)$$

Substituting it into Eq. (11.2.20), followed by both Eq. (11.2.19) and Eq. (11.2.10), we obtain

$$\langle \Phi_{ab}^{rs} | \hat{H} - \mathcal{E} | \Psi \rangle = (2\eta_a^r \eta_b^s + \eta_{ab}^{rs} - \eta_a^r \eta_b^s) E_{\text{corr}} = (\eta_{ab}^{rs} + \eta_a^r \eta_b^s) E_{\text{corr}} \quad (11.2.24)$$

This is Eq. (11.2.13b).

(Q.E.D.)

- The above derivations, with the subsequent results of Eq. (11.2.13a) and Eq. (11.2.13b), highlight the hierarchical constructions of the CC coefficients, with the increasing excitations.
- As mentioned earlier, double excitations are the dominant contributions to correlations; see also Eq. (11.1.25). This concludes that the lowest-order CC approximation is the CCD scheme, where $\hat{T} \approx \hat{T}_2$; i.e.,

$$|\Psi_{\text{CCD}}\rangle = e^{\hat{T}_2} |\Phi\rangle \quad (11.2.25)$$

to be detailed in §11.2.2.

- The next is the CCSD scheme, where $\hat{T} \approx \hat{T}_1 + \hat{T}_2$; i.e.,

$$|\Psi_{\text{CCSD}}\rangle = e^{\hat{T}_1 + \hat{T}_2} |\Phi\rangle \quad (11.2.26)$$

with Eq. (11.2.12) and Eq. (11.2.13) being sufficient. The numerical scaling of CCSD is $\mathcal{O}(N^6)$. In comparison, the HF theory is an $\mathcal{O}(N^4)$ method.

- The CCSD(T) includes further the perturbative treatment on triple excitations. On the other hand, within the triple excitations manifold, those linked single-double pairs are dominant, as mentioned earlier. We can then derive the CC expression of $\langle \Phi_{abc}^{rst} | \hat{H} | \Psi \rangle$, followed by the linked single-double approximation on each triple η_{abc}^{rst} -coefficient. This will give rise to an alternative CCSD(T) formalism, with similar accuracy. Numerically, CCSD(T) is an $\mathcal{O}(N^7)$ -method, feasible for moderate size systems and used as one of the computational benchmarks.

11.2.2 Onset of coupled cluster doubles

A. The CCD amplitude equations

- Let us focus on $|\Psi_{\text{CCD}}\rangle \equiv e^{\hat{T}_2} |\Phi\rangle$. This is the dominant contribution to the ground-state properties and satisfies Eqs. (11.2.12) and (11.2.13) with $\eta_a^r = \eta_b^s = 0$; i.e.,

$$E_{\text{corr}}^{\text{CCD}} = \frac{1}{4} \sum_{abrs} \eta_{ab}^{rs} \langle ab || rs \rangle \quad (11.2.27)$$

and

$$\eta_{ab}^{rs} E_{\text{corr}}^{\text{CCD}} = \langle \Phi_{ab}^{rs} | (\hat{H} - \mathcal{E}) e^{\hat{T}_2} | \Phi \rangle \quad (11.2.28)$$

The CCD amplitudes satisfy

$$\begin{aligned} (E_{\text{corr}}^{\text{CCD}} + \epsilon_{ab}^{rs}) \eta_{ab}^{rs} &= \langle ab || rs \rangle + \sum_{c,t} \langle cs || tb \rangle \eta_{ac}^{rt} \\ &+ \frac{1}{2} \sum_{cd} \langle cd || ab \rangle \eta_{cd}^{rs} + \frac{1}{2} \sum_{tu} \langle rs || tu \rangle \eta_{ab}^{tu} \\ &+ \frac{1}{8} \sum_{cdtu} \langle cd || tu \rangle \eta_{ab}^{rs} \eta_{cd}^{tu} \end{aligned} \quad (11.2.29)$$

with $\epsilon_{ab}^{rs} \equiv \epsilon_a + \epsilon_b - \epsilon_r - \epsilon_s$, Eq. (11.1.34). The last term above arises from $\frac{1}{2} \langle \Phi_{ab}^{rs} | (\hat{H} - \mathcal{E}) \hat{T}_2^2 | \Phi \rangle$ that does not exist in the DCI expression (11.1.35).

B. On the aspect of random phase approximation

• Recall the size extensivity analysis of DCI formalism. In this relation, it suggests a non-truncated CI approximation, including all $2n$ -excitations,

$$|\Psi_{\text{CI}}\rangle \approx \left(1 + \sum_{n=1}^{N/2} \hat{\mathcal{T}}_{2n} \right) |\Phi\rangle \approx e^{\hat{\mathcal{T}}_2} |\Phi\rangle \quad (11.2.30)$$

This is a CCD-like wave function, satisfying the size extensivity, with a random phase approximation (RPA) form, as follows.

• Let us focus on $\hat{\mathcal{T}}_{2n}$, the $2n$ -excitation operator in Eq. (11.2.30), related to the CI expression (11.1.21). To proceed, we recast $\hat{\mathcal{T}}_2$ of Eq. (11.2.2) as

$$\hat{\mathcal{T}}_2 = \frac{1}{(2!)^2} \sum_{abrs} \hat{\tau}_{ab}^{rs}, \quad \text{with} \quad \boxed{\hat{\tau}_{ab}^{rs} \equiv g_{ab}^{rs} \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_b \hat{a}_a} \quad (11.2.31)$$

(a) Consider first [cf. Eq. (11.1.8)]

$$\hat{\mathcal{T}}_4 = \frac{1}{(4!)^2} \sum_{\substack{abcd \\ rstu}} \hat{\tau}_{abcd}^{rstu}, \quad \text{with} \quad \hat{\tau}_{abcd}^{rstu} \equiv g_{abcd}^{rstu} \hat{a}_r^\dagger \hat{a}_s^\dagger \hat{a}_t^\dagger \hat{a}_u^\dagger \hat{a}_d \hat{a}_c \hat{a}_b \hat{a}_a \quad (11.2.32)$$

Adopt now the *pair-factorization ansatz*, in which each $\hat{\tau}_{abcd}^{rstu}$ is approximately factorized into $\frac{1}{2} \binom{4}{2}^2 = 18$ number of *coupled-pair* terms,

$$\begin{aligned} \hat{\tau}_{abcd}^{rstu} &\approx (\hat{\tau}_{ab}^{rs} * \hat{\tau}_{cd}^{tu}) \\ &\equiv \hat{\tau}_{ab}^{rs} \hat{\tau}_{cd}^{tu} + \hat{\tau}_{ac}^{rs} \hat{\tau}_{bd}^{tu} + \hat{\tau}_{ad}^{rs} \hat{\tau}_{bc}^{tu} + \hat{\tau}_{ab}^{tu} \hat{\tau}_{cd}^{rs} + \hat{\tau}_{ac}^{tu} \hat{\tau}_{bd}^{rs} \\ &+ \hat{\tau}_{ad}^{tu} \hat{\tau}_{bc}^{rs} + \hat{\tau}_{ab}^{rt} \hat{\tau}_{cd}^{su} + \hat{\tau}_{ac}^{rt} \hat{\tau}_{bd}^{su} + \hat{\tau}_{ad}^{rt} \hat{\tau}_{bc}^{su} + \hat{\tau}_{ab}^{su} \hat{\tau}_{cd}^{rt} \\ &+ \hat{\tau}_{ac}^{su} \hat{\tau}_{bd}^{rt} + \hat{\tau}_{ad}^{su} \hat{\tau}_{bc}^{rt} + \hat{\tau}_{ab}^{st} \hat{\tau}_{cd}^{ru} + \hat{\tau}_{ac}^{st} \hat{\tau}_{bd}^{ru} + \hat{\tau}_{ad}^{st} \hat{\tau}_{bc}^{ru} \\ &+ \hat{\tau}_{ab}^{ru} \hat{\tau}_{cd}^{st} + \hat{\tau}_{ac}^{ru} \hat{\tau}_{bd}^{st} + \hat{\tau}_{ad}^{ru} \hat{\tau}_{bc}^{st} \end{aligned} \quad (11.2.33)$$

in relation to Eq. (11.2.31). The resulting Eq. (11.2.32) reads

$$\hat{\mathcal{T}}_4 \approx \frac{1}{2!} \hat{\mathcal{T}}_2^2 \quad (11.2.34)$$

Formally, this describes the aforementioned observation that within the quadruple $|\mathbf{Q}\rangle$ -manifold, those linked double-double pairs are the dominant contributions.

(b) Turn to

$$\hat{\mathcal{T}}_{2n} = \frac{1}{[(2n)!]^2} \sum_{\substack{ab \cdots cd \\ rs \cdots tu}} \hat{\tau}_{ab \cdots cd}^{rs \cdots tu} \quad (11.2.35)$$

Each $2n$ -tuple operator, $\hat{\tau}_{ab \cdots cd}^{rs \cdots tu}$, is now factorized into a set of n -double products, $\{\hat{\tau}_{ab}^{rs} * \cdots * \hat{\tau}_{cd}^{tu}\}$, as in line with Eq. (11.2.33). The total number of such terms is

$$\frac{1}{n!} \left[\binom{2n}{2} \binom{2n-2}{2} \cdots \binom{4}{2} \binom{2}{2} \right]^2 = \frac{[(2n)!]^2}{n! 4^n} \quad (11.2.36)$$

The resulting Eq. (11.2.35), in relation to $\mathcal{T}_2$ of Eq. (11.2.31), reads

$$\hat{\mathcal{T}}_{2n} \approx \frac{1}{n!} \hat{\mathcal{T}}_2^n \quad (11.2.37)$$

This gives rise to Eq. (11.2.30), with the RPA form of $|\Psi\rangle \approx e^{\hat{\mathcal{T}}_2} |\Phi\rangle$.

- As a CCD-like wave function, Eqs. (11.2.27) and (11.2.29) remain valid also for $\{g_{ab}^{rs}\}$, as if $\hat{\mathcal{T}}_2$ is replaced by $\hat{T}_2$ in the CC formalism. The above analyse highlights the intimated relations between the truncated CC wave functions and the full CI-based factorization ansatz and the resulting random phase approximations.

11.2.3 Some useful remarks

There are static and dynamic correlation contributions. While the latter is taken care of by the virtual spin-orbitals, the static correlations occur in the degenerate or near-degenerate cases. Evidently, either $|\Psi_{\text{CI}}\rangle$ or $|\Psi_{\text{CC}}\rangle$ is a dynamic correlation theory, with a single-configuration SCF reference $|\Phi\rangle \equiv |\Phi_{\text{HF}}\rangle$ and the resulting canonical HF spin-orbital basis descriptions.

- Multi-configuration SCF reference,

$$|\Phi_{\text{MCSCF}}\rangle = \sum_m c_m |\Phi_m\rangle; \quad m > 1 \quad (11.2.38)$$

would be needed for degenerate or near-degenerate systems, where static correlations are important. Both $\{|\Phi_m\rangle\}$ and $\{c_m\}$ are self-consistent determined and simultaneously optimized via the variation principle.

(a) General speaking, each determinantal $|\Phi_m\rangle$ corresponds to a classical *valence-bond structure* in molecules and/or the *monomer state* in molecular aggregates.

(b) As noticed, $|\Phi_{\text{MCSCF}}\rangle$ appears as a multi-reference CI wave function; Eq. (11.2.38) versus Eq. (11.1.20). Evidently, $|\Phi_{\text{MCSCF}}\rangle \neq |\Phi_{\text{HF}}\rangle$ and $|\Phi_{\text{MCSCF}}\rangle \neq |\Psi_{\text{CI}}\rangle$. While $|\Psi_{\text{CI}}\rangle$ involves only dynamic correlations, $|\Phi_{\text{MCSCF}}\rangle$ engages only static correlations. On the other hand, $|\Psi_{\text{MCSCF}}^{\text{CI}}\rangle$, which includes also the virtual orbitals for excitations, is a theory for *both static and dynamic correlations*. Moreover, the truncated $|\Psi_{\text{MCSCF}}^{\text{CI}}\rangle$ is not size extensive; $|\Psi_{\text{MCSCF}}^{\text{CC}}\rangle$ would be the choice of use.

- Occupied orbitals in the MCSCF-subspace are further divided into active and inactive ones. Those *active orbitals* comprise the complete active-space SCF (CASSCF) wave function, which is also *spin adaptive*, with both the orbitals and configuration expansion coefficients being optimized. Evidently, $|\Phi_{\text{CASSCF}}\rangle$ is more feasible than $|\Phi_{\text{MCSCF}}\rangle$.
- To facilitate computing $|\Phi_{\text{CASSCF}}\rangle$, exploited could also be its matrix product states representation and the related density matrix renormalization group (DMRG) methods; See, for example, Haibo Ma, Ulrich Schollwöck, and Zhigang Shuai, Density Matrix Renormalization Group (DMRG)-based Approaches in Computational Chemistry (Elsevier, 2022).

Chapter 12

Reduced Descriptions and Density Functional Theory

This chapter describes the reduced-particle theories for many-electron systems. While the wave function represents a pure state, reduced theories can in principle be exact and go with mixed states. Note also that reduced density operator description is the standard for open quantum systems, in which the Hamiltonian alone is insufficient. Thermodynamic variables such as *chemical potential* and *temperature* would also be needed. It is witnessed the recent advancement in the general theory of open quantum systems. The future development along this line would also cover the electronic structure systems with strong correlations.

The first topic of this chapter is concerned with the basic theories of reduced density operators in electronic structures. As known, the fermionic operators involve only single-particle and two-particle dynamical properties. In §12.1, after the formal definition of reduced p -particle density operators (RpDOs), we present the R1DO and R2DO formalisms. In particular, the expectation value of electron-electron interaction can be partitioned into the classical Coulomb integral and the quantum exchange-correlation (XC) energy. While the former engages only electron density, the latter depends also on electrons-pair correlation.

The second topic is concerned with the density functional theory (DFT). It exploits the electron density $\rho(\mathbf{r})$ as the basic variable in the variational determination of ground electronic-state energy. Discussed in §12.2 will be the Hohenberg-Kohn theorems for the existence of exact DFT, the issues of representability, the basic features and mathematical skills, followed by the historical model-level descriptions. Finally, we will present in §12.3 the Kohn-Sham self-consistent-orbitals equation for the first-principle DFT implementations.

12.1 Reduced Density Operator Descriptions

12.1.1 *Energy aspect of many-electron systems*

A. Basic properties of reduced density operators

- Consider

$$E = \langle \Psi | \hat{H}_e | \Psi \rangle = \text{Tr}(\hat{H}_e \hat{\rho}_N) \quad (12.1.1)$$

with $\hat{\varrho}_N \equiv |\Psi\rangle\langle\Psi|$, for an N -electron molecular system,

$$\hat{H}_e = (\hat{T}_e + \hat{V}_{en}) + \hat{V}_{ee} = \sum_{i=1}^N \left[\frac{1}{2} \hat{\mathbf{p}}_i^2 + v(\mathbf{r}_i) \right] + \frac{1}{2} \sum_{i,j=1}^N \frac{1}{r_{ij}} \quad (12.1.2)$$

where

$$v(\mathbf{r}) = - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \quad (12.1.3)$$

The determinantal wave function,

$$\Psi(\mathbf{x}^N) = \langle \mathbf{x}^N | \Psi \rangle \equiv \langle \mathbf{x}_1 \cdots \mathbf{x}_N | \Psi \rangle; \quad \begin{cases} \mathbf{x}_j \equiv \mathbf{r}_j s_j \\ s_j = \alpha, \beta \end{cases} \quad (12.1.4)$$

is antisymmetric with respect to the particle permutation.

• While $\hat{\varrho}_N \equiv |\Psi\rangle\langle\Psi|$ is a pure state by nature, Eq. (12.1.1) supports also a mixed-state description, with R1DO and R2DO, where RpDO denotes the reduced p -electron density operator,

$$\hat{\varrho}_p \equiv \binom{N}{p} \text{tr}_{\mathbf{x}_{p+1} \cdots \mathbf{x}_N} \hat{\varrho}_N, \quad \text{with } \text{tr} \hat{\varrho}_p = \binom{N}{p} \quad (12.1.5)$$

The trace runs over the complementary spin-spacial subspace; i.e.,

$$\text{tr}_{\mathbf{x}_{p+1} \cdots \mathbf{x}_N} = \int d\mathbf{x}_{p+1} \cdots d\mathbf{x}_N$$

The resulting RpDO spans in the p -electron subspace and reads

$$\varrho_p(\mathbf{x}^p, \mathbf{x}'^p) \equiv \varrho_p(\mathbf{x}_1 \cdots \mathbf{x}_p, \mathbf{x}'_1 \cdots \mathbf{x}'_p) = \langle \mathbf{x}_1 \cdots \mathbf{x}_p | \hat{\varrho}_p | \mathbf{x}'_1 \cdots \mathbf{x}'_p \rangle \quad (12.1.6)$$

with the orthonormality and completeness relations [cf. Eq. (8.1.10)],

$$\langle \mathbf{x}_1 \cdots \mathbf{x}_p | \mathbf{x}'_1 \cdots \mathbf{x}'_p \rangle = \det |\delta(\mathbf{x}_1 - \mathbf{x}'_1) \cdots \delta(\mathbf{x}_p - \mathbf{x}'_p)| \quad (12.1.7a)$$

$$\frac{1}{p!} \int d\mathbf{x}^p |\mathbf{x}_1 \cdots \mathbf{x}_p\rangle \langle \mathbf{x}_1 \cdots \mathbf{x}_p| = \hat{I}_p \quad (12.1.7b)$$

The factor $p!$ arises from the permutations of p -electrons.

• For the nonrelative energy, Eq. (12.1.1), required are only the spin-traced counterparts to $\varrho_1(\mathbf{x}, \mathbf{x}')$ and $\varrho_2(\mathbf{x}_1 \mathbf{x}_2, \mathbf{x}'_1 \mathbf{x}'_2)$, denoted respectively

$$\rho_1(\mathbf{r}, \mathbf{r}') \equiv \int \varrho_1(\mathbf{r} s, \mathbf{r}' s) ds \quad (12.1.8)$$

and

$$\rho_2(\mathbf{r}_1 \mathbf{r}_2, \mathbf{r}'_1 \mathbf{r}'_2) \equiv \iint \varrho_2(\mathbf{r}_1 s_1 \mathbf{r}_2 s_2, \mathbf{r}'_1 s_1 \mathbf{r}'_2 s_2) ds_1 ds_2 \quad (12.1.9)$$

From Eq. (12.1.5), we have

$$\int \rho_2(\mathbf{r}_1 \mathbf{r}_2, \mathbf{r}'_1 \mathbf{r}_2) d\mathbf{r}_2 = \frac{1}{2}(N-1)\rho_1(\mathbf{r}_1, \mathbf{r}'_1) \quad (12.1.10)$$

• **Electron density and electrons-pair density** are related to

$$\rho(\mathbf{r}) \equiv \rho_1(\mathbf{r}, \mathbf{r}) \quad \text{and} \quad \pi(\mathbf{r}_1, \mathbf{r}_2) \equiv \rho_2(\mathbf{r}_1 \mathbf{r}_2, \mathbf{r}_1 \mathbf{r}_2) \quad (12.1.11)$$

respectively, satisfying [cf. Eq. (12.1.5)]

$$\int \rho(\mathbf{r}) d\mathbf{r} = N \quad \text{and} \quad \iint \pi(\mathbf{r}_1, \mathbf{r}_2) d\mathbf{r}_1 d\mathbf{r}_2 = \binom{N}{2} \quad (12.1.12)$$

(a) Electron possesses spins, so that [cf. Eq. (12.1.11)]

$$\rho(\mathbf{r}) = \sum_{\sigma=\alpha,\beta} \rho^\sigma(\mathbf{r}) \quad \text{and} \quad \pi(\mathbf{r}, \mathbf{r}') = \sum_{\sigma,\sigma'} \pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') \quad (12.1.13)$$

Here, σ and σ' are the spins of two electrons, located at $\mathbf{r} \equiv \mathbf{r}_1$ and $\mathbf{r}' \equiv \mathbf{r}_2$, respectively. We have [cf. Eq. (12.1.10)]

$$\sum_{\sigma'} \int \pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') d\mathbf{r}' = \frac{1}{2}(N-1)\rho^\sigma(\mathbf{r}) \quad (12.1.14)$$

The total number of electron is $N = N_\alpha + N_\beta$. Moreover,

$$\int \rho^\sigma(\mathbf{r}) d\mathbf{r} = N_\sigma \quad (12.1.15)$$

(b) Spin-polarized electrons-pair density counterparts are [cf. Eq. (12.1.12)]

$$\iint \pi^{\sigma\sigma}(\mathbf{r}, \mathbf{r}') d\mathbf{r} d\mathbf{r}' = \frac{1}{2}N_\sigma(N_\sigma - 1) \quad (12.1.16a)$$

$$\iint \pi^{\alpha\beta}(\mathbf{r}, \mathbf{r}') d\mathbf{r} d\mathbf{r}' = \frac{1}{2}N_\alpha N_\beta \quad (12.1.16b)$$

Note that

$$\pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') = \pi^{\sigma'\sigma}(\mathbf{r}', \mathbf{r}) \quad (12.1.17)$$

B. Energetics analysis

• Consider the contributions to the energy expectation value, Eq. (12.1.1), from the different terms of Hamiltonian, Eq. (12.1.2),

$$E = \langle \hat{T}_e \rangle + \langle \hat{V}_{en} \rangle + \langle \hat{V}_{ee} \rangle \quad (12.1.18)$$

The R1DO supports the evaluation of the kinetic energy, with

$$\langle \hat{T}_e \rangle = -\frac{1}{2} \int [\nabla_{\mathbf{r}}^2 \rho_1(\mathbf{r}, \mathbf{r}')]_{\mathbf{r}'=\mathbf{r}} d\mathbf{r} \quad (12.1.19)$$

On the other hand, the potential energy, $\langle \hat{V} \rangle \equiv \langle \hat{V}_{\text{en}} \rangle + \langle \hat{V}_{ee} \rangle$, goes with the electron density and pair density, as follows.

(a) First of all, the electron density $\rho(\mathbf{r})$ determines not only [cf. Eq. (12.1.3)]

$$\langle \hat{V}_{\text{en}} \rangle = \int v(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} \equiv V_{\text{en}}[\rho(\mathbf{r})] \quad (12.1.20)$$

but also the classical Coulomb integral,

$$J \equiv \langle V_{ee} \rangle_{\text{classical}} = \frac{1}{2} \iint \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \equiv J[\rho(\mathbf{r})] \quad (12.1.21)$$

Specified above are two functionals of $\rho(\mathbf{r})$, which respond to

$$\phi(\mathbf{r}) \equiv \frac{\delta V_{\text{en}}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} + \frac{\delta J[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} = v(\mathbf{r}) + \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad (12.1.22)$$

This is the classical *electrostatic Coulomb potential* on individual electron from all other charges, with $v(\mathbf{r})$ being given in Eq. (12.1.3).

(b) As seen from Eq. (12.1.21), the classical Coulomb integral does not take the electron–electron correlation into account. Therefore, $J > \langle \hat{V}_{ee} \rangle$. Their difference is the so-called *exchange–correlation* (XC) energy,

$$\mathcal{K}_{\text{XC}} \equiv J - \langle \hat{V}_{ee} \rangle > 0 \quad (12.1.23)$$

This is the nonclassical contribution, to be elaborated further in §12.1.2. The full electron–electron interaction is related to the pair density via

$$\langle \hat{V}_{ee} \rangle = \frac{1}{2} \iint \frac{\pi(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad (12.1.24)$$

12.1.2 Exchange–correlation formalism

A. Pair correlation function

The most basic two–particle property is the *pair correlation function*:

$$\Pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') \equiv 1 - \frac{2\pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}')}{\rho^\sigma(\mathbf{r}) \rho^{\sigma'}(\mathbf{r}')} \quad (12.1.25)$$

At $\mathbf{r} = \mathbf{r}'$, it goes by the involving edge functions normalization ratios,

$$\Pi^{\sigma\sigma}(\mathbf{r}, \mathbf{r}) = (N_\sigma - 1)/N_\sigma \quad \text{and} \quad \Pi^{\alpha\beta}(\mathbf{r}, \mathbf{r}) = 1 \quad (12.1.26)$$

On the other hand,

$$\Pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') \rightarrow 0, \quad \text{as } |\mathbf{r} - \mathbf{r}'| \rightarrow \infty \quad (12.1.27)$$

where two electrons are no longer correlated.

Remarks: General speaking, the pair correlation $\Pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}')$, Eq. (12.1.15), is closely related to the *copula* function in probability theory and statistics. Copula, named by Sklar in 1959, represents a class of certain well-behaved marginal distributions. This would be one of worthy developments to come: “*Copulas are widely used to describe and model the dependence among random variables in many research fields, such as quantitative finance, earthquake statistics, and cosmological data analysis. It is anticipated that copula may also serve as a powerful tool in quantum chemistry*” (cited from Wang et al. *Chin. J. Chem. Phys.*, **36**, 685–690 (2023); <https://doi.org/10.1063/1674-0068/cjcp2210154>).

B. Exchange–correlation energy via pair correlation function

The electrons–pair density, in terms of pair correlation, reads

$$\pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') = \frac{1}{2} \rho^\sigma(\mathbf{r}) [1 - \Pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}')] \rho^{\sigma'}(\mathbf{r}') \quad (12.1.28)$$

Note that [cf. Eq. (12.1.21)]

$$J \equiv \langle V_{ee} \rangle_{\text{classical}} = \sum_{\sigma, \sigma'} J^{\sigma\sigma'} \quad (12.1.29)$$

with

$$J^{\sigma\sigma'} = \frac{1}{2} \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} \rho^\sigma(\mathbf{r}) \rho^{\sigma'}(\mathbf{r}') \quad (12.1.30)$$

The XC integral in Eq. (12.1.24) is then

$$\mathcal{K}_{\text{XC}}^{\sigma\sigma'} = \frac{1}{2} \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} \rho^\sigma(\mathbf{r}) \Pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') \rho^{\sigma'}(\mathbf{r}') \quad (12.1.31)$$

The total XC energy is a spin–sum property and given by

$$\mathcal{K}_{\text{XC}} = (\mathcal{K}_{\text{XC}}^{\alpha\alpha} + \mathcal{K}_{\text{XC}}^{\beta\beta}) + (\mathcal{K}_{\text{XC}}^{\alpha\beta} + \mathcal{K}_{\text{XC}}^{\beta\alpha}) = K_{\text{X}} + K_{\text{C}} \quad (12.1.32)$$

The last identity is the working definition, for the fact that exchange K_{X} and the correlation K_{C} arise from the parallel–spin pairs and from those of opposite spins, respectively. Moreover, the Fermi hole and Coulomb hole pictures imply that $J > \mathcal{K}_{\text{XC}} > K_{\text{X}}, K_{\text{C}} > 0$.

C. The exchange–correlation hole theorem

- Introduce now the XC hole function,

$$\rho_{\text{XC}}^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') \equiv -\Pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}')\rho^{\sigma'}(\mathbf{r}') \quad (12.1.33)$$

so that the electrons-pair density, Eq. (12.1.28), can be recast as

$$\pi^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') = \frac{1}{2}\rho^\sigma(\mathbf{r})[\rho^{\sigma'}(\mathbf{r}') + \rho_{\text{XC}}^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}')] \quad (12.1.34)$$

Together with Eqs. (12.1.10) and (12.1.11), we obtain

$$\sum_{\sigma'} \int \rho_{\text{XC}}^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}') d\mathbf{r}' = -1 \quad (12.1.35)$$

This is the **XC hole theorem** that holds for all many-electron systems at *any real-space point* $\mathbf{r}$, with any spin, either $\sigma = \alpha$ or β , such that

$$\int \rho_{\text{XC}}^{\alpha\alpha}(\mathbf{r}, \mathbf{r}') d\mathbf{r}' + \int \rho_{\text{XC}}^{\alpha\beta}(\mathbf{r}, \mathbf{r}') d\mathbf{r}' = -1 \quad (12.1.36)$$

(i) The first integral describes the *Fermi hole* and contributes to the exchange. It concerns an electron at any real-space point correlating with another electron of same spin. This integral vanishes in a spin-singlet two-electron system, such as the ground state of H_2 molecule.

(ii) The second integral describes the *Coulomb hole* and contributes to the correlation; i.e., the interaction between two electrons with difference spins. Moreover, as implied in Eq. (8.1.6), the single-determinantal wave function approximation, such as the Hartree-Fock theory, goes with $\rho_{\text{XC}}^{\alpha\beta}(\mathbf{r}, \mathbf{r}')|_{\text{HF}} = 0$ itself without integration.

12.1.3 Reduced density matrixes formalisms

A. Determination of reduced density matrixes

- We can obtain R1DM and R2DM, the canonical matrixes of R1DO and R2DO, by using the second-quantization algebra; see §8.2. Without loss of generality, we focus on R1DM via

$$\langle \hat{a}_k^\dagger \hat{a}_j \rangle = \text{tr}[(\hat{a}_k^\dagger \hat{a}_j) \varrho_1] = \text{tr}[(\hat{a}_k^\dagger |0\rangle \langle 0| \hat{a}_j) \varrho_1] = \text{tr}[|\chi_k\rangle \langle \chi_j| \varrho_1]$$

We obtain

$$\varrho_{jk} \equiv \langle \chi_j | \hat{\varrho}_1 | \chi_k \rangle = \langle \Psi | \hat{a}_k^\dagger \hat{a}_j | \Psi \rangle \quad (12.1.37)$$

- In the following, we elaborate Eq. (12.1.37), with respect to an arbitrary *normalized* N -electron wave function,

$$|\Psi\rangle = \sum_{\mathbf{j}} c_{\mathbf{j}} |\Phi_{\mathbf{j}}\rangle \equiv \sum_{j_1 \cdots j_N} c_{j_1 \cdots j_N} |\chi_{j_1} \cdots \chi_{j_N}\rangle \quad (12.1.38)$$

The resulting Eq. (12.1.37) reads

$$\varrho_{jk} = \sum_{\mathbf{j}, \mathbf{k}} c_{\mathbf{k}}^* c_{\mathbf{j}} \langle \Phi_{\mathbf{k}} | \hat{a}_k^\dagger \hat{a}_j | \Phi_{\mathbf{j}} \rangle, \quad \text{with } j \in \mathbf{j} \text{ and } k \in \mathbf{k} \quad (12.1.39)$$

(a) The basis MOs, $\{|\chi_j(\mathbf{x})\rangle; j = 1, \dots, \mathcal{N}\}$, are *all pre-ordered*, determined by the canonical Hartree–Fock equation (10.2.24). The total $\binom{\mathcal{N}}{N}$ –number of distinct N –electron determinants constitute the Fock–space bases, participating in Eq. (12.1.39), where

$$\langle \Phi_{\mathbf{k}} | \Phi_{\mathbf{j}} \rangle = \delta_{\mathbf{k}\mathbf{j}} = \delta_{k_1 j_1} \cdots \delta_{k_N j_N} \quad (12.1.40)$$

since $j_1 < \dots < j_N$ and $k_1 < \dots < k_N$, as in line with the aforementioned preset ordering.

(b) Set then $j \equiv j_1$ and $k \equiv k_1$, so that $\mathbf{j} \equiv j\mathbf{j}'$ and $\mathbf{k} \equiv k\mathbf{k}'$, where $\mathbf{j}' = j_2 \cdots j_N$ and $\mathbf{k}' = k_2 \cdots k_N$. Together with Eq. (12.1.40), we obtain Eq. (12.1.39) the final expression of

$$\varrho_{jk} = \sum_{\mathbf{j}, \mathbf{k}} c_{\mathbf{k}}^* c_{\mathbf{j}} \langle \Phi_{k\mathbf{k}'} | \hat{a}_k^\dagger \hat{a}_j | \Phi_{j\mathbf{j}'} \rangle = \sum_{\mathbf{j}'} c_{k\mathbf{j}'}^* c_{j\mathbf{j}'} \quad (12.1.41)$$

where $\mathbf{j}' = j_2 \cdots j_N$.

B. The R1DO and natural orbitals

• *Natural spin orbitals*, named by Löwdin and denoted $|\chi_j^{\text{NSO}}\rangle$ below, are the eigenvectors of R1DO:

$$\hat{\varrho}_1 |\chi_j^{\text{NSO}}\rangle = n_j |\chi_j^{\text{NSO}}\rangle \quad \Rightarrow \quad \hat{\varrho}_1 = \sum_j n_j |\chi_j^{\text{NSO}}\rangle \langle \chi_j^{\text{NSO}}| \quad (12.1.42)$$

(a) Together with $\chi_j^{\text{NSO}}(\mathbf{x}) \equiv \langle \mathbf{x} | \chi_j^{\text{NSO}} \rangle$, we obtain

$$\varrho_1(\mathbf{x}, \mathbf{x}') \equiv \langle \mathbf{x} | \hat{\varrho}_1 | \mathbf{x}' \rangle = \sum_j n_j \chi_j^{\text{NSO}}(\mathbf{x}) [\chi_j^{\text{NSO}}(\mathbf{x}')]^* \quad (12.1.43)$$

This is Eq. (12.1.6), with $p = 1$. In comparison, Eq. (12.1.37) specifies the Fock–space spin–orbital representation of R1DO.

(b) The orthonormality and completeness relations are [cf. Eq. (12.1.7)]

$$\langle \chi_k^{\text{NSO}} | \chi_j^{\text{NSO}} \rangle = \delta_{kj} \quad \text{and} \quad \sum_j |\chi_j^{\text{NSO}}\rangle \langle \chi_j^{\text{NSO}}| = \hat{I}_1 \quad (12.1.44)$$

(c) The occupation numbers satisfy

$$0 \leq n_j \leq 1 \quad \text{and} \quad \sum_j n_j = N \quad (12.1.45)$$

(d) Evidently, the pure-state $\hat{\rho}_{1;\text{pure}}$ corresponds to that all nonzero occupation numbers are of the values of $n_j = 1$. This is the case of single-determinantal wave functions, to be elaborated further in §12.2.3.

• Let $|n_j^\sigma\rangle \equiv |\psi_j^\sigma\rangle^{\text{nat}}$ be the spin-polarized spacial component of $|\chi_j^{\text{NSO}}\rangle$. Now in parallel to Eq. (12.1.42), we have

$$\hat{\rho}_1^\sigma |n_j^\sigma\rangle = n_j^\sigma |n_j^\sigma\rangle \quad \Rightarrow \quad \hat{\rho}_1^\sigma = \sum_j n_j^\sigma |n_j^\sigma\rangle \langle n_j^\sigma| \quad (12.1.46)$$

Each $\hat{\rho}_1^\sigma$ remains Hermitian and the positivity, with

$$0 \leq n_j^\sigma \leq 1 \quad \text{and} \quad \text{tr} \hat{\rho}_1^\sigma = \sum_j n_j^\sigma = N_\sigma \quad (12.1.47)$$

As usual, $\rho_1^\sigma(\mathbf{r}, \mathbf{r}') \equiv \langle \mathbf{r} | \hat{\rho}_1^\sigma | \mathbf{r}' \rangle$ and $\rho_1(\mathbf{r}, \mathbf{r}') = \rho_1^\alpha(\mathbf{r}, \mathbf{r}') + \rho_1^\beta(\mathbf{r}, \mathbf{r}')$ that is identical to Eq. (12.1.8). In general, spin-polarized orbitals satisfy

$$\langle n_j^\sigma | n_k^\sigma \rangle = \delta_{jk}, \quad \text{but} \quad \langle n_j^\alpha | n_k^\beta \rangle \neq \delta_{jk} \quad (12.1.48)$$

as Eq. (8.1.37), unless $\hat{\rho}_1^\alpha = \hat{\rho}_1^\beta$ that is adopted in the spin-restricted description, in the absence of spin polarizations.

• **Charge and spin densities:** Equation (12.1.46) gives rise to

$$\rho^\sigma(\mathbf{r}) \equiv \rho_1^\sigma(\mathbf{r}, \mathbf{r}) = \sum_j n_j^\sigma \langle \mathbf{r} | n_j^\sigma \rangle \langle n_j^\sigma | \mathbf{r} \rangle; \quad 0 \leq n_j^\sigma \leq 1 \quad (12.1.49)$$

The electron charge and spin densities are, respectively,

$$\rho(\mathbf{r}) \equiv \rho^\alpha(\mathbf{r}) + \rho^\beta(\mathbf{r}) \quad \text{and} \quad \rho^s(\mathbf{r}) \equiv \rho^\alpha(\mathbf{r}) - \rho^\beta(\mathbf{r}) \quad (12.1.50)$$

(a) In contact to molecular systems, we revisit the above formalism, with the LCAO-MO representation, Eq. (7.2.15),

$$|n_j^\sigma\rangle = \sum_\mu C_{\mu j}^\sigma |\varphi_\mu\rangle \quad \text{and} \quad S_{\mu\nu} \equiv \langle \varphi_\mu | \varphi_\nu \rangle \neq \delta_{\mu\nu} \quad (12.1.51)$$

(b) In the AOs representation, the R1DO $\hat{\rho}_1^\sigma$ is given by

$$P_{\mu\nu}^\sigma \equiv \langle \varphi_\mu | \hat{\rho}_1^\sigma | \varphi_\nu \rangle = \sum_j n_j^\sigma C_{\mu j}^\sigma (C_{\nu j}^\sigma)^* \quad (12.1.52)$$

The resulting Eq. (12.1.49) goes with

$$\rho^\sigma(\mathbf{r}) = \sum_{\mu\nu} P_{\mu\nu}^\sigma \varphi_\mu(\mathbf{r}) \varphi_\nu^*(\mathbf{r}) \quad (12.1.53)$$

Its normalization follows

$$N_\sigma = \int \rho^\sigma(\mathbf{r}) d\mathbf{r} = \sum_{\mu\nu} P_{\mu\nu}^\sigma S_{\nu\mu} = \text{tr}(\mathbf{P}^\sigma \mathbf{S}) = \sum_j n_j^\sigma \quad (12.1.54)$$

(c) The electronic charge density matrix and spin density matrix are

$$\mathbf{P} \equiv \mathbf{P}^\alpha + \mathbf{P}^\beta \quad \text{and} \quad \mathbf{P}^S \equiv \mathbf{P}^\alpha - \mathbf{P}^\beta \quad (12.1.55)$$

respectively. The resulting Eq. (12.1.50) in AOs reads

$$\rho(\mathbf{r}) = \sum_{\mu,\nu} P_{\mu\nu} \varphi_\mu(\mathbf{r}) \varphi_\nu^*(\mathbf{r}) \quad \text{and} \quad \rho^S(\mathbf{r}) = \sum_{\mu,\nu} P_{\mu\nu}^S \varphi_\mu(\mathbf{r}) \varphi_\nu^*(\mathbf{r}) \quad (12.1.56)$$

The spin density $\rho^S(\mathbf{r})$ is concerned with unpaired electrons and related to the Electron Spin Resonance (ESR) spectroscopy.

(d) In Eq. (12.1.56), while $P_{\mu\mu}$ is the charge at the specified AO, the off-diagonal $P_{\mu\nu}$ describes the bond between different AOs pairs. Moreover, the number of electrons associated with the AO of φ_μ is given by $(\mathbf{PS})_{\mu\mu}$. The net charge q_A in atom A , as suggested by Milliken, is then

$$q_A = Z_A - \sum_{\mu \in A} (\mathbf{PS})_{\mu\mu} \quad (\text{Milliken charge}) \quad (12.1.57)$$

C. The R2DO and natural geminals

- Natural spin geminals, named by Löwdin, are the eigenvectors of R2DO,

$$\hat{\varrho}_2 |\vartheta_{jk}^{\text{NSG}}\rangle = g_{jk} |\vartheta_{jk}^{\text{NSG}}\rangle \quad \Rightarrow \quad \hat{\varrho}_2 = \frac{1}{2} \sum_{jk} g_{jk} |\vartheta_{jk}^{\text{NSG}}\rangle \langle \vartheta_{jk}^{\text{NSG}}| \quad (12.1.58)$$

The orthonormality and completeness read [cf. Eq. (12.1.7)]

$$\langle \vartheta_{ij}^{\text{NSG}} | \vartheta_{kl}^{\text{NSG}} \rangle = \delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk} \quad \text{and} \quad \frac{1}{2} \sum_{jk} |\vartheta_{ij}^{\text{NSG}}\rangle \langle \vartheta_{ij}^{\text{NSG}}| = \hat{I}_2 \quad (12.1.59)$$

While $|\vartheta_{kj}^{\text{NSG}}\rangle = -|\vartheta_{jk}^{\text{NSG}}\rangle$, the geminal occupation numbers satisfy $g_{jj} \equiv 0$ and the Schwartz-like inequality,

$$0 \leq g_{jk} = g_{kj} \leq n_j n_k; \quad (j \neq k) \quad (12.1.60)$$

in relation to the product of natural orbital occupation numbers pair. Equation (12.1.6) with $p = 2$ reads

$$\varrho_2(\mathbf{x}_1 \mathbf{x}_2, \mathbf{x}'_1 \mathbf{x}'_2) = \frac{1}{2} \sum_{jk} g_{jk} \langle \mathbf{x}_1 \mathbf{x}_2 | \vartheta_{jk}^{\text{NSG}} \rangle \langle \vartheta_{jk}^{\text{NSG}} | \mathbf{x}'_1 \mathbf{x}'_2 \rangle \quad (12.1.61)$$

- In parallel, arising from Eq. (12.1.58) are four sets of spin-polarized natural geminals, given by

$$\hat{\rho}_2^{\sigma\sigma'} |\theta_{jk}^{\sigma\sigma'}\rangle = g_{ij}^{\sigma\sigma'} |\theta_{jk}^{\sigma\sigma'}\rangle; \quad \sigma, \sigma' = \alpha, \beta \quad (12.1.62)$$

- (a) Equation (12.1.16) amounts to

$$\text{tr} \hat{\rho}_2^{\sigma\sigma} = \frac{1}{2} N_\sigma (N_\sigma - 1) \quad \text{and} \quad \text{tr} \hat{\rho}_2^{\alpha\beta} = \text{tr} \hat{\rho}_2^{\beta\alpha} = \frac{1}{2} N_\alpha N_\beta \quad (12.1.63)$$

with $\hat{\rho}_2^{\alpha\beta} = \hat{\rho}_2^{\beta\alpha}$, as described earlier. Moreover, $N_\alpha + N_\beta = N$ and

$$\text{tr} \hat{\rho}_2 = \frac{1}{2} N_\alpha (N_\alpha - 1) + \frac{1}{2} N_\beta (N_\beta - 1) + N_\alpha N_\beta = \frac{1}{2} N (N - 1) \quad (12.1.64)$$

- (b) The spin-polarized spacial counterparts to Eq. (12.1.61) satisfy

$$\sum_{\sigma'} \int \rho_2^{\sigma\sigma'}(\mathbf{r}_1 \mathbf{r}_2, \mathbf{r}'_1 \mathbf{r}'_2) d\mathbf{r}_2 = \frac{1}{2} (N - 1) \rho_1^\sigma(\mathbf{r}_1, \mathbf{r}'_1) \quad (12.1.65)$$

As specified earlier, $\pi^{\sigma\sigma'}(\mathbf{r}_1, \mathbf{r}_2) \equiv \rho_2^{\sigma\sigma'}(\mathbf{r}_1 \mathbf{r}_2, \mathbf{r}_1 \mathbf{r}_2)$ describes the electrons-pair density. Equation (12.1.14) is reproduced.

Exercises

E12.1.1 By using Eq. (12.1.41), perform the natural orbital analysis on the results of the minimum H₂ model molecule obtained in §10.4.

E12.1.2 Recast the kinetic energy, Eq. (12.1.19), in terms of natural orbitals.

E12.1.3 Express $\text{tr} \hat{\rho}_1^2$ in terms of the natural orbital occupation numbers, $\{n_j\}$, and show that $\text{tr} \hat{\rho}_1^2 \leq \text{tr} \hat{\rho}_1$. Discuss the cases where the equal sign holds.

E12.1.4 Consider the natural geminal occupation number in relation to the product of natural orbital occupation numbers pair. Justify that $g_{jj} = 0$ and $0 \leq g_{jk} = g_{kj} \leq n_j n_k$, Eq. (12.1.60) with $j \neq k$.

E12.1.5 Justify the following expressions:

- (a) A pure-state R1DO goes by $n_j = 0$ or 1; i.e.,

$$\hat{\rho}_{1;\text{pure}}^\sigma = \sum_{a=1}^{N_\sigma} |\psi_a^\sigma\rangle \langle \psi_a^\sigma| \quad (12.1.66)$$

- (b) The corresponding exchange integral is given by

$$K_X^\sigma(\text{pure}) = \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} |\rho_{1;\text{pure}}^\sigma(\mathbf{r}, \mathbf{r}')|^2 \quad (12.1.67)$$

- (c) For close-shell systems, where $N_\alpha = N_\beta = N/2$ we obtain

$$K_X(\text{pure}) = \frac{1}{4} \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} |\rho_{1;\text{pure}}(\mathbf{r}, \mathbf{r}')|^2; \quad (\text{closed-shell}) \quad (12.1.68)$$

E12.1.6 Revisit Eqs. (8.1.5) and (8.1.6), in relation to the basic properties of a single-determinantal wave function, $|\Phi\rangle$, Eq. (8.1.7). Discuss these properties, with respect to

the classical Coulomb integral, J of Eq. (12.1.21), and the exchange–correlation integral, $K_{\text{XC}} \equiv K_{\text{X}} + K_{\text{C}}$, Eq. (12.1.31). In particular, the Hartree–Fock theory goes with $K_{\text{X}}(\text{HF})$ of Eq. (12.1.67) and $K_{\text{C}}(\text{HF}) = 0$. The underlying $\hat{v}^{\text{HF}}(\mathbf{r})$ contains no correlation; see Eq. (10.2.16). It is worth noting that the Kohn–Sham orbital theory adopts an alternative single–determinantal wave function description, with the effective $\hat{v}_{\text{eff}}(\mathbf{r})$ containing both exchange and correlation; see §12.4 for the details.

12.2 Density Functional Theory and Related Issues

Density functional theory (DFT) exploits the electron density function as the basic variable, primary for the *ground-state* energy. The formal DFT is rooted at the Hohenberg–Kohn theorems (1964) and implemented with the Kohn–Sham self-consistent methods (1965) that will be discussed in §12.3. To that end, this section is concerned with the Hohenberg–Kohn theorems (§12.2.1), the representability issues (§12.2.2), the basic skills (§12.2.3) and the historical model-level descriptions (§12.2.4).

References

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12.2.1 Hohenberg–Kohn theorems

- Let us recast the electronic Hamiltonian as (atomic unit)

$$\hat{H} = \hat{T}_e + \hat{V}_{ee} + \hat{V}_{\text{ext}}(\{\mathbf{R}_A\}) \equiv -\frac{1}{2} \sum_{i=1}^N \nabla_i^2 + \frac{1}{2} \sum_{i,j=1}^N \frac{1}{r_{ij}} + \sum_{i=1}^N v(\mathbf{r}_i) \quad (12.2.1)$$

The first-two terms describe the electronic kinetic energy and electron–electron interaction, respectively. The last term refers to be *external*, arising from individual electron in the nuclei-fields, with

$$v(\mathbf{r}) = - \sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \quad (12.2.2)$$

This is the only part making *each individual electronic system distinct*, assuming the number N of electrons is given.

Observations:

(a) All properties of the system are fixed whenever $v(\mathbf{r})$ is given (within a trivial constant). As the electronic Hamiltonian is concerned, $v(\mathbf{r})$ is the external potential and can *include further the electromagnetic fields interactions, if any*, which are also single-electron properties in nature.

(b) In place of the electronic Hamiltonian descriptors, N and $v(\mathbf{r})$, *either* the *ground-state* antisymmetric wave function, $|\Psi(\mathbf{x}^N)\rangle$, *or* the corresponding density, $\rho(\mathbf{r})$, can be the basic variable, as the ground-state energy functional is considered; i.e.,

$$E_0(N) = E[\Psi(\mathbf{x}^N)] = E[\rho(\mathbf{r})] \quad (12.2.3)$$

Here $\mathbf{x}^N \equiv \mathbf{x}_1 \cdots \mathbf{x}_N$, with $\mathbf{x}_j \equiv \mathbf{r}_j \sigma_j$, including spin $\sigma_j = \alpha, \beta$.

(c) For the nondegenerate case, there are one-to-one correspondences between the aforementioned three types of descriptors, denoted as

$$\boxed{v(\mathbf{r}) \leftrightarrow |\Psi(\mathbf{x}^N)\rangle \leftrightarrow \rho(\mathbf{r})} \quad (N \text{ fixed}) \quad (12.2.4)$$

• **The First HK Theorem:** *Existence of $v(\mathbf{r}) = v[\rho(\mathbf{r})]$, where $\rho(\mathbf{r})$ is the density function of ground-state, assuming non-degenerate. In other words, $\rho(\mathbf{r})$ uniquely determines the external potential, $v(\mathbf{r}) = v[\rho(\mathbf{r})]$, that in turn fixes the electronic Hamiltonian $\hat{H}$.*

The proof is simple, by *reductio ad absurdum*, since the theorem is equivalent to the impossibility of $v'(\mathbf{r}) \rightarrow \rho(\mathbf{r})$ and $v(\mathbf{r}) \rightarrow \rho(\mathbf{r})$, if $v'(\mathbf{r}) \neq v(\mathbf{r})$ (*apart from a trivial zero-point energy*) that results in $|\Psi'_0\rangle \neq |\Psi_0\rangle$. Let $E_0 = \langle \Psi_0 | \hat{H} | \Psi_0 \rangle$ and $E'_0 = \langle \Psi'_0 | \hat{H}' | \Psi'_0 \rangle$. Moreover, $\langle \Psi'_0 | \hat{H} | \Psi'_0 \rangle = E'_0 + \langle \Psi'_0 | \hat{V}_{\text{ext}} - \hat{V}'_{\text{ext}} | \Psi'_0 \rangle$, with $\hat{H} - \hat{H}' = \hat{V}_{\text{ext}} - \hat{V}'_{\text{ext}}$ as by Eq. (12.2.1). The variation principle would give rise to

$$E_0 < \langle \Psi'_0 | \hat{H} | \Psi'_0 \rangle = E'_0 + \langle \Psi'_0 | \hat{V}_{\text{ext}} - \hat{V}'_{\text{ext}} | \Psi'_0 \rangle = E'_0 + \int [v(\mathbf{r}) - v'(\mathbf{r})] \rho(\mathbf{r}) d\mathbf{r}$$

Exchanging the roles of primed and unprimed quantities would then be

$$E'_0 < E_0 + \int [v'(\mathbf{r}) - v(\mathbf{r})] \rho(\mathbf{r}) d\mathbf{r}$$

The sum of above two equations results in $E_0 + E'_0 < E'_0 + E_0$ a fallacy. We have thus proved the first HK theorem by *reductio ad absurdum*.

• **The Second HK Theorem** – Energy functional variational principle: *Any trial density, $\tilde{\rho}(\mathbf{r}) \geq 0$, satisfies*

$$E_0 \leq E[\tilde{\rho}(\mathbf{r})] \quad (12.2.5)$$

Deductions and remarks:

(a) The existence of HK functional,

$$F_{\text{HK}}[\rho(\mathbf{r})] \equiv \langle \hat{T}_e \rangle + \langle \hat{V}_{ee} \rangle = T[\rho(\mathbf{r})] + (J[\rho(\mathbf{r})] - \mathcal{K}_{\text{xc}}[\rho(\mathbf{r})]) \quad (12.2.6)$$

for the non-degenerate ground state of any systems. The ground-state electronic energy functional reads

$$E_0[\rho(\mathbf{r})] = \int v(\mathbf{r})\rho(\mathbf{r})d\mathbf{r} + F_{\text{HK}}[\rho(\mathbf{r})] \quad (12.2.7)$$

Formally, $F_{\text{HK}}[\rho(\mathbf{r})]$ is universal, as it is defined without $v(\mathbf{r})$.

(b) As known, any steady-state energy functional, $E[\rho(\mathbf{r})]$, satisfies

$$\delta \left\{ E[\rho(\mathbf{r})] - \mu \left[\int \rho(\mathbf{r})d\mathbf{r} - N \right] \right\} = 0 \quad (12.2.8)$$

The resulting Euler equation via Eqs. (12.2.5)–(12.2.8) reads

$$\mu = \frac{\delta F_{\text{HK}}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} + v(\mathbf{r}) = \frac{\delta T[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} - \frac{\delta \mathcal{K}_{\text{xc}}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} + \phi(\mathbf{r}) \quad (12.2.9)$$

Here, $\phi(\mathbf{r})$ is the classical electrostatic Coulomb potential; cf. Eq. (12.1.22).

(c) The first HK theorem claims the existence of both the kinetic energy functional, $T[\rho(\mathbf{r})]$, and the nonclassical exchange–correlation energy functional, $\mathcal{K}_{\text{xc}}[\rho(\mathbf{r})] \equiv J[\rho(\mathbf{r})] - \langle \hat{V}_{ee} \rangle > 0$ [cf. Eq. (12.1.23)]. Their explicit expressions are unknown. The related issues are discussed as follows.

12.2.2 Issues of representability

A. The v –representability via external potentials

- Consider the first HK theorem, Eq. (12.2.4), with $v(\mathbf{r}) \leftrightarrow \Psi(\mathbf{x}^N) \leftrightarrow \rho(\mathbf{r})$, including all $\tilde{v}(\mathbf{r}) \leftrightarrow \tilde{\Psi}(\mathbf{x}^N) \leftrightarrow \tilde{\rho}(\mathbf{r})$. This is the **v –representability**: *Each individual density function arises from the ground-state antisymmetric wave function of given electron Hamiltonian, specified by an arbitrary external potential $v_{\text{ext}}(\mathbf{r})$.*
- The second HK theorem, Eq. (12.2.5), describes the energy functional variation principle, for those v –representable ground-state density functions. However, it is an open question whether an *arbitrary* given $\rho(\mathbf{r}) \geq 0$, satisfying the basic properties,

$$\int \rho(\mathbf{r})d\mathbf{r} = N, \quad \int |\nabla \rho^{\frac{1}{2}}(\mathbf{r})|^2 d\mathbf{r} < \infty \quad (12.2.10a)$$

and [cf. Eq. (2.3.35)]

$$\int \nabla^2 \rho(\mathbf{r})d\mathbf{r} = 0 \quad (12.2.10b)$$

Even for single-particle systems, there are densities that do not come from a ground-state wave function of any $v(\mathbf{r})$ [H. Englisch and R. Englisch, “Hohenberg–Kohn theorem and non- v –representable densities”, *Physica* **121A**, 253–268 (1983)].

B. The N –representability via N –electron wave functions

- On the other hand, DFT can practically be carried out with those N -representable densities, $\rho(\mathbf{r}) = \rho[\Psi(\mathbf{x}^N)]$, arising from N -electron anti-symmetric wave functions. This requirement is much weaker than the v -representability, since the former is necessary for the latter.
- The N -representable energy functional variation goes *formally* by the Levy's constraint search [M. Levy, "Universal variational functionals of electron densities, first-order density matrix, natural spin-orbitals and solution to the v -representability problem", *Proc. Natl. Acad. Sci. USA* **76**, 6062–6065 (1979)],

$$\begin{aligned}
 E_0 &= \min_{\Psi} \langle \Psi | \hat{H} | \Psi \rangle = \min_{\rho(\mathbf{r})} \left[\min_{\Psi \rightarrow \rho(\mathbf{r})} \langle \Psi | \hat{H} | \Psi \rangle \right] \\
 &= \min_{\rho(\mathbf{r})} \left[F[\rho(\mathbf{r})] + \int v(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} \right]
 \end{aligned} \tag{12.2.11}$$

The involving three steps are as follows:

- (i) Divide the entire N -particle $\{\Psi(\mathbf{x}^N)\}$ -space into regions, such that $\{\Psi_k(\mathbf{x}^N)\} \rightarrow \rho_k(\mathbf{r})$. Each region comprises several distinct wave functions, but all are associated with a same density;
- (ii) Search first the local energy minimum within each region and then the global minimum among all local minima.

Remarks:

- (a) The Levy's constraint search, Eq. (12.2.11), is conceptually useful. It formally circumvents the v -representability problem. The resulting HK functional, $F_{\text{HK}}[\rho(\mathbf{r})] = F[\rho(\mathbf{r})]$, is concerned only with those N -representable densities. The external field $v(\mathbf{r})$ enters through its mean value; see Eq. (12.2.7) and also the above constraint search formulas.
- (b) The *degenerate* ground-state manifold goes by $\{\Psi_{0;p}(\mathbf{x}^N)\} \rightarrow \{\rho_{0;p}(\mathbf{r})\}$. While their energy expectation values are same, the N -representable HK functionals, $F[\rho_{0;p}(\mathbf{r})]$, with $p = 1, \dots, g_0$, are different in general. Further specification goes by the molecular symmetry and/or spin analysis. The resulting N -representable DFT is effectively applicable to the *degenerate ground-state* case.
- (c) The N -representability condition is ensured by any reasonable density [cf. Eq. (12.2.10)]. This offers the basic idea of practical DFT implementations, including the degenerate scenario.
- (d) External electromagnetic field interaction is of one-electron in nature. Adding it to the original $v(\mathbf{r})$, Eq. (12.2.2), does not alter DFT formulations. For example, consider

$$v_{\text{ext}}^{\sigma}(\mathbf{r}) = v(\mathbf{r}) + g_s \mu_B \hat{\mathbf{s}} \cdot \mathbf{B}(\mathbf{r}) \tag{12.2.12}$$

Added is the interaction between the single-electron spin angular momentum, $\hat{\mathbf{s}}$, and the local external magnetic field, $\mathbf{B}(\mathbf{r})$, where μ_B is the Bohr's magneton and g_s is the Landé factor; see Eqs. (6.2.17)–(6.2.19). The dependence of $v_{\text{ext}}^{\sigma}(\mathbf{r})$, Eq. (12.4.17), on spin state $\sigma = \alpha$ or β will be transparent, with $\hat{\mathbf{s}}\mathbf{B}(\mathbf{r}) = \hat{s}_z B_z(\mathbf{r})$.

C. Alternative views on the representability issues

- Turn to Eq. (12.2.9), with focus on the unknown $T[\rho(\mathbf{r})]$ that is related to the reduced density matrix, R1DM via Eq. (12.1.19); i.e.,

$$T[\rho_1] = -\frac{1}{2} \int \nabla_{\mathbf{r}}^2 \rho_1(\mathbf{r}, \mathbf{r}')|_{\mathbf{r}'=\mathbf{r}} d\mathbf{r} \quad (12.2.13)$$

where [cf. Eq. (12.1.49)]

$$\rho_1(\mathbf{r}, \mathbf{r}') = \sum_{j,\sigma} n_j^\sigma \psi_j^\sigma(\mathbf{r}) [\psi_j^\sigma(\mathbf{r}')]^* \quad (12.2.14a)$$

with

$$0 \leq n_j^\sigma \leq 1 \quad \text{and} \quad \sum_{j,\sigma} n_j^\sigma = N \quad (12.2.14b)$$

The *mixed-state* related density function is

$$\rho(\mathbf{r}) = \rho_1(\mathbf{r}, \mathbf{r}) = \sum_{j,\sigma} n_j^\sigma |\psi_j^\sigma(\mathbf{r})|^2 \quad (12.2.15)$$

that in general is not $v(\mathbf{r})$ -representable.

- **Noninteracting system is always v -representable.** This goes by a pure-state R1DM related density,

$$\rho(\mathbf{r}) = \sum_{a,\sigma}^{\text{occ}} |\psi_a^\sigma(\mathbf{r})|^2 = \sum_{a,\sigma}^{\text{occ}} |\chi_a(\mathbf{x})|^2 \quad (12.2.16a)$$

with the N -occupied spin-orbitals comprising

$$|\chi_a(\mathbf{x}_1) \cdots \chi_b(\mathbf{x}_N)\rangle \equiv \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_a(1) & \cdots & \chi_b(1) \\ \cdots & \cdots & \cdots \\ \chi_a(N) & \cdots & \chi_b(N) \end{vmatrix} \quad (12.2.16b)$$

Take this as a *effective reference*, with the $v_{\text{eff}}(\mathbf{r})$ -representability being

$$|\chi_a(\mathbf{x}_1) \cdots \chi_b(\mathbf{x}_N)\rangle \xrightarrow{v_{\text{eff}}(\mathbf{r})} \rho(\mathbf{r}) \leftarrow |\Psi_0(\mathbf{x}^N)\rangle_{\text{exact}} \quad (12.2.17)$$

The last arrow highlights that $\rho(\mathbf{r})$ can in principle be identical to that via the exact ground-state wave function, whenever the single-determinantal reference description is appropriate. In other words, the noninteracting v -representability, Eq. (12.2.17), holds for *non-degenerate* ground states. The currently used Kohn–Sham DFT method exploits the *noninteracting reference system* description; see §12.3.

12.2.3 Onsets of density function description

A. Properties of density function

• Compared to wave function $\Psi(\mathbf{r}^N)$ that is a $3N$ -dimension unmeasurable property, electron density $\rho(\mathbf{r})$ is a three-dimension and measurable property, satisfying Eq. (12.2.10). Other basic considerations are as follows.

(a) Spherical-average of density function around each nucleus satisfies the cusp condition (Kato, 1957),

$$\left. \frac{d\rho(\mathbf{r})}{dr} \right|_{\mathbf{r}=\mathbf{R}_A} = -2Z_A\rho(\mathbf{R}_A) \quad (12.2.18)$$

with Z_A being the nuclear charge.

(b) The asymptotic behavior at atomic/molecular tail is dominant by that of the highest occupied orbital; i.e.,

$$\rho^{\frac{1}{2}}(\mathbf{r}) \Big|_{r \rightarrow \infty} \propto r^{\kappa} e^{-\sqrt{2I_N}r} \quad (12.2.19)$$

where $\kappa \geq 0$, while

$$\begin{aligned} I_N &\equiv E_0(N-1) - E_0(N) > 0 \\ A_N &\equiv E_0(N) - E_0(N+1) > 0 \end{aligned} \quad (12.2.20)$$

are the *ionization energy* and *electron affinity* of the N -electron molecule. The thermodynamic chemical potential amounts to $\mu \approx -(I_N + A_N)/2$.

(c) For chemical reactions or molecular excitation, we will also be interested in the DFT with varying electron number. Consider then a given atomic or molecular species. The ionization energies are of $I_{N+1} \leq I_N$ in general. The corresponding energy $E_0(N)$ is a *convex* function, satisfying

$$E_0(N+1) + E_0(N-1) \geq 2E_0(N) \quad (12.2.21)$$

The **fraction-number** DFT goes by the *linear-segment interpolations*,

$$E_0(N+a) = (1-a)E_0(N) + aE_0(N+1); \quad 0 < a < 1 \quad (12.2.22)$$

This is the Perdew-Parr-Levy-Baldur condition (1982) and equivalent to

$$\partial E_0(N)/\partial N = -A_N \quad (12.2.23)$$

with A_N being the *electron affinity*; cf. Eq. (12.2.20).

B. The Euler-Lagrange equation and skills

• The key quantity in the DFT study is the energy functional,

$$E_0[\rho(\mathbf{r})] = \min E[\rho(\mathbf{r})] \quad (12.2.24)$$

The Lagrangian multiplier associated with the constraint, $\int \rho(\mathbf{r})d\mathbf{r} = N$, is the chemical potential μ in Eq. (12.2.8). The resulting Euler–Lagrange equation reads

$$\mu = \frac{\delta E[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} \quad (\text{Constant}) \quad (12.2.25)$$

for the required electron density, $\rho(\mathbf{r})$.

• **Functional derivatives:** In general, a functional $\mathcal{F}[\rho(x)]$ involves not only $\rho(x)$ itself, but also its derivatives, with the form of

$$\mathcal{F}[\rho(x)] = \int G(\rho, \rho^{(1)}, \dots, \rho^{(n)}; x) dx, \quad \text{where } \rho^{(k)}(x) \equiv \frac{d^k \rho(x)}{dx^k} \quad (12.2.26a)$$

The integrand $G(\rho, \rho^{(1)}, \dots, \rho^{(n)}; x)$ is an analytical function of those variables inside the parentheses. The functional derivative is given by

$$\frac{\delta \mathcal{F}[\rho(x)]}{\delta \rho(x)} = \frac{\partial G}{\partial \rho} - \frac{d}{dx} \frac{\partial G}{\partial \rho^{(1)}} + \dots + (-1)^n \frac{d^n}{dx^n} \frac{\partial G}{\partial \rho^{(n)}} \quad (12.2.26b)$$

The above formula hold for replacing x by $\mathbf{r}$ and d/dx by ∇ , such that

$$\frac{\partial G}{\partial \rho} - \frac{d}{dx} \frac{\partial G}{\partial \rho^{(1)}} \rightarrow \frac{\partial G}{\partial \rho} - \nabla \cdot \frac{\partial G}{\partial (\nabla \rho)} \quad (12.2.27)$$

Consider in the following are two representing functionals.

(a) Let us start with a *local* functional, whose derivative is

$$\frac{\delta}{\delta \rho(\mathbf{r})} \left[\int \rho^a(\mathbf{r}) d\mathbf{r} \right] = a \int \rho^{a-1}(\mathbf{r}) d\mathbf{r} \quad (12.2.28)$$

In particular, for the classical Coulomb integral, Eq. (12.1.21),

$$J[\rho(\mathbf{r})] = \frac{1}{2} \iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad \Rightarrow \quad \frac{\delta J[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \quad (12.2.29)$$

(b) Consider then the lowest *nonlocal* functional,

$$\mathcal{F}[\rho(\mathbf{r})] = \int G(\rho, \nabla \rho) d\mathbf{r} \quad (12.2.30)$$

exemplified with

$$G(\rho, \nabla \rho) = \frac{\nabla \rho(\mathbf{r}) \cdot \nabla \rho(\mathbf{r})}{\rho(\mathbf{r})} \quad (12.2.31)$$

The related two terms in Eq. (12.2.27) are

$$\frac{\partial G}{\partial \rho} = - \frac{\nabla \rho(\mathbf{r}) \cdot \nabla \rho(\mathbf{r})}{\rho^2(\mathbf{r})}$$

and

$$\nabla \cdot \frac{\partial G}{\partial(\nabla \rho)} = \nabla \cdot \left[2 \frac{\nabla \rho(\mathbf{r})}{\rho(\mathbf{r})} \right] = -2 \frac{\nabla \rho(\mathbf{r})}{\rho^2(\mathbf{r})} \nabla \rho(\mathbf{r}) + 2 \frac{\nabla^2 \rho(\mathbf{r})}{\rho(\mathbf{r})}$$

for Eq. (12.2.30) with Eq. (12.2.31). We obtain

$$\frac{\delta}{\delta \rho(\mathbf{r})} \left[\int \frac{|\nabla \rho(\mathbf{r})|^2}{\rho(\mathbf{r})} d\mathbf{r} \right] = \frac{|\nabla \rho(\mathbf{r})|^2}{\rho^2(\mathbf{r})} - 2 \frac{\nabla^2 \rho(\mathbf{r})}{\rho(\mathbf{r})} \quad (12.2.32)$$

• **Use of Euler–Lagrange equation:** As an illustration, consider the shape of maximum enclosed area A in plane by a given length L of string loop. For the setup, let the string be described by radial function, $r(\theta)$, where $\theta \in [0, 2\pi)$, satisfying $r(\theta + 2\pi) = r(\theta)$.

(a) The enclosed loop area functional is

$$A[r(\theta)] = \frac{1}{2} \int_0^{2\pi} d\theta r^2(\theta) \quad (12.2.33)$$

satisfying $\delta A[r(\theta)]/\delta r(\theta) = r(\theta)$ [cf. Eq. (12.2.28)].

(b) The loop perimeter functional is given by

$$P[r(\theta)] = \int_0^{2\pi} d\theta \sqrt{r^2(\theta) + [r'(\theta)]^2} \quad (12.2.34)$$

where $r'(\theta) \equiv dr(\theta)/d\theta$. By using Eq. (12.2.26), with $n = 1$, we obtain

$$\frac{\delta P[r(\theta)]}{\delta r(\theta)} = \frac{r}{\sqrt{r^2 - r'^2}} - \frac{d}{d\theta} \left(\frac{r'}{\sqrt{r^2 - r'^2}} \right) = \frac{r^3 - r^2 r'' + 2r r'^2}{(r^2 - r'^2)^{\frac{3}{2}}} \quad (12.2.35)$$

(c) The task of maximizing the enclosed area is concerned with

$$\frac{\delta}{\delta r} \left\{ A[r] - \mu(P[r] - L) \right\} = 0 \quad \Rightarrow \quad \frac{\delta A[r(\theta)]}{\delta r(\theta)} = \mu \frac{\delta P[r(\theta)]}{\delta r(\theta)} \quad (12.2.36)$$

Here, the Lagrangian multiplier μ is conjugate to the fixed loop length constraint. Apply then $\delta A[r(\theta)]/\delta r(\theta) = r(\theta)$ and Eq. (12.2.35), resulting in

$$r = \mu \frac{r^3 - r^2 r'' + 2r r'^2}{(r^2 - r'^2)^{\frac{3}{2}}} \quad \Rightarrow \quad r(\theta) = \mu = r \quad (\text{constant}) \quad (12.2.37)$$

where $r' = r'' = 0$. The maximum loop area is then $A = \pi r^2$ via Eq. (12.2.33), with the circle shape, surrounded by the given loop length of $P = 2\pi r = L$ via Eq. (12.2.34). These are the solutions to the Euler equation (12.2.36).

12.2.4 Thomas–Fermi and related models

A. Uniform electron gas model

- For electronic structure properties, there had been a long history, with using density as basic variable, but at the status of *models*. The original idea attributes to Thomas (1927), Fermi (1927), and Dirac (1930), exploiting the *uniform electron gas* (UEG) (or the Fermi–gas) model. This is an oversimplified model, but provides a valuable reference for realistic atoms and molecules, where electrons are not uniformly distributed.
- First of all, UEG is a three–dimension noninteracting system. Each molecule moves in a cubic box of volume $L^3 = V$. The related plane waves,

$$\psi_{\mathbf{n}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\mathbf{r}} = \psi_{n_x}(x)\psi_{n_y}(y)\psi_{n_z}(z) \quad (12.2.38)$$

with $\mathbf{k} = 2\pi\mathbf{n}/L$, are subject to the *period boundary condition*,

$$\psi_{n_X}(X + L) = \psi_{n_X}(X); \quad X = x, y, z \quad (12.2.39)$$

The resulting energies are quantized, with $n_X = 0, \pm 1, \pm 2, \dots$ and

$$\epsilon_{\mathbf{n}} = \frac{\hbar^2}{2m_e} \left(\frac{2\pi}{L} \right)^2 (n_x^2 + n_y^2 + n_z^2) = \frac{\hbar^2 \mathbf{k}^2}{2m_e} \quad (12.2.40)$$

- The *reduced one–electron density matrix* via Eq. (12.2.38), with the spin–degeneracy of 2, is given by

$$\rho_1(\mathbf{r}_1, \mathbf{r}_2) = 2 \sum_{\mathbf{n}}^{\text{occ}} \psi_{\mathbf{n}}(\mathbf{r}_1) \psi_{\mathbf{n}}^*(\mathbf{r}_2) = \frac{2}{V} \sum_{\mathbf{k}}^{\text{occ}} e^{i\mathbf{k}(\mathbf{r}_1 - \mathbf{r}_2)} \quad (12.2.41)$$

where $\mathbf{k} \equiv 2\pi\mathbf{n}/L$. The summation runs over all *occupied states*.

(a) In contact with the Fermi distribution function at temperature $T = 0$, only $\epsilon_{\mathbf{k}} \geq k_F^2/(2m_e)$ is occupied, where $k_F \equiv |\mathbf{k}_{\text{occ}}|_{\text{max}}$ is the Fermi wave vector amplitude and $\epsilon_F = k_F^2/(2m_e)$ is the Fermi energy.

(b) The UEG is considered in the thermodynamic limit, with $V \rightarrow \infty$ and $N \rightarrow \infty$, but V/N is finite. While the energies of Eq. (12.2.40) become *continuous*, the corresponding Eq. (12.2.41) reads

$$\rho_1(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{4\pi^3} \int_0^{k_F} dk \int_0^\pi d\theta \int_0^{2\pi} d\phi e^{i\mathbf{k}(\mathbf{r}_1 - \mathbf{r}_2) \cdot \hat{\mathbf{k}}} k^2 \sin \theta \quad (12.2.42)$$

The related electron density is given by

$$\rho(\mathbf{r}) = \rho_1(\mathbf{r}, \mathbf{r}) = \frac{1}{4\pi^3} \int_0^{k_F} dk \int_0^\pi d\theta \int_0^{2\pi} d\phi k^2 \sin \theta = \frac{k_F^3}{3\pi^2} \quad (12.2.43)$$

The resulting Fermi wave vector amplitude is

$$k_F(\mathbf{r}) = [3\pi^2 \rho(\mathbf{r})]^{1/3} \quad (12.2.44)$$

- The kinetic energy is related to

$$T = -\frac{1}{2} \nabla_{\mathbf{r}}^2 \rho_1(\mathbf{r}, \mathbf{r}')|_{\mathbf{r}'=\mathbf{r}} = \frac{1}{8\pi^3} \int_0^{k_F} dk \int_0^\pi d\theta \int_0^{2\pi} d\phi k^4 \sin \theta = \frac{k_F^5}{10\pi^2}$$

Together with Eq. (12.2.44), we obtain

$$T_{\text{TF}}[\rho(\mathbf{r})] \equiv C_F \int \rho^{5/3}(\mathbf{r}) d\mathbf{r}, \quad \text{where } C_F = \frac{3}{10} (3\pi^2)^{2/3} \quad (12.2.45)$$

This is the Thomas–Fermi kinetic energy functional.

B. Dirac's exchange–energy functional with UEG model

- In 1930, Dirac obtained the exchange–energy functional via Eq. (12.2.42) for UEG the expression,

$$K_{\text{x}}^{\text{Dirac}}[\rho(\mathbf{r})] \equiv C_{\text{x}} \int \rho^{4/3}(\mathbf{r}) d\mathbf{r}; \quad C_{\text{x}} = \frac{3}{4} \left(\frac{3}{\pi} \right)^{1/3} \quad (12.2.46)$$

Derivations:

(a) Let us recast $\rho_1(\mathbf{r}_1, \mathbf{r}_2)$ with $\mathbf{r} \equiv \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2)$ and $\mathbf{s} \equiv \mathbf{r}_1 - \mathbf{r}_2$. The latter is conjugate to the Wigner phase–space momentum; see Eq. (3.4.18). Set further $\mathbf{k}\mathbf{s} = ks \cos \theta$. The resulting Eq. (12.2.42) reads

$$\rho_1(\mathbf{r}_1, \mathbf{r}_2) = \rho_1(\mathbf{r}, \mathbf{s}) = \frac{1}{4\pi^3} \int_0^{k_F} dk \int_0^\pi d\theta \int_0^{2\pi} d\phi e^{iks \cos \theta} k^2 \sin \theta \quad (12.2.47)$$

Applying then

$$\begin{aligned} \int_0^\pi e^{iks \cos \theta} \sin \theta d\theta &= \frac{2 \sin(ks)}{ks} \\ \int_0^{k_F} k^2 \frac{\sin(ks)}{ks} dk &= \frac{1}{s^3} [\sin(k_F s) - (k_F s) \cos(k_F s)] \end{aligned} \quad (12.2.48)$$

and Eq. (12.2.43), we obtain Eq. (12.2.47) the expression,

$$\rho_1(\mathbf{r}, \mathbf{s}) = 3\tau^{-3} (\sin \tau - \tau \cos \tau) \rho(\mathbf{r}) \quad (12.2.49a)$$

where [cf. Eq. (12.2.44)]

$$\tau \equiv k_F(\mathbf{r}) s = [3\pi^2 \rho(\mathbf{r})]^{1/3} s \quad (12.2.49b)$$

This is a basic feature of UEG, depending on $\mathbf{s}$ only through its amplitude $s \equiv |\mathbf{s}|$.

(b) Substitute Eq. (12.2.49) into Eq. (12.1.68), resulting in

$$K_X = \frac{1}{4} \iint \frac{|\rho_1(\mathbf{r}, s)|^2}{s} d\mathbf{r} ds = 9\pi \int \frac{\rho^2(\mathbf{r})}{k_F^2(\mathbf{r})} d\mathbf{r} \cdot \int_0^\infty \frac{(\sin \tau - \tau \cos \tau)^2}{\tau^5} d\tau \quad (12.2.50)$$

To proceed, consider first

$$I \equiv 4 \int_0^\infty \frac{(\sin \tau - \tau \cos \tau)^2}{\tau^5} d\tau = 4 \int_0^\infty \frac{A^2(\tau)}{\tau^5} d\tau \quad (12.2.51)$$

where

$$A(\tau) \equiv \sin \tau - \tau \cos \tau$$

Moreover,

$$A'(\tau) \equiv \frac{dA(\tau)}{d\tau} = \tau \sin \tau, \quad \left(\frac{\sin \tau}{\tau} \right)' = -\frac{A(\tau)}{\tau^2} \quad \text{and} \quad \frac{A(\tau)}{\tau^2} \Big|_0^\infty = 0$$

Evaluate then Eq. (12.2.51) via the integral-by-part method, resulting in

$$I = 2 \int_0^\infty \frac{A(\tau)A'(\tau)}{\tau^4} d\tau = 2 \int_0^\infty \frac{A(\tau)}{\tau^2} \frac{\sin \tau}{\tau} d\tau = - \int_0^\infty d \left(\frac{\sin \tau}{\tau} \right)^2 = 1 \quad (12.2.52)$$

Together with [cf. Eq. (12.2.44)]

$$\frac{9\pi}{4} \frac{\rho^2(\mathbf{r})}{k_F^2(\mathbf{r})} = C_X \rho^{\frac{4}{3}}(\mathbf{r}), \quad \text{with} \quad C_X = \frac{9\pi}{4(3\pi^2)^{\frac{2}{3}}} = \frac{3}{4} \left(\frac{3}{\pi} \right)^{\frac{1}{3}} \quad (12.2.53)$$

we obtain Eq. (12.2.50) the result of

$$K_X = C_X \int \rho^{\frac{4}{3}}(\mathbf{r}) d\mathbf{r} \quad (12.2.54)$$

This is Eq. (12.2.46).

(Q.E.D.)

C. Thomas–Fermi–Dirac energy functional

• Consider the energy functional via Eq. (12.2.45) and Eq. (12.2.46), on top of the classical Coulomb interaction, Eqs. (12.1.20) and (12.1.21); i.e.,

$$\begin{aligned} E_{\text{TFD}}[\rho(\mathbf{r})] &= C_F \int \rho^{\frac{5}{3}}(\mathbf{r}) d\mathbf{r} - C_X \int \rho^{\frac{4}{3}}(\mathbf{r}) d\mathbf{r} \\ &\quad + \left[\int v(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} + \frac{1}{2} \iint \frac{\rho(\mathbf{r}_1) \rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|} d\mathbf{r}_1 d\mathbf{r}_2 \right] \end{aligned} \quad (12.2.55)$$

The related chemical potential, Eq. (12.2.15), is

$$\mu_{\text{TFD}} = \frac{\delta E_{\text{TFD}}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} = \frac{5}{3}C_F \rho^{\frac{2}{3}}(\mathbf{r}) - \frac{4}{3}C_X \rho^{\frac{1}{3}}(\mathbf{r}) + \phi(\mathbf{r}) \quad (12.2.56)$$

with $\phi(\mathbf{r})$ of Eq. (12.1.22), the classical electrostatic potential.

D. Weizsacker's gradient correction

• The functional form of Eq. (12.2.55) is an example of the local-density approximation (LDA), arising from the UEG description. Better descriptions should include the effect of inhomogeneity of nuclei in real space. The first attempt was taken by von Weizsacker (1935), with the gradient correction,

$$T_W[\rho(\mathbf{r})] = \frac{\hbar^2}{8m_e} \int \frac{|\nabla \rho(\mathbf{r})|^2}{\rho(\mathbf{r})} d\mathbf{r} \quad (12.2.57)$$

Remarks: This equation is obtained, by considering a *one-electron* system. The ground-state wave function has no nodes, satisfying $\psi_0(\mathbf{r}) = \rho^{\frac{1}{2}}(\mathbf{r}) > 0$ and

$$\nabla \psi_0(\mathbf{r}) = \nabla \rho^{\frac{1}{2}}(\mathbf{r}) = \frac{\nabla \rho(\mathbf{r})}{2\rho^{\frac{1}{2}}(\mathbf{r})} \quad (12.2.58)$$

The kinetic energy is then

$$T_W = \frac{\hbar^2}{2m_e} \int \nabla \psi_0(\mathbf{r}) \cdot \nabla \psi_0(\mathbf{r}) d\mathbf{r} = \frac{\hbar^2}{8m_e} \int \frac{|\nabla \rho(\mathbf{r})|^2}{\rho(\mathbf{r})} d\mathbf{r} \quad (12.2.59)$$

This is Eq. (12.2.57), satisfying [cf. Eq. (12.2.32)]

$$\frac{\delta T_W[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} = \frac{\hbar^2}{8m_e} \left[\frac{|\nabla \rho(\mathbf{r})|^2}{\rho^2(\mathbf{r})} - 2 \frac{\nabla^2 \rho(\mathbf{r})}{\rho(\mathbf{r})} \right] \quad (12.2.60)$$

• In line with Eqs. (12.2.55)–(12.2.60), we have the *Thomas–Fermi–Dirac–Weizsacker energy functional*,

$$E_{\text{TFDW}}[\rho(\mathbf{r})] = E_{\text{TFD}}[\rho(\mathbf{r})] + T_W[\rho(\mathbf{r})] \quad (12.2.61)$$

and the corresponding

$$\mu_{\text{TFDW}} = \frac{\delta E_{\text{TFDW}}}{\delta \rho(\mathbf{r})} = \mu_{\text{TFD}} + \frac{\hbar^2}{8m_e} \left[\frac{|\nabla \rho(\mathbf{r})|^2}{\rho^2(\mathbf{r})} - 2 \frac{\nabla^2 \rho(\mathbf{r})}{\rho(\mathbf{r})} \right] \quad (12.2.62)$$

General speaking, the TFDW performs much better than the TFD. The latter is a local-density approximation, involving the UEG description. At any rate, TFDW is less accurate than the Hartree–Fock (HF) theory.

- *Molecular specification* has to be included in the $v(\mathbf{r})$ -representable density function and the resulting exchange-correlation functional, as discussed before. The first-principle DFT methods described in the coming section are useful balance between accurate and computational cost.

12.3 Kohn–Sham Density Functional Theory

12.3.1 Kohn–Sham orbitals formalism

A. Basic principles and considerations

- The first-principle DFT methods are based on the Kohn–Sham (KS) scheme (1965) that starts with a fictitious *noninteracting reference system*,

$$\hat{H}_{\text{KS}} = \sum_{i=1}^N \hat{h}_s(i), \quad \text{with } \hat{h}_s = -\frac{1}{2} \nabla_{\mathbf{r}}^2 + v_s(\mathbf{r}) \quad (12.3.1)$$

The subscript “s” denotes the effective single-electron descriptions, with the potential energy taking the form of [cf. Eq. (12.1.22)]

$$v_s(\mathbf{r}) \equiv \phi(\mathbf{r}) + v_{\text{XC}}(\mathbf{r}) = \left[v(\mathbf{r}) + \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' \right] + v_{\text{XC}}(\mathbf{r}) \quad (12.3.2)$$

Formally, $v_{\text{XC}}(\mathbf{r})$, defined in Eqs. (12.3.11)–(12.3.14), describes the nonstatic and nonclassical *exchange–correlation* contributions.

- The KS ground-state is a single Slater determinantal wave function,

$$|\Phi_{\text{KS}}\rangle = |\chi_1 \cdots \chi_N\rangle_{\text{KS}}, \quad \text{with } \langle \chi_a(\mathbf{x}) | \chi_b(\mathbf{x}) \rangle = \delta_{ab} \quad (12.3.3)$$

The involving N -number of orbitals satisfy

$$\left[-\frac{1}{2} \nabla_{\mathbf{r}}^2 + v_s(\mathbf{r}) \right] |\chi_a\rangle = \epsilon_a |\chi_a\rangle; \quad a = 1, \cdots, N \quad (12.3.4)$$

- To proceed, we specify Eq. (12.2.17), the $v_s(\mathbf{r})$ -representability description with $v_{\text{eff}}(\mathbf{r}) = v_s(\mathbf{r})$; i.e.,

$$\boxed{|\Psi(\mathbf{x}^N)\rangle \rightarrow \rho(\mathbf{r}) \xleftarrow{v_s(\mathbf{r})} |\chi_1(\mathbf{x}_1) \cdots \chi_N(\mathbf{x}_N)\rangle_{\text{KS}} \equiv |\Phi_{\text{KS}}\rangle} \quad (12.3.5)$$

Denote hereafter $|\Psi(\mathbf{x}^N)\rangle \equiv |\Psi(\mathbf{x}^N)\rangle$. As usual, $\mathbf{x} \equiv \mathbf{r}s$, with $s = \uparrow$ and $\downarrow$.

(a) While $\rho(\mathbf{r})$ arises from the exact ground-state antisymmetric wave function, the second arrow describes the establishment of the effective noninteracting $v_s(\mathbf{r})$ -representability of the density function; i.e.,

$$\rho(\mathbf{r}) = \rho_{\text{KS}}(\mathbf{r}) = \sum_{a=1}^N \int |\chi_a(\mathbf{x})|^2 ds \quad (12.3.6)$$

(b) While $\rho_1(\mathbf{r}, \mathbf{r}')$ via the true $|\Psi(\mathbf{x}^N)\rangle$ is a mixed-state R1DM, the KS-orbitals

counterpart via $|\Phi_{\text{KS}}\rangle$ is a pure-state; i.e.,

$$\rho_{\text{S}}(\mathbf{r}, \mathbf{r}') \equiv \rho_{\text{I}}^{\text{KS}}(\mathbf{r}, \mathbf{r}') = \sum_{a=1}^N \int \chi_a(\mathbf{r}s) \chi_a^*(\mathbf{r}'s) \text{d}s \neq \rho_{\text{I}}(\mathbf{r}, \mathbf{r}') \quad (12.3.7)$$

but $\rho_{\text{S}}(\mathbf{r}, \mathbf{r}) = \rho_{\text{I}}(\mathbf{r}, \mathbf{r}) = \rho(\mathbf{r})$ the true density.

- The KS-reference kinetic energy functional reads

$$T_{\text{KS}}[\rho] \equiv \langle \Phi_{\text{KS}} | \hat{T} | \Phi_{\text{KS}} \rangle = \frac{1}{2} \sum_{a=1}^N \int |\nabla_{\mathbf{r}} \chi_a(\mathbf{x})|^2 d\mathbf{x} < T[\rho] \quad (12.3.8)$$

with $T_{\text{KS}}[\rho] \leq T[\rho] \equiv \langle \Psi | \hat{T} | \Psi \rangle$, as inferred from the aforementioned pure-state versus mixed-state descriptions. In fact, the KS orbitals participating in the single determinantal wave function are subject to

$$T_{\text{KS}}[\rho] = \min_{\Phi \rightarrow \rho(\mathbf{r})} \langle \Phi | \hat{T} | \Phi \rangle \quad (12.3.9)$$

This is the key **principle** of construction the KS orbitals with Eq. (12.3.4) to the required determinant $|\Phi_{\text{KS}}\rangle$; See Eqs. (12.3.17) and (12.3.18) for the further elaborations.

B. Constructions and insights of Eq. (12.3.4)

- To proceed, we recast the Hohenberg–Kohn functional Eq. (12.2.6) as

$$F_{\text{HK}}[\rho] = T[\rho] + V_{ee}[\rho] \equiv T_{\text{KS}}[\rho] + J[\rho] + E_{\text{XC}}[\rho] \quad (12.3.10)$$

- (i) The unknown exchange–correlation energy functional reads

$$E_{\text{XC}}[\rho] \equiv T_{\text{C}}[\rho] - \mathcal{K}_{\text{XC}}[\rho] \quad (12.3.11)$$

with [cf. Eq. (12.3.8)]

$$T_{\text{C}}[\rho] \equiv T[\rho] - T_{\text{KS}}[\rho] \geq 0 \quad (12.3.12)$$

and [cf. Eq. (12.1.23)]

$$\mathcal{K}_{\text{XC}}[\rho] \equiv J[\rho] - V_{ee}[\rho] \geq 0 \quad (12.3.13a)$$

where [cf. Eq. (12.1.21)]

$$J[\rho] \equiv \langle V_{ee} \rangle_{\text{classical}} = \frac{1}{2} \iint \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad (12.3.13b)$$

- (ii) The corresponding exchange–correlation potential is given by

$$v_{\text{XC}}(\mathbf{r}) \equiv \frac{\delta E_{\text{XC}}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} \quad (12.3.14)$$

This is the unknown term in the KS reference potential of Eq. (12.3.2).

- The ground-state energy functional, Eq. (12.2.7) with Eq. (12.3.10), reads

$$E[\rho] = \int v(\mathbf{r})\rho(\mathbf{r})d\mathbf{r} + (T_{\text{KS}}[\rho] + J[\rho] + E_{\text{XC}}[\rho]) \quad (12.3.15)$$

The KS reference of Eq. (12.3.2) completely dictate Euler equation,

$$\mu = \frac{\delta E[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} = \frac{\delta T_{\text{KS}}[\rho(\mathbf{r})]}{\delta \rho(\mathbf{r})} + v_{\text{s}}(\mathbf{r}) \quad (12.3.16)$$

- The KS-DFT is a *variational orbitals* formalism. The KS kinetic energy functional minimization Eq. (12.3.9) goes rather with

$$T_{\text{KS}}[\rho] = \min_{\{\chi_a(\mathbf{x})\} \rightarrow \rho(\mathbf{r})} T_{\text{KS}}[\{\chi_a(\mathbf{x})\}] \quad (12.3.17a)$$

$$T_{\text{KS}}[\{\chi_a(\mathbf{x})\}] \equiv -\frac{1}{2} \sum_{a=1}^N \int \chi_a^*(\mathbf{x}) \nabla_{\mathbf{r}}^2 \chi_a(\mathbf{x}) d\mathbf{x} \quad (12.3.17b)$$

under the constraints $\{\chi_a(\mathbf{x})\} \rightarrow \rho(\mathbf{r})$ and normalization, so that

$$\begin{aligned} & \frac{\delta}{\delta \chi_a^*(\mathbf{x})} \left\{ T_{\text{KS}}[\{\chi_a(\mathbf{x})\}] + \int v_{\text{s}}(\mathbf{r}) \left[\left(\sum_{a=1}^N \int |\chi_a(\mathbf{x})|^2 d\mathbf{s} \right) - \rho(\mathbf{r}) \right] d\mathbf{r} \right\} \\ &= \frac{\delta}{\delta \chi_a^*(\mathbf{x})} \sum_{a=1}^N \epsilon_a \left[\int |\chi_a(\mathbf{x})|^2 d\mathbf{x} - 1 \right] \end{aligned} \quad (12.3.18)$$

The obtained Eq. (12.3.4) is identical to that *minimizes the energy functional* $E[\{\chi_a(\mathbf{x})\} \rightarrow \rho(\mathbf{r})]$. As results, the KS-orbitals together with Eq. (12.3.6), constitute a self-consistent field description.

- The sum of KS-orbital energies is [cf. Eqs. (12.3.2)–(12.3.4)]

$$\begin{aligned} \sum_{a=1}^N \epsilon_a &= \sum_{a=1}^N \langle \chi_a(\mathbf{x}) | -\frac{1}{2} \nabla_{\mathbf{r}}^2 + v_{\text{s}}(\mathbf{r}) | \chi_a(\mathbf{x}) \rangle \\ &= T_{\text{KS}}[\rho(\mathbf{r})] + \int \left[v(\mathbf{r}) + \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + v_{\text{XC}}(\mathbf{r}) \right] \rho(\mathbf{r}) d\mathbf{r} \\ &= T_{\text{KS}}[\rho(\mathbf{r})] + V_{\text{en}}[\rho(\mathbf{r})] + 2J[\rho(\mathbf{r})] + \int v_{\text{XC}}(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} \end{aligned} \quad (12.3.19)$$

with Eqs. (12.1.20) and (12.1.21) being used. On the other hand,

$$\begin{aligned} E[\rho(\mathbf{r})] &= T_{\text{KS}}[\rho(\mathbf{r})] + V_{\text{en}}[\rho(\mathbf{r})] + J[\rho(\mathbf{r})] + E_{\text{XC}}[\rho(\mathbf{r})] \\ &= \sum_{a=1}^N \epsilon_a - J[\rho(\mathbf{r})] + E_{\text{XC}}[\rho(\mathbf{r})] - \int v_{\text{XC}}(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} \end{aligned} \quad (12.3.20)$$

- Note that the potential $v_{\text{XC}}(\mathbf{r}) \equiv v_{\text{XC}}[\rho(\mathbf{r})]$ to $E_{\text{XC}}[\rho(\mathbf{r})]$ is also a density functional [cf. Eq. (12.3.14)], while the exact form of $E_{\text{XC}}[\rho(\mathbf{r})]$ is *unknown*. Moreover, the dependence of $T_{\text{KS}}[\rho(\mathbf{r})]$ on $\rho(\mathbf{r})$ is *indirect* for many-electron systems. The aforementioned Weizsacker expression, $T_{\text{W}}[\rho]$ of Eq. (12.2.57), holds only for one-electron systems.

12.3.2 Related formulations and elaborations

A. Extended KS–DFT formalism

- The extended KS–orbitals possess fractional occupation numbers,

$$0 \leq n_j \leq 1 \quad \text{and} \quad \sum_j n_j = N \quad (12.3.21)$$

as those used in the exact R1DM [cf. Eqs. (12.1.8) and (12.1.43)],

$$\rho_1(\mathbf{r}, \mathbf{r}') = \sum_j n_j \int \chi_j(\mathbf{r}s) \chi_j^*(\mathbf{r}'s) ds \quad (12.3.22)$$

The corresponding Eq. (12.3.9) is related to the exact kinetic energy,

$$\langle \hat{T} \rangle = -\frac{1}{2} \int [\nabla_{\mathbf{r}}^2 \rho_1(\mathbf{r}, \mathbf{r}')]_{\mathbf{r}'=\mathbf{r}} d\mathbf{r} = \min_{\{\chi_j(\mathbf{x})\}} T[\{\chi_j(\mathbf{x})\}; n_j] \quad (12.3.23a)$$

$$T[\{\chi_j(\mathbf{x})\}; \{n_j\}] \equiv -\frac{1}{2} \sum_j n_j \int \chi_j^*(\mathbf{x}) \nabla_{\mathbf{r}}^2 \chi_j(\mathbf{x}) d\mathbf{x} \quad (12.3.23b)$$

While $\{n_j\}$ are fixed parameters, the extended KS orbitals are variational, subject to

$$\rho(\mathbf{r}) = \sum_j n_j \int |\chi_j(\mathbf{x})|^2 ds \quad \text{and} \quad \int |\chi_j(\mathbf{x})|^2 d\mathbf{x} = 1 \quad (12.3.24)$$

Let $v_{\text{eff}}(\mathbf{r})$ and $\{\epsilon'_j\}$ be their Lagrangian multiples, respectively. The minimization equation is similar as Eq. (12.3.18) and results in

$$\left[-\frac{1}{2} \nabla^2 + v_{\text{eff}}(\mathbf{r}) \right] |\chi_j\rangle = \frac{\epsilon'_j}{n_j} |\chi_j\rangle \equiv \epsilon_j |\chi_j\rangle \quad (12.3.25)$$

The above equations give rise to the energy,

$$\begin{aligned} E[\{\chi_j\}; \{n_j\}] &= T[\{\chi_j\}; \{n_j\}] + \int v_{\text{eff}}(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} \\ &= \sum_j n_j \langle \chi_j(\mathbf{x}) | -\frac{1}{2} \nabla_{\mathbf{r}}^2 + v_{\text{eff}}(\mathbf{r}) | \chi_j(\mathbf{x}) \rangle \end{aligned} \quad (12.3.26)$$

and the Janek's theorem,

$$\frac{\partial E}{\partial n_j} = \langle \chi_j(\mathbf{x}) | -\frac{1}{2} \nabla_{\mathbf{r}}^2 + v_{\text{eff}}(\mathbf{r}) | \chi_j(\mathbf{x}) \rangle = \epsilon_j \quad (12.3.27)$$

• Remarks:

- (a) Recall Eq. (12.2.22) the Perdew–Parr–Levy–Baldur condition that energy $E(N+a)$ must be straight line between two integers. Fraction-charge density

functionals would be needed; See for example, Cohen, Mori-Sanchez, and Yang, *Chem. Rev.* **112**, 289 (2012).

(b) Equation (12.3.25) reproduces Eq. (12.3.4), with setting $v_{\text{eff}}(\mathbf{r}) = v_{\text{S}}(\mathbf{r})$ and $n_a = 1$ for those $\chi_j(\mathbf{x}) \in \{\chi_a(\mathbf{x}); a = 1, \dots, N\}$; otherwise $n_j = 0$. The KS orbitals are those N lowest eigenstates of Eq. (12.3.25), having the minimal $T_{\text{KS}} \leq T$. Note that $\{n_j\}$ are not variation parameters; otherwise the kinetic energy

(c) Formally, the KS orbitals are the “*best natural-orbital-approximants*” at the single determinantal $|\Phi(\mathbf{x}^N)\rangle$ -level of description, having the exact $\rho(\mathbf{r})$ via the true ground-state $|\Psi(\mathbf{x}^N)\rangle$, despite that $\rho_{\text{S}}(\mathbf{r}, \mathbf{r}') \neq \rho_1(\mathbf{r}, \mathbf{r}')$, as specified in Eq. (12.3.7).

B. Adiabatic connection formalism

- Let us revisit the unknown property in DFT [cf. Eq. (12.3.10)],

$$E_{\text{XC}}[\rho] = F_{\text{HK}}[\rho] - T_{\text{KS}}[\rho] - J[\rho] \quad (12.3.28)$$

with $F_{\text{HK}}[\rho]$ being the Hohenberg–Kohn functional. Its λ -integral equivalence will be helpful in the later developments of approximation methods.

- To proceed, consider an adiabatic λ -path that smoothly connects between the KS reference system ($\lambda = 0$) and the true system ($\lambda = 1$); i.e.,

$$F_{\lambda=1}[\rho] \equiv F_{\text{HK}}[\rho] \quad \text{and} \quad F_{\lambda=0}[\rho] \equiv T_{\text{KS}}[\rho] \quad (12.3.29)$$

We obtain Eq. (12.3.28) the expression,

$$E_{\text{XC}}[\rho] = \int_0^1 \frac{\partial F_{\lambda}[\rho]}{\partial \lambda} d\lambda - J[\rho] \quad (12.3.30)$$

- Implied above is the **adiabatic connection**, without altering $\rho(\mathbf{r})$. This is a constraint on the variational minimization,

$$F_{\lambda}[\rho] = \min_{\{\Psi_{\lambda}\} \rightarrow \rho(\mathbf{r})} \langle \Psi_{\lambda} | \hat{T} + \lambda \hat{V}_{ee} | \Psi_{\lambda} \rangle \quad (12.3.31)$$

that gives rise to

$$\hat{H}_{\lambda} |\Psi_{\lambda}\rangle = \left[\hat{T} + \lambda \hat{V}_{ee} + \sum_{i=1}^N v_{\lambda}(\mathbf{r}_i) \right] |\Psi_{\lambda}\rangle = E_{\lambda} |\Psi_{\lambda}\rangle \quad (12.3.32)$$

Derivations:

(a) First of all, the last term of $\hat{H}_{\lambda}$ is the effective noninteracting potential, which does not appear in the the Hohenberg–Kohn functional, Eq. (12.3.31). The involving $v_{\lambda}(\mathbf{r})$ must be λ -dependent. This is required by $\{\Psi_{\lambda}\} \rightarrow \rho(\mathbf{r})$, as specified in Eq. (12.3.31), that all $|\Psi_{\lambda}\rangle$ along the *adiabatic* λ -path possess the *same* density $\rho(\mathbf{r})$ of realist systems [cf. Eq. (12.3.33)].

(b) As specified in Eq. (12.3.31), the adiabatic connection constraint requires that all $\{\Psi_\lambda\}$ have the same $\rho(\mathbf{r})$. In other words,

$$\rho(\mathbf{r}) = \langle \Psi_\lambda | \hat{\rho}(\mathbf{r}) | \Psi_\lambda \rangle, \quad \text{with} \quad \hat{\rho}(\mathbf{r}) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \quad (12.3.33)$$

Let $v_\lambda(\mathbf{r})$ be the Lagrangian multiplier, associated with this constraint. Together with $\langle \Psi_\lambda | \Psi_\lambda \rangle = 1$, whose Lagrangian multiplier is set to be E_λ , we obtain Eq. (12.3.31) the minimization equation,

$$\frac{\delta}{\delta \Psi_\lambda^*} \left\{ \langle \Psi_\lambda | \hat{T} + \lambda \hat{V}_{ee} | \Psi_\lambda \rangle + v(\mathbf{r}) \langle \Psi_\lambda | \hat{\rho}(\mathbf{r}) | \Psi_\lambda \rangle \right\} = E_\lambda \frac{\delta \langle \Psi_\lambda | \Psi_\lambda \rangle}{\delta \Psi_\lambda^*} \quad (12.3.34)$$

This gives rise to

$$\left[\hat{T} + \lambda \hat{V}_{ee} + \sum_{i=1}^N v_\lambda(\mathbf{r}_i) \right] | \Psi_\lambda \rangle = E_\lambda | \Psi_\lambda \rangle \quad (12.3.35)$$

This is Eq. (12.3.32).

(Q.E.D.)

- Applying then the Hellmann–Feynman theorem, we obtain

$$\frac{\partial E_\lambda}{\partial \lambda} = \left\langle \Psi_\lambda \left| \frac{\partial \hat{H}_\lambda}{\partial \lambda} \right| \Psi_\lambda \right\rangle = \langle \Psi_\lambda | \hat{V}_{ee} | \Psi_\lambda \rangle + \sum_{i=1}^N \left\langle \Psi_\lambda \left| \frac{\partial v_\lambda(\mathbf{r}_i)}{\partial \lambda} \right| \Psi_\lambda \right\rangle$$

This results in the integrand of Eq. (12.3.30) the expression,

$$\frac{\partial F_\lambda[\rho]}{\partial \lambda} = \langle \Psi_\lambda | \hat{V}_{ee} | \Psi_\lambda \rangle \quad (12.3.36)$$

The final adiabatic connect expression reads

$$E_{\text{XC}}[\rho] = \int_0^1 \langle \Psi_\lambda | \hat{V}_{ee} | \Psi_\lambda \rangle d\lambda - J[\rho] \quad (12.3.37)$$

C. The exchange–correlation virial theorem

- The virial theorem, Eq. (2.3.20), for N -electron systems reads

$$\begin{aligned} 2T[\rho] &= \sum_{i=1}^N \langle \Psi | \mathbf{r}_i \cdot \nabla_i (\hat{V}_{\text{en}} + \hat{V}_{ee}) | \Psi \rangle \\ &= \int \rho(\mathbf{r}) [\mathbf{r} \cdot \nabla v(\mathbf{r})] d\mathbf{r} - \langle \Psi | \hat{V}_{ee} | \Psi \rangle \end{aligned} \quad (12.3.38)$$

While the first identity is general, the second identity holds since $V_{ee}(\{r_{ij}\})$ is a homogeneous function of order -1 .

- In parallel, the KS determinantal $|\Phi_{\text{KS}}\rangle$ from $\hat{H}_{\text{KS}}$, Eq. (12.3.1) with Eq. (12.3.2), satisfies

$$2T_{\text{KS}}[\rho] = \int \rho(\mathbf{r}) \{ \mathbf{r} \cdot \nabla [v(\mathbf{r}) + v_{\text{XC}}(\mathbf{r})] \} d\mathbf{r} - J[\rho] \quad (12.3.39)$$

Together with [cf. Eqs. (12.3.10)–(12.3.13)]

$$\langle \Psi | \hat{V}_{ee} | \Psi \rangle - J[\rho] = U_{\text{C}}[\rho] + E_{\text{X}}[\rho] = E_{\text{XC}}[\rho] - T_{\text{C}}[\rho] \quad (12.3.40)$$

and Eq. (12.3.38), we obtain the **XC virial theorem**,

$$E_{\text{XC}}[\rho] + T_{\text{C}}[\rho] = - \int \rho(\mathbf{r}) [\mathbf{r} \cdot \nabla v_{\text{XC}}(\mathbf{r})] d\mathbf{r} \quad (12.3.41)$$

As usual,

$$E_{\text{XC}}[\rho] = E_{\text{X}}[\rho] + E_{\text{C}}[\rho] \quad \text{and} \quad v_{\text{XC}}(\mathbf{r}) = v_{\text{X}}(\mathbf{r}) + v_{\text{C}}(\mathbf{r}) \quad (12.3.42)$$

D. Other useful relations

- In line with Eqs. (12.3.11)–(12.3.13), we have further

$$E_{\text{C}}[\rho] = T_{\text{C}}[\rho] + U_{\text{C}}[\rho] \quad (12.3.43)$$

This is the *correlation energy* in DFT, including both the kinetic and potential contributions,

$$\begin{aligned} T_{\text{C}}[\rho] &= T[\rho] - T_{\text{KS}}[\rho] > 0 \\ U_{\text{C}}[\rho] &= V_{ee}[\rho] - (J[\rho] + E_{\text{X}}[\rho]) < 0 \end{aligned} \quad (12.3.44)$$

with $T_{\text{KS}}[\rho]$ and $J[\rho]$ being given in Eq. (12.3.8) and Eq. (12.3.13b), respectively.

- It is easy to identify that [cf. Eq. (12.3.26)]

$$\begin{aligned} E_{\text{X}}[\rho] &= \langle \Phi_{\text{KS}}[\rho] | \hat{V}_{ee} | \Phi_{\text{KS}}[\rho] \rangle - J[\rho] \\ U_{\text{C}}[\rho] &= \langle \Psi[\rho] | \hat{V}_{ee} | \Psi[\rho] \rangle - \langle \Phi_{\text{KS}}[\rho] | \hat{V}_{ee} | \Phi_{\text{KS}}[\rho] \rangle \end{aligned} \quad (12.3.45a)$$

and [cf. Eq. (12.3.43)]

$$E_{\text{C}}[\rho] = \langle \Psi[\rho] | \hat{H} | \Psi[\rho] \rangle - \langle \Phi_{\text{KS}}[\rho] | \hat{H} | \Phi_{\text{KS}}[\rho] \rangle \leq 0 \quad (12.3.45b)$$

The inequality follows as if $\Phi_{\text{KS}}[\rho]$ is a trial wave function. Moreover, in relation to Eq. (12.1.23) and Eq. (12.1.32), we have

$$\begin{aligned} E_{\text{X}}[\rho] &= -K_{\text{X}}[\rho] = -(\mathcal{K}_{\text{XC}}^{\alpha\alpha}[\rho] + \mathcal{K}_{\text{XC}}^{\beta\beta}[\rho]) \leq 0 \\ U_{\text{C}}[\rho] &= -K_{\text{C}}[\rho] = -(\mathcal{K}_{\text{XC}}^{\alpha\beta}[\rho] + \mathcal{K}_{\text{XC}}^{\beta\alpha}[\rho]) \leq 0 \end{aligned} \quad (12.3.46)$$

with noting that $E_C = U_C + T_C$, whereas $T_C \simeq -U_C/2$ for many system.

E. Remarks

- Conventional KS–DFT methods are based on approximated $E_{xc}[\rho]$, referred to the density functional approximations (DFAs). The resulting KS–determinants are *not* variational wave functions of electronic Hamiltonian. The evaluated total energy $E[\rho(\mathbf{r})]$ may even be lower than the true ground–state energy.
- Commonly used DFAs cover from the local spin–density approximation (LSDA or LDA), generalized gradient approximation (GGA), to various hybrid functional schemes. These are constructed, more or less in the engineering manner, with the aids of the uniform electron gas models (§12.2.4), the Hartree–Fock exchange [cf. Eq. (12.1.67)], and/or the adiabatic connection formalism with approximate implementations, etc. The related details are well documented; see the standard computational chemistry/material science software manuals and the demonstrative simulations therein.
- To have a true breaking through development, it requires a *rigorous* and *systematic* construction method for $E_{xc}[\rho]$, without invoking approximations. In the coming section, we focus on a recent advancement by the author when he was finishing this book.

12.4 Ab Initio Kohn–Sham Orbitals Formalism

12.4.1 Basic observations and implications

A. The nature of Kohn–Sham determinants

• Before proceed, we would like to clarify some conceptual issues, concerning the KS determinantal $|\Phi_{\text{KS}}(\mathbf{x}^N)\rangle$, with respect to conventional Slater determinant, exemplified with $|\Phi_{\text{HF}}(\mathbf{x}^N)\rangle$ via the Hartree–Fock theory.

(a) First of all, the KS orbital theory is based on $\rho(\mathbf{r}) \equiv \rho(\mathbf{r}; \Psi) = \rho(\mathbf{r}; \Phi_{\text{KS}})$, with $|\Phi_{\text{KS}}(\mathbf{x}^N)\rangle$ evaluated via KS equations acquires the exact electron density. Not only the exchange but also the correlation effects are included. The *correlation-free picture* of conventional Slater determinant, Eq. (8.1.6), would *no longer be valid* for KS determinantal wave functions. The HF counterpart $\rho^{\text{HF}}(\mathbf{r}) = \rho(\mathbf{r}; \Phi_{\text{HF}})$ is an approximate, without correlation.

(b) Physical properties can be largely explained with KS orbitals. Chemical bond is form by doubly occupied KS molecular orbital, rather than the HF counterpart. The spin contamination, which appears in the original single spin–unrestricted Slater determinants discussed in §8.1.3, would also be largely eliminated.

(c) From the aspect of effective single–electron potential, the KS scheme goes with [cf. Eq. (12.3.2) versus Eq. (10.2.16)]

$$v_{\text{eff}}^{\text{KS}}(\mathbf{r}) \equiv v_{\text{S}}(\mathbf{r}) = [\phi(\mathbf{r}) + v_{\text{X}}(\mathbf{r})] + v_{\text{C}}(\mathbf{r}) \equiv v_{\text{eff}}^{\text{HF}}(\mathbf{r}) + v_{\text{C}}(\mathbf{r}) \quad (12.4.1)$$

This would suggest the KS reference system be not fictitious, but effectively take care of correlation, at least for moderately strong correlation systems.

• As known for noninteracting systems, any RpDO is formally a *pure state*, so that an orbital–level description is sufficient. Non-degenerate ground–state density $\rho(\mathbf{r})$ is single–electron $v_{\text{S}}(\mathbf{r})$ –representable. This is in line with the Hohenberg–Kohn theorems and dictates further

$$E_{\text{XC}}[\rho] = E_{\text{X}}[\rho] + U_{\text{C}}[\rho] + T_{\text{C}}[\rho] \quad (12.4.2)$$

(i) While the pure–state results of individual energy components may be approximations, *their sum would in principle be exact, as long as all formulations being established within a unified framework.*

(ii) As long as $E_{\text{X}}[\rho]$ and $U_{\text{C}}[\rho]$ are identified, one could exploit the XC virial theorem, Eq. (12.3.41), to solve the kinetic correction $T_{\text{C}}[\rho]$. In practice, it is the orbital–level virial theorem to be exploited; see Eqs. (12.4.21)–(12.4.23).

• Denote for later use the spin–polarized potentials,

$$v_{\text{X}}^{\sigma}(\mathbf{r}) \equiv \frac{\delta E_{\text{X}}[\rho]}{\delta \rho^{\sigma}(\mathbf{r})}, \quad u_{\text{C}}^{\sigma}(\mathbf{r}) \equiv \frac{\delta U_{\text{C}}[\rho]}{\delta \rho^{\sigma}(\mathbf{r})} \quad \text{and} \quad w_{\text{C}}^{\sigma}(\mathbf{r}) \equiv \frac{\delta T_{\text{C}}[\rho]}{\delta \rho^{\sigma}(\mathbf{r})} \quad (12.4.3)$$

The spin–polarized HF potential, Eq. (10.3.4) with Eq. (10.3.5), reads

$$v_{\text{HF}}^{\sigma}(\mathbf{r}) = v_{\text{J}}^{\sigma}(\mathbf{r}) + v_{\text{X}}^{\sigma}(\mathbf{r}) + v_{\text{J}}^{\bar{\sigma}}(\mathbf{r}) \quad (12.4.4a)$$

with $\bar{\sigma}$ being the opposite spin to σ and $\hat{v}_J^\sigma(\mathbf{r}) = \delta J[\rho]/\delta \rho^\sigma(\mathbf{r})$, satisfying

$$\begin{aligned} \langle \psi_a^\sigma(\mathbf{r}) | \hat{v}_{\text{HF}}^\sigma(\mathbf{r}) | \psi_a^\sigma(\mathbf{r}) \rangle &= \sum_{b=1}^{N_\sigma} [(\psi_a^\sigma \psi_b^\sigma | | \psi_a^\sigma \psi_b^\sigma) - (\psi_a^\sigma \psi_b^\sigma | | \psi_b^\sigma \psi_a^\sigma)] \\ &\quad + \sum_{b=1}^{N_{\bar{\sigma}}} (\psi_a^\sigma \psi_b^{\bar{\sigma}} | | \psi_a^\sigma \psi_b^{\bar{\sigma}}) \end{aligned} \quad (12.4.4b)$$

The notation of two-electron spatial integral, Eq. (10.1.14), is used.

B. The forms of $E_X[\rho]$ and $U_C[\rho]$

Both $E_X[\rho]$ and $U_C[\rho]$ can be expressed in terms of the spin-polarized orbitals $\{\psi_a^\sigma(\mathbf{r})\}$ that form the KS-determinantal $|\Phi_{\text{KS}}\rangle$. As derived soon below, we obtain

$$E_X[\rho] = -\frac{1}{2} \sum_{\sigma} \iint \frac{d\mathbf{r}d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} |\rho_s^\sigma(\mathbf{r}, \mathbf{r}')|^2 \quad (12.4.5a)$$

$$U_C[\rho] = -\text{Re} \iint \frac{d\mathbf{r}d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} [\rho_s^\alpha(\mathbf{r}, \mathbf{r}')]^* \rho_s^\beta(\mathbf{r}, \mathbf{r}') \quad (12.4.5b)$$

with

$$\rho_s^\sigma(\mathbf{r}, \mathbf{r}') \equiv \rho_1^{\sigma;\text{KS}}(\mathbf{r}, \mathbf{r}') = \sum_{a=1}^{N_\sigma} \psi_a^\sigma(\mathbf{r}) [\psi_a^\sigma(\mathbf{r}')]^* \quad (12.4.6)$$

Note that $\rho_s^\sigma(\mathbf{r}, \mathbf{r}') \neq \rho_1^\sigma(\mathbf{r}, \mathbf{r}')$, but $\rho_s^\sigma(\mathbf{r}, \mathbf{r}) = \rho_1^\sigma(\mathbf{r}, \mathbf{r}) = \rho^\sigma(\mathbf{r})$ the true ground-state electron density with the specified spin.

Derivations of Eq. (12.4.5):

(i) As mentioned earlier, any RpDO arising from a single-determinantal wave function is a pure state. While $\rho_s^\sigma(\mathbf{r}, \mathbf{r}')$, Eq. (12.4.6), is via $|\Phi_{\text{KS}}\rangle$, the correlation and exchange between electrons are dictated by the spin-polarized electron-pair density, $\pi_{\text{KS}}^{\sigma\sigma'}(\mathbf{r}_1, \mathbf{r}_2)$, with the Kohn-Sham orbitals framework. This is the diagonal elements of the corresponding R2DO [cf. Eq. (12.1.11) with Eq. (12.1.9)]. That is

$$\pi_{\text{KS}}^{\sigma\sigma'}(\mathbf{r}_1, \mathbf{r}_2) \equiv \rho_{2;\text{KS}}^{\sigma\sigma'}(\mathbf{r}_1\mathbf{r}_2, \mathbf{r}_1\mathbf{r}_2) \quad (12.4.7)$$

by which [cf. Eq. (12.1.24)]

$$\begin{aligned} V_{ee}[\rho] &= \frac{1}{2} \sum_{\sigma, \sigma'} \iint \frac{\pi_{\text{KS}}^{\sigma\sigma'}(\mathbf{r}, \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}d\mathbf{r}' \\ &= \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{a=1}^{N_\sigma} \sum_{b=1}^{N_{\sigma'}} [(\psi_a^\sigma \psi_b^{\sigma'} | | \psi_a^\sigma \psi_b^{\sigma'}) - (\psi_a^\sigma \psi_b^{\sigma'} | | \psi_b^{\sigma'} \psi_a^\sigma)] \end{aligned} \quad (12.4.8)$$

(ii) The first term in the summand brackets reads explicitly

$$(\psi_a^\sigma \psi_b^{\sigma'} || \psi_a^\sigma \psi_b^{\sigma'}) \equiv \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} |\psi_a^\sigma(\mathbf{r})|^2 |\psi_b^{\sigma'}(\mathbf{r}')|^2 \quad (12.4.9a)$$

so that [cf. Eq. (12.1.21)]

$$\frac{1}{2} \sum_{\sigma, \sigma'} \sum_{a=1}^{N_\sigma} \sum_{b=1}^{N_{\sigma'}} (\psi_a^\sigma \psi_b^{\sigma'} || \psi_a^\sigma \psi_b^{\sigma'}) = \frac{1}{2} \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} \rho(\mathbf{r}) \rho(\mathbf{r}') = J[\rho(\mathbf{r})] \quad (12.4.9b)$$

The resulting Eq. (12.4.8) can be arranged as [cf. Eq. (12.3.40)]

$$E_X[\rho] + U_C[\rho] = V_{ee}[\rho] - J[\rho] = -\frac{1}{2} \sum_{\sigma, \sigma'} \sum_{a=1}^{N_\sigma} \sum_{b=1}^{N_{\sigma'}} (\psi_a^\sigma \psi_b^{\sigma'} || \psi_b^{\sigma'} \psi_a^\sigma) \quad (12.4.10)$$

Together with Eq. (12.3.46), we obtain

$$E_X[\rho] = -\frac{1}{2} \sum_{\sigma=\alpha, \beta} \sum_{a,b=1}^{N_\sigma} (\psi_a^\sigma \psi_b^\sigma || \psi_b^\sigma \psi_a^\sigma) \quad (12.4.11a)$$

$$U_C[\rho] = -\text{Re} \sum_{a=1}^{N_\alpha} \sum_{b=1}^{N_\beta} (\psi_a^\alpha \psi_b^\beta || \psi_b^\beta \psi_a^\alpha) \quad (12.4.11b)$$

(iii) Note that [cf. Eq. (10.1.14)]

$$(\psi_a^\sigma \psi_b^{\sigma'} || \psi_b^{\sigma'} \psi_a^\sigma) \equiv \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} [\psi_a^\sigma(\mathbf{r}) \psi_b^{\sigma'}(\mathbf{r}')]^* \psi_b^{\sigma'}(\mathbf{r}) \psi_a^\sigma(\mathbf{r}') \quad (12.4.12)$$

Together with Eq. (12.4.6), we obtain

$$\sum_{a,b} [\psi_a^\sigma(\mathbf{r}) \psi_b^{\sigma'}(\mathbf{r}')]^* \psi_b^{\sigma'}(\mathbf{r}) \psi_a^\sigma(\mathbf{r}') = [\rho_s^\sigma(\mathbf{r}, \mathbf{r}')]^* \rho_s^{\sigma'}(\mathbf{r}, \mathbf{r}') \quad (12.4.13)$$

As results, Eq. (12.4.11) amounts to

$$\begin{aligned} E_X^\sigma[\rho] &= -\frac{1}{2} \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} |\rho_s^\sigma(\mathbf{r}, \mathbf{r}')|^2 \\ U_C^\sigma[\rho] &= -\text{Re} \iint \frac{d\mathbf{r} d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} [\rho_s^\alpha(\mathbf{r}, \mathbf{r}')]^* \rho_s^\beta(\mathbf{r}, \mathbf{r}') \end{aligned} \quad (12.4.14)$$

This is Eqs. (12.4.5). (Q.E.D.)

12.4.2 Spin-polarized Kohn–Sham orbitals

A. The self-consistent field descriptions

- Modern DFT calculations go by the spin-polarized KS orbitals. Each individual $\rho^\sigma(\mathbf{r})$ is $v_{\text{eff}}^\sigma(\mathbf{r})$ -representable, satisfying

$$\rho^\sigma(\mathbf{r}) = \sum_{a=1}^{N_\sigma} |\psi_a^\sigma(\mathbf{r})|^2 \quad \text{and} \quad \rho(\mathbf{r}) = \sum_{\sigma=\alpha,\beta} \rho^\sigma(\mathbf{r}) \quad (12.4.15)$$

As usual, $\langle \psi_j^\sigma | \psi_k^\sigma \rangle = \delta_{jk}$ but $\langle \psi_j^\alpha | \psi_k^\beta \rangle \neq \delta_{jk}$. The spin-polarized orbitals are variational, such that [cf. Eq. (12.4.3)]

$$\frac{\delta U_C[\rho]}{\delta [\psi_a^\sigma(\mathbf{r})]^*} = \frac{\delta U_C[\rho]}{\delta \rho^\sigma(\mathbf{r})} \frac{\partial \rho^\sigma(\mathbf{r})}{\partial [\psi_a^\sigma(\mathbf{r})]^*} = \hat{u}_C^\sigma(\mathbf{r}) \psi_a^\sigma(\mathbf{r}) \quad (12.4.16)$$

On the other hand, Eq. (12.4.11b) gives rise to

$$\langle \psi_a^\sigma(\mathbf{r}) | \hat{u}_C^\sigma(\mathbf{r}) | \psi_a^\sigma(\mathbf{r}) \rangle = - \sum_{b=1}^{N_{\bar{\sigma}}} \langle \psi_a^\sigma \psi_b^{\bar{\sigma}} | | \psi_b^{\bar{\sigma}} \psi_a^\sigma \rangle \quad (12.4.17)$$

Note that as the canonical KS orbital equations in concern, it is sufficient to have the diagonal matrix elements being specified.

- Spin-polarized KS spatial orbitals satisfy [Eq. (10.3.3)]

$$\left[-\frac{1}{2} \nabla^2 + v_{\text{eff}}^\sigma(\mathbf{r}) \right] |\psi_a^\sigma(\mathbf{r})\rangle = \epsilon_a^\sigma |\psi_a^\sigma(\mathbf{r})\rangle; \quad a = 1, \dots, N_\sigma \quad (12.4.18)$$

where

$$v_{\text{eff}}^\sigma(\mathbf{r}) = v_{\text{ext}}^\sigma(\mathbf{r}) + v_{\text{HF}}^\sigma(\mathbf{r}) + v_{\text{non-HF}}^\sigma(\mathbf{r}) \quad (12.4.19)$$

While $v_{\text{HF}}^\sigma(\mathbf{r})$ had been specified as Eq. (12.4.4), the external $v_{\text{ext}}^\sigma(\mathbf{r})$ can have an explicit spin σ -dependence, in the presence of spin-orbit coupling or magnetic field interaction; see Eq. (12.2.12). The non-HF term is

$$v_{\text{non-HF}}^\sigma(\mathbf{r}) = u_C^\sigma(\mathbf{r}) + w_C^\sigma(\mathbf{r}) \quad (12.4.20)$$

with $u_C^\sigma(\mathbf{r})$ being given via Eq. (12.4.17).

- Orbital expression of $w_C^\sigma(\mathbf{r})$ is also needed for the final SCF descriptions. To the end, consider the virial theorem, Eq. (2.3.23) via Eq. (12.4.18),

$$\langle \psi_a^\sigma(\mathbf{r}) | \mathbf{r} \cdot \nabla v_{\text{eff}}^\sigma(\mathbf{r}) | \psi_a^\sigma(\mathbf{r}) \rangle = \langle \nabla \psi_a^\sigma(\mathbf{r}) | \nabla \psi_a^\sigma(\mathbf{r}) \rangle \quad (12.4.21)$$

and recast it as

$$\begin{aligned} \langle \psi_a^\sigma(\mathbf{r}) | \mathbf{r} \cdot \nabla w_C^\sigma(\mathbf{r}) | \psi_a^\sigma(\mathbf{r}) \rangle &= \langle \nabla \psi_a^\sigma(\mathbf{r}) | \nabla \psi_a^\sigma(\mathbf{r}) \rangle \\ &\quad - \langle \psi_a^\sigma(\mathbf{r}) | \mathbf{r} \cdot \nabla \tilde{v}^\sigma(\mathbf{r}) | \psi_a^\sigma(\mathbf{r}) \rangle \end{aligned} \quad (12.4.22)$$

where [cf. Eqs. (12.4.19) and (12.4.20)]

$$\tilde{v}^\sigma(\mathbf{r}) \equiv v_{\text{ext}}^\sigma(\mathbf{r}) + v_{\text{HF}}^\sigma(\mathbf{r}) + u_{\text{C}}^\sigma(\mathbf{r}) \quad (12.4.23)$$

whose orbital expression is known; see Eqs. (12.4.4) and (12.4.17). As results, Eq. (12.4.22) determines $w_{\text{C}}(\mathbf{r})$ up to a constant that is irrelevant to the required KS-SCF orbitals via Eq. (12.4.18). Nevertheless, the boundary value of $w_{\text{C}}(\mathbf{r} \rightarrow \infty) = 0$ is applicable and satisfied via the LCAO-MO implementations.

B. The final formulations

- The LCAO-MOs are [cf. Eq. (10.3.15)]

$$|\psi_a^\sigma(\mathbf{r})\rangle = \sum_{\mu=1}^{\mathcal{N}} C_{\mu a}^\sigma |\varphi_\mu(\mathbf{r})\rangle \quad (12.4.24)$$

We have

$$\langle \psi_a^\sigma(\mathbf{r}) | \psi_b^\sigma(\mathbf{r}) \rangle = \sum_{\mu, \nu} (C_{\mu a}^\sigma)^* C_{\nu b}^\sigma S_{\mu\nu} = \delta_{ab} \quad (12.4.25)$$

- Substitute Eq. (12.4.24) into Eq. (10.3.6), resulting in

$$\sum_{\nu} (h_{\mu\nu} + v_{\mu\nu}^{\sigma;\text{HF}}) C_{\nu j}^\sigma = \epsilon_j^\sigma \sum_{\nu} S_{\mu\nu} C_{\nu j}^\sigma \quad (12.4.26)$$

This is the *Pople-Nesbet equation*, with the details as follows.

(a) While $h_{\mu\nu}$ had been given in Eq. (10.3.13), the Hartree-Fock matrix, Eq. (10.3.4), with Eqs. (10.3.5), (10.3.14) and (12.4.24), reads

$$v_{\mu\nu}^{\sigma;\text{HF}} = \sum_{\kappa, \eta} (P_{\eta\kappa}^\sigma + P_{\eta\kappa}^{\bar{\sigma}}) (\mu\kappa | \nu\eta) - \sum_{\kappa, \eta} P_{\eta\kappa}^\sigma (\mu\kappa | \eta\nu) \quad (12.4.27)$$

The spin-polarized densities are

$$P_{\mu\nu}^\sigma \equiv \langle \varphi_\mu | \hat{\rho}_1^\sigma | \varphi_\nu \rangle \approx \sum_{a=1}^{N_\sigma} \langle \varphi_\mu | \psi_a^\sigma \rangle \langle \psi_a^\sigma | \varphi_\nu \rangle = \sum_{a=1}^{N_\sigma} C_{\mu a}^\sigma (C_{\nu a}^\sigma)^* \quad (12.4.28)$$

(b) The first identity is exact, formally defined via the reduced one-electron density operator, $\hat{\rho}_1^\sigma$, that is in principle a mixed state; see §12.1.1. The second identity exploits the HF theory, which is a pure-state approximation, with

$$\hat{\rho}_1^\sigma(\text{HF}) = \sum_{a=1}^{N_\sigma} |\psi_a^\sigma\rangle \langle \psi_a^\sigma| \quad (12.4.29)$$

resulting from the single-determinantal description. We will revisit the spin-polarized densities, Eq. (12.4.28), via the formally exact $\hat{\rho}_1^\sigma$ in §12.1.